

Optimal Transport for Machine Learners

Compact teaching notes

Gabriel Peyré

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1 Optimal Matching between Point Clouds

1.1 Monge Problem for Discrete Points

Assignment problem. Let $C \in \mathbb{R}^{n \times n}$ be a cost matrix, where $C_{i,j}$ is the cost of pairing source i with target j , and let $\text{Perm}(n)$ denote the set of bijections of $\llbracket n \rrbracket = \{1, \dots, n\}$. The optimal assignment problem is

$$\min_{\sigma \in \text{Perm}(n)} \frac{1}{n} \sum_{i=1}^n C_{i, \sigma(i)}. \quad (1.1)$$

Convex costs on the line.

Proposition 1.1 (Monotone matching on the line). *Assume that $x_1, \dots, x_n$ are pairwise distinct and that $y_1, \dots, y_n$ are pairwise distinct. If $C_{i,j} = h(x_i - y_j)$ for a strictly convex function $h : \mathbb{R} \rightarrow \mathbb{R}$, then the unique optimizer is the order-preserving permutation, characterized by $(x_i - x_{i'})(y_{\sigma(i)} - y_{\sigma(i')}) > 0$ for every $i \neq i'$. In particular, the result applies to $h(s) = |s|^p$ for $p > 1$.*

Proof. A permutation that does not preserve order contains an inversion: after relabeling, $x < x'$ are matched to $y' > y$. Set $d = y' - y > 0$ and $D(s) = \frac{h(s) - h(s-d)}{d}$. Strict convexity of h makes D strictly increasing. Therefore

$$\begin{aligned} h(x - y') + h(x' - y) - h(x - y) - h(x' - y') \\ = d(D(x' - y) - D(x - y)) > 0. \end{aligned}$$

Swapping the two inverted targets strictly lowers the cost. Repeating this exchange eliminates every inversion; the only order-preserving bijection between the two sorted clouds pairs equal ranks. $\square$

If h is convex but not strictly convex, the same exchange inequality is non-strict. The equal-rank matching remains optimal, but other, non-monotone optimizers can coexist. To construct a monotone optimizer, choose sorting permutations σ_X, σ_Y such that $x_{\sigma_X(1)} \leq x_{\sigma_X(2)} \leq \dots$ and $y_{\sigma_Y(1)} \leq y_{\sigma_Y(2)} \leq \dots$ and match $x_{\sigma_X(k)}$ with $y_{\sigma_Y(k)}$. Thus $\sigma = \sigma_Y \circ \sigma_X^{-1}$ is optimal.

Concave costs on the line. $p_1 < q_1 < p_2 < q_2 < \dots < p_N < q_N$, with the opposite orientation handled by exchanging the roles of p and q , Delon, Salomon and Sobolevski define the local indicators

$$\begin{aligned} I_k^p(i) &= c(p_i, q_{i+k}) + \sum_{r=0}^{k-1} c(p_{i+r+1}, q_{i+r}) - \sum_{r=0}^k c(p_{i+r}, q_{i+r}), \\ I_k^q(i) &= c(p_{i+k+1}, q_i) + \sum_{r=1}^k c(p_{i+r}, q_{i+r}) - \sum_{r=0}^k c(p_{i+r+1}, q_{i+r}), \end{aligned}$$

For arbitrary real masses, the stratification argument in the same work gives a larger $O(n^3)$ worst-case bound. Very concave costs also motivate the simpler greedy rule that repeatedly matches the closest red-blue pair. These methods are not generic linear-programming solvers: they exploit the order and concavity specific to the line.

Histogram equalization.

Flat directions for the linear cost.

Optimal transport on the circle. $d_{\mathbb{S}^1}(x, y) := \min_{k \in \mathbb{Z}} |x - y + k|$, $c_p(x, y) := d_{\mathbb{S}^1}(x, y)^p$, $p > 1$.

Proposition 1.2 (Discrete circle transport by a cut). *Assume that the $2n$ points $x_1, \dots, x_n, y_1, \dots, y_n$ are pairwise distinct. Let $x_{(1)}, \dots, x_{(n)}$ and $y_{(1)}, \dots, y_{(n)}$ be fixed cyclic orderings, with indices understood modulo n . For $p > 1$, every optimal assignment for the cost c_p is a cyclic shift $x_{(k)} \mapsto y_{(k+s)}$, $k \in \llbracket n \rrbracket$, $s \in \{0, \dots, n-1\}$.*

$$E_s := \frac{1}{n} \sum_{k=1}^n d_{\mathbb{S}^1}(x_{(k)}, y_{(k+s)})^p$$

over the n shifts. For each optimal assignment there is a cut $\theta \in \mathbb{S}^1 \setminus (\{x_i\}_i \cup \{y_j\}_j)$, crossed by none of its shortest transport arcs, such that the lifted assignment on $(\theta, \theta + 1)$ is the equal-rank matching on the line.

Proof. Fix an optimal assignment σ and let γ_i be an open shortest arc from x_i to $y_{\sigma(i)}$, with either orientation fixed in the antipodal case. The path-uncrossing lemma of Delon, Rabin and Gousseau states that if two arcs of an optimal assignment intersect, then they have the same orientation and neither is strictly contained in the other. Its proof is a two-edge exchange: opposite orientations or strict containment would, by monotonicity and strict convexity of $r \mapsto r^p$, produce a strictly cheaper pairing of the four endpoints.

Suppose that the arcs cover the circle. Since each γ_i is open, every source point lies in another transport arc. Order the sources cyclically. The path-uncrossing lemma then propagates one common orientation: either $x_{(k+1)} \in \gamma_{(k)}$ for every k , or $x_{(k-1)} \in \gamma_{(k)}$ for every k . In the first case, define a new assignment by giving $x_{(k+1)}$ the target previously assigned to $x_{(k)}$; every transport arc becomes strictly shorter. The opposite orientation gives the analogous backward reassignment. Both contradict optimality. Hence some θ lies outside every transport arc.

Cut the circle at such a θ and lift it to $(\theta, \theta + 1)$. Every matched geodesic avoids the cut, so its circular length equals the ordinary distance between the corresponding lifts. Proposition 1.1 now implies that the lifted matching preserves order. Returning to the circle, an order-preserving bijection between two cyclically ordered n -point sets is a cyclic shift. Minimizing over these n shifts therefore recovers the optimum. $\square$

Two-dimensional geometry.

Proposition 1.3 (No proper crossings in Euclidean-distance matchings). *Let $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R}^2$ and let $c(x, y) = \|x - y\|$. If σ is optimal, then two matched segments cannot cross properly: their relative interiors cannot meet at a point where their supporting lines are distinct.*

Proof. Suppose that $[x_i, y_{\sigma(i)}]$ and $[x_j, y_{\sigma(j)}]$ cross properly at z . The triangle inequality along the two reconnected paths gives $\|x_i - y_{\sigma(j)}\| \leq \|x_i - z\| + \|z - y_{\sigma(j)}\|$, and $\|x_j - y_{\sigma(i)}\| \leq \|x_j - z\| + \|z - y_{\sigma(i)}\|$. The sum of the two right-hand sides is exactly the cost of the two original segments. At least one inequality is strict because a proper crossing is non-collinear. Swapping $\sigma(i)$ and $\sigma(j)$ therefore strictly decreases the assignment cost, contradicting optimality. Collinear overlaps are excluded from the statement precisely because the exchange can then have equal cost. $\square$

$$C_n = \frac{1}{n+1} \binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n^{3/2}}}.$$

1.2 Hungarian Algorithm

We now present a first algorithm for solving the optimal matching problem. The Hungarian method gives a self-contained preview of duality by computing an assignment together with an optimality certificate. The auction algorithm is postponed to Section 5.2, after the discrete semi-dual has been introduced.

Hungarian primal-dual method. The Hungarian method constructs a partial matching and a dual certificate together. The certificate can be isolated as follows.

Proposition 1.4 (Dual certificate for an assignment). *If $f_i + g_j \leq C_{i,j}$ for all i, j , then every $\tau \in \text{Perm}(n)$ satisfies $\sum_i C_{i,\tau(i)} \geq \sum_i f_i + \sum_j g_j$. If equality holds on every edge $(i, \sigma(i))$ of a permutation σ , then σ is optimal and (f, g) is dual optimal.*

Proof. Sum $C_{i,\tau(i)} \geq f_i + g_{\tau(i)}$ over i and use that τ permutes the target indices. Tightness along σ makes this lower bound equal to its assignment cost. $\square$

This is the uniform-mass instance of the discrete dual in Section 4.1. The equality graph contains the zero-slack edges (i, j) satisfying $f_i + g_j = C_{i,j}$. By Proposition 1.4, a perfect matching in this graph is optimal. Starting at an unmatched source, an alternating tree follows equality edges to targets and matched edges back to sources. Reaching an unmatched target gives an augmenting path. If no equality edge leaves the reached source set S toward T^c , use

$$\delta = \min_{i \in S, j \notin T} (C_{i,j} - f_i - g_j), \quad f_i \leftarrow f_i + \delta \ (i \in S), \quad g_j \leftarrow g_j - \delta \ (j \in T),$$

which preserves feasibility and creates a new equality edge. Since $|S| = |T| + 1$, this shift increases the dual objective by δ . Section 5.2 contrasts this exact augmenting-tree mechanism with the ε -relaxed auction ascent.

Proposition 1.5 (Correctness and complexity of the Hungarian primal-dual method). *For every $C \in \mathbb{R}^{n \times n}$, the Hungarian method terminates after n augmentation phases and returns an optimal permutation σ . With maintained slacks, it uses $O(n^3)$ arithmetic and comparison operations and $O(n^2)$ storage.*

Proof. Initialize $f_i = \min_j C_{i,j}$ and $g_j = 0$. Feasibility holds and the empty matching uses only equality edges. Before reaching a free target, the tree has $|S| = |T| + 1$, so T omits a target. A dual shift leaves slacks unchanged on $S \times T$, decreases them by δ on $S \times T^c$, and increases them on $S^c \times T$. Hence it preserves feasibility and tight matched edges while creating a new equality edge. The tree therefore reaches a free target in at most n expansions; flipping the resulting alternating path adds one matched pair. Exactly n phases produce a perfect matching σ . Its edges are tight, so Proposition 1.4 proves optimality. Maintaining $\ell_j = \min_{i \in S} (C_{i,j} - f_i - g_j)$ and a minimizing parent costs $O(n)$ per tree expansion: when a source enters S , one compares its row of slacks with the current ℓ_j . Hence one phase costs $O(n^2)$ and all n phases cost $O(n^3)$. $\square$

2 Monge Problem between Measures

2.1 Measures

Histograms.

Definition 2.1 (Probability simplex). The probability simplex of length n is $\Sigma_n := \{a \in \mathbb{R}_+^n ; \sum_{i=1}^n a_i = 1\}$. Its elements are also called probability vectors or histograms.

Discrete measure, empirical measure.

Definition 2.2 (Discrete measure). A discrete measure with weights a and locations $x_1, \dots, x_n \in \mathcal{X}$ is

$$\alpha = \sum_{i=1}^n a_i \delta_{x_i}, \tag{2.1}$$

where δ_x is the Dirac mass at position x . It is a probability measure if $a \in \Sigma_n$, and a positive measure if all weights a_i are nonnegative.

General measures. A Dirac measure δ_x is then defined as $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise, and this extends by linearity for discrete measures of the form (2.1) as

$\alpha(A) = \sum_{x_i \in A} a_i$. We denote by $\mathcal{M}_+(\mathcal{X})$ the cone of positive finite measures on $\mathcal{X}$, characterized by $\alpha(A) \geq 0$ for every Borel set A . The probability measures form the subset $\mathcal{M}_+^1(\mathcal{X}) = \{\alpha \in \mathcal{M}_+(\mathcal{X}) ; \alpha(\mathcal{X}) = 1\}$.

Polish metric spaces.

Definition 2.3 (Polish metric space). A metric space $(\mathcal{X}, d)$ is Polish if it is complete and separable: every Cauchy sequence converges in $\mathcal{X}$, and $\mathcal{X}$ contains a countable dense subset. More generally, a topological space is called Polish if its topology can be induced by some complete separable metric.

$$\mathcal{S} = C([0, 1]; \mathbb{R}^d), \quad d_\infty(\gamma, \eta) := \sup_{t \in [0, 1]} \|\gamma(t) - \eta(t)\|$$

Definition 2.4 (Support of a measure). Let α be a positive Borel measure on a metric space $\mathcal{X}$. Its support is $\text{supp}(\alpha) := \{x \in \mathcal{X} ; \alpha(B(x, r)) > 0 \text{ for every } r > 0\}$. If $\mathcal{X}$ is separable, this is the smallest closed set of full α -mass.

Radon measures. $\langle f, \alpha \rangle := \int f(x) d\alpha(x)$. $\int_{\mathcal{X}} f(x) d\alpha(x) = \sum_{i=1}^n a_i f(x_i)$.

Relative densities.

Definition 2.5 (Relative density). If α is absolutely continuous with respect to a positive reference measure λ , its relative density is the Radon–Nikodym derivative $\rho_\alpha := \frac{d\alpha}{d\lambda}$, $d\alpha(x) = \rho_\alpha(x) d\lambda(x)$. Equivalently, for every nonnegative measurable function h , and hence for every integrable h , $\int_{\mathcal{X}} h(x) d\alpha(x) = \int_{\mathcal{X}} h(x) \rho_\alpha(x) d\lambda(x)$.

Total variation norm.

Definition 2.6 (Total variation). For a finite signed Radon measure α on a compact space $\mathcal{X}$, $\|\alpha\|_{\text{TV}} := \sup_{f \in \mathcal{C}(\mathcal{X})} \{\langle f, \alpha \rangle ; \|f\|_\infty \leq 1\}$.

$|\alpha|(A) := \sup_{A=\cup_i B_i} \sum_i |\alpha(B_i)|$, where the supremum is over finite or countable measurable partitions of A . Thus, if $\alpha = \sum_i a_i \delta_{x_i}$ with distinct atoms, $|\alpha| = \sum_i |a_i| \delta_{x_i}$; if $d\alpha(x) = \rho(x) d\lambda(x)$, then $d|\alpha|(x) = |\rho(x)| d\lambda(x)$.

Proposition 2.7 (Dual and measure definitions of total variation). For a finite signed Radon measure α on a compact space $\mathcal{X}$, $\|\alpha\|_{\text{TV}} = |\alpha|(\mathcal{X})$.

Proof. The inequality $\|\alpha\|_{\text{TV}} \leq |\alpha|(\mathcal{X})$ follows from $|\int f d\alpha| \leq \int |f| d|\alpha| \leq \|f\|_\infty |\alpha|(\mathcal{X})$. For the reverse inequality, write the Jordan decomposition $\alpha = \alpha^+ - \alpha^-$, so that $|\alpha| = \alpha^+ + \alpha^-$. The measurable sign $s = \frac{d\alpha}{d|\alpha|}$ takes values in $\{-1, 1\}$ outside a $|\alpha|$ -null set and satisfies $d\alpha = s d|\alpha|$. By regularity of Radon measures on compact spaces, s can be approximated in $L^1(|\alpha|)$ by continuous functions f_k with $\|f_k\|_\infty \leq 1$. Hence $\int f_k d\alpha \rightarrow \int s d\alpha = |\alpha|(\mathcal{X})$, which proves the equality. The final statement is the Riesz–Markov–Kakutani representation theorem with this norm. $\square$

For two absolutely continuous measures $d\alpha = \rho_\alpha d\lambda$ and $d\beta = \rho_\beta d\lambda$, this gives the concrete formula

$\|\alpha - \beta\|_{\text{TV}} = \int_{\mathcal{X}} |\rho_\alpha(x) - \rho_\beta(x)| d\lambda(x)$. $\alpha = \sum_{i=1}^n \tilde{a}_i \delta_{x_i}$, $\beta = \sum_{j=1}^m \tilde{b}_j \delta_{y_j}$, $a_k := \sum_{i: x_i = z_k} \tilde{a}_i$, $b_k := \sum_{j: y_j = z_k} \tilde{b}_j$. Then $\alpha = \sum_{k=1}^r a_k \delta_{z_k}$ and $\beta = \sum_{k=1}^r b_k \delta_{z_k}$. This convention merges masses located at the same point before comparing the two measures.

$\|\alpha - \beta\|_{\text{TV}} = \sum_{k=1}^r |a_k - b_k|$.

Probabilistic interpretation. Probability measures represent the laws of random variables. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be an abstract probability space and let $\mathcal{X}$ be Polish.

$\alpha(A) = \mathbb{P}(\{\omega \in \Omega ; X(\omega) \in A\})$ for Borel sets $A \subset \mathcal{X}$. For every integrable function f , integration with respect to this law is expectation:

$\int_{\mathcal{X}} f(x) d\alpha(x) = \mathbb{E}[f(X)]$.

2.2 Push-Forward

For a measurable map $T : \mathcal{X} \rightarrow \mathcal{Y}$, the push-forward operator $T_\# : \mathcal{M}(\mathcal{X}) \rightarrow \mathcal{M}(\mathcal{Y})$ records the distribution of the image points. For discrete measures (2.1), the push-forward operation consists simply in moving the positions of all the points in the support of the measure

$T_\# \alpha := \sum_i a_i \delta_{T(x_i)}$.

Definition 2.8 (Push-forward). For a measurable map $T : \mathcal{X} \rightarrow \mathcal{Y}$ and $\alpha \in \mathcal{M}(\mathcal{X})$, the push-forward measure $\beta = T_\# \alpha \in \mathcal{M}(\mathcal{Y})$ is defined, for every measurable set $B \subset \mathcal{Y}$, by

$$\beta(B) = \alpha(\{x \in \mathcal{X} ; T(x) \in B\}). \quad (2.2)$$

Equivalently, for every bounded measurable function $h : \mathcal{Y} \rightarrow \mathbb{R}$,

$$\int_{\mathcal{Y}} h(y) d\beta(y) = \int_{\mathcal{X}} h(T(x)) d\alpha(x). \quad (2.3)$$

Note that $T_\#$ preserves positivity and total mass, so that if $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ then $T_\# \alpha \in \mathcal{M}_+^1(\mathcal{Y})$.

Proposition 2.9 (Push-forward formula for densities). Let $U, V \subset \mathbb{R}^d$ be open, let α and β have Lebesgue densities ρ_α and ρ_β on U and V , and let $T : U \rightarrow V$ be a C^1 diffeomorphism such that $\beta = T_\# \alpha$. Then, for Lebesgue-almost every $x \in U$,

$$\rho_\alpha(x) = |\det(T'(x))| \rho_\beta(T(x)). \quad (2.4)$$

Equivalently, for Lebesgue-almost every $y \in V$, $\rho_\beta(y) = \rho_\alpha(T^{-1}y) \frac{1}{|\det(T'(T^{-1}y))|}$.

Proof. Apply (2.3) to an arbitrary compactly supported continuous test function h and use the change of variables $y = T(x)$, for which $dy = |\det(T'(x))| dx$.

$$\begin{aligned} \int_V h(y) \rho_\beta(y) dy &= \int_V h(y) d\beta(y) = \int_U h(T(x)) d\alpha(x) = \int_U h(T(x)) \rho_\alpha(x) dx \\ &= \int_V h(y) \rho_\alpha(T^{-1}y) \frac{dy}{|\det(T'(T^{-1}y))|}, \end{aligned}$$

Since the identity holds for every such h , uniqueness of Lebesgue densities gives $\rho_\beta(y) = \rho_\alpha(T^{-1}y) \frac{1}{|\det(T'(T^{-1}y))|}$. The equivalent source-variable formula follows by setting $y = T(x)$: $\rho_\alpha(x) = |\det(T'(x))| \rho_\beta(T(x))$ where $T'(x) \in \mathbb{R}^{d \times d}$ is the Jacobian matrix of T . $\square$

2.3 Monge's Formulation

Monge problem. Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, $\beta \in \mathcal{M}_+^1(\mathcal{Y})$ and a nonnegative measurable cost $c : \mathcal{X} \times \mathcal{Y} \rightarrow [0, +\infty]$, the Monge problem is

$$\mathcal{M}_c(\alpha, \beta) := \inf_{T: \mathcal{X} \rightarrow \mathcal{Y} \text{ measurable}} \left\{ \int_{\mathcal{X}} c(x, T(x)) d\alpha(x) ; T_{\#}\alpha = \beta \right\}. \quad (2.5)$$

Proposition 2.10 (Empirical Monge maps and matchings). *Assume that the source atoms $x_1, \dots, x_n$ are distinct and that $\alpha = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$, $\beta = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$. If $T_{\#}\alpha = \beta$, then for each distinct target value z in the support of β , exactly $n\beta(\{z\})$ source atoms are mapped to z . In particular, if the y_j are distinct, then there exists a permutation $\sigma \in \text{Perm}(n)$ such that $T(x_i) = y_{\sigma(i)}$ for all i . Conversely, every such assignment of source atoms to target atoms with the correct masses defines a feasible Monge map on the support of α , and in the distinct-target case $\int_{\mathcal{X}} c(x, T(x)) d\alpha(x) = \frac{1}{n} \sum_{i=1}^n c(x_i, y_{\sigma(i)})$. The values of a feasible map away from $\{x_1, \dots, x_n\}$ may be chosen arbitrarily, provided the resulting extension is measurable. If source locations are repeated, they should first be merged into atoms with larger masses; such atoms cannot be split by a Monge map.*

Proof. Since $T_{\#}\alpha = \beta$, all images $T(x_i)$ must belong to the support of β ; otherwise the push-forward would give positive mass to a point outside that support. For any target atom z , $\beta(\{z\}) = \alpha(T^{-1}(\{z\})) = \frac{1}{n} \# \{i ; T(x_i) = z\}$. This proves the counting statement. If all target atoms have mass $1/n$, each target receives exactly one source atom, which is a permutation. The converse and the cost identity follow by direct substitution. $\square$

Proposition 2.11 (Existence of transport maps from atomless sources). *Let α and β be Borel probability measures on Polish spaces, and assume that α is atomless. Then there exists a measurable map T such that $T_{\#}\alpha = \beta$.*

Proof. A standard measure-isomorphism theorem identifies the atomless standard probability space $(\mathcal{X}, \alpha)$ with $([0, 1], \text{Leb})$, modulo null sets. It is therefore enough to construct a map from $[0, 1]$ to the target law β . Since the target is Polish, it is a standard Borel space; choose a Borel isomorphism i from a full- β Borel subset of $\mathcal{Y}$ onto a Borel subset of $[0, 1]$. The generalized quantile of $i_{\#}\beta$ pushes Lebesgue measure to $i_{\#}\beta$. Composing it with i^{-1} and with the source isomorphism gives the desired measurable map, after arbitrary definitions on the discarded null sets. $\square$

Monge distance. When $\mathcal{X} = \mathcal{Y}$, $1 \leq p < +\infty$ and $c(x, y) = d(x, y)^p$ for a metric d , set $\mathcal{E}_\alpha(T) := \int_{\mathcal{X}} d(x, T(x))^p d\alpha(x)$.

$$\tilde{\mathcal{W}}_p(\alpha, \beta)^p := \inf_{T: \mathcal{X} \rightarrow \mathcal{X} \text{ measurable}} \{ \mathcal{E}_\alpha(T) ; T_{\#}\alpha = \beta \}. \quad (2.6)$$

If the constraint set is empty, then $\tilde{\mathcal{W}}_p(\alpha, \beta) = +\infty$.

Proposition 2.12 (Directed Monge distance). *Let $\mathcal{X} = \mathcal{Y}$ be a Polish metric space and $1 \leq p < +\infty$. On probability measures with finite p -th moment, the quantity $\tilde{\mathcal{W}}_p$ is nonnegative, vanishes only on the diagonal, and satisfies the triangle inequality. It is therefore a directed extended distance: it need not be symmetric and may take the value $+\infty$.*

Proof. Nonnegativity is immediate. If $\tilde{\mathcal{W}}_p(\alpha, \beta) = 0$, choose feasible maps T_k with $\int d(x, T_k(x))^p d\alpha(x) \rightarrow 0$. For every bounded 1-Lipschitz function h ,

$$\left| \int h d\beta - \int h d\alpha \right| = \left| \int h(T_k(x)) - h(x) d\alpha(x) \right| \leq \left(\int d(x, T_k(x))^p d\alpha(x) \right)^{1/p} \rightarrow 0.$$

Since bounded Lipschitz functions separate probability measures on Polish metric spaces, $\alpha = \beta$. Conversely, the identity map shows that $\tilde{\mathcal{W}}_p(\alpha, \alpha) = 0$.

If either $\tilde{\mathcal{W}}_p(\alpha, \gamma)$ or $\tilde{\mathcal{W}}_p(\gamma, \beta)$ is infinite, the claim is automatic. If both are finite, feasible maps $S_{\#}\alpha = \gamma$ and $T_{\#}\gamma = \beta$ exist, so their composition is feasible from α to β ; in particular, the left-hand side is finite. Fix $\varepsilon > 0$ and choose ε -minimizers satisfying $\mathcal{E}_\alpha(S)^{1/p} \leq \tilde{\mathcal{W}}_p(\alpha, \gamma) + \varepsilon$, $\mathcal{E}_\gamma(T)^{1/p} \leq \tilde{\mathcal{W}}_p(\gamma, \beta) + \varepsilon$. The composed map is feasible from α to β , and Minkowski's inequality gives

$$\begin{aligned} \tilde{\mathcal{W}}_p(\alpha, \beta) &\leq \left(\int d(x, T(S(x)))^p d\alpha(x) \right)^{1/p} \\ &\leq \left(\int d(x, S(x))^p d\alpha(x) \right)^{1/p} + \left(\int d(S(x), T(S(x)))^p d\alpha(x) \right)^{1/p} \\ &= \mathcal{E}_\alpha(S)^{1/p} + \mathcal{E}_\gamma(T)^{1/p} \leq \tilde{\mathcal{W}}_p(\alpha, \gamma) + \tilde{\mathcal{W}}_p(\gamma, \beta) + 2\varepsilon. \end{aligned}$$

Letting $\varepsilon \rightarrow 0$ gives the result. $\square$

2.4 Existence and Uniqueness of the Monge Map

Brenier's theorem.

Theorem 2.13 (Brenier). *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ have finite second moments, and assume that α is absolutely continuous with respect to Lebesgue measure. For the quadratic cost $c(x, y) = \|x - y\|^2$, there exists a convex function $\varphi : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ such that $T = \nabla \varphi$, $T_{\#}\alpha = \beta$, and T is the unique optimal Monge map α -almost everywhere. The optimal Kantorovich plan is $(\text{Id}, T)_{\#}\alpha$.*

Proof. Choose optimal c -concave potentials (f, g) for the cost $\|x - y\|^2$ and define $\varphi(x) := \frac{1}{2}\|x\|^2 - \frac{1}{2}f(x)$, $\psi(y) := \frac{1}{2}\|y\|^2 - \frac{1}{2}g(y)$. The c -concavity of f makes φ convex. Dual feasibility is equivalent to $\varphi(x) + \psi(y) \geq \langle x, y \rangle$, and equality holds for almost every pair under any optimal plan. Replacing ψ by the convex conjugate φ^* therefore yields the Fenchel equality $\varphi(x) + \varphi^*(y) = \langle x, y \rangle$ on the optimal relation, hence $y \in \partial\varphi(x)$. Since α has a density and convex functions are differentiable Lebesgue-almost everywhere, $\partial\varphi(x) = \{\nabla\varphi(x)\}$ for α -almost every x . Every optimal plan is therefore concentrated on the same graph, which proves existence and uniqueness of both the optimal plan and the Monge map. $\square$

$$\langle \nabla\varphi(x) - \nabla\varphi(x'), x - x' \rangle \geq 0.$$

Definition 2.14 (Measures not charging hypersurfaces). A Borel measure α on $\mathbb{R}^d$ does not charge hypersurfaces if $\alpha(S) = 0$ for every countably $(d-1)$ -rectifiable set S , i.e. every set that can be covered, up to an $\mathcal{H}^{d-1}$ -null set, by countably many C^1 hypersurfaces.

Radial measures. Radial symmetry gives a useful higher-dimensional case where the Brenier map reduces to a one-dimensional monotone rearrangement. A measure α on $\mathbb{R}^d$ is radial if $Q_\# \alpha = \alpha$ for every orthogonal map Q .

Proposition 2.15 (Optimal transport between radial measures). *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ be radial probability measures with finite second moments, and assume that α is absolutely continuous with respect to Lebesgue measure. Define their radial distribution functions $F_\alpha(r) := \alpha(\{x : \|x\| \leq r\})$, $F_\beta(r) := \beta(\{y : \|y\| \leq r\})$, $r \geq 0$, and let $F_\beta^{-1}(u) := \inf\{r \geq 0 : F_\beta(r) \geq u\}$ be the generalized inverse. Set $\tau(r) := F_\beta^{-1}(F_\alpha(r))$. Then the quadratic Brenier map from α to β is the radial map $T(0) = 0$, $T(x) = \frac{\tau(\|x\|)}{\|x\|}x$ for $x \neq 0$. Moreover, if $\alpha_R = (x \mapsto \|x\|)_\# \alpha$ is the law of the source radius, then $\mathcal{W}_2^2(\alpha, \beta) = \int_0^\infty |r - \tau(r)|^2 d\alpha_R(r)$.*

Proof. Let $\alpha_R = (\|\cdot\|)_\# \alpha$ and $\beta_R = (\|\cdot\|)_\# \beta$ be the radial laws. Since α is absolutely continuous, α_R has no atoms. The map $\tau = F_\beta^{-1} \circ F_\alpha$ is therefore the monotone one-dimensional transport from α_R to β_R . If $X \sim \alpha$ and $R = \|X\|$, then $\tau(R)$ has law β_R . Moreover, by radiality, the conditional direction $X/\|X\|$ given $R = r$ is uniform on the sphere for α_R -a.e. $r > 0$. Thus T changes only the radius, from R to $\tau(R)$, and keeps the uniform angular distribution. It follows that $T_\# \alpha$ is radial with radial law β_R , hence $T_\# \alpha = \beta$.

The function τ is nondecreasing and nonnegative. Define $\psi(r) := \int_0^r \tau(s) ds$ on $\mathbb{R}_+$. Then ψ is convex and nondecreasing, so $\varphi(x) := \psi(\|x\|)$ is convex on $\mathbb{R}^d$. Since α is absolutely continuous, it does not charge the origin nor the radii where the monotone function τ is discontinuous. Thus $\nabla \varphi(x) = \frac{\tau(\|x\|)}{\|x\|}x = T(x)$ for α -a.e. x . The map T is therefore a gradient of a convex potential and pushes α to β . Brenier's theorem identifies it as the unique quadratic optimal Monge map. The displayed cost formula follows by substituting $T(x)$ in $\int \|x - T(x)\|^2 d\alpha(x)$. $\square$

If α is not absolutely continuous, the same radial idea is still useful but has to be interpreted at the level of couplings of the radial variables: atoms on spheres may have to be split, so a Monge map need not exist.

Polar factorization. Brenier's theorem does more than solve one transport problem: it provides a canonical way to extract the “monotone part” of a nondegenerate map. Suppose one starts from a square-integrable deformation $u : \Omega \rightarrow \mathbb{R}^d$, for instance a velocity snapshot, an image deformation or a generative map.

Proposition 2.16 (Polar factorization). *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain endowed with normalized Lebesgue measure λ , and let $u \in L^2(\Omega; \mathbb{R}^d)$. Assume that the image law $\beta = u_\# \lambda$ is absolutely continuous with respect to Lebesgue measure. Then there exist a measure-preserving map $s : \Omega \rightarrow \Omega$ and a convex function φ such that $u = \nabla \varphi \circ s$ λ -a.e. The Brenier factor $\nabla \varphi$ and the rearrangement s are unique λ -almost everywhere.*

Proof. By Brenier's theorem there is a unique convex-gradient map $T = \nabla \varphi$ transporting λ to β . Since β is also absolutely continuous, the reverse Brenier map S transporting β to λ exists, and the two maps are inverse almost everywhere: $T \circ S = \text{Id}$ β -almost everywhere and $S \circ T = \text{Id}$ λ -almost everywhere. Set $s := S \circ u$. Then $s_\# \lambda = S_\#(u_\# \lambda) = S_\# \beta = \lambda$, $T \circ s = T \circ S \circ u = u$ λ -a.e. Thus s is measure preserving and gives the announced factorization. Uniqueness of T follows from Brenier's theorem, and then $s = S \circ u$ is unique almost everywhere. $\square$

The name is meant to echo the usual polar decomposition of matrices. This analogy becomes literal for linear maps under the Gaussian reference measure.

$A = SO$, where $O = S^{-1}A$ is orthogonal when A is invertible. If A is singular, $S^\dagger A$ is the canonical partial isometry on $(\ker A)^\perp$; because A is square, it can be extended arbitrarily to an orthogonal matrix O on all of $\mathbb{R}^d$ while preserving $A = SO$.

Displacement interpolation.

Definition 2.17 (Monge and McCann displacement interpolation). If $T_\# \alpha = \beta$, the Monge interpolation generated by T is the curve $T_t(x) := (1-t)x + tT(x)$, $\alpha_t := (T_t)_\# \alpha$, $t \in [0, 1]$. For the quadratic cost, when T is the optimal Brenier map, this curve is called McCann's displacement interpolation.

Proposition 2.18 (Directed Monge displacement geodesics). *Let $1 \leq p < +\infty$, let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ have finite p -th moments, and let T be an optimal map for $\tilde{\mathcal{W}}_p(\alpha, \beta)$ in (2.6). Set $\alpha_t = (T_t)_\# \alpha$ with $T_t = (1-t)\text{Id} + tT$. Assume that, for every $t < 1$, T_t is one-to-one on a full α -measure Borel set. Its inverse on the image of that set is then measurable and defined α_t -almost everywhere. For $0 \leq s \leq t \leq 1$, $\tilde{\mathcal{W}}_p(\alpha_s, \alpha_t) = (t-s)\tilde{\mathcal{W}}_p(\alpha, \beta)$. Thus $t \mapsto \alpha_t$ is an oriented constant-speed geodesic for the directed Monge distance. In particular, for $p = 2$, this applies to the Brenier map under the hypotheses of Theorem 2.13.*

Proof. The case $s = t$ is trivial, so assume $s < t$. Since $s < 1$, the inverse T_s^{-1} is defined α_s -almost everywhere along the transported particles. Define $S_{s,t} := T_t \circ T_s^{-1}$ α_s -a.e. Then $(S_{s,t})_\# \alpha_s = \alpha_t$, and, using the optimality of T ,

$$\begin{aligned} \tilde{\mathcal{W}}_p(\alpha_s, \alpha_t)^p &\leq \int \|S_{s,t}(z) - z\|^p d\alpha_s(z) \\ &= \int \|T_t(x) - T_s(x)\|^p d\alpha(x) = (t-s)^p \tilde{\mathcal{W}}_p(\alpha, \beta)^p. \end{aligned}$$

The same particle construction gives $\tilde{\mathcal{W}}_p(\alpha, \alpha_s) \leq s\tilde{\mathcal{W}}_p(\alpha, \beta)$. If $t < 1$, the map $T \circ T_t^{-1}$ sends α_t to β and gives $\tilde{\mathcal{W}}_p(\alpha_t, \beta) \leq (1-t)\tilde{\mathcal{W}}_p(\alpha, \beta)$; if $t = 1$, this latter distance is zero. Using the triangle inequality from Proposition 2.12,

$$\tilde{\mathcal{W}}_p(\alpha, \beta) \leq \tilde{\mathcal{W}}_p(\alpha, \alpha_s) + \tilde{\mathcal{W}}_p(\alpha_s, \alpha_t) + \tilde{\mathcal{W}}_p(\alpha_t, \beta) \leq s\tilde{\mathcal{W}}_p(\alpha, \beta) + \tilde{\mathcal{W}}_p(\alpha_s, \alpha_t) + (1-t)\tilde{\mathcal{W}}_p(\alpha, \beta).$$

Hence $\tilde{\mathcal{W}}_p(\alpha_s, \alpha_t) \geq (t-s)\tilde{\mathcal{W}}_p(\alpha, \beta)$, which proves equality. For a Brenier map $T = \nabla \varphi$, the map T_t is the gradient of $x \mapsto (1-t)\frac{\|x\|^2}{2} + t\varphi(x)$, which is $(1-t)$ -strongly convex for every $t < 1$. On the full-measure set where φ is differentiable, this gives $\langle T_t(x) - T_t(y), x - y \rangle \geq (1-t)\|x - y\|^2$, so T_t is injective there. $\square$

Regularity and the Monge–Ampère equation.

Proposition 2.19 (Caffarelli regularity). *Let $\Omega, \Lambda \subset \mathbb{R}^d$ be bounded uniformly convex domains with C^2 boundaries. Let $\alpha = \rho(x)dx$ and $\beta = \eta(y)dy$ be supported on Ω and Λ , respectively, with $0 < m \leq \rho, \eta \leq M < +\infty$. If $\rho \in C^\alpha(\bar{\Omega})$ and $\eta \in C^\alpha(\bar{\Lambda})$ for some $\alpha \in (0, 1)$, then the Brenier potential φ is strictly convex in Ω and belongs to $C_{\text{loc}}^{2,\alpha}(\Omega)$. Under the standard compatibility and higher boundary-regularity assumptions, the same regularity extends to the boundary.*

Proof. The potential solves the Monge–Ampère equation $\det(\nabla^2 \varphi(x)) = \frac{\rho(x)}{\eta(\nabla \varphi(x))}$ in the Alexandrov sense, with second boundary condition $\nabla \varphi(\Omega) = \Lambda$. The density bounds and convexity of the domains give strict convexity and localization of sections. Caffarelli’s interior theory then yields $C_{\text{loc}}^{2,\alpha}$ estimates for φ ; the boundary statement follows from the boundary regularity theory under uniform convexity and compatibility assumptions. $\square$

For sufficiently smooth positive densities and a classical Brenier map, the change-of-variables formula (2.4) gives the Monge–Ampère equation

$$\det(\nabla^2 \varphi(x)) \rho_\beta(\nabla \varphi(x)) = \rho_\alpha(x). \quad (2.7)$$

Proposition 2.20 (Linearization of the Monge–Ampère equation). *Let $\rho_\varepsilon = \rho_0 + \varepsilon r + o(\varepsilon)$ be a smooth perturbation of a positive reference density ρ_0 on a smooth bounded domain Ω , with $\int_\Omega r dx = 0$. Assume that the expansions hold in a topology strong enough to differentiate the change-of-variables identity. If $T_\varepsilon(x) = x + \varepsilon \nabla u(x) + o(\varepsilon)$ transports $\rho_0 dx$ to $\rho_\varepsilon dx$, then, to first order, $-\nabla \cdot (\rho_0 \nabla u) = r$. In particular, when ρ_0 is constant, the linearized equation is $-\Delta u = r/\rho_0$. If the source and target occupy the same fixed domain and no mass crosses its boundary, the accompanying first-order boundary condition is $\rho_0 \partial_n u = 0$ on $\partial\Omega$; together with $\int_\Omega u dx = 0$, it fixes the additive constant.*

Proof. The change-of-variables equation for T_ε is $\rho_0(x) = \rho_\varepsilon(T_\varepsilon(x)) \det(\nabla T_\varepsilon(x))$. Expanding $\rho_\varepsilon(x + \varepsilon \nabla u) = \rho_0(x) + \varepsilon r(x) + \varepsilon \langle \nabla \rho_0, \nabla u \rangle + o(\varepsilon)$ and $\det(\text{Id} + \varepsilon \nabla^2 u) = 1 + \varepsilon \Delta u + o(\varepsilon)$ gives

$$\rho_0 = \rho_0 + \varepsilon (r + \langle \nabla \rho_0, \nabla u \rangle + \rho_0 \Delta u) + o(\varepsilon) = \rho_0 + \varepsilon (r + \nabla \cdot (\rho_0 \nabla u)) + o(\varepsilon).$$

The first-order term must vanish. $\square$

2.5 Beyond the Quadratic Euclidean Cost

$\mathcal{W}_p$ costs. $\nabla f(x) = \nabla_x c(x, T(x))$. $T(x) = x - \|\nabla f(x)\|^{q-2} \nabla f(x)$, $\frac{1}{p} + \frac{1}{q} = 1$.

Squared geodesic distance. $T(x) = \exp_x(-\nabla \varphi(x))$,

Twist condition.

Definition 2.21 (Twist condition). Let $X, Y \subset \mathbb{R}^d$ and let $c \in C^1(X \times Y)$. The cost c satisfies the twist condition from X to Y if, for every fixed $x \in X$, the map $y \in Y \mapsto \nabla_x c(x, y) \in \mathbb{R}^d$ is injective. The dual twist condition is the analogous injectivity of $x \mapsto \nabla_y c(x, y)$ for every fixed y .

Proposition 2.22 (Twist prevents splitting). *Assume that $c \in C^1(X \times Y)$ satisfies the twist condition from X to Y . Suppose that optimal pairs for the cost c admit a source potential f with the following first-order certificate: f is differentiable on a full α -measure subset of X , and whenever x belongs to this differentiability set and y can be paired optimally with x , one has $\nabla f(x) = \nabla_x c(x, y)$. Then, for α -almost every source point x , there is at most one target point y that can be paired optimally with x .*

Proof. Let (f, g) be optimal Kantorovich potentials and let π be an optimal plan. Complementary slackness gives $f(x) + g(y) = c(x, y)$ for π -almost every (x, y) , while dual feasibility gives $f(z) + g(y) \leq c(z, y)$ for all z . Thus, for almost every contact pair (x, y) , the function $z \mapsto c(z, y) - g(y)$ touches f from above at x . At a point where f is differentiable, this gives $\nabla f(x) = \nabla_x c(x, y)$. If the cost is twisted, this equation determines at most one y . Hence no optimal plan can split the mass of such an x between several target points. Since differentiability holds on a full α -measure set, the plan is concentrated on the graph of a measurable map there. $\square$

The quadratic cost satisfies twist since $\nabla_x \|x - y\|^2 = 2(x - y)$; the bilinear cost $c(x, y) = -\langle x, y \rangle$ satisfies twist since $\nabla_x c(x, y) = -y$; more generally $c(x, y) = h(x - y)$ is twisted when h is smooth strictly convex and ∇h is injective. On a Riemannian manifold, the squared geodesic cost is twisted locally away from the cut locus.

Ma–Trudinger–Wang curvature.

Definition 2.23 (Weak MTW condition). Assume that $c \in C^4(X \times Y)$, that $X, Y \subset \mathbb{R}^d$, that c is twisted, and that the mixed Hessian $\nabla_{xy}^2 c(x, y)$ is invertible. For (x, p) in the image of the change of variables $(x, y) \mapsto (x, -\nabla_x c(x, y))$, let $Y(x, p)$ be defined by $-\nabla_x c(x, Y(x, p)) = p$, and set $A_{ij}(x, p) := -\partial_{x_i x_j}^2 c(x, Y(x, p))$. The cost satisfies the weak MTW condition if $\sum_{i,j,k,l} \partial_{p_k p_l}^2 A_{ij}(x, p) \xi_i \xi_j \eta_k \eta_l \geq 0$ whenever $\langle \xi, \eta \rangle = 0$. A strong MTW condition asks for a positive lower bound, proportional to $\|\xi\|^2 \|\eta\|^2$, on the same orthogonal directions.

Proposition 2.24 (MTW condition and regularity). *Assume that c is smooth, twisted and non-degenerate, that the source and target domains satisfy the appropriate mutual c -convexity assumptions, and that the source and target densities are positive and bounded above and below. Under the weak MTW condition, the optimal map is locally Hölder continuous; with the corresponding global domain hypotheses, it is continuous up to the boundary. Under the strong MTW condition and higher smoothness of the data and domains, one obtains higher regularity estimates.*

Proof. This is the regularity theory of Ma–Trudinger–Wang, Trudinger–Wang and Loeper, stated in structural form rather than with its full boundary hypotheses. The c -convex potential solves a generated Jacobian equation. Twist and non-degeneracy turn its optimal relation into a single-valued map, while the weak MTW inequality controls the geometry of contact sets and yields interior Hölder estimates. Strong MTW provides a quantitative curvature lower bound; combined with smooth densities, domain geometry and elliptic estimates, it yields higher regularity. Conversely, Loeper proved that weak MTW is necessary for continuity for arbitrary smooth positive data. $\square$

2.6 One-Dimensional Transport and Quantiles

Cumulative and quantile functions.

Definition 2.25 (Cumulative and quantile functions). For $\alpha \in \mathcal{M}_+^1(\mathbb{R})$, its cumulative distribution function is

$$\mathcal{C}_\alpha(x) := \alpha((-\infty, x]). \quad (2.8)$$

Its generalized inverse, or quantile function, is

$$\mathcal{C}_\alpha^{-1}(r) := \inf \{x \in \mathbb{R} ; \mathcal{C}_\alpha(x) \geq r\}, \quad r \in (0, 1). \quad (2.9)$$

Proposition 2.26 (Quantile push-forward). *One has $(\mathcal{C}_\alpha^{-1})_\# \text{Leb}_{[0,1]} = \alpha$. If α has no atoms, then $(\mathcal{C}_\alpha)_\# \alpha = \text{Leb}_{[0,1]}$.*

Proof. For simplicity, assume first that α has a strictly positive density, so that $\mathcal{C}_\alpha$ is strictly increasing and continuous. Denote $\gamma := (\mathcal{C}_\alpha^{-1})_\# \text{Leb}_{[0,1]}$. It is enough to prove that $\mathcal{C}_\gamma = \mathcal{C}_\alpha$. For every x ,

$$\mathcal{C}_\gamma(x) = \int_0^1 \mathbf{1}_{(-\infty, x]}(\mathcal{C}_\alpha^{-1}(z)) dz = \int_0^1 \mathbf{1}_{[0, \mathcal{C}_\alpha(x)]}(z) dz = \mathcal{C}_\alpha(x),$$

where we used $\mathcal{C}_\alpha^{-1}(z) \leq x$ if and only if $z \leq \mathcal{C}_\alpha(x)$. General measures follow from the same argument with generalized inverses and right-continuity of the cumulative distribution function. If α has no atoms, the probability integral transform gives $(\mathcal{C}_\alpha)_\# \alpha = \text{Leb}_{[0,1]}$. $\square$

1-D Monge solutions.

Theorem 2.27 (One-dimensional Monge solution). *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$, assume that α has no atoms, and let $h : \mathbb{R} \rightarrow \mathbb{R}$ be convex. Consider the cost $c(x, y) = h(x - y)$. Write $q_\alpha := \mathcal{C}_\alpha^{-1}$ and $q_\beta := \mathcal{C}_\beta^{-1}$, and assume that $h(q_\alpha - q_\beta) \in L^1(0, 1)$. Then*

$$T = q_\beta \circ \mathcal{C}_\alpha = \mathcal{C}_\beta^{-1} \circ \mathcal{C}_\alpha \quad (2.10)$$

satisfies $T_\# \alpha = \beta$ and minimizes $\min_{S_\# \alpha = \beta} \int h(x - S(x)) d\alpha(x)$. In particular, $h(t) = |t|^p$ gives the usual one-dimensional Monge map for every $1 \leq p < +\infty$ whenever the source has no atoms.

Proof. By Proposition 2.26, atomlessness of α gives $(\mathcal{C}_\alpha)_\# \alpha = \text{Leb}_{[0,1]}$, while $(q_\beta)_\# \text{Leb}_{[0,1]} = \beta$. Hence $T_\# \alpha = \beta$.

Let S be any admissible map and set $\pi_S = (\text{Id}, S)_\# \alpha$. The one-dimensional uncrossing theorem for relaxed couplings, proved later as Theorem 3.24, shows that the quantile coupling $\pi^* = (q_\alpha, q_\beta)_\# \text{Leb}_{[0,1]}$ is optimal among all couplings. Since α has no atoms, $(\text{Id}, T)_\# \alpha = \pi^*$, and therefore $\int h(x - T(x)) d\alpha(x) = \int h(x - y) d\pi^*(x, y) \leq \int h(x - y) d\pi_S(x, y) = \int h(x - S(x)) d\alpha(x)$. This proves optimality of T among Monge maps. $\square$

For discrete measures, one cannot directly apply the map formula when the source has atoms, but if the measures are uniform on the same number of Dirac masses, then it is exactly the sorting formula of Proposition 1.1.

Proposition 2.28 (One-dimensional Wasserstein formulas). *Let $1 \leq p < +\infty$ and let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$ have finite p -th moments. Then*

$$\mathcal{W}_p(\alpha, \beta)^p = \int_0^1 \left| \mathcal{C}_\alpha^{-1}(r) - \mathcal{C}_\beta^{-1}(r) \right|^p dr = \|\mathcal{C}_\alpha^{-1} - \mathcal{C}_\beta^{-1}\|_{L^p([0,1])}^p. \quad (2.11)$$

For $p = 1$, this is equivalently

$$\mathcal{W}_1(\alpha, \beta) = \|\mathcal{C}_\alpha - \mathcal{C}_\beta\|_{L^1(\mathbb{R})} = \int_{\mathbb{R}} |\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)| dx \quad (2.12)$$

$$= \int_{\mathbb{R}} \left| \int_{-\infty}^x d(\alpha - \beta) \right| dx. \quad (2.13)$$

Proof. The first formula follows from Theorem 3.24: the optimal coupling is obtained by taking the same quantile level r for both measures. For $p = 1$, use the layer-cake identity. If q_α and q_β are the two quantile functions, then $\int_0^1 |q_\alpha(r) - q_\beta(r)| dr = \int_{\mathbb{R}} \lambda(\{r : q_\alpha(r) \leq x < q_\beta(r)\} \cup \{r : q_\beta(r) \leq x < q_\alpha(r)\}) dx$. The measure of the displayed set is exactly $|\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)|$ for almost every x . $\square$

Proposition 2.29 (Bobkov–Ledoux cumulative formula). *Let $\alpha, \beta \in \mathcal{M}_+^1([a, b])$ and write $s_+ := \max(s, 0)$. Then*

$$\mathcal{W}_2^2(\alpha, \beta) = 2 \int_a^b \int_x^b \left[(\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(y))_+ + (\mathcal{C}_\beta(x) - \mathcal{C}_\alpha(y))_+ \right] dy dx. \quad (2.14)$$

Proof. For any $u, v \in [a, b]$, one has $|u - v|^2 = 2 \int_a^b \int_x^b \left[\mathbf{1}_{\{u \leq x \leq y < v\}} + \mathbf{1}_{\{v \leq x \leq y < u\}} \right] dy dx$, because, when $u < v$, the integration domain is the triangle $u \leq x \leq y < v$, whose area is $(v - u)^2/2$, and the case $v < u$ is symmetric. Apply this identity with $u = q_\alpha(r)$ and $v = q_\beta(r)$, where $q_\alpha = \mathcal{C}_\alpha^{-1}$ and $q_\beta = \mathcal{C}_\beta^{-1}$, and integrate with respect to $r \in (0, 1)$. Fubini's theorem applies since the integrand is bounded. For $x \leq y$, the generalized inverse identities give, up to endpoint sets of zero Lebesgue measure, $\lambda\{r : q_\alpha(r) \leq x \text{ and } q_\beta(r) > y\} = (\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(y))_+$, and the same argument with α and β exchanged gives the second positive part. $\square$

$$\mathcal{C}_{\alpha_t}^{-1}(r) = (1 - t)\mathcal{C}_\alpha^{-1}(r) + t\mathcal{C}_\beta^{-1}(r), \quad r \in (0, 1).$$

OT on trees.

Proposition 2.30 (Cumulative formula on a tree). *Let $\mathsf{T} = (V, E, \ell)$ be a finite weighted tree, where each edge e has length $\ell_e > 0$, and let d_{T} be its geodesic distance. Choose a root o . For an edge $e = (u, v)$ oriented away from o , let V_e be the set of vertices in the connected component containing v after removing e . If $\alpha, \beta \in \mathcal{P}(V)$, then*

$$\mathcal{W}_{1, d_{\mathsf{T}}}(\alpha, \beta) = \min_{\pi \in \mathcal{U}(\alpha, \beta)} \sum_{x, y \in V} d_{\mathsf{T}}(x, y) \pi_{xy} = \sum_{e \in E} \ell_e |\alpha(V_e) - \beta(V_e)|. \quad (2.15)$$

The value is computable in $O(|V|)$ arithmetic operations once the signed masses $\alpha - \beta$ are stored on the vertices, by one postorder traversal of the rooted tree.

Proof. Let $\sigma = \alpha - \beta$. Removing an edge e splits the tree into V_e and its complement. For any coupling π , the net amount of mass crossing e from V_e to $V \setminus V_e$ minus the amount crossing in the reverse direction is fixed and equals $\sigma(V_e)$. Hence the total amount of transported mass whose path uses e is at least $|\sigma(V_e)|$. Since every unit of mass crossing e pays the length ℓ_e , summing over edges gives $\sum_{x, y} d_{\mathsf{T}}(x, y) \pi_{xy} = \sum_{e \in E} \ell_e \sum_{x, y: e \in [x, y]} \pi_{xy} \geq \sum_{e \in E} \ell_e |\sigma(V_e)|$, where $[x, y]$ is the unique path between x and y . Process the rooted tree from the leaves to the root. At a vertex $v \neq o$, compute the total signed mass in the subtree below v , namely $F_e = \sigma(V_e)$ for the edge e joining v to its parent. If $F_e > 0$, send F_e units of surplus from this subtree to the parent side; if $F_e < 0$, import $-F_e$ units from the parent side into the subtree. After this operation the subtree has zero net imbalance and can be collapsed into its parent. Continuing upward balances all subtrees because $\sigma(V) = 0$. Decomposing these edge fluxes into source-to-target paths yields a coupling whose traffic across each edge is exactly $|F_e|$, and therefore whose cost is the right-hand side of (2.15). The same postorder pass computes all F_e , proving the complexity claim. $\square$

Triangular rearrangements.

Proposition 2.31 (Knothe–Rosenblatt triangular rearrangement). *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$. Assume that the first marginal of α and, recursively, the one-dimensional conditional source laws used below are atomless almost everywhere. Fix measurable versions of all regular conditional distributions. Then there is a triangular map $T(x_1, \dots, x_d) = (T_1(x_1), T_2(x_1, x_2), \dots, T_d(x_1, \dots, x_d))$ such that $T_{\sharp} \alpha = \beta$ and, for each k , the function $x_k \mapsto T_k(x_1, \dots, x_k)$ is nondecreasing for $\alpha_{<k}$ -almost every $x_{<k} = (x_1, \dots, x_{k-1})$.*

Proof. The construction is recursive. For $k = 1$, let T_1 be the monotone rearrangement between the first marginal of α and the first marginal of β . Suppose that $T_1, \dots, T_{k-1}$ have already been constructed. Write $x_{<k} = (x_1, \dots, x_{k-1})$ and $T_{<k} = (T_1, \dots, T_{k-1})$. By induction, $(T_{<k})_{\sharp} \alpha_{<k} = \beta_{<k}$, where $\alpha_{<k}$ and $\beta_{<k}$ are the first $(k-1)$ -coordinate marginals. Let $\alpha_{x_{<k}}^k$ and $\beta_{y_{<k}}^k$ be regular conditional laws of the k -th coordinate given the previous coordinates. Define $T_k(x_{<k}, \cdot)$ as the one-dimensional monotone rearrangement from $\alpha_{x_{<k}}^k$ to $\beta_{T_{<k}(x_{<k})}^k$.

The map $T_k(x_{<k}, \cdot)$ is nondecreasing by the one-dimensional rearrangement theorem. The chain rule for disintegrations then shows that, after step k , the first k coordinates of $T_{\sharp} \alpha$ have the same law as the first k coordinates of β . At $k = d$ this gives $T_{\sharp} \alpha = \beta$. Target atoms are handled by generalized quantiles. If a source conditional law has atoms that must be split, the same recursive construction defines a triangular Markov kernel rather than a deterministic map. $\square$

Proposition 2.32 (Anisotropic Brenier maps converge to Knothe–Rosenblatt). *Let α, β be compactly supported probability measures on $\mathbb{R}^d$ with densities bounded above and below on their rectangular supports. In particular, the source and target conditional laws entering the Knothe–Rosenblatt construction are atomless almost everywhere. For $\varepsilon > 0$, set $c_{\varepsilon}(x, y) := \sum_{k=1}^d \varepsilon^{k-1} |x_k - y_k|^2$. Let T_{ε} be the Monge map from α to β for the cost c_{ε} , and let T_{KR} be the triangular Knothe–Rosenblatt rearrangement with the coordinate order used above. Then $T_{\varepsilon} \rightarrow T_{\text{KR}}$ in $L^2(\alpha; \mathbb{R}^d)$ as $\varepsilon \rightarrow 0$.*

Proof. We summarize the scale-separated argument. Let $\pi_{\varepsilon} = (\text{Id}, T_{\varepsilon})_{\sharp} \alpha$ and set $I_k(\gamma) := \int |x_k - y_k|^2 d\gamma(x, y)$, $F_{\varepsilon}(\gamma) = \sum_{k=1}^d \varepsilon^{k-1} I_k(\gamma)$. Since the supports are compact, every sequence $\varepsilon \downarrow 0$ has a subsequence for which π_{ε} converges weakly to a coupling π between α and β . The coordinate costs are bounded and continuous on the common compact support. Dividing F_{ε} by its leading coefficient and passing to the limit shows that π minimizes I_1 . The first source marginal is atomless, so the (x_1, y_1) -marginal of π is the unique monotone coupling; equivalently, $y_1 = T_1(x_1)$ under π .

Optimality against the Knothe coupling π_{KR} gives $F_{\varepsilon}(\pi_{\varepsilon}) \leq F_{\varepsilon}(\pi_{\text{KR}})$. Since $I_1(\pi_{\varepsilon})$ is bounded below by the optimal first-coordinate value $I_1(\pi_{\text{KR}})$, cancellation and division by ε yield $\sum_{k=2}^d \varepsilon^{k-2} I_k(\pi_{\varepsilon}) \leq \sum_{k=2}^d \varepsilon^{k-2} I_k(\pi_{\text{KR}})$. Passing to the limit shows that π minimizes I_2 among couplings with the already identified first-coordinate marginal. Disintegration with respect to (x_1, y_1) turns this into the monotone transport between the conditional laws of x_2 and y_2 , so $y_2 = T_2(x_1, x_2)$ under π .

The higher-coordinate induction repeats this comparison after disintegrating over the already identified coordinates. Its technical point is to control the residual suboptimality of the earlier coordinates before dividing by the next power of ε ; the atomlessness of both source and target conditionals makes the preceding monotone couplings invertible almost everywhere. The scale-separated estimate proved in then gives successively $y_k = T_k(x_1, \dots, x_k)$ for every k .

Finally, let $X \sim \alpha$. The graph couplings are the laws of $(X, T_{\varepsilon}(X))$, and they converge weakly to the law of $(X, T_{\text{KR}}(X))$. To turn this into convergence of maps, fix $\zeta > 0$. By Lusin's theorem, there is a compact set K with $\alpha(K) > 1 - \zeta$ on which T_{KR} is continuous. On K , the set $\{(x, y) : x \in K, \|y - T_{\text{KR}}(x)\| \geq \delta\}$ is closed and has zero mass under the limiting graph coupling. Portmanteau's theorem gives

$$\limsup_{\varepsilon \rightarrow 0} \alpha(\{x : x \in K, \|T_{\varepsilon}(x) - T_{\text{KR}}(x)\| \geq \delta\}) = 0.$$

Adding the complement of K and letting $\zeta \rightarrow 0$ proves convergence in α -probability. Since all maps take values in a common compact set, their squared differences are uniformly integrable, and convergence therefore holds in $L^2(\alpha)$. $\square$

2.7 Gaussian Measures and the Bures Metric

One-dimensional Gaussians. Let $\alpha = \mathcal{N}(m_{\alpha}, \sigma_{\alpha}^2)$ and $\beta = \mathcal{N}(m_{\beta}, \sigma_{\beta}^2)$ be two nondegenerate Gaussians on $\mathbb{R}$. Then

$T(x) = \frac{\sigma_{\beta}}{\sigma_{\alpha}}(x - m_{\alpha}) + m_{\beta}$ satisfies $T_{\sharp} \alpha = \beta$. It is the derivative of the convex function

$$\varphi(x) = \frac{\sigma_{\beta}}{2\sigma_{\alpha}}(x - m_{\alpha})^2 + m_{\beta}x,$$

$$\mathcal{W}_2(\alpha, \beta)^2 = \int_{\mathbb{R}} \left(\frac{\sigma_{\beta}}{\sigma_{\alpha}}(x - m_{\alpha}) + m_{\beta} - x \right)^2 d\alpha(x) = (m_{\alpha} - m_{\beta})^2 + (\sigma_{\alpha} - \sigma_{\beta})^2.$$

The formula extends by continuity to Dirac masses, although the affine Monge map itself only pushes a Dirac source to another Dirac. Thus the OT geometry of one-dimensional Gaussians is the Euclidean geometry of the closed half-plane $(m, \sigma) \in \mathbb{R} \times \mathbb{R}_+$.

Multivariate Gaussians.

$$\alpha = \mathcal{N}(m_\alpha, \Sigma_\alpha), \quad \beta = \mathcal{N}(m_\beta, \Sigma_\beta), \quad T(x) = m_\beta + A(x - m_\alpha), \quad (2.16)$$

Proposition 2.33 (Affine push-forward of Gaussians). *One has $T_\# \alpha = \beta$ if and only if*

$$A \Sigma_\alpha A^\top = \Sigma_\beta. \quad (2.17)$$

Proof. An affine function maps a Gaussian to a Gaussian, so the law of $T(X)$ is determined by its mean and covariance. If $X \sim \alpha$ and $Y = T(X)$, then $\mathbb{E}(Y) = m_\beta + A\mathbb{E}(X - m_\alpha) = m_\beta$, and $\mathbb{E}((Y - m_\beta)(Y - m_\beta)^\top) = A\mathbb{E}((X - m_\alpha)(X - m_\alpha)^\top)A^\top = A\Sigma_\alpha A^\top$. Thus $A\Sigma_\alpha A^\top = \Sigma_\beta$ is necessary and sufficient for Y to have the same mean and covariance as β ; since both laws are Gaussian, this is equivalent to equality in distribution. $\square$

Definition 2.34 (Bures metric). For positive semidefinite covariance matrices Σ and Λ , the Bures metric is

$$\mathcal{B}(\Sigma, \Lambda)^2 := \text{tr} \left(\Sigma + \Lambda - 2(\Sigma^{1/2} \Lambda \Sigma^{1/2})^{1/2} \right). \quad (2.18)$$

Proposition 2.35 (Gaussian $\mathcal{W}_2$ formula and Bures covariance term). *Assume that Σ_α and Σ_β are positive definite. The unique symmetric positive-definite solution of $A\Sigma_\alpha A = \Sigma_\beta$ is*

$$A = \Sigma_\alpha^{-1/2} \left(\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2} \right)^{1/2} \Sigma_\alpha^{-1/2}. \quad (2.19)$$

The affine map $T(x) = m_\beta + A(x - m_\alpha)$ is the optimal quadratic-cost transport from $\mathcal{N}(m_\alpha, \Sigma_\alpha)$ to $\mathcal{N}(m_\beta, \Sigma_\beta)$, and

$$\mathcal{W}_2(\alpha, \beta)^2 = \|m_\alpha - m_\beta\|^2 + \mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2, \quad (2.20)$$

where $\mathcal{B}$ is the Bures metric of Definition 2.34.

Proof. Multiplying $A\Sigma_\alpha A = \Sigma_\beta$ on the left and right by $\Sigma_\alpha^{1/2}$ gives $(\Sigma_\alpha^{1/2} A \Sigma_\alpha^{1/2})^2 = \Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2}$. Since A is symmetric positive, $\Sigma_\alpha^{1/2} A \Sigma_\alpha^{1/2}$ is symmetric positive and is therefore the unique positive square root of the right-hand side. Conversely, the matrix in (2.19) is symmetric positive and satisfies the covariance equation.

By Proposition 2.33, this affine map pushes α to β . It is the gradient of a convex quadratic potential, so Brenier's theorem implies optimality. If $X \sim \alpha$, then

$$\begin{aligned} \mathbb{E}\|X - T(X)\|^2 &= \|m_\alpha - m_\beta\|^2 + \mathbb{E}\|(\text{Id} - A)(X - m_\alpha)\|^2 \\ &= \|m_\alpha - m_\beta\|^2 + \text{tr}((\text{Id} - A)\Sigma_\alpha(\text{Id} - A)^\top) \\ &= \|m_\alpha - m_\beta\|^2 + \text{tr}(\Sigma_\alpha) + \text{tr}(A\Sigma_\alpha A) - 2\text{tr}(A\Sigma_\alpha) \\ &= \|m_\alpha - m_\beta\|^2 + \text{tr}(\Sigma_\alpha) + \text{tr}(\Sigma_\beta) - 2\text{tr}((\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2})^{1/2}), \end{aligned}$$

which is the desired expression. The formula for singular covariance matrices follows by adding ηId and letting $\eta \downarrow 0$. $\square$

The same formula can be read without first restricting to Gaussian measures. Let Φ_2 keep only raw second moments.

Proposition 2.36 (Bures metric as a second-moment quotient). *For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, set its raw second-moment matrix to be $\Phi_2(\alpha) := \int_{\mathbb{R}^d} uu^\top d\alpha(u) \in \mathbb{S}_+^d$. For $\Sigma, \Lambda \in \mathbb{S}_+^d$, $\min_{\Phi_2(\alpha)=\Sigma, \Phi_2(\beta)=\Lambda} \mathcal{W}_2(\alpha, \beta)^2 = \mathcal{B}(\Sigma, \Lambda)^2$. In other words, the Bures metric is the Wasserstein quotient distance induced on positive semidefinite matrices by identifying all probability measures with the same raw second-moment matrix.*

Proof. Assume first that Σ and Λ are positive definite. Fix two admissible laws α, β and an arbitrary coupling π between them, and write $K := \int_{\mathbb{R}^d \times \mathbb{R}^d} uv^\top d\pi(u, v)$. Then $\int \|u - v\|^2 d\pi(u, v) = \text{tr}(\Sigma) + \text{tr}(\Lambda) - 2\text{tr}(K)$. The block second-moment matrix

$$\int \begin{pmatrix} u \\ v \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}^\top d\pi(u, v) = \begin{pmatrix} \Sigma & K \\ K^\top & \Lambda \end{pmatrix}$$

is positive semidefinite. The Schur complement gives $R := \Sigma^{-1/2} K \Lambda^{-1/2}$, $RR^\top \preceq \text{Id}$, so $K = \Sigma^{1/2} R \Lambda^{1/2}$ with $\|R\|_{\text{op}} \leq 1$. By duality between the nuclear and operator norms,

$$\text{tr}(K) \leq \sup_{\|R\|_{\text{op}} \leq 1} \text{tr}(\Sigma^{1/2} R \Lambda^{1/2}) = \|\Lambda^{1/2} \Sigma^{1/2}\|_* = \text{tr} \left((\Sigma^{1/2} \Lambda \Sigma^{1/2})^{1/2} \right).$$

For singular matrices, fix $\eta > 0$. Adding ηId to the two diagonal blocks preserves block positivity, so the same Schur-complement argument gives $\text{tr}(K) \leq \text{tr}(((\Sigma + \eta \text{Id})^{1/2} (\Lambda + \eta \text{Id})^{1/2})^{1/2})$. Letting $\eta \downarrow 0$ and using continuity of the matrix square root gives the same trace bound with Σ, Λ . Hence every admissible coupling has cost at least $\mathcal{B}(\Sigma, \Lambda)^2$.

Conversely, choose the centered Gaussian laws $\mathcal{N}(0, \Sigma)$ and $\mathcal{N}(0, \Lambda)$. They satisfy the two second-moment constraints, and Proposition 2.35 gives $\mathcal{W}_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Lambda))^2 = \mathcal{B}(\Sigma, \Lambda)^2$, again by continuity in the singular case. This admissible pair attains the lower bound, proving the claim. $\square$

$$\Sigma = \begin{pmatrix} a & c \\ c & b \end{pmatrix}, \quad t = \frac{a+b}{\sqrt{2}}, \quad u = \frac{a-b}{\sqrt{2}}, \quad v = \sqrt{2}c,$$

$a = \frac{t+u}{\sqrt{2}}, \quad b = \frac{t-u}{\sqrt{2}}, \quad c = \frac{v}{\sqrt{2}}, \quad t^2 - u^2 - v^2 = 2(ab - c^2) = 2\det(\Sigma), \quad t \geq \sqrt{u^2 + v^2}$. Thus the cone of 2×2 covariance matrices is the Lorentz, or ice-cream, cone. The Bures distance is not the ambient Euclidean distance in these coordinates: on the positive definite interior it is the geodesic distance of a smooth nonlinear geometry, whose geodesics are the covariance parts of Gaussian optimal transport rather than the straight chords inherited from $\mathbb{R}^3$.

Proposition 2.37 (Metric and convexity properties of the Bures term). *The function $\mathcal{B}$ is a distance on positive semidefinite covariance matrices. Moreover, $\mathcal{B}^2$ is jointly convex: for $t \in [0, 1]$, $\mathcal{B}^2((1-t)\Sigma_0 + t\Sigma_1, (1-t)\Lambda_0 + t\Lambda_1) \leq (1-t)\mathcal{B}^2(\Sigma_0, \Lambda_0) + t\mathcal{B}^2(\Sigma_1, \Lambda_1)$.*

Proof. The key identity is the Procrustes representation $\mathcal{B}^2(\Sigma, \Lambda) = \min_{Q^\top Q = \text{Id}} \|\Sigma^{1/2} - \Lambda^{1/2}Q\|_F^2$. Indeed, expanding the square gives $\text{tr } \Sigma + \text{tr } \Lambda - 2 \max_{Q^\top Q = \text{Id}} \text{tr}(\Sigma^{1/2}Q^\top \Lambda^{1/2})$, and the orthogonal Procrustes formula identifies the maximum with $\text{tr}((\Sigma^{1/2}\Lambda\Sigma^{1/2})^{1/2})$. Symmetry, positivity and separation follow immediately from this representation. For the triangle inequality, choose almost optimal orthogonal matrices Q_1, Q_2 for (Σ, Λ) and (Λ, Γ) , respectively. Since Q_2Q_1 is admissible for (Σ, Γ) ,

$$\mathcal{B}(\Sigma, \Gamma) \leq \|\Sigma^{1/2} - \Gamma^{1/2}Q_2Q_1\|_F \leq \|\Sigma^{1/2} - \Lambda^{1/2}Q_1\|_F + \|\Lambda^{1/2} - \Gamma^{1/2}Q_2\|_F.$$

Letting the two choices become optimal proves the metric property.

For convexity, use the equivalent factor formulation $\mathcal{B}^2(\Sigma, \Lambda) = \min_{U^\top U = \Sigma, V^\top V = \Lambda} \|U - V\|_F^2$. Here U and V may have any common number of columns. Choose nearly optimal factors (U_0, V_0) and (U_1, V_1) for the two pairs, and define block factors $U_t = [\sqrt{1-t}U_0, \sqrt{t}U_1]$, $V_t = [\sqrt{1-t}V_0, \sqrt{t}V_1]$. Then $U_tU_t^\top = (1-t)\Sigma_0 + t\Sigma_1$ and $V_tV_t^\top = (1-t)\Lambda_0 + t\Lambda_1$, while $\|U_t - V_t\|_F^2 = (1-t)\|U_0 - V_0\|_F^2 + t\|U_1 - V_1\|_F^2$. Taking the infimum over the initial factors proves joint convexity. $\square$

3 Kantorovich Relaxation

3.1 Discrete Relaxation

Mass splitting and couplings. $\alpha = \sum_i a_i \delta_{x_i}$, $\beta = \sum_j b_j \delta_{y_j}$.

Definition 3.1 (Discrete couplings and mass conservation). Admissible couplings are only constrained to satisfy conservation of mass:

$$U(a, b) := \{P \in \mathbb{R}_+^{n \times m}; P\mathbf{1}_m = a \quad \text{and} \quad P^\top \mathbf{1}_n = b\}. \quad (3.1)$$

Equivalently,

$$P\mathbf{1}_m = \left(\sum_j P_{i,j} \right)_i \in \mathbb{R}^n, \quad P^\top \mathbf{1}_n = \left(\sum_i P_{i,j} \right)_j \in \mathbb{R}^m.$$

Definition 3.2 (Discrete product coupling). Given weights $a \in \Sigma_n$ and $b \in \Sigma_m$, the discrete product, or trivial, coupling is $(a \otimes b)_{i,j} := a_i b_j$. It belongs to $U(a, b)$ and corresponds to choosing the source and target labels independently.

Proposition 3.3 (Discrete product optimality is degenerate). *Assume that the zero-mass rows and columns have been removed, so that $a_i > 0$ and $b_j > 0$, and let C be a finite cost matrix. The product plan $a \otimes b$ minimizes $P \mapsto \langle C, P \rangle$ over $U(a, b)$ if and only if every coupling $P \in U(a, b)$ minimizes it.*

Proof. The reverse implication is immediate. Assume conversely that $a \otimes b$ is optimal and let $Q \in U(a, b)$ be arbitrary. Since all entries of $a \otimes b$ are positive, there exists $t > 0$ small enough that $R := (1+t)(a \otimes b) - tQ$ has nonnegative entries. Its row and column sums are $R\mathbf{1}_m = (1+t)a - ta = a$, $R^\top \mathbf{1}_n = (1+t)b - tb = b$, so $R \in U(a, b)$. Moreover, $a \otimes b = \frac{1}{1+t}R + \frac{t}{1+t}Q$. By optimality of $a \otimes b$, both $\langle C, R \rangle$ and $\langle C, Q \rangle$ are at least $\langle C, a \otimes b \rangle$. Taking the scalar product of the convex-combination identity with C gives $\langle C, a \otimes b \rangle = \frac{1}{1+t}\langle C, R \rangle + \frac{t}{1+t}\langle C, Q \rangle$. A convex average of two numbers not smaller than $\langle C, a \otimes b \rangle$ can equal $\langle C, a \otimes b \rangle$ only if both numbers are equal to it. Hence $\langle C, Q \rangle = \langle C, a \otimes b \rangle$, and Q is optimal. Since Q was arbitrary, all couplings are optimal. $\square$

Thus the product plan is mainly a feasibility witness. Except in the degenerate situation where the linear cost is constant on the whole transportation polytope, it is not expected to solve optimal transport.

Linear-programming structure.

$$L_C(a, b) := \min_{P \in U(a, b)} \langle C, P \rangle := \min_{P \in U(a, b)} \sum_{i,j} C_{i,j} P_{i,j}. \quad (3.2)$$

Proposition 3.4 (Joint convexity of discrete OT). *For every fixed cost matrix $C \in \mathbb{R}^{n \times m}$, the map $(a, b) \mapsto L_C(a, b)$ is jointly convex on $\Sigma_n \times \Sigma_m$.*

Proof. For $r \in \{0, 1\}$, let P_r be optimal between a_r and b_r . For $t \in [0, 1]$, the matrix $P_t = (1-t)P_0 + tP_1$ belongs to $U((1-t)a_0 + ta_1, (1-t)b_0 + tb_1)$. Therefore $L_C((1-t)a_0 + ta_1, (1-t)b_0 + tb_1) \leq \langle C, P_t \rangle = (1-t)L_C(a_0, b_0) + tL_C(a_1, b_1)$. $\square$

Proposition 3.5 (Rank-controlled sparse minimizers). *Let $\mathcal{A} : \mathbb{R}^{n \times m} \rightarrow \mathbb{R}^q$ be linear, let $\mathbf{d} \in \mathbb{R}^q$ and $C \in \mathbb{R}^{n \times m}$, and consider*

$$\inf_{P \geq 0, \mathcal{A}(P) \leq \mathbf{d}} \langle C, P \rangle, \quad (3.3)$$

where the inequality is componentwise. If the feasible set is nonempty and the objective is bounded below on it, then (3.3) admits a minimizer P^ satisfying*

$$\#\{(i, j); P_{i,j}^* > 0\} \leq \text{rank}(\mathcal{A}). \quad (3.4)$$

Here $\text{rank}(\mathcal{A}) = \dim(\text{Im } \mathcal{A})$. The same conclusion holds when $\mathcal{A}(P) \leq \mathbf{d}$ is replaced by $\mathcal{A}(P) = \mathbf{d}$.

Proof. The feasible region is a nonempty polyhedron, and the objective is bounded below on it. The attainment theorem for linear programs therefore provides a minimizer. Among all minimizers, choose P^* with the smallest number of positive entries, and denote its support by S .

Suppose that $\#S > \text{rank}(\mathcal{A})$. The restriction of $\mathcal{A}$ to the $\#S$ -dimensional space of matrices supported on S then has a nontrivial kernel. Hence there is a nonzero matrix H , supported on S , such that $\mathcal{A}(H) = 0$. Since P^* is strictly positive on S , both $P^* + tH$ and $P^* - tH$ remain nonnegative for all sufficiently small $t > 0$. Moreover, $\mathcal{A}(P^* \pm tH) = \mathcal{A}(P^*)$, so both perturbations satisfy

exactly the same linear inequalities, or equalities, as P^* . Optimality forces $\langle C, H \rangle = 0$; otherwise one of the two signs would strictly decrease the objective.

Choose $\sigma \in \{-1, 1\}$ such that σH has a negative entry, and set $t_* := \min_{(i,j): \sigma H_{i,j} < 0} \frac{P_{i,j}^*}{-\sigma H_{i,j}} > 0$. Then $P^* + t_* \sigma H$ is nonnegative, feasible, and has the same cost as P^* . At least one of its positive entries has become zero, while no new entry has appeared outside S . This contradicts the minimality of S and proves (3.4). $\square$

Proposition 3.6 (Sparse optimal plans). *Assume $a_i > 0$, $b_j > 0$ and $\sum_i a_i = \sum_j b_j = 1$. The linear program (3.2) admits an optimal coupling with at most $n + m - 1$ nonzero entries.*

Proof. The transportation polytope is nonempty and compact, so the minimum is finite. Apply the equality case of Proposition 3.5 to the marginal operator $\mathcal{A}_{\text{marg}}(P) := (P\mathbf{1}_m, P^\top \mathbf{1}_n) \in \mathbb{R}^{n+m}$. Its rank is $n + m - 1$. Indeed, every matrix satisfies $\sum_i (P\mathbf{1}_m)_i = \sum_j (P^\top \mathbf{1}_n)_j$, so there is one linear dependence among the $n + m$ marginal coordinates. It is the only one: if $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^m$ satisfy $\langle u, P\mathbf{1}_m \rangle + \langle v, P^\top \mathbf{1}_n \rangle = 0$ for every P , then testing this identity on each elementary matrix gives $u_i + v_j = 0$ for every (i, j) . Thus all u_i equal one scalar and all v_j equal its opposite, so the annihilator of $\text{Im}(\mathcal{A}_{\text{marg}})$ is one-dimensional. It follows that $\text{rank}(\mathcal{A}_{\text{marg}}) = n + m - 1$, and Proposition 3.5 supplies the claimed optimal coupling. $\square$

Proposition 3.7 (North-west corner feasible plan). *Let $a \in \mathbb{R}_+^n$ and $b \in \mathbb{R}_+^m$ have the same positive total mass. Starting from $(i, j) = (1, 1)$ with residual masses $r_i = a_i$ and $s_j = b_j$, skip zero residuals, set $P_{ij} = \min(r_i, s_j)$, subtract this value from both residuals, and advance every index whose residual has become zero. Repeat until all residual masses are exhausted.*

Proof. All assignments are nonnegative. At each step, the mass placed in entry (i, j) is subtracted from exactly one current row residual and one current column residual, so no row or column can receive more mass than prescribed. Conversely, an index is advanced only when its residual has been fully filled. When the algorithm stops, the total assigned mass is $\sum_i a_i = \sum_j b_j$, hence all row and column sums are exactly a and b .

Each positive assignment exhausts at least one current row or one current column. Before the final assignment, at most $n - 1$ row advances and $m - 1$ column advances can occur without terminating the construction. Hence the number of positive entries is at most $(n - 1) + (m - 1) + 1 = n + m - 1$. For acyclicity, view the positive support as a bipartite graph. Once a row or column index is advanced, it never appears again, so each new positive edge either starts a new component or attaches at least one new vertex to the component currently being swept. No edge is ever added between two old vertices of the same component, so no cycle can be created. $\square$

One-dimensional cases.

Proposition 3.8 (One-dimensional weighted sweep). *Let $x_1 \leq \dots \leq x_n$ and $y_1 \leq \dots \leq y_m$ be points on the line, and let $c(x, y) = h(x - y)$ with h convex. The north-west corner plan between the sorted weighted atoms is optimal for (3.2).*

Proof. The north-west plan is monotone: if $i < i'$ and $j > j'$, it cannot put positive mass on both (i, j) and (i', j') , because the sweep exhausts rows and columns in increasing order. To prove optimality, start from an arbitrary feasible plan P . If $P_{11} < \min(a_1, b_1)$, then row 1 sends positive mass to some column $j > 1$, while column 1 receives positive mass from some row $i > 1$. Moving $\eta = \min(P_{1j}, P_{i1})$ from $(1, j)$ and $(i, 1)$ to $(1, 1)$ and (i, j) preserves both marginals. Convexity of h gives the Monge inequality $h(x_1 - y_j) + h(x_i - y_1) \geq h(x_1 - y_1) + h(x_i - y_j)$, so this exchange does not increase the cost. Each exchange either removes the chosen entry in row 1 or removes the chosen entry in column 1. Repeating finitely many times forces $P_{11} = \min(a_1, b_1)$, exactly the first north-west assignment. It exhausts row 1, column 1, or both. Removing every exhausted index reduces the problem to a smaller ordered transportation problem. Induction on $n + m$ produces the complete north-west plan without increasing cost. Sorting costs $O(n \log n + m \log m)$ and the sweep uses at most $n + m - 1$ positive assignments. $\square$

Permutation matrices as couplings.

Definition 3.9 (Permutation matrices). For a permutation $\sigma \in \text{Perm}(n)$, its permutation matrix P_σ is

$$(P_\sigma)_{i,j} = \begin{cases} 1 & \text{if } j = \sigma(i) \\ 0 & \text{otherwise.} \end{cases}$$

The set of all permutation matrices is $\mathcal{P}_n^{\text{perm}} := \{P_\sigma ; \sigma \in \text{Perm}(n)\}$.

$$\langle C, P_\sigma/n \rangle = \frac{1}{n} \sum_{i=1}^n C_{i, \sigma(i)}.$$

Definition 3.10 (Birkhoff polytope). The Birkhoff polytope is the convex set of bistochastic matrices $\mathcal{B}_n := \{P \in \mathbb{R}_+^{n \times n} ; P\mathbf{1}_n = \mathbf{1}_n \text{ and } P^\top \mathbf{1}_n = \mathbf{1}_n\}$.

Then $U(\mathbf{1}_n/n, \mathbf{1}_n/n) = \mathcal{B}_n/n$, and permutation couplings are included in this convex relaxation. More precisely, $\mathcal{P}_n^{\text{perm}} = \mathcal{B}_n \cap \{0, 1\}^{n \times n} \subset \mathcal{B}_n$, $\min_{P \in \mathcal{B}_n} \langle C, P \rangle \leq \min_{P \in \mathcal{P}_n^{\text{perm}}} \langle C, P \rangle$.

Definition 3.11 (Extreme points). For a convex set $\mathcal{C}$ in a finite-dimensional vector space, $\text{Extr}(\mathcal{C}) := \{x \in \mathcal{C} ; x = (y + z)/2, y, z \in \mathcal{C} \Rightarrow y = z = x\}$.

Proposition 3.12 (Existence of extreme points). *If $\mathcal{C}$ is a non-empty compact convex subset of a finite-dimensional vector space, then $\text{Extr}(\mathcal{C})$ is non-empty.*

Proof. Among all non-empty faces of $\mathcal{C}$, choose one of minimal affine dimension. If this face contained two distinct points, maximizing a linear functional that is not constant on the face would produce a non-empty proper exposed subface, contradicting minimality. Hence the minimal face is a singleton, and its point is extreme. $\square$

Proposition 3.13 (Linear programs have extreme minimizers). *Let $\mathcal{C}$ be a non-empty compact convex set. For every linear form ℓ , $\text{Extr}(\mathcal{C}) \cap \text{argmin}_{x \in \mathcal{C}} \ell(x) \neq \emptyset$.*

Proof. The set $S = \text{argmin}_{x \in \mathcal{C}} \ell(x)$ is non-empty, compact and convex. By Proposition 3.12, it has an extreme point x . If $x = (y + z)/2$ with $y, z \in \mathcal{C}$, then by linearity and optimality of x , both y and z also minimize ℓ on $\mathcal{C}$, hence $y, z \in S$. Since x is extreme in S , $y = z = x$. Thus x is extreme in $\mathcal{C}$. $\square$

Theorem 3.14 (Birkhoff–von Neumann). *The extreme points of $\mathcal{B}_n$ are exactly the permutation matrices.*

Proof. We first prove that permutation matrices are extreme. Let $P_\sigma \in \mathcal{P}_n^{\text{perm}}$ and assume that $P_\sigma = \frac{Q+R}{2}$ with $Q, R \in \mathcal{B}_n$. Every bistochastic matrix has entries in $[0, 1]$. Since the only extreme points of $[0, 1]$ are 0 and 1, each entry of P_σ fixes the corresponding entries of Q and R : if $(P_\sigma)_{ij} = 0$, then $Q_{ij} = R_{ij} = 0$, while if $(P_\sigma)_{ij} = 1$, then $Q_{ij} = R_{ij} = 1$. Hence $Q = R = P_\sigma$, so P_σ is extreme.

Pick $P \in \mathcal{B}_n \setminus \mathcal{P}_n^{\text{perm}}$. Since an integral bistochastic matrix is necessarily a permutation matrix, P has at least one fractional entry. We shall split $P = \frac{Q+R}{2}$ with $Q, R \in \mathcal{B}_n$ and $Q \neq R$, proving that P is not extreme.

Associate with P the bipartite graph whose left vertices are the rows, whose right vertices are the columns, and whose edges are the fractional entries $0 < P_{ij} < 1$. An entry equal to 1 uses the whole mass of its row and column, so it is isolated in the positive support and does not appear in this fractional graph. If a left vertex i is incident to a fractional edge (i, j_1) , then it must be incident to at least one other fractional edge. Indeed, the row sum is one; after the contribution $P_{i,j_1} \in (0, 1)$, a positive amount $1 - P_{i,j_1}$ remains in the same row, and it cannot be carried by an entry equal to 1. The same argument applies to right vertices, using the column sums. Thus every non-isolated vertex of the fractional graph has degree at least two.

Starting from any fractional edge, one may therefore walk through adjacent fractional edges without immediately backtracking and without getting stuck. Since the graph is finite, some vertex is eventually visited twice; the portion of the walk between the two visits contains a cycle. Choose a shortest such cycle and write it in alternating form $(i_1, j_1, i_2, j_2, \dots, i_p, j_p)$, $i_{p+1} = i_1$, where both (i_s, j_s) and (i_{s+1}, j_s) are fractional for every s . The minimality of the cycle implies that the vertices i_s are all distinct and that the vertices j_s are all distinct. In particular, $0 < P_{i_s, j_s} < 1$ and $0 < P_{i_{s+1}, j_s} < 1$. Define $\varepsilon := \min_{1 \leq s \leq p} \{P_{i_s, j_s}, P_{i_{s+1}, j_s}, 1 - P_{i_s, j_s}, 1 - P_{i_{s+1}, j_s}\}$. All these numbers are positive, so $\varepsilon > 0$. Split the cycle edges into the two alternating families $A := \{(i_s, j_s)\}_{s=1}^p$, $B := \{(i_{s+1}, j_s)\}_{s=1}^p$.

$$Q_{ij} := \begin{cases} P_{ij}, & (i, j) \notin A \cup B, \\ P_{ij} + \varepsilon/2, & (i, j) \in A, \\ P_{ij} - \varepsilon/2, & (i, j) \in B, \end{cases} \quad R_{ij} := \begin{cases} P_{ij}, & (i, j) \notin A \cup B, \\ P_{ij} - \varepsilon/2, & (i, j) \in A, \\ P_{ij} + \varepsilon/2, & (i, j) \in B. \end{cases}$$

By the definition of ε , all modified entries stay in $[0, 1]$, so Q and R are nonnegative. Each row vertex i_s of the cycle is incident to exactly one edge of A and one edge of B ; the $+\varepsilon/2$ and $-\varepsilon/2$ perturbations therefore cancel in that row. The same cancellation holds in each column vertex j_s , and all other rows and columns are unchanged. $Q\mathbf{1}_n = R\mathbf{1}_n = \mathbf{1}_n$, $Q^\top \mathbf{1}_n = R^\top \mathbf{1}_n = \mathbf{1}_n$, so $Q, R \in \mathcal{B}_n$. Finally, $Q \neq R$ because $\varepsilon > 0$ and the cycle is non-empty, while by construction $P = (Q + R)/2$. Thus P is not extreme. $\square$

$$s|I| = \sum_{i \in I} \sum_{j \in N(I)} R_{ij} \leq \sum_{j \in N(I)} \sum_i R_{ij} = s|N(I)|,$$

Corollary 3.15 (Kantorovich for matching). *If $m = n$ and $\mathbf{a} = \mathbf{b} = \mathbf{1}_n/n$, then the discrete Kantorovich problem (3.2) admits an optimal solution of the form P_σ/n . The associated permutation σ solves the assignment problem of Section 1.1.*

Proof. The feasible set is $\mathcal{B}_n/n$. By Proposition 3.13, the linear objective has an optimal extreme point. Since scaling preserves extreme points and Theorem 3.14 identifies the extreme points of $\mathcal{B}_n$, this optimizer is P_σ/n for some permutation σ . Its cost is exactly $n^{-1} \sum_i C_{i, \sigma(i)}$, so σ is an optimal assignment. $\square$

Rational weights. The coupling set $U(\mathbf{a}, \mathbf{b})$ removes the equal-cardinality and equal-weight restrictions of assignments. For rational weights, it can be represented by a larger uniform matching.

Proposition 3.16 (Rational weights as duplicated uniform matching). *Let $\alpha = \sum_{i=1}^n \frac{k_i}{N} \delta_{x_i}$, $\beta = \sum_{j=1}^m \frac{\ell_j}{N} \delta_{y_j}$, $\sum_i k_i = \sum_j \ell_j = N$, with positive integers k_i, ℓ_j . Replace each x_i by k_i identical copies and each y_j by ℓ_j identical copies, producing two uniform N -point clouds. Then the duplicated assignment problem and the discrete Kantorovich problem between α and β have the same optimal value. Moreover, there exists an optimal coupling of the form $P_{ij} = \frac{n_{ij}}{N}$, $n_{ij} \in \mathbb{N}$, $\sum_j n_{ij} = k_i$, $\sum_i n_{ij} = \ell_j$, and couplings whose scaled entries NP_{ij} are integral are exactly the collapsed assignments between duplicated clouds. Fractional optimal couplings may nevertheless coexist when the optimum is degenerate.*

Proof. Any assignment between the duplicated source and target clouds defines integers n_{ij} counting how many copied particles of type x_i are matched to copied particles of type y_j . These counts satisfy $\sum_j n_{ij} = k_i$ and $\sum_i n_{ij} = \ell_j$, and the associated coupling $P_{ij} = n_{ij}/N$ has marginals k_i/N and ℓ_j/N . The assignment cost is $\frac{1}{N} \sum_{i,j} n_{ij} c(x_i, y_j) = \sum_{i,j} P_{ij} c(x_i, y_j)$. Conversely, any nonnegative integer count matrix with those row and column sums can be realized by allocating the k_i copies of each x_i among the target copies according to $(n_{ij})_j$. Scale a feasible coupling by N and write $Q = NP$. The constraints on Q have integer right-hand sides $(k_1, \dots, k_n, \ell_1, \dots, \ell_m)$. After multiplying the target rows by -1 , their coefficient matrix is the oriented node-edge incidence matrix of a bipartite graph and is therefore totally unimodular. Every vertex of this transportation polytope is integral. Since a linear objective attains its minimum at a vertex, an integral optimal Q exists, proving equality of the two optimal values. The argument establishes existence, not integrality of every optimizer; convex combinations of distinct integral optima give fractional optima. $\square$

3.2 Linear-Programming Algorithms

Transportation simplex and network simplex.

Interior-point methods.

$$P_\varepsilon := \underset{\substack{P\mathbf{1}_m = \mathbf{a}, P^\top \mathbf{1}_n = \mathbf{b} \\ P_{ij} > 0}}{\operatorname{argmin}} \langle C, P \rangle - \varepsilon \sum_{i,j} \log P_{ij}, \quad (3.5)$$

where $\varepsilon > 0$ is decreased along the algorithm. The barrier is singular at the boundary, so each iterate stays strictly inside the transportation polytope; as $\varepsilon \downarrow 0$, the central path approaches the set of LP minimizers.

3.3 Relaxation for Arbitrary Measures

Continuous couplings.

Definition 3.17 (Marginals of a joint measure). Let $\pi \in \mathcal{M}_+^1(\mathcal{X} \times \mathcal{Y})$ and let $P_\mathcal{X}(x, y) = x$, $P_\mathcal{Y}(x, y) = y$ be the coordinate projections. The marginals of π are $\pi_1 := (P_\mathcal{X})_\# \pi \in \mathcal{M}_+^1(\mathcal{X})$, $\pi_2 := (P_\mathcal{Y})_\# \pi \in \mathcal{M}_+^1(\mathcal{Y})$. Equivalently, for all bounded

continuous test functions f on $\mathcal{X}$ and g on $\mathcal{Y}$, $\int_{\mathcal{X} \times \mathcal{Y}} f(x) d\pi(x, y) = \int_{\mathcal{X}} f d\pi_1$, $\int_{\mathcal{X} \times \mathcal{Y}} g(y) d\pi(x, y) = \int_{\mathcal{Y}} g d\pi_2$.

$\int_{\mathcal{Y}} d\pi(x, y) = d\alpha(x)$, $\int_{\mathcal{X}} d\pi(x, y) = d\beta(y)$, which is made rigorous by Definition 3.17, or equivalently by the identities $\pi(A \times \mathcal{Y}) = \alpha(A)$ and $\pi(\mathcal{X} \times B) = \beta(B)$ for measurable sets $A \subset \mathcal{X}$ and $B \subset \mathcal{Y}$.

Definition 3.18 (Couplings). Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, the set of couplings between α and β is

$$\mathcal{U}(\alpha, \beta) := \{\pi \in \mathcal{M}_+^1(\mathcal{X} \times \mathcal{Y}) ; \pi_1 = \alpha \text{ and } \pi_2 = \beta\}. \quad (3.6)$$

This is the continuous analogue of the transportation polytope (3.1).

Definition 3.19 (Tensor product and trivial coupling). Given $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, the tensor product coupling $\alpha \otimes \beta$ is the probability measure on $\mathcal{X} \times \mathcal{Y}$ defined by

$$\int_{\mathcal{X} \times \mathcal{Y}} h(x, y) d(\alpha \otimes \beta)(x, y) = \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} h(x, y) d\beta(y) \right) d\alpha(x)$$

for every bounded measurable h . It is also called the trivial coupling because it makes the two coordinates independent.

$$\int_{\mathcal{X} \times \mathcal{Y}} f(x) d(\alpha \otimes \beta)(x, y) = \left(\int_{\mathcal{X}} f(x) d\alpha(x) \right) \left(\int_{\mathcal{Y}} d\beta(y) \right) = \int f d\alpha,$$

Proposition 3.20 (Product optimality is degenerate). Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$. $\alpha \otimes \beta \in \argmin_{\pi \in \mathcal{U}(\alpha, \beta)} \int c d\pi$, every coupling in $\mathcal{U}(\alpha, \beta)$ is optimal. They are also equivalent to the additive decomposition of the cost on the product support, $c(x, y) = u(x) + v(y)$, $u \in \mathcal{C}(\text{supp}(\alpha))$, $v \in \mathcal{C}(\text{supp}(\beta))$.

Proof. If every coupling is optimal, then $\alpha \otimes \beta$ is optimal. Conversely, assume that $\alpha \otimes \beta$ is optimal. We first show that, for every $x_0, x_1 \in \text{supp}(\alpha)$ and $y_0, y_1 \in \text{supp}(\beta)$, $c(x_0, y_0) + c(x_1, y_1) = c(x_0, y_1) + c(x_1, y_0)$. Indeed, if this equality failed, after exchanging y_0 and y_1 if necessary one would have a strict inequality $c(x_0, y_0) + c(x_1, y_1) > c(x_0, y_1) + c(x_1, y_0)$. Failure of the equality forces $x_0 \neq x_1$ and $y_0 \neq y_1$. By continuity, the strict inequality persists with a uniform margin on small neighborhoods U_0, U_1 of x_0, x_1 and V_0, V_1 of y_0, y_1 , chosen disjoint within each pair. Since the four points lie in the supports, these neighborhoods have positive marginal mass. Denote by α_i and β_i the normalized restrictions of α to U_i and of β to V_i , and choose $0 < \lambda \leq \min\{\alpha(U_0)\beta(V_0), \alpha(U_1)\beta(V_1)\}$. The exchanged measure $\tilde{\pi} = \alpha \otimes \beta - \lambda \alpha_0 \otimes \beta_0 - \lambda \alpha_1 \otimes \beta_1 + \lambda \alpha_0 \otimes \beta_1 + \lambda \alpha_1 \otimes \beta_0$ is nonnegative and has the same two marginals as $\alpha \otimes \beta$. The uniform strict inequality on the neighborhoods implies that $\int c d\tilde{\pi} < \int c d(\alpha \otimes \beta)$, contradicting optimality. Fixing any $x_* \in \text{supp}(\alpha)$ and $y_* \in \text{supp}(\beta)$, the equality of cross differences gives, for all $(x, y) \in \text{supp}(\alpha) \times \text{supp}(\beta)$, $c(x, y) = c(x, y_*) + c(x_*, y) - c(x_*, y_*)$. Thus $c = u + v$ on the product support, with $u(x) = c(x, y_*) - c(x_*, y_*)$ and $v(y) = c(x_*, y)$. These functions are continuous on the corresponding supports. Every coupling is concentrated on this product support, so for any $\pi \in \mathcal{U}(\alpha, \beta)$, $\int c d\pi = \int u d\alpha + \int v d\beta$, which depends only on the marginals. Hence all couplings are optimal. $\square$

The tensor product is therefore a trivial feasible coupling, not a typical optimizer. Product optimality means that the cost cannot distinguish between dependences once the marginals are fixed.

If there exists a map $T : \mathcal{X} \rightarrow \mathcal{Y}$ such that $T_{\#}\alpha = \beta$, then the Monge map induces the graph coupling $\pi = (\text{Id}, T)_{\#}\alpha \in \mathcal{U}(\alpha, \beta)$, characterized by

$\int_{\mathcal{X} \times \mathcal{Y}} h(x, y) d\pi(x, y) = \int_{\mathcal{X}} h(x, T(x)) d\alpha(x)$. Applying this identity to $h(x, y) = f(x)$ or $h(x, y) = g(y)$ gives respectively $\pi_1 = \alpha$ and $\pi_2 = \beta$. Thus graph couplings are precisely the Kantorovich representation of deterministic Monge maps.

Continuous Kantorovich problem. For a nonnegative Borel cost $c : \mathcal{X} \times \mathcal{Y} \rightarrow [0, +\infty]$, the discrete Kantorovich problem (3.2) becomes, for arbitrary measures, the minimization of the average cost over all couplings,

$$\mathcal{L}_c(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y). \quad (3.7)$$

Proposition 3.21 (Convexity in the marginals and concavity in the cost). Fix a nonnegative Borel cost c . The extended-valued map $(\alpha, \beta) \mapsto \mathcal{L}_c(\alpha, \beta)$ is jointly convex on pairs of probability measures. Conversely, for fixed marginals (α, β) , the map $c \mapsto \mathcal{L}_c(\alpha, \beta)$ is concave on the cone of nonnegative Borel costs: for any such c_0, c_1 and $t \in [0, 1]$, $\mathcal{L}_{(1-t)c_0 + tc_1}(\alpha, \beta) \geq (1-t)\mathcal{L}_{c_0}(\alpha, \beta) + t\mathcal{L}_{c_1}(\alpha, \beta)$. The inequality is understood in $[0, +\infty]$; in particular, the map is concave on every convex family of costs on which it is finite.

Proof. For joint convexity, let (α_0, β_0) and (α_1, β_1) be two pairs of probability measures. The claim is immediate at $t \in \{0, 1\}$ or if one of the two values on the right-hand side is infinite. Otherwise, for $\eta > 0$, choose $\pi_i \in \mathcal{U}(\alpha_i, \beta_i)$ such that $\int c d\pi_i \leq \mathcal{L}_c(\alpha_i, \beta_i) + \eta$, $i \in \{0, 1\}$. Then $(1-t)\pi_0 + t\pi_1$ couples $(1-t)\alpha_0 + t\alpha_1$ and $(1-t)\beta_0 + t\beta_1$, and hence $\mathcal{L}_c((1-t)\alpha_0 + t\alpha_1, (1-t)\beta_0 + t\beta_1) \leq (1-t)\mathcal{L}_c(\alpha_0, \beta_0) + t\mathcal{L}_c(\alpha_1, \beta_1) + \eta$. Letting $\eta \rightarrow 0$ proves joint convexity.

For concavity, the endpoint cases are immediate, so assume $t \in (0, 1)$ and fix $\pi \in \mathcal{U}(\alpha, \beta)$. Linearity of integration for nonnegative functions gives $\int ((1-t)c_0 + tc_1) d\pi \geq (1-t)\mathcal{L}_{c_0}(\alpha, \beta) + t\mathcal{L}_{c_1}(\alpha, \beta)$. Taking the infimum over π proves the stated inequality. $\square$

Proposition 3.22 (Existence for lower-semicontinuous costs). Assume that $\mathcal{X}$ and $\mathcal{Y}$ are Polish spaces and that $c : \mathcal{X} \times \mathcal{Y} \rightarrow [0, +\infty]$ is lower semicontinuous. Then the Kantorovich problem (3.7) admits at least one minimizer. In particular, this applies to every continuous nonnegative cost on compact metric spaces.

Proof. The constraint set is non-empty because it contains the product coupling $\alpha \otimes \beta$. It is uniformly tight: given $\varepsilon > 0$, choose compact sets $K_{\mathcal{X}} \subset \mathcal{X}$ and $K_{\mathcal{Y}} \subset \mathcal{Y}$ such that $\alpha(K_{\mathcal{X}}) \geq 1 - \varepsilon/2$ and $\beta(K_{\mathcal{Y}}) \geq 1 - \varepsilon/2$. Every $\pi \in \mathcal{U}(\alpha, \beta)$ then satisfies $\pi(K_{\mathcal{X}} \times K_{\mathcal{Y}}) \geq 1 - \varepsilon$. By Prokhorov's theorem, the set of couplings is relatively compact for weak convergence. It is also weakly closed, because the coordinate projections are continuous and therefore preserve the marginal identities in the limit. Hence $\mathcal{U}(\alpha, \beta)$ is weakly compact. Finally, Portmanteau's theorem makes $\pi \mapsto \int c d\pi$ weakly lower semicontinuous. The direct method gives a minimizer. $\square$

For the Wasserstein cost $c(x, y) = d(x, y)^p$ on a Polish metric space, the natural finite-valued domain is

$$\mathcal{P}_p(\mathcal{X}) := \{\alpha \in \mathcal{M}_+^1(\mathcal{X}) : \int d(x, x_0)^p d\alpha(x) < +\infty\},$$

Monge–Kantorovich equivalence.

Corollary 3.23 (Monge–Kantorovich equivalence under Brenier). *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$, assume that α is absolutely continuous with respect to Lebesgue measure, and let $c(x, y) = \|x - y\|^2$. If T is the Brenier map solving Monge’s problem, then $\pi = (\text{Id}, T)_\# \alpha$ is the unique optimal coupling solving the Kantorovich problem. In particular, Monge and Kantorovich costs are the same.*

Proof. The proof of Brenier’s theorem shows that the support of any optimal Kantorovich plan is contained in the subdifferential $\partial\varphi$ of a convex function φ . When α has a density, φ is differentiable α -almost everywhere, so $\partial\varphi(x) = \{\nabla\varphi(x)\}$ for α -almost every x . Thus every optimal coupling is concentrated on the graph of $T = \nabla\varphi$ and must equal $(\text{Id}, T)_\# \alpha$. The graph coupling is feasible and optimal, and the two formulations have the same value. $\square$

Kantorovich solution in 1-D.

Theorem 3.24 (One-dimensional Kantorovich solution). *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$ and let $h : \mathbb{R} \rightarrow [0, +\infty)$ be convex. Consider the cost $c(x, y) = h(x - y)$. Write $q_\alpha := \mathcal{C}_\alpha^{-1}$ and $q_\beta := \mathcal{C}_\beta^{-1}$, and assume that $h(q_\alpha - q_\beta) \in L^1(0, 1)$. Then the quantile coupling $\pi^* = (q_\alpha, q_\beta)_\# \text{Leb}_{[0,1]} = (\mathcal{C}_\alpha^{-1}, \mathcal{C}_\beta^{-1})_\# \text{Leb}_{[0,1]}$ minimizes $\min_{\pi \in \mathcal{U}(\alpha, \beta)} \int h(x - y) d\pi(x, y)$. In particular, $h(t) = |t|^p$ gives the usual one-dimensional optimal coupling for every $p \geq 1$.*

Proof. The push-forward statement $\pi^* \in \mathcal{U}(\alpha, \beta)$ follows from Proposition 2.26.

The key point is the one-dimensional uncrossing inequality. If $x < x'$ and $y > y'$, set $a = x - y$, $\delta = x' - x > 0$ and $\eta = y - y' > 0$. Convexity of h implies that increments are monotone, hence $h(a + \eta) - h(a) \leq h(a + \delta + \eta) - h(a + \delta)$, which is exactly $h(x - y) + h(x' - y') \geq h(x - y') + h(x' - y)$. Thus removing a crossing never increases the cost. For a finite transport matrix on two ordered grids, if $i < i'$ and $j > j'$ carry crossed masses P_{ij} and $P_{i'j'}$, one may move $\theta = \min(P_{ij}, P_{i'j'})$ units of mass from the crossed entries (i, j) , (i', j') to the uncrossed entries (i, j') , (i', j) . The row and column sums are unchanged, and the preceding inequality shows that the cost does not increase. Repeating this elementary uncrossing step yields an ordered plan; on an ordered uniform quantile grid, this is the diagonal plan.

For general measures, apply the same argument in quantile coordinates. Let $\pi \in \mathcal{U}(\alpha, \beta)$. Let U_α and U_β be uniform random variables on $(0, 1)$, and denote by $K_\alpha(x, dr)$ and $K_\beta(y, ds)$ regular conditional laws of U_α given $q_\alpha(U_\alpha) = x$ and of U_β given $q_\beta(U_\beta) = y$. Given $(X, Y) \sim \pi$, draw conditionally $R \sim K_\alpha(X, \cdot)$ and $S \sim K_\beta(Y, \cdot)$. The law γ of (R, S) has uniform marginals and satisfies $\pi = (q_\alpha, q_\beta)_\# \gamma$.

Let $\kappa_M(r) = \max(-M, \min(r, M))$ and set $q_{\alpha, M} = \kappa_M \circ q_\alpha$ and $q_{\beta, M} = \kappa_M \circ q_\beta$. These functions are bounded and nondecreasing. Approximate them almost everywhere by nondecreasing step functions $q_{\alpha, M}^N$ and $q_{\beta, M}^N$ that are constant on the uniform intervals $I_k = ((k-1)/N, k/N]$. The matrix $G_{k\ell}^N = \gamma(I_k \times I_\ell)$ is a coupling of the two uniform histograms. Proposition 3.8, applied to the ordered step values, therefore gives $\int_0^1 h(q_{\alpha, M}^N(r) - q_{\beta, M}^N(r)) dr \leq \int_{(0,1)^2} h(q_{\alpha, M}^N(r) - q_{\beta, M}^N(s)) d\gamma(r, s)$. For fixed M , the arguments of h remain in the compact interval $[-2M, 2M]$. Dominated convergence as $N \rightarrow \infty$ yields the same inequality for $q_{\alpha, M}$ and $q_{\beta, M}$. Finally, κ_M is nondecreasing and 1-Lipschitz. Hence, for every $x, y \in \mathbb{R}$, there exists $t_M(x, y) \in [0, 1]$ such that $\kappa_M(x) - \kappa_M(y) = t_M(x, y)(x - y)$. Convexity and nonnegativity imply $h(t_M(x, y)(x - y)) \leq (1 - t_M(x, y))h(0) + t_M(x, y)h(x - y) \leq h(0) + h(x - y)$. The assumed integrability controls the diagonal term. If the cost of π is finite, the same bound controls the right-hand side; if it is infinite, the desired comparison is immediate. Dominated convergence as $M \rightarrow \infty$ therefore gives $\int_0^1 h(q_\alpha(r) - q_\beta(r)) dr \leq \int h(x - y) d\pi(x, y)$ for every $\pi \in \mathcal{U}(\alpha, \beta)$. $\square$

3.4 c -Cyclical Monotonicity

Support and c -cyclical monotonicity. The support of a coupling is the topological support introduced in Definition 2.4, now applied to a Radon measure on $\mathcal{X} \times \mathcal{Y}$. Thus $(x, y) \in \text{supp}(\pi)$ exactly when every open neighborhood of (x, y) has positive π -mass.

Definition 3.25 (c -cyclical monotonicity). A set $\Gamma \subset \mathcal{X} \times \mathcal{Y}$ is c -cyclically monotone if, for every $k \geq 2$, every finite family $(x_i, y_i)_{i=1}^k \subset \Gamma$ and every permutation σ of $\{1, \dots, k\}$, $\sum_{i=1}^k c(x_i, y_i) \leq \sum_{i=1}^k c(x_i, y_{\sigma(i)})$.

$$\sum_{i=1}^k c(x_i, y_i) \leq \sum_{i=1}^k c(x_i, y_{i+1}), \quad y_{k+1} = y_1.$$

Optimal matching to optimal transport. Let the marginals be uniform on n points, $\alpha = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ and $\beta = \frac{1}{n} \sum_{i=1}^n \delta_{y_i}$. By Corollary 3.15, there exists an optimal plan induced by a permutation. Its support $\Gamma = \{(x_i, y_{\sigma(i)})\}_i$ is c -cyclically monotone: otherwise exchanging the finitely many targets along a violating cycle would lower the matching cost.

Theorem 3.26 (Optimal plans are c -cyclically monotone). *Assume that $c : \mathcal{X} \times \mathcal{Y} \rightarrow [0, +\infty)$ is continuous and that the Kantorovich value is finite. For any optimal plan π solving (3.7), $\text{supp}(\pi)$ is c -cyclically monotone.*

Proof. Suppose that $\text{supp}(\pi)$ is not c -cyclically monotone. Then there exist points $(x_i, y_i)_{i=1}^k$ in the support and a permutation σ such that $\sum_i c(x_i, y_i) > \sum_i c(x_i, y_{\sigma(i)})$. By continuity of c , after shrinking neighborhoods $U_i \ni x_i$ and $V_i \ni y_i$, the same strict inequality holds uniformly for every choice of points in these neighborhoods: $\sum_i c(u_i, v_i) > \sum_i c(u_i, \tilde{v}_{\sigma(i)})$ ($u_i \in U_i$, $v_i \in V_i$, $\tilde{v}_{\sigma(i)} \in V_{\sigma(i)}$). Choose the sets so that $m_i := \pi(U_i \times V_i) > 0$. Taking

$$0 < \lambda \leq \left(\sum_{i=1}^k \frac{1}{m_i} \right)^{-1}$$

ensures that the scaled restrictions $\pi_i = \lambda \frac{\pi|_{U_i \times V_i}}{m_i}$ have common mass λ and satisfy $\sum_i \pi_i \leq \pi$, even if some rectangles overlap. Let $\alpha_i = (P_{\mathcal{X}})_\# \pi_i$ and $\beta_i = (P_{\mathcal{Y}})_\# \pi_i$. Define $\tilde{\pi} = \pi - \sum_i \pi_i + \sum_i \frac{\alpha_i \otimes \beta_{\sigma(i)}}{\lambda}$. The removed and reinserted first marginals are both $\sum_i \alpha_i$, and the removed and reinserted second marginals are both $\sum_i \beta_i$ because σ is a permutation. Hence $\tilde{\pi} \in \mathcal{U}(\alpha, \beta)$. Integrating the uniform strict inequality against the product probability $\otimes_i (\pi_i / \lambda)$ shows that the reinserted crossed terms have strictly smaller cost than the removed diagonal terms. This contradicts the optimality of π . $\square$

Monotonicity. Assume the optimal plan is induced by a measurable map $T : \mathcal{X} \rightarrow \mathcal{Y}$, meaning that $\pi = (\text{Id}, T)_\# \alpha$. There is a set $G \subset \mathcal{X}$ of full α -measure such that $(x, T(x)) \in \text{supp}(\pi)$ for every $x \in G$. For any $x_1, \dots, x_k \in G$, cyclical monotonicity reads $\sum_{i=1}^k c(x_i, T(x_i)) \leq \sum_{i=1}^k c(x_i, T(x_{i+1}))$, $x_{k+1} = x_1$. For $c(x, y) = \frac{1}{2} \|x - y\|^2$, taking $k = 2$ gives, for every $x, y \in G$, $\langle T(x) - T(y), x - y \rangle \geq 0$,

One dimension. $|x - T(x)|^p + |y - T(y)|^p \leq |x - T(y)|^p + |y - T(x)|^p$, For $p > 1$, strict convexity implies that a reversed pair $x < y$ and $T(x) > T(y)$ violates this inequality. Hence every optimal map is nondecreasing on the full-measure transport set, recovering the classical monotone rearrangement.

3.5 Metric Properties: Wasserstein Distances

OT defines a distance.

Lemma 3.27 (Discrete gluing lemma). *Given $(a, b, c) \in \Sigma_n \times \Sigma_p \times \Sigma_m$, let $P \in U(a, b)$ and $Q \in U(b, c)$. Then there exists a nonnegative three-way array $S \in \mathbb{R}_+^{n \times p \times m}$ such that its two-way marginals satisfy $\sum_k S_{i,j,k} = P_{i,j}$ and $\sum_i S_{i,j,k} = Q_{j,k}$. $R_{i,k} := \sum_j S_{i,j,k}$, belongs to $U(a, c)$. For the canonical construction below, this glued coupling is the twisted matrix product $R = P \text{diag}(1/b)Q$, $R_{i,k} = \sum_{j: b_j > 0} \frac{P_{i,j} Q_{j,k}}{b_j}$. In the matrix notation, $1/b_j$ is understood as 0 when $b_j = 0$.*

Proof. One verifies that

$$S_{i,j,k} = \begin{cases} \frac{P_{i,j} Q_{j,k}}{b_j} & \text{if } b_j \neq 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.8)$$

has the required marginals. Indeed, if $b_j \neq 0$, $\sum_k S_{i,j,k} = \sum_k \frac{P_{i,j} Q_{j,k}}{b_j} = \frac{P_{i,j}}{b_j} (Q \mathbb{1}_m)_j = \frac{P_{i,j}}{b_j} b_j$. If $b_j = 0$, then necessarily $P_{i,j} = 0$ and $\sum_k S_{i,j,k} = 0 = P_{i,j}$. The same computation gives the other prescribed marginal:

$$\sum_i S_{i,j,k} = \begin{cases} \frac{Q_{j,k}}{b_j} \sum_i P_{i,j} = Q_{j,k} & \text{if } b_j > 0 \\ 0 = Q_{j,k} & \text{if } b_j = 0. \end{cases}$$

Summing over j then gives the displayed formula for R . Its row and column sums are $\sum_k R_{i,k} = \sum_j P_{i,j} = a_i$, $\sum_i R_{i,k} = \sum_j Q_{j,k} = c_k$, so $R \in U(a, c)$. $\square$

When the cost matrix is the p th power of a distance matrix, the discrete Kantorovich value becomes a metric on histograms.

Definition 3.28 (Discrete Wasserstein distance). Let $D \in \mathbb{R}_+^{n \times n}$ be a distance matrix on $\llbracket n \rrbracket$ and let $p \geq 1$. The discrete p -Wasserstein distance between histograms $a, b \in \Sigma_n$ is

$$W_p(a, b) := L_{D^p}(a, b)^{1/p}. \quad (3.9)$$

It depends on the chosen ground distance D .

Proposition 3.29 (Metric property of discrete Wasserstein distance). *For every distance matrix D on $\llbracket n \rrbracket$, Definition 3.28 defines a distance on Σ_n : W_p is symmetric, positive, $W_p(a, b) = 0$ if and only if $a = b$, and it satisfies the triangle inequality $\forall a, b, c \in \Sigma_n$, $W_p(a, c) \leq W_p(a, b) + W_p(b, c)$.*

Proof. Transposition maps $U(a, b)$ bijectively onto $U(b, a)$ and preserves the cost because D is symmetric. This proves symmetry. If $a = b$, the diagonal coupling $\text{diag}(a)$ has zero cost. Conversely, if an admissible coupling has zero cost, every off-diagonal entry must vanish because $D_{ij} > 0$ for $i \neq j$. The coupling is therefore diagonal, and its two marginals coincide. This proves definiteness.

For the triangle inequality, let $P \in U(a, b)$ and $Q \in U(b, c)$ be optimal. Lemma 3.27 provides $S \in \mathbb{R}_+^{n \times n \times n}$ with these two marginals. Its first-third marginal $R_{ik} = \sum_j S_{ijk}$ belongs to $U(a, c)$. The ground-metric triangle inequality and Minkowski's inequality give

$$\begin{aligned} W_p(a, c) &\leq \left(\sum_{i,j,k} D_{ik}^p S_{ijk} \right)^{1/p} \\ &\leq \left(\sum_{i,j,k} (D_{ij} + D_{jk})^p S_{ijk} \right)^{1/p} \\ &\leq \left(\sum_{i,j,k} D_{ij}^p S_{ijk} \right)^{1/p} + \left(\sum_{i,j,k} D_{jk}^p S_{ijk} \right)^{1/p} \\ &= W_p(a, b) + W_p(b, c). \end{aligned}$$

$\square$

Continuous gluing.

Lemma 3.30 (Gluing lemma). *Let $(\alpha, \beta, \gamma) \in \mathcal{M}_+^1(\mathcal{X}) \times \mathcal{M}_+^1(\mathcal{Y}) \times \mathcal{M}_+^1(\mathcal{Z})$, where $\mathcal{X}, \mathcal{Y}$ and $\mathcal{Z}$ are Polish spaces in the sense of Definition 2.3. Given $\pi \in U(\alpha, \beta)$ and $\xi \in U(\beta, \gamma)$, there exists a probability measure $\sigma \in \mathcal{M}_+^1(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ such that $(P_{\mathcal{X}, \mathcal{Y}})_\# \sigma = \pi$ and $(P_{\mathcal{Y}, \mathcal{Z}})_\# \sigma = \xi$ where $P_{\mathcal{X}, \mathcal{Y}}(x, y, z) = (x, y)$ and $P_{\mathcal{Y}, \mathcal{Z}}(x, y, z) = (y, z)$ are the coordinate projections.*

Proof. Disintegrate the two couplings with respect to their common marginal β : $\pi(dx, dy) = \pi_y(dx) \beta(dy)$, $\xi(dy, dz) = \beta(dy) \xi_y(dz)$. The Polish hypotheses guarantee the existence of the probability kernels $y \mapsto \pi_y$ and $y \mapsto \xi_y$. Define their conditional product by $\int g d\sigma := \int_{\mathcal{Y}} \int_{\mathcal{X}} \int_{\mathcal{Z}} g(x, y, z) \xi_y(dz) \pi_y(dx) \beta(dy)$ for every bounded Borel function g . Taking $g(x, y, z) = h(x, y)$ recovers π , while taking $g(x, y, z) = h(y, z)$ recovers ξ , proving the two marginal identities.

In the discrete setting, $\pi_{y_j} = \sum_i (P_{ij}/b_j) \delta_{x_i}$ and $\xi_{y_j} = \sum_k (Q_{jk}/b_j) \delta_{z_k}$ whenever $b_j > 0$. Hence $\sigma = \sum_{i,j,k} S_{ijk} \delta_{(x_i, y_j, z_k)}$, $S_{ijk} = \frac{P_{ij} Q_{jk}}{b_j}$, with the value zero when $b_j = 0$, exactly as in (3.8). $\square$

Definition 3.31 (Wasserstein distance). Let $(\mathcal{X}, d)$ be a Polish metric space and $p \geq 1$. For $\alpha, \beta \in \mathcal{P}_p(\mathcal{X})$, the p -Wasserstein distance is

$$\mathcal{W}_p(\alpha, \beta) := \mathcal{L}_{d^p}(\alpha, \beta)^{1/p} = \left(\inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\pi(x, y) \right)^{1/p}. \quad (3.10)$$

It depends on the ground distance d .

Proposition 3.32 (Euclidean mean bound and quadratic decomposition). Let $p \geq 1$ and $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$, and denote their means by $m_\alpha := \int_{\mathbb{R}^d} x d\alpha(x)$, $m_\beta := \int_{\mathbb{R}^d} x d\beta(x)$. Then $\|m_\alpha - m_\beta\| \leq \mathcal{W}_p(\alpha, \beta)$. For $p = 2$, define the centered translates $\alpha^0 := (x \mapsto x - m_\alpha)_\# \alpha$, $\beta^0 := (y \mapsto y - m_\beta)_\# \beta$. Then the following exact decomposition holds: $\mathcal{W}_2^2(\alpha, \beta) = \|m_\alpha - m_\beta\|^2 + \mathcal{W}_2^2(\alpha^0, \beta^0)$.

Proof. For every $\pi \in \mathcal{U}(\alpha, \beta)$, $m_\alpha - m_\beta = \int_{\mathbb{R}^d \times \mathbb{R}^d} (x - y) d\pi(x, y)$. Jensen's inequality therefore gives

$$\|m_\alpha - m_\beta\| \leq \left(\int \|x - y\|^p d\pi(x, y) \right)^{1/p}.$$

Taking the infimum over π proves the first claim.

For $p = 2$, set $\pi^0 := ((x, y) \mapsto (x - m_\alpha, y - m_\beta))_\# \pi$. This translation defines a bijection from $\mathcal{U}(\alpha, \beta)$ onto $\mathcal{U}(\alpha^0, \beta^0)$. Writing $u = x - m_\alpha$ and $v = y - m_\beta$, one has $\int \|x - y\|^2 d\pi = \|m_\alpha - m_\beta\|^2 + \int \|u - v\|^2 d\pi^0(u, v)$, because $\int (u - v) d\pi^0(u, v) = 0$. Taking the infimum over the corresponding couplings proves the decomposition. $\square$

Proposition 3.33 (Metric property of the Wasserstein distance). Definition 3.31 defines a distance on $\mathcal{P}_p(\mathcal{X})$: $\mathcal{W}_p$ is symmetric and nonnegative, $\mathcal{W}_p(\alpha, \beta) = 0$ if and only if $\alpha = \beta$, and $\forall (\alpha, \beta, \gamma) \in \mathcal{P}_p(\mathcal{X})^3$, $\mathcal{W}_p(\alpha, \gamma) \leq \mathcal{W}_p(\alpha, \beta) + \mathcal{W}_p(\beta, \gamma)$.

Proof. The value is finite on $\mathcal{P}_p(\mathcal{X})$ because the product coupling has finite p -cost. If $\tau(x, y) = (y, x)$ is the flip map, then $\tau_\# \pi \in \mathcal{U}(\beta, \alpha)$ whenever $\pi \in \mathcal{U}(\alpha, \beta)$, and the transport cost is unchanged. This proves symmetry. The diagonal coupling $(\text{Id}, \text{Id})_\# \alpha$ proves $\mathcal{W}_p(\alpha, \alpha) = 0$.

Conversely, Proposition 3.22 supplies an optimal coupling π . If $\mathcal{W}_p(\alpha, \beta) = 0$, then $d(x, y) = 0$ for π -almost every (x, y) , so $x = y$ almost surely under π . For every bounded Borel function f , $\int f d\alpha = \int f(x) d\pi(x, y) = \int f(y) d\pi(x, y) = \int f d\beta$, and therefore $\alpha = \beta$.

For the triangle inequality, consider optimal couplings $\pi \in \mathcal{U}(\alpha, \beta)$ and $\xi \in \mathcal{U}(\beta, \gamma)$, and glue them into σ by Lemma 3.30. Let $P_{1,3}(x, y, z) = (x, z)$ and define $\eta := (P_{1,3})_\# \sigma$. Then $\eta \in \mathcal{U}(\alpha, \gamma)$. If the two original couplings are induced by maps T_1 and T_2 , this construction is induced by $T_2 \circ T_1$; gluing therefore extends composition of deterministic maps. The triangle inequality and Minkowski's inequality give

$$\begin{aligned} \mathcal{W}_p(\alpha, \gamma) &\leq \left(\int d(x, z)^p d\sigma(x, y, z) \right)^{1/p} \\ &\leq \left(\int (d(x, y) + d(y, z))^p d\sigma(x, y, z) \right)^{1/p} \\ &\leq \left(\int d(x, y)^p d\sigma(x, y, z) \right)^{1/p} + \left(\int d(y, z)^p d\sigma(x, y, z) \right)^{1/p} \\ &= \left(\int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\pi(x, y) \right)^{1/p} + \left(\int_{\mathcal{X} \times \mathcal{X}} d(y, z)^p d\xi(y, z) \right)^{1/p} \\ &= \mathcal{W}_p(\alpha, \beta) + \mathcal{W}_p(\beta, \gamma). \end{aligned}$$

$\square$

Interpolation induced by an optimal plan.

Definition 3.34 ($\mathcal{W}_2$ geodesic induced by an optimal plan). Let $\alpha_0, \alpha_1 \in \mathcal{P}_2(\mathbb{R}^d)$, and let $\pi^* \in \mathcal{U}(\alpha_0, \alpha_1)$ be optimal for $\mathcal{W}_2^2(\alpha_0, \alpha_1)$. For $t \in [0, 1]$, define $e_t(x, y) := (1 - t)x + ty$, $\alpha_t := (e_t)_\# \pi^*$. The curve $(\alpha_t)_{t \in [0, 1]}$ is the displacement, or McCann, $\mathcal{W}_2$ geodesic induced by π^* .

Proposition 3.35 (Optimal-plan interpolation is a $\mathcal{W}_2$ geodesic). Let $(\alpha_t)_{t \in [0, 1]}$ be defined by Definition 3.34. Then, for every $0 \leq s \leq t \leq 1$, $\mathcal{W}_2(\alpha_s, \alpha_t) = (t - s) \mathcal{W}_2(\alpha_0, \alpha_1)$. Thus $t \mapsto \alpha_t$ is a constant-speed geodesic for the metric $\mathcal{W}_2$.

Proof. Push the optimal plan π^* forward by (e_s, e_t) . $\int \|z - z'\|^2 d\gamma_{s,t}(z, z') = \int \|e_t(x, y) - e_s(x, y)\|^2 d\pi^*(x, y) = (t - s)^2 \mathcal{W}_2^2(\alpha_0, \alpha_1)$. Hence $\mathcal{W}_2(\alpha_s, \alpha_t) \leq (t - s) \mathcal{W}_2(\alpha_0, \alpha_1)$. Applying this upper bound to the three pairs $(0, s)$, (s, t) and $(t, 1)$, and using the triangle inequality of Proposition 3.33, gives $\mathcal{W}_2(\alpha_0, \alpha_1) \leq \mathcal{W}_2(\alpha_0, \alpha_s) + \mathcal{W}_2(\alpha_s, \alpha_t) + \mathcal{W}_2(\alpha_t, \alpha_1) \leq \mathcal{W}_2(\alpha_0, \alpha_1)$. All inequalities are therefore equalities, in particular the middle segment has the claimed length. $\square$

General geodesic spaces. For Dirac masses in Euclidean space, the $\mathcal{W}_2$ geodesic from δ_x to δ_y is $t \mapsto \delta_{(1-t)x + ty}$. The same idea extends to any geodesic metric space $(\mathcal{X}, d)$, meaning that each pair of points can be joined by a constant-speed metric geodesic. For each pair (x, y) , one replaces the Euclidean segment by a curve $\gamma^{x,y} : [0, 1] \rightarrow \mathcal{X}$ such that $\gamma_0^{x,y} = x$, $\gamma_1^{x,y} = y$, and $d(\gamma_s^{x,y}, \gamma_t^{x,y}) = |t - s|d(x, y)$.

Comparison with Monge.

Proposition 3.36 (Kantorovich as the plan-space relaxation of Monge). Let $(\mathcal{X}, d)$ be a compact metric space, let $p \geq 1$, and let $\alpha, \beta \in \mathcal{P}(\mathcal{X})$ with α atomless. Then $\mathcal{W}_p(\alpha, \beta)^p = \inf_{T_\# \alpha = \beta} \int_{\mathcal{X}} d(x, T(x))^p d\alpha(x)$. The infimum on the Monge side need not be attained.

Proof. Let $\pi \in \mathcal{U}(\alpha, \beta)$. Choose finite Borel partitions $(A_i)_i$ and $(B_j)_j$ of $\mathcal{X}$ whose cells have diameter at most ε . Set $m_{ij} = \pi(A_i \times B_j)$. Since α is atomless and $\sum_j m_{ij} = \alpha(A_i)$, each A_i can be partitioned into Borel sets A_{ij} with $\alpha(A_{ij}) = m_{ij}$. If $m_{ij} > 0$, then $\beta(B_j) > 0$, and Proposition 2.11 gives a measurable map $T_{ij} : A_{ij} \rightarrow B_j$, $(T_{ij})_{\#} \frac{\alpha|_{A_{ij}}}{m_{ij}} = \frac{\beta|_{B_j}}{\beta(B_j)}$. Define T by $T = T_{ij}$ on A_{ij} , ignoring null cells. Then $T_{\#}\alpha = \beta$, because the contributions landing in B_j sum to $\beta|_{B_j}$, and the graph coupling $(\text{Id}, T)_{\#}\alpha$ has the same masses m_{ij} as π on all rectangles $A_i \times B_j$. Let $h \in \mathcal{C}(\mathcal{X} \times \mathcal{X})$. By uniform continuity on the compact product, the oscillation of h on every rectangle $A_i \times B_j$ goes to zero as $\varepsilon \rightarrow 0$. Since π and $(\text{Id}, T)_{\#}\alpha$ have the same mass on each rectangle, their integrals against h differ by at most this maximal oscillation. Taking partitions with mesh tending to zero therefore gives graph couplings $(\text{Id}, T_k)_{\#}\alpha$ converging weakly to π . Applying this to the continuous function $(x, y) \mapsto d(x, y)^p$ gives the convergence of costs. Define

$$\mathcal{G}(\alpha, \beta) := \{(\text{Id}, T)_{\#}\alpha ; T_{\#}\alpha = \beta\} \subset \mathcal{U}(\alpha, \beta), \quad F_p(\pi) := \int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\pi(x, y),$$

and let G be the Monge graph functional which equals F_p on $\mathcal{G}(\alpha, \beta)$ and $+\infty$ otherwise. The function F_p is weakly continuous on $\mathcal{U}(\alpha, \beta)$ and satisfies $F_p \leq G$. If H is any weakly lower-semicontinuous function with $H \leq G$, then for the graph approximation above, $H(\pi) \leq \liminf_k H((\text{Id}, T_k)_{\#}\alpha) \leq \lim_k F_p((\text{Id}, T_k)_{\#}\alpha) = F_p(\pi)$. Thus F_p is the largest weakly lower-semicontinuous minorant of G , that is, the lower-semicontinuous envelope. Minimizing this envelope over all couplings gives the Kantorovich value, and the density argument gives the same infimum when restricting to graph couplings. $\square$

Corollary 3.37 (Lower-semicontinuous envelope of the Monge p -cost). *Let $(\mathcal{X}, d)$ be a compact metric space and assume that atomless probability measures are dense in $\mathcal{P}(\mathcal{X})$ for the $\mathcal{W}_p$ topology. Then $(\alpha, \beta) \mapsto \mathcal{W}_p(\alpha, \beta)^p$ is the lower-semicontinuous envelope, on $\mathcal{P}(\mathcal{X}) \times \mathcal{P}(\mathcal{X})$ for the product $\mathcal{W}_p$ topology, of the extended Monge cost*

$$\tilde{\mathcal{W}}_p(\alpha, \beta)^p = \inf_{\substack{T: \mathcal{X} \rightarrow \mathcal{X} \\ T_{\#}\alpha = \beta}} \int_{\mathcal{X}} d(x, T(x))^p d\alpha(x),$$

with the convention $\tilde{\mathcal{W}}_p(\alpha, \beta)^p = +\infty$ if no admissible map exists.

Proof. For every admissible map T , the graph plan $(\text{Id}, T)_{\#}\alpha$ is a coupling, hence $\mathcal{W}_p(\alpha, \beta)^p \leq \tilde{\mathcal{W}}_p(\alpha, \beta)^p$. Since $\mathcal{W}_p^p$ is continuous for the product $\mathcal{W}_p$ topology, it is a lower-semicontinuous minorant of the extended Monge cost. Conversely, let H be any lower-semicontinuous minorant of the extended Monge cost. Fix (α, β) and choose atomless $\alpha_k \rightarrow \alpha$ in $\mathcal{W}_p$. By Proposition 3.36, $\tilde{\mathcal{W}}_p(\alpha_k, \beta)^p = \mathcal{W}_p(\alpha_k, \beta)^p$. Therefore $H(\alpha, \beta) \leq \liminf_k H(\alpha_k, \beta) \leq \lim_k \mathcal{W}_p(\alpha_k, \beta)^p = \mathcal{W}_p(\alpha, \beta)^p$. Thus no larger lower-semicontinuous minorant exists. $\square$

3.6 Metric Properties: Topology and Applications

Convergence in law topology. On a bounded metric space, all $\mathcal{W}_p$ distances define the same topology, although they are not equivalent as distances.

Proposition 3.38 (Comparison of Wasserstein distances on bounded spaces). *Let $(\mathcal{X}, d)$ be a bounded Polish metric space, let $1 \leq p \leq q < \infty$, and let $\alpha, \beta \in \mathcal{P}_q(\mathcal{X})$. Then*

$$\mathcal{W}_p(\alpha, \beta) \leq \mathcal{W}_q(\alpha, \beta) \leq \text{diam}(\mathcal{X})^{\frac{q-p}{q}} \mathcal{W}_p(\alpha, \beta)^{\frac{p}{q}}, \quad \text{diam}(\mathcal{X}) := \sup_{x, y \in \mathcal{X}} d(x, y).$$

Proof. The left inequality follows from Jensen's inequality, applied to any coupling π and to the convex function $r \mapsto r^{q/p}$: $(\int d(x, y)^p d\pi(x, y))^{\frac{q}{p}} \leq \int d(x, y)^q d\pi(x, y)$. Taking the infimum gives $\mathcal{W}_p \leq \mathcal{W}_q$. The right inequality follows from $d(x, y)^q \leq \text{diam}(\mathcal{X})^{q-p} d(x, y)^p$ and taking the infimum over couplings once more. $\square$

Definition 3.39 (Weak, or narrow, topology). A sequence $(\alpha_k)_k$ converges weakly, or narrowly, to α in $\mathcal{M}_+^1(\mathcal{X})$, denoted $\alpha_k \rightharpoonup \alpha$, if and only if $\int_{\mathcal{X}} f d\alpha_k \rightarrow \int_{\mathcal{X}} f d\alpha$ for every $f \in \mathcal{C}_b(\mathcal{X})$. On a compact space, $\mathcal{C}_b(\mathcal{X}) = \mathcal{C}(\mathcal{X})$ and this is precisely weak* convergence in the duality between measures and continuous functions. The term weak* is therefore also used informally in the book when no confusion can arise.

In terms of random vectors, if $X_n \sim \alpha_n$ and $X \sim \alpha$ (not necessarily defined on the same probability space), weak convergence corresponds to convergence in law of X_n toward X .

Strong versus weak topology. $\|\alpha - \beta\|_{\text{TV}} = |\alpha - \beta|(\mathcal{X})$,

Proposition 3.40 (Total variation as Wasserstein for the discrete metric). *Let $\mathcal{X}$ be a standard Borel space, let α and β be probability measures on $\mathcal{X}$, and equip $\mathcal{X}$ with the 0/1 cost $d_0(x, x) = 0$ and $d_0(x, y) = 1$ for $x \neq y$. Then, for every $p \geq 1$, $\inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int d_0(x, y)^p d\pi(x, y) = \frac{1}{2} \|\alpha - \beta\|_{\text{TV}}$. Whenever the 0/1 metric is used as the ground metric, the left-hand side is $\mathcal{W}_p(\alpha, \beta)^p$.*

Proof. Let $\lambda = \alpha + \beta$, write $a = d\alpha/d\lambda$ and $b = d\beta/d\lambda$, and define the common part η by $\frac{d\eta}{d\lambda} = \min(a, b)$. Its residuals $\alpha' = \alpha - \eta$ and $\beta' = \beta - \eta$ are mutually singular and have the same mass $r = 1 - \eta(\mathcal{X}) = \frac{1}{2} \int |a - b| d\lambda = \frac{1}{2} \|\alpha - \beta\|_{\text{TV}}$. For any $\pi \in \mathcal{U}(\alpha, \beta)$, define its diagonal submeasure by $\zeta(A) = \pi(\{(x, x) : x \in A\})$. Then $\zeta \leq \alpha$ and $\zeta \leq \beta$, so its density with respect to λ is bounded by $\min(a, b)$. $\int d_0(x, y)^p d\pi(x, y) = 1 - \pi(\{x = y\}) \geq r$. If $r = 0$, the diagonal coupling attains this bound. If $r > 0$, let $\Delta(x) = (x, x)$ and set $\pi^* = \Delta_{\#}\eta + \frac{1}{r} \alpha' \otimes \beta'$. This is a coupling of α and β . Since $\alpha' \perp \beta'$, the product term is concentrated off the diagonal, so its 0/1 cost is exactly its total mass r . Thus π^* attains the lower bound. $\square$

$\|\delta_{x_n} - \delta_x\|_{\text{TV}} = 2$ and $\mathcal{W}_p(\delta_{x_n}, \delta_x) = d(x_n, x)$.

Proposition 3.41 (Wasserstein convergence on Polish spaces). *Let $(\mathcal{X}, d)$ be Polish, $1 \leq p < +\infty$, and $\alpha_k, \alpha \in \mathcal{P}_p(\mathcal{X})$. Then $\mathcal{W}_p(\alpha_k, \alpha) \rightarrow 0$ if and only if $\alpha_k \rightharpoonup \alpha$ and, for one (equivalently every) $x_0 \in \mathcal{X}$, $\int d(x, x_0)^p d\alpha_k(x) \rightarrow \int d(x, x_0)^p d\alpha(x)$. Equivalently, weak convergence holds and $\lim_{R \rightarrow +\infty} \sup_k \int_{\{d(x, x_0) > R\}} d(x, x_0)^p d\alpha_k(x) = 0$.*

Proof. Wasserstein convergence implies weak convergence and moment convergence by the reverse triangle inequality in L^p under a coupling. Conversely, represent weak convergence by random variables $X_k \rightarrow X$ almost surely. Moment convergence makes $d(X_k, x_0)^p$ uniformly integrable, hence $\mathbb{E}d(X_k, X)^p \rightarrow 0$, which bounds $\mathcal{W}_p^p$. Truncating $d(\cdot, x_0)^p$ proves the equivalent tail criterion. $\square$

On compact spaces the moment condition is automatic, so $\mathcal{W}_p$ metrizes weak convergence.

Proposition 3.42 (Comparison with total variation on finite spaces). *Let $(\mathcal{X}, d)$ be a finite metric space and define $d_{\min} := \min_{x \neq y} d(x, y) > 0$, $d_{\max} := \max_{x, y} d(x, y) < +\infty$. Then, for every $\alpha, \beta \in \mathcal{P}(\mathcal{X})$, $\frac{d_{\min}}{2} \|\alpha - \beta\|_{\text{TV}} \leq \mathcal{W}_1(\alpha, \beta) \leq \frac{d_{\max}}{2} \|\alpha - \beta\|_{\text{TV}}$.*

Proof. We denote $d_0(x, y)$ the distance such that $d_0(x, x) = 0$ and $d_0(x, y) = 1$ for $x \neq y$. One has $d_{\min} d_0(x, y) \leq d(x, y) \leq d_{\max} d_0(x, y)$ so that integrating this against any $\pi \in \mathcal{U}(\alpha, \beta)$ and taking the minimum among those π gives the result using Proposition 3.40. $\square$

$$d_{\min} \leq d(x, y) \leq d_{\max}.$$

3.7 Measure-to-Measure Maps on Wasserstein Space

Maps on Wasserstein space. $\Phi : \mathcal{P}_p(\mathcal{X}) \rightarrow \mathcal{P}_p(\mathcal{X})$

Particle-preserving transport representations.

$$\Phi(\alpha) = \Gamma[\alpha]_{\#} \alpha, \quad \Gamma[\alpha] : \mathcal{X} \rightarrow \mathcal{X}. \quad (3.11)$$

$$\alpha = \sum_{i=1}^n a_i \delta_{x_i} \implies \Phi(\alpha) = \sum_{i=1}^n a_i \delta_{\Gamma[\alpha](x_i)}. \quad (3.12)$$

Mass-splitting Markov maps. The opposite case is a stochastic transformation, where one input particle can generate a full output distribution. Let K be a Markov kernel on $\mathcal{X}$, so that $K(y, \cdot)$ is a probability measure for each y .

$$\int_{\mathcal{X}} d(x, x_0)^p K(y, dx) \leq C(1 + d(y, x_0)^p)$$

$$\int_{\mathcal{X}} f(x) d\Psi(\alpha)(x) = \int_{\mathcal{X}} \int_{\mathcal{X}} f(x) K(y, dx) d\alpha(y) \quad (3.13)$$

If, in addition, $\mathcal{X}$ is Polish and $\mathcal{W}_p(K(y, \cdot), K(y', \cdot)) \leq Ld(y, y')$,

Wasserstein stability.

Proposition 3.43 (Wasserstein stability of transport representations). *Let $(\mathcal{X}, d)$ be Polish, let $p \geq 1$, let $E \subset \mathcal{X}$, and assume that, for all $\alpha, \beta \in \mathcal{P}_p(\mathcal{X})$ supported in E , the maps $T[\alpha] : E \rightarrow \mathcal{X}$ satisfy $d(T[\alpha](x), T[\alpha](y)) \leq L_x d(x, y)$ for all $x, y \in E$, and $\sup_{y \in E} d(T[\alpha](y), T[\beta](y)) \leq L_{\text{law}} \mathcal{W}_p(\alpha, \beta)$. Then $\Phi(\alpha) = T[\alpha]_{\#} \alpha$ is $(L_x + L_{\text{law}})$ -Lipschitz on probability measures supported in E : $\mathcal{W}_p(\Phi(\alpha), \Phi(\beta)) \leq (L_x + L_{\text{law}}) \mathcal{W}_p(\alpha, \beta)$.*

Proof. Let π be an optimal coupling between α and β . The measure $(T[\alpha], T[\beta])_{\#} \pi$ is a coupling between $\Phi(\alpha)$ and $\Phi(\beta)$, hence

$$\mathcal{W}_p(\Phi(\alpha), \Phi(\beta)) \leq \left(\int d(T[\alpha](x), T[\beta](y))^p d\pi(x, y) \right)^{1/p}.$$

By the triangle inequality, $d(T[\alpha](x), T[\beta](y)) \leq L_x d(x, y) + L_{\text{law}} \mathcal{W}_p(\alpha, \beta)$. Minkowski's inequality gives the claim. $\square$

$$\mathcal{W}_p(T_{\#} \alpha, T_{\#} \beta) \leq L \mathcal{W}_p(\alpha, \beta).$$

Mean-field attention. Self-attention is a central example because the number of tokens can be large and variable. A token cloud is represented by the empirical law $\alpha = n^{-1} \sum_i \delta_{x_i}$, and a single-head mean-field attention layer is naturally a transport representation.

$$\Gamma_{\theta}[\alpha](x) = \frac{\int e^{(Qx, Kz)} V z d\alpha(z)}{\int e^{(Qx, Kz)} d\alpha(z)}, \quad \theta = (Q, K, V), \quad (3.14)$$

where, for simplicity, the value map V takes values in the same Euclidean feature space. The corresponding measure map is $\text{Att}_{\theta}(\alpha) = (\Gamma_{\theta}[\alpha])_{\#} \alpha$,

Proposition 3.44 (Compact-support attention stability). *Assume $\mathcal{X} = \mathbb{R}^d$. Fix $R > 0$ and let $\mathcal{P}_R$ be the set of probability measures supported in the Euclidean ball $B(0, R)$. For every $p \geq 1$, there is a constant $C_{\theta, p}(R)$ such that $\mathcal{W}_p(\text{Att}_{\theta}(\alpha), \text{Att}_{\theta}(\beta)) \leq C_{\theta, p}(R) \mathcal{W}_p(\alpha, \beta)$, $\alpha, \beta \in \mathcal{P}_R$. Writing $A_R = \|Q\|_{\text{op}} \|K\|_{\text{op}} R^2$, one can take a constant bounded, up to polynomial factors in R and the operator norms of Q, K, V , by $C_{\theta, p}(R) \lesssim e^{2A_R}$. This is the exponential-in-score-radius behavior, equivalently $e^{O(R^2)}$ for dot-product scores on $B(0, R)$, refined in.*

Proof. Write $A_R = \|Q\|_{\text{op}} \|K\|_{\text{op}} R^2$. For $x, z \in B(0, R)$, the attention weight $a_x(z) = e^{(Qx, Kz)}$ satisfies $e^{-A_R} \leq a_x(z) \leq e^{A_R}$. Moreover the functions a_x and $a_x V z$ have Lipschitz constants bounded by e^{A_R} times polynomial factors in R and $\|Q\|_{\text{op}}, \|K\|_{\text{op}}, \|V\|_{\text{op}}$. By Kantorovich–Rubinstein duality,

$$\left| \int a_x d(\alpha - \beta) \right| + \left\| \int a_x V z d(\alpha - \beta) \right\| \leq C_{\theta}(R) e^{A_R} \mathcal{W}_1(\alpha, \beta).$$

Since the denominator in (3.14) is at least e^{-A_R} , the quotient rule gives $\sup_{x \in B(0, R)} \|\Gamma_{\theta}[\alpha](x) - \Gamma_{\theta}[\beta](x)\| \leq C_{\theta}(R) e^{2A_R} \mathcal{W}_1(\alpha, \beta)$. The same differentiation of the quotient with respect to x gives a spatial Lipschitz bound for $x \mapsto \Gamma_{\theta}[\alpha](x)$ on $B(0, R)$, uniformly in $\alpha \in \mathcal{P}_R$. Since $\mathcal{W}_1 \leq \mathcal{W}_p$ on probability measures and the output remains in the ball $B(0, \|V\|_{\text{op}} R)$, Proposition 3.43, applied with $E = B(0, R)$, proves the estimate. $\square$

3.8 Distributional Robustness and $\mathcal{W}_\infty$

DRO ambiguity sets. Wasserstein distances are also used to define ambiguity sets around an empirical law. Given samples z_i and $\hat{\alpha}_n = \frac{1}{n} \sum_i \delta_{z_i}$, a distributionally robust optimization (DRO) problem replaces the empirical risk $\frac{1}{n} \sum_i \ell_\theta(z_i)$ by $\sup_{\beta: \mathcal{W}_p(\beta, \hat{\alpha}_n) \leq \rho} \int \ell_\theta(z) d\beta(z)$,

$$\sup_{\beta: \mathcal{W}_p(\beta, \hat{\alpha}_n)^p \leq \rho^p} \int \ell_\theta d\beta = \inf_{\lambda \geq 0} \lambda \rho^p + \frac{1}{n} \sum_{i=1}^n \sup_z \{ \ell_\theta(z) - \lambda d(z, z_i)^p \}. \quad (3.15)$$

Thus the robust risk is an empirical risk in which each sample is replaced by its worst penalized perturbation. For $p = 1$ and an L_θ -Lipschitz loss, the Kantorovich–Rubinstein dual gives the transparent upper bound

$\sup_{\beta: \mathcal{W}_1(\beta, \hat{\alpha}_n) \leq \rho} \int \ell_\theta d\beta \leq \frac{1}{n} \sum_i \ell_\theta(z_i) + \rho L_\theta$. Applied to $c = d^p$, Proposition 3.21 shows that $(\alpha, \beta) \mapsto \mathcal{W}_p(\alpha, \beta)^p$ is jointly convex. Thus $\mathcal{W}_1$ itself is jointly convex, but $\mathcal{W}_p$ need not be convex for $p > 1$: on the real line,

$$\mathcal{W}_p((1-t)\delta_0 + t\delta_1, \delta_0) = t^{1/p}, \quad \theta \mapsto \sup_{\beta: \mathcal{W}_p(\beta, \hat{\alpha}_n) \leq \rho} \int \ell_\theta d\beta$$

$\mathcal{W}_\infty$ robustness.

$$\mathcal{W}_\infty(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \text{ess sup}_{(x, y) \sim \pi} d(x, y) \quad (3.16)$$

$$\mathcal{W}_\infty(\alpha, \beta) \leq r \iff \text{there exists } \pi \in \mathcal{U}(\alpha, \beta) \text{ supported on } \{(x, y) : d(x, y) \leq r\}.$$

Thus $\mathcal{W}_\infty$ can be approached by threshold search over support-constrained coupling problems.

Proposition 3.45 ($\mathcal{W}_\infty$ robust envelope around an empirical law). *Let $(\mathcal{Z}, d)$ be a Polish metric space. Let $\hat{\alpha} = \sum_{i=1}^n a_i \delta_{z_i}$ with distinct z_i , positive a_i , and $\sum_i a_i = 1$, and assume that the closed balls $\bar{B}(z_i, \rho)$ are compact. Repeated atoms can always be merged to meet this convention. For any real-valued upper-semicontinuous loss ℓ , $\sup_{\beta: \mathcal{W}_\infty(\beta, \hat{\alpha}) \leq \rho} \int \ell(z) d\beta(z) = \sum_{i=1}^n a_i \sup_{z \in \bar{B}(z_i, \rho)} \ell(z)$.*

Proof. If $\mathcal{W}_\infty(\beta, \hat{\alpha}) \leq \rho$, then, by symmetry, there are couplings $\pi_m \in \mathcal{U}(\hat{\alpha}, \beta)$ whose essential displacements are at most $\rho + 1/m$. Since the two marginals are fixed probability measures on a Polish space, $\mathcal{U}(\hat{\alpha}, \beta)$ is tight and closed, hence weakly compact by Prokhorov's theorem. After extraction, $\pi_m \rightharpoonup \pi \in \mathcal{U}(\hat{\alpha}, \beta)$. For every $\eta > 0$, the closed set $F_\eta = \{(x, z) : d(x, z) \leq \rho + \eta\}$ has $\pi_m(F_\eta) = 1$ for all sufficiently large m . Portmanteau's theorem gives $\pi(F_\eta) = 1$. Letting $\eta \downarrow 0$ along a countable sequence gives $\pi(\{(x, z) : d(x, z) \leq \rho\}) = 1$. Disintegrating π with respect to the first marginal gives $\pi = \sum_i a_i \delta_{z_i} \otimes \nu_i$, where each ν_i is supported in the closed ball $\bar{B}(z_i, \rho)$ and $\beta = \sum_i a_i \nu_i$. Hence $\int \ell d\beta = \sum_i a_i \int \ell d\nu_i \leq \sum_i a_i \sup_{\bar{B}(z_i, \rho)} \ell$. The reverse inequality follows by choosing, for each i , a maximizer $z_i^* \in \bar{B}(z_i, \rho)$ and setting $\beta = \sum_i a_i \delta_{z_i^*}$. The coupling $\sum_i a_i \delta_{(z_i, z_i^*)}$ has essential displacement at most ρ , so this β is feasible and attains the displayed value. $\square$

4 Dual Problem

4.1 Discrete dual

The Kantorovich problem (3.2) is a linear program so that one can equivalently compute its value by solving a dual linear program.

Definition 4.1 (Admissible potentials). For a discrete problem with marginal sizes n, m and cost matrix $C \in \mathbb{R}^{n \times m}$, a pair $(f, g) \in \mathbb{R}^n \times \mathbb{R}^m$ is admissible if it lies below the cost:

$$R(C) := \{(f, g) \in \mathbb{R}^n \times \mathbb{R}^m ; \forall (i, j) \in [n] \times [m], f_i + g_j \leq C_{i,j}\}. \quad (4.1)$$

Equivalently, $f \oplus g \leq C$ entrywise. In particular, the feasible set depends on the cost matrix, not on the marginal weights.

Proposition 4.2 (Discrete Kantorovich duality). *One has*

$$L_C(a, b) = \max_{(f, g) \in R(C)} \langle f, a \rangle + \langle g, b \rangle. \quad (4.2)$$

Proof. For completeness, introduce multipliers (f, g) for the two marginal constraints. The Lagrangian is

$$\mathcal{L}(P, f, g) = \langle C, P \rangle + \langle a - P \mathbf{1}_m, f \rangle + \langle b - P^\top \mathbf{1}_n, g \rangle. \quad (4.3)$$

For fixed potentials, the dual function is $\inf_{P \geq 0} \mathcal{L}(P, f, g) = \langle a, f \rangle + \langle b, g \rangle + \inf_{P \geq 0} \langle C - (f \oplus g), P \rangle$. The last infimum is

$$\inf_{P \geq 0} \langle Q, P \rangle = \begin{cases} 0 & \text{if } Q \geq 0, \\ -\infty & \text{otherwise,} \end{cases}$$

and is therefore finite exactly when $(f, g) \in R(C)$. The primal polytope is nonempty because it contains the product coupling $a \otimes b$, and it is compact; hence the primal value is finite and attained. Finite-dimensional linear-programming strong duality then identifies this value with the maximum of the dual function and also gives dual attainment. $\square$

Proposition 4.3 (Discrete complementary slackness). *For every $P \in \mathcal{U}(a, b)$ and $(f, g) \in R(C)$, $\langle C, P \rangle - \langle f, a \rangle - \langle g, b \rangle = \sum_{i,j} P_{i,j} (C_{i,j} - f_i - g_j) \geq 0$. The plan and potentials are respectively primal and dual optimal if and only if*

$$\text{Supp}(P) \subset \{(i, j) \in [n] \times [m] ; f_i + g_j = C_{i,j}\}. \quad (4.4)$$

Proof. The marginal constraints give $\langle f, a \rangle + \langle g, b \rangle = \sum_{i,j} P_{i,j} (f_i + g_j)$, which proves the identity and nonnegativity. If both feasible objects are optimal, their values agree by Proposition 4.2; every nonnegative summand with $P_{i,j} > 0$ must therefore vanish. Conversely, the support condition makes the gap zero, so weak duality forces both objects to be optimal. $\square$

4.2 General formulation

Continuous duality.

Proposition 4.4 (Kantorovich duality). *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$. Then*

$$\mathcal{L}_c(\alpha, \beta) = \max_{(f, g) \in \mathcal{R}(c)} \int_{\mathcal{X}} f(x) d\alpha(x) + \int_{\mathcal{Y}} g(y) d\beta(y), \quad (4.5)$$

where the set of admissible dual potentials is

$$\mathcal{R}(c) := \{(f, g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y}) ; \forall (x, y), f(x) + g(y) \leq c(x, y)\}. \quad (4.6)$$

Here, (f, g) is a pair of continuous functions, often called “Kantorovich potentials”. The same formula extends under the usual lower-semicontinuity and integrability assumptions, replacing maxima by suprema when dual optimizers need not exist.

Proof. Weak duality is immediate: if $f(x) + g(y) \leq c(x, y)$ and $\pi \in \mathcal{U}(\alpha, \beta)$, then $\int f d\alpha + \int g d\beta = \int (f(x) + g(y)) d\pi(x, y) \leq \int c d\pi$. Taking the supremum over admissible potentials and the infimum over couplings gives the first inequality.

For the reverse inequality, write $V = \mathcal{L}_c(\alpha, \beta)$ and consider the convex cone

$$\mathcal{K} := \left\{ \left(\pi_1, \pi_2, \int c d\pi + r \right) : \pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y}), r \geq 0 \right\}$$

in $\mathcal{M}(\mathcal{X}) \times \mathcal{M}(\mathcal{Y}) \times \mathbb{R}$, equipped with the weak topologies on the measure factors. This cone is closed. Indeed, convergence of the first marginal implies a uniform bound on the masses of the underlying nonnegative measures. Compactness of $\mathcal{X} \times \mathcal{Y}$ then gives a convergent subnet $\pi_k \rightarrow \pi$; continuity of marginalization and of $\pi \mapsto \int c d\pi$ identifies the limit with an element of $\mathcal{K}$, while the limiting epigraph remainder remains nonnegative.

For every $t < V$, the point $z_t = (\alpha, \beta, t)$ does not belong to $\mathcal{K}$. Strong separation of this point from the closed convex cone gives a continuous linear functional. Because $\mathcal{K}$ is a cone containing the origin, the functional can be oriented so that, for some $F \in \mathcal{C}(\mathcal{X})$, $G \in \mathcal{C}(\mathcal{Y})$ and $\lambda \in \mathbb{R}$, $\int F d\alpha + \int G d\beta + \lambda t < 0 \leq \int F d\eta + \int G d\zeta + \lambda s$ for all $(\eta, \zeta, s) \in \mathcal{K}$. Testing $(0, 0, r) \in \mathcal{K}$ gives $\lambda \geq 0$, while testing the element generated by $\pi = \delta_{(x, y)}$ and $r = 0$ gives $F(x) + G(y) + \lambda c(x, y) \geq 0$. The multiplier cannot vanish: if $\lambda = 0$, integrating this last inequality against $\alpha \otimes \beta$ would contradict the strict inequality at z_t . Thus $\lambda > 0$, and $f := -F/\lambda$, $g := -G/\lambda$ satisfy $f \oplus g \leq c$ and $\int f d\alpha + \int g d\beta > t$. Letting $t \uparrow V$ proves the reverse inequality.

Take a maximizing sequence (f_k, g_k) . Successively replace it by $\tilde{g}_k(y) = \min_x \{c(x, y) - f_k(x)\}$, $\tilde{f}_k(x) = \min_y \{c(x, y) - \tilde{g}_k(y)\}$. Each replacement preserves feasibility and cannot decrease the objective. Fixing $x_0 \in \mathcal{X}$, use the gauge freedom $(\tilde{f}_k, \tilde{g}_k) \mapsto (\tilde{f}_k - a_k, \tilde{g}_k + a_k)$ to impose $\tilde{f}_k(x_0) = 0$. Uniform continuity of c shows that $\tilde{g}_k$ inherits a common modulus in the second variable and $\tilde{f}_k$ a common modulus in the first. The gauge gives a uniform bound on $\tilde{f}_k$, and the first envelope then gives one on $\tilde{g}_k$. Arzelà–Ascoli yields uniformly convergent subsequences. Their limit remains admissible and attains the dual supremum. $\square$

Complementary slackness.

Proposition 4.5 (Continuous complementary slackness). *For $\pi \in \mathcal{U}(\alpha, \beta)$ and $(f, g) \in \mathcal{R}(c)$, $\int c d\pi - \int f d\alpha - \int g d\beta = \int (c(x, y) - f(x) - g(y)) d\pi(x, y) \geq 0$. The coupling and potentials are respectively primal and dual optimal if and only if this gap vanishes. In that case,*

$$\text{Supp}(\pi) \subset \{(x, y) \in \mathcal{X} \times \mathcal{Y} ; f(x) + g(y) = c(x, y)\}. \quad (4.7)$$

Proof. The marginal identities give the equality, and admissibility gives nonnegativity. A zero gap is equivalent to equality of the feasible primal and dual values, hence to joint optimality by Proposition 4.4. Finally, the slack is continuous and nonnegative. If it were positive at a point of $\text{Supp}(\pi)$, it would be bounded below by a positive constant on a neighborhood of positive π -mass, contradicting its zero integral. $\square$

$s(x, y) := c(x, y) - f(x) - g(y)$. Dual feasibility is the statement $s \geq 0$, while complementary slackness says that an optimal coupling can only live where this slack vanishes.

4.3 c -transforms

Best-response potentials and the c -transform. $\sup_{g \in \mathcal{C}(\mathcal{Y})} \left\{ \int g d\beta ; \forall (x, y), g(y) \leq c(x, y) - f(x) \right\}$. $\forall y \in \mathcal{Y}, g(y) \leq f^c(y)$

Definition 4.6 (c -transform). For a real-valued function $f : \mathcal{X} \rightarrow \mathbb{R}$, its c -transform is the extended-real-valued function

$$\forall y \in \mathcal{Y}, f^c(y) := \inf_{x \in \mathcal{X}} c(x, y) - f(x). \quad (4.8)$$

For a real-valued function $g : \mathcal{Y} \rightarrow \mathbb{R}$, the $\bar{c}$ -transform associated with $\bar{c}(y, x) = c(x, y)$ is $\forall x \in \mathcal{X}, g^{\bar{c}}(x) := \inf_{y \in \mathcal{Y}} c(x, y) - g(y)$.

When $\mathcal{X}, \mathcal{Y}$ are compact, c is continuous and the input potential is continuous, these infima are finite minima and the transformed potential is continuous.

Since β is nonnegative, maximizing $\int g d\beta$ saturates the bound $g \leq f^c$ on $\text{Supp}(\beta)$, equivalently β -almost everywhere.

Proposition 4.7 (c -transforms solve the semi-relaxed problems). *Assume the compactness and continuity hypotheses of Proposition 4.4. For fixed $f \in \mathcal{C}(\mathcal{X})$, the maximizers of the dual objective over all $g \in \mathcal{C}(\mathcal{Y})$ such that $f \oplus g \leq c$ are exactly the functions satisfying $g = f^c$ β -almost everywhere. Equivalently, f^c gives the value of the one-marginal primal problem*

$$\inf_{\substack{\pi \in \mathcal{M}_+^1(\mathcal{X} \times \mathcal{Y}) \\ \pi_2 = \beta}} \int c(x, y) d\pi(x, y) - \int f(x) d\pi_1(x) = \int f^c(y) d\beta(y).$$

Symmetrically, for fixed g , the maximizers over f are the functions satisfying $f = g^{\bar{c}}$ α -almost everywhere.

Proof. The constraint $f(x) + g(y) \leq c(x, y)$ for all x is equivalent, for each fixed y , to $g(y) \leq \inf_x c(x, y) - f(x) = f^c(y)$. Since β is nonnegative, the largest possible value of $\int g d\beta$ is obtained by saturating this pointwise upper bound on the support of β . The proof for $f = g^c$ is identical after exchanging the two marginals.

For the primal formula, disintegrate any feasible π as $\pi(dx, dy) = \pi_y(dx)\beta(dy)$. Then

$$\int c d\pi - \int f d\pi_1 = \int \left(\int (c(x, y) - f(x)) d\pi_y(x) \right) d\beta(y) \geq \int f^c(y) d\beta(y).$$

The argmin of $x \mapsto c(x, y) - f(x)$ is nonempty and compact. The measurable maximum theorem provides a Borel selector $x_*(y)$ from these argmins. The coupling $\pi(dx, dy) = \delta_{x_*(y)}(dx)\beta(dy)$ then attains equality. $\square$

$(f, g) \mapsto (f, f^c), (f, f^c) \mapsto (f^{c\bar{c}}, f^c)$.

Proposition 4.8 (Modulus stability of c -transforms). *Suppose that ω_Y is a modulus such that $|c(x, y) - c(x, y')| \leq \omega_Y(dy(y, y'))$ for all x, y, y' . Then every finite f^c has the same modulus. Symmetrically, a uniform modulus ω_X of c in its first variable is inherited by g^c . In particular, a uniform L -Lipschitz bound on the cost yields an L -Lipschitz bound on the corresponding transform.*

Proof. For each x , set $F_x(y) = c(x, y) - f(x)$ and $F(y) = f^c(y) = \inf_x F_x(y)$. Since all the functions F_x share the modulus ω_Y ,

$$|F(y) - F(y')| = \left| \inf_x F_x(y) - \inf_x F_x(y') \right| \leq \sup_x |F_x(y) - F_x(y')| \leq \omega_Y(dy(y, y')).$$

The proof in the first variable is identical. $\square$

Euclidean case. The Euclidean quadratic cost is the model case where c -transforms become ordinary convex conjugates after removing the quadratic terms. This is the algebraic bridge between Kantorovich duality and Brenier maps.

$\int q(x, y) d\pi(x, y) = \frac{1}{2} \int \|x\|^2 d\alpha(x) + \frac{1}{2} \int \|y\|^2 d\beta(y) - \int \langle x, y \rangle d\pi(x, y)$. $f^c(y) = \inf_x -\langle x, y \rangle - f(x) = -(-f)^*(y)$, $h^*(y) := \sup_x \langle x, y \rangle - h(x)$. On the full Euclidean space, $f^{c\bar{c}} = -(-f)^{**}$; hence c -closed functions are negatives of lower semicontinuous convex functions, and the closure is the smallest upper semicontinuous concave majorant of f . On restricted compact domains, the same statement holds with the additional restriction that the supporting slopes belong to the opposite domain.

Why hard alternating optimization stops.

Proposition 4.9 (Algebra of c -transforms). *Writing $f^{c\bar{c}} := (f^c)^{\bar{c}}$, the following pointwise identities hold whenever the transforms are well defined:*

$$(i) f \leq f' \Rightarrow f^c \geq f'^c, \quad (ii) f^{c\bar{c}} \geq f, \quad (iii) g^{\bar{c}\bar{c}} \geq g, \quad (iv) f^{c\bar{c}c} = f^c.$$

Proof. The order reversal in (i) follows directly from the minus sign in the definition. For every x and y , $f^c(y) = \inf_{x'} \{c(x', y) - f(x')\} \leq c(x, y) - f(x)$, so $c(x, y) - f^c(y) \geq f(x)$. Taking the infimum over y proves (ii), and exchanging the two spaces proves (iii). Finally, (ii) and the order reversal imply $f^c \geq f^{c\bar{c}c}$, whereas (iii) applied to $g = f^c$ gives $f^{c\bar{c}c} \geq f^c$. This proves (iv). $\square$

$$(f, g) \mapsto (f, f^c) \mapsto (f^{c\bar{c}}, f^c) \mapsto (f^{c\bar{c}}, f^{c\bar{c}c}) = (f^{c\bar{c}}, f^c) \dots \quad (4.9)$$

For the bilinear cost $c(x, y) = -xy$ on a compact interval, $f^{c\bar{c}}$ is the smallest concave majorant representable with slopes in the opposite interval. A hard transform removes non-concave oscillations in one step rather than producing a gradual ascent.

5 Semi-discrete and $\mathcal{W}_1$

5.1 Semi-dual

General measure semi-dual. Denote by $\mathcal{E}_0(f, g)$ the full dual objective $\int f d\alpha + \int g d\beta$, extended by $-\infty$ outside $\mathcal{R}(c)$. For fixed g , dual feasibility is equivalent to $f \leq g^c$, so positivity of α gives the unconstrained concave problem

$$\mathcal{L}_c(\alpha, \beta) = \sup_{g \in \mathcal{C}(\mathcal{Y})} \mathcal{E}(g), \quad \mathcal{E}(g) := \mathcal{E}_0(g^c, g) = \int_{\mathcal{X}} g^c d\alpha + \int_{\mathcal{Y}} g d\beta. \quad (5.1)$$

It is invariant under $g \mapsto g + s$.

Discrete semi-dual. For $\alpha = \sum_i a_i \delta_{x_i}$, $\beta = \sum_j b_j \delta_{y_j}$, $C_{ij} = c(x_i, y_j)$, and common mass M , eliminating the source potential gives

$$\mathcal{L}_C(a, b) = \max_{g \in \mathbb{R}^m} \mathcal{E}(g), \quad \mathcal{E}(g) = \mathcal{E}_0(g^{\bar{C}}, g) = \sum_i a_i (g^{\bar{C}})_i + \sum_j b_j g_j, \quad (g^{\bar{C}})_i = \min_j (C_{ij} - g_j). \quad (5.2)$$

The objective is concave, piecewise affine, and gauge invariant. If $\sigma_g(i) \in \arg\min_j (C_{ij} - g_j)$, then $b - \hat{b}(g)$ is a supergradient, where $\hat{b}_j(g) = \sum_{i: \sigma_g(i)=j} a_i$.

5.2 Auction Algorithm

For the square uniform assignment problem, auction is derived from coordinate maximization of the discrete semi-dual, but uses bidder-specific ε -overshoots to avoid jamming at nonsmooth ties. With g denoting the target Kantorovich potential, set

$$(g^{\bar{C}})_i = \min_j (C_{ij} - g_j), \quad \mathcal{E}(g) = \frac{1}{n} \sum_i (g^{\bar{C}})_i + \frac{1}{n} \sum_j g_j.$$

The discrete Laguerre cells are

$$\mathbb{L}_j^D(g) = \{i : C_{ij} - g_j \leq C_{ik} - g_k \forall k\}.$$

They are the restriction of Definition 5.1. At points with unique minimizers, $\partial_{g_j} \mathcal{E}(g) = (1 - |\mathbb{L}_j^D(g)|)/n$.

Bids and certificate. For an unassigned row i , choose its best and second-best targets j_0, j_1 and update

$$g_{j_0} \leftarrow g_{j_0} - [(C_{i,j_1} - g_{j_1}) - (C_{i,j_0} - g_{j_0}) + \varepsilon]. \quad (5.3)$$

Assign row i to j_0 and unassign the previous owner of j_0 , if any. The overshoot need not increase the nonsmooth semi-dual; its invariant is instead

$$P_{ij} = 1 \implies C_{ij} - g_j \leq (g^{\bar{C}})_i + \varepsilon.$$

If $R_C = \max C_{ij} - \min C_{ij}$ and $g = 0$ initially, the method terminates after at most $n(\lfloor R_C/\varepsilon \rfloor + 1)$ bids and

$$0 \leq \frac{1}{n} \langle C, P \rangle - \min_{P' \in \mathcal{P}_n^{\text{perm}}} \frac{1}{n} \langle C, P' \rangle \leq \varepsilon.$$

The second inequality follows by applying Proposition 1.4 to the feasible pair $(g^{\bar{C}}, g)$ defined by the final target potential. Dense implementation costs $O(n^2(1 + R_C/\varepsilon))$. If C is integral, $\varepsilon < 1/n$ gives an exact assignment. Warm-starting auctions while halving ε yields an η -optimal assignment in $O(n^3(1 + \log_+(R_C/\eta)))$ operations.

5.3 Semi-discrete

Discrete target and Laguerre cells. One can adapt Definition 4.6 of the $\bar{c}$ -transform to this setting by restricting the minimization to the finite support $\{y_j\}_{j=1}^m$ of β . Equivalently, one applies the same definition with the discrete target space $\mathcal{Y} = \{y_j\}_{j=1}^m$ and identifies a vector $g \in \mathbb{R}^m$ with the function $g : \mathcal{Y} \rightarrow \mathbb{R}$ defined by $g(y_j) = g_j$:

$$\forall g \in \mathbb{R}^m, \forall x \in \mathcal{X}, \quad g^{\bar{c}}(x) := \min_{j \in [m]} c(x, y_j) - g_j. \quad (5.4)$$

$$\mathcal{L}_c(\alpha, \beta) = \max_{g \in \mathbb{R}^m} \mathcal{E}(g) := \mathcal{E}_0(g^{\bar{c}}, g) = \int_{\mathcal{X}} g^{\bar{c}}(x) d\alpha(x) + \sum_j g_j b_j. \quad (5.5)$$

The objective satisfies $\mathcal{E}(g + s\mathbf{1}) = \mathcal{E}(g)$ for every $s \in \mathbb{R}$. One may therefore impose the gauge $\sum_j g_j = 0$ without changing the problem.

Definition 5.1 (Laguerre cells and power diagrams). For sites $(y_j)_{j=1}^m$ and weights $g \in \mathbb{R}^m$, the Laguerre cell associated with y_j is

$$\mathbb{L}_j(g) := \{x \in \mathcal{X} ; \forall j' \neq j, c(x, y_j) - g_j \leq c(x, y_{j'}) - g_{j'}\}. \quad (5.6)$$

The cells cover $\mathcal{X}$; after arbitrary tie-breaking on common boundaries, they induce a disjoint partition. When $c(x, y) = \|x - y\|^2$, this Laguerre decomposition is also called a power diagram. If g is constant, it reduces to the ordinary Voronoi diagram.

Mass balance.

$$\mathcal{E}(g) = \sum_{j=1}^m \int_{\mathbb{L}_j(g)} (c(x, y_j) - g_j) d\alpha(x) + \langle g, b \rangle. \quad (5.7)$$

Proposition 5.2 (Gradient of the semi-discrete dual). *If the minimizing index in (5.4) is unique for α -almost every x , then $\mathcal{E}$ is differentiable at g and $\forall j \in [m], \quad \nabla \mathcal{E}(g)_j = b_j - \int_{\mathbb{L}_j(g)} d\alpha$.*

Proof. For α -almost every x , the minimizing index in $\min_j c(x, y_j) - g_j$ is unique. If this index is $j(x)$, then the directional derivative in a direction $h \in \mathbb{R}^m$ is

$$\frac{d}{dt} \Big|_{t=0} \min_j (c(x, y_j) - (g_j + th_j)) = -h_{j(x)}.$$

The corresponding difference quotients are bounded by $\|h\|_\infty$. Dominated convergence therefore gives $d\mathcal{E}(g)[h] = -\sum_j h_j \int_{\mathbb{L}_j(g)} d\alpha + \sum_j h_j b_j$, which is the announced gradient formula. $\square$

Stochastic optimization.

$$\mathcal{E}(g) = \int_{\mathcal{X}} E(g, x) d\alpha(x) = \mathbb{E}_X(E(g, X)), \quad E(g, x) := g^{\bar{c}}(x) + \langle g, b \rangle. \quad (5.8)$$

$$\begin{aligned} \nabla_g E(g, x) &= (b_j - \mathbf{1}_{\mathbb{L}_j(g)}(x))_{j=1}^m, \\ g^{(\ell+1)} &:= g^{(\ell)} + \tau_\ell \nabla_g E(g^{(\ell)}, x_\ell). \end{aligned} \quad (5.9)$$

$$j_\ell \in \operatorname{argmin}_j (c(x_\ell, y_j) - g_j^{(\ell)}), \quad g_j^{(\ell+1)} = g_j^{(\ell)} + \tau_\ell (b_j - \mathbf{1}_{\{j=j_\ell\}}).$$

$$\sum_{\ell=0}^{+\infty} \tau_\ell = +\infty, \quad \sum_{\ell=0}^{+\infty} \tau_\ell^2 < +\infty, \quad (5.10)$$

Proposition 5.3 (Averaged stochastic semi-dual rate). *Let g^* maximize $\mathcal{E}$, set $R = \|g^{(0)} - g^*\|_2$, and suppose the stochastic supergradients are conditionally unbiased and satisfy $\|\nabla_g E(g, x)\|_2 \leq G$. For a horizon $L \geq 1$, use the constant step $\tau = R/(G\sqrt{L})$ and the average $\bar{g}_L = L^{-1} \sum_{\ell=0}^{L-1} g^{(\ell)}$. If $R = 0$, the conclusion is immediate; otherwise, $\mathcal{E}(g^*) - \mathbb{E}[\mathcal{E}(\bar{g}_L)] \leq \frac{RG}{\sqrt{L}}$.*

Proof. Conditionally on $g^{(\ell)}$, the stochastic supergradient is unbiased. Concavity and the squared-distance recursion give

$$2\tau \mathbb{E}[\mathcal{E}(g^*) - \mathcal{E}(g^{(\ell)})] \leq \mathbb{E}\|g^{(\ell)} - g^*\|_2^2 - \mathbb{E}\|g^{(\ell+1)} - g^*\|_2^2 + \tau^2 G^2.$$

Summing from 0 to $L - 1$, discarding the final nonnegative squared distance, and using concavity at $\bar{g}_L$ yields

$$\mathcal{E}(g^*) - \mathbb{E}[\mathcal{E}(\bar{g}_L)] \leq \frac{R^2}{2\tau L} + \frac{\tau G^2}{2}.$$

The stated choice of τ gives the result. $\square$

5.4 Optimal Quantization

Free masses and prescribed weights.

$$\mathcal{Q}_m(\alpha) := \min_{Y=(y_j)_{j=1}^m, \mathbf{b} \in \Sigma_m} \mathcal{W}_p \left(\alpha, \sum_{j=1}^m \mathbf{b}_j \delta_{y_j} \right). \quad (5.11)$$

Proposition 5.4 (Quantization rate and curse of dimensionality). *Let α be supported in a bounded subset $\Omega \subset \mathbb{R}^d$ and assume that $\alpha = \rho dx$ with $\rho \leq \rho_+ < +\infty$. Then, for fixed $p \geq 1$, there exist constants $0 < c \leq C < +\infty$ such that $c m^{-1/d} \leq \mathcal{Q}_m(\alpha) \leq C m^{-1/d}$.*

Proof. Enclose Ω in a cube. With $k = \lfloor m^{1/d} \rfloor$, subdivide this cube into $k^d \leq m$ congruent subcubes and place one codepoint in each subcube meeting Ω . Every source point then lies within distance $C/k \leq C' m^{-1/d}$ of a codepoint. The nearest-codepoint coupling proves the upper bound; unused codepoints may be assigned zero mass.

For the lower bound, fix any set Y of at most m codepoints and write $d_Y(x) = \min_j \|x - y_j\|$. If ω_d is the volume of the Euclidean unit ball, then $\alpha(\{d_Y \leq t\}) \leq \rho_+ m \omega_d t^d$. Set $t_0 = (2\rho_+ m \omega_d)^{-1/d}$. Then $\alpha(\{d_Y > t\}) \geq 1/2$ for $0 < t < t_0$, and the layer-cake formula gives $\int d_Y(x)^p d\alpha(x) = \int_0^{+\infty} p t^{p-1} \alpha(\{d_Y > t\}) dt \geq \frac{t_0^p}{2} \simeq c m^{-p/d}$. Taking the p -th root and minimizing over Y proves the lower bound. $\square$

Lloyd algorithm.

Proposition 5.5 (Free masses give Voronoi cells). *For the cost $c(x, y) = d(x, y)^p$, fix distinct codepoints $Y = (y_j)_{j=1}^m$. Duplicate codepoints can be merged beforehand. Minimizing over the weights $\mathbf{b} \in \Sigma_m$ gives*

$$\min_{\mathbf{b} \in \Sigma_m} \mathcal{W}_p^p \left(\alpha, \sum_j \mathbf{b}_j \delta_{y_j} \right) = \int_{\mathcal{X}} \min_{1 \leq j \leq m} c(x, y_j) d\alpha(x).$$

An optimal coupling is induced by sending each x to a nearest codepoint. The corresponding cells are the Voronoi cells $\mathbb{V}_j(Y) := \{x; \forall j', c(x, y_j) \leq c(x, y_{j'})\}$, up to arbitrary tie-breaking on common boundaries.

Proof. For any coupling between α and a measure supported on Y , the conditional destination of a point x belongs to Y , hence its conditional cost is at least $\min_j c(x, y_j)$. Integrating gives the lower bound. Conversely, choose a measurable nearest-codepoint map $T_Y(x) \in \operatorname{argmin}_j c(x, y_j)$, breaking ties measurably, and set $\mathbf{b}_j = \alpha(T_Y^{-1}(y_j))$. Then $(T_Y)_\# \alpha = \sum_j \mathbf{b}_j \delta_{y_j}$ and the induced transport reaches the displayed lower bound. $\square$

$$\mathcal{Q}_m(\alpha)^p = \min_Y \mathcal{F}(Y), \quad \mathcal{F}(Y) := \int_{\mathcal{X}} \min_{1 \leq j \leq m} c(x, y_j) d\alpha(x). \quad y_j \in \operatorname{argmin}_y \int_{\mathbb{V}_j(Y)} c(x, y) d\alpha(x). \quad y_j = \frac{\int_{\mathbb{V}_j(Y)} x d\alpha(x)}{\int_{\mathbb{V}_j(Y)} d\alpha}.$$

Continuous Lloyd flow. $a_j(Y) = \alpha(\mathbb{V}_j(Y))$, $b_j(Y) = \frac{1}{a_j(Y)} \int_{\mathbb{V}_j(Y)} x d\alpha(x)$ $y_j^{(k+1)} = y_j^{(k)} + \tau(b_j(Y^{(k)}) - y_j^{(k)})$, $0 < \tau \leq 1$, $\dot{y}_j(t) = b_j(Y_t) - y_j(t)$. $\nabla_{y_j} \mathcal{F}(Y) = 2 \int_{\mathbb{V}_j(Y)} (y_j - x) d\alpha(x) = 2a_j(Y)(y_j - b_j(Y))$, $\dot{y}_j(t) = -\frac{1}{2a_j(Y_t)} \nabla_{y_j} \mathcal{F}(Y_t)$. $\frac{d}{dt} \mathcal{F}(Y_t) = -2 \sum_j a_j(Y_t) \|b_j(Y_t) - y_j(t)\|^2 \leq 0$. If $\eta_t = \sum_j w_j \delta_{y_j(t)}$ carries fixed positive weights, independent of the Voronoi masses $a_j(Y_t)$, Proposition 13.1 turns this labelled particle ODE into the weak continuity equation $\partial_t \eta_t + \operatorname{div}(v_t \eta_t) = 0$, $v_t(y_j(t)) = b_j(Y_t) - y_j(t)$, in the sense of the measure evolutions of Section 13. The weights w_j in this transport equation are auxiliary weights for the moving labelled particles; they are not the Voronoi masses used to define the quantization energy.

$$\nu_{Y_t} = \sum_j a_j(Y_t) \delta_{y_j(t)}, \quad \partial_t \nu_{Y_t} + \operatorname{div}(v_t \nu_{Y_t}) = \sum_j \dot{a}_j(Y_t) \delta_{y_j(t)}, \quad v_t(y_j(t)) = \dot{y}_j(t).$$

Mean-field limit and ultrafast diffusion. There is nevertheless a precise high-resolution continuum theory when the number m of codepoints tends to infinity. If $\alpha = \rho dx$, the limiting Eulerian variable is the density σ of sites, meaning heuristically that $m\sigma(x)dx$ codepoints lie in dx .

$$\mathcal{G}_\rho(\sigma) := \int_{\Omega} \rho(x) \sigma(x)^{-r} dx, \quad \int_{\Omega} \sigma dx = 1,$$

$$\begin{aligned} m^r \mathcal{F}_p(Y^{(m)}) &\simeq C_{p,d} \mathcal{G}_\rho(\sigma), \\ m^r \mathcal{Q}_m(\alpha)^p &\longrightarrow C_{p,d} \min_{\sigma} \mathcal{G}_\rho(\sigma). \end{aligned}$$

$$\partial_t \sigma = -r \operatorname{div} \left(\sigma \nabla \left(\frac{\rho}{\sigma^{r+1}} \right) \right),$$

$$\partial_t u = -\frac{r+1}{\omega} \operatorname{div}(\omega \nabla(u^{-r})),$$

Quantization with fixed equal weights. $\nu_Y := \frac{1}{m} \sum_{j=1}^m \delta_{y_j}$, $\mathcal{F}_{\text{eq}}(Y) := \frac{1}{2} \mathcal{W}_2^2(\alpha, \nu_Y)$, $\bar{x}_j(Y) := m \int_{C_j(Y)} x d\alpha(x)$. $\nabla_{y_j} \mathcal{F}_{\text{eq}}(Y) = \frac{1}{m} (y_j - \bar{x}_j(Y))$. As long as the optimal matching between two nearby labelled configurations preserves labels, the $\mathcal{W}_2$ metric on equal-weight empirical measures induces the local particle metric $g_Y(U, V) = m^{-1} \sum_j \langle u_j, v_j \rangle$. Hence the associated $\mathcal{W}_2$ gradient flow is the coupled system $\dot{y}_j(t) = \bar{x}_j(Y_t) - y_j(t)$, $j = 1, \dots, m$. Equivalently, ν_{Y_t} satisfies a continuity equation with velocity $v_t(y_j(t)) = \bar{x}_j(Y_t) - y_j(t)$. This is the so-called finite-particle $\mathcal{W}_2$ gradient-flow viewpoint developed more systematically in Chapter 14; the fixed-weight cells are Laguerre cells rather than the free-mass Voronoi cells used by Lloyd's method.

Equal-weight quantization on the line.

Proposition 5.6 (One-dimensional equal-weight quantization). *Let $\alpha \in \mathcal{M}_+^1(\mathbb{R})$ have finite second moment and quantile function $Q = C_\alpha^{-1}$. For*

$$\mathcal{Q}_m^{\text{eq}}(\alpha)^2 := \min_{y_1 \leq \dots \leq y_m} \mathcal{W}_2^2 \left(\alpha, \frac{1}{m} \sum_{i=1}^m \delta_{y_i} \right),$$

set $I_i = ((i-1)/m, i/m]$. Then the sorted minimizer is unique and its i th atom is $y_i^* = m \int_{I_i} Q(u) du$, and $\mathcal{Q}_m^{\text{eq}}(\alpha)^2 = \sum_{i=1}^m \int_{I_i} |Q(u) - y_i^*|^2 du$. If $Q \in C^1([0, 1])$, then $m^2 \mathcal{Q}_m^{\text{eq}}(\alpha)^2 \rightarrow \frac{1}{12} \int_0^1 |Q'(u)|^2 du$.

Proof. After sorting the atoms, the quantile formula for $\mathcal{W}_2$ gives

$$\mathcal{W}_2^2 \left(\alpha, \frac{1}{m} \sum_{i=1}^m \delta_{y_i} \right) = \sum_{i=1}^m \int_{I_i} |Q(u) - y_i|^2 du.$$

The minimization therefore decouples over the intervals I_i , and the best constant approximation of Q on I_i in L^2 is its average on I_i . Since Q is nondecreasing, these interval averages are nondecreasing, so they satisfy the sorting constraint. Strict convexity in each scalar y_i proves uniqueness and gives the displayed energy.

Denote by $\bar{Q}_{I_i} = m \int_{I_i} Q(u) du$ the interval average. If Q is C^1 , set $h = 1/m$ and write $I_i = (a_i, a_i + h]$. Uniform Taylor expansion gives, for $v \in [0, 1]$, $Q(a_i + hv) = Q(a_i) + hvQ'(a_i) + h r_i(v)$, $\max_i \sup_{v \in [0, 1]} |r_i(v)| \rightarrow 0$. Subtracting the average over $v \in [0, 1]$ and integrating gives $\int_{I_i} |Q(u) - \bar{Q}_{I_i}|^2 du = \frac{h^3}{12} |Q'(a_i)|^2 + o(h^3)$, uniformly in i . Summing over i yields a Riemann sum for $\frac{1}{12} \int_0^1 |Q'(u)|^2 du$. $\square$

Thus the common deterministic rule $m^{-1} \sum_i \delta_{Q((i-1/2)/m)}$ should be read as a midpoint approximation of the optimal bin-average formula. It has the same leading squared error under the same smoothness assumptions.

$$\int_{I_i} |Q(u) - z|^2 du = \int_{I_i} |Q(u) - y_i^*|^2 du + \frac{1}{m} |z - y_i^*|^2,$$

Proposition 5.7 (Quantile-process asymptotics for random placement). *Let $\alpha \in \mathcal{M}_+^1(\mathbb{R})$ have quantile $Q \in C^1([0, 1])$, and let $\hat{\alpha}_m = m^{-1} \sum_{i=1}^m \delta_{X_i}$ with X_i i.i.d. with law α . If B denotes the standard Brownian bridge on $[0, 1]$, then $m \mathcal{W}_2^2(\alpha, \hat{\alpha}_m) \xrightarrow{\text{law}} \int_0^1 B(u)^2 |Q'(u)|^2 du$, and, in expectation,*

$$m \mathbb{E}[\mathcal{W}_2^2(\alpha, \hat{\alpha}_m)] \rightarrow \int_0^1 u(1-u) |Q'(u)|^2 du.$$

Proof. Let $\hat{Q}_m$ be the empirical quantile function. The one-dimensional formula gives $\mathcal{W}_2^2(\alpha, \hat{\alpha}_m) = \int_0^1 |\hat{Q}_m(u) - Q(u)|^2 du$. Write $X_i = Q(U_i)$ with U_i i.i.d. uniform on $[0, 1]$, and let $\hat{U}_m^{-1}$ be the empirical quantile function of the U_i . Then $\hat{Q}_m = Q \circ \hat{U}_m^{-1}$. The classical uniform quantile-process theorem gives $\sqrt{m}(\hat{U}_m^{-1} - \text{Id}) \xrightarrow{\text{law}} -B$ in $L^2(0, 1)$. Because Q' is uniformly continuous and $\hat{U}_m^{-1} \rightarrow \text{Id}$ uniformly in probability, the functional delta method gives $\sqrt{m}(\hat{Q}_m - Q) \xrightarrow{\text{law}} -B Q'$ in $L^2(0, 1)$. Since B and $-B$ have the same law, the sign disappears in the squared L^2 norm. The continuous mapping theorem gives the distributional convergence. Standard fourth-moment bounds for the uniform quantile process, together with the boundedness of Q' , give uniform integrability of the squared L^2 norms and hence convergence of expectations. Since $\mathbb{E}[B(u)^2] = u(1-u)$, Fubini's theorem gives the displayed expectation limit. $\square$

5.5 Wasserstein-1 norm

c -transform for $\mathcal{W}_1$. Assume that d is a distance on $\mathcal{X} = \mathcal{Y}$ and take the ground cost $c(x, y) = d(x, y)$. We denote the Lipschitz constant of $f \in \mathcal{C}(\mathcal{X})$ by

Definition 5.8 (Lipschitz constant). For a function $f : \mathcal{X} \rightarrow \mathbb{R}$ on a metric space $(\mathcal{X}, d)$, its Lipschitz constant is

$$\text{Lip}(f) := \sup \left\{ \frac{|f(x) - f(y)|}{d(x, y)} ; x \neq y \right\}. \quad (5.12)$$

The function is 1-Lipschitz when $\text{Lip}(f) \leq 1$.

Proposition 5.9 (c -transforms and 1-Lipschitz functions). *Suppose $\mathcal{X} = \mathcal{Y}$ and $c(x, y) = d(x, y)$. Then there exists g such that $f = g^c$ if and only if $\text{Lip}(f) \leq 1$. Furthermore, if $\text{Lip}(f) \leq 1$, then $f^c = -f$.*

Proof. First suppose $f = g^c$ for some g . For $x, y \in \mathcal{X}$,

$$|f(x) - f(y)| = \left| \inf_z [d(x, z) - g(z)] - \inf_z [d(y, z) - g(z)] \right| \leq \sup_z |d(x, z) - d(y, z)| \leq d(x, y),$$

where the last inequality is the reverse triangle inequality. Thus $\text{Lip}(f) \leq 1$.

If $\text{Lip}(f) \leq 1$, then $f(x) \leq f(y) + d(x, y)$, so $d(x, y) - f(x) \geq -f(y)$ for all x and hence $f^c(y) \geq -f(y)$. Taking $x = y$ gives $f^c(y) \leq -f(y)$. Therefore $f^c = -f$. Applying the same property to $-f$ gives $(-f)^c = f$, so every 1-Lipschitz function is c -concave. $\square$

$$\mathcal{W}_1(\alpha, \beta) = \max_f \left\{ \int_{\mathcal{X}} f d(\alpha - \beta) ; \text{Lip}(f) \leq 1 \right\} =: \|\alpha - \beta\|_{\mathcal{W}_1}. \quad (5.13)$$

For a discrete signed measure $\alpha - \beta = \sum_{k=1}^N r_k \delta_{z_k}$ on distinct sites, with $\sum_k r_k = 0$, (5.13) becomes the finite-dimensional linear program

$$\mathcal{W}_1(\alpha, \beta) = \max_{(f_k)_k} \left\{ \sum_k f_k r_k ; \forall k, \ell, |f_k - f_\ell| \leq d(z_k, z_\ell) \right\}. \quad (5.14)$$

This linear program can be solved by generic interior-point or first-order methods. If N support points are involved, however, it still contains $O(N^2)$ Lipschitz constraints, mirroring the $O(nm)$ coupling variables of the original discrete Kantorovich LP; the dual formulation alone does not remove the all-pairs structure.

$$\mathcal{W}_1(\alpha, \beta) = \max_{(f_k)_k} \left\{ \sum_k f_k r_k ; \forall k, |f_{k+1} - f_k| \leq z_{k+1} - z_k \right\}.$$

$\mathcal{W}_1$ on Euclidean spaces. Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ have finite first moments and set $\xi = \alpha - \beta$. Rademacher's theorem identifies 1-Lipschitz functions with locally Sobolev functions whose Euclidean gradient has norm at most one almost everywhere. Hence

$$\mathcal{W}_1(\alpha, \beta) = \sup_{f \in W_{\text{loc}}^{1,\infty}(\mathbb{R}^d)} \left\{ \int_{\mathbb{R}^d} f d\xi ; \|\nabla f\|_{L^\infty(\mathbb{R}^d)} \leq 1 \right\}. \quad (5.15)$$

The correct flux space is generally not $L^1(\mathbb{R}^d; \mathbb{R}^d)$. Let $\mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)$ denote the finite $\mathbb{R}^d$ -valued Radon measures, write $|s|$ for total variation, and define divergence distributionally by

$$\langle \text{div } s, \varphi \rangle := - \int_{\mathbb{R}^d} \langle \nabla \varphi, ds \rangle, \quad \varphi \in C_c^1(\mathbb{R}^d).$$

Proposition 5.10 (Euclidean Beckmann formula). *For $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ with finite first moments,*

$$\mathcal{W}_1(\alpha, \beta) = \min_{s \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)} \left\{ |s|(\mathbb{R}^d) ; \text{div } s = \alpha - \beta \right\}. \quad (5.16)$$

If $s = w dx$ has a Lebesgue density, its cost reduces to $\int_{\mathbb{R}^d} \|w(x)\|_2 dx$.

Proof. Let s be feasible. For a smooth 1-Lipschitz test function, $\int f d(\alpha - \beta) = - \int \langle \nabla f, ds \rangle \leq |s|(\mathbb{R}^d)$. Truncation followed by mollification approximates normalized Lipschitz functions by smooth test functions without increasing the gradient bound asymptotically. Equation (5.15) therefore gives $\mathcal{W}_1(\alpha, \beta) \leq |s|(\mathbb{R}^d)$.

Conversely, let π be an optimal coupling and define a vector measure by its action on $\zeta \in C_c(\mathbb{R}^d; \mathbb{R}^d)$: $\langle \zeta(z), ds(z) \rangle := \int_{\mathbb{R}^d \times \mathbb{R}^d} \int_0^1 \langle \zeta((1-t)x + ty), y - x \rangle dt d\pi(x, y)$. For $\varphi \in C_c^1(\mathbb{R}^d)$, the fundamental theorem of calculus gives $\langle \text{div } s, \varphi \rangle = - \int_0^1 \langle \nabla \varphi((1-t)x + ty), y - x \rangle dt d\pi(x, y) = \int \varphi d(\alpha - \beta)$. Moreover, $|s|(\mathbb{R}^d) \leq \int \|x - y\|_2 d\pi(x, y) = \mathcal{W}_1(\alpha, \beta)$. Together with the first inequality, this proves equality and attainment. $\square$

Graph distances and Beckmann flows.

Definition 5.11 (Graph geodesic distance). Let $G = (V, E)$ be a connected finite graph with positive edge lengths $(\ell_e)_{e \in E}$. The graph geodesic distance between two vertices is $d_G(i, j) = \min_{\gamma: i \rightsquigarrow j} \sum_{e \in \gamma} \ell_e$. The minimum is over all paths γ joining i to j .

Proposition 5.12 ($\mathcal{W}_1$ and Beckmann flow on a graph). *Let $G = (V, E)$ be a connected finite graph with positive edge lengths $(\ell_e)_{e \in E}$ and graph geodesic distance d_G . For two probability vectors a, b on V , set $r = a - b$ and orient each edge $e = (i, j)$. If $(\nabla_G f)_e = f_j - f_i$, $\text{div}_G = -\nabla_G^*$ are the finite-difference gradient and its negative adjoint, then a positive flow on the oriented edge $i \rightarrow j$ has positive divergence at i and negative divergence at j . With this convention,*

$$\mathcal{W}_{1,G}(a, b) = \max_f \left\{ \sum_{i \in V} f_i r_i ; |f_i - f_j| \leq \ell_e \quad \forall e = (i, j) \right\} = \min_m \left\{ \sum_{e \in E} \ell_e |m_e| ; \text{div}_G m = r \right\}.$$

The vector m_e is an oriented edge flow, and the constraint $\text{div}_G m = r$ is conservation of mass at each vertex.

Proof. The edge constraints imply, by summing along every path and then minimizing over paths, that $|f_i - f_j| \leq d_G(i, j)$ for all vertices. Conversely, any 1-Lipschitz function for d_G satisfies $|f_i - f_j| \leq d_G(i, j) \leq \ell_e$ on every edge. The first equality is therefore the Kantorovich–Rubinstein formula on the metric space (V, d_G) .

For the second equality, write the graph Beckmann problem and dualize its equality constraint with a potential f : $\inf_m \sum_e \ell_e |m_e| + \sup_f \sum_i f_i (r_i - (\text{div}_G m)_i)$. Using $\text{div}_G = -\nabla_G^*$, the coupling term is $\sum_e m_e (\nabla_G f)_e$. The minimization over each scalar flow m_e is finite exactly when $|(\nabla_G f)_e| \leq \ell_e$, and is then equal to zero. The dual problem is therefore precisely the graph Lipschitz dual above. Strong duality holds because this is a finite-dimensional linear program with a nonempty feasible set: connectedness and $\sum_i r_i = 0$ allow the signed surplus to be routed along paths. This proves the graph Beckmann formula. $\square$

6 Divergences and Dual Norms

6.1 Dual norms (Integral Probability Metrics)

Integral probability metrics.

Definition 6.1 (Dual seminorm and integral probability metric). Let B be a nonempty symmetric convex class of measurable functions that are integrable against the measures under consideration. For a signed measure ξ , define the extended dual seminorm

$$\|\xi\|_B := \sup_f \left\{ \int_{\mathcal{X}} f(x) d\xi(x) ; f \in B \right\}. \quad (6.1)$$

It is a norm when it is finite and B separates signed measures. When applied to $\alpha - \beta$ for probability measures, it is called an integral probability metric (IPM); it is a genuine metric precisely when B separates probability measures.

Proposition 6.2 (Metritzation by dual norms). *Assume that $\mathcal{X}$ is compact, that $B = -B$, and that the measures considered are probability measures.*

1. *If every function in $\mathcal{C}(\mathcal{X})$ can be uniformly approximated by elements of $\text{Span}(B)$, then $\|\alpha_n - \alpha\|_B \rightarrow 0$ implies $\alpha_n \rightarrow \alpha$.*
2. *If $B \subset \mathcal{C}(\mathcal{X})$ is compact for $\|\cdot\|_\infty$, then $\alpha_n \rightarrow \alpha$ implies $\|\alpha_n - \alpha\|_B \rightarrow 0$.*

Proof. For the first implication, $\|\alpha_n - \alpha\|_B \rightarrow 0$ and the symmetry of B imply $|\langle f, \alpha_n - \alpha \rangle| \leq \|\alpha_n - \alpha\|_B$ for $f \in B$. If $h = \sum_{j=1}^J c_j f_j \in \text{Span}(B)$, then $|\langle h, \alpha_n - \alpha \rangle| \leq \left(\sum_{j=1}^J |c_j|\right) \|\alpha_n - \alpha\|_B$, so integrals converge for every $h \in \text{Span}(B)$. Let $u \in \mathcal{C}(\mathcal{X})$ and choose such an h with $\|u - h\|_\infty \leq \eta$. Since α_n and α are probabilities, $|\langle u, \alpha_n - \alpha \rangle| \leq |\langle h, \alpha_n - \alpha \rangle| + 2\eta$. Taking the limsup as $n \rightarrow \infty$ and then letting $\eta \rightarrow 0$ gives $\langle u, \alpha_n \rangle \rightarrow \langle u, \alpha \rangle$ for all $u \in \mathcal{C}(\mathcal{X})$, which is weak convergence.

For the second implication, assume $\alpha_n \rightharpoonup \alpha$ and choose a subsequence $(\alpha_{n_k})_k$ such that $\|\alpha_{n_k} - \alpha\|_B \rightarrow \limsup_n \|\alpha_n - \alpha\|_B$. Since B is compact and the map $f \mapsto \langle f, \alpha_{n_k} - \alpha \rangle$ is continuous on B , the supremum is attained by some $f_{n_k} \in B$. Extracting a further subsequence if needed, $f_{n_k} \rightarrow f$ uniformly for some $f \in B$. Then $\langle f_{n_k}, \alpha_{n_k} - \alpha \rangle = \langle f, \alpha_{n_k} - \alpha \rangle + \langle f_{n_k} - f, \alpha_{n_k} \rangle - \langle f_{n_k} - f, \alpha \rangle$. The first term tends to zero by weak convergence and the last two by uniform convergence. Hence every limsup subsequence has limit zero, proving $\|\alpha_n - \alpha\|_B \rightarrow 0$. $\square$

Corollary 6.3 (Wasserstein metrizes weak convergence). *On a compact metric space, $\mathcal{W}_p$ metrizes weak convergence on probability measures for every $p \geq 1$.*

Proof. For $p = 1$, take $B = \{f ; \text{Lip}(f) \leq 1\}$. The span of B contains all Lipschitz functions, and Lipschitz functions are dense in $\mathcal{C}(\mathcal{X})$ on compact metric spaces.

Conversely, constants do not change the pairing with $\alpha_n - \alpha$. Fix $x_0 \in \mathcal{X}$ and normalize potentials by $f(x_0) = 0$. The normalized unit Lipschitz ball is uniformly bounded by $\text{diam}(\mathcal{X})$ and equicontinuous, hence compact in $\|\cdot\|_\infty$ by Arzelà–Ascoli. Proposition 6.2 gives $\mathcal{W}_1(\alpha_n, \alpha) \rightarrow 0$. Proposition 3.38 shows that all $\mathcal{W}_p$ distances induce the same topology on a compact space, so the result follows for every $p \geq 1$. $\square$

6.2 Dual RKHS Norms and Maximum Mean Discrepancies

Definition 6.4 (Positive and conditionally positive kernels). A symmetric function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is positive definite if for every $n \geq 1$, every $x_1, \dots, x_n \in \mathcal{X}$ and every $r \in \mathbb{R}^n$,

$$\sum_{i,j=1}^n r_i r_j k(x_i, x_j) \geq 0. \quad (6.2)$$

It is conditionally positive definite if the same inequality is required only for zero-sum vectors, $\langle r, \mathbb{1}_n \rangle = 0$.

Definition 6.5 (Kernel seminorm and MMD). Let k be positive definite. More generally, let k be conditionally positive definite and restrict attention to signed measures of total mass zero. For a signed measure ξ with finite kernel energy, the associated seminorm is

$$\|\xi\|_k^2 := \iint_{\mathcal{X} \times \mathcal{X}} k(x, y) d\xi(x) d\xi(y). \quad (6.3)$$

For two probability measures, the maximum mean discrepancy associated with k is $\text{MMD}_k(\alpha, \beta) := \|\alpha - \beta\|_k$. It is a genuine distance when the kernel is characteristic, meaning that $\|\alpha - \beta\|_k = 0$ implies $\alpha = \beta$.

$$\tilde{k}(x, y) = k(x, y) - k(x, x_0) - k(x_0, y) + k(x_0, x_0)$$

Proposition 6.6 (Kernel seminorm as an RKHS dual norm). *Let $\mathcal{H}$ be the RKHS with reproducing kernel k , and assume that the kernel mean embedding $m_\xi := \int k(x, \cdot) d\xi(x)$ is well-defined for the signed measure ξ . Then $\|\xi\|_k = \sup_{\|h\|_{\mathcal{H}} \leq 1} \int h(x) d\xi(x)$, so $\|\cdot\|_k$ is the dual seminorm in the sense of (6.1) associated with the RKHS unit ball.*

Proof. By the reproducing property,

$$\int h(x) d\xi(x) = \left\langle h, \int k(x, \cdot) d\xi(x) \right\rangle_{\mathcal{H}} = \langle h, m_\xi \rangle_{\mathcal{H}}.$$

Cauchy–Schwarz gives $\sup_{\|h\|_{\mathcal{H}} \leq 1} \int h d\xi = \|m_\xi\|_{\mathcal{H}}$. Finally, $\|m_\xi\|_{\mathcal{H}}^2 = \iint k(x, y) d\xi(x) d\xi(y)$, which is exactly (6.3). $\square$

Proposition 6.7 (Universal kernels metrize weak convergence). *Assume that $\mathcal{X}$ is compact and that the RKHS generated by the continuous kernel k is dense in $\mathcal{C}(\mathcal{X})$ for the uniform norm. Then*

$$\text{MMD}_k(\alpha_n, \alpha) \rightarrow 0 \iff \alpha_n \rightharpoonup \alpha$$

for probability measures on $\mathcal{X}$.

Proof. If $\text{MMD}_k(\alpha_n, \alpha) \rightarrow 0$, then for every $g \in \mathcal{H}$, $|\int g d(\alpha_n - \alpha)| \leq \|g\|_{\mathcal{H}} \text{MMD}_k(\alpha_n, \alpha) \rightarrow 0$. For any $h \in \mathcal{C}(\mathcal{X})$ and any $\eta > 0$, choose $g \in \mathcal{H}$ with $\|h - g\|_\infty \leq \eta$. Since α_n and α are probabilities,

$$\left| \int h d(\alpha_n - \alpha) \right| \leq 2\eta + \left| \int g d(\alpha_n - \alpha) \right|,$$

and the last term tends to zero. This proves weak convergence. Conversely, if $\alpha_n \rightharpoonup \alpha$, then $\alpha_n \otimes \alpha_n$, $\alpha_n \otimes \alpha$ and $\alpha \otimes \alpha$ converge weakly on the compact product space. Applying this to the continuous bounded function k in the identity $\text{MMD}_k(\alpha_n, \alpha)^2 = \iint k d\alpha_n d\alpha_n - 2 \iint k d\alpha_n d\alpha + \iint k d\alpha d\alpha$ gives convergence to zero. $\square$

In the special case where α is a discrete measure, one thus has the simple expression

$$\begin{aligned} \|\alpha\|_k^2 &= \sum_{i=1}^n \sum_{i'=1}^n a_i a_{i'} k_{i,i'} = \langle \mathbf{ka}, \mathbf{a} \rangle \quad \text{where} \quad k_{i,i'} := k(x_i, x_{i'}). \\ \|\alpha - \beta\|_k^2 &= \sum_{i,i'} a_i a_{i'} k(x_i, x_{i'}) + \sum_{j,j'} b_j b_{j'} k(y_j, y_{j'}) - 2 \sum_{i,j} a_i b_j k(x_i, y_j). \end{aligned} \quad (6.4)$$

6.3 φ -divergences

Definition by density ratios.

Definition 6.8 (Entropy function). A function $\varphi : \mathbb{R} \rightarrow \mathbb{R} \cup \{+\infty\}$ is an entropy function if it is proper, lower semicontinuous and convex, with $\text{dom } \varphi \subset [0, +\infty)$ and $\text{dom } \varphi \cap (0, +\infty) \neq \emptyset$. Its recession slope is

$$\varphi'_\infty = \lim_{s \rightarrow +\infty} \frac{\varphi(s)}{s} \in \mathbb{R} \cup \{+\infty\}.$$

If $\varphi'_\infty = \infty$, then φ grows faster than any linear function and φ is said to be *superlinear*.

Definition 6.9 (φ -Divergences). Let φ be an entropy function and let $\alpha, \beta \in \mathcal{M}_+(\mathcal{X})$. Write the Lebesgue decomposition of α with respect to β as $\alpha = (d\alpha/d\beta)\beta + \alpha^\perp$, where $\alpha^\perp \perp \beta$. The associated divergence is

$$\mathcal{D}_\varphi(\alpha|\beta) := \int_{\mathcal{X}} \varphi\left(\frac{d\alpha}{d\beta}\right) d\beta + \varphi'_\infty \alpha^\perp(\mathcal{X}), \quad (6.5)$$

with the convention $0 \cdot (+\infty) = 0$. It is extended by $+\infty$ whenever either argument is not a nonnegative measure.

$$\alpha = \sum_i a_i \delta_{x_i} \quad \text{and} \quad \beta = \sum_i b_i \delta_{x_i} \quad (6.6)$$

$$\mathcal{D}_\varphi(a|b) = \sum_{i \in \text{Supp}(b)} \varphi\left(\frac{a_i}{b_i}\right) b_i + \varphi'_\infty \sum_{i \notin \text{Supp}(b)} a_i, \quad (6.7)$$

where $\text{Supp}(b) := \{i \in \llbracket n \rrbracket ; b_i \neq 0\}$.

Proposition 6.10 (Basic properties of φ -divergences). *If φ is an entropy function, then $\mathcal{D}_\varphi$ is jointly 1-homogeneous, convex and weak-* lower semicontinuous in (α, β) .*

Proof. Introduce the lower-semicontinuous perspective

$$\forall (u, v) \in (\mathbb{R}_+)^2, \quad \psi(u, v) = \begin{cases} v\varphi(u/v) & \text{if } v > 0 \\ u\varphi'_\infty & \text{if } v = 0. \end{cases}$$

The joint 1-homogeneity follows directly from this formula. To see convexity, first take $v_0, v_1 > 0$ and $\tau \in [0, 1]$. With $\bar{v} = (1 - \tau)v_0 + \tau v_1$ and $\theta_0 = \frac{(1-\tau)v_0}{\bar{v}}$, $\theta_1 = \frac{\tau v_1}{\bar{v}}$, one has $\theta_0 + \theta_1 = 1$ and $\frac{(1-\tau)u_0 + \tau u_1}{\bar{v}} = \theta_0 \frac{u_0}{v_0} + \theta_1 \frac{u_1}{v_1}$. Jensen's inequality for φ , followed by multiplication by $\bar{v}$, proves convexity of ψ when both second arguments are positive. Its recession value at $v = 0$ gives the lower-semicontinuous convex extension to the closed quadrant. In the discrete case, $\mathcal{D}_\varphi(a|b) = \sum_i \psi(a_i, b_i)$, so homogeneity and convexity follow by summation. For general finite measures, representing both arguments with respect to a common dominating measure yields the integral of ψ ; the standard lower-semicontinuity theorem for convex integral functionals gives weak-* lower semicontinuity, with the recession value accounting for singular mass. $\square$

Proposition 6.11 (Non-negativity of φ -divergences). *Assume that φ is normalized by $\varphi(1) = 0$. For probability distributions $(\alpha, \beta) \in \mathcal{M}_+^1(\mathcal{X})$, one has $\mathcal{D}_\varphi(\alpha|\beta) \geq 0$. If φ is strictly convex, then one has $\mathcal{D}_\varphi(\alpha|\beta) = 0$ if and only if $\alpha = \beta$.*

Proof. Let $m = \alpha + \beta$ and write $a = d\alpha/dm$ and $b = d\beta/dm$. Then $a + b = 1$ m -almost everywhere and, using the perspective ψ above, $\mathcal{D}_\varphi(\alpha|\beta) = \int \psi(a, b) dm$. Since $m(\mathcal{X}) = 2$, the measure $m/2$ is a probability. Jensen's inequality, the 1-homogeneity of ψ , and $\varphi(1) = 0$ give $\frac{1}{2} \mathcal{D}_\varphi(\alpha|\beta) \geq \psi\left(\frac{1}{2} \int a dm, \frac{1}{2} \int b dm\right) = \psi(1/2, 1/2) = 0$. If φ is strictly convex, the map $s \mapsto \psi(s, 1-s)$ is strictly convex on its effective domain: distinct points on the line $u + v = 1$ cannot lie on the same ray, which is the only degeneracy of a strict perspective. Equality in Jensen therefore forces $a = 1/2$, hence $a = b$ and $\alpha = \beta$. For arbitrary finite nonnegative measures, $\varphi \geq 0$ also implies $\varphi'_\infty \geq 0$, so every term in (6.5) is nonnegative. $\square$

Classical examples and topology.

Main families of φ -divergences. $\varphi_\gamma(s) = \frac{s^\gamma - \gamma s + \gamma - 1}{\gamma(\gamma - 1)}$ ($\gamma \neq 0, 1$) $\mathfrak{h}(\alpha, \beta) := \mathcal{D}_{\varphi_H}(\alpha|\beta)^{1/2} = \|\sqrt{\rho_\alpha} - \sqrt{\rho_\beta}\|_{L^2(\lambda)}$.

$$\text{JS}(\alpha, \beta)^2 = \frac{1}{2} \text{KL}\left(\alpha \middle| \frac{\alpha + \beta}{2}\right) + \frac{1}{2} \text{KL}\left(\beta \middle| \frac{\alpha + \beta}{2}\right),$$

and satisfies $0 \leq \text{JS}(\alpha, \beta)^2 \leq \log 2$. Its exact entropy generator is

$$\varphi_{\text{JS}}(s) = \frac{1}{2} \left[s \log s - (s + 1) \log \left(\frac{s + 1}{2} \right) \right], \quad \mathcal{D}_{\varphi_{\text{JS}}}(\alpha|\beta) = \text{JS}(\alpha, \beta)^2.$$

Variational dual formula.

Proposition 6.12 (Variational representation of a φ -divergence). *Assume that $\mathcal{X}$ is compact and let $\alpha, \beta \in \mathcal{M}_+(\mathcal{X})$. Define the Legendre transform of the entropy, already extended by $+\infty$ on negative arguments, by $\varphi^*(s) := \sup_{r \geq 0} \{sr - \varphi(r)\}$. Then*

$$\mathcal{D}_\varphi(\alpha|\beta) = \sup_{f \in \mathcal{C}(\mathcal{X})} \left\{ \int_{\mathcal{X}} f d\alpha - \int_{\mathcal{X}} \varphi^*(f) d\beta \right\}. \quad (6.8)$$

Equivalently, the convex conjugate of $\mathcal{D}_\varphi(\cdot|\beta)$ on $\mathcal{M}(\mathcal{X})$ is

$$\forall f \in \mathcal{C}(\mathcal{X}), \quad \mathcal{D}_\varphi^*(f|\beta) = \int_{\mathcal{X}} \varphi^*(f(x)) d\beta(x). \quad (6.9)$$

Proof. First suppose that $\varphi'_\infty = +\infty$. Finite divergence then forces $\alpha = \rho\beta$ for some $\rho \geq 0$, and pointwise convex duality gives

$$\begin{aligned}\mathcal{D}_\varphi^*(f|\beta) &= \sup_{\rho \geq 0} \int_{\mathcal{X}} (f(x)\rho(x) - \varphi(\rho(x)))d\beta(x) \\ &= \int_{\mathcal{X}} \sup_{r \geq 0} (f(x)r - \varphi(r))d\beta(x) = \int_{\mathcal{X}} \varphi^*(f(x))d\beta(x).\end{aligned}$$

For finite φ'_∞ , the effective domain of φ^* is bounded above by φ'_∞ . Hence a singular mass contributes at most $\varphi'_\infty \alpha^\perp(\mathcal{X})$ to the pairing, exactly matching the recession term in (6.5); approximation by continuous test functions recovers this value. The same conjugate formula therefore holds in general. Proposition 6.10 gives convexity and weak-* lower semicontinuity, so Fenchel–Moreau yields (6.8). $\square$

6.4 GANs via Duality

Divergence-based adversarial losses.

$$\min_{\theta} \mathcal{D}_\varphi(\alpha_\theta|\beta) = \min_{\theta} \sup_f \int_{\mathcal{X}} f(x)d\alpha_\theta(x) - \mathcal{D}_\varphi^*(f|\beta) = \min_{\theta} \sup_f \int_{\mathcal{X}} f(g_\theta(z))d\zeta(z) - \frac{1}{m} \sum_{j=1}^m \varphi^*(f(y_j)).$$

$\min_{\theta} \max_{\xi} \int_{\mathcal{Z}} f_{\xi}(g_{\theta}(z))d\zeta(z) - \frac{1}{m} \sum_{j=1}^m \varphi^*(f_{\xi}(y_j))$. For fixed θ , this restriction gives a lower bound on the exact divergence unless the discriminator class is rich enough to attain the supremum. This distinction is essential for empirical data: if β is discrete and α_θ is non-atomic, a superlinear divergence is $+\infty$, whereas its neural restricted objective can remain finite.

$\widehat{\varphi}_{\text{JS}}(s) = s \log s - (s+1) \log \frac{s+1}{2}$, $\widehat{\varphi}_{\text{JS}}^*(u) = -\log(2 - e^u)$, $u < \log 2$. Thus $\mathcal{D}_{\widehat{\varphi}_{\text{JS}}} = 2 \text{JS}^2$, consistently with the normalization above. In practice the min–max problem is solved by alternating stochastic gradient descent/ascent on (θ, ξ) .

Dual norms and integral probability metrics.

$$\min_{\theta} \|\alpha_\theta - \beta\|_B = \min_{\theta} \sup_{f \in B} \int_{\mathcal{X}} f(x)d(\alpha_\theta - \beta)(x) = \min_{\theta} \sup_{f \in B} \int_{\mathcal{Z}} f(g_\theta(z))d\zeta - \frac{1}{m} \sum_{j=1}^m f(y_j).$$

7 Entropic Regularization: Sinkhorn Algorithm

7.1 Entropic Regularization for Discrete Measures

Entropy penalty.

Definition 7.1 (Discrete Shannon–Boltzmann entropy). For a nonnegative matrix P , its Shannon–Boltzmann entropy is $H(P) := -\sum_{i,j} P_{i,j} \log(P_{i,j})$, with the convention $0 \log(0) = 0$.

$$L_C^\varepsilon(a, b) := \min_{P \in U(a, b)} \langle P, C \rangle - \varepsilon H(P). \quad (7.1)$$

Proposition 7.2 (Existence and uniqueness of entropic OT). Assume that a, b are probability histograms and that C is finite. For every $\varepsilon > 0$, problem (7.1) admits a unique minimizer. If all entries of a and b are positive, then this minimizer is positive on every entry.

Proof. The transport polytope $U(a, b)$ is non-empty and compact, and the objective is continuous on it with the convention $0 \log 0 = 0$, so a minimizer exists. The scalar function $r \mapsto r \log r$ is strictly convex on $[0, +\infty)$. On the relative interior, the Hessian $-\partial^2 H(P) = \text{diag}(1/P_{i,j})$ is positive definite. Hence $-H$ is strictly convex on every nontrivial feasible segment, and $\langle P, C \rangle - \varepsilon H(P)$ has at most one minimizer.

If $a_i, b_j > 0$ and a minimizer had $P_{i,j} = 0$, then for small $t > 0$ the perturbation $P_t = (1-t)P + ta \otimes b$ remains feasible. The directional derivative of the entropic part at $t = 0$ is $-\infty$ because the derivative of $r \log r$ at 0 is $-\infty$ along a positive direction. Thus the objective decreases for small t , contradicting optimality. Therefore the minimizer is strictly positive. $\square$

Smoothing effect.

7.2 Sinkhorn’s Algorithm

Proposition 7.3 (Scaling form of entropic OT). P is the unique solution to (7.1) if and only if there exists $(u, v) \in \mathbb{R}_+^n \times \mathbb{R}_+^m$ such that

$$\forall (i, j) \in \llbracket n \rrbracket \times \llbracket m \rrbracket, \quad P_{i,j} = u_i K_{i,j} v_j \quad \text{where} \quad K_{i,j} := e^{-\frac{C_{i,j}}{\varepsilon}}, \quad (7.2)$$

and $P \in U(a, b)$.

Proof. Without loss of generality, we assume $a_i, b_j > 0$ (otherwise, rows or columns with zero mass are fixed to zero and can be removed). By Proposition 7.2, the minimizer is strictly positive on the remaining support.

We can thus ignore the positivity constraint when introducing two dual variables $f \in \mathbb{R}^n, g \in \mathbb{R}^m$ for each marginal constraint so that the Lagrangian of (7.1) reads

$$\mathcal{E}(P, f, g) = \langle P, C \rangle + \varepsilon \sum_{i,j} P_{i,j} \log(P_{i,j}) + \langle f, a - P \mathbf{1}_m \rangle + \langle g, b - P^\top \mathbf{1}_n \rangle.$$

The first-order conditions are

$\frac{\partial \mathcal{E}(P, f, g)}{\partial P_{i,j}} = C_{i,j} + \varepsilon (\log(P_{i,j}) + 1) - f_i - g_j = 0$. Thus $P_{i,j} = e^{(f_i + g_j - C_{i,j})/\varepsilon - 1}$, which has the stated form with the positive scalings $u_i := e^{f_i/\varepsilon - 1}$ and $v_j := e^{g_j/\varepsilon}$. $\square$

$$\text{diag}(\mathbf{u})\mathbf{K}\text{diag}(\mathbf{v})\mathbf{1}_m = \mathbf{a}, \quad \text{and} \quad \text{diag}(\mathbf{v})\mathbf{K}^\top \text{diag}(\mathbf{u})\mathbf{1}_n = \mathbf{b}, \quad (7.3)$$

$$\mathbf{u} \odot (\mathbf{K}\mathbf{v}) = \mathbf{a} \quad \text{and} \quad \mathbf{v} \odot (\mathbf{K}^\top \mathbf{u}) = \mathbf{b} \quad (7.4)$$

where $\odot$ denotes entrywise multiplication.

When $n = m$ and $\mathbf{a}, \mathbf{b}$ are uniform, this is diagonal scaling toward bistochasticity. Proposition 7.2 guarantees a unique scaled matrix $\mathbf{P}$, although the two scaling vectors retain one multiplicative gauge degree of freedom. If some entries of $\mathbf{K}$ vanish, equivalently if $\mathbf{C}$ can take the value $+\infty$, existence and uniqueness require additional support conditions; here we keep $\mathbf{K} > 0$.

$$\mathbf{u}^{(\ell+1)} := \frac{\mathbf{a}}{\mathbf{K}\mathbf{v}^{(\ell)}} \quad \text{and} \quad \mathbf{v}^{(\ell+1)} := \frac{\mathbf{b}}{\mathbf{K}^\top \mathbf{u}^{(\ell+1)}}, \quad (7.5)$$

7.3 Reformulation Using Relative Entropy

Definition 7.4 (Discrete relative entropy). For nonnegative matrices $\mathbf{P}, \mathbf{Q}$ of the same size, the generalized relative entropy, or Kullback–Leibler divergence, is

$$\text{KL}(\mathbf{P}|\mathbf{Q}) := \sum_{i,j} \mathbf{P}_{i,j} \log \left(\frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}} \right) - \mathbf{P}_{i,j} + \mathbf{Q}_{i,j}. \quad (7.6)$$

The convention is $0 \log(0) = 0$, and $\text{KL}(\mathbf{P}|\mathbf{Q}) = +\infty$ if there exists (i, j) such that $\mathbf{Q}_{i,j} = 0$ but $\mathbf{P}_{i,j} \neq 0$.

For the specific case of comparing probability distributions, where $\mathbf{P}$ and $\mathbf{Q}$ have the same total mass, this further simplifies to $\text{KL}(\mathbf{P}|\mathbf{Q}) = \sum_{i,j} \mathbf{P}_{i,j} \log \left(\frac{\mathbf{P}_{i,j}}{\mathbf{Q}_{i,j}} \right)$. For the reference matrix $\mathbf{Q} = \mathbf{1}_{n \times m}$, one has $-\text{KL}(\mathbf{P}|\mathbf{1}_{n \times m}) = \mathbf{H}(\mathbf{P}) + \sum_{i,j} \mathbf{P}_{i,j} - nm$. On fixed-mass couplings the last two terms are constant, so KL regularization with reference $\mathbf{1}_{n \times m}$ is equivalent to subtracting Shannon–Boltzmann entropy. KL is a particular instance of both a φ -divergence (as defined in Section 6.3) and a Bregman divergence; up to standard affine rescalings, it is the canonical overlap between these two families.

Proposition 7.5 (Non-negativity and definiteness of relative entropy). *For all $\mathbf{P}, \mathbf{Q} \in \mathbb{R}_+^{n \times m}$, one has $\text{KL}(\mathbf{P}|\mathbf{Q}) \geq 0$, with equality if and only if $\mathbf{P} = \mathbf{Q}$.*

Proof. Write $\varphi(s) = s \log s - s + 1$. Convexity gives $\varphi(s) \geq \varphi(1) + \varphi'(1)(s - 1) = 0$, and strict convexity gives equality only at $s = 1$. Hence $\text{KL}(\mathbf{P}|\mathbf{Q}) = \sum_{(i,j): \mathbf{Q}_{i,j} > 0} \mathbf{Q}_{i,j} \varphi(\mathbf{P}_{i,j}/\mathbf{Q}_{i,j}) \geq 0$, whenever $\mathbf{P} \ll \mathbf{Q}$ entrywise; otherwise the divergence is $+\infty$. Equality forces $\mathbf{P}_{i,j} = \mathbf{Q}_{i,j}$ wherever $\mathbf{Q}_{i,j} > 0$, while entries with $\mathbf{Q}_{i,j} = 0$ vanish in both matrices. Thus equality is equivalent to $\mathbf{P} = \mathbf{Q}$. $\square$

When $\mathbf{P}$ and $\mathbf{Q}$ are probability matrices and $\mathbf{P} \ll \mathbf{Q}$, the same divergence can also be written

$$\text{KL}(\mathbf{P}|\mathbf{Q}) = \sum_{i,j} \psi(\mathbf{P}_{i,j}/\mathbf{Q}_{i,j})\mathbf{Q}_{i,j}, \quad \psi(s) = s \log s.$$

$$\sum_{i,j} \psi(\mathbf{P}_{i,j}/\mathbf{Q}_{i,j})\mathbf{Q}_{i,j} \geq \psi \left(\sum_{i,j} \mathbf{P}_{i,j} \right) = \psi(1) = 0.$$

$$\min_{\mathbf{P} \in \mathbf{U}(\mathbf{a}, \mathbf{b})} \langle \mathbf{P}, \mathbf{C} \rangle + \varepsilon \text{KL}(\mathbf{P}|\mathbf{a} \otimes \mathbf{b}). \quad (7.7)$$

For the balanced problem with fixed positive marginals, however, the choice of tensor-product reference does not affect the selected coupling: it only adds a constant to the objective, as shown in the following proposition. In particular, (7.7) and (7.1) have the same unique solution.

Proposition 7.6 (Reference measure shift for KL). *After removing zero-mass rows and columns, assume that $\mathbf{a}, \mathbf{a}' \in \Sigma_n$ and $\mathbf{b}, \mathbf{b}' \in \Sigma_m$ have positive entries. For every $\mathbf{P} \in \mathbf{U}(\mathbf{a}, \mathbf{b})$, one has $\text{KL}(\mathbf{P}|\mathbf{a} \otimes \mathbf{b}) = \text{KL}(\mathbf{P}|\mathbf{a}' \otimes \mathbf{b}') - \text{KL}(\mathbf{a}|\mathbf{a}') - \text{KL}(\mathbf{b}|\mathbf{b}')$. In particular, (7.7) and (7.1) have the same unique minimizer.*

Proof. Expanding the logarithm and using the marginal constraints gives

$$\begin{aligned} \text{KL}(\mathbf{P}|\mathbf{a} \otimes \mathbf{b}) &= \text{KL}(\mathbf{P}|\mathbf{a}' \otimes \mathbf{b}') + \sum_i \mathbf{a}_i \log \frac{\mathbf{a}'_i}{\mathbf{a}_i} + \sum_j \mathbf{b}_j \log \frac{\mathbf{b}'_j}{\mathbf{b}_j} \\ &= \text{KL}(\mathbf{P}|\mathbf{a}' \otimes \mathbf{b}') - \text{KL}(\mathbf{a}|\mathbf{a}') - \text{KL}(\mathbf{b}|\mathbf{b}'). \end{aligned}$$

$\square$

The tensor-product reference is nevertheless useful when supports vary, because it makes explicit which entries are allowed to vanish. It is also the normalization that passes cleanly to the continuous formulation below, where the reference measure is $\alpha \otimes \beta$ rather than an ambient Lebesgue measure.

Proposition 7.7 (Convergence with ε). *Assume, after removing zero-mass rows and columns, that $\mathbf{a}$ and $\mathbf{b}$ are positive and that $\mathbf{C}$ is finite. The unique solution $\mathbf{P}_\varepsilon$ of (7.1) converges to the optimal solution with maximal entropy within the set of all optimal solutions of the Kantorovich problem, namely*

$$\mathbf{P}_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} \underset{\mathbf{P}}{\text{argmin}} \{ -\mathbf{H}(\mathbf{P}) ; \mathbf{P} \in \mathbf{U}(\mathbf{a}, \mathbf{b}), \langle \mathbf{P}, \mathbf{C} \rangle = \mathbf{L}_\mathbf{C}(\mathbf{a}, \mathbf{b}) \} \quad (7.8)$$

so that in particular

$$\mathbf{L}_\mathbf{C}^\varepsilon(\mathbf{a}, \mathbf{b}) \xrightarrow{\varepsilon \rightarrow 0} \mathbf{L}_\mathbf{C}(\mathbf{a}, \mathbf{b}).$$

Moreover,

$$\mathbf{P}_\varepsilon \xrightarrow{\varepsilon \rightarrow \infty} \mathbf{a} \otimes \mathbf{b}. \quad (7.9)$$

Proof. **Case $\varepsilon \rightarrow 0$.** We denote P_ℓ the solution of (7.1) for $\varepsilon = \varepsilon_\ell$.

Since $U(a, b)$ is bounded, we can extract a sequence (that we do not relabel for the sake of simplicity) such that $P_\ell \rightarrow P^*$. Since $U(a, b)$ is closed, $P^* \in U(a, b)$. Using the equivalent KL-normalized formulation (7.7), optimality of P and P_ℓ for their respective optimization problems (for $\varepsilon = 0$ and $\varepsilon = \varepsilon_\ell$) gives

$$0 \leq \langle C, P_\ell \rangle - \langle C, P \rangle \leq \varepsilon_\ell (\text{KL}(P|a \otimes b) - \text{KL}(P_\ell|a \otimes b)). \quad (7.10)$$

Since KL is continuous, taking the limit $\ell \rightarrow +\infty$ in this expression shows that $\langle C, P^* \rangle = \langle C, P \rangle$ so that P^* is a feasible point of (7.8). Furthermore, dividing by ε_ℓ in (7.10) and taking the limit shows that $\text{KL}(P^*|a \otimes b) \leq \text{KL}(P|a \otimes b)$, which shows that P^* is a solution of (7.8). Since the solution P_0^* to this program is unique by strict convexity of $\text{KL}(\cdot|a \otimes b)$ on the optimal face, one has $P^* = P_0^*$, and the whole sequence is converging.

Case $\varepsilon \rightarrow +\infty$. Subtracting $\min_{i,j} C_{i,j}$ from the cost changes every feasible objective by the same constant, so it does not change the minimizer. We can therefore assume $C \geq 0$. Evaluating the energy at $a \otimes b$ (which belongs to the constraint set $U(a, b)$), one has $\langle C, P_\varepsilon \rangle + \varepsilon \text{KL}(P_\varepsilon|a \otimes b) \leq \langle C, a \otimes b \rangle + \varepsilon \times 0$ and since $\langle C, P_\varepsilon \rangle \geq 0$, this leads to $\text{KL}(P_\varepsilon|a \otimes b) \leq \varepsilon^{-1} \langle C, a \otimes b \rangle \leq \frac{\|C\|_\infty}{\varepsilon}$ so that $\text{KL}(P_\varepsilon|a \otimes b) \rightarrow 0$ and thus $P_\varepsilon \rightarrow a \otimes b$ since KL is a valid divergence. $\square$

7.4 General Formulation

Measure formulation.

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) := \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \varepsilon \text{KL}(\pi|\alpha \otimes \beta) \quad (7.11)$$

Definition 7.8 (Relative entropy of measures). For finite nonnegative measures π and ξ on $\mathcal{X} \times \mathcal{Y}$, the relative entropy is

$$\text{KL}(\pi|\xi) := \int_{\mathcal{X} \times \mathcal{Y}} \log\left(\frac{d\pi}{d\xi}\right) d\pi + \xi(\mathcal{X} \times \mathcal{Y}) - \pi(\mathcal{X} \times \mathcal{Y}). \quad (7.12)$$

By convention, $\text{KL}(\pi|\xi) = +\infty$ if π is not absolutely continuous with respect to ξ .

For fixed balanced marginals, the specific product reference in the entropy only matters up to additive constants, exactly as in Proposition 7.6, provided the alternative reference marginals are mutually absolutely continuous with α and β . Its support and absolute-continuity structure are nevertheless essential: they determine which couplings have finite entropy.

Probabilistic interpretation.

Definition 7.9 (Mutual information). If $(X, Y) \sim \pi$ have marginals $X \sim \alpha$ and $Y \sim \beta$, the mutual information of the pair is $\mathcal{I}(X, Y) := \text{KL}(\pi|\alpha \otimes \beta)$. It is nonnegative and vanishes if and only if X and Y are independent.

$$\inf_{X \sim \alpha, Y \sim \beta} \mathbb{E}(c(X, Y)) + \varepsilon \mathcal{I}(X, Y).$$

Convergence with ε .

Proposition 7.10 (Convergence with ε for measures). Let $\mathcal{X}, \mathcal{Y}$ be compact metric spaces, let $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$, and let $\alpha \in \mathcal{P}(\mathcal{X})$, $\beta \in \mathcal{P}(\mathcal{Y})$. Assume that finite-entropy couplings are dense in $\mathcal{U}(\alpha, \beta)$ for weak convergence with convergence of the cost integral. For $\varepsilon > 0$, let π_ε be a minimizer of (7.11). Then $\mathcal{L}_c^\varepsilon(\alpha, \beta) \rightarrow \mathcal{L}_c(\alpha, \beta)$ as $\varepsilon \downarrow 0$, and every weak cluster point of $(\pi_\varepsilon)_\varepsilon$ is an optimal plan for the unregularized Kantorovich problem. If the latter optimal plan is unique, then π_ε converges weakly to it. In particular, for $\mathcal{X} = \mathcal{Y} \subset \mathbb{R}^d$, $c(x, y) = \|x - y\|^2$ and α absolutely continuous, $\pi_\varepsilon \rightarrow (\text{Id}, T)_\# \alpha$, where T is the Brenier map. As $\varepsilon \rightarrow +\infty$,

$$\pi_\varepsilon \rightarrow \alpha \otimes \beta \quad \text{in total variation,} \quad \mathcal{L}_c^\varepsilon(\alpha, \beta) \rightarrow \int_{\mathcal{X} \times \mathcal{Y}} c d\alpha d\beta.$$

Proof. Existence follows from compactness of $\mathcal{U}(\alpha, \beta)$ and lower semicontinuity of the relative entropy. For the limit $\varepsilon \downarrow 0$, take any sequence $\varepsilon_k \downarrow 0$ and a weak cluster point π^* of π_{ε_k} . Since $\text{KL}(\pi|\alpha \otimes \beta) \geq 0$ on probability couplings, $\liminf_k \mathcal{L}_c^{\varepsilon_k}(\alpha, \beta) \geq \liminf_k \int c d\pi_{\varepsilon_k} = \int c d\pi^* \geq \mathcal{L}_c(\alpha, \beta)$. Conversely, by the density assumption, for every $\delta > 0$ there exists a coupling π^δ with finite relative entropy and $\int c d\pi^\delta \leq \mathcal{L}_c(\alpha, \beta) + \delta$. Testing (7.11) with π^δ gives $\limsup_k \mathcal{L}_c^{\varepsilon_k}(\alpha, \beta) \leq \int c d\pi^\delta$. Letting $\delta \downarrow 0$ proves convergence of the values and forces $\int c d\pi^* = \mathcal{L}_c(\alpha, \beta)$. Uniqueness of the unregularized optimizer then identifies all cluster points. Brenier's theorem gives the stated quadratic consequence when α is absolutely continuous.

For the limit $\varepsilon \rightarrow +\infty$, subtract a constant from c and assume $c \geq 0$. Since $\alpha \otimes \beta$ is admissible and has zero relative entropy with respect to itself, $\int c d\pi_\varepsilon + \varepsilon \text{KL}(\pi_\varepsilon|\alpha \otimes \beta) \leq \int c d\alpha d\beta$. Hence $\text{KL}(\pi_\varepsilon|\alpha \otimes \beta) \leq \varepsilon^{-1} \int c d\alpha d\beta$. Pinsker's inequality gives total-variation convergence to $\alpha \otimes \beta$, and the convergence of the values follows by continuity of c . $\square$

Large-temperature expansion. When ε is large, the entropy dominates and the optimal plan is a small perturbation of the product coupling. The useful quantity is the component of the cost that cannot be absorbed into row and column potentials.

Proposition 7.11 (Large-temperature expansion). Let $r = \alpha \otimes \beta$ and assume $c \in L^\infty(r)$. Define the product averages $\bar{c}_\mathcal{X}(x) := \int_\mathcal{Y} c(x, y) d\beta(y)$, $\bar{c}_\mathcal{Y}(y) := \int_\mathcal{X} c(x, y) d\alpha(x)$, $\bar{c} := \int_{\mathcal{X} \times \mathcal{Y}} c dr$, and the centered interaction $c_0(x, y) := c(x, y) - \bar{c}_\mathcal{X}(x) - \bar{c}_\mathcal{Y}(y) + \bar{c}$. Assume that the large-temperature branch $p_\varepsilon = d\pi_\varepsilon/dr$, $f_\varepsilon, g_\varepsilon$ admits an expansion to third order in ε^{-1} near 0 in $L^\infty(r)$, with gauge $\int g_\varepsilon d\beta = 0$. Then $p_\varepsilon(x, y) = 1 - \frac{c_0(x, y)}{\varepsilon} + O(\varepsilon^{-2})$ in $L^2(r)$, and

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \bar{c} - \frac{1}{2\varepsilon} \int_{\mathcal{X} \times \mathcal{Y}} c_0(x, y)^2 dr(x, y) + \frac{1}{6\varepsilon^2} \int_{\mathcal{X} \times \mathcal{Y}} c_0(x, y)^3 dr(x, y) + O(\varepsilon^{-3}).$$

Moreover, with $A(x) := \int_\mathcal{Y} c_0(x, y)^2 d\beta(y)$, $B(y) := \int_\mathcal{X} c_0(x, y)^2 d\alpha(x)$, $\sigma^2 := \int_{\mathcal{X} \times \mathcal{Y}} c_0^2 dr$, one has $f_\varepsilon(x) = \bar{c}_\mathcal{X}(x) - \frac{A(x)}{2\varepsilon} + O(\varepsilon^{-2})$, and $g_\varepsilon(y) = \bar{c}_\mathcal{Y}(y) - \bar{c} + \frac{\sigma^2 - B(y)}{2\varepsilon} + O(\varepsilon^{-2})$ in the same gauge.

Proof. The optimality equation can be written $p_\varepsilon(x, y) = \exp\left(\frac{f_\varepsilon(x) + g_\varepsilon(y) - c(x, y)}{\varepsilon}\right)$, with the constraints $\int p_\varepsilon(x, y) d\beta(y) = 1$ and $\int p_\varepsilon(x, y) d\alpha(x) = 1$. Write $p_\varepsilon = 1 + \varepsilon^{-1}q + O(\varepsilon^{-2})$. The linearized marginal constraints impose $\int q(x, y) d\beta(y) = 0$, $\int q(x, y) d\alpha(x) = 0$. Expanding the objective gives

$$\int c p_\varepsilon dr + \varepsilon \int p_\varepsilon \log(p_\varepsilon) dr = \bar{c} + \frac{1}{\varepsilon} \left(\int c q dr + \frac{1}{2} \int q^2 dr \right) + O(\varepsilon^{-2}).$$

Thus q is the minimizer of the quadratic expression over zero-row and zero-column perturbations. Orthogonal projection in $L^2(r)$ gives $q = -c_0$, because c_0 has zero conditional means. This proves the first-order density and value expansions. To identify the next displayed terms, write $p_\varepsilon = 1 + \varepsilon^{-1}q + \varepsilon^{-2}s + O(\varepsilon^{-3})$ and $f_\varepsilon = f_0 + \varepsilon^{-1}f_1 + O(\varepsilon^{-2})$, $g_\varepsilon = g_0 + \varepsilon^{-1}g_1 + O(\varepsilon^{-2})$, where $f_0 = \bar{c}_X$ and $g_0 = \bar{c}_Y - \bar{c}$. Since $p_\varepsilon = \exp\left(-\frac{c_0}{\varepsilon} + \frac{f_1(x) + g_1(y)}{\varepsilon^2} + O(\varepsilon^{-3})\right)$, one has $s = f_1 + g_1 + c_0^2/2$. The second-order marginal constraints and the gauge $\int g_1 d\beta = 0$ give $f_1 = -A/2$ and $g_1 = (\sigma^2 - B)/2$. Finally, $p_\varepsilon \log p_\varepsilon = \varepsilon^{-1}q + \varepsilon^{-2}(s + \frac{1}{2}q^2) + \varepsilon^{-3}(qs - \frac{1}{6}q^3) + O(\varepsilon^{-4})$, and the zero-conditional-mean property of c_0 cancels all additive terms in s . The order- ε^{-2} contribution to the value is therefore $\frac{1}{2} \int c_0^3 dr - \frac{1}{3} \int c_0^3 dr = \frac{1}{6} \int c_0^3 dr$. $\square$

Small-temperature expansion for smooth densities.

Proposition 7.12 (Small-temperature quadratic expansion). *Let $\alpha = \rho_0 dx$ and $\beta = \rho_1 dx$ be probability measures on $\mathbb{R}^d$ with bounded compactly supported densities. Let $\alpha_t = \rho_t dx$ be their quadratic displacement interpolation and assume $\mathcal{I}_{\text{geo}}(\alpha, \beta) := \int_0^1 \int_{\mathbb{R}^d} \|\nabla \log \rho_t(x)\|^2 \rho_t(x) dx dt < +\infty$. For $c(x, y) = \|x - y\|^2$, define $H(\alpha) := \int_{\mathbb{R}^d} \rho_0 \log \rho_0 dx$ and similarly for β . Then, as $\varepsilon \downarrow 0$,*

$$\mathcal{L}_{\|\cdot\|_2}^\varepsilon(\alpha, \beta) = \mathcal{W}_2^2(\alpha, \beta) - \frac{d\varepsilon}{2} \log(\pi\varepsilon) - \frac{\varepsilon}{2} (H(\alpha) + H(\beta)) + \frac{\varepsilon^2}{16} \mathcal{I}_{\text{geo}}(\alpha, \beta) + o(\varepsilon^2).$$

If, in addition, the endpoint densities are smooth and positive on their supports and the quadratic optimal map is a smooth non-degenerate diffeomorphism, let $f_\varepsilon, g_\varepsilon$ be optimal Sinkhorn potentials and f_0, g_0 Kantorovich potentials, normalized by $\int g_\varepsilon d\beta = \int g_0 d\beta = 0$. Then $f_\varepsilon \rightarrow f_0$ and $g_\varepsilon \rightarrow g_0$ locally uniformly on the interiors of the supports. Moreover, the scalar part of the first potential is fixed by

$$\int f_\varepsilon d\alpha = \mathcal{W}_2^2(\alpha, \beta) - \frac{d\varepsilon}{2} \log(\pi\varepsilon) - \frac{\varepsilon}{2} (H(\alpha) + H(\beta)) + \frac{\varepsilon^2}{16} \mathcal{I}_{\text{geo}}(\alpha, \beta) + o(\varepsilon^2).$$

The spatial order- ε corrections follow from the Laplace prefactors in the two soft c -transform equations and depend on the endpoint densities and the Hessian of the quadratic contact phase.

Proof. For the Brownian reference with heat kernel variance T , the dynamic Schrödinger formulation gives

$$TC_T(\alpha, \beta) = \frac{1}{2} \mathcal{W}_2^2(\alpha, \beta) + \frac{T}{2} (H(\alpha) + H(\beta)) + \frac{T^2}{8} \mathcal{I}_{\text{geo}}(\alpha, \beta) + o(T^2),$$

under the stated regularity assumptions. Here $\mathcal{C}_T$ uses entropy relative to the sigma-finite Brownian endpoint measure:

$$\mathcal{C}_T(\alpha, \beta) = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \mathcal{H}(\pi | p_T(x, y) dx dy), \quad p_T(x, y) = (2\pi T)^{-d/2} e^{-\|x - y\|^2 / (2T)},$$

where $\mathcal{H}(\pi | \xi) := \int \log(d\pi/d\xi) d\pi$ for sigma-finite ξ . When ξ is a probability measure, this agrees with Definition 7.8; unlike the generalized KL of that definition, it remains meaningful when the Brownian reference has infinite total mass. Expanding the heat kernel, multiplying by $2T$, and setting $\varepsilon = 2T$ gives $2TC_T(\alpha, \beta) = \mathcal{L}_{\|\cdot\|_2}^\varepsilon(\alpha, \beta) + 2T(H(\alpha) + H(\beta)) + dT \log(2\pi T)$. Equivalently, $\mathcal{L}_{\|\cdot\|_2}^\varepsilon(\alpha, \beta) = 2TC_T(\alpha, \beta) - 2T(H(\alpha) + H(\beta)) - dT \log(2\pi T)$, which is exactly the displayed expansion because $2T = \varepsilon$. At a dual optimum the exponential penalty integrates to zero, so the gauge $\int g_\varepsilon d\beta = 0$ gives $\int f_\varepsilon d\alpha = \mathcal{L}_{\|\cdot\|_2}^\varepsilon(\alpha, \beta)$. This proves the displayed scalar expansion. Local convergence and the spatial corrections follow from Laplace's method applied to the soft c -transform equations; changing the additive gauge redistributes scalar terms between the two potentials. $\square$

7.5 Dual of Sinkhorn

Discrete dual.

Proposition 7.13 (Dual of entropic OT). *The optimal value of (7.7) is*

$$\min_{P \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle P, C \rangle + \varepsilon \text{KL}(P | \mathbf{a} \otimes \mathbf{b}) = \max_{f \in \mathbb{R}^n, g \in \mathbb{R}^m} \langle f, \mathbf{a} \rangle + \langle g, \mathbf{b} \rangle - \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{i,j}}{\varepsilon}\right) a_i b_j + \varepsilon. \quad (7.13)$$

The optimal (f, g) are linked to scalings (u, v) appearing in (7.2) through

$$u_i = a_i e^{f_i/\varepsilon} \quad \text{and} \quad v_j = b_j e^{g_j/\varepsilon}. \quad (7.14)$$

Proof. We introduce Lagrange multipliers and consider $\min_{P \geq 0} \max_{f, g} \langle C, P \rangle + \varepsilon \text{KL}(P | \mathbf{a} \otimes \mathbf{b}) + \langle \mathbf{a} - P\mathbf{1}, f \rangle + \langle \mathbf{b} - P^\top \mathbf{1}, g \rangle$. Finite-dimensional convex duality allows us to exchange the minimum over P with the maximum over (f, g) , giving

$$\max_{f, g} \langle f, \mathbf{a} \rangle + \langle g, \mathbf{b} \rangle + \varepsilon \min_{P \geq 0} \left(\text{KL}(P | \mathbf{a} \otimes \mathbf{b}) - \left\langle \frac{f \oplus g - C}{\varepsilon}, P \right\rangle \right) = \langle f, \mathbf{a} \rangle + \langle g, \mathbf{b} \rangle - \varepsilon \text{KL}^*\left(\frac{f \oplus g - C}{\varepsilon} | \mathbf{a} \otimes \mathbf{b}\right).$$

One concludes by using (6.9) for $\varphi(r) = r \log(r) - r + 1$ $\text{KL}^*(H | \mathbf{a} \otimes \mathbf{b}) = \sum_{i,j} \varphi^*(H_{i,j}) a_i b_j$. Indeed, the scalar maximization $\varphi^*(s) = \sup_{r \geq 0} \{rs - r \log r + r - 1\}$ has first-order condition $s - \log r = 0$, hence $r = e^s$ and $\varphi^*(s) = e^s - 1$. $\square$

Discrete soft c -transforms.

$$\begin{aligned} a_i - e^{\frac{f_i}{\varepsilon}} a_i \sum_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right) b_j &= 0 \\ f_i &= -\varepsilon \log \sum_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right) b_j. \end{aligned}$$

Definition 7.14 (Soft-min and discrete soft c -transform). For $h \in \mathbb{R}^m$ and weights $b \in \Sigma_m$, the weighted soft-min at temperature $\varepsilon > 0$ is $\min_b^\varepsilon(h) := -\varepsilon \log \sum_j e^{-h_j/\varepsilon} b_j$. It converges to $\min_{j: b_j > 0} h_j$ as $\varepsilon \rightarrow 0$; for positive weights this is $\min_j h_j$. Given a cost matrix C , the discrete soft c -transforms are

$$f_i = \min_b^\varepsilon(C_{i,\cdot} - g), \quad g_j = \min_a^\varepsilon(C_{\cdot,j} - f). \quad (7.15)$$

$$\min_b^\varepsilon(h - \text{cst}) = \min_b^\varepsilon(h) - \text{cst}$$

Continuous dual and soft-transforms. For general, not necessarily discrete, measures (α, β) , the KL-regularized problem (7.11) has the concave dual

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \sup_{f \in \mathcal{C}(\mathcal{X}), g \in \mathcal{C}(\mathcal{Y})} \mathcal{D}_\varepsilon(f, g), \quad (7.16)$$

$$\mathcal{D}_\varepsilon(f, g) := \int_{\mathcal{X}} f(x) d\alpha(x) + \int_{\mathcal{Y}} g(y) d\beta(y) - \varepsilon \int_{\mathcal{X} \times \mathcal{Y}} \left(e^{\frac{f(x) + g(y) - c(x, y)}{\varepsilon}} - 1 \right) d\alpha(x) d\beta(y). \quad (7.17)$$

Definition 7.15 (Continuous soft c -transforms). For $f \in \mathcal{C}(\mathcal{X})$ and $g \in \mathcal{C}(\mathcal{Y})$, define

$$f^{c, \varepsilon}(y) := -\varepsilon \log \left(\int_{\mathcal{X}} e^{\frac{f(x) - c(x, y)}{\varepsilon}} d\alpha(x) \right), \quad y \in \mathcal{Y}, \quad (7.18)$$

$$g^{\bar{c}, \varepsilon}(x) := -\varepsilon \log \left(\int_{\mathcal{Y}} e^{\frac{g(y) - c(x, y)}{\varepsilon}} d\beta(y) \right), \quad x \in \mathcal{X}. \quad (7.19)$$

Proposition 7.16 (Existence and uniqueness of entropic dual potentials). Assume that $\mathcal{X} = \text{supp}(\alpha)$ and $\mathcal{Y} = \text{supp}(\beta)$ are compact and that c is continuous. The dual problem (7.16) has solutions, and its set of solutions is $(f^* + \lambda, g^* - \lambda)$, $\lambda \in \mathbb{R}$.

Proof. Normalize potentials by imposing $\int f d\alpha = 0$. Replacing any pair (f, g) by the corresponding soft c -transforms does not decrease the dual objective, because each soft transform is the exact maximizer in one block variable. For transformed potentials, the oscillations are bounded by the oscillation of the cost: $\|f\|_V + \|g\|_V \leq 2(\sup c - \inf c)$. Moreover the modulus of continuity of the soft transforms is controlled by that of c ; for instance $|g^{\bar{c}, \varepsilon}(x) - g^{\bar{c}, \varepsilon}(x')| \leq \sup_y |c(x, y) - c(x', y)|$. After normalization, maximizing sequences are therefore uniformly bounded and equicontinuous. Arzelà–Ascoli gives a uniformly converging subsequence, and continuity of (7.17) gives a maximizer.

For uniqueness, use strict convexity of $H \mapsto \int e^{H/\varepsilon} d(\alpha \otimes \beta)$ on the image of $(f, g) \mapsto H = f \oplus g - c$, modulo constants. If two optimal pairs exist, their midpoint is also optimal; strict convexity forces the two functions $f \oplus g$ to agree $\alpha \otimes \beta$ -almost everywhere. Full support and continuity then imply equality everywhere on $\mathcal{X} \times \mathcal{Y}$, so $f - f'$ is constant and $g - g'$ is the opposite constant. Without replacing the ambient spaces by the marginal supports, uniqueness has this meaning only on those supports; values outside them do not enter the dual objective. $\square$

Neural dual solvers. $\Phi(x) + \Psi(y) \geq \langle x, y \rangle$. For the quadratic cost this is the same statement after subtracting the quadratic terms, using the convexity properties of soft transforms. One can therefore maximize the continuous dual (7.17) over parameterized convex potentials, estimating the integrals by stochastic samples.

7.6 Path-Space Schrödinger Problem

Unregularized path-space transport. Let $\Omega = C([0, 1]; \mathcal{X})$ be a path space and denote by $e_t : \Omega \rightarrow \mathcal{X}$, $e_t(\omega) = \omega_t$ $(e_0)_\# M = \alpha$, $(e_1)_\# M = \beta$

$$\inf_{M \in \mathcal{P}(\Omega)} \left\{ \int_{\Omega} \mathcal{A}(\omega) dM(\omega) ; (e_0)_\# M = \alpha, (e_1)_\# M = \beta \right\}. \quad (7.20)$$

For the quadratic Wasserstein geometry on $\mathbb{R}^d$, one takes

$$\begin{aligned} \mathcal{A}(\omega) &= \begin{cases} \int_0^1 \|\dot{\omega}_t\|^2 dt, & \text{if } \omega \text{ is absolutely continuous,} \\ +\infty, & \text{otherwise.} \end{cases} \\ c_{\mathcal{A}}(x, y) &:= \inf_{\omega \in \Omega} \{ \mathcal{A}(\omega) ; e_0(\omega) = x, e_1(\omega) = y \}. \end{aligned} \quad (7.21)$$

For the quadratic action above, the minimizing path is the straight segment $\omega_t = (1 - t)x + ty$, and $c_{\mathcal{A}}(x, y) = \|x - y\|^2$.

Proposition 7.17 (Endpoint reduction of path-space transport). Assume that minimizing paths in (7.21) can be selected measurably, or more generally that the infimum can be approximated by measurable selections. Then (7.20) has the same value as the Kantorovich problem $\inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{X}} c_{\mathcal{A}}(x, y) d\pi(x, y)$. Moreover, if π^* is an optimal endpoint coupling and $\omega^{x, y}$ is an optimal path from x to y , then $M^* = \int_{\mathcal{X} \times \mathcal{X}} \delta_{\omega^{x, y}} d\pi^*(x, y)$ is an optimal path law.

Proof. Let M be any feasible path law and set $\pi = (e_0, e_1)_\# M$. Then $\pi \in \mathcal{U}(\alpha, \beta)$ and, by the definition of $c_{\mathcal{A}}$, $\int_{\Omega} \mathcal{A}(\omega) dM(\omega) \geq \int_{\mathcal{X} \times \mathcal{X}} c_{\mathcal{A}}(x, y) d\pi(x, y)$. This proves that the path-space value is at least the Kantorovich value. Conversely, given a coupling π and a measurable selection $(x, y) \mapsto \omega^{x,y}$ with action arbitrarily close to $c_{\mathcal{A}}(x, y)$, the mixture of Dirac path laws $M = \int \delta_{\omega^{x,y}} d\pi(x, y)$ has endpoints α and β and action equal, up to the selected approximation error, to $\int c_{\mathcal{A}} d\pi$. Optimizing over π and letting the approximation error vanish proves equality. If exact minimizing paths are selected for an optimal π^* , the displayed M^* is optimal. $\square$

Entropic path-space problem. Let $\mathcal{R}^\varepsilon \in \mathcal{P}(\Omega)$ be a reference path law, for instance a Brownian or Langevin dynamics at noise level ε . Schrödinger's dynamic problem is the entropy projection

$$\text{SB}_\varepsilon(\alpha, \beta) := \inf_{M \in \mathcal{P}(\Omega)} \{ \varepsilon \text{KL}(M|\mathcal{R}^\varepsilon) ; (e_0)_\# M = \alpha, (e_1)_\# M = \beta \}. \quad (7.22)$$

Stochastic-control interpretation. For Brownian references, the entropy projection has a direct control meaning. To fix constants, suppose that $\mathcal{R}^\varepsilon$ is the law of

$$dX_t = \sqrt{\varepsilon} dB_t, \quad X_0 \sim \alpha, dX_t = u_t dt + \sqrt{\varepsilon} dB_t, \quad X_1 \sim \alpha.$$

$$\text{KL}(M|\mathcal{R}^\varepsilon) = \frac{1}{2\varepsilon} \mathbb{E}_M \int_0^1 \|u_t\|^2 dt, \quad \varepsilon \text{KL}(M|\mathcal{R}^\varepsilon) = \frac{1}{2} \mathbb{E}_M \int_0^1 \|u_t\|^2 dt.$$

If the reference initial law is not α , an additional term $\varepsilon \text{KL}(\alpha|\mathcal{R}_0^\varepsilon)$ appears, but it is fixed by the endpoint constraint. Hence, up to this fixed endpoint cost, the Schrödinger bridge can be read as

$$\inf_u \left\{ \frac{1}{2} \mathbb{E} \int_0^1 \|u_t\|^2 dt ; dX_t = u_t dt + \sqrt{\varepsilon} dB_t, \quad X_0 \sim \alpha, X_1 \sim \beta \right\}, \quad u_t^*(x) = \varepsilon \nabla \log h_t(x).$$

Viscous Benamou–Brenier formulations. $\partial_t \rho_t + \text{div}(\rho_t v_t) = \frac{\sigma}{2} \Delta \rho_t$, $\int_0^1 \int \frac{1}{2} \|v_t(x)\|^2 \rho_t(x) dx dt$. $u_t = v_t - \frac{\sigma}{2} \nabla \log \rho_t$, $\partial_t \rho_t + \text{div}(\rho_t u_t) = 0$.

$$\begin{aligned} \int_0^1 \int \frac{1}{2} \|v_t\|^2 \rho_t dx dt &= \int_0^1 \int \left(\frac{1}{2} \|u_t\|^2 + \frac{\sigma^2}{8} \|\nabla \log \rho_t\|^2 \right) \rho_t dx dt \\ &\quad + \frac{\sigma}{2} \left[\int \rho_1 \log \rho_1 dx - \int \rho_0 \log \rho_0 dx \right]. \\ \int_0^1 \int \left(\frac{1}{2} \|u_t(x)\|^2 + \frac{\sigma^2}{8} \|\nabla \log \rho_t(x)\|^2 \right) \rho_t(x) dx dt. \end{aligned}$$

Thus the Schrödinger bridge is a least-action interpolation with both transport kinetic energy and a Fisher-information penalty. If one instead writes the viscous equation with diffusion coefficient $\sigma \Delta \rho_t$, the same formula is obtained after replacing σ above by 2σ , so the Fisher coefficient becomes $\sigma^2/2$.

$$\mathcal{R}^\varepsilon(d\omega) = \int \mathcal{R}^{\varepsilon,x,y}(d\omega) \mathcal{R}_{01}^\varepsilon(dx, dy), \quad \mathcal{R}_{01}^\varepsilon := (e_0, e_1)_\# \mathcal{R}^\varepsilon,$$

where $\mathcal{R}^{\varepsilon,x,y}$ is the reference bridge conditioned on endpoints (x, y) . Similarly, for any feasible M , write $M(d\omega) = \int M^{x,y}(d\omega) \pi(dx, dy)$, $\pi := (e_0, e_1)_\# M$.

Proposition 7.18 (Endpoint reduction of the Schrödinger problem). *Assume that the regular conditional laws above exist and that the relative-entropy chain rule applies, with value $+\infty$ when absolute continuity fails. Then*

$$\text{SB}_\varepsilon(\alpha, \beta) = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \varepsilon \text{KL}(\pi|\mathcal{R}_{01}^\varepsilon). \quad (7.23)$$

For a fixed endpoint coupling π with finite $\text{KL}(\pi|\mathcal{R}_{01}^\varepsilon)$, the minimizing path law is the mixture of reference bridges

$$M^\pi = \int \mathcal{R}^{\varepsilon,x,y} d\pi(x, y). \quad (7.24)$$

Proof. If M has finite relative entropy with respect to $\mathcal{R}^\varepsilon$, then $\pi = (e_0, e_1)_\# M$ is necessarily absolutely continuous with respect to $\mathcal{R}_{01}^\varepsilon$. The chain rule for relative entropy gives

$$\text{KL}(M|\mathcal{R}^\varepsilon) = \text{KL}(\pi|\mathcal{R}_{01}^\varepsilon) + \int_{\mathcal{X} \times \mathcal{X}} \text{KL}(M^{x,y}|\mathcal{R}^{\varepsilon,x,y}) d\pi(x, y). \quad (7.25)$$

The second term is nonnegative and vanishes exactly when $M^{x,y} = \mathcal{R}^{\varepsilon,x,y}$ for π -almost every (x, y) . Thus, once an endpoint coupling π with finite $\text{KL}(\pi|\mathcal{R}_{01}^\varepsilon)$ is fixed, the best path law is (7.24), and the remaining minimization is precisely (7.23). If no such endpoint coupling exists, both sides are $+\infty$. $\square$

Brownian bridges and Sinkhorn couplings.

$$p_\varepsilon(x, y) \propto \exp\left(-\frac{\|x - y\|^2}{\varepsilon}\right).$$

$$\mathcal{R}_{01}^\varepsilon(dx, dy) \propto \exp\left(-\frac{c(x, y)}{\varepsilon}\right) \alpha(dx) \beta(dy).$$

This includes the usual heat-kernel reference after rewriting it with respect to $\alpha \otimes \beta$, whenever the one-time endpoint densities are fixed and mutually absolutely continuous. The additional one-body density factors only add constants under the marginal constraints.

$\varepsilon \text{KL}(\pi|\mathcal{R}_{01}^\varepsilon) = \int c(x, y) d\pi(x, y) + \varepsilon \text{KL}(\pi|\alpha \otimes \beta) + \text{constant}$, where the constant does not depend on π . Hence (7.23) is exactly the continuous Sinkhorn problem (7.11), up to an additive constant in the value.

$$M_\varepsilon^* = \int \mathcal{R}^{\varepsilon,x,y} d\pi_\varepsilon^*(x, y).$$

7.7 Marginal-Dependent Problems

Let $\mathcal{F}$ and $\mathcal{G}$ be proper convex lower semicontinuous functionals on finite nonnegative measures on $\mathcal{X}$ and $\mathcal{Y}$. The unregularized marginal-dependent transport problem is

$$\inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \int c(x, y) d\pi(x, y) + \mathcal{F}(\pi_1) + \mathcal{G}(\pi_2), \quad (7.26)$$

where $\pi_1 = (p_{\mathcal{X}})_{\#} \pi$ and $\pi_2 = (p_{\mathcal{Y}})_{\#} \pi$ are the two marginals of π . Entropic regularization turns this problem into the scaling-friendly form

$$\inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \int c(x, y) d\pi(x, y) + \mathcal{F}(\pi_1) + \mathcal{G}(\pi_2) + \varepsilon \text{KL}(\pi | \alpha \otimes \beta), \quad (7.27)$$

where α and β are reference measures. In this subsection, when the total mass of π is not fixed, the KL term is understood in the generalized sense associated with $\varphi(s) = s \log s - s + 1$.

$$\text{KL}(\pi | \lambda) = \int (\rho \log \rho - \rho + 1) d\lambda$$

$$\inf_{P \in \mathbb{R}_+^{n \times m}} \langle C, P \rangle + F(P \mathbf{1}_m) + G(P^\top \mathbf{1}_n) + \varepsilon \text{KL}(P | a \otimes b). \quad (7.28)$$

$$\inf_{P \geq 0} F(P \mathbf{1}_m) + G(P^\top \mathbf{1}_n) + \varepsilon \text{KL}(P | K).$$

Proposition 7.19 (Dual and scaling for marginal penalties). *Assume $a_i, b_j > 0$, $\varepsilon > 0$, and a Fenchel qualification condition, for instance the existence of a matrix $P > 0$ such that $P \mathbf{1}_m$ and $P^\top \mathbf{1}_n$ belong to the relative interiors of $\text{dom}(F)$ and $\text{dom}(G)$. The Fenchel dual of (7.28) is*

$$\sup_{f \in \mathbb{R}^n, g \in \mathbb{R}^m} -F^*(-f) - G^*(-g) - \varepsilon \sum_{i,j} a_i b_j \left[\exp \left(\frac{f_i + g_j - C_{ij}}{\varepsilon} \right) - 1 \right]. \quad (7.29)$$

Equivalently, up to the additive constant $\varepsilon \sum_{i,j} a_i b_j$, the last term is

$$-\varepsilon \sum_{i,j} K_{ij} \exp \left(\frac{f_i + g_j}{\varepsilon} \right).$$

If $u = e^{f/\varepsilon}$ and $v = e^{g/\varepsilon}$, exact block ascent in the two dual variables is the generalized Sinkhorn cycle

$$\begin{aligned} r &\leftarrow \text{prox}_{F/\varepsilon}^{\text{KL}}(Kv), & u &\leftarrow r \odot (Kv), \\ s &\leftarrow \text{prox}_{G/\varepsilon}^{\text{KL}}(K^\top u), & v &\leftarrow s \odot (K^\top u), \\ P &= \text{diag}(u) K \text{diag}(v), \end{aligned} \quad (7.30)$$

where divisions are entrywise and

$$\text{prox}_h^{\text{KL}}(z) := \underset{r \in \mathbb{R}_+^d}{\text{argmin}} h(r) + \text{KL}(r | z). \quad (7.31)$$

In particular, $\text{prox}_{F/\varepsilon}^{\text{KL}}(z)$ is the minimizer of $F(r) + \varepsilon \text{KL}(r | z)$.

Proof. Introduce independent variables r and s for the two marginals and use dual variables f, g for the constraints $r = P \mathbf{1}_m$ and $s = P^\top \mathbf{1}_n$. The Lagrangian contains $F(r) + \langle f, r \rangle + G(s) + \langle g, s \rangle + \langle C, P \rangle + \varepsilon \text{KL}(P | a \otimes b) - \langle f \oplus g, P \rangle$. Minimizing over r and s gives $-F^*(-f)$ and $-G^*(-g)$. For each scalar entry, the convex conjugate of $p \mapsto C_{ij}p + \varepsilon(p \log(p/(a_i b_j)) - p + a_i b_j)$ is $q \mapsto \varepsilon a_i b_j (e^{(q - C_{ij})/\varepsilon} - 1)$. Minimizing over P with $q = f_i + g_j$ gives (7.29).

For the scaling form, fix $v > 0$, set $\tilde{K} = K \text{diag}(v)$ and $z = \tilde{K} \mathbf{1}_m = Kv$. Updating the row scaling means looking for $P = \text{diag}(u) \tilde{K}$, so its row marginal is $r = P \mathbf{1}_m = u \odot z$. The row-wise chain rule gives $\text{KL}(\text{diag}(u) \tilde{K} | \tilde{K}) = \sum_i (r_i \log(r_i/z_i) - r_i + z_i) = \text{KL}(r | z)$. Thus exact optimization of this block is equivalent to minimizing $F(r) + \varepsilon \text{KL}(r | z)$ over $r \geq 0$, hence $r = \text{prox}_{F/\varepsilon}^{\text{KL}}(z)$ and $u = r \odot z$. The column update is identical with $w = K^\top u$. $\square$

When $F = \iota_{\{a\}}$ and $G = \iota_{\{b\}}$, the first two dual terms are $\langle f, a \rangle + \langle g, b \rangle$, and one recovers the usual entropic dual (7.17). The classical Sinkhorn update is the special case in which the KL proximal maps return the prescribed marginals a and b .

- **Hard marginal constraint.** If $F = \iota_{\{a\}}$, then $\text{prox}_{F/\varepsilon}^{\text{KL}}(z) = a$.
- **KL marginal relaxation.** If $F(r) = \tau \text{KL}(r | a)$ with $\tau > 0$, then, coordinatewise, $\text{prox}_{F/\varepsilon}^{\text{KL}}(z) = z^{\varepsilon/(\tau+\varepsilon)} \odot a^{\tau/(\tau+\varepsilon)}$, and the scaling update becomes $u \leftarrow (a \odot z)^{\tau/(\tau+\varepsilon)}$, the damped scaling of unbalanced Sinkhorn.
- **Pointwise bounds.** If $F = \iota_{\{\ell \leq r \leq u\}}$, then the proximal map is the coordinatewise clipping $\text{prox}_{F/\varepsilon}^{\text{KL}}(z)_i = \min\{u_i, \max\{\ell_i, z_i\}\}$.
- **Total-variation marginal relaxation.** If $F(r) = \tau \|r - a\|_1$ and $\lambda = \tau/\varepsilon$, then the proximal map is coordinatewise

$$\text{prox}_{F/\varepsilon}^{\text{KL}}(z)_i = \begin{cases} z_i e^\lambda, & z_i < a_i e^{-\lambda}, \\ a_i, & a_i e^{-\lambda} \leq z_i \leq a_i e^\lambda, \\ z_i e^{-\lambda}, & z_i > a_i e^\lambda. \end{cases}$$

- **Fixed total mass.** If $F = \iota_{\{\langle r, \mathbf{1} \rangle = m\}}$, then $\text{prox}_{F/\varepsilon}^{\text{KL}}(z) = mz / \langle z, \mathbf{1} \rangle$.

7.8 Heat Kernels and Hopf–Cole Transforms

Geodesics in heat.

$$h_t(x, y) = (4\pi t)^{-d/2} \exp\left(-\frac{\|x - y\|^2}{4t}\right).$$

For $c(x, y) = \|x - y\|^2$, the Sinkhorn kernel satisfies

$$K_\varepsilon(x, y) = e^{-\|x - y\|^2/\varepsilon} = (\pi\varepsilon)^{d/2} h_{\varepsilon/4}(x, y).$$

On a manifold, one therefore replaces dense multiplication by K_ε with the heat operator $H_\varepsilon = e^{-(\varepsilon/4)L}$; Varadhan's formula gives $-\varepsilon \log h_{\varepsilon/4}(x, y) \rightarrow d_M(x, y)^2$ as $\varepsilon \downarrow 0$.

$$H_\varepsilon = \lim_{q \rightarrow \infty} \left(\text{Id} + \frac{\varepsilon}{4q} L \right)^{-q}.$$

For a sparse L_h , factor $\text{Id} + \varepsilon L_h/(4q)$ once by sparse Cholesky and reuse the factor in every Sinkhorn kernel application. This avoids both the dense kernel and all-pairs geodesic distances.

Soft Hopf–Lax and Hopf–Cole. $c(x, y) = \frac{1}{2}\|x - y\|^2$,

$$(-h)^{\bar{c}}(x) = \inf_y \left\{ h(y) + \frac{1}{2}\|x - y\|^2 \right\},$$

where the sign only reflects the convention of Definition 4.6. In the present entropic setting, the parameter of interest is instead the temperature ε .

$$(-h)^{\bar{c}, \varepsilon}(x) = -\varepsilon \log \int \exp\left(-\frac{h(y) + \|x - y\|^2/2}{\varepsilon}\right) dy. \quad (7.32)$$

$$G_\varepsilon(z) := (2\pi\varepsilon)^{-d/2} \exp\left(-\frac{\|z\|^2}{2\varepsilon}\right)$$

$$(-h)^{\bar{c}, \varepsilon}(x) = -\varepsilon \log \left(G_\varepsilon * e^{-h/\varepsilon} \right)(x) - \frac{\varepsilon d}{2} \log(2\pi\varepsilon). \quad (7.33)$$

Thus a soft quadratic c -transform is a Gaussian convolution followed by a logarithm, up to an explicit additive constant independent of x . This is the bridge between soft-minimum operators, heat kernels and entropic transport potentials.

Proposition 7.20 (Soft quadratic c -transform and Legendre approximation). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be such that the integrals below are finite, and introduce the quadratic shift $\text{S}f(y) := f(y) - \frac{1}{2}\|y\|^2$. For $\varepsilon > 0$, define the soft conjugate by applying the soft $\bar{c}$ -transform to the shifted function:*

$$f^{*, \varepsilon}(p) := \frac{1}{2}\|p\|^2 - (-\text{S}f)^{\bar{c}, \varepsilon}(p). \quad (7.34)$$

Then

$$f^{*, \varepsilon}(p) = \varepsilon \log \int_{\mathbb{R}^d} \exp\left(\frac{\langle p, y \rangle - f(y)}{\varepsilon}\right) dy, \quad (7.35)$$

and equivalently

$$f^{*, \varepsilon}(p) = \frac{1}{2}\|p\|^2 + \varepsilon \log \left(G_\varepsilon * e^{-\text{S}f/\varepsilon} \right)(p) + \frac{\varepsilon d}{2} \log(2\pi\varepsilon). \quad (7.36)$$

If, for instance, f is proper, lower semicontinuous and superlinear, then $f^{*, \varepsilon}(p) \rightarrow f^*(p)$ for every p as $\varepsilon \rightarrow 0$.

Proof. The hard identity follows by completing the square:

$$\inf_y \left\{ \text{S}f(y) + \frac{1}{2}\|p - y\|^2 \right\} = \frac{1}{2}\|p\|^2 - \sup_y \{ \langle p, y \rangle - f(y) \} = \frac{1}{2}\|p\|^2 - f^*(p).$$

This explains the shift: it turns the Legendre–Fenchel transform into a quadratic $\bar{c}$ -transform. Replacing the hard transform by its soft version gives (7.34). Expanding the square in (7.32) with the function $\text{S}f$ gives

$$\exp\left(-\frac{\text{S}f(y) + \|p - y\|^2/2}{\varepsilon}\right) = \exp\left(-\frac{\|p\|^2}{2\varepsilon}\right) \exp\left(\frac{\langle p, y \rangle - f(y)}{\varepsilon}\right),$$

which proves (7.35). Formula (7.36) is the Gaussian-convolution identity (7.33) applied to $\text{S}f$. Under the stated superlinearity assumption, the functions $y \mapsto \langle p, y \rangle - f(y)$ have compact upper level sets, so Laplace's principle applied to (7.35) gives the convergence to $f^*(p)$. $\square$

$$\partial_s \varphi_s + \frac{1}{2} \|\nabla \varphi_s\|^2 = \frac{\varepsilon}{2} \Delta \varphi_s, \quad \partial_s v_s + (v_s \cdot \nabla) v_s = \frac{\varepsilon}{2} \Delta v_s.$$

7.9 Other Convex Regularizers

Density-ratio regularization. Let φ be an entropy function in the sense of Definition 6.8, and recall the φ -divergence $\mathcal{D}_\varphi$ from Definition 6.9. For $\varepsilon > 0$, define the φ -regularized transport value.

$$\mathcal{L}_{c,\varphi}^\varepsilon(\alpha, \beta) := \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \varepsilon \mathcal{D}_\varphi(\pi | \alpha \otimes \beta). \quad (7.37)$$

Proposition 7.21 (Dual and density law for φ -regularized OT). *Assume that the Fenchel–Rockafellar qualification condition holds; for example, let the spaces be compact, let c be continuous, and assume that the product coupling belongs to the continuity domain of the divergence term. Then*

$$\mathcal{L}_{c,\varphi}^\varepsilon(\alpha, \beta) = \sup_{f \in \mathcal{C}(\mathcal{X}), g \in \mathcal{C}(\mathcal{Y})} \int f d\alpha + \int g d\beta - \varepsilon \int \varphi^* \left(\frac{f(x) + g(y) - c(x, y)}{\varepsilon} \right) d\alpha(x) d\beta(y). \quad (7.38)$$

If an optimal plan is absolutely continuous, with density $r^* = \frac{d\pi^*}{d(\alpha \otimes \beta)}$, and (f^*, g^*) are optimal potentials, then $\frac{f^*(x) + g^*(y) - c(x, y)}{\varepsilon} \in \partial\varphi(r^*(x, y))$ $\alpha \otimes \beta$ -a.e. In the smooth interior this reads

$$r^*(x, y) = (\varphi')^{-1} \left(\frac{f^*(x) + g^*(y) - c(x, y)}{\varepsilon} \right).$$

Proof. Introduce dual variables (f, g) for the two marginal constraints. For fixed (f, g) , the minimization over π gives

$$\int f d\alpha + \int g d\beta + \inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \left\{ \int (c - (f \oplus g)) d\pi + \varepsilon \mathcal{D}_\varphi(\pi | \alpha \otimes \beta) \right\}.$$

Using the Legendre formula (6.9) for the convex functional $\mathcal{D}_\varphi(\cdot | \alpha \otimes \beta)$, the infimum equals

$$-\varepsilon \int \varphi^* \left(\frac{f(x) + g(y) - c(x, y)}{\varepsilon} \right) d\alpha(x) d\beta(y),$$

which gives (7.38). Equality in the Fenchel inequality is equivalent to the subgradient inclusion $\frac{f^* \oplus g^* - c}{\varepsilon} \in \partial\varphi(r^*)$, and inversion of φ' gives the density law when φ is differentiable and the optimizer is in the interior of its domain. $\square$

For the KL entropy $\varphi(r) = r \log r - r + 1$, one has $\varphi^*(s) = e^s - 1$, and (7.38) reduces exactly to the continuous Sinkhorn dual (7.17). Other choices replace the exponential density law by another scalar transfer function:

$$\begin{aligned} \varphi(r) = r \log r - r + 1 &\Rightarrow r^* = e^s, \\ \varphi(r) = r - \log r - 1 &\Rightarrow r^* = (1 - s)^{-1} \quad (s < 1), \quad s := \frac{f^* \oplus g^* - c}{\varepsilon}, \\ \varphi(r) = \frac{1}{2}(r - 1)^2 &\Rightarrow r^* = (1 + s)_+, \end{aligned}$$

Generalized soft c -transforms.

$$\begin{aligned} g^{\bar{c}, \varepsilon, \varphi}(x) &\in \operatorname{argmin}_{u \in \mathbb{R}} \left\{ \varepsilon \int_{\mathcal{Y}} \varphi^* \left(\frac{u + g(y) - c(x, y)}{\varepsilon} \right) d\beta(y) - u \right\}. \\ \int_{\mathcal{Y}} (\varphi^*)' \left(\frac{u + g(y) - c(x, y)}{\varepsilon} \right) d\beta(y) &= 1. \end{aligned} \quad (7.39)$$

This equation says that the conditional density induced by the current dual score has unit β -mass, hence it is exactly the row-normalization step in dual variables. The update $f^{c, \varepsilon, \varphi}$ on $\mathcal{X}$ is defined symmetrically by exchanging (x, α, f) and (y, β, g) .

Bregman vs. φ -divergence regularization.

Definition 7.22 (Measure Bregman divergence). If Φ is a differentiable convex functional on a convex class of nonnegative measures and ξ is a reference measure in its domain, the measure Bregman divergence generated by Φ is

$$B_\Phi(\pi | \xi) := \Phi(\pi) - \Phi(\xi) - \int \delta\Phi(\xi) d(\pi - \xi), \quad (7.40)$$

where $\delta\Phi(\xi)$ is the first variation and the formula is understood whenever the right-hand side is well-defined.

Proposition 7.23 (Dual comparison: Bregman vs. density-ratio penalties). *Fix the marginals α, β and set $\xi := \alpha \otimes \beta$. Let Φ be a convex Gâteaux-differentiable functional on nonnegative measures on $\mathcal{X} \times \mathcal{Y}$. Its convex conjugate is, for a continuous test function u ,*

$$\Phi^*(u) := \sup_{\pi \geq 0} \left\{ \int u d\pi - \Phi(\pi) \right\}.$$

Define the Bregman-regularized value, using the same product reference as the density-ratio penalty, $\mathcal{L}_{c,\Phi}^\varepsilon(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int c d\pi + \varepsilon B_\Phi(\pi | \xi)$. Assume that Fenchel duality is exact for this constrained problem, as happens in finite-dimensional discretizations and, more generally, under standard compactness and lower-semicontinuity hypotheses. Then

$$\mathcal{L}_{c,\Phi}^\varepsilon(\alpha, \beta) = \sup_{f, g} \int f d\alpha + \int g d\beta - \varepsilon \left[\Phi^* \left(\delta\Phi(\xi) + \frac{f \oplus g - c}{\varepsilon} \right) - \Phi^*(\delta\Phi(\xi)) \right]. \quad (7.41)$$

If (f^*, g^*) and π^* are optimal and the solution is interior, then $\delta\Phi(\pi^*) = \delta\Phi(\xi) + \frac{f^* \oplus g^* - c}{\varepsilon}$. By contrast, the density-ratio formulation (7.37) has the scalar-integral dual (7.38) and the pointwise density law

$$\frac{f^* \oplus g^* - c}{\varepsilon} \in \partial\varphi \left(\frac{d\pi^*}{d(\alpha \otimes \beta)} \right).$$

Proof. Using (7.40), the Bregman-regularized objective can be written, up to a constant independent of π , as $\varepsilon\Phi(\pi) + \int (c - \varepsilon\delta\Phi(\xi))d\pi - \varepsilon\Phi(\xi) + \varepsilon \int \delta\Phi(\xi)d\xi$. Introduce dual potentials (f, g) for the two marginal constraints. The inner minimization over nonnegative measures π gives

$$\inf_{\pi \geq 0} \left\{ \varepsilon\Phi(\pi) + \int (c - (f \oplus g) - \varepsilon\delta\Phi(\xi))d\pi \right\} = -\varepsilon\Phi^* \left(\delta\Phi(\xi) + \frac{f \oplus g - c}{\varepsilon} \right).$$

Fenchel equality at ξ gives $-\Phi(\xi) + \int \delta\Phi(\xi)d\xi = \Phi^*(\delta\Phi(\xi))$, which yields (7.41). Equality in Fenchel's inequality gives the optimality condition for π^* . The density-ratio dual and density law are exactly those of Proposition 7.21. Placing the two formulas side by side shows the structural difference: Bregman regularization translates the reference measure in the dual coordinate $\delta\Phi$, whereas φ -regularization applies a scalar nonlinearity to the density with respect to the moving product reference $\alpha \otimes \beta$. $\square$

When Φ is separable with respect to a fixed dominating measure ξ_0 , say $\Phi(\pi) = \int h(d\pi/d\xi_0)d\xi_0$ with $\xi \ll \xi_0$, the Bregman optimality condition becomes

$$h' \left(\frac{d\pi^*}{d\xi_0} \right) = h' \left(\frac{d\xi}{d\xi_0} \right) + \frac{f^* \oplus g^* - c}{\varepsilon}.$$

This is an additive update in entropy coordinates. The density-ratio formulation instead uses the scalar law associated with φ relative to $\alpha \otimes \beta$.

Proposition 7.24 (KL is the common Bregman and φ case). *Let ω be a finite reference measure on $\mathcal{X}$ with a measurable set A satisfying $0 < \omega(A) < \omega(\mathcal{X})$. Work on probability measures $\alpha = p\omega$ and $\beta = q\omega$ whose densities are bounded above and below away from 0. Assume that Φ is twice Gâteaux differentiable along bounded zero-mass density perturbations, and that $\varphi \in C^2(0, +\infty)$ is convex with $\varphi(1) = 0$. If $B_\Phi(\alpha|\beta) = \mathcal{D}_\varphi(\alpha|\beta)$ for all such α, β , then there exist $\kappa \geq 0$ and $a \in \mathbb{R}$ such that $\varphi(t) = \kappa t \log t + a(t-1)$. Hence the common divergence is $\kappa \text{KL}(\alpha|\beta)$, and $\Phi(p\omega)$ differs from $\kappa \int p \log p d\omega$ by an affine functional on the positive probability simplex.*

Proof. Fix $\beta = q\omega$ and perturb it by $\alpha_t = (q + th)\omega$, where h is bounded, $\int h d\omega = 0$, and t is small enough that $q + th > 0$. Differentiating twice at $t = 0$ gives $D^2\Phi(q)[h, h] = \varphi''(1) \int \frac{h^2}{q} d\omega$. Indeed the left-hand side is the second variation of the Bregman error, while

$$\mathcal{D}_\varphi(\alpha_t|\beta) = \int q \varphi(1 + th/q) d\omega$$

has second derivative $\varphi''(1) \int h^2/q d\omega$. Setting $\kappa = \varphi''(1) \geq 0$, the functional $\Psi(p\omega) = \Phi(p\omega) - \kappa \int p \log p d\omega$ has zero second variation along every zero-mass line segment in the positive simplex. It is therefore affine there, and $B_\Psi = 0$. Thus $B_\Phi = \kappa \text{KL}$. Since $\mathcal{D}_\varphi = \kappa \text{KL}$, the function $r(t) = \varphi(t) - \kappa t \log t$ generates the zero φ -divergence. Testing on densities for which the ratio p/q takes two values $x < 1 < y$, with weights chosen so that the mean ratio is 1, gives $(y-1)r(x) + (1-x)r(y) = 0$. Hence $r(t)/(t-1)$ is constant on $(0, +\infty) \setminus \{1\}$, so $r(t) = a(t-1)$. This proves the claim. $\square$

Thus the two generalizations lead to different duals and different algorithms. Bregman regularization by $B_\Phi(\pi|\xi)$ keeps the projection geometry of Section 8.1: linear costs tilt the reference in dual coordinates and alternating marginal updates are Bregman projections.

7.10 Sinkhorn Divergences

Entropic bias. $\beta_\varepsilon \in \underset{\beta}{\text{argmin}} \mathcal{L}_c^\varepsilon(\alpha, \beta)$

Proposition 7.25 (Large-temperature entropic bias). *Let $\mathcal{X}, \mathcal{Y}$ be compact and assume that c is continuous. Then $\mathcal{L}_c^\varepsilon(\alpha, \beta) \rightarrow \iint c(x, y) d\alpha(x) d\beta(y)$ as $\varepsilon \rightarrow +\infty$.*

Proof. Let $(f_\varepsilon, g_\varepsilon)$ be optimal dual potentials, normalized by $\int g_\varepsilon d\beta = 0$. The soft c -transform equation gives

$$f_\varepsilon(x) = -\varepsilon \log \int \exp \left(\frac{g_\varepsilon(y) - c(x, y)}{\varepsilon} \right) d\beta(y).$$

For bounded c , the oscillations of normalized entropic potentials are bounded uniformly in ε by the oscillation of c . Hence the log-sum-exp expansion is uniform: $f_\varepsilon(x) = -\int (g_\varepsilon(y) - c(x, y)) d\beta(y) + O(\varepsilon^{-1}) = \int c(x, y) d\beta(y) + O(\varepsilon^{-1})$. At optimality the exponential penalty in the dual integrates to zero, so $\mathcal{L}_c^\varepsilon(\alpha, \beta) = \int f_\varepsilon d\alpha + \int g_\varepsilon d\beta = \iint c(x, y) d\alpha(x) d\beta(y) + O(\varepsilon^{-1})$. This proves the limit. $\square$

Thus, at large ε , the raw entropic cost approaches the bilinear cross-energy $(\alpha, \beta) \mapsto \iint c d\alpha d\beta$, not a distance.

Sinkhorn divergences.

Definition 7.26 (Sinkhorn divergence). Let c be a symmetric cost on a common state space. For $\varepsilon > 0$, the debiased Sinkhorn divergence associated with the entropic OT value $\mathcal{L}_c^\varepsilon$ is

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) := \mathcal{L}_c^\varepsilon(\alpha, \beta) - \frac{1}{2} \mathcal{L}_c^\varepsilon(\alpha, \alpha) - \frac{1}{2} \mathcal{L}_c^\varepsilon(\beta, \beta). \quad (7.42)$$

Lemma 7.27 (Entropic dual cost at optimum). *Let $(f_{\alpha, \beta}, g_{\alpha, \beta})$ be optimal dual potentials, normalized arbitrarily. Then*

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) = \langle f_{\alpha, \beta}, \alpha \rangle + \langle g_{\alpha, \beta}, \beta \rangle. \quad (7.43)$$

Proof. We first notice that at optimality, the relation $f_{\alpha, \beta} = -\varepsilon \log \int_{\mathcal{Y}} e^{\frac{g_{\alpha, \beta}(y) - c(x, y)}{\varepsilon}} d\beta(y)$ after taking the exponential, equivalently reads

$$1 = \int_{\mathcal{Y}} e^{\frac{f_{\alpha, \beta}(x) + g_{\alpha, \beta}(y) - c(x, y)}{\varepsilon}} d\beta(y) \implies \int_{\mathcal{X} \times \mathcal{Y}} \left(e^{\frac{f_{\alpha, \beta} \oplus g_{\alpha, \beta} - c}{\varepsilon}} - 1 \right) d(\alpha \otimes \beta) = 0.$$

Substituting this identity in (7.16) gives the result. $\square$

Proposition 7.28 (Asymptotics of Sinkhorn divergences). *Let $\mathcal{X}$ be a compact metric space, let $\alpha, \beta \in \mathcal{P}(\mathcal{X})$, and assume that $c : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_+$ is continuous and symmetric, with $c(x, x) = 0$. Then $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \rightarrow \mathcal{L}_c(\alpha, \beta)$ when $\varepsilon \rightarrow 0$ and*

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \rightarrow \frac{1}{2} \int -c d(\alpha - \beta) \otimes d(\alpha - \beta) \quad \text{when } \varepsilon \rightarrow +\infty.$$

Proof. The discrete convergence result above already gives the correct intuition; we now use the standard continuous argument. **Case $\varepsilon \rightarrow 0$.** The first limit follows from the standard Γ -convergence argument for entropic optimal transport: the entropy term is lower semicontinuous along weakly converging couplings, while any finite-cost coupling can be approximated by couplings with finite entropy. Since $c(x, x) = 0$, the two self-costs in the debiased expression converge to zero, and the cross term converges to $\mathcal{L}_c(\alpha, \beta)$.

Case $\varepsilon \rightarrow +\infty$. We denote by $(f_\varepsilon, g_\varepsilon)$ optimal dual potentials. After normalizing them and using boundedness of c , their oscillations stay uniformly bounded, so the following expansion is uniform. The optimality condition on f_ε (equivalently the Sinkhorn fixed point on f_ε) reads

$$\begin{aligned} f_\varepsilon &= -\varepsilon \log \int \exp \left(\frac{g_\varepsilon(y) - c(\cdot, y)}{\varepsilon} \right) d\beta(y) = -\varepsilon \log \int \left(1 + \frac{g_\varepsilon(y) - c(\cdot, y)}{\varepsilon} + o(1/\varepsilon) \right) d\beta(y) \\ &= -\varepsilon \log \left(1 + \frac{1}{\varepsilon} \int (g_\varepsilon(y) - c(\cdot, y)) d\beta(y) + o(1/\varepsilon) \right) = - \int g_\varepsilon d\beta + \int c(\cdot, y) d\beta(y) + o(1). \end{aligned}$$

Plugging this relation in the dual expression (7.43) $\mathcal{L}_c^\varepsilon(\alpha, \beta) = \int f_\varepsilon d\alpha + \int g_\varepsilon d\beta = \iint c(x, y) d\alpha(x) d\beta(y) + o(1)$. Applying this expansion to (α, β) , (α, α) and (β, β) gives

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \rightarrow \int c d\alpha \otimes d\beta - \frac{1}{2} \int c d\alpha \otimes d\alpha - \frac{1}{2} \int c d\beta \otimes d\beta = -\frac{1}{2} \int c d(\alpha - \beta) \otimes d(\alpha - \beta).$$

□

Proposition 7.29 (Non-negativity of Sinkhorn divergences). *Assume that the entropic dual admits optimal potentials and that the symmetric kernel $k_\varepsilon(x, y) = e^{-c(x, y)/\varepsilon}$ is positive semidefinite. Then $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \geq 0$.*

Proof. In the following, we denote by $(f_{\alpha, \beta}, g_{\alpha, \beta})$ optimal dual potentials for the dual Schrödinger problem between α and β . We denote by $f_{\alpha, \alpha} = g_{\alpha, \alpha}$ (one can assume they are equal by symmetry) the solution for the problem between α and itself.

Using the suboptimal function $(f_{\alpha, \alpha}, g_{\beta, \beta})$ in the dual maximization problem, and using relation (7.43) for the simplified expression of the dual cost, one obtains

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) \geq \langle f_{\alpha, \alpha}, \alpha \rangle + \langle g_{\beta, \beta}, \beta \rangle - \varepsilon \langle e^{\frac{f_{\alpha, \alpha} \oplus g_{\beta, \beta} - c}{\varepsilon}} - 1, \alpha \otimes \beta \rangle$$

Moreover $\langle f_{\alpha, \alpha}, \alpha \rangle = \frac{1}{2} \mathcal{L}_c^\varepsilon(\alpha, \alpha)$, and similarly for β , so the previous inequality equivalently reads

$$\frac{1}{\varepsilon} \bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \geq 1 - \langle e^{\frac{f_{\alpha, \alpha} \oplus g_{\beta, \beta} - c}{\varepsilon}}, \alpha \otimes \beta \rangle = 1 - \langle \tilde{\alpha}, \tilde{\beta} \rangle_{k_\varepsilon}$$

where $\tilde{\alpha} = e^{f_{\alpha, \alpha}/\varepsilon} \alpha$, $\tilde{\beta} = e^{f_{\beta, \beta}/\varepsilon} \beta$ and we introduced the positive-semidefinite kernel pairing $\langle \tilde{\alpha}, \tilde{\beta} \rangle_{k_\varepsilon} := \int k_\varepsilon(x, y) d\tilde{\alpha}(x) d\tilde{\beta}(y)$. The self Sinkhorn fixed point equation, once exponentiated, reads pointwise $e^{f_{\alpha, \alpha}(x)/\varepsilon} \int k_\varepsilon(x, y) d\tilde{\alpha}(y) = 1$ for α -a.e. x , and hence

$$\|\tilde{\alpha}\|_{k_\varepsilon}^2 = \langle \tilde{\alpha}, \tilde{\alpha} \rangle_{k_\varepsilon} = \int e^{f_{\alpha, \alpha}(x)/\varepsilon} \left(\int k_\varepsilon(x, y) d\tilde{\alpha}(y) \right) d\alpha(x) = 1$$

and similarly $\|\tilde{\beta}\|_{k_\varepsilon}^2 = 1$. Therefore, by Cauchy-Schwarz, one has $1 - \langle \tilde{\alpha}, \tilde{\beta} \rangle_{k_\varepsilon} \geq 0$. □

7.11 Complex ε

Measure fixed point. Let $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$. Whenever the following integral operators are well defined, set, for $\varepsilon \in \mathbb{C} \setminus \{0\}$,

$$K_\varepsilon(x, y) := \exp(-c(x, y)/\varepsilon).$$

$$d\pi_\varepsilon(x, y) = u(x) K_\varepsilon(x, y) v(y) d\alpha(x) d\beta(y). \quad (7.44)$$

$$u(x) \int_{\mathcal{Y}} K_\varepsilon(x, y) v(y) d\beta(y) = 1 \quad \text{for } \alpha\text{-a.e. } x, \quad v(y) \int_{\mathcal{X}} K_\varepsilon(x, y) u(x) d\alpha(x) = 1 \quad \text{for } \beta\text{-a.e. } y. \quad (7.45)$$

$$u^{(k+1)}(x) = \left(\int_{\mathcal{Y}} K_\varepsilon(x, y) v^{(k)}(y) d\beta(y) \right)^{-1}, \quad v^{(k+1)}(y) = \left(\int_{\mathcal{X}} K_\varepsilon(x, y) u^{(k+1)}(x) d\alpha(x) \right)^{-1}, \quad (7.46)$$

whenever the denominators do not vanish. If a branch of the logarithm is fixed along the local continuation and $u = e^{f/\varepsilon}$, $v = e^{g/\varepsilon}$, this is equivalently the pair of soft-transform equations

$$f(x) = -\varepsilon \log \int_{\mathcal{Y}} \exp \left(\frac{g(y) - c(x, y)}{\varepsilon} \right) d\beta(y), \quad g(y) = -\varepsilon \log \int_{\mathcal{X}} \exp \left(\frac{f(x) - c(x, y)}{\varepsilon} \right) d\alpha(x).$$

For real positive ε , this is the usual Sinkhorn fixed point. For complex ε , the same formulas should be read as local analytic identities, not as a positivity statement.

Discrete histograms. If $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, set $C_{ij} = c(x_i, y_j)$ and define the complex Gibbs matrix $K_\varepsilon(C)_{ij} := \exp(-C_{ij}/\varepsilon)$. $P_{\varepsilon,ij} = a_i b_j u_i K_\varepsilon(C)_{ij} v_j$, $u_i \sum_j b_j K_\varepsilon(C)_{ij} v_j = 1$, $v_j \sum_i a_i K_\varepsilon(C)_{ij} u_i = 1$. $P_\varepsilon = \text{diag}(u) K_\varepsilon(C) \text{diag}(v)$, $P_\varepsilon \mathbf{1}_m = \mathbf{a}$, $P_\varepsilon^\top \mathbf{1}_n = \mathbf{b}$.

$$\mathbf{u}^{(k+1)} = \mathbf{a} \oslash (K_\varepsilon(C) \mathbf{v}^{(k)}), \quad \mathbf{v}^{(k+1)} = \mathbf{b} \oslash (K_\varepsilon(C)^\top \mathbf{u}^{(k+1)}), \quad (7.47)$$

where $\oslash$ denotes entrywise division. For complex ε , this iteration should be read as a local analytic parametrization of the fixed point, not as a globally convergent algorithm.

$\mathbf{f} = \hat{\mathbf{f}} + \varepsilon \log \mathbf{a}$, $\mathbf{g} = \hat{\mathbf{g}} + \varepsilon \log \mathbf{b}$,

Theorem 7.30 (Carlier's holomorphic continuation of Sinkhorn). *Fix positive histograms $\mathbf{a}^0 \in \Sigma_n$, $\mathbf{b}^0 \in \Sigma_m$, a finite real cost matrix C^0 , and a real temperature $\varepsilon_0 > 0$. Let $(\mathbf{f}^0, \mathbf{g}^0)$ be the absorbed log-scalings at this point, normalized by $\sum_i a_i^0 f_i^0 = 0$. Then there are complex neighborhoods of $(\varepsilon_0, \mathbf{a}^0, \mathbf{b}^0, C^0)$, inside the affine constraint $\sum_i a_i = \sum_j b_j$, and a unique holomorphic map $(\varepsilon, \mathbf{a}, \mathbf{b}, C) \mapsto (\mathbf{f}_\varepsilon, \mathbf{g}_\varepsilon) \in \mathbb{C}^n \times \mathbb{C}^m$ satisfying the fixed gauge $\sum_i a_i^0 f_{\varepsilon,i} = 0$ and solving*

$$P_{\varepsilon,ij} = \exp\left(\frac{f_{\varepsilon,i} + g_{\varepsilon,j} - C_{ij}}{\varepsilon}\right), \quad P_\varepsilon \mathbf{1}_m = \mathbf{a}, \quad P_\varepsilon^\top \mathbf{1}_n = \mathbf{b}.$$

Proof. Consider the holomorphic map which sends $(\mathbf{f}, \mathbf{g}, \varepsilon, \mathbf{a}, \mathbf{b}, C)$ to the n row residuals, the first $m-1$ column residuals, and the gauge condition,

$$R_i = \sum_j \exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon}\right) - a_i, \quad S_j = \sum_i \exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon}\right) - b_j, \quad G = \sum_i a_i^0 f_i,$$

where $1 \leq i \leq n$ and $1 \leq j \leq m-1$. At the real base point the associated coupling P^0 is strictly positive. Let $(\delta \mathbf{f}, \delta \mathbf{g})$ belong to its kernel and set $\delta P_{ij} = \frac{1}{\varepsilon_0} P_{ij}^0 (\delta f_i + \delta g_j)$. The linearized row sums vanish. The first $m-1$ linearized column sums vanish by definition of the residuals, and the last one vanishes because the total row sum equals the total column sum. Therefore

$$0 = \sum_i \delta f_i \sum_j \delta P_{ij} + \sum_j \delta g_j \sum_i \delta P_{ij} = \frac{1}{\varepsilon_0} \sum_{i,j} P_{ij}^0 |\delta f_i + \delta g_j|^2.$$

Since $P_{ij}^0 > 0$, one has $\delta f_i + \delta g_j = 0$ for all i, j . Hence $\delta \mathbf{f}$ is constant and $\delta \mathbf{g}$ is the opposite constant. The gauge derivative $\sum_i a_i^0 \delta f_i = 0$ forces this constant to vanish. The derivative is thus invertible, and the holomorphic implicit-function theorem gives the local holomorphic potentials. The formulas for u_ε , v_ε and P_ε are holomorphic compositions of these potentials. $\square$

8 Entropic Regularization: Convergence

The chapter revisits Sinkhorn convergence through several complementary lenses. Bregman projections explain the alternating-projection geometry, Fortet's order argument gives qualitative fixed-point convergence, and an order-theoretic M-function view-point explains why Sinkhorn-like scaling can still converge even when the equations are no longer the gradient of a convex potential.

8.1 Sinkhorn Convergence: Bregman View

Alternating KL projections.

Definition 8.1 (Bregman divergence). Let Φ be a differentiable strictly convex function on the interior of a convex domain Ω . For $P, Q \in \text{int}(\Omega)$, the Bregman divergence generated by Φ is $B_\Phi(P|Q) := \Phi(P) - \Phi(Q) - \langle \nabla \Phi(Q), P - Q \rangle$. For the negative entropy $\Phi(P) = \sum_{i,j} P_{i,j} \log P_{i,j}$ on the positive orthant, this gives $B_\Phi(P|Q) = \text{KL}(P|Q)$. The same formula extends lower-semicontinuously to $P \geq 0$ when $Q > 0$, with $0 \log 0 = 0$.

Linear tilts and Gibbs references.

Proposition 8.2 (Linear tilts of Bregman penalties). *Let Φ be differentiable and strictly convex, and let B_Φ be its Bregman divergence. Fix a reference point Q in the interior of the domain. Assume that there exists Q^C such that $\nabla \Phi(Q^C) = \nabla \Phi(Q) - C/\varepsilon$. Then, for all P in the domain, $\langle P, C \rangle + \varepsilon B_\Phi(P|Q) = \varepsilon B_\Phi(P|Q^C) + \text{cst}$, where the constant does not depend on P .*

Proof. Subtract the two Bregman divergences: $B_\Phi(P|Q^C) - B_\Phi(P|Q) = \langle \nabla \Phi(Q) - \nabla \Phi(Q^C), P \rangle + \text{cst}$. Using $\nabla \Phi(Q) - \nabla \Phi(Q^C) = C/\varepsilon$ and multiplying by ε gives the claim. $\square$

For the negative entropy $\Phi(P) = \sum_{i,j} P_{i,j} \log P_{i,j}$, one has $B_\Phi = \text{KL}$. Taking $Q = \mathbf{a} \otimes \mathbf{b}$ gives the tilted reference

$K_{\mathbf{a},\mathbf{b}}^\varepsilon := (\mathbf{a} \otimes \mathbf{b}) \odot e^{-C/\varepsilon}$. $\langle P, C \rangle + \varepsilon \text{KL}(P|\mathbf{a} \otimes \mathbf{b}) = \varepsilon \text{KL}(P|K_{\mathbf{a},\mathbf{b}}^\varepsilon) + \text{cst}$. Thus the unique solution P_ε of (7.1) is the KL projection of the tilted Gibbs reference onto $U(\mathbf{a}, \mathbf{b})$:

$$P_\varepsilon = \text{Proj}_{U(\mathbf{a},\mathbf{b})}^{\text{KL}}(K_{\mathbf{a},\mathbf{b}}^\varepsilon) := \underset{P \in U(\mathbf{a},\mathbf{b})}{\text{argmin}} \text{KL}(P|K_{\mathbf{a},\mathbf{b}}^\varepsilon). \quad (8.1)$$

Cyclic projection convergence.

Proposition 8.3 (Cyclic Bregman projections on affine constraints). *Let Φ be a Legendre strictly convex generator on a finite-dimensional convex domain, and let C_1, C_2 be affine constraint sets whose intersection meets the interior of the domain. Define $P_{k+1} = \text{Proj}_{C_2}^{B_\Phi} \text{Proj}_{C_1}^{B_\Phi}(P_k)$, starting from an interior point P_0 . Assume that the projections are well-defined and that the half-step sequence remains in a compact subset of the interior of the domain. Then P_k converges to the Bregman projection of P_0 onto $C_1 \cap C_2$.*

Proof. We first prove the Pythagorean identity used by the projection argument. For three interior points one has the Bregman three-point formula $B_\Phi(Q|P) = B_\Phi(Q|P^+) + B_\Phi(P^+|P) + \langle \nabla \Phi(P^+) - \nabla \Phi(P), Q - P^+ \rangle$. If $P^+ = \text{Proj}_C^{B_\Phi}(P)$ and C is affine, the first-order optimality condition for minimizing $R \mapsto B_\Phi(R|P)$ over $R \in C$ is $\langle \nabla \Phi(P^+) - \nabla \Phi(P), R - P^+ \rangle = 0 \quad \forall R \in C$,

because $R - P^+$ ranges over the tangent linear space of $\mathcal{C}$. Taking $R = Q \in \mathcal{C}$ cancels the last term and gives $B_\Phi(Q|P) = B_\Phi(Q|P^+) + B_\Phi(P^+|P) \quad \forall Q \in \mathcal{C}$.

Let $(Z_\ell)_\ell$ be the half-step sequence obtained by alternating projections onto $\mathcal{C}_1$ and $\mathcal{C}_2$, so that $Z_{2k} = P_k$ and $Z_{2k+2} = P_{k+1}$. Fix $Q \in \mathcal{C}_1 \cap \mathcal{C}_2$. Applying the identity at each half-step gives $B_\Phi(Q|Z_\ell) - B_\Phi(Q|Z_{\ell+1}) = B_\Phi(Z_{\ell+1}|Z_\ell) \geq 0$. Thus $B_\Phi(Q|Z_\ell)$ decreases and the series $\sum_\ell B_\Phi(Z_{\ell+1}|Z_\ell)$ is finite. The compactness assumption gives cluster points. Since the projection drops tend to zero and Φ is strictly convex on compact subsets of the domain, $\|Z_{\ell+1} - Z_\ell\| \rightarrow 0$. Every cluster point of the even subsequence is therefore also a cluster point of the adjacent odd subsequence. Because these two subsequences lie alternately in the closed affine sets $\mathcal{C}_1$ and $\mathcal{C}_2$, every cluster point belongs to $\mathcal{C}_1 \cap \mathcal{C}_2$.

Let $\bar{P}$ be such a cluster point. For each half-step, the dual displacement $\nabla\Phi(Z_{\ell+1}) - \nabla\Phi(Z_\ell)$ belongs to the normal space of the affine set onto which one projects. Telescoping and using the convergence of Z_ℓ gives $\nabla\Phi(\bar{P}) - \nabla\Phi(P_0) \in N_{\mathcal{C}_1} + N_{\mathcal{C}_2} = N_{\mathcal{C}_1 \cap \mathcal{C}_2}$, where the last equality uses that the sets are affine. This is precisely the first-order optimality condition for minimizing $R \mapsto B_\Phi(R|P_0)$ over $R \in \mathcal{C}_1 \cap \mathcal{C}_2$. Thus $\bar{P}$ is the Bregman projection of P_0 onto the intersection. Strict convexity gives uniqueness of this minimizer, so all cluster points coincide and the whole sequence converges. $\square$

$$\mathcal{C}_a^1 := \{P ; P\mathbf{1}_m = a\} \quad \text{and} \quad \mathcal{C}_b^2 := \{P ; P^\top \mathbf{1}_n = b\}$$

$$P^{(\ell+1)} := \text{Proj}_{\mathcal{C}_a^1}^{\text{KL}}(P^{(\ell)}) \quad \text{and} \quad P^{(\ell+2)} := \text{Proj}_{\mathcal{C}_b^2}^{\text{KL}}(P^{(\ell+1)}). \quad (8.2)$$

Row and column scalings.

Proposition 8.4 (KL projections are scalings). *One has $\text{Proj}_{\mathcal{C}_a^1}^{\text{KL}}(P) = \text{diag}\left(\frac{a}{P\mathbf{1}_m}\right)P$ and $\text{Proj}_{\mathcal{C}_b^2}^{\text{KL}}(P) = P \text{diag}\left(\frac{b}{P^\top \mathbf{1}_n}\right)$.*

Proof. Consider the problem along each row or column vector to impose a fixed sum $s \in \mathbb{R}_+ \min_p \{ \text{KL}(p|q) ; \langle p, \mathbf{1} \rangle = s \}$. The Lagrange multiplier equation for this problem reads $\log(p/q) + \lambda \mathbf{1} = 0 \implies p = uq$ where $u = e^{-\lambda} > 0$. The constraint $\langle p, \mathbf{1} \rangle = s$ is equivalent to $\langle uq, \mathbf{1} \rangle = s$, i.e. $u = s / \sum_i q_i$, which gives the desired scaling formula $p = sq / \sum_i q_i$. $\square$

$$P^{(2\ell)} := \text{diag}(u^{(\ell)})K \text{diag}(v^{(\ell)}),$$

$$\begin{aligned} P^{(2\ell+1)} &:= \text{diag}(u^{(\ell+1)})K \text{diag}(v^{(\ell)}) \\ \text{and } P^{(2\ell+2)} &:= \text{diag}(u^{(\ell+1)})K \text{diag}(v^{(\ell+1)}) \end{aligned}$$

8.2 Sinkhorn Convergence: Monotone Point of View

Fortet monotonicity.

Proposition 8.5 (Monotone fixed-point route to Sinkhorn convergence). *Let c be bounded and continuous on compact spaces, and define the unnormalized double Sinkhorn map $\mathcal{A}(f) := (f^{c,\varepsilon})^{\bar{c},\varepsilon}$. The map $\mathcal{A}$ is order preserving and additively homogeneous: $f \leq h \implies \mathcal{A}(f) \leq \mathcal{A}(h)$, $\mathcal{A}(f + \lambda) = \mathcal{A}(f) + \lambda$. Let f^* be an entropic fixed point and suppose that f_0 is a subsolution, meaning $f_0 \leq \mathcal{A}(f_0)$. After subtracting a sufficiently large constant from f_0 , which preserves the subsolution inequality, assume $f_0 \leq f^*$. Then the iterates $f_{k+1} = \mathcal{A}(f_k)$ increase pointwise, remain bounded above by f^* , and converge uniformly to an entropic fixed point. Their classes modulo constants therefore converge to the unique entropic potential class.*

Proof. The soft c -transform is order reversing: if $g \leq g'$, then $-\varepsilon \log \int e^{(g-c)/\varepsilon} d\beta \geq -\varepsilon \log \int e^{(g'-c)/\varepsilon} d\beta$. The composition of two order-reversing transforms is order preserving. Since each one-sided transform is additively anti-homogeneous, their composition is additively homogeneous. The subsolution inequality gives $f_0 \leq f_1$; order preservation then yields $f_k \leq f_{k+1}$ for every k . Likewise, $f_0 \leq f^*$ implies $f_k \leq \mathcal{A}^k(f^*) = f^*$. Thus $(f_k)_k$ converges pointwise. All f_k with $k \geq 1$ are double soft transforms, so their oscillations and moduli of continuity are controlled by those of the bounded continuous cost. Arzelà–Ascoli makes the convergence uniform, and continuity of $\mathcal{A}$ shows that the limit is a fixed point. Proposition 7.16 identifies all continuous fixed points up to constants, while Proposition 7.2 identifies their common coupling. $\square$

Extension beyond KL. $\Phi_\varphi := \varphi^*$.

$$\begin{aligned} g^{\bar{c},\varepsilon,\varphi}(x) &\in \underset{u \in \mathbb{R}}{\text{argmin}} \left\{ \varepsilon \int_{\mathcal{Y}} \Phi_\varphi \left(\frac{u + g(y) - c(x,y)}{\varepsilon} \right) d\beta(y) - u \right\}, \\ 1 &\in \int_{\mathcal{Y}} \partial \Phi_\varphi \left(\frac{g^{\bar{c},\varepsilon,\varphi}(x) + g(y) - c(x,y)}{\varepsilon} \right) d\beta(y), \quad 1 \in \int_{\mathcal{X}} \partial \Phi_\varphi \left(\frac{f(x) + f^{c,\varepsilon,\varphi}(y) - c(x,y)}{\varepsilon} \right) d\alpha(x), \end{aligned}$$

where the integral of intervals is understood in the usual Aumann sense, under the standard qualification allowing subdifferentiation under the integral. When Φ_φ is differentiable and $\rho_\varphi := \Phi'_\varphi$, this becomes the normalization relations

$$\int_{\mathcal{Y}} \rho_\varphi \left(\frac{g^{\bar{c},\varepsilon,\varphi}(x) + g(y) - c(x,y)}{\varepsilon} \right) d\beta(y) = 1, \quad \int_{\mathcal{X}} \rho_\varphi \left(\frac{f(x) + f^{c,\varepsilon,\varphi}(y) - c(x,y)}{\varepsilon} \right) d\alpha(x) = 1.$$

For KL, $\rho_\varphi(s) = e^s$ and these are exactly Sinkhorn row and column normalizations. For Burg or quadratic penalties, the same equations hold with $\rho_\varphi(s) = (1-s)^{-1}$ or $\rho_\varphi(s) = (1+s)_+$, respectively, on the appropriate domains.

Proposition 8.6 (Monotonicity of generalized double soft transforms). *Assume that the scalar minimizer sets defining $g^{\bar{c},\varepsilon,\varphi}$ and $f^{c,\varepsilon,\varphi}$ are nonempty compact intervals, and choose either the minimal or the maximal minimizer in every scalar problem. Then each selected generalized soft c -transform is order reversing and additively anti-homogeneous:*

$$g \leq g' \implies g^{\bar{c},\varepsilon,\varphi} \geq g'^{\bar{c},\varepsilon,\varphi}, \quad (g + \lambda)^{\bar{c},\varepsilon,\varphi} = g^{\bar{c},\varepsilon,\varphi} - \lambda.$$

The same statements hold for $f \mapsto f^{c,\varepsilon,\varphi}$. $\mathcal{A}_\varphi(f) := (f^{c,\varepsilon,\varphi})^{\bar{c},\varepsilon,\varphi}$ is order preserving and additively homogeneous: $f \leq f' \implies \mathcal{A}_\varphi(f) \leq \mathcal{A}_\varphi(f')$, $\mathcal{A}_\varphi(f + \lambda) = \mathcal{A}_\varphi(f) + \lambda$. In particular, this applies without mentioning selections whenever the scalar minimizers are unique.

Proof. Extend φ by $+\infty$ on $(-\infty, 0)$. Since $\Phi_\varphi(s) = \sup_{r \geq 0} \{sr - \varphi(r)\}$, the function Φ_φ is convex and nondecreasing. For fixed x , set

$$H_g^x(u) := \varepsilon \int_{\mathcal{Y}} \Phi_\varphi \left(\frac{u + g(y) - c(x, y)}{\varepsilon} \right) d\beta(y) - u.$$

If $g \leq g'$, write $\Delta(y) = g'(y) - g(y) \geq 0$. The difference

$$H_{g'}^x(u) - H_g^x(u) = \varepsilon \int_{\mathcal{Y}} \left[\Phi_\varphi \left(\frac{u + g(y) + \Delta(y) - c(x, y)}{\varepsilon} \right) - \Phi_\varphi \left(\frac{u + g(y) - c(x, y)}{\varepsilon} \right) \right] d\beta(y)$$

is nondecreasing in u : for every $\delta \geq 0$, the function $s \mapsto \Phi_\varphi(s + \delta) - \Phi_\varphi(s)$ is nondecreasing by convexity of Φ_φ . Let $I = \operatorname{argmin} H_g^x$ and $I' = \operatorname{argmin} H_{g'}^x$. If $a = \min I$ and $a' = \min I'$ with $a' > a$, then the two optimality inequalities $H_g^x(a) \leq H_g^x(a')$, $H_{g'}^x(a') \leq H_{g'}^x(a)$ imply $H_{g'}^x(a') - H_g^x(a') \leq H_g^x(a) - H_g^x(a')$, contradicting monotonicity unless equality holds throughout. In the equality case a is also a minimizer of $H_{g'}^x$, contradicting the definition of $a' = \min I'$. Hence $\min I' \leq \min I$. The same argument with $b = \max I$ and $b' = \max I'$ shows $\max I' \leq \max I$. Thus both extremal selections are order reversing. Finally, $H_{g+\lambda}^x(u) = H_g^x(u + \lambda) + \lambda$, so the minimizer interval is shifted by $-\lambda$, which proves additive anti-homogeneity. The proof for $f \mapsto f^{\bar{c}, \varepsilon, \varphi}$ is identical. Composing two order-reversing anti-homogeneous selections gives an order-preserving additively homogeneous double transform. $\square$

Variation seminorm and topical maps.

Definition 8.7 (Variation seminorm). For a bounded real-valued function h , the variation seminorm is $\|h\|_V := \sup h - \inf h$. It vanishes exactly on constant functions, hence becomes a norm after quotienting by additive constants.

Proposition 8.8 (Topical maps are variation-nonexpansive). Let E be a vector space of real-valued bounded functions, ordered pointwise, and write $\|\cdot\|_V$ for the variation seminorm of Definition 8.7. Let $\mathcal{T} : E \rightarrow E$ be monotone and additively homogeneous, $f \leq g \Rightarrow \mathcal{T}(f) \leq \mathcal{T}(g)$, $\mathcal{T}(f + \lambda) = \mathcal{T}(f) + \lambda \quad \forall \lambda \in \mathbb{R}$. Then $\|\mathcal{T}(f) - \mathcal{T}(g)\|_V \leq \|f - g\|_V$. The same conclusion holds for order-reversing maps satisfying $\mathcal{T}(f + \lambda) = \mathcal{T}(f) - \lambda$.

Proof. Set $a = \inf(f - g)$ and $b = \sup(f - g)$. Then $g + a \leq f \leq g + b$. If $\mathcal{T}$ is order preserving and additively homogeneous, applying $\mathcal{T}$ gives $\mathcal{T}(g) + a \leq \mathcal{T}(f) \leq \mathcal{T}(g) + b$. Hence every value of $\mathcal{T}(f) - \mathcal{T}(g)$ lies in $[a, b]$, so its oscillation is at most $b - a = \|f - g\|_V$. For an order-reversing, additively anti-homogeneous map, the same inequalities give $\mathcal{T}(g) - b \leq \mathcal{T}(f) \leq \mathcal{T}(g) - a$, and the oscillation bound is identical. $\square$

Corollary 8.9 (Soft transforms are nonexpansive). For every $\varepsilon > 0$, the soft c -transforms (7.18)–(7.19) are 1-Lipschitz for the variation seminorm.

Proof. The soft transform is order reversing and satisfies $(g + \lambda)^{\bar{c}, \varepsilon} = g^{\bar{c}, \varepsilon} - \lambda$, and similarly for the other block. Proposition 8.8 applies to each block, and the composition of two 1-Lipschitz maps is 1-Lipschitz. $\square$

Nonexpansiveness beyond KL. $\mathcal{A}_\varphi(f) := (f^{c, \varepsilon, \varphi})^{\bar{c}, \varepsilon, \varphi} \|\mathcal{A}_\varphi(f) - \mathcal{A}_\varphi(h)\|_V \leq \|f - h\|_V$. Thus the Fortet monotonicity mechanism and the basic nonexpansiveness estimate extend to general convex density-ratio penalties. What remains special to KL is the stronger Hilbert-metric contraction of Section 8.5, which uses the strictly positive Gibbs kernel to obtain a linear rate.

8.3 Monotone Clearing Beyond Variational Sinkhorn

Two-block clearing maps. $Q(z) = (Q^\alpha(u, v), Q^\beta(u, v)) \in \mathbb{R}^n \times \mathbb{R}^m$ $Q_\ell(T_\ell(z), z_{-\ell}) = 0, \quad 1 \leq \ell \leq n + m.$

Definition 8.10 (Z-functions and M-functions). Let $D \subset \mathbb{R}^N$ be an order interval. A map $Q : D \rightarrow \mathbb{R}^N$ is diagonally isotone if $Q_\ell(z_\ell, z_{-\ell})$ is nondecreasing in z_ℓ for every ℓ . It is a Z-function if increasing the other coordinates cannot increase the ℓ -th component: $z_{-\ell} \leq z'_{-\ell} \implies Q_\ell(z_\ell, z_{-\ell}) \geq Q_\ell(z_\ell, z'_{-\ell})$. For a C^1 map this means $\partial Q_\ell / \partial z_k \leq 0$ for $k \neq \ell$. An M-function is a Z-function which is inverse isotone: $Q(z) \leq Q(z') \implies z \leq z'$. A vector z is a subsolution if $Q(z) \leq 0$, and a supersolution if $Q(z) \geq 0$.

Theorem 8.11 (Monotone convergence of coordinate clearing). Let $D = [\underline{z}, \bar{z}] \subset \mathbb{R}^N$ be a closed order interval, with $\underline{z}$ a subsolution and $\bar{z}$ a supersolution. Assume that $Q : D \rightarrow \mathbb{R}^N$ is continuous, diagonally isotone and an M-function. Assume also that, for every $z \in D$ and every ℓ , the scalar clearing equation $Q_\ell(\xi, z_{-\ell}) = 0$ has a unique solution $\xi \in [\underline{z}_\ell, \bar{z}_\ell]$. Then $Q(z) = 0$ has a unique solution $z^* \in D$. The Jacobi coordinate-clearing iterates starting from any subsolution in D increase monotonically to z^* , while those starting from any supersolution in D decrease monotonically to z^* .

Proof. Let T be the coordinate-clearing map. If $z \in D$ is a subsolution, then $Q_\ell(z_\ell, z_{-\ell}) \leq 0 = Q_\ell(T_\ell(z), z_{-\ell})$. Diagonal isotonicity and uniqueness of the scalar zero give $z_\ell \leq T_\ell(z)$ for every ℓ , hence $z \leq T(z)$. Since Q is a Z-function, $Q_\ell(T(z)) \leq Q_\ell(T_\ell(z), z_{-\ell}) = 0$, so $T(z)$ is again a subsolution. Since every subsolution lies below every supersolution by inverse isotonicity, $T(z) \leq \bar{z}$, and therefore $T(z) \in D$. The supersolution argument is the same with all inequalities reversed. The lower and upper iterates are thus monotone and trapped in the compact interval D . Their limits exist, and passing to the limit in the identities $Q_\ell(z_\ell^{k+1}, z_{-\ell}^k) = 0$ gives $Q(z^*) = 0$. If z and z' were two zeros in D , then $Q(z) \leq Q(z')$ and $Q(z') \leq Q(z)$; inverse isotonicity gives $z = z'$. $\square$

Proposition 8.12 (Smooth M-matrix certificate). Let $D \subset \mathbb{R}^N$ be a convex order interval and let $Q : D \rightarrow \mathbb{R}^N$ be C^1 . Assume that $DQ(z)$ has nonpositive off-diagonal entries for every $z \in D$. If there exists $\lambda \gg 0$ such that $\lambda^\top DQ(z) \gg 0$ for all $z \in D$, or equivalently

$$\lambda_\ell \frac{\partial Q_\ell}{\partial z_\ell}(z) > \sum_{k \neq \ell} \lambda_k \left| \frac{\partial Q_k}{\partial z_\ell}(z) \right|, \quad 1 \leq \ell \leq N,$$

then Q is an M-function on D .

Proof. The off-diagonal sign gives the Z-property by integrating the partial derivatives along coordinatewise increasing segments. The displayed inequalities imply that each $DQ(z)$ is a strictly weighted column diagonally dominant Z-matrix, hence a nonsingular M-matrix; in particular $DQ(z)^{-1} \geq 0$ for each fixed z . For two points $z, z' \in D$, set $A = \int_0^1 DQ(z' + t(z - z')) dt$. The matrix A has the same Z-sign pattern and the same strict weighted diagonal dominance, so it is again a nonsingular M-matrix and $A^{-1} \geq 0$. Since $Q(z) - Q(z') = A(z - z')$, the implication $Q(z) \leq Q(z')$ gives $z - z' = A^{-1}(Q(z) - Q(z')) \leq 0$. This is inverse isotonicity. The positive diagonal entries also give diagonal isotonicity. $\square$

Sinkhorn as the canonical M_0 -system. $P_{ij} = r_i K_{ij} s_j$, $r_i = e^{u_i}$, $s_j = e^{-v_j}$. $Q_i^\alpha(u, v) = \sum_j K_{ij} e^{u_i - v_j} - a_i$, $Q_j^\beta(u, v) = b_j - \sum_i K_{ij} e^{u_i - v_j}$. $r_i^+ = \frac{a_i}{\sum_j K_{ij} s_j}$, $s_j^+ = \frac{b_j}{\sum_i K_{ij} r_i}$.

$$\frac{\partial Q_i^\alpha}{\partial u_i} = \sum_j P_{ij}, \quad \frac{\partial Q_i^\alpha}{\partial v_j} = -P_{ij}, \quad \frac{\partial Q_j^\beta}{\partial v_j} = \sum_i P_{ij}, \quad \frac{\partial Q_j^\beta}{\partial u_i} = -P_{ij}.$$

8.4 Sinkhorn Convergence: Sublinear Robust Rate

Sinkhorn is cyclic coordinate ascent on the smooth dual objective $\mathcal{D}_\varepsilon$ defined in (7.17); equivalently, it alternates KL projections on the two marginal constraint sets. Statistical convergence with changing empirical marginals is treated separately in Chapter 9.

Proposition 8.13 (Pinsker inequality). *If $p, q \in \Sigma_n$, then $\|p - q\|_1^2 \leq 2 \text{KL}(p|q)$.*

Proof. Let $A = \{i : p_i \geq q_i\}$ and set $a = \sum_{i \in A} p_i$, $b = \sum_{i \in A} q_i$. Then $a - b = \frac{1}{2} \|p - q\|_1$. Applying data processing for relative entropy to the partition (A, A^c) gives $\text{KL}(p|q) \geq a \log \frac{a}{b} + (1 - a) \log \frac{1-a}{1-b}$. For fixed $b \in (0, 1)$, the function $h(a) = a \log \frac{a}{b} + (1 - a) \log \frac{1-a}{1-b} - 2(a - b)^2$ satisfies $h(b) = h'(b) = 0$ and $h''(a) = \frac{1}{a} + \frac{1}{1-a} - 4 \geq 0$, because $a(1 - a) \leq 1/4$. Hence the binary relative entropy is at least $2(a - b)^2 = \frac{1}{2} \|p - q\|_1^2$. The boundary cases follow by approximation. $\square$

Proposition 8.14 (Robust $O(1/k)$ dual rate for discrete Sinkhorn). *Let $a \in \Sigma_n$ and $b \in \Sigma_m$ be positive histograms, let $C \in \mathbb{R}^{n \times m}$ be finite, and write $R := \max_{i,j} C_{ij} - \min_{i,j} C_{ij}$. Initialize $g_0 = 0$, perform complete Sinkhorn dual cycles, and denote the resulting potentials by (f_k, g_k) for $k \geq 1$. Let $\Delta_k := \mathcal{D}_\varepsilon(f^*, g^*) - \mathcal{D}_\varepsilon(f_k, g_k)$ be the dual suboptimality gap for the entropic dual objective (7.17). Then $0 \leq \Delta_k \leq \frac{8R^2}{\varepsilon k}$, $k \geq 1$.*

Proof. Subtracting $\min_{i,j} C_{ij}$ from the cost changes all transport values by the same constant and leaves dual gaps and Sinkhorn iterates unchanged up to gauge. We may therefore assume $0 \leq C_{ij} \leq R$. For a vector h , introduce the quotient norm $|h|_q := \inf_{\lambda \in \mathbb{R}} \|h + \lambda \mathbf{1}\|_\infty = \frac{1}{2} \|h\|_V$. Every potential produced by a soft transform has oscillation at most R . The same is true for optimal transformed potentials, hence $\max\{|f_k|_q, |g_k|_q\} \leq \frac{R}{2}$. This is precisely the block-quotient dual radius required by the robust cyclic-projection theorem of.

To instantiate that theorem, vectorize P and use the constraint operator $A(P) = (P \mathbf{1}_m, P^\top \mathbf{1}_n)$. Every column of A contains two unit entries, so $\|A\|_{1 \rightarrow 1} = 2$. After each exact row or column projection, the associated nonnegative matrix has total mass one, giving the uniform primal bound $X = 1$. Substituting $U = R/2$, $X = 1$, and $\|A\|_{1 \rightarrow 1} = 2$ into the cited theorem yields $\Delta_k \leq \frac{8XU^2 \|A\|_{1 \rightarrow 1}^2}{\varepsilon k} = \frac{8R^2}{\varepsilon k}$.

For completeness, the mechanism behind this substitution is the KL Pythagorean identity. If

$$P_k = (a \otimes b) \odot \exp\left(\frac{f_k \oplus g_k - C}{\varepsilon}\right),$$

then P_k has exact target marginal, hence total mass one, and $\Delta_k = \varepsilon \text{KL}(P^*|P_k)$. A complete cycle gives the exact ascent identity $\mathcal{D}_\varepsilon(f_{k+1}, g_{k+1}) - \mathcal{D}_\varepsilon(f_k, g_k) = \varepsilon [\text{KL}(P^*|P_k) - \text{KL}(P^*|P_{k+1})]$. Pinsker's inequality controls the marginal residual by this KL ascent, while the quotient-radius bound controls the dual gap by the same residual. Their combination is the scalar recursion $\Delta_k - \Delta_{k+1} \geq \frac{\varepsilon}{8R^2} \Delta_k^2$, whose reciprocal telescoping gives the stated $1/k$ estimate. When $R = 0$, one complete cycle is already optimal and the conclusion is immediate. $\square$

Corollary 8.15 (Approximating unregularized OT by regularized dual costs). *Under the assumptions of Proposition 8.14, suppose $nm > 1$. Let $\mathcal{D}_{\varepsilon,k}$ be the KL-normalized entropic dual value after $k \geq 1$ Sinkhorn cycles, and define the entropy-corrected lower bound $L_{\varepsilon,k} := \mathcal{D}_{\varepsilon,k} - \varepsilon H(a) - \varepsilon H(b)$, $H(a) = -\sum_i a_i \log a_i$. Then $0 \leq L_C(a, b) - L_{\varepsilon,k} \leq \varepsilon \log(nm) + \frac{8R^2}{\varepsilon k}$. $\varepsilon = \frac{\delta}{2 \log(nm)}$, $k \geq \frac{32R^2 \log(nm)}{\delta^2}$, produce a lower bound satisfying $0 \leq L_C(a, b) - L_{\varepsilon,k} \leq \delta$.*

Proof. The KL-normalized objective differs from the entropy convention (7.1) by the constant $\varepsilon H(a) + \varepsilon H(b)$ on the transport polytope, because $\text{KL}(P|a \otimes b) = -H(P) + H(a) + H(b)$. Let E_ε be the optimum of the entropy-regularized objective $\langle P, C \rangle - \varepsilon H(P)$. Since $0 \leq H(P) \leq \log(nm)$ for any coupling matrix, $L_C(a, b) - \varepsilon \log(nm) \leq E_\varepsilon \leq L_C(a, b)$. The corrected iterate satisfies $L_{\varepsilon,k} = E_\varepsilon - \Delta_k$, which gives the displayed value bound after applying Proposition 8.14. The stated choices allocate at most $\delta/2$ to each of the regularization and optimization errors. $\square$

The same identities yield computable diagnostics. If $r = P \mathbf{1}_m$, the KL projection onto the row constraint satisfies $\text{KL}(\text{Proj}_{C_a^{\text{KL}}}(P)|P) = \text{KL}(a|r)$; the analogous column identity uses $s = P^\top \mathbf{1}_n$.

8.5 Sinkhorn Convergence: Linear Hilbert Metric Rate

Projective contraction.

Definition 8.16 (Hilbert metric). On $\mathbb{R}_{+,*}^n$, Hilbert's projective metric is

$$\forall (u, u') \in (\mathbb{R}_{+,*}^n)^2, \quad d_{\mathcal{H}}(u, u') := \|\log(u) - \log(u')\|_V. \quad (8.3)$$

Here, for vectors, $\|z\|_V = \max_i z_i - \min_i z_i$.

Proposition 8.17 (Hilbert metric on the projective cone). *The function $d_{\mathcal{H}}$ defines a complete distance on the projective cone $\mathbb{R}_{+,*}^n / \sim$, where $u \sim u'$ means that $u = su'$ for some $s > 0$.*

Proof. The map $u \mapsto \log u$ identifies $\mathbb{R}_{+,*}^n / \sim$ with the quotient vector space $\mathbb{R}^n / \text{Span}(\mathbf{1}_n)$, because multiplying u by $s > 0$ adds the constant vector $\log(s) \mathbf{1}_n$. The variation seminorm $\|z\|_V = \max_i z_i - \min_i z_i$ vanishes exactly on constant vectors, so it induces a norm on this quotient. Symmetry, the triangle inequality and separation therefore follow from the corresponding norm properties. Completeness follows because $\mathbb{R}^n / \text{Span}(\mathbf{1}_n)$ is finite-dimensional and all finite-dimensional normed spaces are complete. $\square$

Theorem 8.18 (Birkhoff contraction theorem). *Let $K \in \mathbb{R}_{+,*}^{n \times m}$. Then, for $(v, v') \in (\mathbb{R}_{+,*}^m)^2$,*

$$d_{\mathcal{H}}(Kv, Kv') \leq \lambda(K) d_{\mathcal{H}}(v, v') \text{ where } \begin{cases} \lambda(K) := \frac{\sqrt{\eta(K)} - 1}{\sqrt{\eta(K)} + 1} < 1 \\ \eta(K) := \max_{i,j,k,\ell} \frac{K_{i,k}K_{j,\ell}}{K_{j,k}K_{i,\ell}}. \end{cases}$$

Proof. The Birkhoff–Hopf theorem states that a positive linear map A contracts Hilbert’s metric by at most $\tanh(\Delta(A)/4)$, where $\Delta(A) := \sup_{u,v>0} d_{\mathcal{H}}(Au, Av)$. This is the classical cone-contraction theorem of Birkhoff. For the matrix K , expand a cross-ratio between output coordinates, $\frac{(Ku)_i(Kv)_j}{(Ku)_j(Kv)_i}$. Every summand in the numerator is bounded by $\eta(K)$ times the corresponding cross-summand in the denominator, so this ratio is at most $\eta(K)$. Conversely, concentrating u and v on two columns approaching a maximizing quadruple attains this bound in the limit. Hence $\Delta(K) = \log \eta(K)$, $\eta(K) = \max_{i,j,k,\ell} \frac{K_{i,k}K_{j,\ell}}{K_{j,k}K_{i,\ell}}$. Therefore $\tanh(\Delta(K)/4) = (\sqrt{\eta(K)} - 1)/(\sqrt{\eta(K)} + 1)$, which is the claimed factor. $\square$

Nested simplex images. $K^{\ell+1}\Sigma_3 = K^\ell(K\Sigma_3) \subseteq K^\ell\Sigma_3$.

$$J_3 := \frac{1}{3} \mathbb{1}_3 \mathbb{1}_3^\top, \quad A := \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \quad K_{\rho,\delta} := \rho I_3 + (1 - \rho)J_3 + \delta A.$$

$\rho = .90$, $\delta = .014$, $K_1 = K_{\rho,0}$, $K_2 = K_{\rho,\delta}$, $K_3 = K_{\rho,-\delta}$. The control K_1 acts as ρI on the zero-sum tangent plane, so its images are parallel homothetic triangles. Since the restriction of A has eigenvalues $\pm i\sqrt{3}$, the two non-Perron eigenvalues of $K_{\rho,\delta}$ are $\rho \pm i\sqrt{3}\delta$.

$$r_i := \max \{|z| : z \in \text{spec}(K_i), z \neq 1\}$$

Theorem 8.19 (Projective linear convergence of Sinkhorn). *Let $v^{(0)} > 0$, define the Sinkhorn iterates by (7.5), and fix any optimal scaling pair (u^*, v^*) . For $\ell \geq 1$, set $P^{(\ell)} := \text{diag}(u^{(\ell)})K \text{diag}(v^{(\ell)})$, and, for $\ell \geq 0$, denote the row-normalized half-step by $P^{(\ell+1/2)} := \text{diag}(u^{(\ell+1)})K \text{diag}(v^{(\ell)})$. Writing $\lambda = \lambda(K)$, one has*

$$d_{\mathcal{H}}(v^{(\ell)}, v^*) \leq \lambda^{2\ell} d_{\mathcal{H}}(v^{(0)}, v^*), \quad d_{\mathcal{H}}(u^{(\ell+1)}, u^*) \leq \lambda^{2\ell+1} d_{\mathcal{H}}(v^{(0)}, v^*). \quad (8.4)$$

Thus the scaling rays converge linearly. After fixing any continuous gauge, their representatives converge, and $P^{(\ell)} \rightarrow P^$ entrywise.*

$$d_{\mathcal{H}}(u^{(\ell)}, u^*) \leq \frac{d_{\mathcal{H}}(P^{(\ell)} \mathbb{1}_m, a)}{1 - \lambda^2} \quad \text{and} \quad d_{\mathcal{H}}(v^{(\ell)}, v^*) \leq \frac{d_{\mathcal{H}}((P^{(\ell+1/2)})^\top \mathbb{1}_n, b)}{1 - \lambda^2}. \quad (8.5)$$

Lastly,

$$\|\log(P^{(\ell)}) - \log(P^*)\|_\infty \leq d_{\mathcal{H}}(u^{(\ell)}, u^*) + d_{\mathcal{H}}(v^{(\ell)}, v^*), \quad (8.6)$$

where P^ is the unique solution of (7.1).*

Proof. Entrywise multiplication by a fixed positive vector and entrywise inversion are isometries of Hilbert’s metric. Therefore the row map $R(v) = a/(Kv)$ and the column map $C(u) = b/(K^\top u)$ are both λ -Lipschitz by Theorem 8.18. Their full-cycle compositions $C \circ R$ and $R \circ C$ are λ^2 -contractions. Since $v^* = C(R(v^*))$ and $u^* = R(C(u^*))$, iteration gives (8.4).

For any contraction F with factor $q < 1$ and fixed point x^* , the triangle inequality gives $d(x, x^*) \leq d(x, Fx) + d(Fx, Fx^*) \leq d(x, Fx) + q d(x, x^*)$, hence $d(x, x^*) \leq d(x, Fx)/(1 - q)$. Apply this with $q = \lambda^2$ to the two full-cycle maps. The scaling identities imply $d_{\mathcal{H}}(u^{(\ell+1)}, u^{(\ell)}) = d_{\mathcal{H}}(a, P^{(\ell)} \mathbb{1}_m)$, and $d_{\mathcal{H}}(v^{(\ell+1)}, v^{(\ell)}) = d_{\mathcal{H}}(b, (P^{(\ell+1/2)})^\top \mathbb{1}_n)$, which proves (8.5).

Finally, write $\xi_i = \log(u_i^{(\ell)}/u_i^*)$ and $\zeta_j = \log(v_j^{(\ell)}/v_j^*)$. Then $\log(P_{ij}^{(\ell)}/P_{ij}^*) = \xi_i + \zeta_j$, $\text{osc}(\xi \oplus \zeta) = \text{osc}(\xi) + \text{osc}(\zeta)$. Both plans have total mass one, so $\sum_{i,j} P_{ij}^* e^{\xi_i + \zeta_j} = 1$. This proves (8.6). $\square$

Nonlinear Sinkhorn images of the simplex.

$$F_u(u) := R(C(u)) = a \oslash [K(b \oslash (K^\top u))].$$

$\widehat{F}_u(p) := \frac{F_u(p)}{\langle F_u(p), \mathbb{1}_3 \rangle}$, $p \in \Sigma_3$. Thus $\widehat{u}^{(\ell)} := u^{(\ell)}/\langle u^{(\ell)}, \mathbb{1}_3 \rangle$ follows $\widehat{u}^{(\ell+1)} = \widehat{F}_u(\widehat{u}^{(\ell)})$ when ℓ indexes complete Sinkhorn cycles. Since $\widehat{F}_u$ maps Σ_3 into itself, its set-valued iterates are nested:

$$\widehat{F}_u^{\ell+1}(\Sigma_3) = \widehat{F}_u^\ell(\widehat{F}_u(\Sigma_3)) \subseteq \widehat{F}_u^\ell(\Sigma_3).$$

$$\text{diam}_{d_{\mathcal{H}}}(\widehat{F}_u^\ell(\Sigma_3)) \leq \lambda(K)^{2(\ell-1)} \text{diam}_{d_{\mathcal{H}}}(\widehat{F}_u(\Sigma_3)), \quad \ell \geq 1.$$

Dual-potential form of the contraction.

$$\|f_\ell - f^*\|_V = \varepsilon d_{\mathcal{H}}(u^{(\ell)}, u^*), \quad \|g_\ell - g^*\|_V = \varepsilon d_{\mathcal{H}}(v^{(\ell)}, v^*).$$

Thus a complete cycle contracts both potential classes at the rate $\lambda(K)^2$. Moreover, the temperature factor must be retained when passing from potentials to densities:

$$\|\log \frac{d\pi_\ell}{d\pi^*}\|_\infty = \frac{1}{\varepsilon} \|(f_\ell - f^*) \oplus (g_\ell - g^*)\|_\infty \leq \frac{\|f_\ell - f^*\|_V + \|g_\ell - g^*\|_V}{\varepsilon}.$$

$$\lambda = \frac{\sqrt{\eta}-1}{\sqrt{\eta}+1} \leq \tanh(R/(2\varepsilon)) < 1. \quad \|f_\ell - f^*\|_V \leq \frac{\varepsilon}{1-\lambda^2} \|\log((P^{(\ell)} \mathbb{1}_m) \oslash a)\|_V, \quad \|g_\ell - g^*\|_V \leq \frac{\varepsilon}{1-\lambda^2} \|\log(((P^{(\ell+1/2)})^\top \mathbb{1}_n) \oslash b)\|_V.$$

8.6 Entropic Optimal Transport between Gaussians

Proposition 8.20 (Quadratic closure of Sinkhorn iterates). *Let $\beta = \mathcal{N}(m_\beta, \Sigma_\beta)$ on $\mathbb{R}^d$ and take $c(x, y) = \|x - y\|^2$. If $g(y)$ is a quadratic polynomial such that the Gaussian integral below is finite, then the soft transform*

$$f(x) = -\varepsilon \log \int \exp\left(\frac{g(y) - \|x - y\|^2}{\varepsilon}\right) d\beta(y)$$

is a quadratic polynomial in x . In particular, starting Sinkhorn from $g_0 = 0$ gives

$$f_1(x) = \frac{\varepsilon}{2} \log \det\left(\text{Id} + \frac{2\Sigma_\beta}{\varepsilon}\right) + \varepsilon \langle x - m_\beta, (\varepsilon \text{Id} + 2\Sigma_\beta)^{-1}(x - m_\beta) \rangle.$$

Proof. The exponent is the sum of a quadratic polynomial in y and the logarithm of the Gaussian density of β . Completing the square in y evaluates the integral as a positive constant times the exponential of a quadratic polynomial in x . Taking $-\varepsilon \log$ therefore gives a quadratic polynomial.

For $g_0 = 0$, let $Y \sim \beta$. The Gaussian identity

$$\mathbb{E} \exp\left(-\frac{\|x - Y\|^2}{\varepsilon}\right) = \det\left(\text{Id} + \frac{2\Sigma_\beta}{\varepsilon}\right)^{-1/2} \exp(-\langle x - m_\beta, (\varepsilon \text{Id} + 2\Sigma_\beta)^{-1}(x - m_\beta) \rangle)$$

gives the displayed expression. $\square$

Proposition 8.21 (Balanced entropic OT between Gaussians). *Let $\alpha = \mathcal{N}(m_\alpha, \Sigma_\alpha)$ and $\beta = \mathcal{N}(m_\beta, \Sigma_\beta)$ with positive-definite covariances, and let $\Sigma_\alpha^{1/2} \Sigma_\beta^{1/2} = U \text{diag}(\sigma_i) V^\top$ be a singular-value decomposition. For the balanced objective $\min_{\pi \in \mathcal{U}(\alpha, \beta)} \int \|x - y\|^2 d\pi(x, y) + \varepsilon \text{KL}(\pi | \alpha \otimes \beta)$, the optimizer is Gaussian with cross-covariance $K_\varepsilon = \Sigma_\alpha^{1/2} U \text{diag}(s_i) V^\top \Sigma_\beta^{1/2}$, $s_i = \frac{\sqrt{\varepsilon^2 + 16\sigma_i^2} - \varepsilon}{4\sigma_i}$. The optimal value is*

$$\|m_\alpha - m_\beta\|^2 + \text{tr}(\Sigma_\alpha) + \text{tr}(\Sigma_\beta) + \sum_i \left(-2\sigma_i s_i - \frac{\varepsilon}{2} \log(1 - s_i^2)\right).$$

As $\varepsilon \downarrow 0$, $s_i \rightarrow 1$ and the full covariance contribution, including the two trace terms, converges to $\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2$.

Proof. Let (X, Y) be any coupling with finite second moments and cross-covariance $K = \mathbb{E}[(X - m_\alpha)(Y - m_\beta)^\top]$. Replacing (X, Y) by the Gaussian vector with the same mean and covariance leaves the quadratic cost unchanged. Since the marginals are fixed, $\text{KL}(\pi | \alpha \otimes \beta) = -h(X, Y) + h(\alpha) + h(\beta)$, where h denotes differential entropy when it is finite. Among laws with a fixed covariance, the Gaussian maximizes entropy; if the entropy is not finite, the relative entropy is already $+\infty$. Thus the Gaussian replacement cannot increase the objective, and it is enough to optimize over Gaussian couplings.

Any such coupling has covariance

$$\begin{pmatrix} \Sigma_\alpha & K \\ K^\top & \Sigma_\beta \end{pmatrix}.$$

Write $K = \Sigma_\alpha^{1/2} S \Sigma_\beta^{1/2}$. The block covariance constraint is equivalent to the singular values of S being at most one, and finite entropy forces them to be strictly smaller than one. The cost depends on K through $\|m_\alpha - m_\beta\|^2 + \text{tr}(\Sigma_\alpha) + \text{tr}(\Sigma_\beta) - 2\text{tr}(K)$, while $\text{KL}(\pi | \alpha \otimes \beta) = -\frac{1}{2} \log \det(\text{Id} - SS^\top)$. By von Neumann's trace inequality, the minimizer aligns S with the singular vectors of $\Sigma_\alpha^{1/2} \Sigma_\beta^{1/2}$, so $S = U \text{diag}(s_i) V^\top$. The problem separates into scalar minimizations $\min_{0 \leq s < 1} -2\sigma_i s - \frac{\varepsilon}{2} \log(1 - s^2)$. The first-order condition is $2\sigma_i = \varepsilon s / (1 - s^2)$, whose positive solution is the displayed s_i . Substitution gives the value formula. Since $s_i \rightarrow 1$ and $\varepsilon \log(1 - s_i^2) \rightarrow 0$ as $\varepsilon \downarrow 0$, the spectral sum converges to $-2 \sum_i \sigma_i$. The identity

$$\sum_i \sigma_i = \text{tr}\left((\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2})^{1/2}\right)$$

then gives the Bures–Wasserstein covariance contribution. $\square$

Corollary 8.22 (Gaussian Sinkhorn divergence and smoothed Bures term). *Let $\alpha = \mathcal{N}(m_\alpha, \Sigma_\alpha)$ and $\beta = \mathcal{N}(m_\beta, \Sigma_\beta)$ have positive-definite covariances. For $r > 0$, define*

$$\tau_\varepsilon(r) := \frac{\sqrt{\varepsilon^2 + 16r^2} - \varepsilon}{4r}, \quad \psi_\varepsilon(r) := -2r \tau_\varepsilon(r) - \frac{\varepsilon}{2} \log(1 - \tau_\varepsilon(r)^2).$$

If $\sigma_i(\Sigma, \Lambda)$ denotes the singular values of $\Sigma^{1/2} \Lambda^{1/2}$ and $\lambda_i(\Sigma)$ the eigenvalues of Σ , then the debiased Sinkhorn divergence (7.42) is $\mathcal{L}_{\|\cdot\|_2}^\varepsilon(\alpha, \beta) = \|m_\alpha - m_\beta\|^2 + \mathcal{B}_\varepsilon(\Sigma_\alpha, \Sigma_\beta)^2$, where the Gaussian covariance contribution is the debiased smoothed Bures term

$$\mathcal{B}_\varepsilon(\Sigma, \Lambda)^2 := \sum_i \psi_\varepsilon(\sigma_i(\Sigma, \Lambda)) - \frac{1}{2} \sum_i \psi_\varepsilon(\lambda_i(\Sigma)) - \frac{1}{2} \sum_i \psi_\varepsilon(\lambda_i(\Lambda)).$$

Moreover $\mathcal{B}_\varepsilon(\Sigma, \Lambda)^2 \rightarrow \mathcal{B}(\Sigma, \Lambda)^2$ as $\varepsilon \downarrow 0$, where $\mathcal{B}$ is the Bures–Wasserstein metric of Proposition 2.35.

Proof. Proposition 8.21 writes the raw entropic value as the squared mean displacement plus trace terms and a spectral sum. With the notation above, the spectral part is exactly $\sum_i \psi_\varepsilon(\sigma_i(\Sigma_\alpha, \Sigma_\beta))$. Applying the same formula to the self-costs (α, α) and (β, β) replaces these singular values by the eigenvalues of Σ_α and Σ_β . In the polarization formula (7.42), the trace terms cancel: $\text{tr} \Sigma_\alpha + \text{tr} \Sigma_\beta - \frac{1}{2}(2\text{tr} \Sigma_\alpha) - \frac{1}{2}(2\text{tr} \Sigma_\beta) = 0$, while the polarization of the squared mean terms leaves exactly $\|m_\alpha - m_\beta\|^2$. Since $\tau_\varepsilon(r) \rightarrow 1$ and $\varepsilon \log(1 - \tau_\varepsilon(r)^2) \rightarrow 0$, one has $\psi_\varepsilon(r) \rightarrow -2r$. The limit is therefore $\text{tr} \Sigma + \text{tr} \Lambda - 2 \sum_i \sigma_i(\Sigma, \Lambda) = \mathcal{B}(\Sigma, \Lambda)^2$, which is the Bures formula (2.18). $\square$

Proposition 8.23 (One-dimensional Gaussian Sinkhorn rate). *Consider $\alpha = \beta = \mathcal{N}(0, 1)$ on $\mathbb{R}$ with $c(x, y) = (x - y)^2$. If a dual potential has the form $g_q(y) = qy^2 + \text{cst}$, then one soft transform has quadratic coefficient $T_\varepsilon(q) = 1 - \frac{1}{1-q+\varepsilon/2}$, $q < 1 + \varepsilon/2$, and one full Sinkhorn cycle acts as $q \mapsto T_\varepsilon(T_\varepsilon(q))$. The fixed point $q_\star = T_\varepsilon(q_\star)$ is determined by $A_\star^2 - \frac{\varepsilon}{2}A_\star - 1 = 0$, $A_\star := 1 - q_\star + \frac{\varepsilon}{2} = \frac{\varepsilon + \sqrt{\varepsilon^2 + 16}}{4}$.*

$$\rho_\varepsilon = A_\star^{-4} = \left(\frac{4}{\varepsilon + \sqrt{\varepsilon^2 + 16}} \right)^4.$$

Proof. Completing the square in

$$\int \exp\left(\frac{qy^2 - (x - y)^2}{\varepsilon}\right) d\mathcal{N}(0, 1)(y)$$

gives the coefficient $T_\varepsilon(q)$. The fixed-point equation $q_\star = 1 - 1/A_\star$, together with $q_\star = 1 + \varepsilon/2 - A_\star$, gives $A_\star^2 - \frac{\varepsilon}{2}A_\star - 1 = 0$. The positive solution is the displayed $A_\star$. Since $T'_\varepsilon(q) = -\frac{1}{(1-q+\varepsilon/2)^2}$, the derivative of the full-cycle map at the fixed point is $T'_\varepsilon(q_\star)^2 = A_\star^{-4}$. $\square$

8.7 Continuous ε -Sinkhorn Flow Parabolic Monge–Ampère limit.

Definition 8.24 (Continuous ε -Sinkhorn flow). Let $\alpha = e^{-F}dx$ and $\beta = e^{-G}dx$ be smooth positive probability densities on the unit-volume flat torus $\mathbb{T}^d$. For the cost $c(x, y) = d_{\mathbb{T}^d}(x, y)^2/2$, a gauge-fixed potential $u_t : \mathbb{T}^d \rightarrow \mathbb{R}$, with $\int_{\mathbb{T}^d} u_t dx = 0$, follows the continuous ε -Sinkhorn flow if

$$\partial_t u_t(x) = \log \det(\text{Id} + \nabla^2 u_t(x)) - G(x + \nabla u_t(x)) + F(x) - \bar{r}_t, \quad (\text{CES})$$

where $x + \nabla u_t(x)$ is understood modulo $\mathbb{Z}^d$, $\text{Id} + \nabla^2 u_t \succ 0$, and $\bar{r}_t$ is the unique scalar preserving the mean-zero gauge.

$\bar{r}_t = \int_{\mathbb{T}^d} [\log \det(\text{Id} + \nabla^2 u_t(x)) - G(x + \nabla u_t(x)) + F(x)] dx$. Its subtraction fixes the additive constant of the potential. Let $\alpha^{(k)}$ and $\beta^{(k)}$ be the positive grid discretizations of α and β , and define $v_k[u](y) := \frac{1}{k} \log \int_{\mathbb{T}^d} e^{-k(c(x, y) + u(x))} d\alpha^{(k)}(x)$, $(S_k u)(x) := \frac{1}{k} \log \int_{\mathbb{T}^d} e^{-k(c(x, y) + v_k[u](y))} d\beta^{(k)}(y)$.

$$\rho_{k, u}(x) := e^{k(S_k u(x) - u(x))} = e^{-ku(x)} \int_{\mathbb{T}^d} \frac{e^{-kc(x, y)}}{\int_{\mathbb{T}^d} e^{-k(c(x', y) + u(x'))} d\alpha^{(k)}(x')} d\beta^{(k)}(y). \quad (8.7)$$

Proposition 8.25 (Scaled log-Sinkhorn limit). *Let $\varepsilon_k = 1/k$, let $\alpha^{(k)}, \beta^{(k)}$ be the regular-grid density discretizations above, and generate the mean-zero iterates with the normalized operator S_k . Define $u^{(k)}(t) := u_{[kt]}^{(k)}$. Assume that, on every compact time interval, $u^{(k)}(t)$ converges smoothly to u_t , that $\text{Id} + \nabla^2 u_t \succ 0$, and that the Laplace expansion below is uniform along the sequence. Then u_t solves the continuous ε -Sinkhorn flow (CES).*

Proof. By (8.7), the normalized update satisfies

$$u_{m+1}^{(k)} - u_m^{(k)} = k^{-1} \left[\log \rho_{k, u_m^{(k)}} - \int_{\mathbb{T}^d} \log \rho_{k, u_m^{(k)}} dx \right],$$

where the subtraction enforces the mean-zero gauge. Berman's discrete Laplace estimate gives $\rho_{k, u}(x) = \det(\text{Id} + \nabla^2 u(x)) e^{F(x) - G(x + \nabla u(x))} (1 + O(k^{-1}))$, uniformly for the smooth potentials under consideration. Taking logarithms gives $\log \rho_{k, u}(x) = \log \det(\text{Id} + \nabla^2 u(x)) - G(x + \nabla u(x)) + F(x) + O(k^{-1})$. Dividing the increment by the time step $1/k$, passing to the smooth limit $u^{(k)}(t) \rightarrow u_t$, and keeping the mean-zero normalization gives exactly (CES). $\square$

Proposition 8.26 (Stationary continuous Sinkhorn potentials). *Assume that a smooth stationary solution of (CES) satisfies $\text{Id} + \nabla^2 u \succ 0$, and define $T(x) = x + \nabla u(x)$ modulo $\mathbb{Z}^d$. Then $T_\# \alpha = \beta$. Conversely, any smooth map of this form satisfying $\text{Id} + \nabla^2 u \succ 0$ and $T_\# \alpha = \beta$ solves the stationary equation, up to the additive gauge of u .*

Proof. At stationarity, the bracket in (CES) is a constant c : $\log \det(\text{Id} + \nabla^2 u(x)) - G(T(x)) + F(x) = c$. Exponentiating, and using $\nabla T = \text{Id} + \nabla^2 u$, gives $e^{-G(T(x))} \det(\nabla T(x)) = e^{-F(x)} e^c$. Because T is homotopic to the identity and $\text{Id} + \nabla^2 u \succ 0$, it is an orientation-preserving local diffeomorphism of degree one; hence it is a diffeomorphism of the torus. Since α and β have total mass one, integration and the change-of-variables formula give $e^c = 1$. The same identity then becomes precisely the Jacobian equation for $T_\#(e^{-F}dx) = e^{-G}dx$. The converse follows by reversing this argument. $\square$

Gaussian closure. $T_t(x) := x + \nabla u_t(x) = q_t + B_t(x - m_\alpha)$, $B_t \in \mathbb{S}_{++}^d$.

$$F(x) = \frac{1}{2} \langle x - m_\alpha, \Sigma_\alpha^{-1}(x - m_\alpha) \rangle + \text{cst}, \quad G(y) = \frac{1}{2} \langle y - m_\beta, \Sigma_\beta^{-1}(y - m_\beta) \rangle + \text{cst},$$

$$\begin{aligned} \dot{B}_t &= \Sigma_\alpha^{-1} - B_t \Sigma_\beta^{-1} B_t, & \dot{q}_t &= -B_t \Sigma_\beta^{-1} (q_t - m_\beta), & \Sigma_t &= B_t \Sigma_\alpha B_t, & \dot{\Sigma}_t &= \dot{B}_t \Sigma_\alpha B_t + B_t \Sigma_\alpha \dot{B}_t, & B \Sigma_\beta^{-1} B &= \Sigma_\alpha^{-1}, & q &= m_\beta, \\ \partial_t u_t(x) &= \log(1 + u_t''(x)) - G(x + u_t'(x)) + F(x) - \bar{r}_t, \end{aligned}$$

9 Statistical Optimal Transport

9.1 Law of Large Numbers and Central Limit Theorem

Empirical laws.

$$\hat{\alpha}_n := \frac{1}{n} \sum_{i=1}^n \delta_{X_i}. \quad (9.1)$$

The ordinary law of large numbers says that empirical averages converge to expectations. In measure language this means that $\hat{\alpha}_n$ converges weakly toward α , because testing $\hat{\alpha}_n$ against a bounded continuous function φ gives the sample average $n^{-1} \sum_i \varphi(X_i)$. **Proposition 9.1** (Empirical law of large numbers in $\mathcal{W}_p$). *Let $(\mathcal{X}, d)$ be a Polish metric space, let $p \geq 1$, and let $\alpha \in \mathcal{P}_p(\mathcal{X})$. Let $(X_i)_{i \geq 1}$ be i.i.d. random variables with law α , and define $\hat{\alpha}_n$ by (9.1). Then*

$$\hat{\alpha}_n \rightharpoonup \alpha \quad \text{and} \quad \mathcal{W}_p(\hat{\alpha}_n, \alpha) \longrightarrow 0$$

almost surely. Moreover, $\mathbb{E} \mathcal{W}_p(\hat{\alpha}_n, \alpha)^p \longrightarrow 0$.

Proof. Fix a reference point $x_0 \in \mathcal{X}$, and write $r(x) = d(x, x_0)$. Since $\mathcal{X}$ is Polish, the weak topology on $\mathcal{P}(\mathcal{X})$ admits a countable convergence-determining class $(\varphi_k)_{k \geq 1} \subset C_b(\mathcal{X})$. For each fixed k , the strong law of large numbers gives $\int \varphi_k d\hat{\alpha}_n = \frac{1}{n} \sum_{i=1}^n \varphi_k(X_i) \longrightarrow \mathbb{E} \varphi_k(X_1) = \int \varphi_k d\alpha$ almost surely. Intersecting the corresponding probability-one events over the countable set of indices gives, on a single event of probability one, convergence against every φ_k . By the convergence-determining property, this is exactly $\hat{\alpha}_n \rightharpoonup \alpha$.

The moment condition $\alpha \in \mathcal{P}_p(\mathcal{X})$ means $\int r^p d\alpha < +\infty$. Applying the strong law again to the integrable random variable $r(X_1)^p$ gives $\int r^p d\hat{\alpha}_n = \frac{1}{n} \sum_{i=1}^n r(X_i)^p \longrightarrow \int r^p d\alpha$ almost surely. Proposition 3.41 states that weak convergence plus convergence of the p -th moments is equivalent to $\mathcal{W}_p$ -convergence on $\mathcal{P}_p(\mathcal{X})$. Hence $\mathcal{W}_p(\hat{\alpha}_n, \alpha) \rightarrow 0$ almost surely.

Set $A_n = \int r^p d\hat{\alpha}_n$ and $M = \int r^p d\alpha$. The triangle inequality through the Dirac mass δ_{x_0} , followed by $(a+b)^p \leq 2^{p-1}(a^p + b^p)$, gives the deterministic bound $\mathcal{W}_p(\hat{\alpha}_n, \alpha)^p \leq 2^{p-1}(A_n + M)$. The family $(A_n)_n$ is uniformly integrable. Indeed, by the de la Vallée-Poussin criterion, choose a convex superlinear function Ψ such that $\mathbb{E} \Psi(r(X_1)^p) < +\infty$; Jensen's inequality gives $\mathbb{E} \Psi(A_n) \leq \frac{1}{n} \sum_{i=1}^n \mathbb{E} \Psi(r(X_i)^p) = \mathbb{E} \Psi(r(X_1)^p)$. Thus $(A_n + M)_n$, and hence $(\mathcal{W}_p(\hat{\alpha}_n, \alpha)^p)_n$, is uniformly integrable. Together with almost-sure convergence to zero, this implies $\mathbb{E} \mathcal{W}_p(\hat{\alpha}_n, \alpha)^p \rightarrow 0$. $\square$

Central-limit fluctuations.

Proposition 9.2 (Berry–Esseen bound in $\mathcal{W}_1$). *Let $(X_i)_{i=1}^n$ be i.i.d. real random variables such that $\mathbb{E} X_i = 0$, $\mathbb{E} X_i^2 = 1$ and $\mathbb{E} |X_i|^3 < +\infty$. If α_n is the law of $n^{-1/2} \sum_i X_i$ and γ is the standard Gaussian law, then $\mathcal{W}_1(\alpha_n, \gamma) \leq \frac{C \mathbb{E} |X_1|^3}{\sqrt{n}}$, where C is a universal constant.*

Proof. By Kantorovich–Rubinstein duality,

$$\mathcal{W}_1(\alpha_n, \gamma) = \sup_{\text{Lip}(h) \leq 1} |\mathbb{E} h(S_n) - \mathbb{E} h(G)|, \quad S_n = n^{-1/2} \sum_i X_i, \quad G \sim \gamma.$$

For each such h , solve Stein's equation $f'_h(x) - x f_h(x) = h(x) - \mathbb{E} h(G)$. The solution satisfies $\|f'_h\|_\infty + \|f_h\|_\infty \leq C$ for a universal constant. Hence $\mathbb{E} h(S_n) - \mathbb{E} h(G) = \mathbb{E} [f'_h(S_n) - S_n f_h(S_n)]$. Write $S_n^{(i)} = S_n - X_i/\sqrt{n}$. Independence and $\mathbb{E} X_i = 0$ give

$$\mathbb{E} [S_n f_h(S_n)] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbb{E} \left[X_i (f_h(S_n^{(i)} + X_i/\sqrt{n}) - f_h(S_n^{(i)})) \right].$$

Taylor's formula with integral remainder and $\mathbb{E} X_i^2 = 1$ imply

$$\left| \mathbb{E} [S_n f_h(S_n)] - \frac{1}{n} \sum_{i=1}^n \mathbb{E} f'_h(S_n^{(i)}) \right| \leq \frac{C \mathbb{E} |X_1|^3}{\sqrt{n}}.$$

Moreover, the Lipschitz bound on f'_h yields

$$\left| \mathbb{E} f'_h(S_n) - \frac{1}{n} \sum_{i=1}^n \mathbb{E} f'_h(S_n^{(i)}) \right| \leq \frac{C \mathbb{E} |X_1|}{\sqrt{n}} \leq \frac{C \mathbb{E} |X_1|^3}{\sqrt{n}},$$

where the last inequality uses $\mathbb{E} X_1^2 = 1$. Sharper constants and higher-order transport-distance refinements are available, but are omitted from the compact handout. $\square$

Empirical-process fluctuations.

Proposition 9.3 (Finite-dimensional empirical-process CLT). *Let $X_1, X_2, \dots$ be i.i.d. with law α , and let $\varphi_1, \dots, \varphi_q \in L^2(\alpha)$. Then*

$$\sqrt{n} \left(\int \varphi_r d(\hat{\alpha}_n - \alpha) \right)_{r=1}^q \Longrightarrow G_\alpha,$$

where G_α is centered Gaussian in $\mathbb{R}^q$ with covariance

$$\mathbb{E} [(G_\alpha)_r (G_\alpha)_s] = \int \left(\varphi_r - \int \varphi_r d\alpha \right) \left(\varphi_s - \int \varphi_s d\alpha \right) d\alpha.$$

Proof. Apply the multivariate central limit theorem to the i.i.d. centered vectors $(\varphi_r(X_i) - \int \varphi_r d\alpha)_{r=1}^q$. $\square$

9.2 Sample Complexity

Unregularized OT.

Proposition 9.4 (Dyadic partition bound for $\mathcal{W}_1$). *Let $\mathcal{Q}_j$ be nested dyadic partitions of $[0, 1]^d$ into half-open cubes of side length 2^{-j} , with cubes on the outer face closed along the boundary of $[0, 1]^d$. For all $\alpha, \beta \in \mathcal{P}([0, 1]^d)$ and every integer $J \geq 0$, $\mathcal{W}_1(\alpha, \beta) \leq \sqrt{d} \sum_{j=0}^{J-1} 2^{-j} \sum_{Q \in \mathcal{Q}_{j+1}} |\alpha(Q) - \beta(Q)| + \sqrt{d} 2^{-J}$.*

Proof. The proof is constructive: one decomposes the two measures into pieces which can be coupled at different spatial scales. Write $\Omega = [0, 1]^d$. We build positive measures $(\alpha_j, \beta_j)_{j=0}^J$ such that $\sum_{j=0}^J \alpha_j = \alpha$, $\sum_{j=0}^J \beta_j = \beta$, $\alpha_j(Q) = \beta_j(Q)$ for every $Q \in \mathcal{Q}_j$. Assuming such a decomposition for a moment, each pair (α_j, β_j) can be coupled inside the cells of $\mathcal{Q}_j$. Indeed, on every cell $Q \in \mathcal{Q}_j$ with positive mass, take the product coupling between the normalized restrictions of α_j and β_j to Q , multiplied by the common mass $\alpha_j(Q) = \beta_j(Q)$. Summing over cells and over j gives a coupling of α and β . Since the diameter of a cell in $\mathcal{Q}_j$ is at most $\sqrt{d} 2^{-j}$, this coupling has cost bounded by $\mathcal{W}_1(\alpha, \beta) \leq \sqrt{d} \sum_{j=0}^J 2^{-j} \alpha_j(\Omega)$.

Start at the finest scale. For $Q \in \mathcal{Q}_J$, define α_J and β_J on Q by keeping the common part of the two masses:

$$\alpha_J(A \cap Q) = \frac{\alpha(Q) \wedge \beta(Q)}{\alpha(Q)} \alpha(A \cap Q), \quad \beta_J(A \cap Q) = \frac{\alpha(Q) \wedge \beta(Q)}{\beta(Q)} \beta(A \cap Q),$$

with the convention that the corresponding term is zero when the denominator vanishes. For $j = J-1, \dots, 1$, assume that the pieces at all finer levels $k > j$ have been constructed and define the residuals $\alpha'_j = \alpha - \sum_{k=j+1}^J \alpha_k$, $\beta'_j = \beta - \sum_{k=j+1}^J \beta_k$. On each $Q \in \mathcal{Q}_j$, set α_j and β_j equal to the common part of α'_j and β'_j inside Q , exactly as above. Finally set $\alpha_0 = \alpha - \sum_{k=1}^J \alpha_k$ and $\beta_0 = \beta - \sum_{k=1}^J \beta_k$. Each step removes only common residual mass, so all measures are positive, and by construction $\alpha_j(Q) = \beta_j(Q)$ on all cells $Q \in \mathcal{Q}_j$.

For notational consistency, set $\alpha'_0 = \alpha_0$, $\beta'_0 = \beta_0$, $\alpha'_J = \alpha$, and $\beta'_J = \beta$. Let $0 \leq j < J$ and $R \in \mathcal{Q}_j$. Since all pieces constructed at levels finer than $j+1$ have equal mass on every child $Q \in \mathcal{Q}_{j+1}$, the residual imbalance on Q just before the matching at level $j+1$ is the original imbalance: $\alpha'_{j+1}(Q) - \beta'_{j+1}(Q) = \alpha(Q) - \beta(Q)$. After removing the common part at level $j+1$, the remaining source residual inside Q has mass $\alpha'_j(Q) = \alpha'_{j+1}(Q) - \alpha_{j+1}(Q) = (\alpha'_{j+1}(Q) - \beta'_{j+1}(Q))_+$. Because α_j is itself a common submeasure of the residual α'_j , the mass matched at level j inside R satisfies

$$\alpha_j(R) = \beta_j(R) \leq \alpha'_j(R) = \sum_{Q \subset R} (\alpha'_{j+1}(Q) - \beta'_{j+1}(Q))_+ \leq \sum_{Q \subset R} |\alpha(Q) - \beta(Q)|,$$

where the sums are over $Q \in \mathcal{Q}_{j+1}$. Summing over $R \in \mathcal{Q}_j$ gives $\alpha_j(\Omega) \leq \sum_{Q \in \mathcal{Q}_{j+1}} |\alpha(Q) - \beta(Q)|$. The last level satisfies $\alpha_J(\Omega) \leq 1$. Substituting these bounds into the cost estimate proves the claim. $\square$

Proposition 9.5 (Empirical OT has intrinsic-dimension value rates). *Let α and β be probability measures supported on $[0, 1]^d$, and let $\hat{\alpha}_n$ and $\hat{\beta}_m$ be independent empirical measures. Then $\mathbb{E} |\mathcal{W}_1(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{W}_1(\alpha, \beta)| \lesssim_d r_d(n) + r_d(m)$, where, for $N \geq 2$,*

$$r_d(N) = \begin{cases} N^{-1/2}, & d = 1, \\ (\log N) N^{-1/2}, & d = 2, \\ N^{-1/d}, & d \geq 3. \end{cases}$$

For $N = 1$, the value of $r_d(1)$ is absorbed in the implicit constant. In particular, for $d \geq 3$, the worst-case one-sample empirical error for exact $\mathcal{W}_1$ has the characteristic high-dimensional scale $N^{-1/d}$. More generally, for $\mathcal{W}_p$ on regular d' -dimensional supports, the corresponding high-dimensional rate is $N^{-1/d'}$ when $d' > 2p$, under the usual volume-growth and moment assumptions. Thus the relevant dimension is the intrinsic dimension of the support rather than necessarily the ambient dimension.

Proof. By the triangle inequality, $|\mathcal{W}_1(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{W}_1(\alpha, \beta)| \leq \mathcal{W}_1(\hat{\alpha}_n, \alpha) + \mathcal{W}_1(\hat{\beta}_m, \beta)$. It is therefore enough to bound $\mathbb{E} \mathcal{W}_1(\hat{\alpha}_N, \alpha)$. Apply Proposition 9.4 to $(\hat{\alpha}_N, \alpha)$. If $Q \in \mathcal{Q}_{j+1}$, then $N \hat{\alpha}_N(Q)$ is a binomial random variable with parameter $\alpha(Q)$, and hence $\mathbb{E} |\hat{\alpha}_N(Q) - \alpha(Q)| \leq \sqrt{\mathbb{E} |\hat{\alpha}_N(Q) - \alpha(Q)|^2} \leq \sqrt{\frac{\alpha(Q)}{N}}$. Summing over the $2^{(j+1)d}$ cells and using the Cauchy-Schwarz inequality gives

$$\sum_{Q \in \mathcal{Q}_{j+1}} \mathbb{E} |\hat{\alpha}_N(Q) - \alpha(Q)| \leq \frac{1}{\sqrt{N}} \sum_{Q \in \mathcal{Q}_{j+1}} \sqrt{\alpha(Q)} \leq \frac{2^{(j+1)d/2}}{\sqrt{N}}.$$

Therefore, for every $J \geq 0$, $\mathbb{E} \mathcal{W}_1(\hat{\alpha}_N, \alpha) \lesssim_d 2^{-J} + \frac{1}{\sqrt{N}} \sum_{j=0}^{J-1} 2^{j(d/2-1)}$. If $d = 1$, the sum is uniformly bounded and we choose J such that $2^{-J} \lesssim N^{-1/2}$. If $d = 2$, the sum is J , and choosing $2^J \simeq N^{1/2}$ gives $(\log N) N^{-1/2}$. If $d \geq 3$, the sum is dominated by its last term; choosing $2^J \simeq N^{1/d}$ gives $2^{-J} + N^{-1/2} 2^{J(d/2-1)} \simeq N^{-1/d}$. This proves the displayed $\mathcal{W}_1$ rates. The extension to a regular d' -dimensional support uses nested partitions whose cells have diameter $O(2^{-j})$ and cardinality $O(2^{jd'})$; the general $\mathcal{W}_p$ statement follows from the standard multiscale estimates of. The elementary dyadic estimate is not sharp for every law in the critical case $d = 2$. The proposition deliberately retains the distribution-free $(\log N) N^{-1/2}$ bound supplied by the dyadic proof. $\square$

Lower bounds and minimax optimality.

Proposition 9.6 (Minimax lower bound for bounded densities). *Let $d \geq 3$. Let $\mathcal{A}$ be the class of probability measures on $[0, 1]^d$ admitting a density ρ with respect to Lebesgue measure such that $\frac{1}{2} \leq \rho(x) \leq \frac{3}{2}$ for a.e. $x \in [0, 1]^d$. There exists $c_d > 0$ such that, for every $N \geq 1$, $\inf_{\tilde{\alpha}_N} \sup_{\alpha \in \mathcal{A}} \mathbb{E}_{\alpha} \mathcal{W}_1(\tilde{\alpha}_N, \alpha) \geq c_d N^{-1/d}$. Here the infimum is over all estimators based on N i.i.d. samples from α . In particular, the high-dimensional rate of Proposition 9.5 is minimax sharp for $\mathcal{W}_1$, even when one allows arbitrary estimators rather than the empirical measure.*

Proof. We use a standard hypercube construction. Choose a smooth function ψ compactly supported in $(0, 1)^d$, with $\int \psi = 0$, $\|\psi\|_\infty \leq 1$, and whose positive and negative parts are separated in the first coordinate. There exists a compactly supported Lipschitz function φ such that $\int \varphi \psi dx > 0$. We extend both functions by zero outside $(0, 1)^d$, and absorb $\text{Lip}(\varphi)$ into the constants below.

Let m be an integer, $h = 1/m$, and partition $[0, 1]^d$ into $M = m^d$ cubes $Q_k = z_k + h[0, 1]^d$. Define the rescaled bumps

$$\psi_k(x) = \psi\left(\frac{x - z_k}{h}\right) \mathbf{1}_{Q_k}(x).$$

For $\theta = (\theta_1, \dots, \theta_M) \in \{-1, 1\}^M$ and a fixed small $\eta > 0$, set $\rho_\theta(x) = 1 + \eta \sum_{k=1}^M \theta_k \psi_k(x)$, $\alpha_\theta = \rho_\theta dx$. For η small enough, all these densities belong to $\mathcal{A}$, and they integrate to one because every ψ_k has zero mean.

If θ and θ' differ on a set S of coordinates, the dual formula for $\mathcal{W}_1$ gives an additive lower bound. Indeed, rescale φ as

$$\varphi_k(x) = h \varphi\left(\frac{x - z_k}{h}\right).$$

Each φ_k is supported in a fixed strict subcube of Q_k . After dividing by this constant, Kantorovich–Rubinstein duality and the choice $s_k = \text{sign}(\theta_k - \theta'_k)$ on S give $\mathcal{W}_1(\alpha_\theta, \alpha_{\theta'}) \geq c \eta h^{d+1} \text{Ham}(\theta, \theta')$, where Ham denotes Hamming distance and $c > 0$ depends only on the chosen bump.

Neighboring vertices of the hypercube are statistically close. If $\theta^{(k)}$ is obtained from θ by flipping only the k -th sign, then the one-sample Kullback–Leibler divergence is bounded by $\text{KL}(\alpha_\theta | \alpha_{\theta^{(k)}}) \leq C \eta^2 h^d$, because the two densities differ only on Q_k , the perturbation has size $O(\eta)$, and the densities are uniformly bounded away from zero. For N independent samples the divergence is multiplied by N . Choose m so that $m^d \simeq N$. Then Nh^d is bounded, and, by taking η sufficiently small, Pinsker's inequality gives a uniform bound $\text{TV}(\alpha_\theta^{\otimes N}, \alpha_{\theta^{(k)}}^{\otimes N}) \leq q < 1$. Assouad's lemma therefore implies

$$\inf_{\alpha_N} \sup_{\theta \in \{-1, 1\}^M} \mathbb{E}_\theta \mathcal{W}_1(\tilde{\alpha}_N, \alpha_\theta) \geq c' \eta M h^{d+1} (1 - q) = c' \eta (1 - q) h \simeq N^{-1/d}.$$

Decreasing the constant handles the finitely many small values of N , and the claim follows. $\square$

Leveraging smoothness.

Proposition 9.7 (Smoothed plug-in rates). *Let $d \geq 3$, $s > 0$, and let $\alpha = \rho dx$, $\beta = \eta dx$ be probability measures on $[0, 1]^d$. Assume that ρ and η are bounded above and below by positive constants and belong to a Hölder–Zygmund, equivalently Besov $B_{\infty, \infty}^s$, ball of radius L . Let $\hat{\alpha}_n$ and $\hat{\beta}_m$ be independent empirical measures. There exist normalized nonnegative density estimators $\tilde{\alpha}_n = \tilde{\rho}_n dx$, $\tilde{\beta}_m = \tilde{\eta}_m dx$, obtained for instance by a projected wavelet estimator at resolution $h_N \simeq N^{-1/(d+2s)}$, such that*

$$\mathbb{E} \left| \mathcal{W}_1(\tilde{\alpha}_n, \tilde{\beta}_m) - \mathcal{W}_1(\alpha, \beta) \right| \leq C \left(n^{-\frac{s+1}{d+2s}} + m^{-\frac{s+1}{d+2s}} \right).$$

The constant depends on d, s, L , on the chosen wavelet system and on the lower and upper density bounds. Thus any positive smoothness improves on the empirical $N^{-1/d}$ exponent when $d > 2$; if s is of order d , the exponent is of order $1/3$, and as $s \rightarrow \infty$ it approaches the parametric exponent $1/2$.

Proof sketch. It is enough, by the triangle inequality, to bound $\mathbb{E} \mathcal{W}_1(\tilde{\alpha}_N, \alpha)$. For densities bounded below and above, $\mathcal{W}_1$ is controlled by a negative smoothness norm of the density difference; in a wavelet notation one may write, schematically, $\mathcal{W}_1(\tilde{\alpha}_N, \alpha) \lesssim \|\tilde{\rho}_N - \rho\|_{B_{1,1}^{-1}}$. Truncating the empirical wavelet expansion at scale J , or equivalently smoothing at bandwidth $h = 2^{-J}$, gives two terms. The deterministic approximation error of an s -smooth density is $O(2^{-J(s+1)})$ in the negative B^{-1} norm, while the stochastic contribution of the retained coefficients is $N^{-1/2} \sum_{j \leq J} 2^{j(d/2-1)} \simeq N^{-1/2} 2^{J(d/2-1)}$ ($d \geq 3$). Balancing these two quantities gives $2^J \simeq N^{1/(d+2s)}$ and hence the rate $N^{-(s+1)/(d+2s)}$. Projecting the linear wavelet estimate onto the cone of probability densities in the same negative Besov norm preserves the order of this bound. Applying the same estimate to both marginals proves the displayed two-sample bound. The full minimax theorem, including $\mathcal{W}_p$, Besov classes, lower bounds and endpoint logarithms in dimensions 1 and 2, is proved in. $\square$

MMD.

Proposition 9.8 (MMD has a parametric value rate). *Let k be a bounded positive definite kernel with RKHS $\mathcal{H}_k$, and define $\text{MMD}_k(\alpha, \beta) = \left\| \int k(x, \cdot) d(\alpha - \beta)(x) \right\|_{\mathcal{H}_k}$. If $\hat{\alpha}_n$ and $\hat{\beta}_m$ are independent empirical measures, then*

$$\mathbb{E} \left| \text{MMD}_k(\hat{\alpha}_n, \hat{\beta}_m) - \text{MMD}_k(\alpha, \beta) \right| \leq \kappa \left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}} \right)$$

when $k(x, x) \leq \kappa^2$.

Proof. Let $\Phi(x) = k(x, \cdot)$ be the feature map and $m_\alpha = \mathbb{E} \Phi(X)$. The reverse triangle inequality for the RKHS norm gives $\left| \text{MMD}_k(\hat{\alpha}_n, \hat{\beta}_m) - \text{MMD}_k(\alpha, \beta) \right| \leq \text{MMD}_k(\hat{\alpha}_n, \alpha) + \text{MMD}_k(\hat{\beta}_m, \beta)$. Independence cancels the cross terms after taking the squared norm and expectation, giving $\mathbb{E} \text{MMD}_k(\hat{\alpha}_n, \alpha)^2 = \frac{1}{n} \mathbb{E} \|\Phi(X) - m_\alpha\|_{\mathcal{H}_k}^2 = \frac{1}{n} \left(\mathbb{E} k(X, X) - \mathbb{E} k(X, X') \right)$. The same estimate applies to $\hat{\beta}_m$, and Jensen's inequality together with $k(x, x) \leq \kappa^2$ gives the displayed bound. $\square$

Entropic OT.

Proposition 9.9 (Fixed-temperature Sinkhorn divergence has a parametric rate). *Let $c(x, y) = \|x - y\|^2/2$, and assume that α and β are σ^2 -subgaussian probability measures on $\mathbb{R}^d$, in the sense that $\int e^{\|x\|^2/(2d\sigma^2)} d\alpha(x) \leq 2$, $\int e^{\|y\|^2/(2d\sigma^2)} d\beta(y) \leq 2$. Set*

$$q_d := \left\lceil \frac{5d}{2} \right\rceil + 6, \quad b_d := \left\lceil \frac{5d}{4} \right\rceil + 3, \quad \Lambda_{d,\sigma}(\varepsilon) := \varepsilon \left(1 + \frac{\sigma^{q_d}}{\varepsilon^{b_d}} \right).$$

Let $\hat{\alpha}_n$ and $\hat{\beta}_m$ be independent empirical measures. Then, for every $\varepsilon > 0$,

$$\mathbb{E} \left| \bar{\mathcal{L}}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \right| \leq C_d \Lambda_{d,\sigma}(\varepsilon) \left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}} \right).$$

In particular, for fixed $\varepsilon > 0$ the empirical fluctuation is parametric, while for small ε the prefactor deteriorates only polynomially in $1/\varepsilon$, not as $e^{C/\varepsilon}$. Compactly supported measures are covered by this statement after a common translation of the two supports, with σ controlled by the diameter of their union.

Proof. The original proof of Genevay–Chizat–Bach–Cuturi–Peyré also reduces the problem to an empirical process indexed by entropic dual potentials, but it bounds these potentials on a compact domain in a way that produces an exponential factor $e^{D^2/\varepsilon}$. The refinement of Mena–Niles-Weed keeps the same reduction and replaces this step by polynomial Hölder estimates for the quadratic-cost Sinkhorn potentials.

For probability measures α_0, β_0 , write the dual objective, following (7.17), as

$$\mathcal{A}_\varepsilon^{\alpha_0, \beta_0}(u, v) := \int u d\alpha_0 + \int v d\beta_0 - \varepsilon \iint \left(\exp \left(\frac{u(x) + v(y) - c(x, y)}{\varepsilon} \right) - 1 \right) d\alpha_0(x) d\beta_0(y).$$

Following the extension argument of Mena–Niles-Weed, choose optimal potentials in the normalized class for which the Sinkhorn equations hold pointwise on $\mathbb{R}^d$. Thus, at an optimal pair (u, v) ,

$$\int \exp \left(\frac{u(x) + v(y) - c(x, y)}{\varepsilon} \right) d\beta_0(y) = 1 \quad \text{for all } x.$$

Equivalently, the potentials are fixed points of the soft c -transform, but only the dual normalization is needed below. Let (u, v) and (u_n, v_n) be optimal dual potentials for (α_0, β_0) and (α_n, β_0) . By optimality,

$$\mathcal{A}_\varepsilon^{\alpha_0, \beta_0}(u_n, v_n) \leq \mathcal{A}_\varepsilon^{\alpha_0, \beta_0}(u, v), \quad \mathcal{A}_\varepsilon^{\alpha_n, \beta_0}(u, v) \leq \mathcal{A}_\varepsilon^{\alpha_n, \beta_0}(u_n, v_n).$$

Hence the difference of the two optimal values is bounded by evaluating the two objectives at either pair of potentials. For the pair (u, v) associated with (α_0, β_0) , the preceding Sinkhorn equation cancels the exponential term in the change of the first marginal: $\mathcal{A}_\varepsilon^{\alpha_0, \beta_0}(u, v) - \mathcal{A}_\varepsilon^{\alpha_n, \beta_0}(u, v) = \int u d(\alpha_0 - \alpha_n)$. The same identity holds for (u_n, v_n) , using its normalization with respect to β_0 .

$$|\mathcal{L}_c^\varepsilon(\alpha_n, \beta_0) - \mathcal{L}_c^\varepsilon(\alpha_0, \beta_0)| \leq 2 \sup_{w \in \mathcal{F}_{\varepsilon, \sigma}} \left| \int w d(\alpha_n - \alpha_0) \right|,$$

where $\mathcal{F}_{\varepsilon, \sigma}$ denotes the class of first-marginal entropic potentials arising from pairs of σ^2 -subgaussian marginals.

For $\varepsilon = 1$, Mena–Niles-Weed prove that the normalized quadratic-cost potentials belong to a polynomial-growth Hölder class. More precisely, after subtracting the quadratic part $x \mapsto \|x\|^2/2$, these potentials have derivatives up to order $s_0 = \lceil d/2 \rceil + 1$ whose local Hölder norms are bounded by polynomial functions of the subgaussian proxy. Denote by $\mathcal{F}_{\sigma, L}^s$ such a class, where L is a random envelope controlled by the empirical subgaussian moment. Its empirical L_2 -covering numbers satisfy a bound of the form $\log N(\delta, \mathcal{F}_{\sigma, L}^s, L_2(\alpha_n)) \leq C_d L^{d/(2s)} \delta^{-d/s} (1 + \sigma^{2d})$. This holds for $s > s_0 - 1$, with $\mathbb{E}L \leq C$ after the normalization of the subgaussian moment. Since $d/(2s_0) < 1$, Dudley's entropy integral is finite, and the standard Rademacher symmetrization and chaining bound yields $\mathbb{E} \sup_{w \in \mathcal{F}_{1, \sigma}} \left| \int w d(\hat{\alpha}_n - \alpha) \right| \leq C_d (1 + \sigma^{q_d}) n^{-1/2}$. This is exactly the step which avoids the exponential $e^{D^2/\varepsilon}$ factor: one controls the regularity class of the dual potentials polynomially instead of bounding their sup-norm through the worst value of the Gibbs kernel.

The general ε case follows by scaling. If $T_\varepsilon(x) = x/\sqrt{\varepsilon}$, $\alpha_0^\varepsilon = (T_\varepsilon)_\# \alpha_0$, and $\beta_0^\varepsilon = (T_\varepsilon)_\# \beta_0$, then $\mathcal{L}_c^\varepsilon(\alpha_0, \beta_0) = \varepsilon \mathcal{L}_c^1(\alpha_0^\varepsilon, \beta_0^\varepsilon)$. The push-forward of a σ^2 -subgaussian law by T_ε is (σ^2/ε) -subgaussian, so the standard-deviation proxy σ is replaced by $\sigma/\sqrt{\varepsilon}$.

$$\varepsilon C_d \left(1 + \left(\frac{\sigma}{\sqrt{\varepsilon}} \right)^{q_d} \right) n^{-1/2} \leq C_d \varepsilon \left(1 + \frac{\sigma^{q_d}}{\varepsilon^{b_d}} \right) n^{-1/2},$$

where $b_d = \lceil q_d/2 \rceil = \lceil 5d/4 \rceil + 3$ is the rounded exponent used in. Thus $\mathbb{E} |\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \beta) - \mathcal{L}_c^\varepsilon(\alpha, \beta)| \leq C_d \Lambda_{d,\sigma}(\varepsilon) n^{-1/2}$. Changing both marginals is handled by the triangle inequality,

$$\left| \mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{L}_c^\varepsilon(\alpha, \beta) \right| \leq \left| \mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{L}_c^\varepsilon(\alpha, \hat{\beta}_m) \right| + \left| \mathcal{L}_c^\varepsilon(\alpha, \hat{\beta}_m) - \mathcal{L}_c^\varepsilon(\alpha, \beta) \right|.$$

Empirical measures drawn from a subgaussian law have such a random proxy, with moments controlled by the population proxy. Integrating over this proxy gives the Mena–Niles-Weed two-sample extension. The argument does not require the two empirical marginals to have the same sample size, so the two contributions scale as $n^{-1/2}$ and $m^{-1/2}$.

Finally, use the definition $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) = \mathcal{L}_c^\varepsilon(\alpha, \beta) - \frac{1}{2} \mathcal{L}_c^\varepsilon(\alpha, \alpha) - \frac{1}{2} \mathcal{L}_c^\varepsilon(\beta, \beta)$. The cross term is controlled by the two-marginal estimate. For the self term, use the telescoping inequality

$$|\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\alpha}_n) - \mathcal{L}_c^\varepsilon(\alpha, \alpha)| \leq |\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\alpha}_n) - \mathcal{L}_c^\varepsilon(\alpha, \hat{\alpha}_n)| + |\mathcal{L}_c^\varepsilon(\alpha, \hat{\alpha}_n) - \mathcal{L}_c^\varepsilon(\alpha, \alpha)|.$$

Each term is a one-marginal perturbation in the same pathwise sense: the other marginal may itself be empirical, but it is part of the common random subgaussian class. Thus the self term $\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\alpha}_n) - \mathcal{L}_c^\varepsilon(\alpha, \alpha)$ is bounded at the same order as the cross term involving α , and similarly for β . Summing these three contributions gives the displayed estimate, up to changing C_d . $\square$

Sample complexity of estimating OT maps.

$$\begin{aligned}
u_{n,m}^\varepsilon(x) &:= (g_{n,m}^\varepsilon)^{\bar{c},\varepsilon}(x) = -\varepsilon \log \sum_j b_j \exp\left(\frac{g_{n,m,j}^\varepsilon - c_2(x, Y_j)}{\varepsilon}\right), & T_{n,m}^\varepsilon(x) &:= x - \nabla u_{n,m}^\varepsilon(x). \\
\varphi_{n,m}^\varepsilon(x) &:= \frac{1}{2}\|x\|^2 - u_{n,m}^\varepsilon(x) = \varepsilon \log \sum_j b_j \exp\left(\frac{\langle x, Y_j \rangle + g_{n,m,j}^\varepsilon - \frac{1}{2}\|Y_j\|^2}{\varepsilon}\right), & T_{n,m}^\varepsilon &= \nabla \varphi_{n,m}^\varepsilon. \\
T_{n,m}^\varepsilon(x) &= \sum_j \omega_j^\varepsilon(x) Y_j, & \omega_j^\varepsilon(x) &= \frac{b_j \exp((g_{n,m,j}^\varepsilon - c_2(x, Y_j))/\varepsilon)}{\sum_k b_k \exp((g_{n,m,k}^\varepsilon - c_2(x, Y_k))/\varepsilon)}. \tag{9.2}
\end{aligned}$$

$\omega_j^\varepsilon(X_i) = \frac{P_{ij}^\varepsilon}{a_i}$, $T_{n,m}^\varepsilon(X_i) = \frac{1}{a_i} \sum_j P_{ij}^\varepsilon Y_j$. Thus the extrapolated soft-transform map coincides on the empirical support with the barycentric projection (11.43) of the optimal entropic plan. At $\varepsilon = 0$, the hard c -transform is generally nondifferentiable: away from Laguerre boundaries its gradient selects one target site, whereas on a boundary its subdifferential contains several sites.

Proposition 9.10 (Consistency of empirical barycentric maps). *Let $\Omega \subset \mathbb{R}^d$ be compact, let $\alpha, \beta \in \mathcal{P}_2(\Omega)$, and assume that α is absolutely continuous. Let φ be a Brenier potential from α to β for the cost $c_2(x, y) = \|x - y\|^2/2$, and choose a bounded Borel selection $T(x) \in \partial\varphi(x)$ on Ω . At every differentiability point of φ , this selection necessarily equals $\nabla\varphi(x)$, so it is a representative of the unique Brenier map. Let $\hat{\alpha}_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$, $\hat{\beta}_m = \frac{1}{m} \sum_{j=1}^m \delta_{Y_j}$ satisfy $\hat{\alpha}_n \rightarrow \alpha$, $\hat{\beta}_m \rightarrow \beta$ in $\mathcal{W}_2$. Let $\varepsilon_{n,m} \geq 0$ satisfy $\varepsilon_{n,m} \rightarrow 0$ and $\varepsilon_{n,m} \log(\min\{n, m\}) \rightarrow 0$. For each n, m , let $P_{n,m}^\varepsilon$ be an optimal coupling between $\hat{\alpha}_n$ and $\hat{\beta}_m$, with $\varepsilon = \varepsilon_{n,m}$: unregularized if $\varepsilon = 0$, entropic if $\varepsilon > 0$. Its barycentric projection, defined on the support of $\hat{\alpha}_n$ by $\bar{T}_{n,m}^\varepsilon(X_i) := n \sum_j P_{ij}^\varepsilon Y_j$, satisfies $\int \|\bar{T}_{n,m}^\varepsilon(x) - T(x)\|^2 d\hat{\alpha}_n(x) = \frac{1}{n} \sum_i \|\bar{T}_{n,m}^\varepsilon(X_i) - T(X_i)\|^2 \rightarrow 0$ as $n, m \rightarrow +\infty$. If β is also absolutely continuous, the analogous backward barycentric projections converge in $L^2(\hat{\beta}_m)$ to the Brenier map $S: \beta \rightarrow \alpha$.*

Proof. The assumptions imply $\mathcal{L}_{c_2}(\hat{\alpha}_n, \hat{\beta}_m) \rightarrow \mathcal{L}_{c_2}(\alpha, \beta)$. If $\varepsilon = 0$, $P_{n,m}^\varepsilon$ is exactly optimal. If $\varepsilon > 0$, compare its entropic objective with an unregularized optimal empirical plan $Q_{n,m}$. Regarding $Q_{n,m}$ as the joint law of two uniform discrete indices, its relative entropy with respect to their product law is their mutual information. Hence $\text{KL}(Q_{n,m} | \hat{\alpha}_n \otimes \hat{\beta}_m) \leq \min\{\log n, \log m\}$. It follows that $\int c_2 dP_{n,m}^\varepsilon \leq \mathcal{L}_{c_2}(\hat{\alpha}_n, \hat{\beta}_m) + \varepsilon \log(\min\{n, m\})$. Hence the transport costs of $P_{n,m}^\varepsilon$ converge to the population optimum. By compactness, every weak limit point of the plans $P_{n,m}^\varepsilon$ has marginals α, β and optimal quadratic cost. Brenier uniqueness therefore forces the whole sequence to converge weakly to $(\text{Id}, T)_\# \alpha$.

Disintegrate $P_{n,m}^\varepsilon$ with respect to its first marginal and denote its barycentric projection by $\bar{T}_{n,m}^\varepsilon$. Jensen's inequality gives $\int \|\bar{T}_{n,m}^\varepsilon(x) - T(x)\|^2 d\hat{\alpha}_n(x) \leq \iint \|y - T(x)\|^2 dP_{n,m}^\varepsilon(x, y)$. At every differentiability point of φ , the subdifferential is the singleton $\{\nabla\varphi(x)\}$. Since the target is compact, the relevant subgradients are bounded, and the closed-graph property of $\partial\varphi$ implies continuity of the selected gradient at such a point. Convex functions are differentiable Lebesgue-almost everywhere; because α is absolutely continuous, the bounded function $(x, y) \mapsto \|y - T(x)\|^2$ is continuous $(\text{Id}, T)_\# \alpha$ -almost everywhere. The portmanteau theorem for bounded functions whose discontinuity set has zero limiting mass gives $\iint \|y - T(x)\|^2 dP_{n,m}^\varepsilon(x, y) \rightarrow \int \|T(x) - T(x)\|^2 d\alpha(x) = 0$. The proof for the backward map is identical when the optimal plan is also induced by the Brenier map from β to α . $\square$

$$\mathbb{E} \|\hat{T}_n - T_0\|_{L^2(\alpha)}^2 \lesssim \left(n^{-\frac{2s}{2(s-1)+d}} \vee n^{-1} \right) \quad \text{up to logarithmic factors.}$$

$$\mathbb{E} \|T_{\varepsilon, (n, n)} - T_0\|_{L^2(\alpha)}^2 \lesssim \varepsilon^{-d/2} \log(n) n^{-1/2} + \varepsilon^{(s+1)/2} + \varepsilon^2 \mathcal{I}_0(\alpha, \beta),$$

where $\mathcal{I}_0(\alpha, \beta)$ is the Fisher-information quantity appearing in Proposition 7.12. The first term is the available sampling bound for the entropic map, whereas the last two terms are regularization-bias terms that vanish only as $\varepsilon \rightarrow 0$.

$$\mathbb{E} \|T_{\varepsilon, (n, n)} - T_0\|_{L^2(\alpha)}^2 \lesssim (1 + \mathcal{I}_0(\alpha, \beta)) n^{-\frac{s+1}{2(d+s+1)}} \log n.$$

Sliced Wasserstein. For a direction $\theta \in \mathbb{S}^{d-1}$, let $P_\theta(x) = \langle \theta, x \rangle$ be the one-dimensional projection and let σ denote the normalized surface measure on the sphere. For $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$, the sliced Wasserstein distance is

$$\text{SW}_p(\alpha, \beta)^p := \int_{\mathbb{S}^{d-1}} \mathcal{W}_p((P_\theta)_\# \alpha, (P_\theta)_\# \beta)^p d\sigma(\theta).$$

Theorem 9.11 (Dimension-free empirical rate for sliced Wasserstein). *Let $p \geq 1$, let $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$, and let $\hat{\alpha}_n, \hat{\beta}_m$ be independent empirical measures. Assume that all one-dimensional projections satisfy the uniform empirical bounds*

$$\sup_{\theta \in \mathbb{S}^{d-1}} \mathbb{E} \mathcal{W}_p((P_\theta)_\# \hat{\alpha}_n, (P_\theta)_\# \alpha)^p \leq \frac{A_\alpha}{n^{p/2}}, \quad \sup_{\theta \in \mathbb{S}^{d-1}} \mathbb{E} \mathcal{W}_p((P_\theta)_\# \hat{\beta}_m, (P_\theta)_\# \beta)^p \leq \frac{A_\beta}{m^{p/2}}.$$

Then $\mathbb{E} |\text{SW}_p(\hat{\alpha}_n, \hat{\beta}_m) - \text{SW}_p(\alpha, \beta)| \leq \frac{A_\alpha^{1/p}}{\sqrt{n}} + \frac{A_\beta^{1/p}}{\sqrt{m}}$. In particular, for $p = 1$ and supports contained in a Euclidean ball of radius R , the assumption holds with a constant depending only on R . For $p = 2$, the same $n^{-1/2}$ rate for SW_2 holds whenever the projected laws satisfy a uniform one-dimensional quantile-process bound. A sufficient condition is that the projected laws have positive densities $h_{\alpha, \theta}, h_{\beta, \theta}$, with quantile functions $Q_{\alpha, \theta}, Q_{\beta, \theta}$, and obey

$$\sup_{\theta \in \mathbb{S}^{d-1}} \int_0^1 \frac{u(1-u)}{h_{\alpha, \theta}(Q_{\alpha, \theta}(u))^2} + \frac{u(1-u)}{h_{\beta, \theta}(Q_{\beta, \theta}(u))^2} du < +\infty.$$

This is the uniform projected version of the one-dimensional J_2 condition of Bobkov and Ledoux.

Proof. The triangle inequality for $\mathcal{SW}_p$ gives $\left| \mathcal{SW}_p(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{SW}_p(\alpha, \beta) \right| \leq \mathcal{SW}_p(\hat{\alpha}_n, \alpha) + \mathcal{SW}_p(\hat{\beta}_m, \beta)$. By the definition above, Jensen's inequality and the assumed one-dimensional bounds,

$$\mathbb{E} \mathcal{SW}_p(\hat{\alpha}_n, \alpha) \leq \left(\int_{\mathbb{S}^{d-1}} \mathbb{E} \mathcal{W}_p((P_\theta)_\# \hat{\alpha}_n, (P_\theta)_\# \alpha)^p d\sigma(\theta) \right)^{1/p} \leq \frac{A_\alpha^{1/p}}{\sqrt{n}},$$

and the same argument applies to β . For $p = 1$, compact support gives the displayed one-dimensional empirical bound by the Dvoretzky–Kiefer–Wolfowitz inequality and the identity $\mathcal{W}_1(\alpha', \beta') = \int_{\mathbb{R}} |F_{\alpha'} - F_{\beta'}| ds$. For $p = 2$, the displayed J_2 condition gives $\mathbb{E} \mathcal{W}_2^2(\hat{\gamma}_n, \gamma) = O(n^{-1})$, uniformly over the projected laws. $\square$

9.3 Bias and Variance of OT

$$B_{n,m}(\mathcal{V}) := \mathbb{E} \mathcal{V}(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{V}(\alpha, \beta) \quad Z_{n,m}(\mathcal{V}) := \mathcal{V}(\hat{\alpha}_n, \hat{\beta}_m) - \mathbb{E} \mathcal{V}(\hat{\alpha}_n, \hat{\beta}_m),$$

Literature map. The shape of the limit depends sharply on the analytic regularity of the transport value. For exact OT on a finite or countable space, the map from empirical weights to optimal cost is convex and piecewise affine.

Bias versus centered fluctuation.

$$\mathcal{V}(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{V}(\alpha, \beta) = \underbrace{\mathbb{E} \mathcal{V}(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{V}(\alpha, \beta)}_{\text{bias}} + \underbrace{\mathcal{V}(\hat{\alpha}_n, \hat{\beta}_m) - \mathbb{E} \mathcal{V}(\hat{\alpha}_n, \hat{\beta}_m)}_{\text{centered fluctuation}}$$

Proposition 9.12 (Finite-space bias and CLT for exact OT). *Let $\mathbf{a} \in \Sigma_n$ and $\mathbf{b} \in \Sigma_m$ have positive entries, and let $\mathbf{C} \in \mathbb{R}^{n \times m}$. Recall the discrete OT value $L_C(\mathbf{a}, \mathbf{b}) := \min_{\mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle \mathbf{C}, \mathbf{P} \rangle$. Let $\hat{\mathbf{a}}_N$ be the empirical histogram of N independent samples from $\mathbf{a}$, while $\mathbf{b}$ is fixed. Denote by $\mathcal{D}^*(\mathbf{a}, \mathbf{b}) := \operatorname{argmax}_{f_i + g_j \leq C_{ij}} \langle \mathbf{f}, \mathbf{a} \rangle + \langle \mathbf{g}, \mathbf{b} \rangle$ the set of optimal dual vectors, modulo the gauge $(\mathbf{f}, \mathbf{g}) \mapsto (\mathbf{f} + \lambda \mathbf{1}, \mathbf{g} - \lambda \mathbf{1})$. If $G_{\mathbf{a}}$ is a centered Gaussian vector with covariance $\mathbb{E} G_{\mathbf{a}} G_{\mathbf{a}}^\top = \operatorname{diag}(\mathbf{a}) - \mathbf{a} \mathbf{a}^\top$, then*

$$\sqrt{N} (L_C(\hat{\mathbf{a}}_N, \mathbf{b}) - L_C(\mathbf{a}, \mathbf{b})) \implies \sup_{(\mathbf{f}, \mathbf{g}) \in \mathcal{D}^*(\mathbf{a}, \mathbf{b})} \langle \mathbf{f}, G_{\mathbf{a}} \rangle.$$

Moreover, $\sqrt{N} (\mathbb{E} L_C(\hat{\mathbf{a}}_N, \mathbf{b}) - L_C(\mathbf{a}, \mathbf{b})) \rightarrow \mathbb{E} \sup_{(\mathbf{f}, \mathbf{g}) \in \mathcal{D}^*(\mathbf{a}, \mathbf{b})} \langle \mathbf{f}, G_{\mathbf{a}} \rangle$, and the rescaled variance converges to the variance of the same limit. If the source dual potential $\mathbf{f}^*$ is unique up to constants, the limit is Gaussian with variance

$$\sigma_{\text{OT}}^2 = \sum_i a_i \left((\mathbf{f}^*)_i - \sum_k a_k (\mathbf{f}^*)_k \right)^2.$$

In this differentiable case, the first-order bias vanishes: $\mathbb{E} L_C(\hat{\mathbf{a}}_N, \mathbf{b}) - L_C(\mathbf{a}, \mathbf{b}) = o(N^{-1/2})$.

Proof. The multinomial central limit theorem gives $\sqrt{N}(\hat{\mathbf{a}}_N - \mathbf{a}) \implies G_{\mathbf{a}}$, $G_{\mathbf{a}} \in \{\xi : \langle \mathbf{1}, \xi \rangle = 0\}$. The dual formulation writes the map $\mathbf{a}' \mapsto L_C(\mathbf{a}', \mathbf{b})$ as the supremum, over the feasible polyhedron $f_i + g_j \leq C_{ij}$, of the affine functions $\mathbf{a}' \mapsto \langle \mathbf{f}, \mathbf{a}' \rangle + \langle \mathbf{g}, \mathbf{b} \rangle$. Its directional derivative at $\mathbf{a}$, in any tangent direction $\mathbf{h}$ with $\langle \mathbf{1}, \mathbf{h} \rangle = 0$, is therefore $D_{\mathbf{a}} L_C(\mathbf{a}, \mathbf{b})[\mathbf{h}] = \sup_{(\mathbf{f}, \mathbf{g}) \in \mathcal{D}^*(\mathbf{a}, \mathbf{b})} \langle \mathbf{f}, \mathbf{h} \rangle$, which is the finite-dimensional form of Danskin's theorem already used in Proposition 11.21. The directional delta method gives the displayed distributional limit. After fixing a gauge, the relevant dual face is bounded, so the support functions above grow at most linearly in the multinomial fluctuation. Uniform integrability then yields convergence of the first two moments, hence the bias and variance statements. If $\mathbf{f}^*$ is unique modulo constants, the support function reduces to the linear form $\langle \mathbf{f}^*, G_{\mathbf{a}} \rangle$. Its expectation is zero, and the covariance $\operatorname{diag}(\mathbf{a}) - \mathbf{a} \mathbf{a}^\top$ gives exactly the displayed centered second moment of $\mathbf{f}^*$. $\square$

$$r_{N,M} (L_C(\hat{\mathbf{a}}_N, \hat{\mathbf{b}}_M) - L_C(\mathbf{a}, \mathbf{b})) \implies \sup_{(\mathbf{f}, \mathbf{g}) \in \mathcal{D}^*(\mathbf{a}, \mathbf{b})} \left(\sqrt{1 - \lambda} \langle \mathbf{f}, G_{\mathbf{a}} \rangle + \sqrt{\lambda} \langle \mathbf{g}, G_{\mathbf{b}} \rangle \right),$$

$$\frac{1}{N} \sum_i a_i (f_i^* - \bar{f}^*)^2 + \frac{1}{M} \sum_j b_j (g_j^* - \bar{g}^*)^2, \quad \bar{f}^* = \sum_i a_i f_i^*, \quad \bar{g}^* = \sum_j b_j g_j^*.$$

Proposition 9.13 (Finite-space bias and CLT for entropic OT). *Fix $\varepsilon > 0$, positive histograms $\mathbf{a}, \mathbf{b}$, and a finite cost matrix $\mathbf{C}$. Consider the finite-dimensional KL-normalized entropic value $\mathcal{V}_{C, \varepsilon}(\mathbf{a}, \mathbf{b}) := \min_{\mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle \mathbf{C}, \mathbf{P} \rangle + \varepsilon \operatorname{KL}(\mathbf{P} | \mathbf{a} \otimes \mathbf{b})$. This is the discrete version of the value $\mathcal{V}_{C, \varepsilon}$ used in Proposition 11.22. Let $(\mathbf{f}_\varepsilon, \mathbf{g}_\varepsilon)$ be normalized entropic dual potentials for $\mathcal{V}_{C, \varepsilon}(\mathbf{a}, \mathbf{b})$. If $\hat{\mathbf{a}}_N$ is the empirical histogram of N samples from $\mathbf{a}$, then*

$$\sqrt{N} (\mathcal{V}_{C, \varepsilon}(\hat{\mathbf{a}}_N, \mathbf{b}) - \mathcal{V}_{C, \varepsilon}(\mathbf{a}, \mathbf{b})) \implies \mathcal{N}(0, \sigma_{\varepsilon, \mathbf{a}}^2),$$

and

$$\mathbb{E} \mathcal{V}_{C, \varepsilon}(\hat{\mathbf{a}}_N, \mathbf{b}) - \mathcal{V}_{C, \varepsilon}(\mathbf{a}, \mathbf{b}) = O(N^{-1}), \quad N \operatorname{Var} (\mathcal{V}_{C, \varepsilon}(\hat{\mathbf{a}}_N, \mathbf{b})) \rightarrow \sigma_{\varepsilon, \mathbf{a}}^2,$$

where

$$\sigma_{\varepsilon, \mathbf{a}}^2 = \sum_i a_i \left((\mathbf{f}_\varepsilon)_i - \sum_k a_k (\mathbf{f}_\varepsilon)_k \right)^2.$$

For two independent empirical histograms, the first-order asymptotic variance is

$$\frac{\sigma_{\varepsilon, \mathbf{a}}^2}{N} + \frac{\sigma_{\varepsilon, \mathbf{b}}^2}{M}, \quad \sigma_{\varepsilon, \mathbf{b}}^2 = \sum_j b_j \left((\mathbf{g}_\varepsilon)_j - \sum_\ell b_\ell (\mathbf{g}_\varepsilon)_\ell \right)^2.$$

Proof. For positive histograms and fixed $\varepsilon > 0$, the entropic problem has a unique optimizer, and the KL-normalized value is smooth on a neighborhood of (a, b) inside the relative interiors of the simplices. Proposition 11.22 gives $D_a \mathcal{V}_{C, \varepsilon}(a, b)[h] = \langle f_\varepsilon, h \rangle$. Applying the ordinary delta method to the multinomial CLT $\sqrt{N}(\hat{a}_N - a) \Rightarrow G_a$ gives the one-sample limit $\langle f_\varepsilon, G_a \rangle$, whose variance is the displayed covariance formula. To justify the bias expansion, fix a relative-interior neighborhood U of a on which the Hessian is bounded. A multinomial concentration bound gives $\mathbb{P}(\hat{a}_N \notin U) \leq C e^{-cN}$. On U , Taylor's formula has a centered linear term and a quadratic remainder bounded by $C \|\hat{a}_N - a\|^2$, whose expectation is $O(N^{-1})$. On the exponentially unlikely complement, the entropic value remains bounded by a constant depending only on C and ε ; the natural definition on a boundary face simply discards zero-weight atoms. This proves the $O(N^{-1})$ bias. The same expansion, together with bounded moments of the multinomial fluctuation, gives convergence of the rescaled variance. The two-sample formula follows by applying the argument independently to the two marginals. $\square$

For the entropy-only convention L_C^ε used in Section 7.1, one has, for positive histograms, $L_C^\varepsilon(a, b) = \mathcal{V}_{C, \varepsilon}(a, b) - \varepsilon H(a) - \varepsilon H(b)$. Thus the marginal entropy derivatives described after Proposition 11.22 must be added to the potentials: the derivative with respect to the source marginal is represented on the simplex tangent space by $f_\varepsilon + \varepsilon \log a$, up to an irrelevant additive constant.

Three error terms when entropy estimates exact OT. If the target is the exact cost $\mathcal{L}_c(\alpha, \beta)$ but the statistic uses the entropic value $\mathcal{L}_c^\varepsilon$, three errors should be kept separate:

$$\begin{aligned} \mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{L}_c(\alpha, \beta) &= \underbrace{\mathcal{L}_c^\varepsilon(\alpha, \beta) - \mathcal{L}_c(\alpha, \beta)}_{\text{regularization bias}} + \underbrace{\mathbb{E} \mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{L}_c^\varepsilon(\alpha, \beta)}_{\text{statistical bias}} \\ &\quad + \underbrace{\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \mathbb{E} \mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m)}_{\text{centered fluctuation}}. \end{aligned}$$

The same decomposition applies to the debiased value $\tilde{\mathcal{L}}_c^\varepsilon$ by replacing $\mathcal{L}_c^\varepsilon$ everywhere by $\tilde{\mathcal{L}}_c^\varepsilon$. Although algebraic, this decomposition is statistically important and is usually read after choosing $\varepsilon = \varepsilon_n$.

What changes in continuous spaces. The finite-dimensional formulas above are not merely toy models; they are the cleanest template for the general mechanism. Whenever an OT value is Hadamard differentiable and admits a unique sufficiently regular dual potential f^* , the empirical-process CLT and the delta method give a first-order Gaussian limit with one-sample variance $\text{Var}_\alpha(f^*(X))$.

9.4 Sketching Sinkhorn in Linear Time

PSD kernels and ordinary feature sketches. The RKHS/MMD section, Section 6.2, already used positive semidefinite kernels to define Hilbertian discrepancies between measures. Recall that a symmetric kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is positive semidefinite if, for every finite family $(x_i)_{i=1}^n$, the Gram matrix $(k(x_i, x_j))_{ij}$ is positive semidefinite.

$$k(x, y) = \int_Z \langle \varphi(x, z), \varphi(y, z) \rangle_{\mathbb{R}^p} d\xi(z), \quad \varphi : \mathcal{X} \times Z \rightarrow \mathbb{R}^p, \quad \tilde{k}_r(x, y) = \frac{1}{r} \sum_{\ell=1}^r \langle \varphi(x, z_\ell), \varphi(y, z_\ell) \rangle_{\mathbb{R}^p}.$$

Application to Sinkhorn kernels. $\alpha = \sum_{i=1}^n a_i \delta_{x_i}$, $\beta = \sum_{j=1}^m b_j \delta_{y_j}$, $K_{ij} = k_\varepsilon(x_i, y_j) := e^{-c(x_i, y_j)/\varepsilon}$. $u = \frac{a}{Kv}$, $v = \frac{b}{K^\top u}$, $(\Phi_X)_{i, (\ell, q)} = r^{-1/2} \varphi_q(x_i, z_\ell)$, $(\Phi_Y)_{j, (\ell, q)} = r^{-1/2} \varphi_q(y_j, z_\ell)$. $\tilde{K}_{ij} = (\Phi_X \Phi_Y^\top)_{ij} = \frac{1}{r} \sum_{\ell=1}^r \langle \varphi(x_i, z_\ell), \varphi(y_j, z_\ell) \rangle_{\mathbb{R}^p} \simeq K_{ij}$. $\tilde{K}v = \Phi_X(\Phi_Y^\top v)$, $\tilde{K}^\top u = \Phi_Y(\Phi_X^\top u)$.

Proposition 9.14 (Finite positivity and logarithmic accuracy). *Let $\varepsilon > 0$ and $K_{ij} = \mathbb{E} h_{ij}(Z) > 0$ for $1 \leq i \leq n$, $1 \leq j \leq m$, where Z has law ξ and $|h_{ij}(Z)| \leq M$ almost surely. Set $c_{ij} = -\varepsilon \log K_{ij}$, and let $\tilde{K}_{ij} = \frac{1}{r} \sum_{\ell=1}^r h_{ij}(Z_\ell)$, $Z_1, \dots, Z_r \sim \xi$. Then, for every $\eta > 0$, $\mathbb{P}(\max_{i,j} |\tilde{K}_{ij} - K_{ij}| > \eta) \leq 2nm \exp(-\frac{r\eta^2}{2M^2})$. In particular, if $\kappa := \min_{i,j} K_{ij} > 0$, $\eta < \kappa$, and the event above does not occur, then $\tilde{K}$ is entrywise positive. If moreover $\eta \leq \kappa/2$, then the sketched cost $\tilde{c}_{ij} = -\varepsilon \log \tilde{K}_{ij}$ satisfies $\max_{i,j} |\tilde{c}_{ij} - c_{ij}| \leq \frac{2\varepsilon\eta}{\kappa}$. If, more sharply, the relative variables $\zeta_{ij}(Z) := h_{ij}(Z)/K_{ij}$ satisfy $|\zeta_{ij}(Z)| \leq \psi$ almost surely, then for every $0 < \delta \leq 1/2$, $\mathbb{P}(\max_{i,j} |\frac{\tilde{K}_{ij}}{K_{ij}} - 1| > \delta) \leq 2nm \exp(-\frac{r\delta^2}{2\psi^2})$. On the complementary event, $\tilde{K}_{ij} > 0$ and $\max_{i,j} |\tilde{c}_{ij} - c_{ij}| \leq 2\varepsilon\delta$.*

Proof. For a fixed pair (i, j) , Hoeffding's inequality applied to bounded variables in $[-M, M]$ gives $\mathbb{P}(|\tilde{K}_{ij} - K_{ij}| > \eta) \leq 2 \exp(-\frac{r\eta^2}{2M^2})$. A union bound over the nm entries gives the first claim. If $\max_{i,j} |\tilde{K}_{ij} - K_{ij}| \leq \eta < \kappa$, then $\tilde{K}_{ij} \geq \kappa - \eta > 0$.

Writing $\tilde{K}_{ij} = K_{ij}(1 + s_{ij})$, one has $|s_{ij}| \leq \eta/\kappa \leq 1/2$, hence $|\log(1 + s_{ij})| \leq 2|s_{ij}|$, which proves the absolute logarithmic bound. Applying the same Hoeffding argument to $\zeta_{ij}(Z)$, whose expectation is 1, gives the relative estimate. On the corresponding event, $\tilde{K}_{ij} = K_{ij}(1 + s_{ij})$ with $|s_{ij}| \leq \delta \leq 1/2$, so positivity and the bound $\varepsilon |\log(1 + s_{ij})| \leq 2\varepsilon\delta$ follow. $\square$

$$r_*(X, \eta, \varepsilon_0) \lesssim (\eta R^2 + \log(n/\varepsilon_0))^d$$

Positive features and total positivity.

$$\text{DNN}_n = \{A \in \mathbb{R}^{n \times n} ; A \succeq 0, A_{ij} \geq 0\}, \quad \text{CP}_n = \{BB^\top ; B \in \mathbb{R}_+^{n \times q} \text{ for some } q \geq 1\}.$$

$$k(x, y) = \int_Z \langle \varphi(x, z), \varphi(y, z) \rangle_{\mathbb{R}^p} d\xi(z), \quad \varphi(x, z) \in \mathbb{R}_+^p,$$

Proposition 9.15 (Basic closure of totally positive kernels). *Nonnegative sums and pointwise products of totally positive kernels are totally positive. The same holds for measurable nonnegative mixtures whenever the resulting kernel is finite. In particular, positive autocorrelations $k(x, y) = \int g(u - x)g(u - y) du$, $g \geq 0$, and all nonnegative mixtures of such kernels are totally positive.*

Proof. For sums, concatenate the nonnegative features and rescale them by the square roots of the nonnegative coefficients. For products, tensor the nonnegative features:

$$\langle \varphi_1(x, z_1), \varphi_1(y, z_1) \rangle \langle \varphi_2(x, z_2), \varphi_2(y, z_2) \rangle = \langle \varphi_1(x, z_1) \otimes \varphi_2(x, z_2), \varphi_1(y, z_1) \otimes \varphi_2(y, z_2) \rangle.$$

The tensor feature remains entrywise nonnegative. A measurable mixture is obtained by adjoining the mixing parameter to the latent variable; equivalently, on each finite point set it is a limit of conic combinations in the closed completely positive cone. The autocorrelation formula is already a positive feature representation with latent variable u and scalar feature $g(u - x)$. $\square$

Proposition 9.16 (Generalized Gaussian positive features). *Let $0 < p \leq 2$, $\varepsilon > 0$, and $k_{p,\varepsilon}(x, y) = \exp\left(-\frac{\|x-y\|^p}{\varepsilon}\right)$, $x, y \in \mathbb{R}^d$. Set $a = p/2 \in (0, 1]$. Let S_a be the positive a -stable random variable normalized by $\mathbb{E}(e^{-tS_a}) = e^{-t^a}$, $t \geq 0$, and set $\Lambda = \varepsilon^{-1/a}S_a$. Denote by $\nu_{a,\varepsilon}$ the law of Λ . Then $\exp\left(-\frac{t^a}{\varepsilon}\right) = \int_0^{+\infty} e^{-\lambda t} d\nu_{a,\varepsilon}(\lambda)$, $t \geq 0$. Equivalently, for a nonnegative random variable $\Lambda \sim \nu_{a,\varepsilon}$, $k_{p,\varepsilon}(x, y) = \mathbb{E}_\Lambda \exp(-\Lambda\|x-y\|^2)$. Fix a centering point $x_0 \in \mathbb{R}^d$, let $Z \sim \mathcal{N}(0, \text{Id}_d)$, independent of Λ , and set $q_x(\Lambda) = \sqrt{2\Lambda}(x - x_0)$, $\varphi_p(x, \Lambda, Z) = \exp(\langle Z, q_x(\Lambda) \rangle - \|q_x(\Lambda)\|^2)$. Then $k_{p,\varepsilon}(x, y) = \mathbb{E}_{\Lambda, Z}(\varphi_p(x, \Lambda, Z)\varphi_p(y, \Lambda, Z))$. Thus $k_{p,\varepsilon}$ is totally positive in the complete-positive sense. In particular, if $(\Lambda_\ell, Z_\ell)_{\ell=1}^r$ are i.i.d. copies of (Λ, Z) , then $k_{p,\varepsilon}(x, y) = \sum_{\ell=1}^r \varphi_\ell(x)\varphi_\ell(y)$, $\varphi_\ell(x) = r^{-1/2}\varphi_p(x, \Lambda_\ell, Z_\ell)$, is an unbiased nonnegative feature sketch of $k_{p,\varepsilon}$. For $p = 2$, the mixing law reduces to the Dirac mass $\nu_{1,\varepsilon} = \delta_{1/\varepsilon}$, which recovers the usual Gaussian feature formula.*

Proof. The function $s \mapsto e^{-s^a}$ is completely monotone for $0 < a \leq 1$. Bernstein's theorem therefore gives the Laplace-transform representation of a positive a -stable random variable S_a . The scaling $\Lambda = \varepsilon^{-1/a}S_a$ gives $\mathbb{E}e^{-t\Lambda} = \exp\left(-\frac{t^a}{\varepsilon}\right)$, which is the displayed representation by $\nu_{a,\varepsilon}$. Applying this identity with $t = \|x-y\|^2$ gives the Gaussian-mixture formula. Conditionally on $\Lambda = \lambda$, the Gaussian moment-generating function gives

$$\begin{aligned} \mathbb{E}_Z(\varphi_p(x, \lambda, Z)\varphi_p(y, \lambda, Z)) &= \exp(-\|q_x(\lambda)\|^2 - \|q_y(\lambda)\|^2) \mathbb{E}_Z(\exp(\langle Z, q_x(\lambda) + q_y(\lambda) \rangle)) \\ &= \exp\left(-\frac{1}{2}\|q_x(\lambda) - q_y(\lambda)\|^2\right) = \exp(-\lambda\|x-y\|^2). \end{aligned}$$

Taking the expectation over Λ gives $k_{p,\varepsilon}$. Since φ_p is positive, this is a positive-feature representation; hence every finite Gram matrix is completely positive. The Monte-Carlo sketch is just the empirical average of this representation. $\square$

$$k_\lambda(x, y) = \lambda + \cos^2\left(\frac{x-y}{2}\right), \quad \lambda > 0. \quad \langle H, K \rangle = \frac{21}{4} - \frac{5\sqrt{5}}{2} < 0.$$

Positive sketches for Sinkhorn. $k_{p,\varepsilon}(x, y) = \exp\left(-\frac{\|x-y\|^p}{\varepsilon}\right)$. For $p \geq 1$, this is the usual p -Wasserstein power cost; for $0 < p < 1$, it should be read simply as a concave power transport cost. The factor $1/p$, often used in the definition of $\mathcal{W}_p^p$, only rescales ε .

$$\varphi_\ell(x) = r^{-1/2}\varphi_p(x, \Lambda_\ell, Z_\ell), \quad 1 \leq \ell \leq r.$$

$$\tilde{k}_{p,\varepsilon}(x, y) = \sum_{\ell=1}^r \varphi_\ell(x)\varphi_\ell(y) \geq 0, \quad \mathbb{E}[\tilde{k}_{p,\varepsilon}(x, y)] = k_{p,\varepsilon}(x, y).$$

$$\tilde{P} = \text{diag}(u)\Phi_X\Phi_Y^\top \text{diag}(v) = LR^\top, \quad L = \text{diag}(u)\Phi_X, \quad R = \text{diag}(v)\Phi_Y.$$

Connection with linear attention.

10 Generalized Wasserstein Distances

10.1 Unbalanced OT

Relaxed formulation. For nonnegative measures $(\alpha, \beta) \in \mathcal{M}_+(\mathcal{X}) \times \mathcal{M}_+(\mathcal{Y})$, a generic relaxed formulation is

$$\text{UW}_c(\alpha, \beta) = \inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta), \quad (10.1)$$

where ψ_1, ψ_2 are convex entropy functions. Exact conservation $(\pi_1, \pi_2) = (\alpha, \beta)$ is replaced by a cost for changing the marginals. $\text{UW}_{c,\tau}(\alpha, \beta) = \inf_{\pi \geq 0} \int c d\pi + \tau \mathcal{D}_{\bar{\psi}_1}(\pi_1|\alpha) + \tau \mathcal{D}_{\bar{\psi}_2}(\pi_2|\beta)$. These figures should be read as pictures of a single relaxed plan π : the same nonnegative measure determines both the transported coupling and its two relaxed marginals. The small- τ result below formalizes the opposite regime, where transport becomes negligible compared with local mass variation.

Proposition 10.1 (Small-transport-scale limit for marginal penalties). *Assume that α, β are finite measures on a compact metric space $\mathcal{X}$, that c is continuous, $c \geq 0$, and $c(x, y) = 0$ if and only if $x = y$. Assume also that the marginal divergences are nonnegative, weak-* lower semicontinuous, and have weak-* compact sublevel sets on $\mathcal{M}_+(\mathcal{X})$. Then*

$$\lim_{\tau \downarrow 0} \frac{1}{\tau} \text{UW}_{c,\tau}(\alpha, \beta) = \inf_{\chi \in \mathcal{M}_+(\mathcal{X})} \mathcal{D}_{\bar{\psi}_1}(\chi|\alpha) + \mathcal{D}_{\bar{\psi}_2}(\chi|\beta).$$

The right-hand side is the infimal gluing divergence obtained by matching the two measures through a common zero-transport marginal χ . In the dominated case, if $\alpha = a\lambda$, $\beta = b\lambda$, and the minimizing common marginal is absolutely continuous, $\chi = r\lambda$, this divergence decouples pointwise as

$$\int \mathfrak{m}_{\bar{\psi}_1, \bar{\psi}_2}(a(x), b(x)) d\lambda(x), \quad \mathfrak{m}_{\bar{\psi}_1, \bar{\psi}_2}(a, b) := \inf_{r \geq 0} a\bar{\psi}_1(r/a) + b\bar{\psi}_2(r/b),$$

with the usual recession conventions when $a = 0$ or $b = 0$. For superlinear entropies, and in particular for KL, finite energy forces this dominated form. Thus, when $\bar{\psi}_1 = \bar{\psi}_2$ is the KL entropy, $\inf_{\chi \in \mathcal{M}_+(\mathcal{X})} \text{KL}(\chi|\alpha) + \text{KL}(\chi|\beta) = \int (\sqrt{a} - \sqrt{b})^2 d\lambda$. Thus the KL marginal relaxation contains the squared Hellinger distance as its local mass-variation limit.

Proof. For the upper bound, restrict to diagonal plans $\pi = (\text{Id}, \text{Id})_\# \chi$, whose transport cost is zero and whose two marginals are both χ .

For the lower bound, let $\tau_n \downarrow 0$ and let π_n be almost minimizing plans with bounded scaled values $\tau_n^{-1} \text{UW}_{c,\tau_n}(\alpha, \beta)$. Since the divergences are nonnegative, $\int c d\pi_n = O(\tau_n)$, hence $\int c d\pi_n \rightarrow 0$. The bounded scaled values also put the two marginals in compact divergence sublevel sets. Since a coupling has the same total mass as each marginal, the couplings are tight on $\mathcal{X} \times \mathcal{X}$. Up to subsequences, $\pi_n \rightharpoonup \pi_0$. Lower semicontinuity of the transport cost yields $\int c d\pi_0 = 0$, so π_0 is concentrated on the diagonal. Its two marginals are therefore equal to a common measure χ . Lower semicontinuity of the marginal divergences gives $\liminf_n \frac{1}{\tau_n} \text{UW}_{c,\tau_n}(\alpha, \beta) \geq \mathcal{D}_{\bar{\psi}_1}(\chi|\alpha) + \mathcal{D}_{\bar{\psi}_2}(\chi|\beta)$, and optimizing over χ gives the lower bound.

In the dominated case, writing $\chi = r\lambda$ gives $\mathcal{D}_{\bar{\psi}_1}(\chi|\alpha) + \mathcal{D}_{\bar{\psi}_2}(\chi|\beta) = \int a\bar{\psi}_1(r/a) + b\bar{\psi}_2(r/b)d\lambda$, so the minimization over χ decouples into the scalar envelope $\mathfrak{m}_{\bar{\psi}_1, \bar{\psi}_2}$. For KL, no singular part is admissible when α and β are dominated by λ . The pointwise objective is $r \log(r/a) - r + a + r \log(r/b) - r + b$. Its optimality condition is $\log(r/a) + \log(r/b) = 0$, hence $r = \sqrt{ab}$, and the minimum is $a + b - 2\sqrt{ab} = (\sqrt{a} - \sqrt{b})^2$. $\square$

Proposition 10.2 (Dual of unbalanced optimal transport). *Under the usual Fenchel–Rockafellar qualification assumptions, one has equality between (10.1) and $\text{UW}_c(\alpha, \beta) = \sup_{f \oplus g \leq c} -\mathcal{D}_{\psi_1}^*(-f|\alpha) - \mathcal{D}_{\psi_2}^*(-g|\beta)$.*

Proof. Use the variational formula (6.9) for the dual of a divergence and introduce the marginal variables through continuous potentials: $\inf_{\pi \geq 0} \sup_{f, g} \int c d\pi + \int -f d\pi_1 + \int -g d\pi_2 - \mathcal{D}_{\psi_1}^*(-f|\alpha) - \mathcal{D}_{\psi_2}^*(-g|\beta)$. Exchanging the infimum and the supremum gives $\sup_{f, g} -\mathcal{D}_{\psi_1}^*(-f|\alpha) - \mathcal{D}_{\psi_2}^*(-g|\beta) + \inf_{\pi \geq 0} \int (c - (f \oplus g)) d\pi$. The last infimum is 0 when $f \oplus g \leq c$ and $-\infty$ otherwise, which gives the displayed dual. $\square$

Reverse and homogeneous formulations.

$$\begin{aligned} & \int c(x, y) d\pi(x, y) + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta) \\ &= \int \left(c(x, y) + \psi_1\left(\frac{d\pi_1}{d\alpha}(x)\right) \frac{d\alpha}{d\pi_1}(x) + \psi_2\left(\frac{d\pi_2}{d\beta}(y)\right) \frac{d\beta}{d\pi_2}(y) \right) d\pi(x, y). \end{aligned}$$

$$L_c(r, s) := c + r\psi_1(1/r) + s\psi_2(1/s), \quad (10.2)$$

with the usual recession convention at $r = 0$ or $s = 0$. If $\alpha = F\pi_1 + \alpha^\perp$ and $\beta = G\pi_2 + \beta^\perp$ are the Lebesgue decompositions of the reference marginals with respect to the transported marginals, then the reverse formulation reads $\text{UW}_c(\alpha, \beta) = \inf_{\pi \geq 0} \int L_{c(x, y)}(F(x), G(y)) d\pi(x, y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y})$.

$$H_c(r, s) := \inf_{\theta > 0} \theta L_c(r/\theta, s/\theta), \quad (10.3)$$

$$\text{HW}_c(\alpha, \beta) = \inf_{\pi \geq 0} \int H_{c(x, y)}(F(x), G(y)) d\pi(x, y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y}). \quad (10.4)$$

$$\begin{aligned} \text{HW}_c(\alpha, \beta) &= \inf_{\substack{\lambda \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y}) \\ u, v \geq 0}} \left\{ \int_{\mathcal{X} \times \mathcal{Y}} H_{c(x, y)}(u(x, y), v(x, y)) d\lambda(x, y) \right. \\ &\quad \left. \text{subject to } (p_1)_\#(u\lambda) = \alpha, \quad (p_2)_\#(v\lambda) = \beta \right\}. \end{aligned} \quad (10.5)$$

Proposition 10.3 (Homogenization does not change the unbalanced cost). *Assume that the entropy functions are proper, lower semicontinuous and convex, with the usual recession extensions at zero. Then $\text{HW}_c(\alpha, \beta) = \text{UW}_c(\alpha, \beta)$.*

Proof. Since $H_c(r, s) \leq L_c(r, s)$ by choosing the scale equal to one, every competitor in the reverse formulation gives a homogeneous competitor of no larger cost. Hence $\text{HW}_c \leq \text{UW}_c$.

For the converse, use the semi-coupling representation (10.5) and fix a feasible triple (λ, u, v) . For a positive measurable scale $\theta(x, y)$, set $\tilde{\pi} = \theta\lambda$. Disintegrate λ with respect to its first spatial marginal. If $\bar{u}(x)$ and $\bar{\theta}(x)$ denote the corresponding conditional averages of u and θ , then $\alpha = \bar{u}(p_1)_\# \lambda$ and $\tilde{\pi}_1 = \bar{\theta}(p_1)_\# \lambda$. Convexity of the perspective $(a, m) \mapsto a\psi_1(m/a)$ and conditional Jensen give $\mathcal{D}_{\psi_1}(\tilde{\pi}_1|\alpha) \leq \int u \psi_1(\theta/u) d\lambda$. The same argument on the second marginal yields the analogous bound with (v, ψ_2) . Therefore

$$\int c d\tilde{\pi} + \mathcal{D}_{\psi_1}(\tilde{\pi}_1|\alpha) + \mathcal{D}_{\psi_2}(\tilde{\pi}_2|\beta) \leq \int [c\theta + u\psi_1(\theta/u) + v\psi_2(\theta/v)] d\lambda.$$

The pointwise infimum of the integrand over $\theta > 0$ is precisely $H_c(u, v)$. A measurable η -minimizing choice of θ exists because this is a lower-semicontinuous normal integrand; the recession conventions cover $u = 0$ or $v = 0$. Letting $\eta \downarrow 0$ and then minimizing over (λ, u, v) gives $\text{UW}_c \leq \text{HW}_c$. $\square$

Conic lifting. Assume now that $\mathcal{X} = \mathcal{Y}$ and $\psi_1 = \psi_2 = \psi$. The last formulation lifts the problem to the cone space $\mathfrak{C}[\mathcal{X}] := (\mathcal{X} \times \mathbb{R}_+)/\sim$, where all points $(x, 0)$ are identified at the apex. For an exponent $p \geq 1$, define

$$\Delta((x, r), (y, s)) := H_{c(x, y)}(r^p, s^p)^{1/p}.$$

- $\mathcal{D}_\psi = \text{KL}$, $p = 2$, and $c(x, y) = -\log \cos^2(d(x, y) \wedge \pi/2)$ give the Hellinger–Kantorovich or Wasserstein–Fisher–Rao cone metric $\Delta((x, r), (y, s))^2 = r^2 + s^2 - 2rs \cos(d(x, y) \wedge \pi/2)$.
- $\mathcal{D}_\psi = \text{KL}$, $p = 2$, and $c(x, y) = d(x, y)^2$ give the Gaussian Hellinger formula $\Delta((x, r), (y, s))^2 = r^2 + s^2 - 2rs e^{-d(x, y)^2/2}$. It is a cone metric whenever the Gaussian kernel $k(x, y) = e^{-d(x, y)^2/2}$ is positive definite. This holds, for example, when $(\mathcal{X}, d)$ is a subset of a Hilbert space: if $k(x, y) = \langle \Phi(x), \Phi(y) \rangle$ with $\|\Phi(x)\| = 1$, then the displayed quantity is $\|r\Phi(x) - s\Phi(y)\|^2$. For a general metric space, positive definiteness of this kernel is an additional hypothesis.
- $\mathcal{D}_\psi = \text{TV}$, $p = 1$, and $c(x, y) = d(x, y)$ give the partial-transport cone cost $\Delta((x, r), (y, s)) = r + s - (r \wedge s)(2 - d(x, y))_+$.

For a finite measure η on the cone, define its weighted base projection $\mathbf{P}_p \eta \in \mathcal{M}_+(\mathcal{X})$ by

$$\int_{\mathcal{X}} \varphi(x) d(\mathbf{P}_p \eta)(x) = \int_{\mathfrak{C}[\mathcal{X}]} \varphi(x) r^p d\eta(x, r) \text{ for every bounded Borel } \varphi. \text{ This is well defined at the apex because the weight } r^p \text{ vanishes there.}$$

$$\text{CW}(\alpha, \beta) = \inf_{\gamma \in \mathcal{M}_+(\mathfrak{C}[\mathcal{X}]^2)} \left\{ \int \Delta((x, r), (y, s))^p d\gamma; \mathbf{P}_p \gamma_1 = \alpha, \quad \mathbf{P}_p \gamma_2 = \beta \right\}.$$

Theorem 10.4 (Cone formulation of unbalanced OT). *Assume that $\mathcal{X}$ is a compact metric space, that $(r, s) \mapsto H_{c(x, y)}(r, s)$ is convex for every (x, y) , and that $(x, y, r, s) \mapsto H_{c(x, y)}(r, s)$ is lower semicontinuous. Then $\text{UW}_c(\alpha, \beta) = \text{HW}_c(\alpha, \beta) = \text{CW}(\alpha, \beta)$. If, in addition, Δ is a continuous distance on $\mathfrak{C}[\mathcal{X}]$, then $\text{CW}^{1/p}$ is a distance between nonnegative finite measures.*

Proof. The equality $UW = HW$ is Proposition 10.3.

Let (λ, u, v) be feasible in (10.5). Define the cone coupling $\gamma = ((x, y) \mapsto ((x, u(x, y)^{1/p}), (y, v(x, y)^{1/p})))_{\#} \lambda$. Then $P_p \gamma_1 = \alpha$ and $P_p \gamma_2 = \beta$ by the weighted marginal constraints. Moreover $\int \Delta((x, r), (y, s))^p d\gamma = \int H_{c(x, y)}(u(x, y), v(x, y)) d\lambda(x, y)$. Taking the infimum gives $CW \leq HW$.

Conversely, let γ be feasible for CW . Choose arbitrary representatives of cone points at the apex. This does not affect the weighted constraints, and it does not change the cost because, by the recession convention, $H_{c(x, y)}(0, s)$ and $H_{c(x, y)}(r, 0)$ do not depend on the arbitrary spatial representative attached to the zero radius. Let λ be the projection of γ onto the spatial variables (x, y) , and disintegrate $\gamma = \gamma_{x, y} d\lambda(x, y)$ with respect to this projection. Set $u(x, y) = \int r^p d\gamma_{x, y}(r, s)$, $v(x, y) = \int s^p d\gamma_{x, y}(r, s)$. The cone marginal constraints imply $(p_1)_{\#}(u\lambda) = \alpha$ and $(p_2)_{\#}(v\lambda) = \beta$. Since H_c is convex in its two arguments, as a perspective of the convex reverse cost L_c , Jensen's inequality gives, for λ -a.e. (x, y) , $H_{c(x, y)}(u(x, y), v(x, y)) \leq \int H_{c(x, y)}(r^p, s^p) d\gamma_{x, y}(r, s)$. After integration and using $\Delta((x, r), (y, s))^p = H_{c(x, y)}(r^p, s^p)$, this yields $HW \leq CW$. Hence $UW = HW = CW$.

Assume now that Δ is a distance. The nontrivial point in the triangle inequality is that two cone plans sharing the same weighted base projection need not have the same ordinary cone marginal. The radial homogeneity removes this obstruction. Indeed, common radial rescaling $((x, r), (y, s)) \mapsto ((x, r/\vartheta), (y, s/\vartheta))$, $\gamma \mapsto \vartheta^p \gamma$, preserves both weighted projections and the p -action. After adding mass at the cone apex, the normalization lemma for homogeneous lifts therefore lets one rescale two almost optimal plans for (α, β) and (β, ζ) so that their ordinary middle cone marginal is the same lift $\eta_\beta = M\delta_o + (x \mapsto (x, 1))_{\#} \beta$. Here $M < +\infty$ is chosen large enough for both plans; additional mass at (o, o) changes neither the weighted projections nor the action. The ordinary gluing lemma now produces a joint cone plan (Z_0, Z_1, Z_2) . Since Δ is a metric, Minkowski's inequality gives

$$\left(\int \Delta(Z_0, Z_2)^p \right)^{1/p} \leq \left(\int \Delta(Z_0, Z_1)^p \right)^{1/p} + \left(\int \Delta(Z_1, Z_2)^p \right)^{1/p}.$$

Taking infima proves the triangle inequality. This normalization-and-gluing argument is the metric engine of the homogeneous cone construction.

For completeness, normalize each non-apex pair by taking $\vartheta = (r^p + s^p)^{1/p}$. The transformed radii then satisfy $r^p + s^p = 1$, while the transformed plan has total mass $\alpha(\mathcal{X}) + \beta(\mathcal{X})$; apex mass can again be fixed without changing the problem. Thus minimizing plans may be restricted to a weakly compact family with bounded radii. Compactness of $\mathcal{X}$ and lower semicontinuity of the cone cost give an optimal plan. If $CW(\alpha, \beta) = 0$, such a plan is concentrated on the diagonal of the cone. There $x = y$ whenever the common radius is positive and $r = s$, so its two weighted base projections coincide and $\alpha = \beta$. Nonnegativity and symmetry are immediate, completing the metric proof. $\square$

Entropic KL relaxation.

$$\begin{aligned} \inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} & \int c d\pi + \mathcal{D}_{\psi_1}(\pi_1 | \alpha) + \mathcal{D}_{\psi_2}(\pi_2 | \beta) + \varepsilon \mathcal{D}_\varphi(\pi | \alpha \otimes \beta). \\ \sup_{f, g} & -\mathcal{D}_{\psi_1}^*(-f | \alpha) - \mathcal{D}_{\psi_2}^*(-g | \beta) - \varepsilon \mathcal{D}_\varphi^*\left(\frac{f \oplus g - c}{\varepsilon} \middle| \alpha \otimes \beta\right). \end{aligned} \quad (10.6)$$

For $\mathcal{D}_\varphi = \text{KL}$, the primal-dual relation is $d\pi = e^{(f \oplus g - c)/\varepsilon} d\alpha d\beta$. If, in addition, $\mathcal{D}_{\psi_1} = \mathcal{D}_{\psi_2} = \tau \text{KL}$, the dual specializes to

$$\begin{aligned} \sup_{f, g} & -\tau \int (e^{-f/\tau} - 1) d\alpha - \tau \int (e^{-g/\tau} - 1) d\beta - \varepsilon \iint (e^{(f \oplus g - c)/\varepsilon} - 1) d\alpha d\beta, \\ f & \leftarrow \omega g^{\bar{c}, \varepsilon}, \quad g \leftarrow \omega f^{c, \varepsilon}, \quad \omega := \frac{\tau}{\tau + \varepsilon}, \end{aligned}$$

where the soft c -transforms are those of Definition 7.15. Equivalently,

$$\begin{aligned} f(x) &= -\frac{\tau\varepsilon}{\tau + \varepsilon} \log \int_{\mathcal{Y}} \exp\left(\frac{g(y) - c(x, y)}{\varepsilon}\right) d\beta(y), \\ g(y) &= -\frac{\tau\varepsilon}{\tau + \varepsilon} \log \int_{\mathcal{X}} \exp\left(\frac{f(x) - c(x, y)}{\varepsilon}\right) d\alpha(x). \\ u_i &\leftarrow \left(\frac{a_i}{(Kv)_i}\right)^\omega, \quad v_j \leftarrow \left(\frac{b_j}{(K^\top u)_j}\right)^\omega, \quad P = \text{diag}(u)K \text{diag}(v). \end{aligned}$$

Metric contraction of the damped updates. For balanced entropic OT, potentials are only defined up to the gauge $(f, g) \sim (f + \lambda, g - \lambda)$, so Hilbert's projective metric is natural. The KL marginal penalties break this invariance: under the usual finite-dual assumptions, the dual potentials are unique up to α - and β -null sets, and the natural distance is the ordinary L^∞ distance on potentials.

$d_T((u, v), (u', v')) := \max\{\|\log u - \log u'\|_\infty, \|\log v - \log v'\|_\infty\}$,

Proposition 10.5 (Linear contraction of unbalanced Sinkhorn). *Assume that the damped soft-transform map sends bounded potentials to bounded potentials; for instance, this holds when c is bounded. Let $\omega = \tau/(\tau + \varepsilon) < 1$, and define $d_\infty((f, g), (f', g')) := \max\{\|f - f'\|_\infty, \|g - g'\|_\infty\}$. The damped soft-transform map*

$$\mathcal{T}_{\text{GS}}(f, g) := (\tilde{f}, \tilde{g}), \quad \tilde{f} = \omega g^{\bar{c}, \varepsilon}, \quad \tilde{g} = \omega f^{c, \varepsilon},$$

is a contraction: $d_\infty(\mathcal{T}_{\text{GS}}(f, g), \mathcal{T}_{\text{GS}}(f', g')) \leq \omega d_\infty((f, g), (f', g'))$. (f^, g^*) and the iterates satisfy $d_\infty((f^{(k)}, g^{(k)}), (f^*, g^*)) \leq \omega^k d_\infty((f^{(0)}, g^{(0)}), (f^*, g^*))$. Equivalently, for $u^{(k)} = e^{f^{(k)}/\varepsilon}$ and $v^{(k)} = e^{g^{(k)}/\varepsilon}$, $d_T((u^{(k)}, v^{(k)}), (u^*, v^*)) \leq \omega^k d_T((u^{(0)}, v^{(0)}), (u^*, v^*))$.*

Proof. The soft c -transform is 1-Lipschitz in L^∞ . Indeed, if $\|g - g'\|_\infty \leq \delta$, then $g - \delta \leq g' \leq g + \delta$. Since $g \mapsto g^{\tilde{c}, \varepsilon}$ is order reversing and satisfies $(g + \lambda)^{\tilde{c}, \varepsilon} = g^{\tilde{c}, \varepsilon} - \lambda$, this implies $g^{\tilde{c}, \varepsilon} - \delta \leq (g')^{\tilde{c}, \varepsilon} \leq g^{\tilde{c}, \varepsilon} + \delta$. Hence $\|g^{\tilde{c}, \varepsilon} - (g')^{\tilde{c}, \varepsilon}\|_\infty \leq \|g - g'\|_\infty$, and the same argument applies to $f \mapsto f^{\tilde{c}, \varepsilon}$.

Let $(\tilde{f}, \tilde{g}) = \mathcal{T}_{\text{GS}}(f, g)$ and $(\tilde{f}', \tilde{g}') = \mathcal{T}_{\text{GS}}(f', g')$. With $D = d_\infty((f, g), (f', g'))$, the previous Lipschitz estimate gives $\|\tilde{f} - \tilde{f}'\|_\infty \leq \omega \|g - g'\|_\infty \leq \omega D$, and then $\|\tilde{g} - \tilde{g}'\|_\infty \leq \omega \|\tilde{f} - \tilde{f}'\|_\infty \leq \omega^2 D \leq \omega D$. This proves the contraction. The Banach fixed-point theorem gives existence, uniqueness and the displayed linear rate. $\square$

WFR between unnormalized Gaussians. $\alpha = r_0 \mathcal{N}(m_0, \Sigma_0)$, $\beta = r_1 \mathcal{N}(m_1, \Sigma_1)$, $r_0, r_1 > 0$, $\Sigma_0, \Sigma_1 \succ 0$, where r_i is the total mass, $m_i \in \mathbb{R}^d$ is the mean and Σ_i is the covariance. For a common marginal-relaxation strength $\tau > 0$, define

$$\text{GHK}_\tau^2(\alpha, \beta) := \inf_{\pi \in \mathcal{M}_+(\mathbb{R}^d \times \mathbb{R}^d)} \left\{ \int \|x - y\|^2 d\pi + \tau \text{KL}(\pi_1 | \alpha) + \tau \text{KL}(\pi_2 | \beta) \right\}. \quad (10.7)$$

$$\begin{aligned} \widehat{\Sigma}_i &= \text{Id} + \frac{2}{\tau} \Sigma_i, & X_\tau &= \Sigma_0 + \Sigma_1 + \frac{\tau}{2} \text{Id}, \\ J_\tau &= (\widehat{\Sigma}_0 \widehat{\Sigma}_1)^{1/2} \left[\text{Id} - \frac{2}{\tau} (\Sigma_0 \widehat{\Sigma}_0^{-1} \widehat{\Sigma}_1^{-1} \Sigma_1)^{1/2} \right], & \mathbf{a}_\tau &= \frac{\exp\left(-\frac{1}{4} \|m_0 - m_1\|_{X_\tau^{-1}}^2\right)}{\sqrt{\det J_\tau}}. \end{aligned}$$

Proposition 10.6 (Gaussian Hellinger–Kantorovich endpoint). *With the preceding notation,*

$$\text{GHK}_\tau^2(\alpha, \beta) = \tau (r_0 + r_1 - 2\sqrt{r_0 r_1} \mathbf{a}_\tau). \quad (10.8)$$

Proof. Write $\alpha = r_0 \bar{\alpha}$, $\beta = r_1 \bar{\beta}$ with normalized Gaussian $\bar{\alpha}, \bar{\beta}$. If a plan of mass $M > 0$ is written as $M\bar{\pi}$ with normalized marginals $\bar{\alpha}, \bar{\beta}$, then

$$\text{KL}(M\bar{\alpha} \mid r_0 \bar{\alpha}) = M \text{KL}(\bar{\alpha} \mid \bar{\alpha}) + M \log(M/r_0) - M + r_0.$$

Hence, with $\mathcal{A} = \mathcal{W}_2^2(\tilde{\alpha}, \tilde{\beta}) + \tau \text{KL}(\tilde{\alpha} \mid \bar{\alpha}) + \tau \text{KL}(\tilde{\beta} \mid \bar{\beta})$, minimizing in M gives $M = \sqrt{r_0 r_1} e^{-\mathcal{A}/(2\tau)}$ and value $\tau(r_0 + r_1) - 2\tau M$. Gelbrich's inequality and moment matching show that replacing each normalized marginal by the Gaussian with the same mean and covariance cannot increase $\mathcal{A}$. The mean stationarity equations give a minimum $\frac{\tau}{2} \|m_0 - m_1\|_{X_\tau^{-1}}^2$. For the covariances, set $\eta = 2/\tau$, $A_i = \Sigma_i^{-1}$, $C_i = A_i + \eta \text{Id}$, and let L be the positive solution of $LC_1 L = C_0$. With $P = [A_0 + \eta(\text{Id} - L)]^{-1}$ and $Q = LPL$, the covariance stationarity equations hold. Their traces cancel the Bures term against the trace terms in the two Gaussian KL divergences, leaving

$$\mathcal{A}_{\text{cov}} = \frac{\tau}{2} \log \frac{\det \Sigma_0 \det \Sigma_1}{\det P \det Q} = \tau \log \det J_\tau.$$

Thus $M = \sqrt{r_0 r_1} \mathbf{a}_\tau$, which gives (10.8). Quadratic KL-dual potentials obtained from the density ratios of these adjusted Gaussian marginals satisfy $\varphi(x) + \psi(y) \leq \|x - y\|^2$, with equality on the Gaussian optimal transport graph, and certify global optimality. $\square$

$$\text{UW}_{\tau, \varepsilon}^{\text{KL}}(\alpha, \beta) := \inf_{\pi \in \mathcal{M}_+(\mathbb{R}^d \times \mathbb{R}^d)} \left\{ \int \|x - y\|^2 d\pi + \tau \text{KL}(\pi_1 | \alpha) + \tau \text{KL}(\pi_2 | \beta) + \varepsilon \text{KL}(\pi | \alpha \otimes \beta) \right\}. \quad (10.9)$$

$$\begin{aligned} \omega &= \frac{\tau}{\tau + 2\sigma^2}, & \ell &= \sigma^2 + \frac{\tau}{2}, & X &= \Sigma_0 + \Sigma_1 + \ell \text{Id}, \\ \tilde{\Sigma}_i &= \frac{\tau}{2} (\text{Id} - \ell(\Sigma_i + \ell \text{Id})^{-1}), & C &= \left(\frac{1}{\omega} \tilde{\Sigma}_0 \tilde{\Sigma}_1 + \frac{\sigma^4}{4} \text{Id} \right)^{1/2} - \frac{\sigma^2}{2} \text{Id}. \end{aligned}$$

$$\begin{aligned} M_{\varepsilon, \tau} &= \sigma^{\frac{d\sigma^2}{\tau + \sigma^2}} \left[r_0 r_1 \det(C) \left(\frac{\det(\tilde{\Sigma}_0 \tilde{\Sigma}_1)^\omega}{\det(\Sigma_0 \Sigma_1)} \right)^{1/2} \right]^{\frac{1}{1+\omega}} \\ &\quad \times \frac{\exp\left(-\frac{\|m_0 - m_1\|_{X^{-1}}^2}{2(1+\omega)}\right)}{\sqrt{\det\left(C - \frac{2}{\tau} \tilde{\Sigma}_0 \tilde{\Sigma}_1\right)}}. \end{aligned} \quad (10.10)$$

Proposition 10.7 (Entropic KL–UOT between unnormalized Gaussians). *For the preceding parameters, the unique optimizer of (10.9) is a scaled Gaussian plan of total mass $M_{\varepsilon, \tau}$, and the optimal value is*

$$\text{UW}_{\tau, \varepsilon}^{\text{KL}}(\alpha, \beta) = \tau(r_0 + r_1) + \varepsilon r_0 r_1 - (\varepsilon + 2\tau) M_{\varepsilon, \tau}. \quad (10.11)$$

Proof. The KL primal–dual relation below (10.6) turns quadratic dual potentials into the exponential of a quadratic form, hence into a scaled Gaussian plan. The damped soft transforms preserve this quadratic class. Solving their matrix fixed point gives the matrices X , $\tilde{\Sigma}_i$ and C above; evaluating the Gaussian normalizing constant gives (10.10). Finally, integrating the two marginal optimality relations and the plan optimality relation cancels all moment terms and leaves $\tau(r_0 + r_1) + 2\sigma^2 r_0 r_1 - 2(\sigma^2 + \tau) M_{\varepsilon, \tau}$, which is (10.11) because $\varepsilon = 2\sigma^2$. $\square$

Partial optimal transport. $0 \leq m \leq \min\{\alpha(\mathcal{X}), \beta(\mathcal{Y})\}$,

$$\text{POT}_m(\alpha, \beta) := \inf_{\substack{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y}) \\ \pi_1 \leq \alpha, \pi_2 \leq \beta \\ \pi(\mathcal{X} \times \mathcal{Y}) = m}} \int c \, d\pi.$$

Thus only a submeasure of α is transported onto a submeasure of β ; the remaining mass is left unmatched. The corresponding Lagrangian form is obtained by adding total-variation penalties.

$\inf_{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y})} \int c \, d\pi + \lambda \|\alpha - \pi_1\|_{\text{TV}} + \lambda \|\beta - \pi_2\|_{\text{TV}}$. Assume $c \geq 0$. Allowing transported marginals larger than the available marginals does not improve the value: excess transported mass can be trimmed, which decreases the transport cost and cannot increase the sum of the two total-variation penalties. Hence an optimal plan may be chosen with $\pi_1 \leq \alpha$ and $\pi_2 \leq \beta$. If $m = \pi(\mathcal{X} \times \mathcal{Y})$, the penalty then reduces to

$$\lambda(\alpha(\mathcal{X}) - m) + \lambda(\beta(\mathcal{Y}) - m),$$

Proposition 10.8 (TV penalization selects fixed-mass partial OT). *Assume that $\mathcal{X}, \mathcal{Y}$ are compact metric spaces, that c is continuous and nonnegative, and set $A = \alpha(\mathcal{X})$, $B = \beta(\mathcal{Y})$, $M = \min(A, B)$. For $\lambda \geq 0$, define the TV-penalized partial value*

$$\text{POT}_\lambda^{\text{TV}}(\alpha, \beta) := \inf_{\substack{\pi \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y}) \\ \pi_1 \leq \alpha, \pi_2 \leq \beta}} \left\{ \int c \, d\pi + \lambda(A + B - 2\pi(\mathcal{X} \times \mathcal{Y})) \right\}.$$

Then

$$\text{POT}_\lambda^{\text{TV}}(\alpha, \beta) = \lambda(A + B) + \inf_{0 \leq m \leq M} \{ \text{POT}_m(\alpha, \beta) - 2\lambda m \}.$$

If π_λ minimizes $\text{POT}_\lambda^{\text{TV}}$ and $m_\lambda = \pi_\lambda(\mathcal{X} \times \mathcal{Y})$, then m_λ minimizes $m \mapsto \text{POT}_m(\alpha, \beta) - 2\lambda m$ and π_λ minimizes $\text{POT}_{m_\lambda}(\alpha, \beta)$. Conversely, if 2λ is a subgradient at m of the convex function $m \mapsto \text{POT}_m(\alpha, \beta)$ extended by $+\infty$ outside $[0, M]$, then every minimizer of $\text{POT}_m(\alpha, \beta)$ is a minimizer of $\text{POT}_\lambda^{\text{TV}}(\alpha, \beta)$. In particular, every prescribed mass $m \in [0, M]$ is selected by at least one multiplier $\lambda \geq 0$, possibly together with a whole interval of masses when the value function has a flat exposed face.

Proof. Write $V(m) = \text{POT}_m(\alpha, \beta)$. Compactness and continuity give an optimizer for every $m \in [0, M]$: the admissible subcouplings form a weakly compact set and the cost is continuous. The function V is convex on $[0, M]$: if π_0, π_1 are admissible competitors of masses m_0, m_1 , then $(1-t)\pi_0 + t\pi_1$ is admissible for mass $(1-t)m_0 + tm_1$ and has the corresponding convex-combination cost. It is also nondecreasing because any competitor of mass $m_1 > m_0$ can be rescaled by m_0/m_1 to give a competitor of mass m_0 with no larger cost. Finally, V is Lipschitz. If $0 \leq m_0 < m_1 \leq M$ and π_0 is optimal at mass m_0 , both residual measures $\alpha - (\pi_0)_1$ and $\beta - (\pi_0)_2$ have mass at least $m_1 - m_0$. Choose submeasures of each residual with this common mass and couple them. Adding the resulting plan to π_0 gives an admissible plan of mass m_1 at an additional cost bounded by $\|c\|_\infty(m_1 - m_0)$.

For any admissible π in the definition of $\text{POT}_\lambda^{\text{TV}}$, let $m = \pi(\mathcal{X} \times \mathcal{Y})$. The plan is admissible for $V(m)$, hence its objective is at least $V(m) + \lambda(A + B - 2m)$. Taking the infimum over π gives the lower bound by the displayed right-hand side. Conversely, for each m and each minimizing sequence for $V(m)$, the same plans are admissible in $\text{POT}_\lambda^{\text{TV}}$ and have objective tending to $V(m) + \lambda(A + B - 2m)$. Taking the infimum over m gives the reverse inequality.

The optimizer statement follows from the same inequalities. If π_λ minimizes the penalized problem with mass m_λ , then equality in the lower bound forces both $V(m_\lambda) - 2\lambda m_\lambda$ to be minimal over $[0, M]$ and π_λ to be optimal at fixed mass m_λ . Conversely, $2\lambda \in \partial V(m)$ means $V(m') - 2\lambda m' \geq V(m) - 2\lambda m$ for all $m' \in [0, M]$. Thus any minimizer of $V(m)$ attains the penalized infimum. Because the extended function is proper, convex, lower semicontinuous and Lipschitz on its effective interval, its subdifferential is nonempty at every $m \in [0, M]$ after adding the normal cone of the interval. Monotonicity of V ensures that a subgradient can be chosen nonnegative, hence equal to 2λ for some $\lambda \geq 0$. $\square$

Proposition 10.9 (Fixed total mass reduces partial OT to balanced OT). *Let $(\mathcal{X}, d)$ be a metric space, let $p \geq 1$, set $c = d^p$, and fix $m > 0$. On the class of nonnegative measures with total mass exactly m and finite p th moment, $\mathcal{M}_m^p(\mathcal{X}) := \{\alpha \in \mathcal{M}_+(\mathcal{X}) ; \alpha(\mathcal{X}) = m, \int d(x_0, x)^p d\alpha(x) < +\infty\}$ for one, hence any, base point x_0 , the quantity $d_m(\alpha, \beta) := \text{POT}_m(\alpha, \beta)^{1/p}$ is a distance. More precisely, if $\bar{\alpha} = \alpha/m$ and $\bar{\beta} = \beta/m$, then $d_m(\alpha, \beta) = m^{1/p} \mathcal{W}_p(\bar{\alpha}, \bar{\beta})$. If instead POT_m is considered on a larger class of measures with mass at least m , then it is not a distance in general for $m > 0$: different measures can share a common submeasure of mass m , giving zero partial cost.*

Proof. If $\alpha(\mathcal{X}) = \beta(\mathcal{X}) = m$ and π is admissible in $\text{POT}_m(\alpha, \beta)$, then $\pi_1 \leq \alpha$ and both measures have mass m , hence $\pi_1 = \alpha$. Similarly $\pi_2 = \beta$. Thus the admissible plans are exactly the ordinary couplings between α and β . Dividing such a coupling by m gives a probability coupling between $\bar{\alpha}$ and $\bar{\beta}$, and conversely multiplying any coupling of $\bar{\alpha}, \bar{\beta}$ by m gives a coupling of α, β . Therefore $\text{POT}_m(\alpha, \beta) = m \mathcal{W}_p(\bar{\alpha}, \bar{\beta})^p$. The metric properties of d_m follow from those of $\mathcal{W}_p$ in Proposition 3.33.

For the last claim, take three distinct points x, y, z in a metric space and a number $r > 0$. The measures $\alpha = m\delta_x + r\delta_y$ and $\beta = m\delta_x + r\delta_z$ are different but the partial plan $m\delta_{(x,x)}$ is admissible and has zero cost. Hence $\text{POT}_m(\alpha, \beta) = 0$ although $\alpha \neq \beta$. $\square$

Thus the TV-penalized formulation is the Lagrangian envelope of constrained partial OT: increasing λ penalizes unmatched mass more strongly and therefore selects larger transported masses, while small λ makes deletion and creation cheaper. As the prescribed transported mass decreases, the plan keeps only the least costly facing portions of the two marginals.

10.2 Sliced Wasserstein Distances

One-dimensional projections.

Spherical averaging.

Definition 10.10 (Sliced Wasserstein distance). Let $p \geq 1$, let $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$, and let σ be the uniform probability measure on the sphere $\mathbb{S}^{d-1}$. The sliced p -Wasserstein distance is

$$\text{SW}_p(\alpha, \beta)^p := \int_{\mathbb{S}^{d-1}} \mathcal{W}_p((P_\theta)_\# \alpha, (P_\theta)_\# \beta)^p d\sigma(\theta). \quad (10.12)$$

Proposition 10.11 (Metric properties of sliced Wasserstein). *For $p \geq 1$, SW_p is a distance on $\mathcal{P}_p(\mathbb{R}^d)$. Moreover, SW_p metrizes weak convergence together with convergence of the p th moment. Finally,*

$$\text{SW}_p(\alpha, \beta)^p \leq \kappa_{d,p} \mathcal{W}_p(\alpha, \beta)^p, \quad \kappa_{d,p} := \int_{\mathbb{S}^{d-1}} |\theta_1|^p d\sigma(\theta) = \frac{\Gamma(d/2)\Gamma((p+1)/2)}{\sqrt{\pi}\Gamma((d+p)/2)}.$$

In particular, $\kappa_{d,2} = 1/d$. Conversely, if $d \geq 2$, $R > 0$, and $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$ are supported in $B_R := \{x \in \mathbb{R}^d; \|x\| \leq R\}$, then there is a constant $C_{d,p}$ depending only on d and p such that Bonnotte's historical estimate reads $\mathcal{W}_p(\alpha, \beta)^p \leq C_{d,p} R^{p-\frac{1}{d+1}} \text{SW}_p(\alpha, \beta)^{\frac{1}{d+1}}$. The sharp reverse estimate for $p = 1$ also gives, for every $p \geq 1$, constants $\tilde{C}_{d,p}, \tilde{c}_{d,p} > 0$ such that

$$\mathcal{W}_p(\alpha, \beta) \leq \tilde{C}_{d,p} R^{1-\frac{1}{pd}} \text{SW}_p(\alpha, \beta)^{\frac{1}{pd}}, \quad \text{equivalently} \quad \text{SW}_p(\alpha, \beta) \geq \tilde{c}_{d,p} R \left(\frac{\mathcal{W}_p(\alpha, \beta)}{R} \right)^{pd}.$$

For $p = 1$, the exponent $1/d$ in this last comparison is sharp.

Proof. Non-negativity and symmetry follow from the one-dimensional Wasserstein distance. For the triangle inequality, apply the triangle inequality of $\mathcal{W}_p$ for each direction θ and then Minkowski's inequality in $L^p(\mathbb{S}^{d-1})$.

If $\text{SW}_p(\alpha, \beta) = 0$, then $(P_\theta)_\# \alpha = (P_\theta)_\# \beta$ for almost every direction. By continuity of characteristic functions this holds for all directions, and the Cramér–Wold theorem implies $\alpha = \beta$. This proves separation.

Let π be any coupling between α and β . Projecting π gives an admissible coupling between the corresponding one-dimensional projections, hence $\mathcal{W}_p((P_\theta)_\# \alpha, (P_\theta)_\# \beta)^p \leq \int_{\mathbb{R}^d \times \mathbb{R}^d} |\langle \theta, x - y \rangle|^p d\pi(x, y)$. Rotational invariance of σ gives $\int_{\mathbb{S}^{d-1}} |\langle \theta, z \rangle|^p d\sigma(\theta) = \kappa_{d,p} \|z\|^p$. Integrating the first inequality in θ and optimizing over π proves the stated direct comparison. The beta-integral formula for the first coordinate of a uniform point on the sphere gives the displayed value of $\kappa_{d,p}$; in particular, $\kappa_{d,2} = 1/d$.

To prove the topological statement, first suppose that $\mathcal{W}_p(\alpha_n, \alpha) \rightarrow 0$. The inequality $\text{SW}_p \leq \mathcal{W}_p$ immediately gives $\text{SW}_p(\alpha_n, \alpha) \rightarrow 0$. Conversely, suppose that $\text{SW}_p(\alpha_n, \alpha) \rightarrow 0$. The spherical identity

$$\int_{\mathbb{S}^{d-1}} \int_{\mathbb{R}^d} |\langle \theta, x \rangle|^p d\alpha_n(x) d\sigma(\theta) = \kappa_{d,p} \int_{\mathbb{R}^d} \|x\|^p d\alpha_n(x), \quad \kappa_{d,p} > 0,$$

and the triangle inequality in $L^p(\mathbb{S}^{d-1} \times (0, 1))$ show that the p th moments of (α_n) are uniformly bounded. Hence the sequence is tight. From every subsequence one can extract a further subsequence for which the projected $\mathcal{W}_p$ distances converge to zero for σ -a.e. θ . Every weak limit therefore has the same one-dimensional projections as α , so Cramér–Wold identifies it with α . Thus $\alpha_n \rightarrow \alpha$. Finally, let $Q_{n,\theta}$ and Q_θ be the quantile functions of the projected measures. The one-dimensional quantile formula gives $\text{SW}_p(\alpha_n, \alpha) = \|Q_{n,\theta} - Q_\theta\|_{L^p(\mathbb{S}^{d-1} \times (0,1))}$. Hence the corresponding L^p norms converge. By the spherical identity, their p th powers are respectively $\kappa_{d,p} \int \|x\|^p d\alpha_n(x)$ and $\kappa_{d,p} \int \|x\|^p d\alpha(x)$. Thus the p th moments converge. Weak convergence plus convergence of these moments is equivalent to convergence in $\mathcal{W}_p$.

The reverse estimates on compact sets are deeper and use the Radon structure of slices. Bonnotte's Lemma 5.1.4 proves $\mathcal{W}_1(\alpha, \beta) \leq C_d R^{\frac{d}{d+1}} \text{SW}_1(\alpha, \beta)^{\frac{1}{d+1}}$ by mollifying a Kantorovich–Rubinstein test function, representing the regularized test through one-dimensional projections, bounding the resulting action by SW_1 , and optimizing the smoothing scale. On B_R , $\mathcal{W}_p^p \leq (2R)^{p-1} \mathcal{W}_1$, $\text{SW}_1 \leq \text{SW}_p$, where the first inequality follows by evaluating the p -cost on a $\mathcal{W}_1$ -optimal coupling, and the second follows from the monotonicity of one-dimensional Wasserstein distances and Hölder's inequality in θ . These inequalities give Bonnotte's stated general- p estimate. The sharper $p = 1$ comparison theorem of Carlier, Figalli, Mérigot and Wang replaces $1/(d+1)$ by the optimal exponent $1/d$. $\mathcal{W}_p^p \leq C_{d,p} R^{p-\frac{1}{d}} \text{SW}_p^{\frac{1}{d}}$, which is equivalent to the final two-sided formulation in the statement. This supplies a lower bound on SW_p in terms of $\mathcal{W}_p$ for every p , but the asserted sharpness concerns only $p = 1$. $\square$

Proposition 10.12 (First-order comparison: strictness and atomic equality). *Let $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$ be absolutely continuous, and let $\varphi \in C^2(\mathbb{R}^d)$ be such that $\nabla \varphi \in L^2(\alpha; \mathbb{R}^d)$ and $\nabla^2 \varphi$ is bounded. For $|t|$ small enough, set $T_t(x) := x + t \nabla \varphi(x)$, $\alpha_t := (T_t)_\# \alpha$. Then T_t is the quadratic Brenier map from α to α_t for $|t|$ small enough, and $\mathcal{W}_2(\alpha, \alpha_t)^2 = t^2 \int_{\mathbb{R}^d} \|\nabla \varphi(x)\|^2 d\alpha(x)$. Moreover, $\text{SW}_2(\alpha, \alpha_t)^2 \leq \frac{t^2}{d} \int_{\mathbb{R}^d} \|\nabla \varphi(x)\|^2 d\alpha(x)$. The upper bound can be strict at order t^2 . Indeed, let $d \geq 2$, let $\alpha = \mathcal{N}(0, \text{Id})$, and take $\varphi(x) = \frac{1}{2} x^\top A x$ for a symmetric matrix A that is not a scalar multiple of Id . Then*

$$\lim_{t \rightarrow 0} \frac{\text{SW}_2(\alpha, \alpha_t)^2}{t^2} = \frac{\text{tr}(A)^2 + 2 \text{tr}(A^2)}{d(d+2)} < \frac{\text{tr}(A^2)}{d} = \frac{1}{d} \lim_{t \rightarrow 0} \frac{\mathcal{W}_2(\alpha, \alpha_t)^2}{t^2}.$$

Separately, equality holds at first order along every finite atomic curve with fixed weights. More precisely, let $a_i > 0$ satisfy $\sum_i a_i = 1$, let $x_1, \dots, x_n$ be pairwise distinct, and let $v_i \in \mathbb{R}^d$. If $\eta = \sum_{i=1}^n a_i \delta_{x_i}$ and $\eta_t = \sum_{i=1}^n a_i \delta_{x_i + tv_i}$, then $\lim_{t \rightarrow 0} \frac{\text{SW}_2(\eta, \eta_t)^2}{t^2} = \frac{1}{d} \sum_{i=1}^n a_i \|v_i\|^2 = \frac{1}{d} \lim_{t \rightarrow 0} \frac{\mathcal{W}_2(\eta, \eta_t)^2}{t^2}$.

Proof. The map T_t is the gradient of $\psi_t(x) := \frac{1}{2} \|x\|^2 + t\varphi(x)$. Since $\nabla^2 \varphi$ is bounded, $\nabla^2 \psi_t = \text{Id} + t \nabla^2 \varphi$ is positive semidefinite for $|t|$ small enough. Hence ψ_t is convex and $T_t = \nabla \psi_t$ is a Brenier map. Its graph is cyclically monotone, so the Brenier optimality criterion gives the optimality of $(\text{Id}, T_t)_\# \alpha$, and $\mathcal{W}_2(\alpha, \alpha_t)^2 = \int \|x - T_t(x)\|^2 d\alpha(x) = t^2 \int \|\nabla \varphi(x)\|^2 d\alpha(x)$. For a fixed direction θ , the coupling $(P_\theta, P_\theta \circ T_t)_\# \alpha$ is admissible between $(P_\theta)_\# \alpha$ and $(P_\theta)_\# \alpha_t$. Therefore $\mathcal{W}_2((P_\theta)_\# \alpha, (P_\theta)_\# \alpha_t)^2 \leq t^2 \int |\langle \theta, \nabla \varphi(x) \rangle|^2 d\alpha(x)$.

Integrating over the sphere and using $\int_{\mathbb{S}^{d-1}} |\langle \theta, w \rangle|^2 d\sigma(\theta) = \frac{1}{d} \|w\|^2$ gives the sliced estimate.

For the Gaussian example, $T_t = \text{Id} + tA$ and, for every $\theta \in \mathbb{S}^{d-1}$, the projected source and target are centered one-dimensional Gaussians with standard deviations 1 and $\|(\text{Id} + tA)\theta\|$. $\lim_{t \rightarrow 0} \frac{1}{t^2} \mathcal{W}_2((P_\theta)_\# \alpha, (P_\theta)_\# \alpha_t)^2 = (\theta^\top A \theta)^2$. The spherical fourth-moment identity gives $\int_{\mathbb{S}^{d-1}} (\theta^\top A \theta)^2 d\sigma(\theta) = \frac{\text{tr}(A)^2 + 2 \text{tr}(A^2)}{d(d+2)}$. On the other hand, the Brenier formula above gives $\mathcal{W}_2(\alpha, \alpha_t)^2 = t^2 \text{tr}(A^2)$.

The strict inequality follows from $\text{tr}(A)^2 < d \text{tr}(A^2)$ when A is symmetric and non-scalar.

Finally, consider the atomic curve. For all sufficiently small t , the diagonal coupling between x_i and $x_i + tv_i$ is optimal for $\mathcal{W}_2$. Indeed, writing $c_{ij}(t) = \|x_i - x_j - tv_j\|^2$, the discrete dual potentials $f_i = 0$ and $g_j = c_{jj}(t)$ are feasible for small t ,

because $c_{jj}(t) < c_{ij}(t)$ whenever $i \neq j$, and they certify the diagonal plan. Thus $\mathcal{W}_2(\eta, \eta_t)^2 = t^2 \sum_{i=1}^n a_i \|v_i\|^2$. For every θ outside the finite union of great spheres $\{|\langle \theta, x_i - x_j \rangle| = 0\}$, the projected atoms are distinct and their order is unchanged for sufficiently small t . The one-dimensional monotone coupling then pairs each projected atom with its perturbation, so $\lim_{t \rightarrow 0} \frac{1}{t^2} \mathcal{W}_2((P_\theta)_\# \eta, (P_\theta)_\# \eta_t)^2 = \sum_{i=1}^n a_i |\langle \theta, v_i \rangle|^2$. The diagonal projected coupling provides the same expression as an integrable upper bound for every t . Dominated convergence and the spherical second-moment identity therefore give the asserted sliced limit. $\square$

Intrinsic sliced length.

$$\ell_{\text{SW}_2}(\alpha, \beta) := \inf_{\substack{\gamma_0=\alpha, \gamma_1=\beta \\ \gamma \text{ SW}_2\text{-absolutely continuous}}} \int_0^1 |\dot{\gamma}_t|_{\text{SW}_2} dt, \quad (10.13)$$

where $|\dot{\gamma}_t|_{\text{SW}_2}$ is the metric derivative. Park and Slepčev prove that the infimum is attained, so $(\mathcal{P}_2(\mathbb{R}^d), \ell_{\text{SW}_2})$ is a geodesic space.

$$\text{SW}_2 \leq \ell_{\text{SW}_2} \leq d^{-1/2} \mathcal{W}_2. \quad \ell_{\text{SW}_2}(\alpha, \alpha_t) < d^{-1/2} \mathcal{W}_2(\alpha, \alpha_t). \quad |\dot{\alpha}_t^\delta|_{\mathcal{W}_2} = 1, \quad |\dot{\alpha}_t^\delta|_{\text{SW}_2} \leq C_d(t + \delta). \quad \limsup_{h \downarrow 0} \frac{\ell_{\text{SW}_2}(\alpha_0^\delta, \alpha_h^\delta)}{\mathcal{W}_2(\alpha_0^\delta, \alpha_h^\delta)} \leq C_d \delta.$$

$$\inf_{\substack{\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d) \\ \alpha \neq \beta}} \frac{\ell_{\text{SW}_2}(\alpha, \beta)}{\mathcal{W}_2(\alpha, \beta)} = 0. \quad (10.14)$$

Thus ℓ_{SW_2} and $\mathcal{W}_2$ are not globally bi-Lipschitz equivalent, even after inserting the factor $d^{-1/2}$.

$$\frac{\text{SW}_2(\eta, \beta)}{\mathcal{W}_2(\eta, \beta)} \rightarrow \frac{1}{\sqrt{d}}, \quad \frac{\ell_{\text{SW}_2}(\eta, \beta)}{\mathcal{W}_2(\eta, \beta)} \rightarrow \frac{1}{\sqrt{d}} \quad \text{as } \mathcal{W}_\infty(\eta, \beta) \rightarrow 0,$$

Sliced Wasserstein between Gaussians. For $\alpha = \mathcal{N}(m_\alpha, \Sigma_\alpha)$ and $\beta = \mathcal{N}(m_\beta, \Sigma_\beta)$,

$$\begin{aligned} \text{SW}_2(\alpha, \beta)^2 &= \frac{\|m_\alpha - m_\beta\|^2}{d} + \int_{\mathbb{S}^{d-1}} \left(\sqrt{\theta^\top \Sigma_\alpha \theta} - \sqrt{\theta^\top \Sigma_\beta \theta} \right)^2 d\sigma(\theta) \\ &= \frac{\|m_\alpha - m_\beta\|^2 + \text{tr}(\Sigma_\alpha) + \text{tr}(\Sigma_\beta)}{d} - 2 \int_{\mathbb{S}^{d-1}} \sqrt{(\theta^\top \Sigma_\alpha \theta)(\theta^\top \Sigma_\beta \theta)} d\sigma(\theta). \end{aligned} \quad (10.15)$$

One always has $\text{SW}_2^2 \leq \mathcal{W}_2^2/d$. For positive-definite covariances, equality holds exactly when they are proportional. More generally, if $\Sigma_\alpha = s^2 \Sigma$ and $\Sigma_\beta = t^2 \Sigma$, then $\text{SW}_2^2 = \mathcal{W}_2^2/d = d^{-1}[\|m_\alpha - m_\beta\|^2 + (s - t)^2 \text{tr}(\Sigma)]$. Thus two isotropic covariances saturate the bound, but one isotropic covariance does so only when the other is also isotropic. For $\Sigma_\alpha = a^2 uu^\top$, $\Sigma_\beta = b^2 vv^\top$, and $\chi = |\langle u, v \rangle|$,

$$\text{SW}_2^2 = \frac{1}{d} \left[\|m_\alpha - m_\beta\|^2 + a^2 + b^2 - \frac{4ab}{\pi} \left(\sqrt{1 - \chi^2} + \chi \arcsin \chi \right) \right], \quad \mathcal{W}_2^2 = \|m_\alpha - m_\beta\|^2 + a^2 + b^2 - 2ab\chi.$$

L^q -sliced, max-sliced and subspace variants.

Definition 10.13 (L^q -sliced and subspace-sliced Wasserstein). Let $p \in [1, \infty)$, $q \in [1, \infty]$, and $1 \leq k \leq d$. Denote by $\text{St}(d, k) := \{U \in \mathbb{R}^{d \times k} ; U^\top U = \text{Id}_k\}$ the Stiefel manifold, equipped with its normalized invariant probability measure $\sigma_{d,k}$. For $q < \infty$, define

$$\text{SW}_{p,q,k}(\alpha, \beta) := \left(\int_{\text{St}(d,k)} \mathcal{W}_p((U^\top)_\# \alpha, (U^\top)_\# \beta)^q d\sigma_{d,k}(U) \right)^{1/q}, \quad (10.16)$$

and extend the definition to $q = \infty$ by $\text{SW}_{p,\infty,k}(\alpha, \beta) := \sup_{U \in \text{St}(d,k)} \mathcal{W}_p((U^\top)_\# \alpha, (U^\top)_\# \beta)$. The supremum is attained because the Stiefel manifold is compact and the projected Wasserstein distance depends continuously on U . Moreover, the integrand is unchanged under $U \mapsto UQ$ for $Q \in \text{O}(k)$, because $Q^\top$ is an isometry of $\mathbb{R}^k$. Thus the construction depends only on the subspace spanned by the columns of U : the Stiefel manifold is a convenient parametrization of the corresponding Grassmann manifold. We use the abbreviations $\text{SW}_{p,q} := \text{SW}_{p,q,1}$, $\text{SW}_p = \text{SW}_{p,p}$, $\text{MaxSW}_{p,k} := \text{SW}_{p,\infty,k}$, $\text{MaxSW}_p = \text{SW}_{p,\infty}$. Thus $k = 1$ gives line-based slicing, $q = \infty$ gives max-slicing, and $k = d$ gives $\text{SW}_{p,q,d} = \mathcal{W}_p$ for every q . Quadratic max-slicing is useful when only a small set of directions carries the discrepancy, for instance in generative modeling. For $p = 2$, the k -dimensional max-sliced case is the projection-robust Wasserstein distance, which is related to the subspace-robust distances studied later in Section 10.5.

Proposition 10.14 (Basic bounds for sliced variants). Let $p \in [1, \infty)$, $1 \leq q \leq r \leq \infty$, and $1 \leq k \leq \ell \leq d$. Each $\text{SW}_{p,q,k}$ is a finite distance on $\mathcal{P}_p(\mathbb{R}^d)$ and metrizes the same topology as $\mathcal{W}_p$. For every $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$,

$$\text{SW}_{p,q,k}(\alpha, \beta) \leq \text{SW}_{p,r,k}(\alpha, \beta) \leq \mathcal{W}_p(\alpha, \beta), \quad (10.17)$$

$$\text{SW}_{p,q,k}(\alpha, \beta) \leq \text{SW}_{p,q,\ell}(\alpha, \beta) \leq \mathcal{W}_p(\alpha, \beta). \quad (10.18)$$

For a unit vector $e \in \mathbb{S}^{d-1}$, set

$$\kappa_{d,k,p} := \int_{\text{St}(d,k)} \|U^\top e\|^p d\sigma_{d,k}(U) = \frac{\Gamma(d/2)\Gamma((k+p)/2)}{\Gamma(k/2)\Gamma((d+p)/2)}.$$

This constant is independent of e , satisfies $\kappa_{d,d,p} = 1$ and $\kappa_{d,1,p} = \kappa_{d,p}$ from Proposition 10.11, and gives the sharper bounds

$$\text{SW}_{p,q,k}(\alpha, \beta) \leq \begin{cases} \kappa_{d,k,p}^{1/p} \mathcal{W}_p(\alpha, \beta), & 1 \leq q \leq p, \\ \kappa_{d,k,p}^{1/q} \mathcal{W}_p(\alpha, \beta), & p \leq q < \infty, \\ \mathcal{W}_p(\alpha, \beta), & q = \infty. \end{cases} \quad (10.19)$$

Moreover,

$$\text{SW}_p(\alpha, \beta) \leq \text{SW}_{p,p,k}(\alpha, \beta) \leq \kappa_{d,k,p}^{1/p} \mathcal{W}_p(\alpha, \beta). \quad (10.20)$$

In particular, $\kappa_{d,k,2} = k/d$ and $\text{SW}_{2,2,k}^2 \leq (k/d) \mathcal{W}_2^2$. If $q \geq p$, the compact-support reverse estimates in Proposition 10.11 remain valid with $\text{SW}_{p,q,k}$ in place of SW_p .

Proof. Write $D_U(\alpha, \beta) := \mathcal{W}_p((U^\top)_\# \alpha, (U^\top)_\# \beta)$. Non-negativity and symmetry are inherited from $\mathcal{W}_p$. Minkowski's inequality in $L^q(\sigma_{d,k})$, or the supremum triangle inequality when $q = \infty$, proves the triangle inequality. The reverse triangle inequality for $\mathcal{W}_p$ shows that $U \mapsto D_U(\alpha, \beta)$ is continuous. Indeed, for $U_n \rightarrow U$,

$$\mathcal{W}_p((U_n^\top)_\# \alpha, (U^\top)_\# \alpha) \leq \|U_n - U\|_{\text{op}} \left(\int \|x\|^p d\alpha(x) \right)^{1/p},$$

and likewise for β . If $\text{SW}_{p,q,k}(\alpha, \beta) = 0$, then $D_U = 0$ for all U : this is immediate for $q = \infty$, while for finite q it follows from continuity and the full support of $\sigma_{d,k}$. For any $\theta \in \mathbb{S}^{d-1}$, choose a frame U whose first column is θ . Equality of the two k -dimensional projected measures then implies equality of their first-coordinate marginals, hence $(P_\theta)_\# \alpha = (P_\theta)_\# \beta$. The Cramér–Wold theorem gives $\alpha = \beta$, proving separation.

Every $U^\top$ is 1-Lipschitz, so $D_U \leq \mathcal{W}_p(\alpha, \beta)$. Monotonicity of L^q norms on a probability space proves (10.17). To compare projection dimensions, let $V \in \text{St}(d, \ell)$ and $R \in \text{St}(\ell, k)$. Since $(VR)^\top = R^\top V^\top$ and $R^\top$ is 1-Lipschitz, $D_{VR}(\alpha, \beta) \leq D_V(\alpha, \beta)$. Under the product of the invariant measures, VR is uniformly distributed on $\text{St}(d, k)$. Integration proves (10.18) for finite q ; for $q = \infty$, extend each k -frame to an ℓ -frame and use the same contraction.

For the sharper constants, let $\pi \in \Gamma(\alpha, \beta)$. Then $D_U(\alpha, \beta)^p \leq \int \|U^\top(x - y)\|^p d\pi(x, y)$. Integration in U , followed by minimization in π , gives $\|D_\cdot\|_{L^p(\sigma_{d,k})}^p \leq \kappa_{d,k,p} \mathcal{W}_p(\alpha, \beta)^p$. If $q \leq p$, use $\|D_\cdot\|_{L^q} \leq \|D_\cdot\|_{L^p}$. If $p \leq q < \infty$, use $D_U \leq \mathcal{W}_p$ to obtain $\int D_U^q d\sigma_{d,k}(U) \leq \mathcal{W}_p(\alpha, \beta)^{q-p} \int D_U^p d\sigma_{d,k}(U)$. This proves (10.19). The beta-integral for the p th power of the length of a random k -dimensional projection gives the displayed formula for $\kappa_{d,k,p}$.

Finally, let $\omega \in \mathbb{S}^{k-1}$. Since $P_{U\omega} = P_\omega \circ U^\top$, projection contraction gives $\mathcal{W}_p((P_{U\omega})_\# \alpha, (P_{U\omega})_\# \beta) \leq D_U(\alpha, \beta)$. Averaging its p th power first over ω and then over U makes $U\omega$ uniform on $\mathbb{S}^{d-1}$ and proves the first inequality in (10.20). For $q \geq p$, (10.17) gives $\text{SW}_p \leq \text{SW}_{p,p,k} \leq \text{SW}_{p,q,k}$, which transfers the stated reverse estimates.

For $\eta \in \mathcal{P}_p(\mathbb{R}^d)$, set $M_p(\eta) := (\int \|x\|^p d\eta(x))^{1/p}$. Since the only coupling between $(U^\top)_\# \eta$ and δ_0 sends every point to the origin, $D_U(\eta, \delta_0)^p = \int \|U^\top x\|^p d\eta(x)$, $\int D_U(\eta, \delta_0)^p d\sigma_{d,k}(U) = \kappa_{d,k,p} M_p(\eta)^p$. Also $D_U(\eta, \delta_0) \leq M_p(\eta)$.

$$\text{SW}_{p,q,k}(\eta, \delta_0) \geq \begin{cases} \kappa_{d,k,p}^{1/q} M_p(\eta), & 1 \leq q \leq p, \\ \kappa_{d,k,p}^{1/p} M_p(\eta), & p \leq q \leq \infty. \end{cases} \quad (10.21)$$

Indeed, the second line follows from monotonicity of L^q norms. For $q \leq p$ and $M_p(\eta) > 0$, the pointwise bound $D_U \leq M_p(\eta)$ gives $D_U^q \geq M_p(\eta)^{q-p} D_U^p$; the zero-moment case is immediate.

Suppose now that $\text{SW}_{p,q,k}(\alpha_n, \alpha) \rightarrow 0$. The triangle inequality and (10.21) imply $\sup_n M_p(\alpha_n) < \infty$. Write $D_{n,U} := D_U(\alpha_n, \alpha)$. The reverse triangle inequality and the coupling of each point with itself give $|D_{n,U} - D_{n,V}| \leq \|U - V\|_{\text{op}}(M_p(\alpha_n) + M_p(\alpha))$, so $(D_{n,\cdot})_n$ is equicontinuous on the compact Stiefel manifold. For $q < \infty$, equicontinuity and the full support of $\sigma_{d,k}$ upgrade $\|D_{n,\cdot}\|_{L^q} \rightarrow 0$ to $\|D_{n,\cdot}\|_{L^\infty} \rightarrow 0$; for $q = \infty$ this is immediate. Given $\theta \in \mathbb{S}^{d-1}$, choose a frame U whose first column is θ . Projection contraction then yields $\mathcal{W}_p((P_\theta)_\# \alpha_n, (P_\theta)_\# \alpha) \leq D_{n,U}$, uniformly in θ . Hence $\text{SW}_p(\alpha_n, \alpha) \rightarrow 0$, and Proposition 10.11 gives $\mathcal{W}_p(\alpha_n, \alpha) \rightarrow 0$. The converse follows directly from $\text{SW}_{p,q,k} \leq \mathcal{W}_p$. $\square$

Min-SW lifted transport plans.

$$\langle x_{\sigma_\theta(1)}, \theta \rangle < \dots < \langle x_{\sigma_\theta(n)}, \theta \rangle, \quad \langle y_{\tau_\theta(1)}, \theta \rangle < \dots < \langle y_{\tau_\theta(n)}, \theta \rangle.$$

$$\pi_\theta = \frac{1}{n} \sum_{i=1}^n \delta_{(x_{\sigma_\theta(i)}, y_{\tau_\theta(i)})} \in \mathcal{U}(\alpha, \beta). \quad (10.22)$$

$$\alpha_\theta = (P_\theta)_\# \alpha, \quad \beta_\theta = (P_\theta)_\# \beta, \quad \varpi_\theta = (q_{\alpha_\theta}, q_{\beta_\theta})_\# \text{Leb}_{[0,1]}$$

$$\mathcal{C}_\theta(\alpha, \beta) := \{\pi \in \mathcal{U}(\alpha, \beta) : (P_\theta, P_\theta)_\# \pi = \varpi_\theta\}. \quad (10.23)$$

$$\alpha(dx) = \int_{\mathbb{R}} \alpha^s(dx) d\alpha_\theta(s), \quad \beta(dy) = \int_{\mathbb{R}} \beta^t(dy) d\beta_\theta(t)$$

$$\pi_\theta^0(dx, dy) = \int_{\mathbb{R}^2} \alpha^s(dx) \beta^t(dy) d\varpi_\theta(s, t) \quad (10.24)$$

$$\text{Min-SW}_2(\alpha, \beta)^2 := \min_{\theta \in \mathbb{S}^{d-1}} \min_{\pi \in \mathcal{C}_\theta(\alpha, \beta)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 d\pi(x, y). \quad (10.25)$$

Proposition 10.15 (Min-SW bounds, exactness and metric status). *For $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2(\alpha, \beta) \leq \text{Min-SW}_2(\alpha, \beta) \leq \inf_{z \in \mathbb{R}^d} \left(2 \int \|x - z\|^2 d\alpha(x) + 2 \int \|y - z\|^2 d\beta(y) \right)^{1/2}. \quad (10.26)$$

If both supports lie in a compact set K , the last term can be replaced by $\text{diam}(K)$. Moreover, $\text{Min-SW}_2(\alpha, \beta) = \mathcal{W}_2(\alpha, \beta)$ if and only if some quadratic optimal plan $\pi^ \in \mathcal{U}(\alpha, \beta)$ satisfies $(P_\theta, P_\theta)_\# \pi^* = \varpi_\theta$ for some θ . Thus, for equal-weight point clouds, equality holds whenever an optimal assignment is induced by sorting both clouds along one direction. In particular, it holds for two uniform n -point measures when $d \geq 2n - 1$ and the concatenated family $x_1, \dots, x_n, y_1, \dots, y_n$ is in general position. Here general position means that, for each $1 \leq k \leq d$, no subset of $k + 2$ points from this family lies in a k -dimensional affine subspace. The function Min-SW_2 is non-negative, symmetric and separates measures, but it does not satisfy the triangle inequality in general. It is therefore a separating symmetric discrepancy, not a distance.*

Proof. Every $\pi \in \mathcal{C}_\theta(\alpha, \beta)$ is an admissible coupling, which proves the first inequality in (10.26). For every $z \in \mathbb{R}^d$, the pointwise bound $\|x - y\|^2 \leq 2\|x - z\|^2 + 2\|y - z\|^2$ proves the second one after minimizing over z , while $\|x - y\| \leq \text{diam}(K)$ gives the compact-support version. Equality with $\mathcal{W}_2$ holds precisely when the constrained set for some direction contains a quadratic optimal plan. Under the stated dimension and general-position hypotheses, every ordering of the concatenated $2n$ -point family is induced by some linear projection. One may therefore choose an ordering that realizes an optimal assignment; this is the projection-order theorem proved in.

Symmetry follows by transposing the constrained plans, and separation follows from the lower bound by $\mathcal{W}_2$. To see that the triangle inequality can fail, let $\alpha_X, \alpha_Y, \alpha_Z$ be the uniform laws on the rows of

$$X = \begin{pmatrix} -4 & -4 \\ 0 & 3 \\ 1 & -2 \\ 4 & -4 \end{pmatrix}, \quad Z = \begin{pmatrix} -3 & 2 \\ 2 & 2 \\ 0 & 0 \\ 2 & -4 \end{pmatrix}, \quad Y = \frac{X + Z}{2}.$$

A direct enumeration of the $4!$ assignments shows that the identity assignment is quadratically optimal for both (X, Y) and (Y, Z) . It is induced by sorting along directions proportional to $(2, -1)$ and $(3, 1)$, respectively. Hence $\text{Min-SW}_2(\alpha_X, \alpha_Y)^2 = \text{Min-SW}_2(\alpha_Y, \alpha_Z)^2 = \frac{51}{16}$. For (X, Z) , the projected orders change only across the ten lines orthogonal to pairwise differences of rows of X or Z . Since the cost is linear in the plan, it suffices to enumerate the resulting angular cells and, on their boundaries, the compatible extreme permutations. $\frac{1}{4}\{63, 67, 73, 79, 91, 95, 105, 131, 135\}$. Thus $\text{Min-SW}_2(\alpha_X, \alpha_Z)^2 = 63/4$, and consequently $\text{Min-SW}_2(\alpha_X, \alpha_Z) = \frac{\sqrt{63}}{2} > \frac{\sqrt{51}}{2} = \text{Min-SW}_2(\alpha_X, \alpha_Y) + \text{Min-SW}_2(\alpha_Y, \alpha_Z)$. $\square$

10.3 Quotient Wasserstein and Wasserstein–Procrustes Quotient distances.

Definition 10.16 (Quotient Wasserstein distance). Let a group $\mathcal{G}$ act on a metric space $(\mathcal{X}, d)$ by isometries, and assume that the action preserves finite p th moments. Write $g_\# \alpha$ for the push-forward of a measure α by $g \in \mathcal{G}$, and $[\alpha] := \{g_\# \alpha : g \in \mathcal{G}\}$ for its orbit. The quotient p -Wasserstein distance between the orbits of $\alpha, \beta \in \mathcal{P}_p(\mathcal{X})$ is

$$\mathcal{W}_{p, \mathcal{G}}([\alpha], [\beta]) := \inf_{g, h \in \mathcal{G}} \mathcal{W}_p(g_\# \alpha, h_\# \beta) = \inf_{g \in \mathcal{G}} \mathcal{W}_p(\alpha, g_\# \beta).$$

Proposition 10.17 (Metric property on the quotient). If $\mathcal{G}$ acts by isometries and preserves $\mathcal{P}_p(\mathcal{X})$, then $\mathcal{W}_{p, \mathcal{G}}$ is a pseudo-distance on $\mathcal{P}_p(\mathcal{X})$. If the infimum is attained between every pair of orbits, then it is a distance on the orbit space $\mathcal{P}_p(\mathcal{X})/\mathcal{G}$. This applies in particular to continuous actions of compact groups.

Proof. Isometry of the action gives $\mathcal{W}_p(g_\# \alpha, g_\# \beta) = \mathcal{W}_p(\alpha, \beta)$, which proves the second formula in Definition 10.16. Non-negativity and symmetry are inherited from $\mathcal{W}_p$. For the triangle inequality, choose $g_1, g_2 \in \mathcal{G}$. Then

$$\mathcal{W}_p(\alpha, (g_1)_\# \beta) + \mathcal{W}_p(\beta, (g_2)_\# \gamma) = \mathcal{W}_p(\alpha, (g_1)_\# \beta) + \mathcal{W}_p((g_1)_\# \beta, (g_1 g_2)_\# \gamma).$$

Applying the triangle inequality for $\mathcal{W}_p$ and then taking infima over g_1, g_2 gives the result. If the infimum is attained and the quotient distance is zero, there exist $g, h \in \mathcal{G}$ such that $\mathcal{W}_p(g_\# \alpha, h_\# \beta) = 0$, hence $g_\# \alpha = h_\# \beta$ and therefore $[\alpha] = [\beta]$. For a continuous action of a compact group, the objective is continuous on a compact parameter set, so attainment follows. Without attainment, the construction is naturally a metric only after identifying distinct orbits at zero orbit distance. $\square$

Rigid motions and Wasserstein–Procrustes. $\inf_{R \in \text{O}(d), t \in \mathbb{R}^d, \pi \in \mathcal{U}(\alpha, \beta)} \int \|Rx + t - y\|^2 d\pi(x, y)$.

Proposition 10.18 (Wasserstein–Procrustes upper certificate). Let $p \geq 1$ and $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$. Equip both copies of $\mathbb{R}^d$ with the Euclidean distance and take $\Delta(u, v) = |u - v|$ in (12.9). Then

$$\mathcal{GW}((\mathbb{R}^d, \|\cdot\|, \alpha), (\mathbb{R}^d, \|\cdot\|, \beta)) \leq 2 \mathcal{W}_{p, \text{E}(d)}([\alpha], [\beta]), \quad (10.27)$$

where the quotient distance on the right is defined in Definition 10.16.

Proof. Fix $g, h \in \text{E}(d)$. For every $\pi \in \mathcal{U}(\alpha, \beta)$, the push-forward $\tilde{\pi} = (g \times h)_\# \pi$, where $(g \times h)(x, y) = (g(x), h(y))$, belongs to $\mathcal{U}(g_\# \alpha, h_\# \beta)$. Since rigid motions preserve Euclidean distances, $\|g(x) - g(x')\| = \|x - x'\|$ and $\|h(y) - h(y')\| = \|y - y'\|$. The GW objective therefore has the same value at π and $\tilde{\pi}$. Applying the same argument to g^{-1} and h^{-1} shows that this correspondence is bijective, hence $\mathcal{GW}((\mathbb{R}^d, \|\cdot\|, \alpha), (\mathbb{R}^d, \|\cdot\|, \beta)) = \mathcal{GW}((\mathbb{R}^d, \|\cdot\|, g_\# \alpha), (\mathbb{R}^d, \|\cdot\|, h_\# \beta))$. Proposition 12.9, applied to the aligned measures on the same Euclidean ground space, now gives $\mathcal{GW}((\mathbb{R}^d, \|\cdot\|, \alpha), (\mathbb{R}^d, \|\cdot\|, \beta)) \leq 2 \mathcal{W}_p(g_\# \alpha, h_\# \beta)$. Taking the infimum over $g, h \in \text{E}(d)$ and using Definition 10.16 proves (10.27). $\square$

Thus a good Wasserstein–Procrustes registration forces small GW distortion. The converse need not hold: GW may be small because of an intrinsic correspondence that is not induced by an ambient rigid motion.

$$P^{(k)} \in \arg\min_{P \in \text{U}(a, b)} \sum_{i, j} P_{ij} \|R^{(k)} x_i + t^{(k)} - y_j\|^2. \quad (10.28)$$

$$(R^{(k+1)}, t^{(k+1)}) \in \arg\min_{R \in \text{O}(d), t \in \mathbb{R}^d} \sum_{i, j} P_{ij}^{(k)} \|R x_i + t - y_j\|^2, \quad (10.29)$$

Proposition 10.19 (Rigid update for a fixed coupling). Let $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, and fix $P \in \text{U}(a, b)$. Define $\bar{x} = \sum_i a_i x_i$, $\bar{y} = \sum_j b_j y_j$, $M_P = \sum_{i, j} P_{ij} (y_j - \bar{y})(x_i - \bar{x})^\top$. If $M_P = U \Sigma V^\top$ is a singular value decomposition, then the minimizers over the full orthogonal group of $\min_{R \in \text{O}(d), t \in \mathbb{R}^d} \sum_{i, j} P_{ij} \|R x_i + t - y_j\|^2$ are given by $R^* = UV^\top$ and $t^* = \bar{y} - R^* \bar{x}$, up to the usual non-uniqueness when M_P is rank deficient. If one imposes $R \in \text{SO}(d)$, set $D = \text{diag}(1, \dots, 1, \det(UV^\top))$, $R^* = UDV^\top$, $t^* = \bar{y} - R^* \bar{x}$.

Proof. For fixed R , differentiating with respect to t gives $t = \bar{y} - R\bar{x}$. Substituting this value centers the variables and leaves $\sum_{i,j} P_{ij} \|R(x_i - \bar{x}) - (y_j - \bar{y})\|^2$. The two quadratic terms independent of R are fixed, so the problem is equivalent to maximizing $\sum_{i,j} P_{ij} \langle R(x_i - \bar{x}), y_j - \bar{y} \rangle = \text{tr}(R^\top U \Sigma V^\top) \leq \text{tr}(\Sigma)$ for $R \in O(d)$, with equality for $R = UV^\top$. Under the constraint $R \in SO(d)$, the same argument applies with the additional determinant constraint on $U^\top R V$, which gives the displayed diagonal correction. $\square$

10.4 Vector Quantiles and Linear Optimal Transport

Vector quantiles. Assume that α_{ref} is absolutely continuous. For a target law α with finite second moment, its vector quantile relative to α_{ref} is the Brenier map

$$T_\alpha = \nabla \varphi_\alpha, \quad (T_\alpha)_\# \alpha_{\text{ref}} = \alpha, \quad \min_{T_\# \alpha_{\text{ref}} = \alpha} \int \|x - T(x)\|^2 d\alpha_{\text{ref}}(x).$$

Linearized Wasserstein coordinates.

$$\alpha \mapsto T_\alpha - \text{Id} \in L^2(\alpha_{\text{ref}}; \mathbb{R}^d), \quad \text{LOT}_{\alpha_{\text{ref}}}(\alpha, \beta) = \|T_\alpha - T_\beta\|_{L^2(\alpha_{\text{ref}})}. \quad (10.30)$$

$\text{LOT}_{\alpha_{\text{ref}}}(\alpha_{\text{ref}}, \alpha) = \|T_\alpha - \text{Id}\|_{L^2(\alpha_{\text{ref}})} = \mathcal{W}_2(\alpha_{\text{ref}}, \alpha)$. For two arbitrary targets, the coupling $(T_\alpha, T_\beta)_\# \alpha_{\text{ref}}$ is admissible but not generally optimal, so $\text{LOT}_{\alpha_{\text{ref}}}$ is a tangent-space approximation of the Wasserstein geometry. Uniqueness of the Brenier maps shows that $\text{LOT}_{\alpha_{\text{ref}}}$ is itself a distance on the target measures for which these maps are defined.

$\bar{T} = \sum_s \lambda_s T_{\alpha_s}$, $\bar{\alpha}_{\text{LOT}} = \bar{T}_\# \alpha_{\text{ref}}$. This is exact in one dimension, where quantile functions linearize $\mathcal{W}_2$, and it is especially useful when many barycenters with changing weights must be evaluated quickly.

Proposition 10.20 (Quantitative stability of linear OT). *Let $X, Y \subset \mathbb{R}^d$ be compact convex sets, assume that X has positive Lebesgue measure, and let α_{ref} be normalized Lebesgue measure on X . Then there exists a constant $C = C(d, X, Y)$ such that, for all $\alpha, \beta \in \mathcal{P}(Y)$,*

$$\mathcal{W}_2(\alpha, \beta) \leq \text{LOT}_{\alpha_{\text{ref}}}(\alpha, \beta) \quad \text{and} \quad \text{LOT}_{\alpha_{\text{ref}}}(\alpha, \beta) \leq C \mathcal{W}_1(\alpha, \beta)^{2/15}.$$

Proof. The first inequality is immediate: $(T_\alpha, T_\beta)_\# \alpha_{\text{ref}}$ is a feasible coupling between α and β . The second inequality is the quantitative stability theorem of M  rigot, Delalande and Chazal. Its proof first controls the difference of the two Kantorovich potentials in terms of $\mathcal{W}_1(\alpha, \beta)$, then combines convexity and interpolation estimates to control the gradients in $L^2(\alpha_{\text{ref}})$; the resulting dimension-independent H  lder exponent is $2/15$. In one dimension, quantile functions make the embedding isometric and the exponent improves to one. In several dimensions, the theorem is global but only H  lder; it should not be read as a global Lipschitz estimate in $\mathcal{W}_2$. $\square$

Principal components in linear OT coordinates. $z_i := T_{\alpha_i} - \text{Id}$, $\bar{z} := \frac{1}{N} \sum_{i=1}^N z_i$. $\mathcal{C}_{\text{LOT}} h := \frac{1}{N} \sum_{i=1}^N \langle z_i - \bar{z}, h \rangle_{L^2(\alpha_{\text{ref}})} (z_i - \bar{z})$, $e_k = \frac{1}{\sqrt{N \lambda_k}} \sum_{i=1}^N v_i^{(k)} (z_i - \bar{z})$, $a_{i,k} := \langle z_i - \bar{z}, e_k \rangle_{L^2(\alpha_{\text{ref}})}$. $T_a(x) := x + \bar{z}(x) + \sum_{k=1}^m a_k e_k(x)$, $\alpha_a := (T_a)_\# \alpha_{\text{ref}}$.

10.5 Spectral and Robust Wasserstein Distances

Definition 10.21 (Monotone spectral gauge). A monotone spectral gauge on positive semidefinite matrices is a convex, positively 1-homogeneous map $\gamma : \mathcal{S}_+^d \rightarrow \mathbb{R}_+$ such that $\gamma(M) = 0$ only for $M = 0$, $\gamma(QMQ^\top) = \gamma(M)$ for every orthogonal matrix Q , and $0 \preceq M \preceq N \implies \gamma(M) \leq \gamma(N)$. For $1 \leq q \leq +\infty$, the Schatten gauge

$$\gamma_q(M) := \|M\|_{S_q} = \begin{cases} \left(\sum_{i=1}^d \lambda_i(M)^q \right)^{1/q}, & 1 \leq q < +\infty, \\ \lambda_{\max}(M), & q = +\infty, \end{cases}$$

is a monotone spectral gauge. The cases $q = 1$, $q = 2$ and $q = +\infty$ are respectively the trace, Frobenius and spectral gauges.

Definition 10.22 (Spectral Wasserstein distance). Let γ be a monotone spectral gauge. For a coupling $\pi \in \mathcal{U}(\alpha, \beta)$, define its displacement covariance $M_\pi := \int (x - y)(x - y)^\top d\pi(x, y)$. The spectral Wasserstein distance associated with γ is

$$\mathcal{W}_\gamma(\alpha, \beta)^2 := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \gamma(M_\pi). \quad (10.31)$$

The special case $\gamma(M) = \text{tr}(M)$ gives the usual quadratic Wasserstein distance $\mathcal{W}_2$. The spectral gauge $\gamma(M) = \lambda_{\max}(M)$ instead measures the worst transported variance direction.

$$\mathcal{W}_{2,A}(\alpha, \beta)^2 := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int (x - y)^\top A (x - y) d\pi(x, y) = \mathcal{W}_2((A^{1/2})_\# \alpha, (A^{1/2})_\# \beta)^2. \quad (10.32)$$

$$\mathcal{B}_\gamma := \{A \succeq 0; \text{tr}(AM) \leq \gamma(M) \text{ for all } M \succeq 0\}, \quad (10.33)$$

$\mathcal{B}_{\gamma_q} = \{A \succeq 0; \|A\|_{S_{q^*}} \leq 1\}$, $\frac{1}{q} + \frac{1}{q^*} = 1$, with the usual endpoint conventions. Thus the trace gauge has polar set $\{A : 0 \preceq A \preceq \text{Id}\}$, the Frobenius gauge is self-polar on $\mathcal{S}_+^d$, and the spectral gauge has polar set $\{A \succeq 0 : \text{tr}(A) \leq 1\}$.

Proposition 10.23 (Robust representation and metric equivalence). *Assume, for simplicity, that the measures are compactly supported and that γ is closed and finite on the positive semidefinite cone. Then $\mathcal{W}_\gamma(\alpha, \beta)^2 = \sup_{A \in \mathcal{B}_\gamma} \mathcal{W}_{2,A}(\alpha, \beta)^2$. If there exist constants $0 < a \leq b < +\infty$ such that $a\text{Id} \in \mathcal{B}_\gamma$ and $\mathcal{B}_\gamma \subset \{A : 0 \preceq A \preceq b\text{Id}\}$, equivalently $a \text{tr}(M) \leq \gamma(M) \leq b \text{tr}(M)$ ($M \succeq 0$), then $\sqrt{a} \mathcal{W}_2(\alpha, \beta) \leq \mathcal{W}_\gamma(\alpha, \beta) \leq \sqrt{b} \mathcal{W}_2(\alpha, \beta)$. The robust representation proves that $\mathcal{W}_\gamma$ is a distance: the supremum of pseudodistances gives symmetry and the triangle inequality, while the lower comparison with $\mathcal{W}_2$ gives definiteness. When γ is the restriction of a norm to the positive semidefinite cone, these bounds hold automatically in finite dimension, so $\mathcal{W}_\gamma$ is equivalent to $\mathcal{W}_2$ on measures with finite second moments.*

Proof. Using the polar representation of γ , $\mathcal{W}_\gamma(\alpha, \beta)^2 = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \sup_{A \in \mathcal{B}_\gamma} \text{tr}(AM_\pi)$. The coupling set is convex and compact for weak convergence under compact support. Since γ is a finite gauge on a finite-dimensional cone and vanishes only at the

origin, it is equivalent to the trace norm on the slice $\text{tr}(M) = 1$, so $\mathcal{B}_\gamma$ is convex and compact. The map $(\pi, A) \mapsto \text{tr}(AM_\pi)$ is affine in each variable and continuous. Sion's minimax theorem gives

$$\inf_{\pi} \sup_{A \in \mathcal{B}_\gamma} \text{tr}(AM_\pi) = \sup_{A \in \mathcal{B}_\gamma} \inf_{\pi} \text{tr}(AM_\pi) = \sup_{A \in \mathcal{B}_\gamma} \mathcal{W}_{2,A}(\alpha, \beta)^2.$$

For fixed $A \succeq 0$, $\mathcal{W}_{2,A}$ is the Wasserstein pseudodistance associated with the seminorm $x \mapsto \|A^{1/2}x\|$. Since all terms are nonnegative, the robust identity also gives $\mathcal{W}_\gamma(\alpha, \beta) = \sup_{A \in \mathcal{B}_\gamma} \mathcal{W}_{2,A}(\alpha, \beta)$. A supremum of pseudodistances is symmetric and satisfies the triangle inequality.

If $a\text{Id} \in \mathcal{B}_\gamma$ and $A \preceq b\text{Id}$ for all $A \in \mathcal{B}_\gamma$, then $a\mathcal{W}_2(\alpha, \beta)^2 = \mathcal{W}_{2,a\text{Id}}(\alpha, \beta)^2 \leq \mathcal{W}_\gamma(\alpha, \beta)^2 \leq b\mathcal{W}_2(\alpha, \beta)^2$, which proves definiteness and equivalence with $\mathcal{W}_2$. The equivalence between these operator bounds and $a\text{tr}(M) \leq \gamma(M) \leq b\text{tr}(M)$ follows directly from the polar formula. In finite dimension, any norm restricted to the positive semidefinite cone is equivalent to the trace norm on that cone. The finite-second-moment case follows by truncation when these norm-equivalence bounds hold. $\square$

Definition 10.24 (Subspace robust Wasserstein). For $1 \leq k \leq d$, the Paty–Cuturi subspace robust Wasserstein distance is

$$\text{SRW}_{2,k}(\alpha, \beta) := \sup_{U^\top U = \text{Id}_k} \mathcal{W}_2((U^\top)_\# \alpha, (U^\top)_\# \beta) = \sup_{P^2 = P = P^\top, \text{tr}(P) = k} \mathcal{W}_{2,P}(\alpha, \beta).$$

$\gamma_k(M) = \sum_{\ell=1}^k \lambda_\ell(M)$, $\mathcal{B}_{\gamma_k} = \{A; 0 \preceq A \preceq \text{Id}, \text{tr}(A) \leq k\}$. Thus $k = d$ gives $\gamma_d(M) = \text{tr}(M)$ and recovers $\mathcal{W}_2$. The convex hull of rank- k projectors is $\{A; 0 \preceq A \preceq \text{Id}, \text{tr}(A) = k\}$, and, since $M \succeq 0$, the associated support function is the same Ky Fan gauge. Thus $\mathcal{W}_{\gamma_k}$ is the convexified spectral counterpart of $\text{SRW}_{2,k}$, while $\text{SRW}_{2,k}$ keeps the original non-convex rank constraint.

$$\text{SRW}_{2,k}(\alpha, \beta) \leq \mathcal{W}_{\gamma_k}(\alpha, \beta) \leq \mathcal{W}_2(\alpha, \beta), \quad \sqrt{\frac{k}{d}} \mathcal{W}_2(\alpha, \beta) \leq \mathcal{W}_{\gamma_k}(\alpha, \beta),$$

$$\left\{ \frac{k}{d} \text{Id} \right\} \cup \{P; P^2 = P = P^\top, \text{tr}(P) = k\} \subset \mathcal{B}_{\gamma_k} \subset \{A; 0 \preceq A \preceq \text{Id}\}.$$

For $k = 1$, $\gamma_1(M) = \lambda_{\max}(M)$ and $\mathcal{B}_{\gamma_1} = \{A \succeq 0; \text{tr}(A) \leq 1\}$.

10.6 Conditional Wasserstein Distances

Definition 10.25 (Conditional couplings and conditional OT). Let (S, λ) be a standard Borel probability space of conditions and let (Ω, d) be a Polish metric space. For $p \geq 1$, denote by $\mathcal{P}_{p,\lambda}(S \times \Omega)$ the set of probability measures on $S \times \Omega$ whose first marginal is λ and whose disintegration $\alpha(ds, dx) = \alpha_s(dx)\lambda(ds)$ satisfies $\int_S \int_\Omega d(x, x_0)^p d\alpha_s(x) d\lambda(s) < +\infty$ for some, hence every, $x_0 \in \Omega$. For $\alpha, \beta \in \mathcal{P}_{p,\lambda}(S \times \Omega)$, a conditional coupling is a measure $\Pi(ds, dx, dy) = \pi_s(dx, dy)\lambda(ds)$ such that $\pi_s \in \mathcal{U}(\alpha_s, \beta_s)$ for λ -a.e. s . The set of such couplings is denoted $\mathcal{U}_\lambda(\alpha, \beta)$. Let $(s, x, y) \mapsto c_s(x, y)$ be jointly Borel, nonnegative, and lower semicontinuous in (x, y) for every s . The conditional OT value is

$$\begin{aligned} \mathcal{L}_c^\lambda(\alpha, \beta) &:= \inf_{\Pi \in \mathcal{U}_\lambda(\alpha, \beta)} \int_S \int_{\Omega \times \Omega} c_s(x, y) d\pi_s(x, y) d\lambda(s) \\ &= \int_S \inf_{\pi_s \in \mathcal{U}(\alpha_s, \beta_s)} \int_{\Omega \times \Omega} c_s(x, y) d\pi_s(x, y) d\lambda(s). \end{aligned} \quad (10.34)$$

Definition 10.26 (Conditional Wasserstein distance). For the fiber-independent cost $c_s(x, y) = c_p(x, y) := d(x, y)^p$, define

$$\mathcal{W}_{p,\lambda}(\alpha, \beta) := \left(\mathcal{L}_{c_p}^\lambda(\alpha, \beta) \right)^{1/p} = \left(\int_S \mathcal{W}_p^p(\alpha_s, \beta_s) d\lambda(s) \right)^{1/p}. \quad (10.35)$$

Thus $\mathcal{L}_c^\lambda$ is the general conditional Kantorovich value, whereas $\mathcal{W}_{p,\lambda}$ is its metric specialization to the constant family $c_s = c_p = d^p$, after taking the p th root. Equivalently, $\mathcal{W}_{p,\lambda}(\alpha, \beta)^p = \mathcal{L}_{c_p}^\lambda(\alpha, \beta)$.

Proposition 10.27 (Metric property). For $p \geq 1$, $\mathcal{W}_{p,\lambda}$ is a distance on $\mathcal{P}_{p,\lambda}(S \times \Omega)$. Moreover, this metric space is Polish.

Proof. Non-negativity and symmetry follow from the corresponding properties of $\mathcal{W}_p$ on each fiber. If $\mathcal{W}_{p,\lambda}(\alpha, \beta) = 0$, then $\mathcal{W}_p(\alpha_s, \beta_s) = 0$ for λ -a.e. s , hence $\alpha_s = \beta_s$ for λ -a.e. s and the disintegrated measures α and β coincide. For the triangle inequality, let $\gamma \in \mathcal{P}_{p,\lambda}(S \times \Omega)$. Since $\mathcal{W}_p(\alpha_s, \gamma_s) \leq \mathcal{W}_p(\alpha_s, \beta_s) + \mathcal{W}_p(\beta_s, \gamma_s)$ for a.e. s , Minkowski's inequality in $L^p(S, \lambda)$ gives $\mathcal{W}_{p,\lambda}(\alpha, \gamma) \leq \mathcal{W}_{p,\lambda}(\alpha, \beta) + \mathcal{W}_{p,\lambda}(\beta, \gamma)$. To see completeness and separability, identify a measure with the λ -a.e. equivalence class of its measurable disintegration map $s \mapsto \alpha_s \in \mathcal{P}_p(\Omega)$. Formula (10.35) is the corresponding $L^p(S, \lambda; \mathcal{P}_p(\Omega))$ metric. Since $(\mathcal{P}_p(\Omega), \mathcal{W}_p)$ is Polish, the standard L^p space of measurable metric-valued maps is Polish as well. $\square$

Proposition 10.28 (Conditional geodesics). Assume that (Ω, d) is a geodesic Polish space and let $\alpha_0, \alpha_1 \in \mathcal{P}_{p,\lambda}(S \times \Omega)$. For λ -a.e. s , choose a constant-speed $\mathcal{W}_p$ geodesic $t \mapsto \alpha_{t,s}$ between $\alpha_{0,s}$ and $\alpha_{1,s}$, measurably in s , and set $\alpha_t(ds, dx) = \alpha_{t,s}(dx)\lambda(ds)$. Then $t \mapsto \alpha_t$ is a constant-speed geodesic for $\mathcal{W}_{p,\lambda}$.

Proof. For $0 \leq r \leq t \leq 1$, the fiberwise geodesic property gives $\mathcal{W}_p(\alpha_{r,s}, \alpha_{t,s}) = (t - r) \mathcal{W}_p(\alpha_{0,s}, \alpha_{1,s})$ for λ -a.e. s . Integrating the p th power over S gives $\mathcal{W}_{p,\lambda}(\alpha_r, \alpha_t) = (t - r) \mathcal{W}_{p,\lambda}(\alpha_0, \alpha_1)$. Hence the conditional curve has constant speed and realizes the distance between its endpoints. In Euclidean space one may take, for each s , an optimal plan $\pi_s \in \mathcal{U}(\alpha_{0,s}, \alpha_{1,s})$ and set $\alpha_{t,s} = ((1 - t)p_1 + tp_2)_\# \pi_s$, where p_1, p_2 are the two coordinate projections. On a general geodesic space, the same construction uses a measurable family of optimal dynamical plans. Such a family can be selected measurably in the Polish setting; when geodesics are non-unique, different selections can produce different conditional geodesics. $\square$

11 Generalized OT Problems

11.1 OT Barycenters

Fréchet means. The natural formulation is a Fréchet-mean problem on the space of probability measures: the unknown is the barycenter measure itself, and its support is not prescribed. It uses the continuous Kantorovich value $\mathcal{L}_c$ defined in (3.7).

Definition 11.1 (Optimal-transport barycenter). Given input measures $(\beta_s)_{s=1}^S$ on a space $\mathcal{X}$ and weights $\lambda \in \Sigma_S$, discard any zero-weight inputs. An *optimal-transport barycenter* of this weighted family is any solution of

$$\min_{\alpha \in \mathcal{M}_+^1(\mathcal{X})} \sum_{s=1}^S \lambda_s \mathcal{L}_c(\alpha, \beta_s). \quad (11.1)$$

Fixed-support restriction. $\beta_s = \sum_{j=1}^{n_s} b_{s,j} \delta_{x_{s,j}}$, $\mathbf{b}_s = (b_{s,j})_{j=1}^{n_s} \in \Sigma_{n_s}$. $(C_s)_{ij} = c(y_i, x_{s,j}) \in \mathbb{R}^{n \times n_s}$. Thus its value is $\mathcal{L}_{C_s}(\mathbf{a}, \mathbf{b}_s)$, in the notation of the discrete Kantorovich problem (3.2).

Definition 11.2 (Fixed-support discrete OT barycenter). With the candidate sites and cost matrices above, a *fixed-support discrete OT barycenter* is a measure $\alpha^* = \sum_i a_i^* \delta_{y_i}$ whose weight vector $\mathbf{a}^* = (a_i^*)_{i=1}^n$ solves

$$\min_{\mathbf{a} \in \Sigma_n} \sum_{s=1}^S \lambda_s \mathcal{L}_{C_s}(\mathbf{a}, \mathbf{b}_s), \quad (11.2)$$

$$\min_{Y=(y_i)_{i=1}^n \in (\mathbb{R}^d)^n} \sum_{s=1}^S \lambda_s \mathcal{L}_{C_s(Y)}(\mathbf{a}, \mathbf{b}_s), \quad [C_s(Y)]_{ij} = \|y_i - x_{s,j}\|^2.$$

For fixed Y , the objective is the convex problem (11.2) in $\mathbf{a}$; its dependence on Y is generally nonconvex, so the joint problem is nonconvex. Thus choosing the support is itself a significant computational part of the barycenter problem, rather than a harmless preprocessing step.

Proposition 11.3 (Two-measure barycenters are Wasserstein geodesics). *Let $\beta_0, \beta_1 \in \mathcal{P}_2(\mathbb{R}^d)$ and $t \in [0, 1]$. For $S = 2$, $c(x, y) = \|x - y\|^2$ and weights $(1 - t, t)$, the minimizers of the barycenter problem (11.1) are exactly the time- t points of constant-speed $\mathcal{W}_2$ geodesics from β_0 to β_1 . The minimum value is $t(1 - t) \mathcal{W}_2^2(\beta_0, \beta_1)$. In particular, if π is any optimal coupling between β_0 and β_1 , then $\alpha_t := ((x, y) \mapsto (1 - t)x + ty)_{\#} \pi$ is a barycenter. If β_0 has a density, the barycenter is unique and $\alpha_t = ((1 - t)\text{Id} + tT)_{\#} \beta_0$, where T is the Brenier map from β_0 to β_1 .*

Proof. The endpoint cases $t \in \{0, 1\}$ are immediate, so assume $0 < t < 1$. For any candidate $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, set $A = \mathcal{W}_2(\beta_0, \alpha)$, $B = \mathcal{W}_2(\alpha, \beta_1)$, $D = \mathcal{W}_2(\beta_0, \beta_1)$. The identity $(1 - t)A^2 + tB^2 = t(1 - t)(A + B)^2 + ((1 - t)A - tB)^2$ and the triangle inequality $A + B \geq D$ give $(1 - t)\mathcal{W}_2^2(\alpha, \beta_0) + t\mathcal{W}_2^2(\alpha, \beta_1) \geq t(1 - t)D^2$. For the interpolation α_t induced by an optimal coupling, Proposition 3.35 gives $A = tD$ and $B = (1 - t)D$, so the lower bound is attained.

Conversely, every minimizer must attain equality in both inequalities above. Hence $A + B = D$ and $(1 - t)A = tB$, which imply $A = tD$ and $B = (1 - t)D$. Parameterize a constant-speed geodesic from β_0 to α on $[0, t]$ and one from α to β_1 on $[t, 1]$; both pieces then have speed D . Their concatenation is globally geodesic: any shorter path between points on opposite pieces, combined with the remaining subsegments, would contradict $D = \mathcal{W}_2(\beta_0, \beta_1)$. Thus the concatenated geodesic passes through α at time t . Finally, if β_0 has a density, Brenier's theorem (Theorem 2.13) gives the unique optimal coupling $(\text{Id}, T)_{\#} \beta_0$, and the displayed McCann interpolation follows from Definition 2.17. $\square$

Proposition 11.4 (Mean and support of quadratic Wasserstein barycenters). *Let $\beta_1, \dots, \beta_S \in \mathcal{P}_2(\mathbb{R}^d)$, let $\lambda \in \Sigma_S$, and let α^* be a barycenter for the squared Euclidean cost. Define*

$$K := \overline{\text{conv}} \left(\bigcup_{\{s: \lambda_s > 0\}} \text{supp}(\beta_s) \right).$$

Then $\text{supp}(\alpha^) \subset K$, $\int_{\mathbb{R}^d} x \, d\alpha^*(x) = \sum_{s=1}^S \lambda_s \int_{\mathbb{R}^d} x \, d\beta_s(x)$.*

Proof. Write $m_\alpha = \int x \, d\alpha(x)$ and let α^0 be its centered translate, as in Proposition 3.32. That proposition gives, for every candidate α , $\sum_{s=1}^S \lambda_s \mathcal{W}_2^2(\alpha, \beta_s) = \sum_{s=1}^S \lambda_s \|m_\alpha - m_{\beta_s}\|^2 + \sum_{s=1}^S \lambda_s \mathcal{W}_2^2(\alpha^0, \beta_s^0)$. For a fixed centered law α^0 , the second sum is independent of m_α , whereas the first is uniquely minimized at $m_\alpha = \sum_s \lambda_s m_{\beta_s}$. Translating α^* without changing its centered law therefore proves the mean identity.

For the support statement, let P_K be the Euclidean metric projection onto the closed convex set K . For every $y \in K$, the projection inequality gives $\|x - y\|^2 \geq \|P_K(x) - y\|^2 + \|x - P_K(x)\|^2$. For each s with $\lambda_s > 0$, let π_s be an optimal coupling between α^* and β_s . Since $\beta_s(K) = 1$, pushing π_s forward by (P_K, Id) and integrating the preceding inequality yields $\mathcal{W}_2^2((P_K)_{\#} \alpha^*, \beta_s) \leq \mathcal{W}_2^2(\alpha^*, \beta_s) - \int_{\mathbb{R}^d} \text{dist}(x, K)^2 \, d\alpha^*(x)$. After multiplication by λ_s and summation, barycenter optimality forces the last integral to vanish. Hence $\alpha^*(K) = 1$, and the closedness of K gives $\text{supp}(\alpha^*) \subset K$. $\square$

If all input supports are compact, their finite union is compact and so is its convex hull; in that case the closure in the definition of K is unnecessary.

One-dimensional case.

Proposition 11.5 (Quantile barycenters on the line). *For $\mathcal{X} = \mathbb{R}$ and $c(x, y) = |x - y|^2$, the quantile function of a Wasserstein barycenter is the weighted average of the input quantile functions: $\mathcal{C}_{\alpha^*}^{-1}(r) = \sum_{s=1}^S \lambda_s \mathcal{C}_{\beta_s}^{-1}(r)$, $r \in (0, 1)$.*

Proof. The one-dimensional formula (2.11) gives $\sum_s \lambda_s \mathcal{W}_2^2(\alpha, \beta_s) = \int_0^1 \sum_s \lambda_s \left| \mathcal{C}_\alpha^{-1}(r) - \mathcal{C}_{\beta_s}^{-1}(r) \right|^2 \, dr$. The minimization decouples pointwise in r . For each fixed r , the minimizer of $z \mapsto \sum_s \lambda_s |z - \mathcal{C}_{\beta_s}^{-1}(r)|^2$ is the weighted average $\sum_s \lambda_s \mathcal{C}_{\beta_s}^{-1}(r)$. This function is nondecreasing because it is a positive weighted sum of nondecreasing quantile functions, hence it is a valid quantile function. $\square$

Gaussian case.

Proposition 11.6 (Nondegenerate Gaussian inputs remain Gaussian). *Let the positive-weight inputs be $\beta_s = \mathcal{N}(m_s, \Sigma_s)$, with $\Sigma_s \succeq 0$, and assume that at least one Σ_s is positive definite. The quadratic Wasserstein barycenter is unique and has the form $\alpha^* = \mathcal{N}(m, \Sigma)$, where $m = \sum_s \lambda_s m_s$. Its positive-definite covariance Σ is the unique minimizer of the Bures objective*

$$\mathcal{E}(X) := \sum_s \lambda_s \mathcal{B}(X, \Sigma_s)^2. \quad (11.3)$$

For $X \in \mathbb{S}_{++}^d$, define

$$R_s(X) := \left(X^{1/2} \Sigma_s X^{1/2} \right)^{1/2}, \quad T_s(X) := X^{-1/2} R_s(X) X^{-1/2}, \quad (11.4)$$

$$\bar{T}(X) := \sum_s \lambda_s T_s(X), \quad \Psi_\lambda(X) := \sum_s \lambda_s R_s(X). \quad (11.5)$$

Here $T_s(X)$ is the positive-semidefinite linear optimal map from $\mathcal{N}(0, X)$ to $\mathcal{N}(0, \Sigma_s)$. The covariance Σ is equivalently characterized by either of the stationarity equations

$$\bar{T}(\Sigma) = \text{Id} \quad \Longleftrightarrow \quad \Psi_\lambda(\Sigma) = \Sigma. \quad (11.6)$$

In dimension one, if $\sigma_s = \sqrt{\Sigma_s}$ denotes the input standard deviation, then the barycenter standard deviation is $\sigma = \sum_s \lambda_s \sigma_s$.

Proof. Let $\text{Gauss}(\alpha)$ denote the Gaussian measure with the same mean and covariance as α . For every competitor $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$ and every Gaussian input β_s , the Gelbrich contraction of Theorem 15.14 gives $\mathcal{W}_2^2(\text{Gauss}(\alpha), \beta_s) = \mathcal{W}_2^2(\text{Gauss}(\alpha), \text{Gauss}(\beta_s)) \leq \mathcal{W}_2^2(\alpha, \beta_s)$. Summing with weights λ_s shows that moment-matched Gaussian projection cannot increase the barycenter objective. Since a barycenter exists, projecting any minimizer therefore produces a Gaussian barycenter. The input with positive-definite covariance is absolutely continuous; the uniqueness criterion stated after Definition 11.1 then implies that the barycenter itself is this Gaussian measure.

For a Gaussian candidate, formula (2.20) separates the objective as

$$\sum_s \lambda_s \mathcal{W}_2^2(\mathcal{N}(m, \Sigma), \mathcal{N}(m_s, \Sigma_s)) = \sum_s \lambda_s \|m - m_s\|^2 + \sum_s \lambda_s \mathcal{B}(\Sigma, \Sigma_s)^2.$$

The first term is uniquely minimized at $m = \sum_s \lambda_s m_s$. The second is the Bures barycenter problem. Uniqueness of the Wasserstein barycenter makes its covariance minimizer unique, and the presence of a positive-definite input makes this minimizer positive definite.

First suppose that $\Sigma_s \succ 0$. Cyclicity of the trace gives $\text{tr } R_s(X) = \text{tr} \left(\Sigma_s^{1/2} X \Sigma_s^{1/2} \right)^{1/2}$. For a symmetric perturbation Z , differentiation of the trace of the matrix square root yields

$$D[\text{tr } R_s(X)] [Z] = \frac{1}{2} \text{tr}(T_s(X) Z).$$

$$D\mathcal{E}(X)[Z] = \text{tr}((\text{Id} - \bar{T}(X))Z). \quad (11.7)$$

The same identity for singular Σ_s follows by replacing Σ_s with $\Sigma_s + \delta \text{Id}$ and letting $\delta \downarrow 0$. Since the minimizer lies in $\mathbb{S}_{++}^d$, first-order optimality is $\bar{T}(\Sigma) = \text{Id}$. Finally, definitions (11.4)–(11.5) give $\Psi_\lambda(X) = X^{1/2} \bar{T}(X) X^{1/2}$, so $\bar{T}(\Sigma) = \text{Id}$ is equivalent to $\Psi_\lambda(\Sigma) = \Sigma$. Conversely, a positive-definite solution of this equation is stationary for the convex objective (11.3), hence is its unique minimizer. In dimension one, $\mathcal{B}(\sigma^2, \sigma_s^2)^2 = (\sigma - \sigma_s)^2$, whose minimizer is $\sigma = \sum_s \lambda_s \sigma_s$. $\square$

Remark 11.7 (Forward KL barycenter of Gaussian laws). Assume that every $\Sigma_s \succ 0$. The forward KL barycenter $\alpha_{\text{KL}} = \arg\min_\alpha \sum_s \lambda_s \text{KL}(\alpha \mid \beta_s)$ is $\mathcal{N}(m_{\text{KL}}, \Sigma_{\text{KL}})$, where

$$\Sigma_{\text{KL}} = \left(\sum_s \lambda_s \Sigma_s^{-1} \right)^{-1}, \quad m_{\text{KL}} = \Sigma_{\text{KL}} \sum_s \lambda_s \Sigma_s^{-1} m_s.$$

Thus its covariance is the weighted harmonic mean and, unlike for the Wasserstein barycenter, its mean depends on the covariances. This follows by normalizing the geometric mean of the Gaussian densities. The KL direction matters: reverse-KL averaging gives the mixture $\sum_s \lambda_s \beta_s$, generally not a Gaussian.

$x \mapsto m_s + T_s(X)(x - m)$.

$$\mathcal{K}_1(X) := \bar{T}(X) X \bar{T}(X) = X^{-1/2} \Psi_\lambda(X)^2 X^{-1/2}. \quad (11.8)$$

$$B_\eta(X) := (1 - \eta) \text{Id} + \eta \bar{T}(X)$$

$$\mathcal{K}_\eta(X) := B_\eta(X) X B_\eta(X). \quad (11.9)$$

Proposition 11.8 (Convergence of the Gaussian transport fixed-point iteration). *Assume that $\lambda_s > 0$, $\Sigma_s \succeq 0$, and that at least one Σ_s is positive definite. Fix $\eta \in (0, 1]$ and $X_0 \succ 0$, and iterate $X_{\ell+1} = \mathcal{K}_\eta(X_\ell)$. Then every X_ℓ is positive definite and $X_\ell \rightarrow \Sigma$, where Σ is the covariance of the Gaussian barycenter in Proposition 11.6. More precisely, with $\mathcal{R}(X) := \text{tr}((\text{Id} - \bar{T}(X))X(\text{Id} - \bar{T}(X)))$, one has*

$$\mathcal{E}(X_{\ell+1}) \leq \mathcal{E}(X_\ell) - \eta(2 - \eta) \mathcal{R}(X_\ell), \quad (11.10)$$

and therefore $\sum_{\ell \geq 0} \mathcal{R}(X_\ell) < +\infty$. Equivalently, $\mathcal{B}(X_\ell, X_{\ell+1})^2 = \eta^2 \mathcal{R}(X_\ell)$.

Proof. Fix $X \succ 0$, abbreviate $T_s = T_s(X)$, $\bar{T} = \bar{T}(X)$ and $B = B_\eta(X)$, and let $Z \sim \mathcal{N}(0, X)$. Since $T_s Z \sim \mathcal{N}(0, \Sigma_s)$, the pair $(BZ, T_s Z)$ is an admissible coupling between $\mathcal{N}(0, \mathcal{K}_\eta(X))$ and $\mathcal{N}(0, \Sigma_s)$. For every $z \in \mathbb{R}^d$, the Euclidean variance decomposition around $\bar{T}z = \sum_s \lambda_s T_s z$ gives

$$\begin{aligned} \sum_s \lambda_s \|Bz - T_s z\|^2 &= \sum_s \lambda_s \|\bar{T}z - T_s z\|^2 + (1 - \eta)^2 \|z - \bar{T}z\|^2, \\ \sum_s \lambda_s \|z - T_s z\|^2 &= \sum_s \lambda_s \|\bar{T}z - T_s z\|^2 + \|z - \bar{T}z\|^2. \end{aligned}$$

Integrating and bounding each Wasserstein cost by the admissible coupling $(BZ, T_s Z)$ proves (11.10). Moreover, $B \succ 0$: this is immediate when $\eta < 1$, and for $\eta = 1$ it follows because the weighted sum $\bar{T}$ contains at least one positive-definite term. Hence $z \mapsto Bz$ is the optimal map from $\mathcal{N}(0, X)$ to $\mathcal{N}(0, \mathcal{K}_\eta(X))$, which proves the displayed identity for the Bures step length.

Set $\delta(X) := \det(X)^{1/(2d)}$, $c_0 := \sum_s \lambda_s \det(\Sigma_s)^{1/(2d)} > 0$. The Minkowski determinant inequality, applied to the positive-semidefinite matrices Id and $T_s(X)$ with weights $1 - \eta$ and $\eta \lambda_s$, gives

$$\begin{aligned} \delta(\mathcal{K}_\eta(X)) &= \delta(X) \det(B_\eta(X))^{1/d} \\ &\geq (1 - \eta)\delta(X) + \eta \sum_s \lambda_s \delta(X) \det(T_s(X))^{1/d} \\ &= (1 - \eta)\delta(X) + \eta c_0, \end{aligned}$$

where the last identity uses $T_s(X) X T_s(X) = \Sigma_s$. It follows that $\delta(X_\ell) \geq \min\{\delta(X_0), c_0\} > 0$ for every ℓ .

The decreasing energy also bounds the traces. Indeed, the Schatten Cauchy–Schwarz inequality in the Bures formula gives $\mathcal{B}(X, \Sigma_s)^2 \geq \left(\sqrt{\text{tr } X} - \sqrt{\text{tr } \Sigma_s}\right)^2$, so (11.10) bounds $\text{tr } X_\ell$ uniformly. A uniform trace bound together with the determinant lower bound places all iterates in a compact subset of $\mathbb{S}_{++}^d$.

Summing (11.10) gives $\mathcal{R}(X_\ell) \rightarrow 0$. Every subsequence therefore has a further subsequence converging to some $X_\infty \succ 0$. Continuity of the matrix square root and inverse on the preceding compact set yields $\mathcal{R}(X_\infty) = 0$, hence $\bar{T}(X_\infty) = \text{Id}$. Proposition 11.6 identifies X_∞ with the unique covariance Σ . Since every cluster point is Σ , the full sequence converges to Σ . The undamped case $\eta = 1$ is the Gaussian specialization of Theorem 4.2 of Álvarez-Esteban et al.; the argument above also proves the damped extension. $\square$

Sinkhorn for barycenters.

$$\min_{a \in \Sigma_n} \sum_{s=1}^S \lambda_s L_{C_s}^\varepsilon(a, b_s) \quad (11.11)$$

$$\min_{(P_s)_s} \left\{ \varepsilon \sum_s \lambda_s \text{KL}(P_s | K_s) ; \forall s, P_s^\top \mathbf{1}_n = b_s, \quad P_1 \mathbf{1}_{n_1} = \dots = P_S \mathbf{1}_{n_S} \right\}, \quad (11.12)$$

where $K_s := e^{-C_s/\varepsilon}$. The barycenter a is implicitly encoded in the common row marginal $a = P_1 \mathbf{1}_{n_1} = \dots = P_S \mathbf{1}_{n_S}$.

$$P_s = \text{diag}(u_s) K_s \text{diag}(v_s), \quad (11.13)$$

$$\forall s \in \llbracket 1, S \rrbracket, \quad v_s^{(\ell+1)} := \frac{b_s}{K_s^\top u_s^{(\ell)}}, \quad (11.14)$$

$$\forall s \in \llbracket 1, S \rrbracket, \quad u_s^{(\ell+1)} := \frac{a^{(\ell+1)}}{K_s v_s^{(\ell+1)}}, \quad (11.15)$$

$$\text{where } a^{(\ell+1)} := \prod_s (K_s v_s^{(\ell+1)})^{\lambda_s}. \quad (11.16)$$

$g_s^+ = \varepsilon \log \left(\frac{b_s}{K_s^\top e^{f_s/\varepsilon}} \right)$, $q_s := K_s e^{g_s^+/\varepsilon}$, $a^+ = \prod_s q_s^{\lambda_s}$, $f_s^+ = \varepsilon \log \left(\frac{a^+}{q_s} \right)$. Thus a complete generalized Sinkhorn cycle is exact two-block coordinate ascent on the dual, or equivalently alternating minimization of its negative. Indeed, the constraint follows from $\log a^+ = \sum_s \lambda_s \log q_s$, while the first-order conditions require $e^{f_s^+/\varepsilon} q_s = a^+$ for every s .

Proposition 11.9 (Dual of entropic barycenters). *Under the preceding positivity assumptions, the optimal scalings in (11.13) can be written as $(u_s, v_s) = (e^{f_s/\varepsilon}, e^{g_s/\varepsilon})$, where $(f_s, g_s)_s$ solve the dual problem*

$$\max_{(f_s, g_s)_s} \left\{ \sum_s \lambda_s \left(\langle g_s, b_s \rangle - \varepsilon \langle K_s e^{g_s/\varepsilon}, e^{f_s/\varepsilon} \rangle + \varepsilon \sum_{i,j} (K_s)_{i,j} \right) ; \sum_s \lambda_s f_s = 0 \right\}. \quad (11.17)$$

Proof. Introduce Lagrange multipliers in (11.12):

$$\min_{(P_s)_s, a} \max_{(f_s, g_s)_s} \sum_s \lambda_s \left(\varepsilon \text{KL}(P_s | K_s) + \langle a - P_s \mathbf{1}_{n_s}, f_s \rangle + \langle b_s - P_s^\top \mathbf{1}_n, g_s \rangle \right).$$

The explicit constraint $a \in \Sigma_n$ may be dropped here: nonnegativity of the couplings, together with $P_s \mathbf{1}_{n_s} = a$ and $P_s^\top \mathbf{1}_n = b_s$, already forces $a \in \Sigma_n$. We may therefore minimize the Lagrangian over $a \in \mathbb{R}^n$. Strong duality holds, so one can exchange the

minimum and maximum. Finiteness of the minimum with respect to $\mathbf{a}$ gives the vector constraint $\sum_s \lambda_s \mathbf{f}_s = 0$, while minimization with respect to $\mathbf{P}_s$ gives the Legendre transform of $\text{KL}(\cdot | \mathbf{K}_s)$:

$$\max_{(\mathbf{f}_s, \mathbf{g}_s)_s} \sum_s \lambda_s \left[\langle \mathbf{g}_s, \mathbf{b}_s \rangle - \varepsilon \text{KL}^* \left(\frac{\mathbf{f}_s \oplus \mathbf{g}_s}{\varepsilon} \middle| \mathbf{K}_s \right) \right], \quad \sum_s \lambda_s \mathbf{f}_s = 0.$$

The separable conjugate is

$$\text{KL}^*(\mathbf{U} | \mathbf{K}) = \sum_{i,j} \mathbf{K}_{i,j} (e^{\mathbf{U}_{i,j}} - 1), \quad (11.18)$$

because for $k > 0$, $\sup_{r \geq 0} ur - (r \log(r/k) - r + k) = k(e^u - 1)$, and the case $k = 0$ follows by lower semicontinuity. Substituting this conjugate gives (11.17), including its additive constant. The coordinate maximization in $\mathbf{g}_s$ gives (11.14); the block maximization in all $(\mathbf{f}_s)_s$ under $\sum_s \lambda_s \mathbf{f}_s = 0$ gives the weighted geometric mean (11.16) and then (11.15). $\square$

Wasserstein-over-Wasserstein and barycenters. $\int_{\mathcal{P}(\Omega)} \mathcal{L}_c(\beta_0, \alpha) d\mathfrak{A}(\alpha) < +\infty$, $\mathcal{B}_c(\mathfrak{A}) := \text{Argmin}_{\beta \in \mathcal{P}(\Omega)} \int_{\mathcal{P}(\Omega)} \mathcal{L}_c(\beta, \alpha) d\mathfrak{A}(\alpha)$.
 $\tilde{\alpha}_{\mathfrak{A}} := \text{argmin}_{\beta \in \mathcal{P}(\Omega)} \int_{\mathcal{P}(\Omega)} \mathcal{L}_c(\beta, \alpha) d\mathfrak{A}(\alpha)$,

Proposition 11.10 (Law of large numbers for barycenters over measures). *Let Ω be a compact metric space, let $c : \Omega \times \Omega \rightarrow \mathbb{R}$ be continuous, and let $\mathfrak{A} \in \mathcal{P}(\mathcal{P}(\Omega))$. Let $\alpha_1, \alpha_2, \dots$ be independent random probability measures with common law $\mathfrak{A}$, and set $\hat{\mathfrak{A}}_p := \frac{1}{p} \sum_{i=1}^p \delta_{\alpha_i} \in \mathcal{P}(\mathcal{P}(\Omega))$. Then, almost surely,*

$$\hat{\mathfrak{A}}_p \rightarrow \mathfrak{A} \quad \text{in } \mathcal{P}(\mathcal{P}(\Omega)), \quad (11.19)$$

$$\bar{\alpha}_{\hat{\mathfrak{A}}_p} \rightarrow \bar{\alpha}_{\mathfrak{A}} \quad \text{in } \mathcal{P}(\Omega). \quad (11.20)$$

Moreover, if $\mathcal{B}_c(\mathfrak{A}) = \{\tilde{\alpha}_{\mathfrak{A}}\}$ is a singleton and if $\tilde{\alpha}_{\hat{\mathfrak{A}}_p} \in \mathcal{B}_c(\hat{\mathfrak{A}}_p)$ is any empirical barycenter, then, almost surely,

$$\tilde{\alpha}_{\hat{\mathfrak{A}}_p} \rightarrow \tilde{\alpha}_{\mathfrak{A}} \quad \text{in } \mathcal{P}(\Omega). \quad (11.21)$$

Proof. Set $K = \mathcal{P}(\Omega)$. Since Ω is compact metric, K is compact metric for weak convergence, and so is $\mathcal{P}(K)$. The space $\mathcal{C}(K)$ is separable for the uniform norm. Applying the scalar strong law to a countable dense family of test functions and then using uniform approximation gives, almost surely, for every $\Phi \in \mathcal{C}(K)$, $\int \Phi(\alpha) d\hat{\mathfrak{A}}_p(\alpha) = \frac{1}{p} \sum_{i=1}^p \Phi(\alpha_i) \rightarrow \int \Phi(\alpha) d\mathfrak{A}(\alpha)$. This is exactly (11.19).

For the collapsed mixtures, take $f \in \mathcal{C}(\Omega)$ and define $\Phi_f(\alpha) = \int_{\Omega} f d\alpha$. This function is continuous on $\mathcal{P}(\Omega)$. Hence

$$\int_{\Omega} f d\bar{\alpha}_{\hat{\mathfrak{A}}_p} = \int_{\mathcal{P}(\Omega)} \Phi_f(\alpha) d\hat{\mathfrak{A}}_p(\alpha) \rightarrow \int_{\mathcal{P}(\Omega)} \Phi_f(\alpha) d\mathfrak{A}(\alpha) = \int_{\Omega} f d\bar{\alpha}_{\mathfrak{A}},$$

which proves (11.20).

Since c is continuous on the compact product $\Omega \times \Omega$, the value $(\beta, \alpha) \mapsto \mathcal{L}_c(\beta, \alpha)$ is continuous on K^2 . Therefore the map $\beta \mapsto h_{\beta}$, where $h_{\beta}(\alpha) = \mathcal{L}_c(\beta, \alpha)$, is continuous from the compact space K to $\mathcal{C}(K)$. Its image $\mathcal{H} := \{h_{\beta} : \beta \in K\}$ is compact in $\mathcal{C}(K)$. The convergence of $\hat{\mathfrak{A}}_p$ to $\mathfrak{A}$ is uniform over $\mathcal{H}$: given $\eta > 0$, cover $\mathcal{H}$ by finitely many η -balls in $\|\cdot\|_{\infty}$, use weak convergence for the centers, and bound the error on the balls by the total variation of $\hat{\mathfrak{A}}_p - \mathfrak{A}$. Hence the empirical objectives $F_p(\beta) := \int \mathcal{L}_c(\beta, \alpha) d\hat{\mathfrak{A}}_p(\alpha)$ converge uniformly on K to $F(\beta) := \int \mathcal{L}_c(\beta, \alpha) d\mathfrak{A}(\alpha)$. Let $\beta_p \in \mathcal{B}_c(\hat{\mathfrak{A}}_p)$. Compactness gives a subsequence $\beta_{p_k} \rightharpoonup \beta$. Uniform convergence, continuity of F , and optimality of β_{p_k} give, for any $\gamma \in K$, $F(\beta) = \lim_k F_{p_k}(\beta_{p_k}) \leq \lim_k F_{p_k}(\gamma) = F(\gamma)$, so $\beta \in \mathcal{B}_c(\mathfrak{A})$. If this set is the singleton $\{\tilde{\alpha}_{\mathfrak{A}}\}$, every converging subsequence has the same limit, and therefore the whole sequence $\tilde{\alpha}_{\hat{\mathfrak{A}}_p}$ converges to $\tilde{\alpha}_{\mathfrak{A}}$, proving (11.21). $\square$

Thus, (11.19) is the classical law of large numbers on Wasserstein space, and (11.20) is its linear image under the collapse map. By contrast, (11.21) is nonlinear: it recomputes a Wasserstein barycenter from the empirical law over measures.

Toward central limit theorems on Wasserstein space. There are nevertheless important settings where such results can be proved. In one dimension, the quantile representation linearizes $\mathcal{W}_2$, so barycenter fluctuations can be studied through empirical averages of quantile functions.

Sliced and Radon barycenters. Slicing gives a scalable surrogate for high-dimensional barycenters by applying the one-dimensional quantile formula of Proposition 11.5 in every projection direction. For measures on $\mathbb{R}^d$, define the sliced barycenter by replacing $\mathcal{W}_2$ with SW_2 , introduced in Definition 10.10 and interpreted through the Radon transform in Section 10.2:

$$\min_{\alpha \in \mathcal{M}_+^1(\mathbb{R}^d)} \sum_{s=1}^S \lambda_s \text{SW}_2(\alpha, \beta_s)^2 = \min_{\alpha \in \mathcal{M}_+^1(\mathbb{R}^d)} \int_{\mathbb{S}^{d-1}} \sum_{s=1}^S \lambda_s \mathcal{W}_2((P_{\theta})_{\#} \alpha, (P_{\theta})_{\#} \beta_s)^2 d\sigma(\theta). \\ \min_{(\gamma_{\theta})_{\theta}} \int_{\mathbb{S}^{d-1}} \sum_{s=1}^S \lambda_s \mathcal{W}_2(\gamma_{\theta}, \beta_{s,\theta})^2 d\sigma(\theta), \quad \beta_{s,\theta} := (P_{\theta})_{\#} \beta_s.$$

$\mathcal{C}_{\gamma_{\theta}}^{-1}(r) = \sum_{s=1}^S \lambda_s \mathcal{C}_{\beta_{s,\theta}}^{-1}(r)$, $r \in [0, 1]$. For two inputs β_0, β_1 and interpolation weight $t \in [0, 1]$, it is useful to make this directionwise construction explicit. Define

$$Q_i(\theta, r) := \mathcal{C}_{\beta_{i,\theta}}^{-1}(r), \quad Q_t(\theta, r) := (1-t)Q_0(\theta, r) + tQ_1(\theta, r), \quad \gamma_{t,\theta} := (Q_t(\theta, \cdot))_{\#} \text{Leb}_{[0,1]}. \quad (11.22)$$

Thus Q_t is the quantile field of the independently interpolated projected laws $(\gamma_{t,\theta})_{\theta}$. When these laws have densities, we denote them by $h_t(\theta, \cdot)$.

Let $h(\theta, t)$ denote a density of γ_{θ} . We use the one-dimensional Fourier transform in the variable t ,

$$\hat{h}(\theta, \omega) := \int_{\mathbb{R}} e^{-i\omega t} h(\theta, t) dt, \quad h(\theta, t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\omega t} \hat{h}(\theta, \omega) d\omega, \quad (11.23)$$

Proposition 11.11 (Radon least-squares pseudoinverse). *Let $d \geq 2$, let σ be the uniform probability measure on $\mathbb{S}^{d-1}$, and let $R\rho(\theta, t) = \int_{\{x: \langle \theta, x \rangle = t\}} \rho(x) d\mathcal{H}^{d-1}(x)$ be the density Radon transform. Write $\mathcal{D}(R) := \{\rho \in L^2(\mathbb{R}^d) : R\rho \in L^2(\mathbb{S}^{d-1} \times \mathbb{R}, d\sigma dt)\}$. Let $h \in L^2(\mathbb{S}^{d-1} \times \mathbb{R}, d\sigma dt)$ and assume that the Fourier expressions below define an element of $\mathcal{D}(R)$. Then the unique solution of*

$$\min_{\rho \in \mathcal{D}(R)} \int_{\mathbb{S}^{d-1}} \int_{\mathbb{R}} |R\rho(\theta, t) - h(\theta, t)|^2 dt d\sigma(\theta) \quad (11.24)$$

is $\rho^\dagger = R^\dagger h$, where

$$R^\dagger h(x) = \frac{|\mathbb{S}^{d-1}|}{2(2\pi)^d} \int_{\mathbb{S}^{d-1}} \int_{\mathbb{R}} e^{i\omega \langle \theta, x \rangle} |\omega|^{d-1} \widehat{h}(\theta, \omega) d\omega d\sigma(\theta). \quad (11.25)$$

Thus $R^\dagger h$ is the density of the pseudoinverse reconstruction, which is a priori a signed measure. If $h = R\rho$ is Radon-consistent and ρ is sufficiently regular, then $R^\dagger R\rho = \rho$.

Proof. For the d -dimensional Fourier transform, use the compatible convention

$$\widehat{\rho}(\xi) = \int_{\mathbb{R}^d} e^{-i\langle \xi, x \rangle} \rho(x) dx, \quad \rho(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{i\langle \xi, x \rangle} \widehat{\rho}(\xi) d\xi.$$

The Fourier-slice theorem gives

$$\widehat{R\rho}(\theta, \omega) = \widehat{\rho}(\omega\theta). \quad (11.26)$$

Applying the one-dimensional Plancherel identity in t rewrites the objective in (11.24), up to the positive factor $1/(2\pi)$, as $\int_{\mathbb{S}^{d-1}} \int_{\mathbb{R}} |\widehat{\rho}(\omega\theta) - \widehat{h}(\theta, \omega)|^2 d\omega d\sigma(\theta)$. Every $\xi \neq 0$ has the two signed-polar representations $(\theta, \omega) = (\xi/\|\xi\|, \|\xi\|)$ and $(-\xi/\|\xi\|, -\|\xi\|)$. The pointwise least-squares minimizer is therefore

$$\widehat{\rho}^\dagger(\xi) = \frac{1}{2} \left(\widehat{h} \left(\frac{\xi}{\|\xi\|}, \|\xi\| \right) + \widehat{h} \left(-\frac{\xi}{\|\xi\|}, -\|\xi\| \right) \right). \quad (11.27)$$

This also proves uniqueness under the stated existence assumptions. Applying the inverse d -dimensional Fourier transform to (11.27) and using signed polar coordinates yields (11.25); the factor $|\mathbb{S}^{d-1}|/2$ accounts for the normalization of σ and the two signed representations of each nonzero frequency. Finally, if $h = R\rho$, then (11.26) makes both terms in (11.27) equal to $\widehat{\rho}(\xi)$, proving $R^\dagger R\rho = \rho$. $\square$

$$m_\Omega(\omega) := |\omega|^{d-1} \chi(\omega/\Omega), \quad R_\Omega^\dagger h(x) := \frac{|\mathbb{S}^{d-1}|}{2(2\pi)^d} \int_{\mathbb{S}^{d-1}} \int_{\mathbb{R}} e^{i\omega \langle \theta, x \rangle} m_\Omega(\omega) \widehat{h}(\theta, \omega) d\omega d\sigma(\theta), \quad (11.28)$$

where χ is an even low-pass window with $\chi(0) = 1$. For the two-input path (11.22), choose $\eta_t \geq 0$ so that the positive part below has nonzero mass, and define the nonnegative, unit-mass reconstruction

$$A_t(x) := \frac{((R_\Omega^\dagger h_t)(x) - \eta_t)_+}{\int_{\mathbb{R}^d} ((R_\Omega^\dagger h_t)(z) - \eta_t)_+ dz}. \quad (11.29)$$

For the endpoints, set $A_i = \rho_i$ when $\beta_i = \rho_i dx$, $i \in \{0, 1\}$. The resulting regularized density is generally only a least-squares approximation to the independently averaged slices. With finitely many directions, the sampled Radon operator is also non-injective, so the pseudoinverse reconstruction and the constrained sliced barycenter may differ.

11.2 Multimarginal OT

Definition and basic structure. The multi-marginal formulation replaces a coupling between two measures by a joint distribution with several prescribed marginals. Given measures $(\alpha_s)_{s=1}^S$ on spaces $(\mathcal{X}_s)_{s=1}^S$ and a lower-semicontinuous cost $c : \mathcal{X}_1 \times \dots \times \mathcal{X}_S \rightarrow \mathbb{R} \cup \{+\infty\}$ bounded from below, the problem reads

$\inf_{\pi \in \mathcal{U}(\alpha_1, \dots, \alpha_S)} \int_{\mathcal{X}_1 \times \dots \times \mathcal{X}_S} c(x_1, \dots, x_S) d\pi(x_1, \dots, x_S)$, where $\mathcal{U}(\alpha_1, \dots, \alpha_S)$ is the set of probability measures whose s -th marginal is α_s . This is still a linear program in the discrete setting, but its ambient tensor has size $\prod_s n_s$.

Monge structure and splitting-set twist.

Definition 11.12 (Twist on splitting sets). Fix $x_1 \in \mathcal{X}_1$. A set $M \subset \mathcal{X}_2 \times \dots \times \mathcal{X}_S$ is a c -splitting set at x_1 if there exist functions $u_s : \mathcal{X}_s \rightarrow \mathbb{R} \cup \{-\infty\}$, for $s = 2, \dots, S$, such that $\sum_{s=2}^S u_s(x_s) \leq c(x_1, x_2, \dots, x_S)$ for all $(x_2, \dots, x_S)$, with equality on M . Assume c is differentiable in x_1 . The cost is twisted on splitting sets if, for every x_1 and every c -splitting set M at x_1 , the map $(x_2, \dots, x_S) \mapsto \nabla_{x_1} c(x_1, x_2, \dots, x_S)$ is injective on M .

Proposition 11.13 (Multi-marginal Monge structure). *Assume that $\mathcal{X}_s \subset \mathbb{R}^d$, that c is continuous and differentiable with respect to x_1 , and that c is twisted on splitting sets. Suppose that Kantorovich dual maximizers $(\varphi_s)_{s=1}^S$ exist, that α_1 is absolutely continuous, and that φ_1 is differentiable α_1 -a.e. Then every optimal plan $\pi^* \in \mathcal{U}(\alpha_1, \dots, \alpha_S)$ is concentrated on the graph of maps $\pi^* = (\text{Id}, T_2, \dots, T_S)_\# \alpha_1$, $(T_s)_\# \alpha_1 = \alpha_s$. In particular, under these hypotheses the optimizer is unique.*

Proof. Let $(\varphi_s)_{s=1}^S$ be optimal dual potentials. Complementary slackness gives a Borel contact set Γ of full π^* -measure on which $\sum_s \varphi_s(x_s) = c(x_1, \dots, x_S)$. After disintegrating with respect to the first marginal, fix a point x_1 where φ_1 is differentiable and where the conditional plan is concentrated on the fiber $M(x_1) = \{(x_2, \dots, x_S) : (x_1, x_2, \dots, x_S) \in \Gamma\}$. Indeed, set $u_2 = \varphi_2 + \varphi_1(x_1)$ and $u_s = \varphi_s$ for $s \geq 3$. Dual feasibility gives $\sum_{s=2}^S u_s(x_s) \leq c(x_1, x_2, \dots, x_S)$, and equality holds on $M(x_1)$. If $(x_2, \dots, x_S) \in M(x_1)$, the function $z \mapsto c(z, x_2, \dots, x_S) - \sum_{s=2}^S \varphi_s(x_s)$ touches φ_1 from above at $z = x_1$. Differentiating at this contact point gives $\nabla \varphi_1(x_1) = \nabla_{x_1} c(x_1, x_2, \dots, x_S)$. All points in the fiber therefore have the same value of $\nabla_{x_1} c$. Twist on splitting sets makes the fiber a singleton for α_1 -a.e. x_1 . Disintegrating π^* with respect to its first marginal gives Dirac conditional measures, hence measurable maps $(T_2, \dots, T_S)$. If π^1 and π^2 are two optimal plans, their average is also optimal. The conditional measure of this average over x_1 is the average of the two Dirac conditionals, and it must again be a Dirac mass by the preceding argument. Hence the two Dirac masses coincide for α_1 -a.e. x_1 , proving uniqueness. $\square$

Coulomb cost and density-functional theory. $c_{\text{Coul}}(x_1, \dots, x_N) := \sum_{1 \leq i < j \leq N} \frac{1}{\|x_i - x_j\|}, \quad \nabla_{x_1} c_{\text{Coul}}(x_1, \dots, x_N) = -\sum_{j=2}^N \frac{x_1 - x_j}{\|x_1 - x_j\|^3}$. For $N = 2$, the map from x_2 to this vector is injective, so the ordinary two-marginal twist argument can be recovered under the required existence, duality and differentiability hypotheses. For $N \geq 3$, however, the displayed total force does not by itself determine the entire tuple $(x_2, \dots, x_N)$; twist on splitting sets, and hence a Monge representation, is not automatic.

If ρ is an electron density with $\int_{\mathbb{R}^3} \rho(x) dx = N$ and $\alpha = \rho/N$ is the associated probability density, the strictly-correlated-electrons relaxation of density-functional theory is the equal-marginal problem

$$V_{\text{ee}}^{\text{SCE}}[\rho] := \inf_{\pi \in \mathcal{U}(\alpha, \dots, \alpha)} \int_{(\mathbb{R}^3)^N} c_{\text{Coul}}(x_1, \dots, x_N) d\pi(x_1, \dots, x_N).$$

$$\pi = (\text{Id}, T_2, \dots, T_N)_\# \alpha, \quad (T_i)_\# \alpha = \alpha,$$

Proposition 11.14 (Cyclic co-motion plans). *Let $\alpha \in \mathcal{P}(\mathbb{R}^3)$ and let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be measurable with $T_\# \alpha = \alpha$ and $T^N = \text{Id}$ α -a.e. Set $T^0 = \text{Id}$ and $\pi_T = (T^0, T^1, \dots, T^{N-1})_\# \alpha$. Then $\pi_T \in \mathcal{U}(\alpha, \dots, \alpha)$. If $R_\sigma(x_1, \dots, x_N) = (x_{\sigma(1)}, \dots, x_{\sigma(N)})$ for $\sigma \in \text{Perm}(N)$, then $\bar{\pi}_T := \frac{1}{N!} \sum_{\sigma \in \text{Perm}(N)} (R_\sigma)_\# \pi_T$ is a symmetric admissible plan and*

$$\int c_{\text{Coul}} d\bar{\pi}_T = \int c_{\text{Coul}} d\pi_T = \int_{\mathbb{R}^3} \sum_{0 \leq i < j \leq N-1} \frac{1}{\|T^i(x) - T^j(x)\|} d\alpha(x).$$

Proof. Since $T_\# \alpha = \alpha$, all iterates T^i preserve α , so every marginal of π_T is α . The plan $\bar{\pi}_T$ is an average of coordinate permutations of π_T , hence has the same marginals and is invariant under coordinate permutations. The Coulomb cost is symmetric in its arguments, so its integral is unchanged by each R_σ . The last identity follows by evaluating c_{Coul} on the graph $(x, T(x), \dots, T^{N-1}(x))$. $\square$

Proposition 11.13 explains the general mechanism that can force graph solutions, while Proposition 11.14 records the additional equal-marginal symmetry used by co-motion maps. Thus the DFT problem is both a central application and a warning that multi-marginal OT is richer than a naive deterministic matching problem.

Multi-marginal formulation of barycenters. $c_{\text{bar}}(x_1, \dots, x_S) = \min_{x \in \mathbb{R}^d} \sum_{s=1}^S \lambda_s \|x - x_s\|^2$.

Proposition 11.15 (Multi-marginal formula for quadratic barycenters). *Let $\beta_1, \dots, \beta_S \in \mathcal{P}_2(\mathbb{R}^d)$ and $\lambda \in \Sigma_S$. Define $B(x_1, \dots, x_S) = \sum_{s=1}^S \lambda_s x_s$, $c_{\text{bar}}(x_1, \dots, x_S) = \min_x \sum_s \lambda_s \|x - x_s\|^2$. If π^* solves the multi-marginal OT problem with marginals $(\beta_s)_s$ and cost c_{bar} , then $\alpha^* = B_\# \pi^*$ is a Wasserstein barycenter. Conversely, every barycenter is obtained this way from an optimal multi-marginal plan.*

Proof. For any candidate barycenter α and couplings $\pi_s \in \mathcal{U}(\alpha, \beta_s)$, glue the couplings along their common α marginal to obtain a joint law of $(X, Y_1, \dots, Y_S)$. Conditioning on $(Y_s)_s$ and minimizing over X gives $\sum_s \lambda_s \mathbb{E} \|X - Y_s\|^2 \geq \mathbb{E} \min_x \sum_s \lambda_s \|x - Y_s\|^2 = \mathbb{E} c_{\text{bar}}(Y_1, \dots, Y_S)$. Taking the infimum over the couplings gives that the barycenter value is at least the multi-marginal value. Conversely, from an optimal multi-marginal plan π^* , set $X = B(Y_1, \dots, Y_S)$. The couplings between X and each Y_s are feasible for the barycenter problem and attain exactly the multi-marginal cost, proving equality and the formula. If α^* is any barycenter, choose optimal couplings between α^* and each β_s and glue them along the common α^* marginal. Since the barycenter and multi-marginal values are equal, the conditional minimization inequality above must be an equality. Thus $X = B(Y_1, \dots, Y_S)$ almost surely for the induced optimal multi-marginal plan, and $\alpha^* = B_\# \pi^*$. $\square$

Corollary 11.16 (Sparse discrete barycenters). *Let $\beta_s = \sum_{i_s=1}^{n_s} b_{s,i_s} \delta_{x_{s,i_s}}$, $s = 1, \dots, S$, be discrete input measures. There exists a quadratic Wasserstein barycenter α^* supported on weighted averages $\sum_s \lambda_s x_{s,i_s}$ of one support point from each input and satisfying $\# \text{supp}(\alpha^*) \leq \sum_{s=1}^S n_s - S + 1$.*

Proof. Write a discrete multi-marginal plan as a nonnegative tensor $P = (P_{i_1, \dots, i_S})$ and let $\mathcal{A}_{\text{marg}}$ collect its S marginals: $(\mathcal{A}_{\text{marg}} P)_{s, i_s} = \sum_{(i_r)_{r \neq s}} P_{i_1, \dots, i_S}$. After vectorizing P , Proposition 3.5 applies to the multi-marginal linear program. Its constraint operator has rank $\text{rank}(\mathcal{A}_{\text{marg}}) = \sum_{s=1}^S n_s - S + 1$. Indeed, a family of vectors $u_s \in \mathbb{R}^{n_s}$ belongs to the annihilator of its image precisely when $\sum_{s=1}^S u_{s, i_s} = 0$ for every $(i_1, \dots, i_S)$. Varying one index at a time shows that every u_s is constant, say $u_s = c_s \mathbf{1}_{n_s}$, and the remaining condition is $\sum_s c_s = 0$. The annihilator therefore has dimension $S - 1$, which proves the rank formula.

The rank-controlled sparsity result therefore supplies an optimal multi-marginal tensor P^* with at most $\sum_s n_s - S + 1$ positive entries. Proposition 11.15 gives the barycenter $\alpha^* = B_\# P^*$. Each positive tensor entry produces at most one atom under B , and collisions can only reduce the support size. This proves the claim. $\square$

Entropic regularization of multi-marginal OT. $\inf_{\pi \in \mathcal{U}(\alpha_1, \dots, \alpha_S)} \int c d\pi + \varepsilon \text{KL}(\pi | \alpha_1 \otimes \dots \otimes \alpha_S)$.

$$d\pi^*(x_1, \dots, x_S) = \exp\left(\frac{\sum_s f_s(x_s) - c(x_1, \dots, x_S)}{\varepsilon}\right) \prod_s d\alpha_s(x_s),$$

Treewidth and graphical structure. The important exception is when the cost factors over a sparse interaction graph. Let $G = (V, E)$ be a finite undirected graph with $V = \{1, \dots, S\}$ and suppose, for simplicity, that

$$c(x_1, \dots, x_S) = \sum_{(r,s) \in E} c_{r,s}(x_r, x_s). \quad (11.30)$$

Definition 11.17 (Tree decomposition and treewidth). A *tree decomposition* of a finite graph $G = (V, E)$ is a tree $\mathcal{T} = (Q, F)$ together with bags $B_q \subseteq V$, $q \in Q$, such that

1. $\bigcup_{q \in Q} B_q = V$;
2. for every edge $(r, s) \in E$, some bag contains both r and s ;
3. for every vertex $s \in V$, the nodes $\{q \in Q : s \in B_q\}$ form a connected subtree of $\mathcal{T}$.

The width of this decomposition is $\max_{q \in Q} |B_q| - 1$, and the *treewidth* $\text{tw}(G)$ is the minimum width over all tree decompositions.

Junction-tree contractions inside Sinkhorn.

$$K_{i_r, i_s}^{r,s} = \exp\left(-\frac{C_{i_r, i_s}^{r,s}}{\varepsilon}\right), \quad h_s(i_s) = (\mathbf{a}_s)_{i_s} (u_s)_{i_s}.$$

$$P_{i_1, \dots, i_S} = \prod_{s \in V} h_s(i_s) \prod_{(r,s) \in E} K_{i_r, i_s}^{r,s}. \quad (11.31)$$

When G is a tree, let $N_G(r)$ denote the neighbors of r . The directed sum-product messages satisfy

$$m_{r \rightarrow s}(i_s) = \sum_{i_r=1}^{n_r} K_{i_r, i_s}^{r,s} h_r(i_r) \prod_{q \in N_G(r) \setminus \{s\}} m_{q \rightarrow r}(i_r). \quad (11.32)$$

$$(\hat{\mathbf{a}}_s)_{i_s} = h_s(i_s) \prod_{r \in N_G(s)} m_{r \rightarrow s}(i_s). \quad (11.33)$$

For the block s currently selected by the cyclic scaling algorithm, the exact coordinate update is then

$$u_s \leftarrow u_s \odot \frac{\mathbf{a}_s}{\widehat{\mathbf{a}}_s},$$

with componentwise products and quotients. This changes only the unary factor h_s .

For a general tree decomposition, choose one host bag for each unary factor, assign each edge factor in (11.31) to a bag containing both endpoints, and denote the product assigned to bag B_q by $\psi_q(i_{B_q})$. For adjacent bags $q, q' \in Q$, set $\mathcal{S}_{q,q'} = B_q \cap B_{q'}$. Junction-tree messages take the form

$$M_{q \rightarrow q'}(i_{\mathcal{S}_{q,q'}}) = \sum_{i_{B_q \setminus \mathcal{S}_{q,q'}}} \psi_q(i_{B_q}) \prod_{\ell \in N_{\mathcal{T}}(q) \setminus \{q'\}} M_{\ell \rightarrow q}(i_{\mathcal{S}_{\ell,q}}). \quad (11.34)$$

$$\mathbf{b}_q(i_{B_q}) = \psi_q(i_{B_q}) \prod_{\ell \in N_{\mathcal{T}}(q)} M_{\ell \rightarrow q}(i_{\mathcal{S}_{\ell,q}}). \quad (11.35)$$

For any $s \in B_q$, summing $\mathbf{b}_q$ over $B_q \setminus \{s\}$ gives $\hat{\mathbf{a}}_s$; the running-intersection property ensures that every bag containing s gives the same result.

Redundant adjacent bags may be contracted, so one can assume that the decomposition is reduced. If its width is w , every bag contains at most $w + 1$ variables and every separator contains at most w .

$$O(|Q| n^{w+1}) \quad \text{time}, \quad O(|Q| n^w) \quad \text{message memory}.$$

$$O\left(\sum_{q \in Q} (1 + \deg_{\mathcal{T}}(q)) \prod_{s \in B_q} n_s\right),$$

$$O\left(\sum_{(q,q') \in F} \prod_{s \in \mathcal{S}_{q,q'}} n_s\right)$$

11.3 Low-Rank Optimal Transport

Definition 11.18 (Low-rank factored couplings). Let $\mathbf{a} \in \Sigma_n$, $\mathbf{b} \in \Sigma_m$ and let $r \geq 1$. A rank- r factored coupling is a triple (Q, R, g) such that $Q \in \mathbb{R}_+^{n \times r}$, $R \in \mathbb{R}_+^{m \times r}$, $g \in \Sigma_r$, with $Q\mathbf{1}_r = \mathbf{a}$, $R\mathbf{1}_r = \mathbf{b}$, $Q^\top \mathbf{1}_n = R^\top \mathbf{1}_m = g$. It induces the coupling

$$P(Q, R, g) = Q \text{diag}(g)^{-1} R^\top, \quad P_{i,j} = \sum_{k=1}^r \frac{Q_{i,k} R_{j,k}}{g_k}, \quad (11.36)$$

where columns with $g_k = 0$ are discarded before applying the formula.

Proposition 11.19 (Factored couplings and nonnegative rank). For every feasible triple (Q, R, g) in Definition 11.18, the matrix $P(Q, R, g)$ belongs to $\mathbf{U}(\mathbf{a}, \mathbf{b})$ and has nonnegative rank at most r . Conversely, every coupling $P \in \mathbf{U}(\mathbf{a}, \mathbf{b})$ with nonnegative rank at most r admits a representation of the form (11.36), after possibly deleting zero latent components.

Proof. The marginal constraints follow by direct summation: $P\mathbf{1}_m = Q \text{diag}(g)^{-1} R^\top \mathbf{1}_m = Q \text{diag}(g)^{-1} g = Q\mathbf{1}_r = \mathbf{a}$, and similarly $P^\top \mathbf{1}_n = \mathbf{b}$. Since P is a sum of r nonnegative rank-one matrices $Q_{:,k} R_{:,k}^\top / g_k$, its nonnegative rank is at most r .

Conversely, suppose $P = UV^\top$ with $U \in \mathbb{R}_+^{n \times r}$ and $V \in \mathbb{R}_+^{m \times r}$. Set $q_k = \sum_i U_{i,k}$ and $s_k = \sum_j V_{j,k}$. Components with $q_k s_k = 0$ do not contribute and can be removed. Define $g_k = q_k s_k$, $Q_{i,k} = U_{i,k} s_k$ and $R_{j,k} = V_{j,k} q_k$. Since P has total mass one, $g \in \Sigma_r$. Moreover $Q\mathbf{1}_r = P\mathbf{1}_m = \mathbf{a}$, $R\mathbf{1}_r = P^\top \mathbf{1}_n = \mathbf{b}$, and both column marginals are equal to g . Finally, $\sum_k \frac{Q_{i,k} R_{j,k}}{g_k} = \sum_k \frac{U_{i,k} s_k V_{j,k} q_k}{q_k s_k} = P_{i,j}$. $\square$

For a cost matrix $C \in \mathbb{R}^{n \times m}$, the low-rank constrained OT value is therefore

$$\min_{(Q,R,g)} \sum_{k=1}^r \frac{Q_{:,k}^\top C R_{:,k}}{g_k} \quad \text{subject to Definition 11.18.}$$

$$\min_{\substack{Q \mathbf{1}_r = \mathbf{a}, \quad Q^\top \mathbf{1}_n = g \\ R \mathbf{1}_r = \mathbf{b}, \quad R^\top \mathbf{1}_m = g}} \sum_{k=1}^r \frac{Q_{:,k}^\top C R_{:,k}}{g_k} + \varepsilon \text{KL}(Q|\mathbf{a} \otimes g) + \varepsilon \text{KL}(R|\mathbf{b} \otimes g), \quad (11.37)$$

where Q, R are nonnegative. For fixed g , this differs only by constants from adding $-\varepsilon(\text{H}(Q) + \text{H}(R))$ to the factorized transport cost.

11.4 Capacity-Constrained Optimal Transport

Let $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, $\beta \in \mathcal{M}_+^1(\mathcal{Y})$ and let $u : \mathcal{X} \times \mathcal{Y} \rightarrow [0, +\infty]$ be a measurable capacity. The capacity-constrained transport value is

$$\mathcal{L}_c^u(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \left\{ \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) ; \quad \pi \ll \alpha \otimes \beta, \quad \frac{d\pi}{d(\alpha \otimes \beta)}(x, y) \leq u(x, y) \right\}. \quad (11.38)$$

The product coupling $\alpha \otimes \beta$ is feasible whenever $u \geq 1$. Thus the constraint is not meant to promote the product plan; rather, it prevents the optimizer from using any pair more than the prescribed density ratio.

For discrete measures $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, a capacity is an upper matrix $\bar{P} \in \mathbb{R}_+^{n \times m}$. The density-ratio discretization of (11.38) is $\bar{P}_{i,j} = u_{i,j} a_i b_j$, and the finite-dimensional problem is the linear program

$$\min_{P \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle C, P \rangle \quad \text{subject to} \quad 0 \leq P_{i,j} \leq \bar{P}_{i,j} \quad \forall (i, j). \quad (11.39)$$

Proposition 11.20 (Feasibility of a capped transport polytope). *The constraints in (11.39) are feasible if and only if*

$$a(I) + b(J) - 1 \leq \bar{P}(I, J) \quad \text{for every } I \subset \{1, \dots, n\}, \quad J \subset \{1, \dots, m\}. \quad (11.40)$$

Proof. If P is feasible, then $P(I, J) = a(I) - P(I, J^c) \geq a(I) - b(J^c) = a(I) + b(J) - 1$. Since $P(I, J) \leq \bar{P}(I, J)$, condition (11.40) is necessary.

For sufficiency, build a flow network with capacity a_i from the source to row vertex i , capacity $\bar{P}_{ij}$ from row i to column j , and capacity b_j from column j to the sink. A cut containing row set I and column set J^c has capacity $1 - a(I) + \bar{P}(I, J) + 1 - b(J)$. Condition (11.40) says that every such cut has capacity at least one. The max-flow/min-cut theorem therefore gives a unit flow, and its row-to-column edge values form the required matrix P . $\square$

$$\min_{P \in \mathcal{U}(\mathbf{a}, \mathbf{b}), 0 \leq P \leq \bar{P}} \langle C, P \rangle + \varepsilon \text{KL}(P|\mathbf{a} \otimes \mathbf{b}). \quad (11.41)$$

On the active support, all iterates and correction factors are positive, so every division in the KL–Dykstra iteration is well defined. KL–Dykstra convergence on finite-dimensional affine and box constraints then shows that, when the capped polytope is nonempty, the iterates converge to the unique minimizer of (11.41).

11.5 Metric Learning and Inverse OT

11.5.1 Differentiating OT losses

First variations of OT values.

Proposition 11.21 (First variations of unregularized OT). *Let $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$, where $\mathcal{X}, \mathcal{Y}$ are compact metric spaces, and let $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$. Define $\mathcal{V}_c(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y)$. Let $\mathcal{O}_c(\alpha, \beta)$ be the set of optimal couplings and let $\mathcal{D}_c(\alpha, \beta)$ be the set of optimal dual potentials (f, g) with $f(x) + g(y) \leq c(x, y)$. If χ is a signed measure with $\chi(\mathcal{X}) = 0$ and $\alpha_t = \alpha + t\chi$ is a probability measure for $0 \leq t \leq t_0$, then*

$$\left. \frac{d}{dt} \right|_{t=0^+} \mathcal{V}_c(\alpha_t, \beta) = \sup_{(f, g) \in \mathcal{D}_c(\alpha, \beta)} \int_{\mathcal{X}} f(x) d\chi(x).$$

If $h \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$ and $c_t = c + th$, then

$$\left. \frac{d}{dt} \right|_{t=0^+} \mathcal{V}_{c_t}(\alpha, \beta) = \inf_{\pi \in \mathcal{O}_c(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} h(x, y) d\pi(x, y).$$

In particular, if the normalized optimal potential f^* and the optimal plan π^* are unique, then $\frac{\delta \mathcal{V}_c}{\delta \alpha} = f^*$, $\frac{\delta \mathcal{V}_c}{\delta \beta} = \pi^*$. The second identity means that the first variation with respect to the function c is represented by the optimal measure π^* on $\mathcal{X} \times \mathcal{Y}$.

Proof. Kantorovich duality writes $\mathcal{V}_c(\alpha, \beta) = \sup_{f+g \leq c} \int f d\alpha + \int g d\beta$. This is a supremum of affine functions of α , so Danskin's theorem gives the one-sided directional derivative as the supremum over active maximizers, namely the optimal dual potentials. The condition $\chi(\mathcal{X}) = 0$ makes the formula independent of the additive gauge $(f, g) \mapsto (f + \lambda, g - \lambda)$.

For the cost variable, $\mathcal{V}_{c_t}(\alpha, \beta) = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int c d\pi + t \int h d\pi$ is an infimum of affine functions of t . Danskin's theorem for a minimum gives the right directional derivative as the infimum of $\int h d\pi$ over the active minimizers. If the active dual potential or coupling is unique, the corresponding directional derivative is linear in the perturbation, which is the displayed first variation. $\square$

$$f^* \in \partial_{\mathbf{a}} \text{L}_C(\mathbf{a}, \mathbf{b}), \quad P^* \in \partial_C^{\text{sup}} \text{L}_C(\mathbf{a}, \mathbf{b}).$$

Proposition 11.22 (First variations of entropic OT). *Under the compact-space and continuous-cost assumptions of Proposition 11.21, let $\varepsilon > 0$ and define the KL-normalized entropic value $\mathcal{V}_{c, \varepsilon}(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int c d\pi + \varepsilon \text{KL}(\pi|\alpha \otimes \beta)$. Choose optimal entropic potentials $(f_\varepsilon, g_\varepsilon)$ in soft-transform form, so that the following density has marginals α and β : $d\pi_\varepsilon(x, y) =$*

$\exp\left(\frac{f_\varepsilon(x)+g_\varepsilon(y)-c(x,y)}{\varepsilon}\right) d\alpha(x)d\beta(y)$. This defines the unique optimal coupling. For the same perturbations $\alpha_t = \alpha + t\chi$ and $c_t = c + th$ as in Proposition 11.21,

$$\frac{d}{dt}\Big|_{t=0+} \mathcal{V}_{c,\varepsilon}(\alpha_t, \beta) = \int f_\varepsilon d\chi, \quad \frac{d}{dt}\Big|_{t=0+} \mathcal{V}_{c_t,\varepsilon}(\alpha, \beta) = \int h d\pi_\varepsilon.$$

Equivalently, $\frac{\delta \mathcal{V}_{c,\varepsilon}}{\delta \alpha} = f_\varepsilon$, $\frac{\delta \mathcal{V}_{c,\varepsilon}}{\delta c} = \pi_\varepsilon$. In finite dimension, for positive histograms, these are ordinary derivatives on the relative interior of the simplices.

Proof. The cost derivative follows directly from the primal envelope theorem, because the entropic optimizer is unique. For the measure derivative, use the dual formula (7.17). At an optimal pair, the soft-transform equations imply the row normalization $\int_Y \exp\left(\frac{f_\varepsilon(x)+g_\varepsilon(y)-c(x,y)}{\varepsilon}\right) d\beta(y) = 1$ for all x . Differentiating the dual objective with respect to α at fixed optimal potentials gives $\int f_\varepsilon d\chi - \varepsilon \int_{\mathcal{X} \times \mathcal{Y}} (e^{(f_\varepsilon(x)+g_\varepsilon(y)-c(x,y))/\varepsilon} - 1) d\chi(x)d\beta(y)$. The second term vanishes by the previous normalization identity, leaving $\int f_\varepsilon d\chi$. The gauge ambiguity of $(f_\varepsilon, g_\varepsilon)$ again disappears because $\chi(\mathcal{X}) = 0$. $\square$

For a finite-dimensional parametrization c_θ or C_θ , the entropic formula gives the backpropagation rule $\partial_\theta \mathcal{V}_{c_\theta,\varepsilon} = \int \partial_\theta c_\theta(x,y) d\pi_\varepsilon(x,y)$, where π_ε is the entropic optimizer. For the unregularized value $\mathcal{V}_{c_\theta}$, uniqueness of the optimal plan π^* gives $\partial_\theta \mathcal{V}_{c_\theta} = \int \partial_\theta c_\theta d\pi^*$. $L_C^\varepsilon(a,b) = \mathcal{V}_{C,\varepsilon}(a,b) - \varepsilon H(a) - \varepsilon H(b)$,

11.5.2 Inverse Optimal Transport

$\inf_{\pi \in \mathcal{U}(\alpha,\beta)} \int c(x,y) d\pi(x,y)$. A useful statistical methodology is to measure the suboptimality of the observed plan through a Fenchel–Young loss. Write the score as $s = -c$ and define the convex regularized prediction value $G_\varepsilon(s) = \sup_{\pi \in \mathcal{U}(\alpha,\beta)} \int s d\pi - \varepsilon \text{KL}(\pi | \alpha \otimes \beta)$. $\mathcal{L}_\varepsilon(c; \hat{\pi}) = G_\varepsilon(-c) + G_\varepsilon^*(\hat{\pi}) + \int c d\hat{\pi}$ is nonnegative by Fenchel’s inequality and vanishes exactly when $\hat{\pi} \in \partial G_\varepsilon(-c)$, i.e. when $\hat{\pi}$ satisfies the regularized optimality conditions for c . Entropic regularization is important here because it makes the forward map smoother and provides gradients with respect to c , at the price of a bias that must be controlled statistically.

In the discrete unregularized case, this loss reduces to the optimality gap of the observed coupling. For $\hat{P} \in \mathcal{U}(a,b)$ and a cost matrix C , denote it by

$$\mathcal{L}_{\text{IOT}}(C; \hat{P}) = \langle C, \hat{P} \rangle - \min_{P \in \mathcal{U}(a,b)} \langle C, P \rangle. \quad C_\theta = \sum_{r=1}^R \theta_r C^{(r)}, \quad \theta \in \Theta, \text{ where } \Theta \text{ is convex and the matrices } C^{(r)} \text{ encode features,}$$

graph distances or a Mahalanobis parameterization.

$c_A(x,y) = \langle Ax, y \rangle$, $A \in \mathbb{R}^{d \times d}$. For empirical measures $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$ and $\beta = \frac{1}{n} \sum_j \delta_{y_j}$, this gives the cost matrix

$C(A)_{i,j} = \langle Ax_i, y_j \rangle$. For a fixed matrix A , the forward prediction is the optimal face

$$\mathcal{P}_A := \operatorname{argmin}_{P \in \mathcal{U}(\mathbb{1}_n/n, \mathbb{1}_n/n)} \langle C(A), P \rangle. \quad \|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\langle x, y \rangle, \quad A_t = -\operatorname{diag}(1+t, 1-t), \quad -1 \leq t \leq 1,$$

$$\mathcal{L}_{\text{IOT}}(C(A_t); \hat{P}) = \langle C(A_t), \hat{P} \rangle - \min_{P \in \mathcal{U}(\mathbb{1}_n/n, \mathbb{1}_n/n)} \langle C(A_t), P \rangle, \quad C(A_t)_{i,j} = \langle A_t x_i, y_j \rangle.$$

The comparison between $n = 10$ and $n = 100$ illustrates an important statistical effect: as the number of sampled points grows, the flat zero region can shrink to a sharper piecewise-affine minimum. A finite-sample linear-programming gap remains polyhedral and therefore has no classical second derivative inside its cells.

Proposition 11.23 (Convex dual-gap formulation of inverse OT). *Let $\hat{P} \in \mathcal{U}(a,b)$ be an observed coupling and let C_θ depend affinely on $\theta \in \Theta$, where Θ is convex. The condition that $\hat{P}$ is optimal for the cost C_θ is equivalent to the existence of dual potentials (f,g) such that $f_i + g_j \leq (C_\theta)_{i,j}$ and $\sum_{i,j} \hat{P}_{i,j}((C_\theta)_{i,j} - f_i - g_j) = 0$.*

$$\min_{\theta \in \Theta, f, g} R(\theta) + \lambda \sum_{i,j} \hat{P}_{i,j}((C_\theta)_{i,j} - f_i - g_j) \quad \text{subject to} \quad f_i + g_j \leq (C_\theta)_{i,j} \quad \forall i, j. \quad (11.42)$$

is convex. It is nontrivial only if Θ imposes a cost normalization or otherwise excludes the zero-cost and marginal-gauge degeneracies.

Proof. For a fixed cost C_θ , Kantorovich duality gives $\min_{P \in \mathcal{U}(a,b)} \langle C_\theta, P \rangle = \max_{f_i + g_j \leq (C_\theta)_{i,j}} \langle f, a \rangle + \langle g, b \rangle$. Since $\hat{P}$ has marginals (a,b) , every dual feasible pair satisfies $\langle C_\theta, \hat{P} \rangle - \langle f, a \rangle - \langle g, b \rangle = \sum_{i,j} \hat{P}_{i,j}((C_\theta)_{i,j} - f_i - g_j) \geq 0$. It vanishes if and only if $\hat{P}$ reaches the dual value and is therefore optimal. If C_θ is affine, $\lambda \geq 0$, and Θ and R are convex, the constraints and objective in (11.42) are convex. If $C_\theta = 0$ is allowed and minimizes R , then $(C, f, g) = (0, 0, 0)$ is a trivial minimizer; this proves the need for the normalization stated above. $\square$

$$\hat{P}_{i,j} \approx a_i b_j \exp\left(\frac{f_i + g_j - (C_\theta)_{i,j}}{\varepsilon}\right),$$

11.6 Weak Optimal Transport

Barycentric projection of a coupling.

Definition 11.24 (Barycentric projection of a coupling). Let $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$ and let $\pi \in \mathcal{U}(\alpha, \beta)$. Disintegrate π with respect to its first marginal as $\pi(dx, dy) = \pi_x(dy)\alpha(dx)$. Since β has finite first moment, the conditional mean is finite for α -a.e. x , and the barycentric projection of π is the map

$$\bar{T}_\pi(x) := \int_{\mathbb{R}^d} y d\pi_x(y), \quad \bar{\beta}_\pi := (\bar{T}_\pi)_\# \alpha. \quad (11.43)$$

Proposition 11.25 (Barycentric projection of a quadratic optimal plan). *Let $\pi \in \mathcal{U}(\alpha, \beta)$ be optimal for the quadratic cost $\|x - y\|^2$ between $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$, and define $\bar{T}_\pi$ and $\bar{\beta}_\pi$ by (11.43). Then $(\text{Id}, \bar{T}_\pi)_\# \alpha$ is an optimal coupling between α and $\bar{\beta}_\pi$. Equivalently, $\bar{T}_\pi$ is a quadratic optimal transport map from α to the projected target $\bar{\beta}_\pi$.*

Proof. By Theorem 3.26, π is concentrated on a c -cyclically monotone set Γ for $c(x, y) = \|x - y\|^2$. For the quadratic cost, and since it is enough to check cyclic permutations, this means that every finite cycle $(x_i, y_i)_{i=1}^m \subset \Gamma$ satisfies $\sum_{i=1}^m \langle x_i, y_i \rangle \geq \sum_{i=1}^m \langle x_i, y_{i+1} \rangle$, $y_{m+1} = y_1$. After changing the disintegration on an α -negligible set, π_x is supported on the section $\Gamma_x = \{y; (x, y) \in \Gamma\}$ for α -a.e. x . Choose $x_1, \dots, x_m$ in this full-measure set and independently sample $Y_i \sim \pi_{x_i}$. Applying the cyclic inequality to (x_i, Y_i) and taking expectations gives $\sum_{i=1}^m \langle x_i, \bar{T}_\pi(x_i) \rangle \geq \sum_{i=1}^m \langle x_i, \bar{T}_\pi(x_{i+1}) \rangle$. Thus $(\text{Id}, \bar{T}_\pi)_\# \alpha$ is concentrated on a cyclically monotone graph. By the cyclic-monotonicity characterization of quadratic optimality, this plan is optimal between its two marginals, namely α and β_π . $\square$

Weak transport costs.

$$\text{WOT}_C(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \int C(x, \pi_x) d\alpha(x). \quad (11.44)$$

Proposition 11.26 (Weak Kantorovich duality). *Let $\mathcal{X}$ and $\mathcal{Y}$ be compact metric spaces, and let $C : \mathcal{X} \times \mathcal{P}(\mathcal{Y}) \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper, jointly lower semicontinuous, bounded from below and convex in its second argument. Fix $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$, and assume that $\text{WOT}_C(\alpha, \beta) < +\infty$. For $g \in \mathcal{C}(\mathcal{Y})$, define the weak C -transform*

$$g^C(x) := \inf_{\nu \in \mathcal{P}(\mathcal{Y})} \left\{ C(x, \nu) - \int g(y) d\nu(y) \right\}.$$

Then

$$\text{WOT}_C(\alpha, \beta) = \sup_{g \in \mathcal{C}(\mathcal{Y})} \left\{ \int g^C(x) d\alpha(x) + \int g(y) d\beta(y) \right\}.$$

When $C(x, \nu) = \int c(x, y) d\nu(y)$, this reduces to the usual Kantorovich dual with $g^C(x) = \inf_y (c(x, y) - g(y))$.

Proof. For any coupling π and any $g \in \mathcal{C}(\mathcal{Y})$, the definition of g^C gives $C(x, \pi_x) \geq g^C(x) + \int g(y) d\pi_x(y)$. After integration with respect to α , the second term becomes $\int g d\beta$ because the second marginal of π is β . This proves weak duality.

For the converse, lift a kernel $x \mapsto \pi_x$ to the measure $\alpha(dx) \delta_{\pi_x}(d\nu)$ on $\mathcal{X} \times \mathcal{P}(\mathcal{Y})$. Relaxing this graph constraint gives probability measures P whose first marginal is α and whose intensity satisfies $\int_{\mathcal{X} \times \mathcal{P}(\mathcal{Y})} \nu dP(x, \nu) = \beta$. This relaxation does not change the value: disintegrating $P(dx, d\nu) = P_x(d\nu) \alpha(dx)$ and replacing P_x by its barycenter $\bar{\nu}_x = \int \nu dP_x(\nu)$ preserves the intensity constraint and cannot increase the objective, by convexity of $C(x, \cdot)$. The relaxed feasible set is compact, and the integral functional is lower semicontinuous. Fenchel–Rockafellar duality for the affine intensity constraint therefore has no gap; its continuous multiplier is $g \in \mathcal{C}(\mathcal{Y})$. Minimization in ν gives $g^C(x)$, while the constraint contributes $\int g d\beta$. This is the weak-cost analogue of Proposition 4.2; see for the Polish-space formulation. $\square$

Proposition 11.27 (Barycentric weak transport is weaker than $\mathcal{W}_2$). *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and define $C_{\text{bar}}(x, \nu) = \|x - \int y d\nu(y)\|^2$. Equivalently, for a coupling π , the integrand is $\|x - \bar{T}_\pi(x)\|^2$. Then $\text{WOT}_{C_{\text{bar}}}(\alpha, \beta) \leq \mathcal{W}_2^2(\alpha, \beta)$.*

Proof. Let π be any coupling and disintegrate it as $\pi_x \alpha$. By Jensen's inequality, $\|x - \bar{T}_\pi(x)\|^2 \leq \int \|x - y\|^2 d\pi_x(y)$. Integrating in x gives $\int C_{\text{bar}}(x, \pi_x) d\alpha(x) \leq \int \|x - y\|^2 d\pi(x, y)$. Taking the infimum over π proves the claim. $\square$

The barycentric cost is the canonical example to keep in mind: admissibility still constrains the full conditional laws to have second marginal β , but the objective only charges the displacement from x to $\bar{T}_\pi(x)$ and ignores the conditional variance around this barycenter.

11.6.1 Martingale Optimal Transport

Definition 11.28 (Martingale couplings and martingale OT). Let $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$. A coupling $\pi \in \mathcal{U}(\alpha, \beta)$ is a martingale coupling if, for the disintegration $\pi(dx, dy) = \pi_x(dy) \alpha(dx)$,

$$\int_{\mathbb{R}^d} y d\pi_x(y) = x \quad \text{for } \alpha\text{-a.e. } x. \quad (11.45)$$

Equivalently, $\bar{T}_\pi = \text{Id}$ in (11.43). The set of such couplings is denoted $\mathcal{U}_{\text{mart}}(\alpha, \beta) := \{\pi \in \mathcal{U}(\alpha, \beta) ; \bar{T}_\pi = \text{Id} \text{ } \alpha\text{-a.e.}\}$.

For a cost c , the martingale OT problem minimizes $\int c d\pi$ over $\mathcal{U}_{\text{mart}}(\alpha, \beta)$, with value $+\infty$ when the admissible set is empty. The terminology comes from probability: if $(X, Y) \sim \pi$, then (11.45) is exactly $\mathbb{E}[Y|X] = X$. Hence martingale OT is a Kantorovich problem with the usual two marginal constraints plus a barycentric constraint on the conditional laws.

Stochastic orders.

Proposition 11.29 (Strassen's theorem for stochastic order). *Let $\alpha, \beta \in \mathcal{P}(\mathbb{R})$ and define the classical stochastic order by*

$$\alpha \preceq_{\text{st}} \beta \iff \int \varphi d\alpha \leq \int \varphi d\beta \quad \text{for every bounded increasing } \varphi.$$

Then $\alpha \preceq_{\text{st}} \beta$ if and only if there exists $\pi \in \mathcal{U}(\alpha, \beta)$ concentrated on $\{(x, y) ; x \leq y\}$.

Proof. A coupling supported on $x \leq y$ immediately gives the integral inequality for every increasing φ . Conversely, testing approximate indicators of intervals $(t, +\infty)$ gives $F_\alpha(t) \geq F_\beta(t)$ for all continuity points of the distribution functions. If U is uniform on $(0, 1)$ and $F_\alpha^{-1}, F_\beta^{-1}$ are the generalized quantiles, then $X = F_\alpha^{-1}(U)$ and $Y = F_\beta^{-1}(U)$ have laws α and β , and the inequality between distribution functions gives $X \leq Y$ almost surely. $\square$

Convex order and martingale feasibility. For martingale OT, the pointwise order constraint is replaced by a barycentric constraint on conditional laws. The corresponding order is the convex order. For $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$,

$$\alpha \preceq_{\text{cx}} \beta \iff \int \varphi d\alpha \leq \int \varphi d\beta \quad \text{for every convex } \varphi \text{ for which both integrals are defined.}$$

For measures with finite first moments, it is enough to test continuous convex functions with at most linear growth. Testing affine functions gives equality of means, while the remaining convex tests say that β is more spread out than α . Strassen's martingale theorem says that this spread condition is exactly what is needed to realize β from α by mean-preserving randomization.

Theorem 11.30 (Strassen's martingale theorem). *Let $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$. Then $\alpha \preceq_{\text{cx}} \beta \iff \mathcal{U}_{\text{mart}}(\alpha, \beta) \neq \emptyset$. Equivalently, β is above α in convex order if and only if there exists a coupling $\pi(dx, dy) = \pi_x(dy)\alpha(dx)$ such that $\int y d\pi_x(y) = x$ for α -almost every x .*

Proof. If π is a martingale coupling, then Jensen's inequality gives, for every convex φ ,

$$\varphi(x) = \varphi\left(\int y d\pi_x(y)\right) \leq \int \varphi(y) d\pi_x(y),$$

and integration in x gives $\int \varphi d\alpha \leq \int \varphi d\beta$.

Conversely, assume $\alpha \preceq_{\text{cx}} \beta$. First suppose that both measures are supported in a compact convex set K . Separate β from the closed convex set of probability measures on K that are reachable from α by martingale kernels supported in K . If β were not in this set, a continuous separating function $\psi : K \rightarrow \mathbb{R}$ would satisfy

$$\int \psi d\beta > \sup \left\{ \int \psi d\eta : \eta \in \mathcal{P}(K), \mathcal{U}_{\text{mart}}(\alpha, \eta) \neq \emptyset \right\}.$$

For a fixed $x \in K$, optimizing over probability measures on K with barycenter x gives the concave envelope $\text{conc}_K \psi(x)$; hence the right-hand side is $\int \text{conc}_K \psi d\alpha$. Since convex order is equivalently the reverse inequality for concave functions, $\int \text{conc}_K \psi d\alpha \geq \int \text{conc}_K \psi d\beta \geq \int \psi d\beta$, a contradiction. For general measures in $\mathcal{P}_1(\mathbb{R}^d)$, the same separation argument is carried out in the $\mathcal{W}_1$ topology, whose continuous test functions have at most linear growth. Compactness is replaced by tightness and uniform integrability of first moments; these properties also make the set of attainable second marginals closed and preserve the martingale constraint under limits. This is the standard extension in Strassen's theorem. $\square$

Proposition 11.31 (Brenier–Strassen projection formula). *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and let C_{bar} be the barycentric quadratic cost of Proposition 11.27. Then*

$$\text{WOT}_{C_{\text{bar}}}(\alpha, \beta) = \inf_{\eta \in \mathcal{P}_2(\mathbb{R}^d) : \eta \preceq_{\text{cx}} \beta} \mathcal{W}_2^2(\alpha, \eta). \quad (11.46)$$

Proof. Let (X, Y) have law $\pi \in \mathcal{U}(\alpha, \beta)$, set $Z = \mathbb{E}[Y | X]$, and denote its law by η . Conditional Jensen gives $\mathbb{E}[\varphi(Z)] \leq \mathbb{E}[\varphi(Y)]$ for every convex φ , so $\eta \preceq_{\text{cx}} \beta$. Since (X, Z) couples α and η , $\mathcal{W}_2^2(\alpha, \eta) \leq \mathbb{E}\|X - Z\|^2 = \int C_{\text{bar}}(x, \pi_x) d\alpha(x)$. Taking the infimum over π proves that the left-hand side of (11.46) is no smaller than the right-hand side.

Conversely, fix $\eta \preceq_{\text{cx}} \beta$. By Theorem 11.30, there is a martingale coupling from η to β . Glue it to an optimal coupling (X, Z) between α and η , conditionally independently given Z . The resulting variables satisfy $\mathbb{E}[Y | Z] = Z$, hence $\mathbb{E}[Y | X] = \mathbb{E}[Z | X]$. Conditional Jensen therefore gives $\mathbb{E}\|X - \mathbb{E}[Y | X]\|^2 = \mathbb{E}\|X - \mathbb{E}[Z | X]\|^2 \leq \mathbb{E}\|X - Z\|^2 = \mathcal{W}_2^2(\alpha, \eta)$. Taking the infimum over η proves the reverse inequality. $\square$

$$\mathcal{N}(m, \Sigma_0) \preceq_{\text{cx}} \mathcal{N}(m, \Sigma_1) \iff \Sigma_1 - \Sigma_0 \succeq 0.$$

12 Beyond Comparing Measures

12.1 Vector and Matrix-Valued Measures

Positive vector-valued measures.

Definition 12.1 (Positive vector-valued measure). A positive $\mathbb{R}_+^m$ -valued measure on $\mathcal{X}$ is a tuple $\alpha = (\alpha^1, \dots, \alpha^m) \in \mathcal{M}_+(\mathcal{X}; \mathbb{R}_+^m)$, where each component α^k is a nonnegative finite measure.

For the dynamic formulas below, assume that $\mathcal{X}$ is either $\mathbb{R}^d$, the flat torus, or a bounded convex domain in $\mathbb{R}^d$ equipped with no-flux boundary conditions. First suppose that the endpoints and the curve have densities.

$$\mathcal{W}_\Phi^2(\alpha_0, \alpha_1) := \inf_{u, V} \int_0^1 \int_{\mathcal{X}} \Phi(u_t(x), V_t(x)) dx dt \quad (12.1)$$

$$\partial_t u_t + \nabla_x \cdot V_t = 0, \quad (\nabla_x \cdot V_t)^k = \sum_{\ell=1}^d \partial_{x_\ell} V_{t,\ell}^k. \quad (12.2)$$

Proposition 12.2 (Diagonal positive vector Benamou–Brenier). *Assume that $\alpha_0^k, \alpha_1^k \in \mathcal{M}_+(\mathcal{X})$ have finite second moments and the same mass m_k for every k . For the diagonal mobility $\mathbf{M}_{\text{diag}}$, the value of (12.1) is*

$$\mathcal{W}_{\text{diag}}^2(\alpha_0, \alpha_1) = \sum_{k: m_k > 0} m_k \mathcal{W}_2^2\left(\frac{\alpha_0^k}{m_k}, \frac{\alpha_1^k}{m_k}\right),$$

with the convention that zero-mass channels contribute zero.

Proof. For $\mathbf{M}_{\text{diag}}(u) = \text{diag}(u_1, \dots, u_m)$, the action separates as $\sum_{\ell=1}^d V_\ell^\top \mathbf{M}_{\text{diag}}(u)^\dagger V_\ell = \sum_{k=1}^m \frac{|V^k|^2}{u^k}$, where $V^k = (V_1^k, \dots, V_d^k)$ is the spatial momentum of channel k , and the scalar perspective convention is used. The constraint (12.2) also separates into $\partial_t u^k + \nabla \cdot V^k = 0$. The minimization therefore splits into m independent scalar Benamou–Brenier problems. If $m_k = 0$, nonnegativity and conservation force the whole channel to vanish. If $m_k > 0$, normalizing $\rho_t^k = u_t^k/m_k$ and $p_t^k = V_t^k/m_k$ factors the channel action as $m_k \int |p_t^k|^2 / \rho_t^k$, hence the scalar value is $m_k \mathcal{W}_2^2(\alpha_0^k/m_k, \alpha_1^k/m_k)$. Summing over the channels proves the claim. $\square$

Positive matrix-valued measures.

Definition 12.3 (Positive matrix-valued measure). Write $\mathbb{S}^m$ for real symmetric matrices and $\mathbb{S}_+^m$ for the positive semidefinite cone. A positive $\mathbb{S}_+^m$ -valued measure is a countably additive map from Borel sets of $\mathcal{X}$ to $\mathbb{S}_+^m$; we denote the cone of such measures by $\mathcal{M}_+(\mathcal{X}; \mathbb{S}_+^m)$.

$$\mathcal{W}_{\text{mat}}^2(\mathcal{A}_0, \mathcal{A}_1) := \inf_{A, P} \int_0^1 \int_{\mathcal{X}} \sum_{\ell=1}^d \text{tr}(P_{t,\ell}^\top A_t^\dagger P_{t,\ell}) dx dt \quad (12.3)$$

$$\partial_t A_t + \nabla_x \cdot P_t = 0, \quad \nabla_x \cdot P_t = \sum_{\ell=1}^d \partial_{x_\ell} P_{t,\ell}. \quad (12.4)$$

Proposition 12.4 (Diagonal matrix subproblem). Assume that the endpoints are diagonal in a fixed orthonormal basis, $\mathcal{A}_i = \text{diag}(\alpha_i^1, \dots, \alpha_i^m)$, $i = 0, 1$, and that $\alpha_0^k(\mathcal{X}) = \alpha_1^k(\mathcal{X}) = m_k$ for every k . If one restricts the admissible curves in (12.3) to remain diagonal in that basis, $A_t = \text{diag}(u_t^1, \dots, u_t^m)$, $P_{t,\ell} = \text{diag}(V_{t,\ell}^1, \dots, V_{t,\ell}^m)$, then the value of this restricted matrix problem is

$$\sum_{k:m_k > 0} m_k \mathcal{W}_2^2\left(\frac{\alpha_0^k}{m_k}, \frac{\alpha_1^k}{m_k}\right),$$

with zero contribution from zero-mass channels. Thus the commuting matrix submodel is exactly the diagonal positive vector-valued Benamou–Brenier model of Proposition 12.2.

Proof. The continuity equation (12.4) is diagonal entry by diagonal entry and gives $\partial_t u^k + \nabla \cdot V^k = 0$. Moreover, $\sum_{\ell=1}^d \text{tr}(P_{t,\ell}^\top A_t^\dagger P_{t,\ell}) = \sum_{k=1}^m \frac{|V_t^k|^2}{u_t^k}$ with the same scalar perspective convention as before. The admissible set and the action are therefore exactly those of the diagonal vector model. $\square$

12.2 Wasserstein over Wasserstein

Proposition 12.5 (Wasserstein spaces as ground spaces). Let $1 \leq p < \infty$. If $(\mathcal{X}, d)$ is Polish, then $(\mathcal{P}_p(\mathcal{X}), \mathcal{W}_p)$ is Polish. If $\mathcal{X}$ is compact, then $\mathcal{P}_p(\mathcal{X}) = \mathcal{P}(\mathcal{X})$ and this space is compact for $\mathcal{W}_p$.

Proof. This is a standard structural theorem for Wasserstein spaces. For completeness, extract from a $\mathcal{W}_p$ -Cauchy sequence a subsequence $(\alpha_k)_k$ such that $\sum_k \mathcal{W}_p(\alpha_k, \alpha_{k+1}) < \infty$. Gluing optimal couplings of consecutive terms produces random variables $(X_k)_k$ with laws α_k and $\sum_k \|d(X_k, X_{k+1})\|_{L^p} < \infty$. Hence X_k converges almost surely and in L^p to an $\mathcal{X}$ -valued random variable; its law is the $\mathcal{W}_p$ limit of the subsequence, and the Cauchy property gives convergence of the original sequence. Separability follows by approximating measures with finitely supported measures on a countable dense subset and rational weights. If $\mathcal{X}$ is compact, Prokhorov's theorem gives compactness of $\mathcal{P}(\mathcal{X})$ for weak convergence, and Proposition 3.41 identifies this topology with the $\mathcal{W}_p$ topology. $\square$

Fix $1 \leq p < \infty$. We denote elements of $\mathcal{P}_p(\mathcal{X})$ by $\alpha, \beta, \dots$. Elements of $\mathcal{P}_p(\mathcal{P}_p(\mathcal{X}))$ are denoted by fraktur letters, for instance $\mathfrak{A}, \mathfrak{B}$; they are probability laws over probability measures, or random probability measures. A basic parametric example is obtained from a measurable family $(\alpha_\zeta)_{\zeta \in Z}$ and a probability law γ on the parameter space:

$$\mathfrak{A} = (\zeta \mapsto \alpha_\zeta)_\# \gamma. \quad (12.5)$$

$\int_Z \mathcal{W}_p(\alpha_\zeta, \delta_{x_0})^p d\gamma(\zeta) < \infty$. If $\gamma = \sum_{i=1}^K a_i \delta_{\alpha_{\zeta_i}}$, then $\mathfrak{A} = \sum_{i=1}^K a_i \delta_{\alpha_{\zeta_i}}$.

Definition 12.6 (Collapsed, or barycentric, mixture). For $\mathfrak{A} \in \mathcal{P}(\mathcal{P}(\mathcal{X}))$, the collapsed, or barycentric, mixture associated with $\mathfrak{A}$ is the measure $\bar{\alpha}_{\mathfrak{A}}$ defined by

$$\int_{\mathcal{X}} f(x) d\bar{\alpha}_{\mathfrak{A}}(x) = \int_{\mathcal{P}(\mathcal{X})} \left(\int_{\mathcal{X}} f(x) d\alpha(x) \right) d\mathfrak{A}(\alpha), \quad (12.6)$$

for bounded continuous f .

$$\int_{\mathcal{X}} d(x, x_0)^p d\bar{\alpha}_{\mathfrak{A}}(x) = \int_{\mathcal{P}_p(\mathcal{X})} \mathcal{W}_p(\alpha, \delta_{x_0})^p d\mathfrak{A}(\alpha).$$

$$\mathbb{W}_p^p(\mathfrak{A}, \mathfrak{B}) := \inf_{\Pi \in \mathcal{U}(\mathfrak{A}, \mathfrak{B})} \int_{\mathcal{P}_p(\mathcal{X}) \times \mathcal{P}_p(\mathcal{X})} \mathcal{W}_p^p(\alpha, \beta) d\Pi(\alpha, \beta), \quad \mathfrak{A}, \mathfrak{B} \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{X})). \quad (12.7)$$

For $p = 2$, Gaussian mixtures provide a particularly explicit example with two levels of geometry. A mixture $\sum_i a_i \mathcal{N}(m_i, \Sigma_i)$ can either be viewed as the collapsed measure on $\mathcal{X}$, or as the component law

$\mathfrak{A} = \sum_i a_i \delta_{\mathcal{N}(m_i, \Sigma_i)}$, $\mathfrak{B} = \sum_j b_j \delta_{\mathcal{N}(n_j, \Lambda_j)}$, $C_{ij} = \|m_i - n_j\|^2 + \mathcal{B}(\Sigma_i, \Lambda_j)^2$. Let $\mathbf{P}^*$ be an optimal coupling between the weights a and b . If A_{ij} denotes the Brenier linear part from Σ_i to Λ_j , then each active component pair is interpolated by

$$m_{ij,t} = (1-t)m_i + tn_j, \quad \Sigma_{ij,t} = ((1-t)\text{Id} + tA_{ij})\Sigma_i((1-t)\text{Id} + tA_{ij})^\top, \quad \bar{\alpha}_t = \sum_{i,j} \mathbf{P}_{ij}^* \mathcal{N}(m_{ij,t}, \Sigma_{ij,t}).$$

Proposition 12.7 (Collapsing is non-expansive). Let $(\mathcal{X}, d)$ be Polish, let $1 \leq p < \infty$, and let $\mathfrak{A}, \mathfrak{B} \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{X}))$. For the collapsed mixtures defined by (12.6), $\mathcal{W}_p(\bar{\alpha}_{\mathfrak{A}}, \bar{\beta}_{\mathfrak{B}}) \leq \mathbb{W}_p(\mathfrak{A}, \mathfrak{B})$.

Proof. Fix $\Pi \in \mathcal{U}(\mathfrak{A}, \mathfrak{B})$ and $\eta > 0$. A standard measurable-selection theorem for optimal transport on Polish spaces gives a measurable kernel $(\alpha, \beta) \mapsto \pi_{\alpha,\beta} \in \mathcal{U}(\alpha, \beta)$ such that $\int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\pi_{\alpha,\beta}(x, y) \leq \mathcal{W}_p(\alpha, \beta)^p + \eta$. Integrating this kernel against Π gives a coupling $\bar{\pi}$ between $\bar{\alpha}_{\mathfrak{A}}$ and $\bar{\beta}_{\mathfrak{B}}$. Its cost satisfies $\int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\bar{\pi}(x, y) \leq \int_{\mathcal{P}_p(\mathcal{X})^2} \mathcal{W}_p^p(\alpha, \beta) d\Pi(\alpha, \beta) + \eta$. Letting $\eta \downarrow 0$, taking the infimum over Π , and then taking the p -th root proves the claim. $\square$

12.3 Gromov–Wasserstein

Discrete formulation. Optimal transport needs a ground cost C to compare histograms (a, b) , and thus cannot be used directly if the histograms are not defined on the same underlying space, or if one cannot pre-register these spaces to define a ground cost. To address this issue, one can instead use a weaker requirement: two matrices $D \in \mathbb{R}^{n \times n}$ and $D' \in \mathbb{R}^{m \times m}$ are available and represent relationships between the points on which the histograms are defined. A typical scenario is when these matrices are powers of distance matrices.

$$\begin{aligned} \mathcal{E}_{D,D'}(P) &:= \sum_{i,j,i',j'} \Delta(D_{i,i'}, D'_{j,j'})^p P_{i,j} P_{i',j'}, \\ \text{GW}((a, D), (b, D'))^p &:= \min_{P \in \mathcal{U}(a,b)} \mathcal{E}_{D,D'}(P), \end{aligned} \quad (12.8)$$

where $p \geq 1$ and Δ is a distance on $\mathbb{R}$, typically $\Delta(u, v) = |u - v|$.

When the matrices D, D' are genuine distance matrices, the general construction below shows that GW satisfies the triangle inequality and defines a distance between metric spaces equipped with a probability distribution, up to measure-preserving isometries.

General setting.

Definition 12.8 (Metric-measure space). A metric-measure space is a triple $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$, where $(\mathcal{X}, d_{\mathcal{X}})$ is Polish and α is a Borel probability measure on $\mathcal{X}$. For a finite p -Gromov–Wasserstein distance, one additionally assumes the finite-size condition $\int_{\mathcal{X} \times \mathcal{X}} d_{\mathcal{X}}(x, x')^p d\alpha(x) d\alpha(x') < \infty$.

The general setting corresponds to computing couplings between metric-measure spaces $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$, where the distance and the measure are both part of the data. Compactness is often assumed below to avoid additional tightness and integrability arguments.

$$\mathcal{GW}(\mathbb{X}, \mathbb{Y})^p := \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X}^2 \times \mathcal{Y}^2} \Delta(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y'))^p d\pi(x, y) d\pi(x', y'). \quad (12.9)$$

Proposition 12.9 (Fixed-space GW is controlled by Wasserstein). *Let $\alpha, \beta \in \mathcal{P}_p(\mathcal{X})$ be probability measures on the same metric space $(\mathcal{X}, d_{\mathcal{X}})$, and take $\Delta(u, v) = |u - v|$ in (12.9). Then $\mathcal{GW}((\mathcal{X}, d_{\mathcal{X}}, \alpha), (\mathcal{X}, d_{\mathcal{X}}, \beta)) \leq 2\mathcal{W}_p(\alpha, \beta)$. The Wasserstein distance on the right is computed with the same ground metric $d_{\mathcal{X}}$. The factor 2 comes from the normalization in (12.9); with the alternative convention $\Delta(u, v) = |u - v|/2$, the displayed constant becomes 1.*

Proof. Let π be any coupling between α and β . For two independent pairs $(X, Y), (X', Y') \sim \pi$, the reverse triangle inequality gives $|d_{\mathcal{X}}(X, X') - d_{\mathcal{X}}(Y, Y')| \leq d_{\mathcal{X}}(X, Y) + d_{\mathcal{X}}(X', Y')$. Taking the L^p norm and using Minkowski gives a bound by $2(\int d_{\mathcal{X}}(x, y)^p d\pi)^{1/p}$. Optimizing over π proves the claim. $\square$

Definition 12.10 (Isometric metric-measure spaces). Two metric-measure spaces $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$ are isometric if there exists a bijection $\varphi : \text{supp}(\alpha) \rightarrow \text{supp}(\beta)$ such that $\varphi_{\#}\alpha = \beta$ and $d_{\mathcal{Y}}(\varphi(x), \varphi(x')) = d_{\mathcal{X}}(x, x')$ for all $x, x' \in \text{supp}(\alpha)$.

Theorem 12.11 (Gromov–Wasserstein metric modulo isometries). *For compact metric-measure spaces, $p \geq 1$ and $\Delta(u, v) = |u - v|$, GW defines a distance up to measure-preserving isometries.*

Proof. Compactness of the supports makes the coupling set compact and the distortion functional continuous, so an optimal coupling exists. If φ is a measure-preserving isometry, the graph coupling $(\text{Id}, \varphi)_{\#}\alpha$ has zero distortion; in particular, $\mathcal{GW}(\mathbb{X}, \mathbb{X}) = 0$. Symmetry and non-negativity are immediate.

If $\mathcal{GW}(\mathbb{X}, \mathbb{Y}) = 0$ and π is an optimal plan, then $d_{\mathcal{X}}(x, x') = d_{\mathcal{Y}}(y, y')$ holds $\pi \otimes \pi$ -almost everywhere. By continuity, this equality holds on $\text{supp}(\pi)^2$. $d_{\pi}((x, y), (x', y')) := \frac{1}{2}d_{\mathcal{X}}(x, x') + \frac{1}{2}d_{\mathcal{Y}}(y, y')$. The first projection $\psi : (x, y) \mapsto x$ is measure-preserving. For $((x, y), (x', y')) \in \text{supp}(\pi)^2$, $d_{\mathcal{X}}(\psi(x, y), \psi(x', y')) = d_{\mathcal{X}}(x, x') = d_{\mathcal{Y}}(y, y') = d_{\pi}((x, y), (x', y'))$, so ψ is an isometry and therefore injective. To see surjectivity onto $\text{supp}(\alpha)$, take $x \in \text{supp}(\alpha)$. Since $\psi_{\#}\pi = \alpha$, there is a sequence $(x_k, y_k) \in \text{supp}(\pi)$ with $x_k \rightarrow x$. The equality of distances on $\text{supp}(\pi)$ makes $(y_k)_k$ Cauchy, and compactness gives a convergent subsequence with limit $(x, y) \in \text{supp}(\pi)$. The same argument for the second projection shows that the support space is also isometric to $\mathbb{Y}$.

For the triangle inequality, let π be an optimal coupling between $\mathbb{X}$ and $\mathbb{Y}$, and ξ an optimal coupling between $\mathbb{Y}$ and $\mathbb{Z} = (Z, d_Z, \gamma)$. By the gluing lemma, take σ on $\mathcal{X} \times \mathcal{Y} \times Z$ whose $(\mathcal{X}, \mathcal{Y})$ and $(\mathcal{Y}, Z)$ marginals are π and ξ . Let $\rho = (P_{\mathcal{X}, Z})_{\#}\sigma$, and write $\bar{\sigma} = \sigma \otimes \sigma$ for the product law of two independent triples (x, y, z) and (x', y', z') . Then ρ is feasible between $\mathbb{X}$ and $\mathbb{Z}$, and

$$\begin{aligned} \mathcal{GW}(\mathbb{X}, \mathbb{Z}) &\leq \left(\int |d_{\mathcal{X}}(x, x') - d_Z(z, z')|^p d\bar{\sigma} \right)^{1/p} \\ &\leq \left(\int |d_{\mathcal{X}}(x, x') - d_{\mathcal{Y}}(y, y')|^p d\bar{\sigma} \right)^{1/p} + \left(\int |d_{\mathcal{Y}}(y, y') - d_Z(z, z')|^p d\bar{\sigma} \right)^{1/p} \\ &= \mathcal{GW}(\mathbb{X}, \mathbb{Y}) + \mathcal{GW}(\mathbb{Y}, \mathbb{Z}), \end{aligned}$$

where the second inequality uses the pointwise triangle inequality followed by Minkowski's inequality. This proves the triangle inequality and completes the metric proof. $\square$

Proposition 12.12 (Marginal stability and empirical GW rates). *Let $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$ be compact metric-measure spaces, with $\Delta(u, v) = |u - v|$ and exponent $p \geq 1$. For any probability measures $\tilde{\alpha}$ on $\mathcal{X}$ and $\tilde{\beta}$ on $\mathcal{Y}$, set $\tilde{\mathbb{X}} = (\mathcal{X}, d_{\mathcal{X}}, \tilde{\alpha})$, $\tilde{\mathbb{Y}} = (\mathcal{Y}, d_{\mathcal{Y}}, \tilde{\beta})$. Then, deterministically, $|\mathcal{GW}(\tilde{\mathbb{X}}, \tilde{\mathbb{Y}}) - \mathcal{GW}(\mathbb{X}, \mathbb{Y})| \leq 2\mathcal{W}_p^{\mathcal{X}}(\tilde{\alpha}, \alpha) + 2\mathcal{W}_p^{\mathcal{Y}}(\tilde{\beta}, \beta)$, where $\mathcal{W}_p^{\mathcal{X}}$ and $\mathcal{W}_p^{\mathcal{Y}}$ denote Wasserstein distances computed with $d_{\mathcal{X}}$ and $d_{\mathcal{Y}}$. If $\tilde{\alpha} = \hat{\alpha}_n$ and $\tilde{\beta} = \hat{\beta}_m$ are independent empirical measures built from iid samples with laws α and β , and if $\hat{\mathbb{X}}_n = (\mathcal{X}, d_{\mathcal{X}}, \hat{\alpha}_n)$, $\hat{\mathbb{Y}}_m = (\mathcal{Y}, d_{\mathcal{Y}}, \hat{\beta}_m)$, then $\mathbb{E}|\mathcal{GW}(\hat{\mathbb{X}}_n, \hat{\mathbb{Y}}_m) - \mathcal{GW}(\mathbb{X}, \mathbb{Y})| \leq 2\mathbb{E}\mathcal{W}_p^{\mathcal{X}}(\hat{\alpha}_n, \alpha) +$*

$2\mathbb{E} \mathcal{W}_p^\mathcal{Y}(\hat{\beta}_m, \beta)$. In particular, if $\mathcal{X}$ and $\mathcal{Y}$ are regular compact subsets of Euclidean spaces of intrinsic dimensions $d_\mathcal{X}$ and $d_\mathcal{Y}$, and the high-dimensional Wasserstein regime $d_\mathcal{X}, d_\mathcal{Y} > 2p$ applies, then $\mathbb{E} \left| \mathcal{GW}(\hat{\mathbb{X}}_n, \hat{\mathbb{Y}}_m) - \mathcal{GW}(\mathbb{X}, \mathbb{Y}) \right| = O(n^{-1/d_\mathcal{X}} + m^{-1/d_\mathcal{Y}})$. For $p = 1$ on subsets of $[0, 1]^d$, the rates, including the low-dimensional logarithmic cases, are those of Proposition 9.5.

Proof. The triangle inequality of Theorem 12.11 gives $\mathcal{GW}(\tilde{\mathbb{X}}, \tilde{\mathbb{Y}}) - \mathcal{GW}(\mathbb{X}, \mathbb{Y}) \leq \mathcal{GW}(\tilde{\mathbb{X}}, \mathbb{X}) + \mathcal{GW}(\mathbb{Y}, \tilde{\mathbb{Y}})$. Exchanging the two pairs gives the same bound for the opposite difference, hence the absolute value. The two terms on the right compare metric-measure spaces with the same underlying metric and only different measures, so Proposition 12.9 bounds them by $2\mathcal{W}_p^\mathcal{X}(\tilde{\alpha}, \alpha)$ and $2\mathcal{W}_p^\mathcal{Y}(\tilde{\beta}, \beta)$. Substituting $\tilde{\alpha} = \hat{\alpha}_n$ and $\tilde{\beta} = \hat{\beta}_m$ and taking expectations gives the second display. The Euclidean rate then follows from the empirical Wasserstein rates recalled in Section 9.2. $\square$

Proposition 12.13 (Gromov–Wasserstein geodesics). *Let $\mathbb{X}_0 = (\mathcal{X}_0, d_{\mathcal{X}_0}, \alpha_0)$ and $\mathbb{X}_1 = (\mathcal{X}_1, d_{\mathcal{X}_1}, \alpha_1)$ be compact metric-measure spaces, take $\Delta(u, v) = |u - v|$ in (12.9), and let π^* be an optimal coupling. Define, on $Z = \mathcal{X}_0 \times \mathcal{X}_1$, $d_t((x_0, x_1), (x'_0, x'_1)) := (1-t)d_{\mathcal{X}_0}(x_0, x'_0) + td_{\mathcal{X}_1}(x_1, x'_1)$, $\mathbb{X}_t = (Z, d_t, \pi^*)$. For $0 < t < 1$, d_t is a metric. At $t = 0$ and $t = 1$, one quotients Z by the zero-distance relation associated with d_t . Then $t \mapsto \mathbb{X}_t$ is a constant-speed geodesic: $\mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) = |t - s| \mathcal{GW}(\mathbb{X}_0, \mathbb{X}_1) \quad \forall s, t \in [0, 1]$.*

Proof. Write $D = \mathcal{GW}(\mathbb{X}_0, \mathbb{X}_1)$. For $s < t$, couple $\mathbb{X}_s$ and $\mathbb{X}_t$ by the diagonal coupling induced by the identity on Z and the measure π^* . For two independent points $z = (x_0, x_1)$ and $z' = (x'_0, x'_1)$ sampled from π^* , $d_t(z, z') - d_s(z, z') = (t - s)(d_{\mathcal{X}_1}(x_1, x'_1) - d_{\mathcal{X}_0}(x_0, x'_0))$. Using this feasible coupling gives $\mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) \leq (t - s)D$. The same construction with the projections from Z to $\mathcal{X}_0$ and $\mathcal{X}_1$ gives $\mathcal{GW}(\mathbb{X}_0, \mathbb{X}_t) \leq tD$ and $\mathcal{GW}(\mathbb{X}_t, \mathbb{X}_1) \leq (1 - t)D$. The triangle inequality for $\mathcal{GW}$ then yields, for $0 \leq s \leq t \leq 1$, $D \leq \mathcal{GW}(\mathbb{X}_0, \mathbb{X}_s) + \mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) + \mathcal{GW}(\mathbb{X}_t, \mathbb{X}_1) \leq sD + \mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) + (1 - t)D$, so $\mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) \geq (t - s)D$. This proves equality. $\square$

Proposition 12.14 (Mémoli profile lower bound). *Let $\mathbb{X} = (\mathcal{X}, d_\mathcal{X}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_\mathcal{Y}, \beta)$ be compact metric-measure spaces and take $\Delta(u, v) = |u - v|$ in (12.9), with the same exponent $p \geq 1$. For each $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, define the distance-profile measures on $\mathbb{R}_+$ by $\alpha_x := (d_\mathcal{X}(x, \cdot))_\# \alpha$, $\beta_y := (d_\mathcal{Y}(y, \cdot))_\# \beta$. Let $\mathfrak{D}_\mathbb{X} = (x \mapsto \alpha_x)_\# \alpha$ and $\mathfrak{D}_\mathbb{Y} = (y \mapsto \beta_y)_\# \beta$, which are probability measures on $\mathcal{P}_p(\mathbb{R}_+)$. Then $\mathbb{W}_p(\mathfrak{D}_\mathbb{X}, \mathfrak{D}_\mathbb{Y}) \leq \mathcal{GW}(\mathbb{X}, \mathbb{Y})$. Here $\mathbb{W}_p$ is the Wasserstein-over-Wasserstein distance (12.7): its ground space is $(\mathcal{P}_p(\mathbb{R}_+), \mathcal{W}_p)$.*

Proof. Fix any $\pi \in \mathcal{U}(\alpha, \beta)$. It induces a coupling $(x, y) \mapsto (\alpha_x, \beta_y)$ between $\mathfrak{D}_\mathbb{X}$ and $\mathfrak{D}_\mathbb{Y}$, hence $\mathbb{W}_p(\mathfrak{D}_\mathbb{X}, \mathfrak{D}_\mathbb{Y})^p \leq \int_{\mathcal{X} \times \mathcal{Y}} \mathcal{W}_p(\alpha_x, \beta_y)^p d\pi(x, y)$. For fixed (x, y) , the map $(x', y') \mapsto (d_\mathcal{X}(x, x'), d_\mathcal{Y}(y, y'))$ pushes the same coupling π to a coupling between α_x and β_y . Therefore $\mathcal{W}_p(\alpha_x, \beta_y)^p \leq \int_{\mathcal{X} \times \mathcal{Y}} |d_\mathcal{X}(x, x') - d_\mathcal{Y}(y, y')|^p d\pi(x', y')$. Integrating in (x, y) gives $\mathbb{W}_p(\mathfrak{D}_\mathbb{X}, \mathfrak{D}_\mathbb{Y})^p \leq \int_{\mathcal{X}^2 \times \mathcal{Y}^2} |d_\mathcal{X}(x, x') - d_\mathcal{Y}(y, y')|^p d\pi(x, y) d\pi(x', y')$. Taking the infimum over π and then the p -th root proves the claim. $\square$

Relation with Wasserstein–Procrustes.

$$\mathbb{W}_p(\mathfrak{D}_\mathbb{X}, \mathfrak{D}_\mathbb{Y}) \leq \mathcal{GW}(\mathbb{X}, \mathbb{Y}) \leq 2 \mathcal{W}_{p, \mathbb{E}(d)}([\alpha], [\beta]), \quad (12.10)$$

where $\mathbb{X} = (\mathbb{R}^d, \|\cdot\|, \alpha)$ and $\mathbb{Y} = (\mathbb{R}^d, \|\cdot\|, \beta)$. The left term is intrinsic and inexpensive, while the right term is an extrinsic rigid-registration certificate.

Entropic regularization and iterative solver. For the common squared distortion $\Delta(u, v)^2 = (u - v)^2$, one often seeks a stationary point of the entropic relaxation

$$\min_{P \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \mathcal{E}_{D, D'}(P) - \varepsilon H(P). \quad (12.11)$$

$$C(P) := D^{\odot 2} \mathbf{a} \mathbf{1}_m^\top + \mathbf{1}_n (D'^{\odot 2} \mathbf{b})^\top - 2D P D'^\top, \quad \nabla \mathcal{E}_{D, D'}(P) = 2C(P). \quad (12.12)$$

$P^{(k+1)} = \operatorname{argmin}_{P \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle C(P^{(k)}), P \rangle - \frac{\varepsilon}{2} H(P)$. The factor $\varepsilon/2$ is essential because (12.12) is one half of the gradient of the quadratic

term. Each update is an ordinary entropic OT problem and can therefore be solved with Sinkhorn iterations.

Adding features.

$$\text{FGW}_{\lambda, p}((\mathbf{a}, D), (\mathbf{b}, D'))^p := \min_{P \in \mathcal{U}(\mathbf{a}, \mathbf{b})} (1 - \lambda) \sum_{i, j} M_{ij} P_{ij} + \lambda \sum_{i, j, i', j'} \Delta(D_{ii'}, D'_{jj'})^p P_{ij} P_{i'j'}.$$

Hausdorff and Gromov–Hausdorff viewpoints. If A, B are compact subsets of a common metric space (Z, d_Z) , their Hausdorff distance is

$$d_H^Z(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} d_Z(a, b), \sup_{b \in B} \inf_{a \in A} d_Z(a, b) \right\}.$$

$$d_{\text{GH}}(\mathcal{X}, \mathcal{Y}) = \inf_{Z, \varphi, \psi} d_H^Z(\varphi(\mathcal{X}), \psi(\mathcal{Y})).$$

12.4 Quantum Optimal Transport

Finite-dimensional states and couplings.

Definition 12.15 (Hermitian and density matrices). Let $\mathbb{H}_n$ be the real vector space of $n \times n$ Hermitian matrices, $\mathbb{H}_n^+ = \{A \in \mathbb{H}_n; A \succeq 0\}$, $\mathbb{H}_n^{+,1} = \{A \in \mathbb{H}_n^+; \operatorname{tr}(A) = 1\}$. Elements of $\mathbb{H}_n^{+,1}$ are density matrices.

A joint quantum state between $\mathbb{C}^n$ and $\mathbb{C}^m$ is a matrix $T \in \mathbb{H}_{nm}^+$ acting on $\mathbb{C}^n \otimes \mathbb{C}^m$. Its marginals are the partial traces, defined by duality through

$$\operatorname{tr}(F \operatorname{Tr}_B T) = \operatorname{tr}((F \otimes \operatorname{Id}_m)T), \quad \operatorname{tr}(G \operatorname{Tr}_A T) = \operatorname{tr}((\operatorname{Id}_n \otimes G)T), \quad (12.13)$$

for all $F \in \mathbb{H}_n$ and $G \in \mathbb{H}_m$. Thus $\operatorname{Tr}_B(T) \in \mathbb{H}_n^+$ and $\operatorname{Tr}_A(T) \in \mathbb{H}_m^+$ play exactly the role of the two marginals of a classical coupling.

Definition 12.16 (Finite-dimensional quantum OT). Let $A \in \mathbb{H}_n^{+,1}$, $B \in \mathbb{H}_m^{+,1}$ and let $C \in \mathbb{H}_{nm}$ be a Hermitian cost observable. The quantum OT value is the semidefinite program

$$\text{QOT}_C(A, B) := \min_{T \in \mathbb{H}_{nm}^+} \{ \text{tr}(CT) : \text{Tr}_B(T) = A, \text{Tr}_A(T) = B \}. \quad (12.14)$$

Proposition 12.17 (Quantum Kantorovich duality). For $A \in \mathbb{H}_n^{+,1}$ and $B \in \mathbb{H}_m^{+,1}$, the dual of (12.14) is

$$\text{QOT}_C(A, B) = \sup_{F \in \mathbb{H}_n, G \in \mathbb{H}_m} \{ \text{tr}(FA) + \text{tr}(GB) : F \otimes \text{Id}_m + \text{Id}_n \otimes G \preceq C \}. \quad (12.15)$$

There is no duality gap. If A and B are positive definite, the supremum is attained. For singular marginals, the primal reduces to the tensor product of their supports; on this reduced space the corresponding dual maximum is attained, while the full-space formula above is always valid as a supremum.

Proof. Introduce Hermitian Lagrange multipliers F and G for the two marginal constraints. The Lagrangian is

$$\text{tr}(CT) + \text{tr}(F(A - \text{Tr}_B T)) + \text{tr}(G(B - \text{Tr}_A T)) = \text{tr}(FA) + \text{tr}(GB) + \text{tr}((C - F \otimes \text{Id}_m - \text{Id}_n \otimes G)T),$$

where (12.13) was used in the last equality. Minimizing over $T \succeq 0$ gives a finite lower bound if and only if $C - F \otimes \text{Id}_m - \text{Id}_n \otimes G \succeq 0$, in which case the infimum in T is 0. When $A, B \succ 0$, the coupling $A \otimes B$ is strictly feasible, so Slater's theorem gives equality of primal and dual values and dual attainment.

For singular marginals, let P_A and P_B be their support projections. If T is feasible, then $\text{tr}((\text{Id}_n - P_A) \otimes \text{Id}_m)T = \text{tr}((\text{Id}_n - P_A)A) = 0$. Positivity implies that T vanishes on $(\text{supp } A)^\perp \otimes \mathbb{C}^m$; the analogous argument for B yields $T = (P_A \otimes P_B)T(P_A \otimes P_B)$. Thus the primal is exactly the problem obtained by compressing A , B and C to $\text{supp } A \otimes \text{supp } B$. The reduced marginals are positive definite, so Slater applies there. Equivalently, the unreduced dual values approach the reduced maximum after assigning sufficiently negative potentials on the orthogonal complements. This proves the stated supremum formula without a gap. $\square$

Entropic regularization and Bregman iterations.

Definition 12.18 (von Neumann quantum entropy). For a density matrix or positive semidefinite matrix T , the shifted von Neumann entropy functional used here is

$$H(T) = \text{tr}(T(\log T - \text{Id})), \quad \nabla H(T) = \log T,$$

with the convention $0 \log 0 = 0$ on eigenvalues. This is the convex negative quantum entropy; on trace-one states it differs from the physical entropy $-\text{tr}(T \log T)$ by a sign and an additive constant.

For $\varepsilon > 0$ define

$$\text{QOT}_C^\varepsilon(A, B) = \min_{T \succeq 0} \{ \text{tr}(CT) + \varepsilon H(T) : \text{Tr}_B(T) = A, \text{Tr}_A(T) = B \}. \quad (12.16)$$

Proposition 12.19 (Entropic quantum OT duality). Assume $A \succ 0$, $B \succ 0$ and $\varepsilon > 0$. Then (12.16) has a unique positive minimizer. Its dual is

$$\text{QOT}_C^\varepsilon(A, B) = \max_{F \in \mathbb{H}_n, G \in \mathbb{H}_m} \left\{ \text{tr}(FA) + \text{tr}(GB) - \varepsilon \text{tr} \exp \left(\frac{F \otimes \text{Id}_m + \text{Id}_n \otimes G - C}{\varepsilon} \right) \right\}. \quad (12.17)$$

At optimality, primal and dual variables are linked by the Gibbs formula

$$T_e(F, G) = \exp \left(\frac{F \otimes \text{Id}_m + \text{Id}_n \otimes G - C}{\varepsilon} \right), \quad (12.18)$$

with $\text{Tr}_B(T_e) = A$ and $\text{Tr}_A(T_e) = B$.

Proof. The feasible set is compact and nonempty, and it contains the positive definite point $A \otimes B$. The trace entropy is strictly convex on positive semidefinite matrices, hence the regularized primal has a unique minimizer. This minimizer is positive definite: if it had a non-trivial kernel, moving toward $A \otimes B$ would have entropy directional derivative $-\infty$ at the boundary while changing the linear cost only at finite rate. Slater's condition justifies the Lagrange dual computation. The Fenchel identity $\sup_{T \succeq 0} \text{tr}(YT) - \varepsilon H(T) = \varepsilon \text{tr} \exp(Y/\varepsilon)$ is the matrix analogue of the scalar exponential conjugacy. Applying it to the Lagrangian of (12.16), with $Y = F \otimes \text{Id}_m + \text{Id}_n \otimes G - C$, gives (12.17). The stationarity condition of this Fenchel identity gives (12.18); differentiating the dual objective with respect to F and G yields the two marginal equations. $\square$

$$D_H(T|K) = \text{tr}(T(\log T - \log K) - T + K).$$

$$\mathcal{M}_A = \{T \succeq 0 : \text{Tr}_B(T) = A\}, \quad \mathcal{M}_B = \{T \succeq 0 : \text{Tr}_A(T) = B\}.$$

Proposition 12.20 (Exact Bregman projections). Assume $A, B \succ 0$ and let $K = \exp(-C/\varepsilon)$. The minimizer of (12.16) is equivalently the minimizer of $D_H(T|K)$ over $\mathcal{M}_A \cap \mathcal{M}_B$. Moreover, if a current positive definite matrix has Gibbs form $T_e(F, G)$, then its Bregman projection onto $\mathcal{M}_A$ has the form $T_e(F^+, G)$, where F^+ is chosen so that $\text{Tr}_B T_e(F^+, G) = A$. The projection onto $\mathcal{M}_B$ is analogous. Thus, when each one-block marginal equation is solved exactly, alternating Bregman projections are equivalent to alternating block maximization of the dual (12.17).

Proof. Since $\log K = -C/\varepsilon$, the identity $\text{tr}(CT) + \varepsilon H(T) = \varepsilon D_H(T|K) - \varepsilon \text{tr}(K)$ holds, so the primal minimizer is the constrained Bregman projection of K up to an additive constant. For the projection of a positive definite matrix S onto $\mathcal{M}_A$, the affine set contains the positive definite point $A \otimes \text{Id}_m/m$; the entropy derivative $\log T - \log S$ is singular at the boundary, so the projection lies in the interior of the positive cone. Its Lagrangian first variation is $\log T - \log S - \Lambda \otimes \text{Id}_m = 0$ for a Hermitian multiplier Λ . Hence $T = \exp(\log S + \Lambda \otimes \text{Id}_m)$. If $S = T_e(F, G)$, this is again of the form $T_e(F + \varepsilon \Lambda, G)$. The multiplier is fixed by the marginal equation $\text{Tr}_B(T) = A$. The same argument applies to $\mathcal{M}_B$. Finally, the first-order optimality condition for maximizing (12.17) over one block is exactly the corresponding marginal equation, so the Bregman and block-dual views coincide. $\square$

$$\text{Tr}_B T_e(F, G) = A, \quad \text{Tr}_A T_e(F, G) = B$$

Gurvits scaling and quantum Sinkhorn.

$$T_s(F, G) = \exp\left(\frac{Z}{2\varepsilon}\right) \exp(-C/\varepsilon) \exp\left(\frac{Z}{2\varepsilon}\right) = (U \otimes V)K(U \otimes V), \quad Z = F \otimes \text{Id}_m + \text{Id}_n \otimes G, \quad (12.19)$$

where $U = \exp(F/(2\varepsilon))$, $V = \exp(G/(2\varepsilon))$ and $K = \exp(-C/\varepsilon)$. If $[Z, C] = 0$, then $T_s(F, G) = T_e(F, G)$; otherwise this is a Strang-type symmetric surrogate.

Fix a Choi convention and let $\mathcal{K} : \mathbb{H}_m \rightarrow \mathbb{H}_n$ be the completely positive map represented by the positive Choi matrix K ; let $\mathcal{K}^*$ be its Hilbert–Schmidt adjoint. Up to the transpose dictated by the chosen Choi convention, the marginal equations for the symmetric coupling take the operator-scaling form

$$U \mathcal{K}(V^2) U = A, \quad V \mathcal{K}^*(U^2) V = B.$$

$$\begin{aligned} R_V &= \mathcal{K}(V^2), \quad U \leftarrow R_V^{-1/2} (R_V^{1/2} A R_V^{1/2})^{1/2} R_V^{-1/2}, \\ S_U &= \mathcal{K}^*(U^2), \quad V \leftarrow S_U^{-1/2} (S_U^{1/2} B S_U^{1/2})^{1/2} S_U^{-1/2}. \end{aligned} \quad (12.20)$$

These inverse square roots are well-defined when $K \succ 0$ and $U, V, A, B \succ 0$. This is Gurvits/operator scaling with prescribed targets.

12.5 Dynamic Time Warping

Ordered alignments versus transport couplings.

Discrete variational problem. Let $x = (x_i)_{i=1}^n$ and $y = (y_j)_{j=1}^m$ be two sequences in a feature space $\mathcal{Z}$, and set $C_{ij} = c(x_i, y_j)$ for a nonnegative cost $c : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}_+$. A warping path is a sequence $\omega = ((i_\ell, j_\ell))_{\ell=1}^L$ with $(i_\ell, j_\ell) \in \{(1, 0), (0, 1), (1, 1)\}$. Denote the set of such paths by $\Omega_{n,m}$ and the corresponding incidence matrix by $(A_\omega)_{ij} = \mathbf{1}_{\{(i,j) \in \omega\}}$. Its length satisfies $\max\{n, m\} \leq L \leq n + m - 1$, its total mass is $\sum_{ij} (A_\omega)_{ij} = L$, and its two marginals are precisely the row and column visit counts.

Definition 12.21 (Dynamic time warping). The DTW value associated with the feature cost c is

$$\text{DTW}_c(x, y) := \min_{\omega \in \Omega_{n,m}} \sum_{\ell=1}^L c(x_{i_\ell}, y_{j_\ell}) = \min_{\omega \in \Omega_{n,m}} \langle A_\omega, C \rangle. \quad (12.21)$$

Dynamic programming.

Proposition 12.22 (DTW dynamic-programming recurrence). Set $D_{0,0} = 0$, $D_{i,0} = D_{0,j} = +\infty$ for $i, j > 0$, and, for $1 \leq i \leq n$, $1 \leq j \leq m$, define

$$D_{ij} = C_{ij} + \min\{D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1}\}. \quad (12.22)$$

Then $D_{nm} = \text{DTW}_c(x, y)$. Storing one minimizing predecessor at every cell and backtracking from (n, m) recovers an optimal path. The computation takes $O(nm)$ time and $O(nm)$ memory; if only the value is required, two rows give $O(\min\{n, m\})$ memory.

Proof. Every admissible path reaching (i, j) enters it from exactly one of $(i-1, j)$, $(i, j-1)$ or $(i-1, j-1)$. Removing the last cell therefore leaves an admissible path to one of these predecessors, while appending (i, j) to any predecessor path gives an admissible path to (i, j) . Minimizing over the three mutually exhaustive last steps proves (12.22) by induction on $i + j$. The complexity follows because each of the nm cells performs constant work. $\square$

Continuous time warping. For simplicity, let $x, y : [0, 1] \rightarrow \mathcal{Z}$ use normalized clocks, and let $\Gamma_\uparrow$ contain pairs (φ, ψ) of absolutely continuous, nondecreasing surjections of $[0, 1]$ onto itself. Equivalently, the endpoint conditions are $\varphi(0) = \psi(0) = 0$ and $\varphi(1) = \psi(1) = 1$. The continuous DTW functional is

$$\text{CDTW}_c(x, y) := \inf_{(\varphi, \psi) \in \Gamma_\uparrow} \int_0^1 c(x(\varphi(s)), y(\psi(s))) (\dot{\varphi}(s) + \dot{\psi}(s)) ds. \quad (12.23)$$

Soft-DTW and the Sinkhorn analogy.

$$\text{softmin}_\varepsilon(r_1, \dots, r_q) = -\varepsilon \log \left(\sum_{k=1}^q e^{-r_k/\varepsilon} \right), \quad (12.24)$$

$$D_{ij}^\varepsilon = C_{ij} + \text{softmin}_\varepsilon(D_{i-1,j}^\varepsilon, D_{i,j-1}^\varepsilon, D_{i-1,j-1}^\varepsilon), \quad \text{sDTW}_{c,\varepsilon}(x, y) = D_{nm}^\varepsilon. \quad (12.25)$$

$$\text{sDTW}_{c,\varepsilon}(x, y) = -\varepsilon \log \sum_{\omega \in \Omega_{n,m}} \exp\left(-\frac{\langle A_\omega, C \rangle}{\varepsilon}\right). \quad (12.26)$$

$$\text{sDTW}_{c,\varepsilon}(x, y) = \min_{q \in \Delta(\Omega_{n,m})} \left\{ \sum_{\omega} q_\omega \langle A_\omega, C \rangle - \varepsilon H(q) \right\}. \quad (12.27)$$

$$\mathbb{P}_\varepsilon(\omega) = \frac{\exp(-\langle A_\omega, C \rangle/\varepsilon)}{\sum_{\omega'} \exp(-\langle A_{\omega'}, C \rangle/\varepsilon)}, \quad E_\varepsilon := \nabla_C \text{sDTW}_{c,\varepsilon}(x, y). \quad (12.28)$$

$$\text{DTW}_c(x, y) - \varepsilon \log |\Omega_{n,m}| \leq \text{sDTW}_{c,\varepsilon}(x, y) \leq \text{DTW}_c(x, y), \quad (12.29)$$

because the partition sum is bounded below by its largest term and above by $|\Omega_{n,m}|$ times that term. Hence $\text{sDTW}_{c,\varepsilon} \rightarrow \text{DTW}_c$ as $\varepsilon \rightarrow 0$.

$$E_\varepsilon = \mathbb{E}_{\omega \sim \mathbb{P}_\varepsilon} [A_\omega]. \quad (12.30)$$

Thus $(E_\varepsilon)_{ij}$ is the probability that a Gibbs path visits cell (i, j) . The matrix E_ε is a diffuse expected alignment and converges, when the hard optimum is unique, to its path-incidence matrix.

$$\overline{\text{sDTW}}_{c,\varepsilon}(x, y) = \text{sDTW}_{c,\varepsilon}(x, y) - \frac{1}{2} \text{sDTW}_{c,\varepsilon}(x, x) - \frac{1}{2} \text{sDTW}_{c,\varepsilon}(y, y). \quad (12.31)$$

13 Dynamic Optimal Transport

13.1 Evolutions over the Space of Measures

Lagrangian and Eulerian descriptions.

$$\frac{dx(t)}{dt} = v_t(x(t)), \quad (13.1)$$

$$\frac{\partial \alpha_t}{\partial t} + \operatorname{div}(v_t \alpha_t) = 0. \quad (13.2)$$

$$\int_0^1 \int_{\mathbb{R}^d} (\partial_t \varphi(t, x) + \langle v_t(x), \nabla_x \varphi(t, x) \rangle) d\alpha_t(x) dt = 0. \quad (13.3)$$

This equation is obtained from (13.2) by integration by parts. Hence, for smooth positive densities, the classical and weak formulations are equivalent; the weak formulation is useful because it still makes sense for discrete measures whose particles evolve according to (13.1).

Proposition 13.1 (Lagrangian flows solve the continuity equation). *Let $(T_t)_{t \in [0,1]}$ be a C^1 family of diffeomorphisms of $\mathbb{R}^d$ and define $\alpha_t = (T_t)_\# \alpha_0$. Assume that the derivatives below are integrable with respect to α_0 , and define the Eulerian velocity field by $v_t(T_t(y)) = \partial_t T_t(y)$. Then (α_t, v_t) solves the continuity equation in the weak sense (13.3). In particular, if $\alpha_0 = \frac{1}{n} \sum_i \delta_{x_i(0)}$ is empirical, then $\alpha_t = \frac{1}{n} \sum_i \delta_{x_i(t)}$ is empirical as well, with particle velocities $\dot{x}_i(t) = v_t(x_i(t))$.*

Proof. Let $\varphi \in C_c^1((0,1) \times \mathbb{R}^d)$. Since $\alpha_t = (T_t)_\# \alpha_0$, $\frac{d}{dt} \int \varphi(t, x) d\alpha_t(x) = \frac{d}{dt} \int \varphi(t, T_t(y)) d\alpha_0(y)$. The chain rule gives

$$\frac{d}{dt} \int \varphi(t, T_t(y)) d\alpha_0(y) = \int (\partial_t \varphi(t, T_t(y)) + \langle \nabla_x \varphi(t, T_t(y)), \partial_t T_t(y) \rangle) d\alpha_0(y).$$

Using the definition of v_t and the push-forward relation, this equals

$$\int (\partial_t \varphi(t, x) + \langle \nabla_x \varphi(t, x), v_t(x) \rangle) d\alpha_t(x).$$

Integrating in time and using the boundary values of φ gives (13.3). $\square$

From measure evolutions to vector fields. For a given evolution $(\alpha_t)_t$, there are typically infinitely many velocity fields v_t satisfying

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0. \quad (13.4)$$

$\mathcal{H}_\alpha = \{v \in L^2(\alpha; \mathbb{R}^d) ; \operatorname{div}(\alpha v) = 0 \text{ in distributions}\}.$

Dacorogna–Moser inversion.

$$v_t = -\frac{1}{\rho_t} \nabla \Delta^{-1}(\partial_t \rho_t), \quad (13.5)$$

$\alpha_t = (1-t)\alpha_0 + t\alpha_1 = \rho_t dx$, $\rho_t = (1-t)\rho_0 + t\rho_1$. $\operatorname{div} w = \rho_0 - \rho_1$, $w \cdot n = 0$ on $\partial\Omega$, for instance $w = -\nabla \varphi$ with $\Delta \varphi = \rho_1 - \rho_0$ and Neumann boundary condition. Then $v_t = \frac{w}{\rho_t}$ satisfies $\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0$. The flow $\partial_t T_t = v_t \circ T_t$, $T_0 = \operatorname{Id}$, therefore transports $\rho_0 dx$ onto $\rho_t dx$, and T_1 solves the prescribed-Jacobian problem $\rho_1(T_1(x)) \det(\nabla T_1(x)) = \rho_0(x)$.

Least-square inversion and gradient structure.

$$\min_v \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 d\alpha_t(x) dt \quad \text{subject to} \quad \partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0. \quad (13.6)$$

Proposition 13.2 (Least-square velocities are gradients). *Assume that $\alpha_t = \rho_t dx$ is a smooth positive density curve, that $\partial_t \rho_t$ has zero integral, and that boundary terms vanish. The minimizer of (13.6), if it exists, is a gradient field $v_t = \nabla \varphi_t$, where φ_t , unique up to an additive constant on each connected component, solves the weighted Poisson equation*

$$-\operatorname{div}(\rho_t \nabla \varphi_t) = \partial_t \rho_t, \quad v_t = -\nabla \Delta_{\alpha_t}^{-1}(\partial_t \alpha_t), \quad \Delta_{\alpha_t} \varphi = \operatorname{div}(\alpha_t \nabla \varphi). \quad (13.7)$$

Proof. Introduce a Lagrange multiplier φ_t for the continuity equation. The constrained problem has the formal saddle formulation

$$\min_v \max_\varphi \int_0^1 \left[\frac{1}{2} \int_{\mathbb{R}^d} \|v_t(x)\|^2 d\alpha_t(x) + \int_{\mathbb{R}^d} \varphi_t(x) (\operatorname{div}(\alpha_t v_t)(x) + \partial_t \alpha_t(x)) dx \right] dt.$$

Integrating by parts in the divergence term gives, for each t ,

$$\int \left(\frac{1}{2} \|v_t\|^2 - \langle \nabla \varphi_t, v_t \rangle \right) d\alpha_t + \int \varphi_t \partial_t \alpha_t.$$

The pointwise minimizer in v_t is therefore $v_t = \nabla \varphi_t$. Substituting this into the constraint $\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0$ gives the weighted Poisson equation in (13.7). The inverse notation is just a shorthand for solving this equation on zero-mean right-hand sides, modulo additive constants. $\square$

13.2 Benamou–Brenier dynamic formulation of OT

Benamou–Brenier formulation.

Theorem 13.3 (Benamou–Brenier). *For probability measures $\alpha_0, \alpha_1 \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{(\alpha_t, v_t)} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 d\alpha_t(x) dt, \quad (13.8)$$

where the infimum is over (α_t, v_t) solving $\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0$ with $\alpha_{t=0} = \alpha_0$ and $\alpha_{t=1} = \alpha_1$. If α_0 has a density and T is the optimal Monge map $T_{\#}\alpha_0 = \alpha_1$, a minimizing curve is

$$\alpha_t = ((1-t)\text{Id} + tT)_{\#}\alpha_0, \quad v_t((1-t)x + tT(x)) = T(x) - x \quad \text{for } \alpha_0\text{-a.e. } x \text{ and a.e. } t \in (0, 1). \quad (13.9)$$

Proof. For the inequality “dynamic $\leq$ static”, assume first that a Monge map T exists and define (α_t, v_t) by (13.9). Since the Lagrangian velocity $T(x) - x$ is independent of t , $\int_0^1 \int \|v_t\|^2 d\alpha_t dt = \int \|T(x) - x\|^2 d\alpha_0(x)$, so the dynamic cost is no larger than the static Monge cost. Without a Monge map, the same construction is made with an optimal coupling π : sample $(X, Y) \sim \pi$ and move along the straight path $\gamma_{X,Y}(t) = (1-t)X + tY$. This path measure has action $\int \|x - y\|^2 d\pi(x, y)$; projecting path velocities onto their conditional mean at time t gives an admissible Eulerian velocity with no larger action, so the dynamic value is no larger than the Kantorovich value.

Conversely, for a smooth deterministic path, take the flow T_t defined by $\dot{T}_t = v_t \circ T_t$ and $T_0 = \text{Id}$. Then $\alpha_t = (T_t)_{\#}\alpha_0$ and $(T_1)_{\#}\alpha_0 = \alpha_1$. Jensen’s inequality gives $\|T_1(x) - x\|^2 \leq \int_0^1 \|v_t(T_t(x))\|^2 dt$. After integration with respect to α_0 , the Monge cost is bounded above by the dynamic action. For general finite-energy solutions of the continuity equation, the superposition principle lifts the curve to a probability measure on absolutely continuous paths; applying Jensen’s inequality pathwise gives a coupling of the endpoints whose quadratic cost is no larger than the action. Thus the Kantorovich value is bounded above by the dynamic value. $\square$

Convex moment-based reformulation. Although (13.8) is not jointly convex in (α_t, v_t) , it becomes convex after replacing velocities by momenta. Given a velocity $v \in L^2(\alpha; \mathbb{R}^d)$, define the momentum

$$\omega := \alpha v, \quad \omega(A) = \int_A v(x) d\alpha(x),$$

$$J(a, m) := \begin{cases} \|m\|^2/a, & a > 0, \\ 0, & a = 0 \text{ and } m = 0, \\ +\infty, & a = 0 \text{ and } m \neq 0, \end{cases} \quad (a, m) \in [0, +\infty) \times \mathbb{R}^d. \quad (13.10)$$

$$\mathbb{J}(\alpha, \omega) := \int J\left(\frac{d\alpha}{d\lambda}(x), \frac{d\omega}{d\lambda}(x)\right) d\lambda(x). \quad (13.11)$$

$$\mathbb{J}(\alpha, \omega) < +\infty \iff \omega = v\alpha \text{ with } v \in L^2(\alpha; \mathbb{R}^d), \quad \mathbb{J}(\alpha, \omega) = \int \|v\|^2 d\alpha.$$

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \alpha_t + \text{div } \omega_t = 0 \\ \alpha_{t=0} = \alpha_0, \alpha_{t=1} = \alpha_1}} \int_0^1 \mathbb{J}(\alpha_t, \omega_t) dt. \quad (13.12)$$

$$\inf_{\substack{\partial_t \rho_t + \text{div } m_t = 0 \\ \rho_{t=0} d\mathbf{x} = \alpha_0, \rho_{t=1} d\mathbf{x} = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} J(\rho_t(x), m_t(x)) dx dt = \inf_{\substack{\partial_t \rho_t + \text{div } m_t = 0 \\ \rho_{t=0} d\mathbf{x} = \alpha_0, \rho_{t=1} d\mathbf{x} = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} \frac{\|m_t(x)\|^2}{\rho_t(x)} dx dt,$$

Dual Hamilton–Jacobi formulation. The momentum formulation also has a useful dual. It turns the least-action problem into a Hamilton–Jacobi subsolution inequality for a scalar potential, with equality on the part of space-time actually visited by the optimal curve.

Proposition 13.4 (Dual Benamou–Brenier problem). *Assume, for simplicity, that the densities are smooth, compactly supported, and that boundary terms vanish. Then the convex dynamic value (13.12) has the dual formulation*

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \sup_{\varphi} \left\{ \int_{\mathbb{R}^d} \varphi_1 d\alpha_1 - \int_{\mathbb{R}^d} \varphi_0 d\alpha_0 ; \partial_t \varphi_t + \frac{1}{4} \|\nabla \varphi_t\|^2 \leq 0 \right\}, \quad (13.13)$$

where the supremum is over smooth test potentials $\varphi : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}$. In the general metric-measure statement, the same formula is understood after relaxation: φ is a Hamilton–Jacobi subsolution, for instance in the viscosity sense, and finite-dimensional discretizations follow directly from Fenchel–Rockafellar duality. If (ρ, m) and φ are smooth primal and dual optimizers, then

$$m_t = \frac{\rho_t}{2} \nabla \varphi_t, \quad \partial_t \varphi_t + \frac{1}{4} \|\nabla \varphi_t\|^2 = 0 \quad \text{on } \{\rho_t > 0\}. \quad (13.14)$$

Equivalently, the optimal Eulerian velocity is $v_t = m_t/\rho_t = \nabla \varphi_t/2$.

Proof. Let (ρ, m) satisfy the continuity equation and let φ be smooth. Multiplying the constraint by φ and integrating by parts gives

$$\int_{\mathbb{R}^d} \varphi_1 d\alpha_1 - \int_{\mathbb{R}^d} \varphi_0 d\alpha_0 = \int_0^1 \int_{\mathbb{R}^d} (\rho_t \partial_t \varphi_t + \langle m_t, \nabla \varphi_t \rangle) dx dt.$$

If $\partial_t \varphi_t + \|\nabla \varphi_t\|^2/4 \leq 0$, Young’s inequality yields, pointwise, $\rho \partial_t \varphi + \langle m, \nabla \varphi \rangle \leq -\frac{\rho}{4} \|\nabla \varphi\|^2 + \langle m, \nabla \varphi \rangle \leq J(\rho, m)$, with the perspective convention (13.10). Hence the dual objective of every feasible φ is no larger than the action of every feasible primal pair.

Conversely, introduce φ as a Lagrange multiplier for $\partial_t \rho + \text{div } m = 0$. After integration by parts, and up to the fixed endpoint contribution in (13.13), the pointwise minimization over (ρ, m) contains $J(\rho, m) - \rho \partial_t \varphi - \langle m, \nabla \varphi \rangle$. For fixed $\rho > 0$, its minimum over m is attained at $m = \rho \nabla \varphi/2$ and equals $-\rho(\partial_t \varphi + \|\nabla \varphi\|^2/4)$. The recession cases in (13.10) give the same conclusion when $\rho = 0$. Thus the infimum over $\rho \geq 0$ is finite exactly when the Hamilton–Jacobi inequality holds, in which case the pointwise infimum is 0. Fenchel–Rockafellar duality then gives no duality gap. Equality in the two pointwise inequalities gives (13.14). $\square$

$\frac{d}{dt}\varphi_t(\gamma(t)) = \partial_t\varphi_t(\gamma(t)) + \langle \nabla\varphi_t(\gamma(t)), \dot{\gamma}(t) \rangle \leq \|\dot{\gamma}(t)\|^2$, where the last inequality uses $\partial_t\varphi + \|\nabla\varphi\|^2/4 \leq 0$ and Young's inequality. Integrating and minimizing over curves gives

$\varphi_1(y) - \varphi_0(x) \leq \|x - y\|^2$. Thus $(-\varphi_0, \varphi_1)$ is a feasible static Kantorovich dual pair for the quadratic cost. At optimality the inequality is saturated on the endpoint pairs connected by the primal characteristics, which is the Eulerian version of complementary slackness.

Proximal splitting. The convex momentum formulation also explains the original Benamou–Brenier solver. Suppressing discretization indices, write $U = (\rho, m)$, where m is the density of the momentum measure.

$$\mathcal{F}(U) = \int_0^1 \int_{\mathbb{R}^d} J(\rho_t(x), m_t(x)) dx dt, \quad \mathcal{C} = \{(\rho, m) ; \partial_t \rho + \operatorname{div} m = 0, \rho_0 dx = \alpha_0, \rho_1 dx = \alpha_1\},$$

$\min_U \mathcal{F}(U) + \mathcal{G}(U)$.

$$\operatorname{prox}_{\tau\mathcal{F}}(\bar{U}) = \operatorname{argmin}_U \frac{1}{2}\|U - \bar{U}\|_{\mathbb{H}}^2 + \tau\mathcal{F}(U), \quad \operatorname{prox}_{\tau\mathcal{G}}(\bar{U}) = \operatorname{argmin}_{U \in \mathcal{C}} \frac{1}{2}\|U - \bar{U}\|_{\mathbb{H}}^2.$$

For the standard product metric, the first map is local in (t, x) : it is the proximal operator of the convex perspective J . The second map is the orthogonal projection onto the affine set defined by the divergence equation and endpoint constraints. Douglas–Rachford then alternates these two simple operations:

$$\begin{aligned} U^{k+1} &= \operatorname{prox}_{\tau\mathcal{F}}(Z^k), \\ \tilde{U}^{k+1} &= \operatorname{prox}_{\tau\mathcal{G}}(2U^{k+1} - Z^k), \\ Z^{k+1} &= Z^k + \tilde{U}^{k+1} - U^{k+1}. \end{aligned}$$

Path-space formulation. Let $\mathcal{S} = C([0, 1]; \mathbb{R}^d)$ be the space of continuous paths endowed with the uniform topology. For $t \in [0, 1]$ define the evaluation map $e_t : \mathcal{S} \rightarrow \mathbb{R}^d$, $e_t(\gamma) = \gamma(t)$.

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{M \in \mathcal{P}(\mathcal{S})} \left\{ \int_{\mathcal{S}} \int_0^1 \|\dot{\gamma}(t)\|^2 dt dM(\gamma) ; (e_0)_\# M = \alpha_0, (e_1)_\# M = \alpha_1 \right\}.$$

$(e_t, \dot{e}_t)_\# M^*(dx, dq) = \alpha_t(dx) \delta_{v_t^*(x)}(dq)$, where v_t^* is the optimal velocity field in the Benamou–Brenier formulation. Hence M^* concentrates on straight-line geodesics and, for a.e. t , assigns exactly one direction at α_t -a.e. spatial point.

13.3 Generalized Dynamic Wasserstein Distances

The quadratic Benamou–Brenier formula is only one instance of a broader fixed-mass dynamic language. The objects introduced here are metric: they specify admissible curves, tangent variables and path energies.

Path actions. In the mass-preserving Euclidean setting, the basic input is an instantaneous action $\mathbb{A}(\alpha, w)$, where α is the current measure and w is an admissible velocity representative. When this action is normalized as a squared infinitesimal speed, it generates the length-space value

$$D_{\mathbb{A}}^2(\alpha_0, \alpha_1) = \inf_{\alpha_t, v_t} \left\{ \int_0^1 \mathbb{A}(\alpha_t, v_t) dt : \partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0, \alpha_{t=0} = \alpha_0, \alpha_{t=1} = \alpha_1 \right\}. \quad (13.15)$$

Quadratic, or Riemannian, tangent actions.

$$\mathbb{A}(\alpha; w, z) = \langle Q_\alpha w, z \rangle_{L^2(\alpha)}, \quad \mathbb{A}(\alpha, w) = \langle Q_\alpha w, w \rangle_{L^2(\alpha)},$$

$$D_Q^2(\alpha_0, \alpha_1) = D_{\mathbb{A}}^2(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0 \\ \alpha_{t=0} = \alpha_0, \alpha_{t=1} = \alpha_1}} \int_0^1 \langle Q_{\alpha_t} v_t, v_t \rangle_{L^2(\alpha_t)} dt. \quad (13.16)$$

The usual $\mathcal{W}_2$ geometry corresponds to $Q_\alpha = \operatorname{Id}$ in this simplified notation. Thus Q_α records how the chosen geometry deforms the Euclidean $L^2(\alpha)$ tangent norm: no deformation for $\mathcal{W}_2$, and a nontrivial tensor for generalized Riemannian geometries.

Local velocity actions. $A : [0, +\infty) \times \mathbb{R}^d \rightarrow [0, +\infty]$, $(a, w) \mapsto A(a, w)$, where $a \in \mathbb{R}_+$ is a density value and $w \in \mathbb{R}^d$ is a velocity value, and defines

$$\mathbb{A}(\alpha, w) = \int A\left(\frac{d\alpha}{d\lambda}(x), w(x)\right) d\lambda(x). \quad (13.17)$$

For a fixed reference λ , this covers density-dependent mobilities and congestion constraints. If, in addition, A is positively 1-homogeneous in its first variable, $A(\eta a, w) = \eta A(a, w)$ for $\eta \geq 0$, then the same formula is intrinsic: replacing λ by another dominating measure gives the same value. The usual Benamou–Brenier action is the model case

$$A_2(a, w) = a\|w\|^2, \quad \mathbb{A}(\alpha, w) = \int \|w\|^2 d\alpha. \quad (13.18)$$

Homogeneous momentum actions.

$$J_A(a, m) := \begin{cases} A(a, m/a), & a > 0, \\ 0, & a = 0 \text{ and } m = 0, \\ +\infty, & a = 0 \text{ and } m \neq 0, \end{cases} \quad (13.19)$$

$$\mathbb{J}_{A, \lambda}(\alpha, \omega) := \int J_A\left(\frac{d\alpha}{d\lambda}, \frac{d\omega}{d\lambda}\right) d\lambda, \quad (13.20)$$

$$J_2(a, m) = \frac{\|m\|^2}{a},$$

Proposition 13.5 (Concave mobilities give convex momentum actions). *Let $I \subset [0, +\infty)$ be a convex interval, let $\theta : I \rightarrow [0, +\infty)$ be concave, and let $L : \mathbb{R}^d \rightarrow [0, +\infty]$ be convex with $L(0) = 0$. Define, on the set where $\theta(a) > 0$,*

$$J_{\theta,L}(a, m) := \theta(a)L\left(\frac{m}{\theta(a)}\right), \quad A_{\theta,L}(a, w) := J_{\theta,L}(a, aw) = \theta(a)L\left(\frac{aw}{\theta(a)}\right), \quad (13.21)$$

and extend $J_{\theta,L}$ to the boundary by lower semicontinuity. Then $J_{\theta,L}$ is convex in (a, m) . In particular, this single construction contains $A(a, w) = aL(w)$ by taking $\theta(a) = a$, and the concave-mobility quadratic action $A(a, w) = \frac{a^2\|w\|^2}{\theta(a)}$ by taking $L(u) = \|u\|^2$.

Proof. The perspective $P(s, m) = sL(m/s)$ of a convex function is convex on $s > 0$. Since $L(0) = 0$, it is also nonincreasing in the scalar s for fixed m : if $s_2 \geq s_1 > 0$, then $L(m/s_2) = L((s_1/s_2)(m/s_1)) \leq (s_1/s_2)L(m/s_1)$, hence $P(s_2, m) \leq P(s_1, m)$. Let $a_\zeta = (1 - \zeta)a_0 + \zeta a_1$ and $m_\zeta = (1 - \zeta)m_0 + \zeta m_1$. By concavity of θ , $\theta(a_\zeta) \geq (1 - \zeta)\theta(a_0) + \zeta\theta(a_1)$. Using the monotonicity in s , and then the convexity of P , gives

$$J_{\theta,L}(a_\zeta, m_\zeta) = P(\theta(a_\zeta), m_\zeta) \leq P((1 - \zeta)\theta(a_0) + \zeta\theta(a_1), m_\zeta) \leq (1 - \zeta)J_{\theta,L}(a_0, m_0) + \zeta J_{\theta,L}(a_1, m_1).$$

The boundary cases follow by the chosen lower-semicontinuous extension. $\square$

Proposition 13.6 (Homogeneous dynamic actions define distances). *Fix a reference measure λ , omitted from the notation only in the intrinsic case where $\mathbb{J}_{A,\lambda}$ does not depend on λ . Assume that the momentum perspective J_A defined in (13.19) is lower semicontinuous, convex in (a, m) , even in m , and satisfies $J_A(a, 0) = 0$. Assume moreover that, for some $r > 1$, $J_A(a, \xi m) = |\xi|^r J_A(a, m)$ for all admissible a , all m , and all $\xi \in \mathbb{R}$, and that $J_A(a, m) = 0$ if and only if $m = 0$. Equivalently, the evenness, homogeneity and nondegeneracy assumptions translate, away from zero density, into $A(a, -w) = A(a, w)$, $A(a, \xi w) = |\xi|^r A(a, w)$, $A(a, w) = 0 \iff w = 0$ ($a > 0$). Define, on every fixed-mass class,*

$$D_{A,\lambda}(\alpha_0, \alpha_1) := \inf_{\substack{\partial_t \alpha_t + \operatorname{div} \omega_t = 0 \\ \alpha_t = 0 = \alpha_0, \quad \alpha_t = 1 = \alpha_1}} \left(\int_0^1 \mathbb{J}_{A,\lambda}(\alpha_t, \omega_t) dt \right)^{1/r}.$$

Assume finally that the relaxed dynamic problem is sequentially closed and attains its infimum whenever the value is finite. Then $D_{A,\lambda}$ is an extended distance on each finite-action component: it is symmetric, satisfies the triangle inequality, and $D_{A,\lambda}(\alpha_0, \alpha_1) = 0$ only when $\alpha_0 = \alpha_1$. In the intrinsic case, we write D_A .

Proof. The constant curve $\alpha_t = \alpha_0$, $\omega_t = 0$, gives $D_{A,\lambda}(\alpha_0, \alpha_0) = 0$. Conversely, if $D_{A,\lambda}(\alpha_0, \alpha_1) = 0$, attainment provides a relaxed zero-action minimizer, which satisfies $\int_0^1 \mathbb{J}_{A,\lambda}(\alpha_t, \omega_t) dt = 0$. Since $\mathbb{J}_{A,\lambda} \geq 0$, this implies $\omega_t = 0$ for a.e. t . The continuity equation then gives $\partial_t \alpha_t = 0$ in the sense of distributions, so $\alpha_0 = \alpha_1$.

Symmetry follows by time reversal. If $(\alpha_t, \omega_t)_{t \in [0,1]}$ connects α_0 to α_1 , then $\tilde{\alpha}_t = \alpha_{1-t}$, $\tilde{\omega}_t = -\omega_{1-t}$ connects α_1 to α_0 , satisfies the same continuity equation, and has the same action because J_A is even in m .

Let (α^1, ω^1) connect α_0 to α_1 with action E_1 , and let (α^2, ω^2) connect α_1 to α_2 with action E_2 . For $\tau \in (0, 1)$, concatenate the two curves after the time changes $s = t/\tau$ and $s = (t - \tau)/(1 - \tau)$:

$$(\alpha_t, \omega_t) = \begin{cases} (\alpha_{t/\tau}^1, \tau^{-1}\omega_{t/\tau}^1), & 0 \leq t \leq \tau, \\ (\alpha_{(t-\tau)/(1-\tau)}^2, (1-\tau)^{-1}\omega_{(t-\tau)/(1-\tau)}^2), & \tau \leq t \leq 1. \end{cases}$$

The rescaled curve is admissible from α_0 to α_2 . By r -homogeneity in the momentum variable, its action is $\tau^{1-r}E_1 + (1-\tau)^{1-r}E_2$. Writing $X = E_1^{1/r}$ and $Y = E_2^{1/r}$, the optimal allocation is $\tau = X/(X + Y)$, with the evident limiting convention if $X + Y = 0$, and gives

$$\inf_{\tau \in (0,1)} \{ \tau^{1-r}E_1 + (1-\tau)^{1-r}E_2 \} = (X + Y)^r.$$

Hence $D_{A,\lambda}(\alpha_0, \alpha_2) \leq E_1^{1/r} + E_2^{1/r}$. Taking E_1 and E_2 arbitrarily close to the two optimal action values yields $D_{A,\lambda}(\alpha_0, \alpha_2) \leq D_{A,\lambda}(\alpha_0, \alpha_1) + D_{A,\lambda}(\alpha_1, \alpha_2)$, with the usual extended-real convention when one of the distances is infinite. $\square$

Concave-mobility actions.

$$J_\theta(a, m) := \begin{cases} \|m\|^2/\theta(a), & \theta(a) > 0, \\ 0, & \theta(a) = 0 \text{ and } m = 0, \\ +\infty, & \theta(a) = 0 \text{ and } m \neq 0. \end{cases}$$

$A_\theta(a, w) = J_\theta(a, aw) = \frac{a^2\|w\|^2}{\theta(a)}$. $\mathbb{A}_{\theta,\lambda}(\alpha, w) = \int A_\theta(a(x), w(x)) d\lambda(x) = \int \frac{a(x)^2}{\theta(a(x))} \|w(x)\|^2 d\lambda(x)$, $\mathbb{A}_{\theta,\lambda}(\alpha, w) = \int \frac{a(x)}{\theta(a(x))} \|w(x)\|^2 d\alpha(x)$. Hence this is a local Riemannian case, in the sense of (13.16), whenever the multiplier $a(x)/\theta(a(x))$ is finite and positive. The associated tensor is the multiplication operator

$$(Q_{\theta,\lambda,\alpha} w)(x) = \frac{a(x)}{\theta(a(x))} w(x), \quad \alpha = a\lambda, \quad (13.22)$$

defined α -a.e. on the set where $\theta(a) > 0$. Except for linear mobilities $\theta(a) = ca$, and in particular the normalized case $\theta(a) = a$ which recovers $\mathcal{W}_2$, the pointwise velocity action $A_\theta(a, w)$ is not positively 1-homogeneous in the density variable a .

$\mathcal{W}_{\theta,\lambda}(\alpha_0, \alpha_1) := D_{\mathbb{A}_{\theta,\lambda}}(\alpha_0, \alpha_1)$, where the subscript λ recalls that the action is measured through the density $a = d\alpha/d\lambda$. Equivalently, because this action is quadratic in the momentum, $\mathcal{W}_{\theta,\lambda}^2$ is the path value (13.15) with $\mathbb{A} = \mathbb{A}_{\theta,\lambda}$.

Proposition 13.7 (Concave-mobility dynamic distances). *For a fixed reference measure λ , and under the standard compactness and lower-semicontinuity hypotheses ensuring existence of relaxed minimizers, $\mathcal{W}_{\theta,\lambda} = D_{\mathbb{A}_{\theta,\lambda}}$ is an extended distance on each finite-action component contained in $\{\alpha : \alpha \ll \lambda\}$. In particular, whenever the relevant component has finite pairwise distances, $\mathcal{W}_{\theta,\lambda}$ is a genuine metric there.*

Proof. Proposition 13.5, applied with $L(u) = \|u\|^2$, gives the lower-semicontinuous convex momentum density J_θ . It is even in m , satisfies $J_\theta(a, 0) = 0$, and obeys $J_\theta(a, \xi m) = |\xi|^2 J_\theta(a, m)$. Moreover $J_\theta(a, m) = 0$ if and only if $m = 0$, with the boundary convention used in its definition. For the fixed reference λ , the hypotheses of Proposition 13.6 therefore hold with $r = 2$; the existence assumption is supplied by the compactness hypothesis in the statement. That proposition gives symmetry, separation and the triangle inequality for $D_{A_\theta, \lambda}$, hence for $W_{\theta, \lambda}$. $\square$

Dynamic spectral Wasserstein distances. Let γ be a monotone spectral gauge on $\mathbb{S}_+^d$. For a probability measure α and a velocity field $v \in L^2(\alpha; \mathbb{R}^d)$, define the spectral tangent action

$$\mathbb{A}_\gamma(\alpha, v) := \gamma\left(\int v(x)v(x)^\top d\alpha(x)\right). \quad (13.23)$$

$$W_{\gamma, \text{dyn}}^2 := D_{\mathbb{A}_\gamma}^2. \quad (13.24)$$

$$\mathbb{J}_\gamma(\alpha, \omega) = \gamma\left(\int \left(\frac{d\omega}{d\alpha}\right) \left(\frac{d\omega}{d\alpha}\right)^\top d\alpha\right),$$

$$\mathbb{J}_\gamma(\rho, m) = \gamma\left(\int \frac{m(x)m(x)^\top}{\rho(x)} dx\right).$$

$$A_{\text{lin}}(a, w) = a w^\top G w, \quad J_{\text{lin}}(a, m) = \frac{m^\top G m}{a},$$

Proposition 13.8 (Static/dynamic equality for spectral Wasserstein). *Let γ be a monotone spectral gauge and let $\alpha_0, \alpha_1 \in \mathcal{P}_2(\mathbb{R}^d)$. Then $W_{\gamma, \text{dyn}}^2(\alpha_0, \alpha_1) = W_\gamma^2(\alpha_0, \alpha_1)$. The common value defines a distance on $\mathcal{P}_2(\mathbb{R}^d)$; the triangle inequality is established by the robust representation in Proposition 10.23.*

Proof. We first prove $W_{\gamma, \text{dyn}}^2 \leq W_\gamma^2$. Let $\pi \in \mathcal{U}(\alpha_0, \alpha_1)$ and let $(X, Y) \sim \pi$. Set $Z_t = (1-t)X + tY$ and $\alpha_t = (Z_t)_\# \pi$. Define the Eulerian velocity as the conditional mean $v_t(z) := \mathbb{E}[Y - X \mid Z_t = z]$. Then (α_t, v_t) solves the continuity equation. If $M_\pi := \int (x - y)(x - y)^\top d\pi(x, y)$, $C_t := \int v_t(z)v_t(z)^\top d\alpha_t(z)$, conditional Jensen gives $C_t \preceq M_\pi$. Indeed, for every direction $u \in \mathbb{R}^d$,

$$u^\top C_t u = \mathbb{E}[\mathbb{E}[\langle u, Y - X \rangle \mid Z_t]^2] \leq \mathbb{E}[(\langle u, Y - X \rangle)^2] = u^\top M_\pi u.$$

By the Loewner monotonicity of γ , $\int_0^1 \gamma(C_t) dt \leq \gamma(M_\pi)$. Taking the infimum over π gives the first inequality.

Conversely, let (α_t, v_t) be an admissible finite-action competitor. Since γ is a finite positive gauge on the finite-dimensional cone $\mathbb{S}_+^d$, it is equivalent to the trace on this cone; hence finite spectral action implies the usual finite kinetic energy needed for the superposition principle. Thus there exists a probability law η on absolutely continuous paths ω such that $(e_t)_\# \eta = \alpha_t$ and $\dot{\omega}_t = v_t(\omega_t)$ for a.e. t . Let $\pi = (e_0, e_1)_\# \eta$ be the induced endpoint coupling. With $C_t = \int v_t v_t^\top d\alpha_t$, Jensen's inequality along each path gives

$$(\omega_1 - \omega_0)(\omega_1 - \omega_0)^\top = \left(\int_0^1 \dot{\omega}_t dt\right) \left(\int_0^1 \dot{\omega}_t dt\right)^\top \preceq \int_0^1 \dot{\omega}_t \dot{\omega}_t^\top dt.$$

After integration over η , $M_\pi \preceq \int_0^1 C_t dt$. Monotonicity of γ , followed by convexity, yields

$$\gamma(M_\pi) \leq \gamma\left(\int_0^1 C_t dt\right) \leq \int_0^1 \gamma(C_t) dt.$$

Since π is admissible for the static problem, $W_\gamma^2(\alpha_0, \alpha_1)$ is bounded by the action of the dynamic competitor. Taking the infimum over all competitors gives the reverse inequality.

The crucial assumption is the monotonicity of γ : both parts of the proof produce Loewner-order comparisons between covariance matrices, and these comparisons imply inequalities between actions only for monotone gauges. $\square$

Kernelized Benamou–Brenier distances. Let k be a positive definite kernel on $\mathbb{R}^d$ with scalar RKHS $\mathcal{H}_k$. The vector-valued RKHS is

$$\mathcal{H}_k^d := \mathcal{H}_k \times \cdots \times \mathcal{H}_k, \quad \|v\|_{\mathcal{H}_k^d}^2 := \sum_{\ell=1}^d \|v_\ell\|_{\mathcal{H}_k}^2.$$

$$\mathbb{A}_k(\alpha, v) := \|v\|_{\mathcal{H}_k^d}^2, \quad W_k^2 := D_{\mathbb{A}_k}^2, \quad (13.25)$$

where the general distance formula (13.15) is understood with the restricted admissible tangent class $v_t \in \mathcal{H}_k^d$. The action itself is independent of α ; the measure only enters through the continuity equation $\partial_t \alpha_t + \text{div}(\alpha_t v_t) = 0$, which says how the common smooth velocity field moves all particles.

Proposition 13.9 (Kernelized dynamic distance). *Assume that the kernel has uniformly bounded diagonal, $\kappa_k^2 := \sup_{x \in \mathbb{R}^d} k(x, x) < +\infty$. Then the quantity W_k defined by (13.25) is an extended distance on probability measures. More precisely, it is symmetric, satisfies the triangle inequality, and $W_k(\alpha_0, \alpha_1) = 0$ implies $\alpha_0 = \alpha_1$. It can take the value $+\infty$ between different finite-action components.*

Proof. The constant curve gives zero self-distance. To prove separation without assuming existence of a minimizing curve, use the RKHS evaluation bound $\|v(x)\| \leq \kappa_k \|v\|_{\mathcal{H}_k^d}$. For every $\varphi \in C_c^1(\mathbb{R}^d)$ and every admissible curve of action E , the weak continuity equation and Cauchy–Schwarz give

$$\begin{aligned} \left| \int \varphi d(\alpha_1 - \alpha_0) \right| &\leq \int_0^1 \int \|\nabla \varphi(x)\| \|v_t(x)\| d\alpha_t(x) dt \\ &\leq \kappa_k \|\nabla \varphi\|_\infty \left(\int_0^1 \|v_t\|_{\mathcal{H}_k^d}^2 dt \right)^{1/2} = \kappa_k \|\nabla \varphi\|_\infty \sqrt{E}. \end{aligned}$$

Taking the infimum over admissible curves shows that $\mathcal{W}_k(\alpha_0, \alpha_1) = 0$ forces equality of the integrals of every compactly supported smooth test function, hence $\alpha_0 = \alpha_1$.

Symmetry follows by time reversal and by replacing v_t with $-v_{1-t}$. For the triangle inequality, concatenate an almost optimal curve from α_0 to α_1 of action E_1 with one from α_1 to α_2 of action E_2 . Compressing the first curve into a time interval of length τ multiplies its action by $1/\tau$, and compressing the second into length $1 - \tau$ multiplies its action by $1/(1 - \tau)$. Optimizing over $\tau \in (0, 1)$ gives $(\sqrt{E_1} + \sqrt{E_2})^2$, and taking infima yields the triangle inequality. $\square$

One should read $\mathcal{W}_k$ as an extended distance on finite-action components, not as a replacement for $\mathcal{W}_2$ on all of $\mathcal{P}_2(\mathbb{R}^d)$.

13.4 Nonlocal Wasserstein Distances

$$\theta(a, b) := \begin{cases} \frac{a - b}{\log a - \log b}, & a \neq b, \\ a, & a = b, \end{cases} \quad (13.26)$$

Continuum jump kernels.

$$\int_{\mathcal{X} \times \mathcal{X}} \Phi(x, y) \mathbf{J}(\mathrm{d}x, \mathrm{d}y) := \int_{\mathcal{X}} \left(\int_{\mathcal{X}} \Phi(x, y) K(x, \mathrm{d}y) \right) \mathbf{m}(\mathrm{d}x). \quad (13.27)$$

$\bar{\nabla}\varphi(x, y) := \varphi(y) - \varphi(x)$ for the nonlocal gradient, and use the logarithmic mean θ defined in (13.26). A curve $\alpha_t = \rho_t \mathbf{m}$ driven by an antisymmetric velocity $v_t(x, y) = -v_t(y, x)$ satisfies the nonlocal continuity equation if, for all test functions φ ,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int \varphi \mathrm{d}\alpha_t = \frac{1}{2} \iint \bar{\nabla}\varphi(x, y) v_t(x, y) \theta(\rho_t(x), \rho_t(y)) \mathbf{J}(\mathrm{d}x, \mathrm{d}y). \quad (13.28)$$

$$\mathbb{A}_K(\alpha, v) := \frac{1}{2} \iint |v(x, y)|^2 \theta(\rho(x), \rho(y)) \mathbf{J}(\mathrm{d}x, \mathrm{d}y),$$

$$\mathcal{W}_K^2(\alpha_0, \alpha_1) := \inf_{\rho_t, v_t} \int_0^1 \mathbb{A}_K(\alpha_t, v_t) \mathrm{d}t, \quad (13.29)$$

where the infimum is over all curves solving (13.28) with endpoints α_0, α_1 .

Proposition 13.10 (Metric properties of nonlocal transport). *Under standard regularity and irreducibility assumptions on the jump kernel, $\mathcal{W}_K$ is an extended distance on the set of probability measures absolutely continuous with respect to $\mathbf{m}$, and finite-distance pairs are connected by constant-speed geodesics.*

Proof. We use the analytic compactness and lower-semicontinuity theorem for the logarithmic-mean action. In particular, action-bounded sequences of admissible curves are compact for the narrow topology, the weak nonlocal continuity equation is closed under this convergence, and the action is lower semicontinuous. These facts are the nonlocal analogues of the compactness theorem used in the Benamou–Brenier formulation.

Nonnegativity is immediate from the definition of $\mathbb{A}_K(\alpha, v)$. If $\alpha_0 = \alpha_1$, the constant curve $\rho_t = \rho_0$, $v_t = 0$, is admissible and has zero action, hence $\mathcal{W}_K(\alpha_0, \alpha_0) = 0$.

Symmetry follows by time reversal. If (ρ_t, v_t) transports α_0 to α_1 , set $\tilde{\rho}_t = \rho_{1-t}$, $\tilde{v}_t = -v_{1-t}$. For every test function φ , differentiating $\int \varphi \tilde{\rho}_t \mathrm{d}\mathbf{m}$ and using (13.28) shows that $(\tilde{\rho}_t, \tilde{v}_t)$ solves the same continuity equation from α_1 to α_0 . Since the action is quadratic in v , $\mathbb{A}_K(\tilde{\alpha}_t, \tilde{v}_t) = \mathbb{A}_K(\alpha_{1-t}, v_{1-t})$, so the action is unchanged.

For the triangle inequality, let (ρ_t^0, v_t^0) connect α_0 to α_1 with action E_0 , and let (ρ_t^1, v_t^1) connect α_1 to α_2 with action E_1 . For $0 < \zeta < 1$, concatenate the two curves after the rescaling

$$(\rho_t, v_t) = \begin{cases} (\rho_{t/\zeta}^0, \zeta^{-1} v_{t/\zeta}^0), & 0 \leq t \leq \zeta, \\ (\rho_{(t-\zeta)/(1-\zeta)}^1, (1-\zeta)^{-1} v_{(t-\zeta)/(1-\zeta)}^1), & \zeta < t \leq 1. \end{cases}$$

The factors ζ^{-1} and $(1-\zeta)^{-1}$ are exactly those needed for the weak continuity equation after changing time. Since $v \mapsto \mathbb{A}_K(\alpha, v)$ is quadratic, $\int_0^1 \mathbb{A}_K(\alpha_t, v_t) \mathrm{d}t = \frac{E_0}{\zeta} + \frac{E_1}{1-\zeta}$. Optimizing in ζ , or taking $\zeta = \sqrt{E_0}/(\sqrt{E_0} + \sqrt{E_1})$ when both actions are positive, gives a concatenated action $(\sqrt{E_0} + \sqrt{E_1})^2$. Taking infima over the two curves yields $\mathcal{W}_K(\alpha_0, \alpha_2) \leq \mathcal{W}_K(\alpha_0, \alpha_1) + \mathcal{W}_K(\alpha_1, \alpha_2)$.

If $\mathcal{W}_K(\alpha_0, \alpha_1) = 0$, choose admissible curves with actions tending to zero. By the compactness and lower-semicontinuity theorem quoted above, a subsequence converges to an admissible curve (ρ_t, v_t) of zero action. Hence $v_t = 0$ for $\theta(\rho_t(x), \rho_t(y)) \mathbf{J}(\mathrm{d}x, \mathrm{d}y) \mathrm{d}t$ -a.e. (t, x, y) . Substituting in the weak continuity equation gives $\frac{\mathrm{d}}{\mathrm{d}t} \int \varphi \mathrm{d}\alpha_t = 0$ for every admissible test function φ . The irreducibility/separation assumption ensures that these test functions determine the measure, so α_t is constant and $\alpha_0 = \alpha_1$.

Finally, if $\mathcal{W}_K(\alpha_0, \alpha_1) < +\infty$, the same direct-method compactness applied to a minimizing sequence gives a minimizer of (13.29). Reparametrizing this minimizing curve by metric arclength gives a constant-speed curve. Equivalently, for every $0 \leq s < t \leq 1$, the restriction of the minimizer to $[s, t]$, rescaled to unit time, is admissible between α_s and α_t ; the triangle inequality and equality of the total minimal action force $\mathcal{W}_K(\alpha_s, \alpha_t) = (t-s)\mathcal{W}_K(\alpha_0, \alpha_1)$ after this constant-speed parametrization. Thus finite-distance pairs are joined by constant-speed geodesics. The consequences for entropy dynamics and fractional PDE examples are developed in Section 14.6. $\square$

For a fixed jump kernel, this geometry is genuinely nonlocal and does not coincide with ordinary $\mathcal{W}_2$. The local metric is nevertheless recovered in a small-jump limit. More explicitly, on $\mathcal{X} = \mathbb{R}^d$, let $\eta(z) = \bar{\eta}(\|z\|)$ be a nonnegative radial profile with $M_2(\eta) := \int_{\mathbb{R}^d} \|z\|^2 \eta(z) \mathrm{d}z \in (0, +\infty)$,

$$K_\varepsilon(x, \mathrm{d}y) := \eta_\varepsilon(y - x) \mathrm{d}y, \quad \eta_\varepsilon(z) := \varepsilon^{-d} \eta(z/\varepsilon). \quad (13.30)$$

$$\int \|y - x\|^2 K_\varepsilon(x, \mathrm{d}y) = \varepsilon^2 M_2(\eta), \quad \int (y - x)(y - x)^\top K_\varepsilon(x, \mathrm{d}y) = \varepsilon^2 \frac{M_2(\eta)}{d} \mathrm{Id}. \quad (13.31)$$

$$\widehat{K}_\varepsilon := \frac{2d}{\varepsilon^2 M_2(\eta)} K_\varepsilon, \quad \int (y - x)(y - x)^\top \widehat{K}_\varepsilon(x, \mathrm{d}y) = 2\mathrm{Id}.$$

$$\mathcal{W}_{\widehat{K}_\varepsilon} = \varepsilon \sqrt{\frac{M_2(\eta)}{2d}} \mathcal{W}_{K_\varepsilon} \xrightarrow{\varepsilon \rightarrow 0} \mathcal{W}_2.$$

Discrete Wasserstein distances on Markov chains. $J_{ij} := \pi_i K_{ij} = \pi_j K_{ji}$

$$\Sigma_n := \left\{ a \in \mathbb{R}_+^n : \sum_i a_i = 1 \right\}.$$

$\rho_i(a) := \frac{a_i}{\pi_i}$, $a_i = \pi_i \rho_i(a)$. The logarithmic mean θ defined in (13.26) is the mobility selected so that the entropy calculus later recovers exactly the Markov evolution; see Proposition 14.33. The identity $\theta(a, b)(\log a - \log b) = a - b$ converts entropy gradients into density differences along graph edges.

$\bar{\nabla} \psi(i, j) := \psi_j - \psi_i$.

$$(\mathcal{K}_\rho \psi)_i := \sum_j K_{ij} \theta(\rho_i, \rho_j) (\psi_i - \psi_j), \quad (\mathcal{L}_a \psi)_i := \pi_i (\mathcal{K}_{\rho(a)} \psi)_i. \quad (13.32)$$

$$\mathbb{A}_K(a, \psi) := \frac{1}{2} \sum_{i,j} J_{ij} \theta(\rho_i(a), \rho_j(a)) (\bar{\nabla} \psi(i, j))^2. \quad (13.33)$$

$$\mathcal{W}_K^2(a_0, a_1) := \inf_{a_t, \psi_t} \int_0^1 \mathbb{A}_K(a_t, \psi_t) dt, \quad \dot{a}_t + \mathcal{L}_{a_t} \psi_t = 0, \quad (13.34)$$

13.5 Dynamic Unbalanced Wasserstein Distances

Balance equation and tangent variables.

$$\partial_t \rho_t + \nabla \cdot m_t = s_t, \quad \int_0^1 \int \left(\frac{\|m_t\|^2}{\rho_t} + \kappa^2 \frac{s_t^2}{\rho_t} \right) dx dt,$$

Reaction–transport action.

$$A_\kappa(a, w, g) := a(\|w\|^2 + \kappa^2 g^2). \quad (13.35)$$

$$J_\kappa(a, m, r) := \begin{cases} \frac{\|m\|^2 + \kappa^2 r^2}{a}, & a > 0, \\ 0, & a = 0, m = 0, r = 0, \\ +\infty, & a = 0, (m, r) \neq (0, 0). \end{cases} \quad (13.36)$$

$$\mathbb{J}_\kappa(\alpha, \omega, \sigma) := \int J_\kappa \left(\frac{d\alpha}{d\lambda}, \frac{d\omega}{d\lambda}, \frac{d\sigma}{d\lambda} \right) d\lambda. \quad (13.37)$$

Static and dynamic viewpoints. The dynamic balance-equation formula is not a separate object from static unbalanced OT: it is the least-action representation of the same cone distance. To make the normalization explicit, define on the cone $\mathfrak{C}[\mathbb{R}^d]$ the squared cost

$$\Delta_\kappa((x, r), (y, s))^2 := 4\kappa^2 \left[r^2 + s^2 - 2rs \cos \left(\frac{\|x - y\|}{2\kappa} \wedge \frac{\pi}{2} \right) \right]. \quad (13.38)$$

The radii encode masses through the weighted projection P_2 defined in Section 10.1. Accordingly, set

$$CW_\kappa(\alpha_0, \alpha_1) := \inf_{\substack{\gamma \in \mathcal{M}_+(\mathfrak{C}[\mathbb{R}^d]^2) \\ P_2 \gamma_1 = \alpha_0, P_2 \gamma_2 = \alpha_1}} \int \Delta_\kappa((x, r), (y, s))^2 d\gamma(x, r, y, s). \quad (13.39)$$

For $\kappa = 1/2$, this is exactly the normalization of the Hellinger–Kantorovich cone cost used in Section 10.1.

Proposition 13.11 (Static/dynamic equivalence for unbalanced OT). *For nonnegative finite measures α_0, α_1 on $\mathbb{R}^d$, the dynamic value*

$$WFR_\kappa^2(\alpha_0, \alpha_1) := \inf_{\substack{\partial_t \alpha_t + \nabla \cdot \omega_t = \sigma_t \\ \alpha_{t=0} = \alpha_0, \alpha_{t=1} = \alpha_1}} \int_0^1 \mathbb{J}_\kappa(\alpha_t, \omega_t, \sigma_t) dt \quad (13.40)$$

equals the static cone value (13.39). Hence $WFR_\kappa = CW_\kappa^{1/2}$ is the geodesic distance generated by the balance-equation least-action problem.

Proof. The cone construction turns variation of mass into radial motion and spatial transport into angular motion on $\mathfrak{C}[\mathbb{R}^d]$. The normalization can be checked on a smooth cone path. If its base position is x_t , its cone radius is r_t , and the projected mass is $a_t = r_t^2$, then its relative growth rate is $g_t = \dot{a}_t/a_t = 2\dot{r}_t/r_t$. The infinitesimal energy induced by (13.38) is therefore

$$4\kappa^2 \dot{r}_t^2 + r_t^2 \|\dot{x}_t\|^2 = a_t (\|\dot{x}_t\|^2 + \kappa^2 g_t^2),$$

which is precisely the velocity action (13.35).

Applying the Benamou–Brenier theorem on the cone to the lifted endpoint measures now gives a dynamic least-action problem whose static value is CW_κ . Projecting a cone curve back to the base space with the squared cone radius as weight produces a measure curve α_t , a spatial flux measure ω_t , and a source measure σ_t satisfying the balance equation. The cone kinetic energy decomposes exactly into the three-variable perspective action $\mathbb{J}_\kappa$ in (13.40). Conversely, any finite-action triple $(\alpha_t, \omega_t, \sigma_t)$ can be lifted, after relaxation, to a cone curve whose radial and spatial velocities realize the source and flux with the same action. This is the standard static/dynamic identification for the Hellinger–Kantorovich and Wasserstein–Fisher–Rao metrics.

The two infima are therefore equal; lower semicontinuity gives the general finite-measure statement from the smooth positive case. $\square$

Balanced versus unbalanced interpolations.

13.6 Variational Mean Field Games

From individual control to a population equilibrium.

$$\inf_{\gamma(0)=x} \left\{ \int_0^1 \left(\frac{1}{2} \|\dot{\gamma}(t)\|^2 + g(\rho_t(\gamma(t))) \right) dt + \Psi(\gamma(1)) \right\}. \quad (13.41)$$

A Benamou–Brenier planning problem.

$$g(r) = G'(r) \quad (13.42)$$

$$\inf_{(\rho_t, v_t)} \left\{ \int_0^1 \int_{\mathbb{R}^d} \left(\frac{1}{2} \rho_t(x) \|v_t(x)\|^2 + G(\rho_t(x)) \right) dx dt + \int_{\mathbb{R}^d} \Psi(x) \rho_1(x) dx \right\}, \quad (13.43)$$

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0, \quad \rho_{t=0} dx = \alpha_0. \quad (13.44)$$

Unlike in (13.8), the final measure is not prescribed. The terminal potential Ψ selects desirable final states softly, while the integral of $G(\rho_t)$ penalizes crowded intermediate configurations.

Convex momentum formulation. As in the Benamou–Brenier problem, the density–velocity integrand and the constraint in (13.43)–(13.44) are not jointly convex. Set $\omega_t = \alpha_t v_t$, or $m_t = \rho_t v_t$ when densities exist, and define the congestion functional

$$\mathcal{C}_G(\alpha) := \begin{cases} \int_{\mathbb{R}^d} G(\rho(x)) dx, & \alpha = \rho dx, \\ +\infty, & \text{otherwise.} \end{cases} \quad (13.45)$$

$$\inf_{(\alpha_t, \omega_t)} \left\{ \int_0^1 \left(\frac{1}{2} \mathbb{J}(\alpha_t, \omega_t) + \mathcal{C}_G(\alpha_t) \right) dt + \int_{\mathbb{R}^d} \Psi d\alpha_1 \right\}, \quad \partial_t \alpha_t + \operatorname{div} \omega_t = 0, \quad \alpha_{t=0} = \alpha_0. \quad (13.46)$$

$$\frac{1}{2} J(\rho_t, m_t) + G(\rho_t) = \frac{\|m_t\|^2}{2\rho_t} + G(\rho_t), \quad (13.47)$$

Optimality system and game interpretation.

Proposition 13.12 (Potential MFG system). *Assume that an optimizer of (13.46) has a smooth positive density ρ , that G is differentiable, and set $g = G'$. Then there is a value function u such that*

$$\begin{cases} -\partial_t u + \frac{1}{2} \|\nabla u\|^2 = g(\rho), & u_{t=1} = \Psi, & m = -\rho \nabla u. \\ \partial_t \rho - \operatorname{div}(\rho \nabla u) = 0, & \rho_{t=0} dx = \alpha_0, \end{cases} \quad (13.48)$$

Proof. Use $-u$ as the multiplier for $\partial_t \rho + \operatorname{div} m = 0$. After integration by parts, the space–time Lagrangian density is $\frac{\|m\|^2}{2\rho} + G(\rho) + \rho \partial_t u + m \cdot \nabla u$. Stationarity in m gives $m = -\rho \nabla u$. Stationarity in ρ , followed by this substitution, gives the first equation in (13.48). The free terminal density contributes $(\Psi - u_1)\rho_1$, hence $u_1 = \Psi$ under the positivity assumption. Primal feasibility gives the second equation. $\square$

Hard congestion through a bottleneck.

$$G_\kappa(r) = \iota_{[0, \kappa]}(r) = \begin{cases} 0, & 0 \leq r \leq \kappa, \\ +\infty, & \text{otherwise.} \end{cases} \quad (13.49)$$

$$\Gamma_\eta(\rho_1) = \frac{\eta}{2} \int_{\Omega} |\rho_1(x) - \rho_\star(x)|^2 dx, \quad (13.50)$$

where $\rho_\star$ is a desired terminal density.

The role of endpoint relaxation depends on the geometry. If two fixed endpoint densities on $\mathbb{R}^d$ satisfy a spatially constant cap, their ordinary $\mathcal{W}_2$ geodesic satisfies the same cap by Proposition 14.8, so the cap does not alter that geodesic.

Computational consequence.

14 Wasserstein Gradient Flows

14.1 Minimizing Movements and Wasserstein Gradients

$$\alpha_{t+\tau} \in \operatorname{argmin}_{\alpha \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ \frac{1}{2\tau} \mathcal{W}_2(\alpha_t, \alpha)^2 + f(\alpha) \right\}. \quad (14.1)$$

Euclidean gradient flows. If we restrict (14.1) to finite dimensions and assume $\alpha_t = \delta_{x(t)}$ and $\alpha = \delta_x$ (single Dirac measures), this matches the implicit Euler scheme:

$$x(t+\tau) \in \operatorname{argmin}_x \left\{ \frac{1}{2\tau} \|x - x(t)\|^2 + h(x) \right\},$$

where $h(x) = f(\delta_x)$. When h is differentiable and the minimizer is unique, the solution is characterized by the implicit Euler formula

$$x(t+\tau) = (\operatorname{Id} + \tau \nabla h)^{-1}(x(t)). \quad x(t+\tau) = (\operatorname{Id} - \tau \nabla h)(x(t)) = x(t) - \tau \nabla h(x(t)).$$

$$\dot{x}(t) = -\nabla h(x(t)). \quad (14.2)$$

Wasserstein gradient formula. $f((1-\tau)\alpha + \tau\beta) = f(\alpha + \tau\eta) = f(\alpha) + \tau \int [\delta f(\alpha)](x) d\eta(x) + o(\tau)$.

Definition 14.1 (Wasserstein gradient). Assume that f admits a smooth first variation $\delta f(\alpha)$. In the smooth formal calculus on $\mathcal{P}_2(\mathbb{R}^d)$, the Wasserstein gradient of f at α is the gradient vector field $\nabla_{\mathcal{W}}f(\alpha) = \nabla_x \delta f(\alpha)$.

$$\frac{\partial \alpha_t}{\partial t} + \operatorname{div}(-\nabla_{\mathcal{W}}f(\alpha_t)\alpha_t) = 0. \quad (14.3)$$

Proposition 14.2 (Formal Wasserstein gradient). Assume that f admits a smooth first variation $\delta f(\alpha)$ and that α has a smooth positive density. Let v be a sufficiently regular velocity field, compactly supported or satisfying a no-flux boundary condition. For perturbations generated by $(\operatorname{Id} + \tau v)_\# \alpha$, the differential of f is $\frac{d}{d\tau}|_{\tau=0} f((\operatorname{Id} + \tau v)_\# \alpha) = \int \langle \nabla \delta f(\alpha)(x), v(x) \rangle d\alpha(x)$. Hence, for the Riemannian metric $\|v\|_{L^2(\alpha)}^2 = \int \|v\|^2 d\alpha$, the Wasserstein gradient is the vector field $\nabla_{\mathcal{W}}f(\alpha) = \nabla \delta f(\alpha)$.

Proof. The push-forward expansion gives, in the sense of distributions, $(\operatorname{Id} + \tau v)_\# \alpha = \alpha - \tau \operatorname{div}(\alpha v) + o(\tau)$. Using the definition of the first variation, $f((\operatorname{Id} + \tau v)_\# \alpha) = f(\alpha) - \tau \int \delta f(\alpha) \operatorname{div}(\alpha v) dx + o(\tau)$. An integration by parts, with either compact support or vanishing boundary flux, gives $-\int \delta f(\alpha) \operatorname{div}(\alpha v) dx = \int \langle \nabla \delta f(\alpha), v \rangle d\alpha$. By definition of the Riesz representative for the $L^2(\alpha)$ metric, this representative is $\nabla \delta f(\alpha)$. $\square$

From the JKO step to the velocity field. A first-order expansion of the JKO step explains why (14.3) uses the vector field $\nabla_{\mathcal{W}}f(\alpha)$. This is a formal local argument: assume that α_t is absolutely continuous and that, for the nearby competitors selected by the JKO step, the optimal map from α_t can be written as $\operatorname{Id} + \tau v$.

$$\mathcal{W}_2^2(\alpha_t, (\operatorname{Id} + \tau v)_\# \alpha_t) = \tau^2 \|v\|_{L^2(\alpha_t)}^2 \cdot \min_v \frac{\tau}{2} \|v\|_{L^2(\alpha_t)}^2 + f((\operatorname{Id} + \tau v)_\# \alpha_t),$$

$$(\operatorname{Id} + \tau v)_\# \alpha_t = \alpha_t - \tau \operatorname{div}(v \alpha_t) + o(\tau),$$

$$f((\operatorname{Id} + \tau v)_\# \alpha_t) = f(\alpha_t) + \tau \int \langle \nabla_x \delta f(\alpha_t)(x), v(x) \rangle d\alpha_t(x) + o(\tau)$$

$$\min_v f(\alpha_t) + \tau \int \left[\frac{1}{2} \|v(x)\|^2 + \langle \nabla_{\mathcal{W}}f(\alpha_t)(x), v(x) \rangle \right] d\alpha_t(x) + o(\tau).$$

Metric derivative and curves of maximal slope. The same descent principle admits a coordinate-free formulation, which is the right language for limits of JKO schemes. Let $(\mathcal{X}, d)$ be a metric space and let $x : [0, T] \rightarrow \mathcal{X}$ be an absolutely continuous curve.

$$|\dot{x}_t| := \lim_{h \rightarrow 0} \frac{d(x_{t+h}, x_t)}{|h|}$$

$$|\partial f|(x) := \limsup_{\substack{y \rightarrow x \\ y \neq x}} \frac{(f(x) - f(y))_+}{d(x, y)}.$$

When this slope is a strong upper gradient for f , meaning that $|f(y_t) - f(y_s)| \leq \int_s^t |\partial f|(y_r) |\dot{y}_r| dr$ along absolutely continuous curves y_r , an absolutely continuous curve x_t is called a curve of maximal slope if it satisfies the energy-dissipation inequality

$$f(x_t) + \frac{1}{2} \int_s^t |\dot{x}_r|^2 dr + \frac{1}{2} \int_s^t |\partial f|^2(x_r) dr \leq f(x_s), \quad 0 \leq s < t.$$

Discrete evolutions. If f admits a first variation whose spatial gradient is well defined at empirical measures, the characteristic form of (14.3) preserves the number and weights of the atoms. For equal weights, write $\alpha_t = \frac{1}{n} \sum_i \delta_{x_i(t)}$. The particles $X(t) := (x_i(t))_i$ evolve according to the coupled ODEs

$$\dot{x}_i(t) = -n \nabla_{x_i} F(X(t)), \quad (14.4)$$

where $F(X) := f(\frac{1}{n} \sum_i \delta_{x_i})$ and the factor n comes from the empirical Wasserstein metric $\frac{1}{n} \sum_i \|\dot{x}_i\|^2$.

Linear functionals.

Shannon neg-entropy.

Interaction energies.

$$f(\alpha) := \iint k(x, y) d\alpha(x) d\alpha(y). \quad (14.5)$$

For a symmetric kernel k :

$\delta f(\alpha)(x) = 2 \int k(x, y) d\alpha(y)$, $\nabla_{\mathcal{W}}f(\alpha)(x) = 2 \int \nabla_x k(x, y) d\alpha(y)$. For $\alpha_0 = \frac{1}{n} \sum_i \delta_{x_i}$, the flow (14.3) implies particles $(x_i(t))_i$ obey:

$\dot{x}_i(t) = -\frac{2}{n} \sum_j \nabla_x k(x_i(t), x_j(t))$. If k is positive definite, or more generally conditionally positive definite on signed measures of zero total mass as for the energy-distance kernel $k(x, y) = -\|x - y\|$, and one minimizes the squared kernel discrepancy to a teacher distribution β , then

$$\|\alpha - \beta\|_k^2 = \iint k d\alpha d\alpha - 2 \int \left(\int k(x, y) d\beta(y) \right) d\alpha(x) + \text{constant}.$$

Thus MMD-type training energies are exactly an interaction energy plus a linear potential; the teacher distribution appears through the potential $x \mapsto -2 \int k(x, y) d\beta(y)$. The corresponding empirical Wasserstein gradient flow is

$$\dot{x}_i(t) = -\frac{2}{n} \sum_j \nabla_x k(x_i(t), x_j(t)) + 2 \int \nabla_x k(x_i(t), y) d\beta(y).$$

Stochastic particles and McKean–Vlasov limits. $dX_t = b(X_t)dt + \sqrt{2}\sigma dB_t$, and the one-particle law $\alpha_t = \rho_t dx$ directly satisfies the linear Fokker–Planck equation

$\partial_t \rho_t = -\operatorname{div}(b \rho_t) + \sigma^2 \Delta \rho_t$. $dX_t^n(t) = b(X_t^n(t), \alpha_t^n)dt + \sqrt{2}\sigma dB_t(t)$, $\alpha_t^n = \frac{1}{n} \sum_{i=1}^n \delta_{X_t^n(t)}$. For finite n , the empirical law α_t^n is itself random. Under suitable Lipschitz, growth and chaotic-initialization assumptions, propagation of chaos states that finitely many particles become asymptotically independent as $n \rightarrow \infty$, all with the same deterministic law $\rho_t dx$; equivalently, the empirical measure α_t^n converges in probability to this law.

$\partial_t \rho_t = -\operatorname{div}(b(x, \rho_t) \rho_t) + \sigma^2 \Delta \rho_t$. $b(x, \rho) = -\nabla \frac{\delta \mathcal{E}}{\delta \rho}(x)$, $\mathcal{E}(\rho) + \sigma^2 \int \rho \log \rho dx$. For a prescribed target $\beta = \rho_\beta dx$, take $b = \sigma^2 \nabla \log \rho_\beta$.

The SDE, the Fokker–Planck equation, and the deterministic continuity equation with score velocity are then three representations of the same gradient flow:

$$\partial_t \rho_t = \sigma^2 \operatorname{div} \left(\rho_t \nabla \log \frac{\rho_t}{\rho_\beta} \right), \quad v_t = \sigma^2 (\nabla \log \rho_\beta - \nabla \log \rho_t). \quad (14.6)$$

Gradient flows under density constraints.

$$f_{\rho_{\max}}(\rho dx) = \int_{\Omega} h(x) \rho(x) dx + \iota_{\mathcal{K}_{\rho_{\max}}}(\rho dx), \quad \mathcal{K}_{\rho_{\max}} = \{\rho dx \in \mathcal{P}_2(\Omega) ; 0 \leq \rho \leq \rho_{\max} \text{ a.e.}\}, \quad (14.7)$$

where $\Omega \subset \mathbb{R}^d$ is a convex domain and $\mathcal{K}_{\rho_{\max}}$ is assumed nonempty; if Ω has finite volume, this requires $\rho_{\max}|\Omega| \geq 1$. The indicator term has no classical first variation; it contributes the normal cone to $\mathcal{K}_{\rho_{\max}}$ in the Wasserstein first-order condition.

$$\alpha^{k+1} \in \operatorname{argmin}_{\alpha \in \mathcal{K}_{\rho_{\max}}} \frac{1}{2\tau} \mathcal{W}_2^2(\alpha^k, \alpha) + \int_{\Omega} h d\alpha. \quad (14.8)$$

$$\partial_t \rho_t = \operatorname{div}(\rho_t \nabla(h + p_t)), \quad 0 \leq \rho_t \leq \rho_{\max}, \quad p_t \geq 0, \quad p_t(\rho_{\max} - \rho_t) = 0. \quad (14.9)$$

The pressure vanishes in the unsaturated region and acts only where $\rho_t = \rho_{\max}$, pushing mass away from congested zones so that the density cap remains satisfied.

Multi-species gradient flows. Several applications involve several densities living on the same physical space: chemical species, color channels, cell populations or competing phases. Let $\Omega \subset \mathbb{R}^d$ be the physical domain and assume that the component masses $\mathbf{m} = (m_1, \dots, m_p)$ are positive and fixed.

$$\mathcal{P}_{2,\mathbf{m}}(\Omega; \mathbb{R}_+^p) = \left\{ \alpha = (\alpha_1, \dots, \alpha_p) ; \alpha_i \in \mathcal{M}_+(\Omega), \alpha_i(\Omega) = m_i, \int_{\Omega} \|x\|^2 d\alpha_i(x) < +\infty \right\}. \quad (14.10)$$

$$\mathcal{W}_{2,\oplus}^2(\alpha, \eta) = \sum_{i=1}^p m_i \mathcal{W}_2^2\left(\frac{\alpha_i}{m_i}, \frac{\eta_i}{m_i}\right). \quad (14.11)$$

$$\alpha^{k+1} \in \operatorname{argmin}_{\alpha \in \mathcal{P}_{2,\mathbf{m}}(\Omega; \mathbb{R}_+^p)} \frac{1}{2\tau} \mathcal{W}_{2,\oplus}^2(\alpha^k, \alpha) + f(\alpha). \quad (14.12)$$

For smooth densities and first variations $\varphi_i = \delta f / \delta \rho_i$, this yields

$$\partial_t \rho_i = \operatorname{div}(\rho_i \nabla \varphi_i), \quad \varphi_i = \frac{\delta f}{\delta \rho_i}, \quad i = 1, \dots, p. \quad (14.13)$$

$$\mathcal{C}_\beta = \left\{ \alpha \in \mathcal{P}_{2,\mathbf{m}}(\Omega; \mathbb{R}_+^p) ; \sum_{i=1}^p \alpha_i = \beta \right\}. \quad (14.14)$$

$$\alpha^{k+1} \in \operatorname{argmin}_{\alpha \in \mathcal{C}_\beta} \frac{1}{2\tau} \mathcal{W}_{2,\oplus}^2(\alpha^k, \alpha) + f(\alpha). \quad (14.15)$$

$$\partial_t \rho_i = \operatorname{div}(\rho_i \nabla(\varphi_i - \lambda_t)), \quad \operatorname{div}(b \nabla \lambda_t) = \operatorname{div} \left(\sum_{i=1}^p \rho_i \nabla \varphi_i \right). \quad (14.16)$$

For the separable Shannon entropy $f(\alpha) = \sum_i \int_{\Omega} \rho_i \log \rho_i dx$, this becomes

$$\partial_t \rho_i = \Delta \rho_i - \operatorname{div}(\rho_i \nabla \lambda_t), \quad \operatorname{div}(b \nabla \lambda_t) = \Delta b. \quad (14.17)$$

When b is constant, the pressure can be chosen constant and the system reduces to independent heat equations which nevertheless preserve the pointwise sum $\sum_i \rho_i = b$. This product geometry should be contrasted with the vector-valued Benamou–Brenier distances of Section 12.1, where the metric itself can couple the components through a non-diagonal mobility.

14.2 Geodesic Convexity and Convergence

Geodesics and convexity. $\alpha_t = ((1-t)P_0 + tP_1)_{\#} \pi^*$, $t \in [0, 1]$, where $P_0(x, y) = x$ and $P_1(x, y) = y$. If the optimal plan is induced by a Brenier map T , this reduces to $((1-t)\operatorname{Id} + tT)_{\#} \alpha_0$.

Definition 14.3 (Geodesic convexity). A functional f on $\mathcal{P}_2(\mathbb{R}^d)$ is geodesically convex if for every $\mathcal{W}_2$ geodesic $(\alpha_t)_t$, $f(\alpha_t) \leq (1-t)f(\alpha_0) + tf(\alpha_1)$. It is λ -geodesically convex if the right-hand side is improved by $-\frac{\lambda}{2}t(1-t)\mathcal{W}_2^2(\alpha_0, \alpha_1)$.

Proposition 14.4 (Geodesic convexity of linear and quadratic energies). 1. If h is convex, then the linear energy $\alpha \mapsto \int h d\alpha$ is geodesically convex; if h is λ -strongly convex, it is λ -geodesically convex. Conversely, if this linear energy is geodesically convex for all Dirac endpoints, then h is convex.

2. More generally, let $k \geq 2$ and let $H_k : (\mathbb{R}^d)^k \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex. Then the polynomial energy $\alpha \mapsto \int_{(\mathbb{R}^d)^k} H_k(x_1, \dots, x_k) d\alpha(x_1) \cdots d\alpha(x_k)$ is geodesically convex whenever the integral is well defined. If H_k is λ -strongly convex on the product space, this energy is $k\lambda$ -geodesically convex. In particular, the quadratic interaction $\alpha \mapsto \frac{1}{2} \iint W(x, y) d\alpha(x) d\alpha(y)$ is geodesically convex when W is convex on $\mathbb{R}^d \times \mathbb{R}^d$, and it is λ -geodesically convex when W is λ -strongly convex there.

Proof. Let π be an optimal coupling between α_0 and α_1 , and set $X_t = (1-t)X_0 + tX_1$ under $(X_0, X_1) \sim \pi$. Convexity of h gives $h(X_t) \leq (1-t)h(X_0) + th(X_1)$, and strong convexity gives the additional term $-\frac{\lambda}{2}t(1-t)\|X_0 - X_1\|^2$. Integrating proves the first claim. Necessity follows by testing on Dirac masses: the geodesic between δ_x and δ_y is $\delta_{(1-t)x+ty}$, so geodesic convexity of $\int h d\alpha$ gives the usual convexity inequality for h .

For the polynomial statement, take k independent copies $(X_0^\ell, X_1^\ell)_{\ell=1}^k$ of the same optimal coupling and define $X_t^\ell = (1-t)X_0^\ell + tX_1^\ell$. Convexity of H_k on the product space gives $H_k(X_t^1, \dots, X_t^k) \leq (1-t)H_k(X_0^1, \dots, X_0^k) + tH_k(X_1^1, \dots, X_1^k)$. Integrating over the product coupling proves the claim. If H_k is λ -strongly convex, the additional term is $-\frac{\lambda}{2}t(1-t) \sum_{\ell=1}^k \mathbb{E} \|X_0^\ell - X_1^\ell\|^2 = -\frac{k\lambda}{2}t(1-t)\mathcal{W}_2^2(\alpha_0, \alpha_1)$, which proves the stated constant. For the quadratic interaction, the prefactor $1/2$ cancels the factor $k = 2$, leaving the constant λ . $\square$

Proposition 14.5 (One-dimensional interactions and energy-distance MMD). *Let $\varphi : \mathbb{R}_+ \rightarrow \mathbb{R}$ be continuous with at most quadratic growth, and define on $\mathcal{P}_2(\mathbb{R})$ $\mathcal{I}_\varphi(\alpha) := \frac{1}{2} \iint \varphi(|x-y|) d\alpha(x) d\alpha(y)$. Then $\mathcal{I}_\varphi$ is geodesically convex for $\mathcal{W}_2$ on $\mathcal{P}_2(\mathbb{R})$ if and only if φ is convex on $\mathbb{R}_+$. Fix also $\beta \in \mathcal{P}_2(\mathbb{R})$. For the conditionally positive kernel $k(x, y) = -|x-y|$ from Definition 6.4, the squared MMD of Definition 6.5, equivalently the squared energy distance, $\mathcal{E}_\beta(\alpha) := \text{MMD}_k^2(\alpha, \beta) = \iint -|x-y| d(\alpha-\beta)(x) d(\alpha-\beta)(y)$, is geodesically convex as a function of α .*

Proof. Let Q_0, Q_1 be the quantile functions of α_0, α_1 , and let $Q_t = (1-t)Q_0 + tQ_1$ be the quantile function of the one-dimensional $\mathcal{W}_2$ geodesic α_t . For $r > s$, monotonicity gives $Q_i(r) - Q_i(s) \geq 0$ for $i = 0, 1$, and hence $Q_t(r) - Q_t(s) = (1-t)(Q_0(r) - Q_0(s)) + t(Q_1(r) - Q_1(s))$. Using the symmetry of the double integral and the quantile parametrization, one can write $\mathcal{I}_\varphi(\alpha_t) = \int_0^1 \int_0^r \varphi(Q_t(r) - Q_t(s)) ds dr$. Convexity of φ on $\mathbb{R}_+$ gives the desired convexity after integration.

Conversely, test geodesic convexity on two-point measures $\alpha_i = \frac{1}{2}(\delta_0 + \delta_{a_i})$, with $a_i \geq 0$. Their monotone geodesic is $\alpha_t = \frac{1}{2}(\delta_0 + \delta_{(1-t)a_0 + ta_1})$. Since $\mathcal{I}_\varphi(\alpha_t) = \frac{1}{4}\varphi(0) + \frac{1}{4}\varphi((1-t)a_0 + ta_1)$, and the identical $\frac{1}{4}\varphi(0)$ terms cancel from the geodesic-convexity inequality, one obtains the ordinary convexity inequality for φ on $\mathbb{R}_+$.

For the energy distance, the function $\varphi(r) = -r$ is affine on $\mathbb{R}_+$, so the self-interaction $-\iint |x-y| d\alpha(x) d\alpha(y)$ is affine along one-dimensional Wasserstein geodesics even though $z \mapsto -|z|$ is not convex on $\mathbb{R}$. Write Q_β for the quantile function of β . Expanding the MMD and dropping the term depending only on β gives

$$\mathcal{E}_\beta(\alpha) = 2 \int_0^1 \int_0^1 |Q_\alpha(r) - Q_\beta(s)| dr ds - \int_0^1 \int_0^1 |Q_\alpha(r) - Q_\alpha(s)| dr ds + \text{constant}.$$

Here the constant is independent of α . The positive cross-distance term is convex in Q_α , since the absolute value is convex and Q_t is affine in t . The self-distance term is affine along quantile geodesics because $\int_0^1 \int_0^1 |Q_t(r) - Q_t(s)| dr ds = 2 \int_0^1 \int_0^r (Q_t(r) - Q_t(s)) ds dr$. Thus $\alpha \mapsto \mathcal{E}_\beta(\alpha)$ is geodesically convex. $\square$

$\text{MMD}_k^2(\alpha, \beta) = \iint k d\alpha d\alpha - 2 \iint k d\alpha d\beta + \text{constant}$

Theorem 14.6 (McCann displacement convexity for internal energies). *Let $g : [0, +\infty) \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex with $g(0) = 0$, and define $\psi(r) = r^d g(r^{-d})$, $r > 0$. If ψ is convex and nonincreasing on $(0, +\infty)$, then the internal energy*

$$\mathcal{U}_g(\alpha) = \begin{cases} \int_{\mathbb{R}^d} g(\rho(x)) dx, & \alpha = \rho dx, \\ +\infty, & \text{otherwise,} \end{cases}$$

is geodesically convex on $\mathcal{P}_2(\mathbb{R}^d)$.

Proof. It is enough to prove the inequality for smooth positive compactly supported densities; the general case follows by approximation and lower semicontinuity. Let $T = \nabla\varphi$ be the Brenier map from $\alpha_0 = \rho_0 dx$ to $\alpha_1 = \rho_1 dx$, set $T_t = (1-t)\text{Id} + tT$, and denote $J_t(x) = \det((1-t)\text{Id} + t\nabla T(x))$. The interpolation satisfies $\alpha_t = (T_t)_\# \alpha_0$ and $\rho_t(T_t(x))J_t(x) = \rho_0(x)$. By concavity of $\det^{1/d}$ on positive semidefinite matrices, $J_t(x)^{1/d} \geq (1-t) + t \det(\nabla T(x))^{1/d}$. Introduce $s_t(x) = (J_t(x)/\rho_0(x))^{1/d}$. Since $\rho_1(T(x)) \det \nabla T(x) = \rho_0(x)$, this gives $s_t(x) \geq (1-t)\rho_0(x)^{-1/d} + t\rho_1(T(x))^{-1/d}$. Using the change of variables $y = T_t(x)$, one obtains $\int g(\rho_t(y)) dy = \int \rho_0(x) \psi(s_t(x)) dx$. Because ψ is nonincreasing and convex, $\psi(s_t(x)) \leq (1-t)\psi(\rho_0(x)^{-1/d}) + t\psi(\rho_1(T(x))^{-1/d})$. Multiplying by ρ_0 and integrating gives $\mathcal{U}_g(\alpha_t) \leq (1-t)\mathcal{U}_g(\alpha_0) + t\mathcal{U}_g(\alpha_1)$. $\square$

Proposition 14.7 (Geodesic convexity of power divergences). *For $m > 0$, define*

$$\varphi_m(r) = \begin{cases} \frac{r^m - mr + m - 1}{m(m-1)}, & m \neq 1, \\ r \log r - r + 1, & m = 1. \end{cases}$$

Then the following geodesic-convexity statements hold.

1. *If $\Omega \subset \mathbb{R}^d$ is convex, $\beta = b \mathbf{1}_\Omega dx$, and $m \geq 1 - 1/d$, then $\alpha \mapsto \mathcal{D}_{\varphi_m}(\alpha|\beta)$ is geodesically convex on $\mathcal{P}_2(\Omega)$.*
 2. *If $\beta = Z^{-1}e^{-V} dx$ on $\mathbb{R}^d$, V is convex, and $m \geq 1$, then $\alpha \mapsto \mathcal{D}_{\varphi_m}(\alpha|\beta)$ is geodesically convex on $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$.*
 3. *In the KL case $m = 1$, if V is λ -strongly convex, then $\alpha \mapsto \text{KL}(\alpha|\beta) = \mathcal{D}_{\varphi_1}(\alpha|\beta)$ is λ -geodesically convex.*
- All statements use the convention that the divergence is $+\infty$ when α is not absolutely continuous with respect to β .*

Proof. For the flat reference case, write $\alpha = \rho dx$. Up to affine terms in ρ , which only add constants on probability measures, the integrand $b\varphi_m(\rho/b)$ has non-affine part $b^{1-m}\rho^m/(m(m-1))$ for $m \neq 1$, and $\rho \log \rho$ for $m = 1$. For $m = 1$, McCann's criterion gives $\psi(r) = -d \log r$. For $m \neq 1$, the transformed non-affine part is, up to the irrelevant factor b^{1-m} , $\psi_m(r) = \frac{r^{d(1-m)}}{m(m-1)}$. If $m > 1$, this function is convex and nonincreasing. If $0 < m < 1$, it is convex and nonincreasing exactly when $d(1-m) \leq 1$, i.e. $m \geq 1 - 1/d$. This proves the first claim by Theorem 14.6.

Assume next that $m > 1$ and V is convex. Up to constants, $\mathcal{D}_{\varphi_m}(\alpha|\beta) = \frac{Z^{m-1}}{m(m-1)} \int \rho(x)^m e^{(m-1)V(x)} dx$. Let $T = \nabla\varphi$ be the Brenier map from $\alpha_0 = \rho_0 dx$ to α_1 , set $T_t = (1-t)\text{Id} + tT$, and write $J_t(x) = \det((1-t)\text{Id} + t\nabla T(x))$. Along the geodesic $\alpha_t = (T_t)_\# \alpha_0$,

$$\int \rho_t^m e^{(m-1)V} dx = \int \rho_0(x)^m \exp((m-1)(V(T_t(x)) - \log J_t(x))) dx.$$

For each x , the function $t \mapsto V(T_t(x))$ is convex, and $t \mapsto -\log J_t(x)$ is convex because $\log \det$ is concave on positive definite matrices. Since $m-1 > 0$, the exponential of their sum is convex in t . Integrating gives geodesic convexity. The nonsmooth case follows by approximation.

Finally, for $m = 1$, $\text{KL}(\alpha|\beta) = \int \rho \log \rho dx + \int V d\alpha + \log Z$. The entropy term is 0-geodesically convex by Theorem 14.6. The linear potential term is λ -geodesically convex when V is λ -strongly convex, by Proposition 14.4. Taking $\lambda = 0$ also covers the $m = 1$ case in the second claim. $\square$

Geodesically convex constraints.

Proposition 14.8 (Geodesic convexity of density caps). *Assume that $\Omega \subset \mathbb{R}^d$ is convex and let $\rho_{\max} > 0$. The set $\mathcal{K}_{\rho_{\max}}$ in (14.7) is geodesically convex in $\mathcal{P}_2(\Omega)$: if $\alpha_0, \alpha_1 \in \mathcal{K}_{\rho_{\max}}$, then the constant-speed $\mathcal{W}_2$ geodesic $(\alpha_t)_{t \in [0,1]}$ between them also belongs to $\mathcal{K}_{\rho_{\max}}$.*

Proof. Assume first that the endpoints have smooth positive densities $\alpha_i = \rho_i dx$ with $\rho_i \leq \rho_{\max}$, and let $T = \nabla \varphi$ be the Brenier map from α_0 to α_1 . Since Ω is convex, $T_t = (1-t)\text{Id} + tT$ maps Ω into Ω . The interpolation is $\alpha_t = (T_t)_\# \alpha_0$ and $\rho_t(T_t(x)) = \frac{\rho_0(x)}{\det((1-t)\text{Id} + t\nabla T(x))}$. The concavity of $\det^{1/d}$ on positive semidefinite matrices and the identity $\rho_1(T(x)) \det \nabla T(x) = \rho_0(x)$ give $\rho_t(T_t(x))^{-1/d} \geq (1-t)\rho_0(x)^{-1/d} + t\rho_1(T(x))^{-1/d} \geq \rho_{\max}^{-1/d}$. Hence $\rho_t \leq \rho_{\max}$ for all t . The general case follows by approximation and lower semicontinuity of the L^∞ bound, as in McCann's displacement-convexity theory. $\square$

Dissipation and finite metric length. $\frac{d}{dt}h(x_t) = -\|\nabla h(x_t)\|^2 = -\|\dot{x}_t\|^2$. Thus, for $0 \leq s < t$, Cauchy's inequality gives

$$\int_s^t \|\dot{x}_r\| dr \leq \sqrt{t-s} \left(\int_s^t \|\dot{x}_r\|^2 dr \right)^{1/2} = \sqrt{t-s} (h(x_s) - h(x_t))^{1/2}. \quad (14.18)$$

Proposition 14.9 (Finite metric length from dissipation). *Let $(\alpha_r)_{r \in [0,T]}$ be an absolutely continuous curve in $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$. Assume that, for all $0 \leq s < t \leq T$, it satisfies the exact energy-dissipation identity $f(\alpha_s) - f(\alpha_t) = \int_s^t |\dot{\alpha}_r|_{\mathcal{W}_2}^2 dr$. Then*

$$\text{Length}_{\mathcal{W}_2}((\alpha_r)_{r \in [s,t]}) = \int_s^t |\dot{\alpha}_r|_{\mathcal{W}_2} dr \leq \sqrt{t-s} (f(\alpha_s) - f(\alpha_t))^{1/2}. \quad (14.19)$$

In the smooth Wasserstein gradient-flow setting with velocity $v_r = -\nabla_{\mathcal{W}} f(\alpha_r)$, the dissipation identity reads equivalently $f(\alpha_s) - f(\alpha_t) = \int_s^t |\dot{\alpha}_r|_{\mathcal{W}_2}^2 dr = \int_s^t \int \|v_r(x)\|^2 d\alpha_r(x) dr$. If one assumes only the metric energy-dissipation inequality for a curve of maximal slope, then the same length estimate holds with the right-hand side multiplied by $\sqrt{2}$.

Proof. Cauchy's inequality gives

$$\int_s^t |\dot{\alpha}_r|_{\mathcal{W}_2} dr \leq \sqrt{t-s} \left(\int_s^t |\dot{\alpha}_r|_{\mathcal{W}_2}^2 dr \right)^{1/2}.$$

The exact energy-dissipation identity substitutes the last integral by $f(\alpha_s) - f(\alpha_t)$, which proves (14.19). In the metric theory of curves of maximal slope, the energy-dissipation inequality gives instead $\frac{1}{2} \int_s^t |\dot{\alpha}_r|_{\mathcal{W}_2}^2 dr \leq f(\alpha_s) - f(\alpha_t)$, and the same argument yields the additional factor $\sqrt{2}$. $\square$

$\mathcal{W}_2(\alpha_s, \alpha_t) \leq \text{Length}_{\mathcal{W}_2}((\alpha_r)_{r \in [s,t]}) \leq \sqrt{f(\alpha_0) - m_T} \sqrt{t-s}$, with an additional factor $\sqrt{2}$ under the energy-dissipation inequality. Thus an energy-dissipating trajectory is $1/2$ -Hölder continuous as a curve in Wasserstein space on every time interval where the energy remains bounded from below.

Proposition 14.10 (Energy decay for convex Wasserstein flows). *Assume formally that f is geodesically convex, admits a smooth first variation, and has a minimizer α^* . Let $(\alpha_t)_t$ be a smooth solution of the Wasserstein gradient flow $\partial_t \alpha_t + \text{div}(\alpha_t v_t) = 0$, $v_t = -\nabla_{\mathcal{W}} f(\alpha_t)$. Then $\frac{d}{dt}f(\alpha_t) = -\int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2 d\alpha_t(x) \leq 0$. If T_t is the optimal map from α_t to α^* , then $f(\alpha_t) - f(\alpha^*) \leq -\frac{d}{dt} \frac{1}{2} \mathcal{W}_2^2(\alpha_t, \alpha^*)$, and consequently $f(\alpha_t) - f(\alpha^*) \leq \frac{\mathcal{W}_2^2(\alpha_0, \alpha^*)}{2t}$.*

Proof. The chain rule and Proposition 14.2 give $\frac{d}{dt}f(\alpha_t) = \int \langle \nabla_{\mathcal{W}} f(\alpha_t)(x), v_t(x) \rangle d\alpha_t(x) = -\int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2 d\alpha_t(x)$. Geodesic convexity along the geodesic $((1-s)\text{Id} + sT_t)_\# \alpha_t$ gives $f(\alpha^*) - f(\alpha_t) \geq \int \langle \nabla_{\mathcal{W}} f(\alpha_t)(x), T_t(x) - x \rangle d\alpha_t(x)$. Since $v_t = -\nabla_{\mathcal{W}} f(\alpha_t)$, this reads $f(\alpha_t) - f(\alpha^*) \leq \int \langle v_t(x), T_t(x) - x \rangle d\alpha_t(x)$. $\frac{d}{dt} \frac{1}{2} \mathcal{W}_2^2(\alpha_t, \alpha^*) = \int \langle x - T_t(x), v_t(x) \rangle d\alpha_t(x)$, which proves the differential inequality. Integrating it from 0 to t and using the monotonicity of $s \mapsto f(\alpha_s)$ gives

$$t(f(\alpha_t) - f(\alpha^*)) \leq \int_0^t (f(\alpha_s) - f(\alpha^*)) ds \leq \frac{1}{2} \mathcal{W}_2^2(\alpha_0, \alpha^*).$$

$\square$

Convergence: the Wasserstein-PL viewpoint. $|\partial f|^2(\alpha) := \int \|\nabla_{\mathcal{W}} f(\alpha)(x)\|^2 d\alpha(x)$

Definition 14.11 (Wasserstein-Polyak-Łojasiewicz inequality). Let $f_\star := \inf_{\alpha \in \mathcal{P}_2(\mathbb{R}^d)} f(\alpha) > -\infty$. The functional f satisfies the Wasserstein-PL inequality with constant $\kappa > 0$ if

$$|\partial f|^2(\alpha) \geq 2\kappa(f(\alpha) - f_\star) \quad (14.20)$$

for every α in the domain of f . In the smooth Otto notation this reads $\int \|\nabla_{\mathcal{W}} f(\alpha)\|^2 d\alpha \geq 2\kappa(f(\alpha) - f_\star)$.

Theorem 14.12 (Wasserstein-PL convergence). *Assume that $f_\star > -\infty$, that f satisfies the Wasserstein-PL inequality (14.20) with constant $\kappa > 0$, and that $(\alpha_t)_{t \geq 0}$ is a global Wasserstein gradient flow satisfying the full energy-dissipation identity*

$$\frac{d}{dt}f(\alpha_t) = -|\dot{\alpha}_t|^2 = -|\partial f|^2(\alpha_t) \quad \text{for a.e. } t > 0. \quad (14.21)$$

Set $E(t) := f(\alpha_t) - f_\star$. Then, for $0 \leq s \leq t$, $E(t) \leq e^{-2\kappa(t-s)}E(s)$, and the length of the flow segment satisfies $\text{Length}_{\mathcal{W}_2}((\alpha_r)_{r \in [s,t]}) \leq \sqrt{\frac{2}{\kappa}}(\sqrt{E(s)} - \sqrt{E(t)})$. If, in addition, f is lower semicontinuous for $\mathcal{W}_2$ -convergence, then α_t converges in $\mathcal{W}_2$ to a minimizer $\alpha_\infty \in \text{Argmin } f$, and $\mathcal{W}_2(\alpha_t, \alpha_\infty) \leq \sqrt{\frac{2}{\kappa}}\sqrt{E(t)} \leq \sqrt{\frac{2}{\kappa}}\sqrt{E(0)}e^{-\kappa t}$. In particular, $\text{dist}_{\mathcal{W}_2}(\alpha_t, \text{Argmin } f) \leq \sqrt{\frac{2}{\kappa}}\sqrt{E(0)}e^{-\kappa t}$.

Proof. The energy estimate is immediate from (14.21) and (14.20): $E'(t) = -|\partial f|^2(\alpha_t) \leq -2\kappa E(t)$ for almost every t . Gronwall's lemma gives $E(t) \leq e^{-2\kappa(t-s)} E(s)$.

The distance estimate uses the same two identities in a slightly different way. On the set where $E(r) > 0$, the PL inequality gives $|\partial f|(\alpha_r) \geq \sqrt{2\kappa E(r)}$. Using the full dissipation identity and $|\dot{\alpha}_r| = |\partial f|(\alpha_r)$,

$$\begin{aligned} \text{Length}_{\mathcal{W}_2}((\alpha_r)_{r \in [s, t]}) &= \int_s^t |\dot{\alpha}_r| dr = \int_s^t |\partial f|(\alpha_r) dr \\ &= \int_s^t \frac{|\partial f|^2(\alpha_r)}{|\partial f|(\alpha_r)} dr \leq \frac{1}{\sqrt{2\kappa}} \int_s^t \frac{|\partial f|^2(\alpha_r)}{\sqrt{E(r)}} dr \\ &= \frac{1}{\sqrt{2\kappa}} \int_s^t \frac{-E'(r)}{\sqrt{E(r)}} dr = \sqrt{\frac{2}{\kappa}} (\sqrt{E(s)} - \sqrt{E(t)}). \end{aligned}$$

If E vanishes on a subinterval, then the flow is stationary there by (14.21), so the same estimate follows by applying the argument on the part where $E > 0$. Since Wasserstein distance is bounded by metric length, the same upper bound controls $\mathcal{W}_2(\alpha_s, \alpha_t)$. Letting $t \rightarrow +\infty$, the energy estimate gives $E(t) \rightarrow 0$, and the length estimate shows that the tail of $(\alpha_t)_t$ is Cauchy. Since $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is complete, there is a limit α_∞ . Lower semicontinuity yields $f(\alpha_\infty) \leq \liminf_{t \rightarrow +\infty} f(\alpha_t) = f_\star$, hence $\alpha_\infty \in \text{Argmin } f$. Sending the endpoint of the length estimate to $+\infty$ with $s = t$ fixed gives $\mathcal{W}_2(\alpha_t, \alpha_\infty) \leq \sqrt{\frac{2}{\kappa}} \sqrt{E(t)}$. The distance-to-the-set bound follows because α_∞ is one minimizer. $\square$

Geodesic strong convexity.

Proposition 14.13 (Strong geodesic convexity implies Wasserstein–PL). *Assume that f is λ -geodesically convex with $\lambda > 0$, that $f_\star$ is attained at some $\alpha^\star$, and that the Wasserstein slope is finite at α . Then $|\partial f|^2(\alpha) \geq 2\lambda(f(\alpha) - f_\star)$. Thus positive Wasserstein strong convexity implies the Wasserstein–PL inequality with the same constant.*

Proof. Fix α and a minimizer $\alpha^\star$. Set $g = f(\alpha) - f_\star$, $r = \mathcal{W}_2(\alpha, \alpha^\star)$, $a = |\partial f|(\alpha)$. If $r = 0$, then $\alpha = \alpha^\star$ and there is nothing to prove. Otherwise let $(\alpha_s)_{s \in [0, 1]}$ be a constant-speed geodesic from α to $\alpha^\star$. By λ -geodesic convexity, $f(\alpha_s) \leq (1-s)f(\alpha) + sf_\star - \frac{\lambda}{2}s(1-s)r^2$. Therefore $f(\alpha) - f(\alpha_s) \geq sg + \frac{\lambda}{2}s(1-s)r^2$. Dividing by $\mathcal{W}_2(\alpha, \alpha_s) = sr$ and letting $s \downarrow 0$ in the definition of the descending metric slope yields $a \geq \frac{g}{r} + \frac{\lambda}{2}r$. Equivalently, $g \leq ar - \lambda r^2/2$. Maximizing the right-hand side over $r \geq 0$ gives $g \leq a^2/(2\lambda)$, which is the desired PL inequality. $\square$

Wasserstein Kurdyka–Łojasiewicz inequalities.

Definition 14.14 (Wasserstein Kurdyka–Łojasiewicz inequality). Let $f_\star = \inf f$, let $E(\alpha) = f(\alpha) - f_\star$, and fix $c > 0$, $\theta \in [0, 1)$. We say that f satisfies a power Wasserstein–KL inequality on a set $\mathcal{U} \subset \mathcal{P}_2(\mathbb{R}^d)$ if $|\partial f|(\alpha) \geq cE(\alpha)^\theta$, $\alpha \in \mathcal{U}$, $0 < E(\alpha) < +\infty$. Equivalently, the desingularizing function $\psi(s) = \frac{s^{1-\theta}}{c(1-\theta)}$ satisfies $\psi'(E(\alpha))|\partial f|(\alpha) \geq 1$. The Wasserstein–PL inequality (14.20) is the special case $\theta = 1/2$ with $c = \sqrt{2\kappa}$. The regime $\theta > 1/2$ gives sublinear rates, while $\theta = 1/2$ is exactly the exponential PL regime.

Theorem 14.15 (Sublinear convergence under Wasserstein–KL). *Let $(\alpha_t)_{t \geq 0}$ be a Wasserstein gradient flow of f satisfying the full energy-dissipation identity (14.21). Assume that $f_\star > -\infty$, that $E(t) = f(\alpha_t) - f_\star$ is finite, and that the trajectory stays in a region where the Wasserstein–KL inequality of Definition 14.14 holds with $\theta \in (1/2, 1)$. If $E(0) > 0$, then*

$$E(t) \leq (E(0)^{1-2\theta} + c^2(2\theta-1)t)^{-1/(2\theta-1)}.$$

Moreover, for every $t \geq 0$, $\text{Length}_{\mathcal{W}_2}((\alpha_s)_{s \geq t}) \leq \frac{E(t)^{1-\theta}}{c(1-\theta)}$. If f is lower semicontinuous on $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, then α_t converges to some $\alpha_\infty \in \text{Argmin } f$ and

$$\mathcal{W}_2(\alpha_t, \alpha_\infty) \leq \frac{E(t)^{1-\theta}}{c(1-\theta)} = O\left(t^{-(1-\theta)/(2\theta-1)}\right).$$

Proof. If E reaches zero at some time, then the flow is already at a global minimizer. Indeed, the integrated energy-dissipation identity forces both $|\dot{\alpha}_t|$ and $|\partial f|(\alpha_t)$ to vanish afterwards. We therefore argue on intervals where $E > 0$. For a.e. t , $-E'(t) = |\partial f|^2(\alpha_t) = |\dot{\alpha}_t|^2$. The Wasserstein–KL inequality yields $E'(t) \leq -c^2 E(t)^{2\theta}$. Since $1-2\theta < 0$, $\frac{d}{dt} E(t)^{1-2\theta} = (1-2\theta)E(t)^{-2\theta} E'(t) \geq c^2(2\theta-1)$, and integration from 0 to t gives the announced polynomial bound.

For the length estimate, use again $|\dot{\alpha}_t| = |\partial f|(\alpha_t)$. For $T > t$,

$$\begin{aligned} \int_t^T |\dot{\alpha}_s| ds &= \int_t^T \frac{|\partial f|^2(\alpha_s)}{|\partial f|(\alpha_s)} ds \leq \frac{1}{c} \int_t^T -E'(s) E(s)^{-\theta} ds \\ &= \frac{E(t)^{1-\theta} - E(T)^{1-\theta}}{c(1-\theta)} \leq \frac{E(t)^{1-\theta}}{c(1-\theta)}. \end{aligned}$$

Letting $T \rightarrow +\infty$ shows that the tail has finite length. Hence $(\alpha_t)_t$ is Cauchy in $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, which is complete, and converges to some α_∞ . Since the energy bound gives $E(t) \rightarrow 0$, lower semicontinuity gives $f(\alpha_\infty) = f_\star$. The distance to α_∞ is bounded by the remaining tail length, which proves the last estimate. $\square$

Proposition 14.16 (Powers of PL energies are KL). *Let $g \geq 0$ satisfy $\inf g = 0$ and the Wasserstein–PL inequality $|\partial g|^2(\alpha) \geq 2\kappa g(\alpha)$ on a region where $g > 0$. Fix $r \geq 1$ and set $f(\alpha) = g(\alpha)^r$. Whenever the metric-slope chain rule $|\partial(g^r)|(\alpha) = rg(\alpha)^{r-1}|\partial g|(\alpha)$ holds, f satisfies the Wasserstein–KL inequality $|\partial f|(\alpha) \geq r\sqrt{2\kappa} f(\alpha)^{1-1/(2r)}$. Thus f has KL exponent $\theta = 1 - 1/(2r)$. For $r = 1$ this is PL; for $r > 1$, Theorem 14.15 gives $f(\alpha_t) = O(t^{-r/(r-1)})$ along the f -gradient flow, under its hypotheses.*

Proof. By the chain rule and the PL inequality for g , $|\partial f| = rg^{r-1}|\partial g| \geq r\sqrt{2\kappa} g^{r-1/2} = r\sqrt{2\kappa} f^{1-1/(2r)}$. The exponent and rate follow by substituting $\theta = 1 - 1/(2r)$ into Theorem 14.15. $\square$

Convexity and curvature. The same language is not restricted to subsets of $\mathbb{R}^d$. If $(\mathcal{X}, d, \mathbf{m})$ is a geodesic metric-measure space, $\mathcal{W}_2$ geodesics can be defined by transporting each pair of endpoints along metric geodesics, or more intrinsically by dynamical optimal plans on path space, as discussed in Section 3.5.

$$\text{Ent}_{\mathbf{m}}(\alpha) := \begin{cases} \int_{\mathcal{X}} \rho \log \rho \, d\mathbf{m}, & \text{if } \alpha = \rho \mathbf{m}, \\ +\infty, & \text{otherwise.} \end{cases}$$

Theorem 14.17 (Ricci curvature and entropy convexity). *Let (M, g) be a smooth compact connected Riemannian manifold without boundary, and let $\mathbf{m} = \text{vol}_g$. For $\lambda \in \mathbb{R}$, the lower Ricci bound $\text{Ric}_g \geq \lambda g$ holds if and only if $\text{Ent}_{\mathbf{m}}$ is λ -geodesically convex on $(\mathcal{P}_2(M), \mathcal{W}_2)$.*

14.3 Functional Inequalities via Optimal Transport

Optimal transport proves inequalities by turning comparison into geometry. One transports a density by a monotone or Brenier map, interpolates along the resulting rays, and converts concavity of a Jacobian determinant or convexity of an energy into an integral estimate.

Brunn–Minkowski and isoperimetry.

Proposition 14.18 (Brunn–Minkowski and Euclidean isoperimetry). *Let $A, B \subset \mathbb{R}^d$ be bounded Borel sets with positive Lebesgue measure. For $t \in [0, 1]$, set $(1-t)A + tB := \{(1-t)x + ty : x \in A, y \in B\}$. Then $\text{Vol}((1-t)A + tB)^{1/d} \geq (1-t)\text{Vol}(A)^{1/d} + t\text{Vol}(B)^{1/d}$. $\text{Per}(A) \geq d\omega_d^{1/d}\text{Vol}(A)^{(d-1)/d}$. The constant is sharp, with equality for Euclidean balls.*

Proof. We first give the argument for regular sets, the general Borel case following by approximation from inside and outside. Let α and β be the uniform probability measures on A and B , and let $T = \nabla\varphi$ be the Brenier map with $T_{\#}\alpha = \beta$. The interpolating map $T_t = (1-t)\text{Id} + tT$ sends α to a measure α_t supported in $(1-t)A + tB$. At points where T is differentiable, $\det DT_t = \det((1-t)\text{Id} + tDT)$. Since DT is symmetric positive semidefinite and $\det^{1/d}$ is concave on the positive semidefinite cone, $\det(DT_t)^{1/d} \geq (1-t) + t\det(DT)^{1/d}$. The change-of-variables identity for the uniform measures gives $\det(DT) = \text{Vol}(B)/\text{Vol}(A)$ almost everywhere. If ρ_t denotes the density of α_t , a second application of the same formula gives $\rho_t(T_t(x))^{-1/d} = \text{Vol}(A)^{1/d} \det(DT_t(x))^{1/d} \geq c_t$, where $c_t = (1-t)\text{Vol}(A)^{1/d} + t\text{Vol}(B)^{1/d}$. Thus $\rho_t \leq c_t^{-d}$ almost everywhere. Since α_t has total mass one and is supported in $(1-t)A + tB$, this gives $\text{Vol}((1-t)A + tB) \geq c_t^d$, which is the displayed Brunn–Minkowski inequality. For isoperimetry, use the homogeneity of Brunn–Minkowski and apply it to A and εB_1 . Writing $A + \varepsilon B_1$ for the parallel set gives $\text{Vol}(A + \varepsilon B_1)^{1/d} \geq \text{Vol}(A)^{1/d} + \varepsilon\omega_d^{1/d}$. Using $\text{Vol}(A + \varepsilon B_1) = \text{Vol}(A) + \varepsilon \text{Per}(A) + o(\varepsilon)$ and differentiating at $\varepsilon = 0$ yields the claimed perimeter bound. Balls attain equality, so the constant is optimal. $\square$

Prékopa–Leindler.

Proposition 14.19 (Prékopa–Leindler inequality). *Let $u, v, w : \mathbb{R}^d \rightarrow [0, +\infty)$ be integrable, and let $t \in [0, 1]$. Assume that $w((1-t)x + ty) \geq u(x)^{1-t}v(y)^t$ for all $x, y \in \mathbb{R}^d$. Then*

$$\int_{\mathbb{R}^d} w \geq \left(\int_{\mathbb{R}^d} u \right)^{1-t} \left(\int_{\mathbb{R}^d} v \right)^t.$$

Proof. The endpoint cases are immediate, so fix $t \in (0, 1)$. Set $U = \int u$ and $V = \int v$. If $U = 0$ or $V = 0$, the conclusion is trivial; hence assume $U, V > 0$. Let $\alpha = (u/U)\text{dx}$ and $\beta = (v/V)\text{dx}$, and let T be the Brenier map from α to β . Put $T_t = (1-t)\text{Id} + tT$. On a smooth positive regularization, T_t is the gradient of a strictly convex function and is therefore one-to-one almost everywhere. The area formula gives $\int w \geq \int w(T_t(x)) \det(DT_t(x))\text{dx}$. For each eigenvalue $s \geq 0$ of DT , weighted arithmetic–geometric mean gives $(1-t) + ts \geq s^t$. Hence $\det((1-t)\text{Id} + tDT) \geq \det(DT)^t$. $\int w \geq \int u(x)^{1-t}v(T(x))^t \det(DT(x))^t\text{dx}$. Since $T_{\#}\alpha = \beta$, the Monge–Ampère identity gives $\det(DT(x)) = \frac{u(x)V}{v(T(x)U)}$ almost everywhere. Substitution yields

$$\int w \geq \left(\frac{V}{U} \right)^t \int u(x)\text{dx} = U^{1-t}V^t.$$

Approximation removes the smoothness and positivity assumptions. $\square$

Logarithmic Sobolev inequalities.

$$\mathcal{I}(\alpha|\beta) := \int \|\nabla \log h\|^2 \, d\alpha = \int \frac{\|\nabla h\|^2}{h} \, d\beta, \quad (14.22)$$

$$\text{KL}(\alpha|\beta) \leq \frac{1}{2\lambda} \mathcal{I}(\alpha|\beta) \quad \text{for all probability measures } \alpha. \quad (14.23)$$

Proposition 14.20 (Curvature implies logarithmic Sobolev). *Let $\beta = Z^{-1}e^{-V}\text{dx}$ be a smooth probability measure on $\mathbb{R}^d$. If $\nabla^2 V \succeq \lambda \text{Id}$ for some $\lambda > 0$, then β satisfies (14.23) with constant λ .*

Proof. Apply the HWI inequality of Proposition 14.26. Under the curvature assumption, for every absolutely continuous α , $\text{KL}(\alpha|\beta) \leq \mathcal{W}_2(\alpha, \beta) \sqrt{\mathcal{I}(\alpha|\beta)} - \frac{\lambda}{2} \mathcal{W}_2^2(\alpha, \beta)$. For fixed $s = \sqrt{\mathcal{I}(\alpha|\beta)}$, maximize the right-hand side over $r = \mathcal{W}_2(\alpha, \beta) \geq 0$. The elementary inequality $rs - \lambda r^2/2 \leq s^2/(2\lambda)$ gives (14.23). Approximation yields the standard nonsmooth statement. This is the Otto–Villani transport route from the Bakry–Emery curvature bound to logarithmic Sobolev; the original Γ_2 calculus gives the same criterion directly. $\square$

Proposition 14.21 (Logarithmic Sobolev inequality implies entropy decay). *Let $\beta = e^{-V}\text{dx}/Z$ be a smooth probability measure on $\mathbb{R}^d$ satisfying the logarithmic Sobolev inequality (14.23) with constant $\lambda > 0$. Let α_t solve the Wasserstein gradient flow of $f(\alpha) = \text{KL}(\alpha|\beta)$, namely $\partial_t \rho_t = \nabla \cdot (\rho_t \nabla V) + \Delta \rho_t$, $\alpha_t = \rho_t \text{dx}$. If $\text{KL}(\alpha_0|\beta) < +\infty$, then $\text{KL}(\alpha_t|\beta) \leq e^{-2\lambda t} \text{KL}(\alpha_0|\beta)$. Moreover, if $\alpha_0, \beta \in \mathcal{P}_2(\mathbb{R}^d)$, then $\mathcal{W}_2(\alpha_t, \beta) \leq \sqrt{\frac{2}{\lambda} \text{KL}(\alpha_0|\beta)} e^{-\lambda t}$.*

Proof. The first variation of f is $\log(\rho/\rho_\beta)$, up to an additive constant. Along the smooth Wasserstein gradient flow, $\frac{d}{dt} \text{KL}(\alpha_t|\beta) = -\int \|\nabla \log \frac{d\alpha_t}{d\beta}\|^2 d\alpha_t = -\mathcal{I}(\alpha_t|\beta)$. The logarithmic Sobolev inequality bounds the Fisher information below by $2\lambda \text{KL}(\alpha_t|\beta)$. Hence $\frac{d}{dt} \text{KL}(\alpha_t|\beta) \leq -2\lambda \text{KL}(\alpha_t|\beta)$, and Gronwall's lemma gives the entropy estimate. The Wasserstein estimate follows from the length estimate in the Wasserstein–PL theorem: apply Theorem 14.12 to $f = \text{KL}(\cdot|\beta)$, whose unique minimizer is β , and use the logarithmic Sobolev inequality as the corresponding Wasserstein–PL inequality. $\square$

Gaussian logarithmic Sobolev inequality.

Proposition 14.22 (Gaussian logarithmic Sobolev inequality). *Let γ_d be the standard Gaussian measure on $\mathbb{R}^d$. If $\rho > 0$ is smooth, has sufficient decay, and $\int \rho d\gamma_d = 1$, then $\text{Ent}_{\gamma_d}(\rho) := \int \rho \log \rho d\gamma_d \leq \frac{1}{2} \int \frac{\|\nabla \rho\|^2}{\rho} d\gamma_d$.*

Proof. Let $\alpha = \rho\gamma_d$, and let $T = \nabla\varphi$ be the Brenier map with $T_\# \alpha = \gamma_d$. Write $T(x) = x + \xi(x)$. The Monge–Ampère equation reads $\rho(x)e^{-\|x\|^2/2} = e^{-\|T(x)\|^2/2} \det(DT(x))$. Thus $\log \rho(x) = -\langle x, \xi(x) \rangle - \frac{1}{2} \|\xi(x)\|^2 + \log \det(\text{Id} + D\xi(x))$. Since DT is positive semidefinite, $\log \det(\text{Id} + D\xi) \leq \text{tr}(D\xi) = \text{div } \xi$. Multiplying by ρ and integrating with respect to γ_d gives $\text{Ent}_{\gamma_d}(\rho) \leq \int \rho(\text{div } \xi - \langle x, \xi \rangle) d\gamma_d - \frac{1}{2} \int \rho \|\xi\|^2 d\gamma_d$. Gaussian integration by parts gives $\int \rho(\text{div } \xi - \langle x, \xi \rangle) d\gamma_d = -\int \langle \nabla \rho, \xi \rangle d\gamma_d$. Completing the square pointwise, $-\langle \nabla \rho, \xi \rangle - \frac{1}{2} \rho \|\xi\|^2 \leq \frac{1}{2} \frac{\|\nabla \rho\|^2}{\rho}$. This proves the inequality. The general Sobolev-class statement follows by approximation. $\square$

$$\rho_a(x) = \exp\left(\langle a, x \rangle - \frac{1}{2} \|a\|^2\right), \quad a \in \mathbb{R}^d,$$

$\int \rho_a \log \rho_a d\gamma_d = \frac{1}{2} \|a\|^2$, $\frac{1}{2} \int \frac{\|\nabla \rho_a\|^2}{\rho_a} d\gamma_d = \frac{1}{2} \|a\|^2$. Thus equality is attained along translations, and the sharp logarithmic Sobolev constant of the standard Gaussian is $\lambda = 1$. Proposition 14.21 then gives the classical exponential convergence of the Ornstein–Uhlenbeck flow.

Poincaré as a linearized Wasserstein–PL inequality. $\beta_\varepsilon = (1 + \varepsilon\xi)\beta$, $\beta_0 = \beta$.

$$\mathbf{H}_\beta^f(\xi) := \lim_{\varepsilon \rightarrow 0} \frac{2(f(\beta_\varepsilon) - f(\beta))}{\varepsilon^2}, \quad \mathbf{D}_\beta^f(\xi) := \lim_{\varepsilon \rightarrow 0} \frac{|\partial f|^2(\beta_\varepsilon)}{\varepsilon^2},$$

Proposition 14.23 (Linearizing a Wasserstein–PL inequality). *Let β be a minimizer of an energy f , and assume that f satisfies $|\partial f|^2(\alpha) \geq 2\kappa(f(\alpha) - f(\beta))$ in a neighborhood of β . Let ξ be a smooth bounded perturbation with $\int \xi d\beta = 0$, and set $\beta_\varepsilon = (1 + \varepsilon\xi)\beta$. If $\mathbf{H}_\beta^f(\xi)$ and $\mathbf{D}_\beta^f(\xi)$ exist, then $\kappa \mathbf{H}_\beta^f(\xi) \leq \mathbf{D}_\beta^f(\xi)$.*

Proof. Apply the PL inequality to β_ε : $|\partial f|^2(\beta_\varepsilon) \geq 2\kappa(f(\beta_\varepsilon) - f(\beta))$. Divide by ε^2 and let $\varepsilon \rightarrow 0$. The definitions of $\mathbf{H}_\beta^f$ and $\mathbf{D}_\beta^f$ give the claim. $\square$

For $f(\alpha) = \text{KL}(\alpha|\beta)$, Proposition 14.23 says precisely that logarithmic Sobolev linearizes to Poincaré. Indeed, for $h_\varepsilon = 1 + \varepsilon\xi$,

$$\text{KL}(h_\varepsilon\beta|\beta) = \frac{\varepsilon^2}{2} \int \xi^2 d\beta + o(\varepsilon^2), \quad \mathcal{I}(h_\varepsilon\beta|\beta) = \varepsilon^2 \int \|\nabla \xi\|^2 d\beta + o(\varepsilon^2).$$

Proposition 14.24 (Poincaré from logarithmic Sobolev). *Let β be a smooth probability measure on $\mathbb{R}^d$. Assume that β satisfies the logarithmic Sobolev inequality with constant $\lambda > 0$,*

$$\text{KL}(\alpha|\beta) \leq \frac{1}{2\lambda} \mathcal{I}(\alpha|\beta), \quad \mathcal{I}(\alpha|\beta) := \int \|\nabla \log \frac{d\alpha}{d\beta}\|^2 d\alpha.$$

Then, for every smooth ξ with $\int \xi d\beta = 0$, $\lambda \int \xi^2 d\beta \leq \int \|\nabla \xi\|^2 d\beta$. In particular, if $d\beta = Z^{-1}e^{-V}dx$ and $\nabla^2 V \succeq \lambda \text{Id}$, so that β is λ -strongly log-concave, then the conclusion holds by the Bakry–Emery criterion.

Proof. For $h_\varepsilon = 1 + \varepsilon\xi$, with $|\varepsilon|$ small enough, one has $\beta_\varepsilon = h_\varepsilon\beta$ and $\text{KL}(\beta_\varepsilon|\beta) = \int h_\varepsilon \log h_\varepsilon d\beta = \frac{\varepsilon^2}{2} \int \xi^2 d\beta + o(\varepsilon^2)$, because $\int \xi d\beta = 0$. The Fisher information expands as

$$\mathcal{I}(\beta_\varepsilon|\beta) = \int \|\nabla \log h_\varepsilon\|^2 h_\varepsilon d\beta = \varepsilon^2 \int \|\nabla \xi\|^2 d\beta + o(\varepsilon^2).$$

Inserting these expansions in the logarithmic Sobolev inequality and letting $\varepsilon \rightarrow 0$ gives the Poincaré inequality. The last assertion is the usual Bakry–Emery implication from λ -convexity of V to the logarithmic Sobolev inequality. $\square$

$$L_\beta \xi := \Delta \xi - \langle \nabla V, \nabla \xi \rangle = \rho_\beta^{-1} \nabla \cdot (\rho_\beta \nabla \xi). \quad \int \|\nabla \xi\|^2 d\beta = \langle \xi, -L_\beta \xi \rangle_{L^2(\beta)}.$$

$$\|\dot{h}\|_{-1,\beta}^2 := \inf_v \left\{ \int \|v\|^2 d\beta : \dot{h} = -\rho_\beta^{-1} \nabla \cdot (\rho_\beta v) \right\}.$$

Talagrand's transport-entropy inequality. The distance estimate in Theorem 14.12 already reveals the general principle: when $\text{KL}(\cdot|\beta)$ satisfies logarithmic Sobolev with constant λ , the entropy flow yields the transport-entropy bound $\mathcal{W}_2^2(\alpha, \beta) \leq 2 \text{KL}(\alpha|\beta)/\lambda$.

Proposition 14.25 (Gaussian T_2 inequality). *For every $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, $\frac{1}{2} \mathcal{W}_2^2(\alpha, \gamma_d) \leq \text{KL}(\alpha|\gamma_d)$, with the convention that the right-hand side is $+\infty$ when $\alpha \not\ll \gamma_d$.*

Proof. If $\text{KL}(\alpha|\gamma_d) = +\infty$, the claim is immediate. Assume first that $\alpha = \rho\gamma_d$ with ρ smooth and positive. Let $T = \nabla\varphi$ be the Brenier map with $T_\# \alpha = \gamma_d$, and write $T(x) = x + \xi(x)$. The change of variables gives $\rho(T(x))e^{-\|T(x)\|^2/2} \det(DT(x)) = e^{-\|x\|^2/2}$. Hence $\log \rho(T(x)) = \langle x, \xi(x) \rangle + \frac{1}{2} \|\xi(x)\|^2 - \log \det(\text{Id} + D\xi(x))$. Using again $\log \det(\text{Id} + D\xi) \leq \text{div } \xi$ and integrating with respect to γ_d , $\text{KL}(\alpha|\gamma_d) \geq \frac{1}{2} \int \|\xi\|^2 d\gamma_d + \int (\langle x, \xi \rangle - \text{div } \xi) d\gamma_d$. The last integral vanishes by Gaussian integration by parts. Since T is optimal, $\int \|\xi\|^2 d\gamma_d = \mathcal{W}_2^2(\gamma_d, \alpha)$, which proves the claim. Approximation gives the general case. Equality holds for translations $\alpha = \mathcal{N}(a, \text{Id})$, so the constant is sharp. $\square$

The HWI bridge.

Proposition 14.26 (Otto–Villani HWI inequality). *Let $\beta = \rho_\beta dx = e^{-V} dx / Z$ be a smooth probability measure in $\mathcal{P}_2(\mathbb{R}^d)$, and assume that $\nabla^2 V \succeq \lambda \text{Id}$ for some $\lambda \in \mathbb{R}$. For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, define the relative Fisher information by $\mathcal{I}(\alpha|\beta) := \int \|\nabla \log \frac{\rho_\alpha}{\rho_\beta}\|^2 d\alpha$, when $\alpha = \rho_\alpha dx$ and the expression is finite, and set $\mathcal{I}(\alpha|\beta) = +\infty$ otherwise. Then $\text{KL}(\alpha|\beta) \leq \mathcal{W}_2(\alpha, \beta) \sqrt{\mathcal{I}(\alpha|\beta)} - \frac{\lambda}{2} \mathcal{W}_2^2(\alpha, \beta)$.*

Proof. If $\mathcal{I}(\alpha|\beta) = +\infty$, there is nothing to prove. Assume first that ρ_α is smooth and positive. Let T be the Brenier map from α to β , set $v = T - \text{Id}$, and let $\alpha_t = ((1-t)\text{Id} + tT)_\# \alpha$. By Theorem 14.6 and Proposition 14.4, $f(\eta) = \text{KL}(\eta|\beta)$ is λ -geodesically convex. Thus the one-dimensional function $e(t) = f(\alpha_t)$ satisfies $e(1) \geq e(0) + e'(0) + \frac{\lambda}{2} \mathcal{W}_2^2(\alpha, \beta)$. Since $e(1) = f(\beta) = 0$, it remains to compute the initial derivative. The continuity equation for the geodesic gives $\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0$ and $v_0 = v$. Hence $e'(0) = \int \langle \nabla \log \frac{\rho_\alpha}{\rho_\beta}, v \rangle d\alpha$. Therefore $\text{KL}(\alpha|\beta) \leq - \int \langle \nabla \log \frac{\rho_\alpha}{\rho_\beta}, T - \text{Id} \rangle d\alpha - \frac{\lambda}{2} \mathcal{W}_2^2(\alpha, \beta)$. Cauchy–Schwarz and $\int \|T - \text{Id}\|^2 d\alpha = \mathcal{W}_2^2(\alpha, \beta)$ give the announced bound. The general case follows by standard approximation and lower semicontinuity of entropy, Fisher information, and $\mathcal{W}_2$. $\square$

When $\lambda > 0$, the HWI inequality is a bridge from curvature to PL. It is not itself a PL inequality because it still contains the distance term $\mathcal{W}_2(\alpha, \beta)$. Optimizing the right-hand side, or simply using Young’s inequality $rs - \lambda r^2/2 \leq s^2/(2\lambda)$, gives the logarithmic Sobolev estimate $\text{KL}(\alpha|\beta) \leq \frac{1}{2\lambda} \mathcal{I}(\alpha|\beta)$.

14.4 Training Two-Layer MLPs as Wasserstein Flows

Wasserstein training of two-layer MLPs. $x = (u, v) \in \mathbb{R}^d \times \mathbb{R}^{d'}$, where u is the inner weight and v is the outer vector weight. For a scalar nonlinearity σ , define the vector-valued feature $\psi(x, z) = v \sigma(\langle u, z \rangle) \in \mathbb{R}^{d'}$. $G_X(z) = \frac{1}{n} \sum_{i=1}^n \psi(x_i, z)$, $G_\alpha(z) = \int \psi(x, z) d\alpha(x)$, $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$. Let ζ be a probability distribution on data-label pairs $(z, y) \in \mathbb{R}^d \times \mathbb{R}^{d'}$. The population risk is $f(\alpha) = \int \ell(G_\alpha(z), y) d\zeta(z, y)$,

$$\dot{x}_i = -n \nabla_{x_i} F(X), \quad F(X) = f\left(\frac{1}{n} \sum_i \delta_{x_i}\right),$$

which is the gradient flow of $F(X) = f(\alpha_X)$ for this Wasserstein particle metric, equivalently Euclidean gradient descent with the time scale multiplied by n . It gives a particle discretization of (14.3).

Assume that ℓ is differentiable in its first variable. The first variation is

$$\delta f(\alpha)(x) = \int \langle \nabla_1 \ell(G_\alpha(z), y), \psi(x, z) \rangle d\zeta(z, y), \quad (14.24)$$

$\nabla_{\mathcal{W}} f(\alpha)(x) = \nabla_x \delta f(\alpha)(x) = \int [D_x \psi(x, z)]^\top \nabla_1 \ell(G_\alpha(z), y) d\zeta(z, y)$. For the squared Euclidean loss $\ell(s, y) = \frac{1}{2} \|s - y\|^2$, the energy is the sum of a quadratic interaction and a linear potential:

$$f(\alpha) = \frac{1}{2} \iint k(x, x') d\alpha(x) d\alpha(x') + \int g(x) d\alpha(x) + \frac{1}{2} \int \|y\|^2 d\zeta(z, y), \quad (14.25)$$

$$k(x, x') = \int \langle \psi(x, z), \psi(x', z) \rangle d\zeta(z, y), \quad g(x) = - \int \langle y, \psi(x, z) \rangle d\zeta(z, y). \quad (14.26)$$

$$\delta f(\alpha)(x) = \int k(x, x') d\alpha(x') + g(x), \quad \nabla_{\mathcal{W}} f(\alpha)(x) = \int \nabla_x k(x, x') d\alpha(x') + \nabla_x g(x).$$

Classical convexity and stationarity. $f(\alpha) = \frac{1}{2} \iint k(x, x') d\alpha(x) d\alpha(x') + \int V(x) d\alpha(x) + C$, $Q((1-s)\alpha + s\beta) \leq (1-s)Q(\alpha) + sQ(\beta)$, $Q(\alpha) = \frac{1}{2} \iint k d\alpha d\alpha$.

Proposition 14.27 (Affine convexity and stationary positive densities). *Let $\Omega \subset \mathbb{R}^d$ be connected, let $f = Q + \int V d\alpha + C$ be as above on $\mathcal{P}(\Omega)$, and assume that Q is classically convex. Suppose that a Wasserstein gradient flow α_t of f converges to $\alpha_\infty = \rho_\infty dx$, where $\rho_\infty > 0$ almost everywhere on Ω , and that $\inf_{t \geq 0} f(\alpha_t) > -\infty$. Assume that $\delta f(\alpha_\infty) \in C^1(\Omega)$ and that the squared Wasserstein slope is lower semicontinuous along the convergence, in the sense that*

$$\int_\Omega \|\nabla \delta f(\alpha_\infty)\|^2 d\alpha_\infty \leq \liminf_{n \rightarrow \infty} \int_\Omega \|\nabla \delta f(\alpha_{t_n})\|^2 d\alpha_{t_n}$$

whenever $t_n \rightarrow \infty$. Then α_∞ is a global minimizer of f .

Proof. The energy-dissipation identity and the lower bound on f along the convergent trajectory imply $\int_0^\infty \int_\Omega \|\nabla \delta f(\alpha_t)\|^2 d\alpha_t dt < +\infty$. Hence there exists $t_n \rightarrow \infty$ for which the inner integral tends to zero. The assumed lower semicontinuity gives $\int_\Omega \|\nabla \delta f(\alpha_\infty)\|^2 d\alpha_\infty = 0$. Because $\rho_\infty > 0$ almost everywhere and $\nabla \delta f(\alpha_\infty)$ is continuous, this gradient vanishes throughout Ω . Connectedness then implies that $\delta f(\alpha_\infty)$ is constant. $\int_\Omega \delta f(\alpha_\infty) d(\beta - \alpha_\infty) = 0$. Classical convexity of f in the affine variable α gives the subgradient inequality $f(\beta) \geq f(\alpha_\infty) + \int \delta f(\alpha_\infty) d(\beta - \alpha_\infty) = f(\alpha_\infty)$. Thus no competitor has smaller energy. Strict positivity is essential to this argument: without it, stationarity controls the first variation only on the support explored by the limiting measure. For square-loss two-layer mean-field models, (14.25) is exactly of this quadratic-plus-linear form, and positive semidefiniteness of the induced kernel k is the classical convexity assumption. $\square$

Proposition 14.28 (Formal global optimality for two-homogeneous mean-field flows). *Assume that the feature is positively two-homogeneous in the neuron variable, $\psi(\lambda x, z) = \lambda^2 \psi(x, z)$ ($\lambda > 0$), and that $f(\alpha) = J(G_\alpha)$ with J convex and differentiable as a functional of the predictor. Let α be a smooth stationary point of the Wasserstein flow, so that $\nabla_x \delta f(\alpha)(x) = 0$ on $\text{supp}(\alpha)$. Assume also full directional support: for every unit direction ω , the support of α intersects the ray $\{\lambda \omega : \lambda > 0\}$. Then α is a global minimizer of f over the mean-field model class.*

Proof. Choose the first-variation representative

$$h_\alpha(x) = \delta f(\alpha)(x) = \langle \nabla J(G_\alpha), \psi(x, \cdot) \rangle_\rho.$$

Additive constants in first variations do not affect the Wasserstein gradient or first-order inequalities between probability measures; this normalization is useful because it inherits the homogeneity of ψ . By two-homogeneity of ψ , one has $h_\alpha(\lambda x) = \lambda^2 h_\alpha(x)$, $\lambda > 0$. Taking $x = 0$ and any $\lambda \neq 1$ in this identity gives $h_\alpha(0) = 0$. We first record that stationarity forces h_α to vanish on $\text{supp}(\alpha)$.

Indeed, if $x = r\omega \in \text{supp}(\alpha)$ with $r > 0$ and $\|\omega\| = 1$, then $0 = \langle \nabla h_\alpha(r\omega), \omega \rangle = \frac{d}{ds} h_\alpha(s\omega) \Big|_{s=r} = 2r h_\alpha(\omega)$, so $h_\alpha(\omega) = 0$, and hence

$h_\alpha(x) = r^2 h_\alpha(\omega) = 0$. The case $x = 0$ has already been covered, so $h_\alpha = 0$ on $\text{supp}(\alpha)$ and $\int h_\alpha d\alpha = 0$.

Suppose that a competitor β satisfies $f(\beta) < f(\alpha)$. Since $G_\beta - G_\alpha = \int \psi(x, \cdot) d(\beta - \alpha)(x)$, convexity of J gives $f(\beta) - f(\alpha) \geq \int h_\alpha(x) d(\beta - \alpha)(x)$. The left-hand side is negative, while $\int h_\alpha d\alpha = 0$, so $\int h_\alpha d\beta < 0$. Since $h_\alpha(0) = 0$, this strict negativity must occur away from the origin: there exists a nonzero point $x = r\omega$ with $r > 0$, $\|\omega\| = 1$, and $h_\alpha(x) < 0$. Equivalently $h_\alpha(\omega) = h_\alpha(x)/r^2 < 0$. By full directional support, there is $r_\omega > 0$ such that $r_\omega \omega \in \text{supp}(\alpha)$. At this support point, $\langle \nabla h_\alpha(r_\omega \omega), \omega \rangle = 2r_\omega h_\alpha(\omega) < 0$, which contradicts the stationarity condition $\nabla h_\alpha = 0$ on $\text{supp}(\alpha)$. Thus no competitor has smaller risk.

The rigorous Chizat–Bach theorem replaces the full directional support assumption by propagation and overparameterization hypotheses ensuring that any negative first-variation direction is represented by the evolving support. The same radial-derivative contradiction, applied after this support-propagation step, then rules out non-optimal stationary limits. $\square$

14.5 Generalized Dynamic Wasserstein Flows

Generalized Wasserstein flows.

$$\alpha_{t+\tau} \in \operatorname{argmin}_\alpha \frac{1}{2\tau} d^2(\alpha_t, \alpha) + f(\alpha). \quad (14.27)$$

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t v_{\alpha_t}) = 0, \quad (14.28)$$

The metric ingredient needed below is the squared-speed action $\mathbb{A}(\alpha, w)$ introduced in Section 13.3. Its length distance is already defined in (13.15); the present section only uses the local cost in the infinitesimal variational step defined below.

$$\mathbb{A}_p(\alpha, w) = \left(\int \|w\|^p d\alpha \right)^{2/p}.$$

$$A_2(a, w) = a\|w\|^2, \quad J_2(a, m) = \frac{\|m\|^2}{a}, \quad \mathbb{D}_\mathbb{A} = \mathcal{W}_2.$$

Definition 14.29 (Penalized minimization oracle). Let $\mathbb{A}(\alpha, \cdot)$ be the local squared-speed action at α . For a cotangent field u for which the pairing below is finite, usually $u = \nabla_{\mathcal{W}} f(\alpha)$, the penalized minimization oracle associated with this action is

$$\operatorname{PMO}_{\mathbb{A}, \alpha}(u) \in \operatorname{argmin}_w \left\{ \int \langle u(x), w(x) \rangle d\alpha(x) + \frac{1}{2} \mathbb{A}(\alpha, w) \right\}, \quad (14.29)$$

where the minimization is over admissible finite-action velocity representatives for the chosen geometry. In Hilbertian examples $u \in L^2(\alpha; \mathbb{R}^d)$; for $\mathcal{W}_p$, the natural dual space is $L^q(\alpha; \mathbb{R}^d)$, with $q = p/(p-1)$.

For an energy f , the formal small-step expansion of (14.27), when the distance is generated by the squared-speed action $\mathbb{A}$ and the JKO displacement admits an admissible finite-action tangent representative, selects the PMO direction

$$v_\alpha = \operatorname{PMO}_{\mathbb{A}, \alpha}(\nabla_{\mathcal{W}} f(\alpha)). \quad (14.30)$$

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t \operatorname{PMO}_{\mathbb{A}, \alpha_t}(\nabla_{\mathcal{W}} f(\alpha_t))) = 0. \quad (14.31)$$

$$\frac{d}{dt} f(\alpha_t) = \langle \nabla_{\mathcal{W}} f(\alpha_t), v_t \rangle_{\alpha_t} = -\mathbb{A}(\alpha_t, v_t), \quad v_t = \operatorname{PMO}_{\mathbb{A}, \alpha_t}(\nabla_{\mathcal{W}} f(\alpha_t)).$$

For $\mathcal{W}_2$, $\mathbb{A}(\alpha, w) = \int \|w\|^2 d\alpha$ and $\operatorname{PMO}_{\mathbb{A}, \alpha}(u) = -u$. Hence (14.31) reduces to the standard Wasserstein gradient-flow equation

$$\partial_t \alpha_t = \operatorname{div}(\alpha_t \nabla_{\mathcal{W}} f(\alpha_t)),$$

In the restricted Riemannian case of (13.16), where

$$\begin{aligned} \mathbb{A}(\alpha, w) &= \langle Q_\alpha w, w \rangle_{L^2(\alpha)}, \\ \operatorname{PMO}_{\mathbb{A}, \alpha}(u) &= -Q_\alpha^{-1} u, \quad v_\alpha = -Q_\alpha^{-1} \nabla_{\mathcal{W}} f(\alpha). \end{aligned} \quad (14.32)$$

$$\mathbb{A}_p(\alpha, w) = \left(\int \|w\|^p d\alpha \right)^{2/p}.$$

$$\operatorname{PMO}_{\mathbb{A}_p, \alpha}(u)(x) = -U^{2-q} \|u(x)\|^{q-2} u(x), \quad (14.33)$$

The examples below are organized by the same dictionary. Concave-mobility flows use $\mathbb{A}_{\theta, \lambda}(\alpha, v)$ from Section 13.3; spectral flows use $\mathbb{A}_\gamma(\alpha, v)$ from (13.23); kernelized Benamou–Brenier flows use $\mathbb{A}_k(\alpha, v) = \|v\|_{\mathcal{H}_k^d}^2$ with a restricted RKHS tangent space.

General mobility flows. Let λ be Lebesgue measure, write $\alpha = \rho\lambda$, and let $u = \nabla(\delta f/\delta\rho)$. For the velocity action $\mathbb{A}_{\theta,\lambda}$, the PMO introduced in Definition 14.29 is the pointwise minimizer of $\int \langle u, w \rangle \rho dx + \frac{1}{2} \int \frac{\rho}{\theta(\rho)} \|w\|^2 \rho dx$,

$$\text{PMO}_{\mathbb{A}_{\theta,\lambda},\rho\lambda}(u) = -\frac{\theta(\rho)}{\rho} u. \quad (14.34)$$

$$\partial_t \rho_t = \nabla \cdot \left(\theta(\rho_t) \nabla \frac{\delta f}{\delta \rho}(\rho_t) \right). \quad (14.35)$$

$$f(\rho) = \int U(\rho(x)) dx + \int V(x) \rho(x) dx,$$

$$\partial_t \rho = \nabla \cdot (\theta(\rho) (U''(\rho) \nabla \rho + \nabla V)).$$

Dynamic spectral Wasserstein flows. Specializing Definition 14.29 to the spectral action $\mathbb{A}_\gamma$, the descent direction associated with a field $g \in L^2(\alpha; \mathbb{R}^d)$ is simply $\text{PMO}_{\mathbb{A}_\gamma,\alpha}(g)$. The field g should be read as the Wasserstein gradient $g = \nabla_{\mathcal{W}} f(\alpha) = \nabla \delta f(\alpha)$.

Proposition 14.30 (Polar formula for the spectral direction). *Let $S_\alpha(g) := \int g(x)g(x)^\top d\alpha(x)$. Assume that the inverse-trace problem admits a positive definite minimizer*

$$A_\alpha^* \in \operatorname{argmin}_{A \in \mathcal{B}_\gamma \cap \mathbb{S}_{++}^d} \operatorname{tr}(A^{-1} S_\alpha(g)).$$

Then

$$\text{PMO}_{\mathbb{A}_\gamma,\alpha}(g)(x) = -(A_\alpha^*)^{-1} g(x) \quad (14.36)$$

is the spectral PMO direction. If the minimizer lies on the boundary of $\mathcal{B}_\gamma$, the same formula is understood by a limiting argument along positive definite minimizers, or equivalently by replacing the inverse by the corresponding inverse on the range of $S_\alpha(g)$.

Proof. Using the polar formula $\gamma(M) = \sup_{A \in \mathcal{B}_\gamma} \operatorname{tr}(AM)$, the PMO objective (14.29) with $\mathbb{A} = \mathbb{A}_\gamma$ can be written as

$$\sup_{A \in \mathcal{B}_\gamma} \left\{ \int \langle g, v \rangle d\alpha + \frac{1}{2} \int v^\top A v d\alpha \right\}.$$

For a fixed $A \succ 0$, the quadratic lower bound $\int \langle g, v \rangle d\alpha + \frac{1}{2} \int v^\top A v d\alpha$ is minimized by $v_A = -A^{-1}g$, with value $-\frac{1}{2} \operatorname{tr}(A^{-1} S_\alpha(g))$. Let $v_\star = -(A_\alpha^*)^{-1}g$ and $M_\star = \int v_\star(x)v_\star(x)^\top d\alpha(x) = (A_\alpha^*)^{-1} S_\alpha(g) (A_\alpha^*)^{-1}$. Since A_α^* minimizes $A \mapsto \operatorname{tr}(A^{-1} S_\alpha(g))$ over the convex polar set, its first-order optimality condition gives $\operatorname{tr}(A M_\star) \leq \operatorname{tr}(A_\alpha^* M_\star)$ for all $A \in \mathcal{B}_\gamma$. Thus A_α^* is active in the polar representation of $\gamma(M_\star)$, and $\gamma(M_\star) = \operatorname{tr}(A_\alpha^* M_\star) = \operatorname{tr}((A_\alpha^*)^{-1} S_\alpha(g))$. For any v , the polar formula gives $\int \langle g, v \rangle d\alpha + \frac{1}{2} \mathbb{A}_\gamma(\alpha, v) \geq \int \langle g, v \rangle d\alpha + \frac{1}{2} \int v^\top A_\alpha^* v d\alpha$, and the right-hand side is minimized at $v_\star$. The activity identity above shows that equality holds at $v_\star$, hence $v_\star$ is the spectral PMO direction. The boundary case follows by approximation with positive definite matrices in the polar set. $\square$

For the trace gauge, $\mathcal{B}_\gamma = \{A; 0 \preceq A \preceq \operatorname{Id}\}$ and the minimizer is $A_\alpha^* = \operatorname{Id}$. For the operator gauge $\gamma(M) = \lambda_{\max}(M)$, one has $\mathcal{B}_\gamma = \{A \succeq 0; \operatorname{tr}(A) \leq 1\}$; if $S_\alpha(g) \succ 0$, then

$$A_\alpha^* = \frac{S_\alpha(g)^{1/2}}{\operatorname{tr}(S_\alpha(g)^{1/2})}, \quad \text{PMO}_{\mathbb{A}_\gamma,\alpha}(g)(x) = -\operatorname{tr}(S_\alpha(g)^{1/2})^{-1} S_\alpha(g)^{1/2} g(x).$$

Proposition 13.8 identifies the static spectral distance $\mathcal{W}_\gamma$ with the length distance generated by $\mathbb{A}_\gamma$. The infinitesimal descent direction is therefore defined directly by the general PMO problem (14.29) specialized to this action.

Proposition 14.31 (Formal normalized spectral gradient flow). *Assume that f has a smooth first variation and that the spectral PMO is attained along a smooth curve. The formal gradient flow generated by the dynamic spectral action $\mathbb{A}_\gamma$ solves*

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t \text{PMO}_{\mathbb{A}_\gamma,\alpha_t}(\nabla_{\mathcal{W}} f(\alpha_t))) = 0. \quad (14.37)$$

Equivalently, if A_t^* solves the inverse-trace problem for $S_t = \int \nabla_{\mathcal{W}} f(\alpha_t)(x) \nabla_{\mathcal{W}} f(\alpha_t)(x)^\top d\alpha_t(x)$, then the velocity is $v_t(x) = -(A_t^*)^{-1} \nabla_{\mathcal{W}} f(\alpha_t)(x)$. Moreover, along smooth solutions, $\frac{d}{dt} f(\alpha_t) = -\mathbb{A}_\gamma(\alpha_t, v_t)$. If the minimizing movements (14.27) for $d = \mathcal{W}_\gamma$ converge to a smooth curve with this minimal-action metric derivative, that limit satisfies the same equation.

Proof. The action-based steepest-descent rule (14.30) gives $v_t = \text{PMO}_{\mathbb{A}_\gamma,\alpha_t}(\nabla_{\mathcal{W}} f(\alpha_t))$. Proposition 14.30 gives the explicit inverse-trace formula. Finally, the variation formula gives $\frac{d}{dt} f(\alpha_t) = \int \langle \nabla_{\mathcal{W}} f(\alpha_t), v_t \rangle d\alpha_t$. Since v_t minimizes a 2-homogeneous penalized objective, the one-dimensional rescaling $s \mapsto s v_t$ has derivative zero at $s = 1$, hence $\int \langle \nabla_{\mathcal{W}} f(\alpha_t), v_t \rangle d\alpha_t + \mathbb{A}_\gamma(\alpha_t, v_t) = 0$, which proves the dissipation identity. $\square$

14.6 Nonlocal Wasserstein Flows

Nonlocal Wasserstein flows and fractional PDEs.

Proposition 14.32 (Entropy gradient flow for reversible jump kernels). *Under the regularity and irreducibility assumptions used in Proposition 13.10, the Markov semigroup generated by the closure of $L\psi(x) = \int (\psi(y) - \psi(x)) K(x, dy)$ is the gradient flow, for the distance $\mathcal{W}_K$ defined in (13.29), of the relative entropy $\operatorname{Ent}_{\mathbf{m}}(\alpha) = \int \rho \log \rho d\mathbf{m}$, $\alpha = \rho \mathbf{m}$. Equivalently, in weak form, the entropy flow is*

$$\partial_t \rho_t = L \rho_t. \quad (14.38)$$

When K is singular, the integral defining L is understood in the principal-value sense.

Proof idea. The key identity is the logarithmic-mean relation $\theta(a, b)(\log a - \log b) = a - b$. The first variation of $\operatorname{Ent}_{\mathbf{m}}$ is $\log \rho + 1$. With the sign convention of (13.28), the steepest descent velocity is $v(x, y) = -\bar{\nabla}(\log \rho + 1)(x, y) = \log \rho(x) - \log \rho(y)$. Hence $\theta(\rho(x), \rho(y))v(x, y) = \rho(x) - \rho(y)$. Substituting this in (13.28) and using the symmetry of J gives $\frac{d}{dt} \int \varphi \rho d\mathbf{m} = \int \varphi(x) \int (\rho(y) - \rho(x)) K(x, dy) \mathbf{m}(dx)$, which is the weak form of $\partial_t \rho = L \rho$. The metric properties needed to justify this computation as a gradient-flow identity are the distance and geodesic results recalled in Proposition 13.10. $\square$

$$K_s(x, dy) = c_{d,s} \frac{dy}{|x-y|^{d+s}}, \quad 0 < s < 2. \quad L_s \psi(x) = \text{p.v.} \int_{\mathbb{R}^d} (\psi(y) - \psi(x)) \frac{c_{d,s}}{|x-y|^{d+s}} dy = -(-\Delta)^{s/2} \psi(x),$$

$$\partial_t \rho_t = -(-\Delta)^{s/2} \rho_t. \quad (14.39)$$

More generally, when a target-dependent jump kernel K_β satisfies detailed balance with a desired invariant measure $\beta = \rho_\beta \mathbf{m}$, the same construction can be applied to $r_t = d\alpha_t/d\beta$. The gradient flow of $\text{KL}(\alpha|\beta)$ is then $\partial_t r_t = L_\beta r_t$.

Discrete Wasserstein flows on Markov chains. $\text{Ent}_\pi(\rho) := \sum_i \pi_i \rho_i \log \rho_i \cdot \frac{\rho^{k+1} - \rho^k}{\tau} = -\mathcal{K}_{\rho^{k+1}} \log \rho^{k+1} + o(1) = K \rho^{k+1} + o(1)$. Here $(K\rho)_i = \sum_j K_{ij}(\rho_j - \rho_i)$. Under smoothness and strict positivity, $\rho^{k+1} = \rho^k + O(\tau)$, so the last expression also equals $K\rho^k + O(\tau)$.

Proposition 14.33 (Entropy gradient flow of a reversible Markov chain). *Let K be reversible with invariant law π . The gradient flow of Ent_π for the metric $\mathcal{W}_K$ is the forward equation of the Markov chain,*

$$\dot{\rho}_i(t) = \sum_j K_{ij}(\rho_j(t) - \rho_i(t)). \quad (14.40)$$

Equivalently, for the masses $a_i(t) = \pi_i \rho_i(t)$, this is $\dot{a}_i(t) = \sum_j (a_j(t) K_{ji} - a_i(t) K_{ij})$.

Proof. The first variation of the entropy, with respect to the weighted pairing $\sum_i \pi_i \xi_i \varphi_i$, is $\log \rho_i + 1$. Constants do not contribute to $\mathcal{K}_\rho$, hence the metric gradient-flow equation is $\dot{\rho} = -\mathcal{K}_\rho \log \rho$. Using the identity $\theta(a, b)(\log a - \log b) = a - b$, one obtains, componentwise, $\dot{\rho}_i = -\sum_j K_{ij} \theta(\rho_i, \rho_j)(\log \rho_i - \log \rho_j) = \sum_j K_{ij}(\rho_j - \rho_i)$, which is the density form of the Kolmogorov forward equation. Multiplying by π_i and using detailed balance gives the equation for the probability masses. $\square$

14.7 Dynamic Unbalanced OT and WFR Flows

The generalized dynamic Wasserstein flows of Section 14.5 are primarily mass-preserving: their tangent vectors are generated by continuity equations and are therefore velocity fields advecting the measure. Unbalanced transport changes the state space from probabilities to positive finite measures, where descent may both move and create or remove mass.

Definition 14.34 (WFR minimizing movement). Let $f : \mathcal{M}_+(\mathbb{R}^d) \rightarrow \mathbb{R} \cup \{+\infty\}$ be an energy on positive finite measures and let WFR_κ be the Wasserstein–Fisher–Rao distance with growth scale $\kappa > 0$, defined by the dynamic balance-equation formulation (13.40) and its static equivalence in Proposition 13.11. The unbalanced JKO step is

$$\alpha^{k+1} \in \underset{\alpha \in \mathcal{M}_+(\mathbb{R}^d)}{\text{argmin}} \frac{1}{2\tau} \text{WFR}_\kappa^2(\alpha^k, \alpha) + f(\alpha). \quad (14.41)$$

The continuous-time limit, when it exists, is called the WFR_κ gradient flow of f .

$$\partial_t \rho_t + \text{div}(\rho_t v_t) = g_t \rho_t,$$

Proposition 14.35 (Formal WFR gradient-flow PDE). *Assume that f has a smooth first variation $\varphi_t = \delta f(\alpha_t)$ and that $\alpha_t = \rho_t dx$ is smooth and positive. The formal WFR_κ gradient flow of f is*

$$\partial_t \rho_t = \text{div}(\rho_t \nabla \varphi_t) - \frac{1}{\kappa^2} \varphi_t \rho_t. \quad (14.42)$$

Along such a smooth solution,

$$\frac{d}{dt} f(\alpha_t) = - \int \left(\|\nabla \varphi_t\|^2 + \frac{1}{\kappa^2} \varphi_t^2 \right) \rho_t dx \leq 0. \quad (14.43)$$

Proof. For an infinitesimal tangent pair (v, g) , the density variation is $\zeta = -\text{div}(\rho v) + g\rho$. The first variation of f is $\int \varphi \zeta = \int \langle \nabla \varphi, v \rangle \rho dx + \int \varphi g \rho dx$, after integration by parts. For the inner product $\langle (v, g), (\tilde{v}, \tilde{g}) \rangle_\rho = \int \langle v, \tilde{v} \rangle \rho dx + \kappa^2 \int g \tilde{g} \rho dx$, the Riesz representative is $(\nabla \varphi, \varphi/\kappa^2)$. Steepest descent therefore uses $(v, g) = (-\nabla \varphi, -\varphi/\kappa^2)$, which gives (14.42). Substituting this pair in the first-variation formula gives

$$\frac{d}{dt} f(\alpha_t) = - \int \left(\|\nabla \varphi_t\|^2 + \frac{1}{\kappa^2} \varphi_t^2 \right) \rho_t dx,$$

which is (14.43). $\square$

14.8 Conditional Wasserstein Training of Infinite ResNets

Finite-depth finite-width ResNets.

$$z_{r+1} = z_r + \tau G_{r,n_r}(z_r), \quad G_{r,n_r}(z) := \frac{1}{n_r} \sum_{i=1}^{n_r} \psi(\theta_{r,i}, z), \quad z_0 = z, \quad r = 0, \dots, L-1. \quad (14.44)$$

$$\begin{aligned} f_{L,n}((\theta_{r,i})_{r,i}) &= \int \ell(z_L^\theta(z), y) d\zeta(z, y). \quad \alpha_r = \frac{1}{n_r} \sum_{i=1}^{n_r} \delta_{\theta_{r,i}}, \quad G_{r,n_r}(z) = \int_\Theta \psi(\theta, z) d\alpha_r(\theta). \quad \alpha^{L,n}(ds, d\theta) = \\ &= \frac{1}{L} \sum_{r=0}^{L-1} \delta_{s_r}(ds) \alpha_r(d\theta), \quad s_r = r/L, \end{aligned}$$

Finite-depth mean-field limit. $z_{r+1} = z_r + \tau G_{\alpha_r}(z_r)$, $z_0 = z$, $G_{\alpha_r}(z) := \int_\Theta \psi(\theta, z) d\alpha_r(\theta)$. $f_L(\alpha_0, \dots, \alpha_{L-1}) = \int \ell(z_L^\alpha(z), y) d\zeta(z, y)$.

Continuous-depth conditional geometry. $\alpha(ds, d\theta) = \alpha_s(d\theta)ds$, or, equivalently, a probability measure on $[0, 1] \times \Theta$ whose first marginal is Lebesgue measure. The general conditional Wasserstein distance is defined in Section 10.6.

$$\mathcal{W}_{2,\lambda}^2(\alpha, \beta) = \int_0^1 \mathcal{W}_2^2(\alpha_s, \beta_s) ds. \quad (14.45)$$

Infinite-depth mean-field ResNet.

$$\partial_s z_s = G_{\alpha_s}(z_s) := \int_{\Theta} \psi(\theta, z_s) d\alpha_s(\theta), \quad z_0 = z. \quad (14.46)$$

$$f_{\text{Res}}(\alpha) = \int \ell(z_1^\alpha(z), y) d\zeta(z, y).$$

Conditional Wasserstein gradient flow.

$$\left. \frac{d}{d\varepsilon} f(\alpha + \varepsilon \xi) \right|_{\varepsilon=0} = \int_S \int_{\Theta} \frac{\delta f}{\delta \alpha_s}(\alpha)(\theta) d\xi_s(\theta) d\lambda(s).$$

$$\partial_t \alpha_{t,s} + \text{div}_{\theta}(\alpha_{t,s} v_{t,s}) = 0, \quad v_{t,s}(\theta) = -\nabla_{\theta} \frac{\delta f}{\delta \alpha_s}(\alpha_t)(\theta), \quad (14.47)$$

for λ -a.e. condition s , whenever the first variation is differentiable in the parameter variable. In the ResNet case one takes $f = f_{\text{Res}}$ and $S = [0, 1]$.

Particle discretization and layerwise matching. $\alpha_r = \frac{1}{n_r} \sum_{i=1}^{n_r} \delta_{\theta_{r,i}}, \quad \beta_r = \frac{1}{n_r} \sum_{i=1}^{n_r} \delta_{\theta'_{r,i}}, \quad \mathcal{W}_{2,\lambda_L}^2(\alpha^{L,n}, \beta^{L,n}) = \frac{1}{L} \sum_{r=0}^{L-1} \mathcal{W}_2^2(\alpha_r, \beta_r) \cdot \frac{1}{L} \sum_{r=0}^{L-1} \frac{1}{n_r} \sum_{i=1}^{n_r} \|\theta_{r,i} - \theta'_{r,i}\|^2,$

14.9 Second-Order Momentum Flows

Finite-dimensional momentum. Let $F : \mathbb{R}^N \rightarrow \mathbb{R}$ be a smooth finite-dimensional objective. With step size $h > 0$ and damping $\lambda_{\text{fric}} > 0$, one convenient velocity form is

$$S^{k+1} = (1 - \lambda_{\text{fric}} h) S^k - h \nabla F(X^k), \quad X^{k+1} = X^k + h S^{k+1}. \quad (14.48)$$

If S^k approximates $\dot{X}(kh)$, this is a semi-implicit, or symplectic-Euler, discretization of the first-order system: the velocity update is explicit, while the position uses the newly updated velocity. The limiting system is

$$\dot{X}(t) = S(t), \quad \dot{S}(t) = -\lambda_{\text{fric}} S(t) - \nabla F(X(t)), \quad (14.49)$$

$$\ddot{X}(t) + \lambda_{\text{fric}} \dot{X}(t) + \nabla F(X(t)) = 0. \quad (14.50)$$

$$Y^k = X^k + \theta_k(X^k - X^{k-1}), \quad X^{k+1} = Y^k - h \nabla F(Y^k), \quad (14.51)$$

$$\ddot{X}(t) + \frac{r}{t} \dot{X}(t) + \nabla F(X(t)) = 0. \quad (14.52)$$

Empirical Wasserstein lift.

$$\alpha_X := \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad F(X) := f(\alpha_X). \quad (14.53)$$

$$\|\dot{X}\|_n^2 := \frac{1}{n} \sum_{i=1}^n \|\dot{x}_i\|^2,$$

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} F(x_1, \dots, x_i + \varepsilon \xi, \dots, x_n) = \frac{1}{n} \langle \nabla \delta f(\alpha_X)(x_i), \xi \rangle.$$

Thus, if $\nabla^{(n)} F$ denotes the gradient for the metric $\|\cdot\|_n$, then

$$(\nabla^{(n)} F(X))_i = n \nabla_{x_i} F(X) = \nabla \delta f(\alpha_X)(x_i) = \nabla_{\mathcal{W}} f(\alpha_X)(x_i). \quad (14.54)$$

$$v_{\alpha}(x) := -\nabla_{\mathcal{W}} f(\alpha)(x) = -\nabla \delta f(\alpha)(x). \quad (14.55)$$

$$\dot{x}_i(t) = s_i(t), \quad \dot{s}_i(t) = -\lambda_{\text{fric}} s_i(t) + v_{\alpha_t}(x_i(t)), \quad \alpha_t = \frac{1}{n} \sum_{i=1}^n \delta_{x_i(t)}. \quad (14.56)$$

$$\ddot{x}_i(t) + \lambda_{\text{fric}} \dot{x}_i(t) = v_{\alpha_t}(x_i(t)) = -\nabla_{\mathcal{W}} f(\alpha_t)(x_i(t)).$$

Proposition 14.36 (Discrete mechanical energy dissipation). *Assume that f and the particle solution are smooth enough to justify differentiating the identities above along (14.56). Define $\mathcal{H}_n(X, S) := F(X) + \frac{1}{2n} \sum_{i=1}^n \|s_i\|^2$, $S = (s_1, \dots, s_n)$. Then along (14.56),*

$$\frac{d}{dt} \mathcal{H}_n(X(t), S(t)) = -\frac{\lambda_{\text{fric}}}{n} \sum_{i=1}^n \|s_i(t)\|^2 \leq 0. \quad (14.57)$$

Proof. By (14.54) and the definition $v_{\alpha} = -\nabla_{\mathcal{W}} f(\alpha)$, one has $\nabla_{x_i} F(X) = -v_{\alpha_X}(x_i)/n$. Hence $\frac{d}{dt} F(X(t)) = \sum_{i=1}^n \langle \nabla_{x_i} F(X(t)), s_i(t) \rangle = -\frac{1}{n} \sum_{i=1}^n \langle v_{\alpha_t}(x_i(t)), s_i(t) \rangle$. On the other hand, $\frac{d}{dt} \frac{1}{2n} \sum_{i=1}^n \|s_i(t)\|^2 = \frac{1}{n} \sum_{i=1}^n \langle s_i(t), -\lambda_{\text{fric}} s_i(t) + v_{\alpha_t}(x_i(t)) \rangle$. The force terms cancel, leaving (14.57). $\square$

Phase-space formulation.

Proposition 14.37 (Phase-space Liouville formulation). *Let v_α be smooth enough that (14.56) has a classical solution. Define the empirical phase-space measure $\eta_t^n := \frac{1}{n} \sum_{i=1}^n \delta_{(x_i(t), s_i(t))} \in \mathcal{P}(\mathbb{R}_x^d \times \mathbb{R}_s^d)$, $\alpha_t^n = (\pi_x)_\# \eta_t^n$. Then η_t^n solves, in the sense of distributions,*

$$\partial_t \eta_t^n + \nabla_x \cdot (s \eta_t^n) + \nabla_s \cdot ((-\lambda_{\text{fric}} s + v_{\alpha_t^n}(x)) \eta_t^n) = 0. \quad (14.58)$$

If, as $n \rightarrow \infty$, the empirical phase-space laws converge to a smooth density $\eta_t(x, s)$ and the nonlinear fields converge accordingly, the limiting kinetic equation is

$$\partial_t \eta_t + \nabla_x \cdot (s \eta_t) + \nabla_s \cdot ((-\lambda_{\text{fric}} s + v_{\alpha_t}(x)) \eta_t) = 0, \quad \alpha_t = (\pi_x)_\# \eta_t. \quad (14.59)$$

Proof. Let $\varphi \in C_c^\infty(\mathbb{R}_x^d \times \mathbb{R}_s^d)$. Along the particle system,

$$\frac{d}{dt} \int \varphi d\eta_t^n = \frac{1}{n} \sum_{i=1}^n [\langle \nabla_x \varphi(x_i, s_i), s_i \rangle + \langle \nabla_s \varphi(x_i, s_i), -\lambda_{\text{fric}} s_i + v_{\alpha_t^n}(x_i) \rangle].$$

Rewriting the right-hand side as an integral against η_t^n and integrating by parts gives (14.58). Passing formally to the mean-field limit gives (14.59). $\square$

$$k(x, y) = -\|x - y\|, \quad f(\alpha) = \text{MMD}_k^2(\alpha, \beta) = \iint k(x, y) d(\alpha - \beta)(x) d(\alpha - \beta)(y). \quad v_\alpha(x) = 2 \int \frac{x-y}{\|x-y\|} d(\alpha - \beta)(y).$$

Entropy without an empirical energy. For the numerical illustration, we add the quadratic confinement $V(x) = \|x\|^2/2$. In one dimension the overdamped equation becomes the Ornstein–Uhlenbeck Fokker–Planck equation $\partial_t \rho_t = \partial_{xx} \rho_t + \partial_x(x \rho_t)$, whose stationary law is the centered Gaussian $\mathcal{N}(0, 1)$. The Newton lift is discretized directly in phase space: for a density $\eta_t(x, s)$ with spatial marginal $\rho_t(x) = \int \eta_t(x, s) ds$, it reads $\partial_t \eta_t + \partial_x(s \eta_t) + \partial_s((-\partial_x \log \rho_t(x) - x) \eta_t) = 0$.

15 Generative Models via Transportation

15.1 Generative Models via Flow Matching

Stochastic interpolant. The word “stochastic” can hide two different levels of randomness. We first use the simpler one: after drawing a latent variable $U \sim \pi$, the path $t \mapsto I_t(U)$ is deterministic and differentiable.

We assume first that α_t is obtained by pushing a latent distribution $\pi \in \mathcal{P}(\mathbb{R}^{d'})$ through a time-dependent map $I_t : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$; the latent dimension d' may be larger than the data dimension d , i.e.

$$\forall t \in [0, 1], \quad \alpha_t := (I_t)_\# \pi. \quad (15.1)$$

If $\pi = \alpha \otimes \beta$ and $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$, $\beta = \frac{1}{m} \sum_j \delta_{y_j}$, then α_t consists of $n \times m$ Dirac masses

$\alpha_t = \frac{1}{nm} \sum_{i,j} \delta_{I_t(x_i, y_j)}$. If $\pi = (\text{Id}, T)_\# \alpha$ is a Brenier-type coupling, then $\alpha_t = ((1-t)\text{Id} + tT)_\# \alpha$ is the so-called McCann OT interpolation.

Flow matching formula.

Proposition 15.1 (Flow matching vector field). *For each fixed t , assume $\partial_t I_t \in L^2(\pi; \mathbb{R}^d)$. Consider the flow-matching problem over measurable fields $v_t : \mathbb{R}^d \rightarrow \mathbb{R}^d$*

$$\min_{v_t} \int_{\mathbb{R}^{d'}} \|v_t(I_t(u)) - [\partial_t I_t](u)\|^2 d\pi(u). \quad (15.2)$$

Its minimizer is characterized α_t -almost everywhere by the conditional expectation

$$v_t(z) = \mathbb{E}_{u \sim \pi}([\partial_t I_t](u) \mid z = I_t(u)). \quad (15.3)$$

Then the pair (α_t, v_t) satisfies the continuity equation (13.2).

Proof. We first recall the two equivalent ways of writing the interpolated measure. Formally, one may write $\alpha_t(z) = \int_{\mathbb{R}^{d'}} \delta(z - I_t(u)) d\pi(u)$, while the rigorous meaning is that, for every smooth test function φ ,

$$\int_{\mathbb{R}^d} \varphi(z) d\alpha_t(z) = \int_{\mathbb{R}^{d'}} \varphi(I_t(u)) d\pi(u). \quad (15.4)$$

The minimizer in (15.2) is the orthogonal projection in $L^2(\pi; \mathbb{R}^d)$ of the latent velocity $\partial_t I_t(u)$ onto the closed subspace of functions that depend on u only through $I_t(u)$. This projection is the conditional expectation (15.3). Formally, this can be read as $v_t(z) = \frac{1}{\alpha_t(z)} \int_{\mathbb{R}^{d'}} \delta(z - I_t(u)) [\partial_t I_t](u) d\pi(u)$, and rigorously it means that, for every smooth test vector field m ,

$$\int \langle m(z), v_t(z) \rangle d\alpha_t(z) = \int \langle m(I_t(u)), [\partial_t I_t](u) \rangle d\pi(u). \quad (15.5)$$

The weak form of $\partial_t \alpha_t + \text{div}(\alpha_t v_t) = 0$ is that, for every smooth scalar test function φ ,

$$\frac{d}{dt} \int \varphi(z) d\alpha_t(z) - \int \langle v_t(z), \nabla \varphi(z) \rangle d\alpha_t(z) = 0. \quad (15.6)$$

Using (15.4) and differentiating under the integral sign gives

$$\frac{d}{dt} \int \varphi(z) d\alpha_t(z) = \int \langle \nabla \varphi(I_t(u)), [\partial_t I_t](u) \rangle d\pi(u). \quad (15.7)$$

On the other hand, applying (15.5) with $m = \nabla \varphi$ gives

$$\int \langle v_t(z), \nabla \varphi(z) \rangle d\alpha_t(z) = \int \langle \nabla \varphi(I_t(u)), [\partial_t I_t](u) \rangle d\pi(u). \quad (15.8)$$

Comparing (15.7) and (15.8) yields (15.6), which is the desired continuity equation. $\square$

The conditional expectation in (15.3) has a simple measure-theoretic meaning. Let $\alpha_t = (I_t)_\# \pi$ and define the vector-valued flux measure ω_t on $\mathbb{R}^d$ by

$\int_{\mathbb{R}^d} \langle \psi(z), d\omega_t(z) \rangle := \int_{\mathbb{R}^d} \langle \psi(I_t(u)), [\partial_t I_t](u) d\pi(u) \rangle d\omega_t(z) = v_t(z) d\alpha_t(z)$, $v_t = \frac{d\omega_t}{d\alpha_t}$. In the language of Lebesgue decomposition, ω_t has only an absolutely continuous part with respect to α_t and no singular part; the conditional expectation is precisely its density. This agrees with the flux notation used in the dynamic formulation of Section 13.2.

$v_t(z) = \int_{\{I_t(u)=z\}} [\partial_t I_t](u) d\pi_{t,z}(u)$. Thus the solution of (15.2) is the conditional expectation of the velocities $\partial_t I_t$: intuitively, $v_t(z)$ is the average velocity of all trajectories passing through z .

For the exact field v_t , integrating the ODE $\dot{x} = v_t(x)$ defines a transport map T_t . If v_t is regular enough, or more generally if the continuity equation has a unique solution for this velocity, then $(T_t)_\# \alpha_0 = \alpha_t$.

Connection with diffusion models. In the special case where $I_t(x, y) = (1-t)x + ty$ is a linear interpolation and $\pi = \alpha \otimes \beta$, the curve α_t is a convolution of rescaled versions of α_0 and α_1 . The flow-matching problem (15.2) becomes

$$\min_{(v_t)_t} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|v_t((1-t)x + ty) - (y-x)\|^2 d\alpha_0(x) d\alpha_1(y).$$

Proposition 15.2 (Tweedie identity). *Let W be a random vector in $\mathbb{R}^d$ with law $\beta \in \mathcal{P}_1(\mathbb{R}^d)$. For $\sigma > 0$, observe $Z = W + \sigma \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \text{Id})$ is independent of W . Denote by ρ_σ the smooth positive density of $\beta_\sigma = \beta * \mathcal{N}(0, \sigma^2 \text{Id})$, which is the law of Z . Then the conditional mean admits the following everywhere-defined version: $\mathbb{E}[W \mid Z = z] = z + \sigma^2 \nabla \log \rho_\sigma(z)$ for all $z \in \mathbb{R}^d$.*

Proof. Let φ_σ be the $\mathcal{N}(0, \sigma^2 \text{Id})$ density. Bayes' rule gives $\mathbb{E}[W \mid Z = z] = \frac{1}{\rho_\sigma(z)} \int_{\mathbb{R}^d} w \varphi_\sigma(z-w) d\beta(w)$. Differentiating the Gaussian convolution under the integral sign and using $\nabla_z \varphi_\sigma(z-w) = -\sigma^{-2}(z-w) \varphi_\sigma(z-w)$ yields $\nabla \rho_\sigma(z) = \int \nabla_z \varphi_\sigma(z-w) d\beta(w) = -\sigma^{-2}(z - \mathbb{E}[W \mid Z = z]) \rho_\sigma(z)$. Rearranging finishes the proof. $\square$

Proposition 15.3 (Gaussian-endpoint flow-matching field). *Let $X \sim \alpha$ and $Y \sim \mathcal{N}(0, \text{Id})$ be independent. For $t \in (0, 1)$ set $Z_t = (1-t)X + tY$, $\alpha_t = \text{Law}(Z_t) = \rho_t dx$. The regression minimizer $v^* : \mathbb{R}^d \times (0, 1) \rightarrow \mathbb{R}^d$ of $\min_v \int_0^1 \int_{\mathbb{R}^d \times \mathbb{R}^d} |y-x-v((1-t)x+ty, t)|^2 d\alpha(x) d\mathcal{N}(y) dt$ is $v^*(x, t) = -\frac{1}{1-t}x - \frac{t}{1-t} \nabla \log \rho_t(x)$ ($x \in \mathbb{R}^d$, $t \in (0, 1)$). In particular, for each $t \in (0, 1)$ this field is a gradient field,*

$$v^*(\cdot, t) = -\nabla \left(\frac{\|\cdot\|^2}{2(1-t)} + \frac{t}{1-t} \log \rho_t \right).$$

Proof. Fix $t \in (0, 1)$ and write $W = (1-t)X$, $\sigma = t$, so that $Z_t = W + \sigma Y$ matches the setting of Proposition 15.2. Conditional expectations satisfy $v^*(z, t) = \mathbb{E}[Y - X \mid Z_t = z] = \frac{1}{t} \mathbb{E}[Z_t - W \mid Z_t = z] - \frac{1}{1-t} \mathbb{E}[W \mid Z_t = z]$. Applying Proposition 15.2 to $\mathbb{E}[W \mid Z_t = z]$ and noting $\mathbb{E}[Y \mid Z_t = z] = -t \nabla \log \rho_t(z)$ gives the claimed formula. $\square$

When is the induced map optimal? Integrating the learned velocity gives a deterministic map from α_0 to α_1 , but this map is not automatically the Brenier optimal map. It is optimal only in special cases where the accumulated flow remains the gradient of a convex potential.

Proposition 15.4 (Gaussian flow matching and optimality). *Let $\Sigma_0, \Sigma_1 \succ 0$ and let $X_0 \sim \mathcal{N}(0, \Sigma_0)$ and $X_1 \sim \mathcal{N}(0, \Sigma_1)$ be independent. Consider the linear flow-matching interpolation $Z_t = (1-t)X_0 + tX_1$, $\alpha_t = \text{Law}(Z_t) = \mathcal{N}(0, \Sigma_t)$, where*

$$\Sigma_t = (1-t)^2 \Sigma_0 + t^2 \Sigma_1. \quad (15.9)$$

Then the exact flow-matching velocity is affine, $v_t(z) = A_t z$, with

$$A_t = (t\Sigma_1 - (1-t)\Sigma_0) \Sigma_t^{-1}. \quad (15.10)$$

The induced flow map T_t^{FM} from α_0 to α_t is

$$T_t^{\text{FM}} = \Sigma_0^{1/2} \left((1-t)^2 \text{Id} + t^2 \Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2} \right)^{1/2} \Sigma_0^{-1/2}. \quad (15.11)$$

In particular,

$$T_1^{\text{FM}} = \Sigma_0^{1/2} (\Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2})^{1/2} \Sigma_0^{-1/2}. \quad (15.12)$$

This terminal map coincides with the quadratic optimal transport map

$$T^{\text{OT}} = \Sigma_0^{-1/2} (\Sigma_0^{1/2} \Sigma_1 \Sigma_0^{1/2})^{1/2} \Sigma_0^{-1/2} \quad (15.13)$$

if and only if $\Sigma_0 \Sigma_1 = \Sigma_1 \Sigma_0$.

Proof. The conditional-expectation formula gives $v_t(z) = \mathbb{E}[X_1 - X_0 \mid Z_t = z]$. Since all variables are jointly Gaussian, this conditional expectation is linear and $v_t(z) = \text{Cov}(X_1 - X_0, Z_t) \text{Cov}(Z_t)^{-1} z = (t\Sigma_1 - (1-t)\Sigma_0) \Sigma_t^{-1} z$, which proves (15.10). To solve the characteristic equation, whiten the source by setting $C = \Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2}$, $\tilde{Z}_t = \Sigma_0^{-1/2} Z_t$. In these coordinates the source covariance is Id and $\tilde{\Sigma}_t = (1-t)^2 \text{Id} + t^2 C$. Because Id and C commute, the affine flow map in whitened coordinates is simply $\tilde{T}_t = \tilde{\Sigma}_t^{1/2}$. Indeed, $\frac{d}{dt} \tilde{\Sigma}_t^{1/2} = (tC - (1-t)\text{Id}) \tilde{\Sigma}_t^{-1/2}$, which is exactly the equation $\dot{\tilde{T}}_t = \tilde{A}_t \tilde{T}_t$ with $\tilde{T}_0 = \text{Id}$. Returning to the original coordinates gives (15.11), and $t = 1$ gives (15.12).

Both T_1^{FM} and T^{OT} push $\mathcal{N}(0, \Sigma_0)$ to $\mathcal{N}(0, \Sigma_1)$. The Brenier map between nondegenerate Gaussians is the unique symmetric positive definite linear map with this property. Hence $T_1^{\text{FM}} = T^{\text{OT}}$ if and only if T_1^{FM} is symmetric positive definite. The map T_1^{FM} is similar to $C^{1/2}$, so if it is symmetric then it is automatically positive definite. Since $C^{1/2}$ is symmetric positive definite, $(T_1^{\text{FM}})^\top = \Sigma_0^{-1/2} C^{1/2} \Sigma_0^{1/2}$. Thus symmetry of T_1^{FM} is equivalent to $\Sigma_0 C^{1/2} = C^{1/2} \Sigma_0$, hence to $\Sigma_0 C = C \Sigma_0$ by functional calculus. Multiplying this identity on the left and right by $\Sigma_0^{1/2}$ gives $\Sigma_0 \Sigma_1 = \Sigma_1 \Sigma_0$. Conversely, if Σ_0 and Σ_1 commute, they are orthogonally co-diagonalizable, and both (15.12) and (15.13) reduce in that basis to the diagonal map with entries $\sqrt{\lambda_{1,k}/\lambda_{0,k}}$. This proves the equivalence. $\square$

Variations on the interpolant. $v_t(z) = \lambda'(t) v_{\lambda(t)}^{\text{lin}}(z)$, $v_r^{\text{lin}}(z) = \mathbb{E}[Y - X \mid (1-r)X + rY = z]$. $Z_t = a_t X + b_t Y$, $Y \sim \mathcal{N}(0, \sigma^2 \text{Id})$, then both the mixture centers and the component variances are changed. Writing p_t for the density of Z_t and $s_t = \nabla \log p_t$, Tweedie's formula gives, away from times where $a_t = 0$,

$$v_t(z) = a'_t \mathbb{E}[X \mid Z_t = z] + b'_t \mathbb{E}[Y \mid Z_t = z] = \frac{a'_t}{a_t} z + \left(\frac{a'_t b'_t}{a_t} - b'_t b_t \right) \sigma^2 s_t(z).$$

For the linear bridge, $a_t = 1 - t$ and $b_t = t$, this recovers the formula above. For the variance-preserving Ornstein–Uhlenbeck noising used in diffusion models,

$a_\tau = e^{-\tau}$, $b_\tau = \sqrt{1 - e^{-2\tau}}$, one obtains the forward probability-flow velocity $v_\tau(z) = -z - \sigma^2 \nabla \log p_\tau(z)$. Sampling integrates this ODE backward in τ .

$$a_t = (1 - t)(1 - 2t), \quad b_t = t,$$

15.2 One-Step Generative Models

One-step models via parameter-domain discrepancy flows. The most direct one-step strategy is to train the generator parameters themselves by descending a discrepancy between generated and data distributions. Let ζ be a simple latent distribution, let $f_\theta : \mathcal{Z} \rightarrow \mathcal{X}$ be a neural generator, and let β be the data distribution.

$(f_\theta)_\# \zeta = \beta$, or, in practice, such that the generated law is close to β . Given a discrepancy $\mathcal{D}$ on probability measures, define

$$\mathcal{E}_\beta(\gamma) := \mathcal{D}(\gamma, \beta), \quad H_\beta(\theta) := \mathcal{E}_\beta((f_\theta)_\# \zeta) = \mathcal{D}((f_\theta)_\# \zeta, \beta). \quad (15.14)$$

$$\dot{\theta}_t = -\nabla_\theta H_\beta(\theta_t). \quad (15.15)$$

$\alpha_t := (f_{\theta_t})_\# \zeta$, $X_t = f_{\theta_t}(Z)$, $Z \sim \zeta$. If $t \mapsto \theta_t$ is smooth and f_θ is differentiable with respect to θ , then

$\dot{X}_t = \partial_\theta f_{\theta_t}(Z) \dot{\theta}_t$. For α_t -a.e. x , let $\eta_{t,x}$ be the disintegration of the latent law ζ with respect to the map f_{θ_t} . Thus $\eta_{t,x}$ is supported on the fiber $\{z; f_{\theta_t}(z) = x\}$, and for every bounded measurable ψ ,

$$\begin{aligned} \int \psi(z) d\zeta(z) &= \int_{\mathcal{X}} \left(\int \psi(z) d\eta_{t,x}(z) \right) d\alpha_t(x). \\ v_t(x) &= \int \partial_\theta f_{\theta_t}(z) \dot{\theta}_t d\eta_{t,x}(z) = - \int \partial_\theta f_{\theta_t}(z) \nabla_\theta H_\beta(\theta_t) d\eta_{t,x}(z). \end{aligned} \quad (15.16)$$

$$\frac{d}{dt} \int \varphi(x) d\alpha_t(x) = \int \langle \nabla \varphi(x), v_t(x) \rangle d\alpha_t(x),$$

$$\partial_t \alpha_t + \text{div}(\alpha_t v_t) = 0. \quad (15.17)$$

The velocity (15.16) depends on the parameter value θ_t and on the latent disintegration, not only on the measure α_t . If the parametrization is non-identifiable, two different parameters can generate the same measure while inducing different velocities.

$$-\nabla_\gamma \mathcal{E}_\beta(\gamma) \big|_{\gamma=\alpha_t} \cdot \hat{\beta}_n = \frac{1}{n} \sum_{i=1}^n \delta_{Y_i}, \quad Y_i \stackrel{\text{i.i.d.}}{\sim} \beta,$$

$$\mathbb{E} \left[\nabla_\theta H_{\hat{\beta}_n}(\theta) \right] \neq \nabla_\theta H_\beta(\theta).$$

One-step model using Wasserstein flow of discrepancy. Let ζ be a simple latent distribution and let $\alpha_\theta = (G_\theta)_\# \zeta$ be the model distribution. Assume that the target data distribution is β . The Wasserstein-flow construction chooses a discrepancy $\mathcal{E}_\beta(\alpha)$,

$$\partial_t \alpha_t + \text{div}(\alpha_t w_t) = 0, \quad w_t(x) = -\nabla \delta_\alpha \mathcal{E}_\beta(\alpha_t)(x). \quad (15.18)$$

$$\min_{\eta} \int \|U_\eta(x) - w_t(x)\|^2 d\alpha_t(x). \quad (15.19)$$

$$\alpha_\theta^+ = (\text{Id} + \tau U_{\eta_t})_\# \alpha_\theta, \quad \text{or equivalently} \quad G_\theta^+(z) = G_\theta(z) + \tau U_{\eta_t}(G_\theta(z)).$$

Sliced-Wasserstein flow. $\mathcal{E}_\beta(\alpha) = \frac{1}{2} \text{SW}_2^2(\alpha, \beta)$, with SW_2 defined in Section 10.2. With the continuity-equation convention $\partial_t \alpha_t + \text{div}(\alpha_t w_t) = 0$, the descent velocity points from the current projected law toward the target projected law.

$$w_{\text{SW}}[\alpha, \beta](x) = \int_{\mathbb{S}^{d-1}} (T_\theta(P_\theta x) - P_\theta x) \theta d\sigma(\theta). \quad (15.20)$$

The sign follows from the one-dimensional identity: the first variation of $\frac{1}{2} \mathcal{W}_2^2(\alpha, \beta)$ has spatial derivative $s - T(s)$, so the descent velocity is $T(s) - s$, and composing with P_θ lifts it in the direction θ . Thus empirical implementations only require one-dimensional sorting along sampled directions, followed by averaging the lifted projected displacements.

Stein variational gradient descent. For the formal density-level calculation, assume $\alpha = \rho_\alpha dx$ and $\beta = \rho_\beta dx$ have smooth positive densities, and let v be a smooth compactly supported vector field. For the perturbation $\alpha_\varepsilon = (\text{Id} + \varepsilon v)_\# \alpha$, integration by parts gives the Stein first variation

$$\begin{aligned} \frac{d}{d\varepsilon} \text{KL}(\alpha_\varepsilon | \beta) \bigg|_{\varepsilon=0} &= - \int (\langle \nabla \log \rho_\beta(x), v(x) \rangle + \text{div} v(x)) d\alpha(x). \\ v_\alpha^{\text{SVGD}}(x) &= \int \left(k(y, x) \nabla \log \rho_\beta(y) + \nabla_y k(y, x) \right) d\alpha(y). \end{aligned} \quad (15.21)$$

$$\partial_t \alpha_t + \text{div}(\alpha_t v_{\alpha_t}^{\text{SVGD}}) = 0. \quad (15.22)$$

For particles $\alpha_n^\ell = n^{-1} \sum_i \delta_{X_i^\ell}$, this becomes the explicit update

$$X_i^{\ell+1} = X_i^\ell + \tau \frac{1}{n} \sum_{j=1}^n \left(k(X_j^\ell, X_i^\ell) \nabla \log \rho_\beta(X_j^\ell) + \nabla_{X_j} k(X_j^\ell, X_i^\ell) \right). \quad (15.23)$$

Self-corrected drifting fields.

$$B_\varepsilon[\nu](x) := \frac{\int (y - x) K_\varepsilon(x, y) d\nu(y)}{\int K_\varepsilon(x, y) d\nu(y)}. \quad (15.24)$$

For the Gaussian kernel $K_\varepsilon(x, y) = \exp(-\|x - y\|^2/(2\varepsilon))$, this normalized field is a score of a smoothed density:

$$B_\varepsilon[\nu](x) = \varepsilon \nabla \log \left(\int K_\varepsilon(x, y) d\nu(y) \right). \quad (15.25)$$

$$u_t(x) = B_\varepsilon[\beta](x) - B_\varepsilon[\alpha_t](x) = \varepsilon \nabla \log \frac{\int K_\varepsilon(x, y) d\beta(y)}{\int K_\varepsilon(x, y) d\alpha_t(y)}. \quad (15.26)$$

The first term pulls samples toward data, while the second term corrects self-attraction and prevents all particles from collapsing onto the same high-density region. For a fixed reference measure, $B_\varepsilon[\nu]$ is precisely the Gaussian mean-shift displacement in (15.33): it moves x toward the local kernel barycenter of ν .

Proposition 15.5 (Instantaneous gradient representation of drifting). *Let $\alpha_t = \rho_t dx$ be a smooth curve of probability measures with positive densities, and let $u_t = \nabla \varphi_t$ be a smooth time-dependent gradient field. Define the semi-relaxed functional*

$$\mathcal{R}_t(\alpha|\alpha_t) := - \int \varphi_t(x) d\alpha(x) + \int \varphi_t(x) d\alpha_t(x). \quad (15.27)$$

Here α_t and φ_t are frozen when taking the first variation with respect to the first argument α . Then the continuity equation $\partial_t \alpha_t + \operatorname{div}(\alpha_t u_t) = 0$ is the formal Wasserstein gradient descent of the frozen time-dependent functional $\alpha \mapsto \mathcal{R}_t(\alpha|\alpha_t)$.

Proof. Since α_t and φ_t are fixed in the variation with respect to α , the first variation is $\delta_\alpha \mathcal{R}_t(\alpha|\alpha_t)(x) = -\varphi_t(x)$. By Proposition 14.2, $\nabla_{\mathcal{W}} \mathcal{R}_t(\alpha|\alpha_t) = \nabla \delta_\alpha \mathcal{R}_t(\alpha|\alpha_t) = -\nabla \varphi_t = -u_t$. The Wasserstein gradient-descent velocity is the negative of this gradient, namely u_t . Substituting this velocity in the continuity equation gives the claimed flow. $\square$

15.3 Moment Measures

Definition 15.6 (Moment measure of a convex function). Let $u : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper lower-semicontinuous convex function such that $Z_u := \int_{\mathbb{R}^d} e^{-u(x)} dx < +\infty$. The normalized log-concave measure associated with u is $\eta_u := Z_u^{-1} e^{-u(x)} dx$. Whenever ∇u is defined η_u -almost everywhere, the moment measure of u is $\mathfrak{M}(u) := (\nabla u)_\# \eta_u$.

$\int y d\mathfrak{M}(u)(y) = Z_u^{-1} \int \nabla u(x) e^{-u(x)} dx = -Z_u^{-1} \int \nabla(e^{-u(x)}) dx = 0$. Thus moment measures are necessarily centered. The nonsmooth theory uses essentially continuous convex functions: these are lower-semicontinuous convex functions whose set of discontinuity points has zero $\mathcal{H}^{d-1}$ measure.

Theorem 15.7 (Cordero-Erausquin-Klartag). *Let $\alpha \in \mathcal{P}_1(\mathbb{R}^d)$ be a probability measure. There exists an essentially continuous convex function u such that $\alpha = \mathfrak{M}(u)$ if and only if $\int y d\alpha(y) = 0$ and $\operatorname{supp}(\alpha)$ is not contained in a hyperplane. Under these conditions, u is unique up to translations of the argument and additive constants.*

Optimal-transport variational formulation.

$$\mathcal{C}_\alpha(\eta) := \sup_{\pi \in \mathcal{U}(\eta, \alpha)} \int_{\mathbb{R}^d \times \mathbb{R}^d} x \cdot y d\pi(x, y). \quad (15.28)$$

$$\mathcal{C}_\alpha(\eta) = \inf_{v \text{ convex}} \left\{ \int v(x) d\eta(x) + \int v^*(y) d\alpha(y) \right\}, \quad (15.29)$$

where the infimum is over convex functions for which both integrals are well defined and v^* is the Legendre transform. If $\eta, \alpha \in \mathcal{P}_2(\mathbb{R}^d)$, then

$$\mathcal{C}_\alpha(\eta) = \frac{1}{2} \int \|x\|^2 d\eta(x) + \frac{1}{2} \int \|y\|^2 d\alpha(y) - \frac{1}{2} \mathcal{W}_2^2(\eta, \alpha). \quad (15.30)$$

$$\min_{\eta \in \mathcal{P}_1(\mathbb{R}^d)} \{ \mathcal{H}(\eta) + \mathcal{C}_\alpha(\eta) \}, \quad \mathcal{H}(r dx) := \int r(x) \log r(x) dx, \quad (15.31)$$

Proposition 15.8 (Variational characterization of moment measures). *Let $\alpha \in \mathcal{P}_1(\mathbb{R}^d)$ be centered and not supported on a hyperplane. Then the minimization problem (15.31) admits a solution, unique up to translations. Every minimizer is a probability measure with log-concave density of the form $\eta = Z_u^{-1} e^{-u} dx$, where u is convex, essentially continuous, and satisfies the moment-measure equation $\alpha = (\nabla u)_\# \eta$. Conversely, any essentially continuous convex u satisfying this equation yields a global minimizer. If $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, the objective $\eta \mapsto \mathcal{H}(\eta) + \mathcal{C}_\alpha(\eta)$ is displacement convex along $\mathcal{W}_2$ geodesics of absolutely continuous measures in $\mathcal{P}_2(\mathbb{R}^d)$ whenever the terms are finite. When merely $\alpha \in \mathcal{P}_1(\mathbb{R}^d)$, the maximal-correlation term remains convex along such finite-second-moment geodesics by approximation; existence and uniqueness on the full $\mathcal{P}_1$ domain follow from the lower-semicontinuity argument of Santambrogio.*

Proof. The existence proof has two ingredients. First, since α is centered, translating η does not change either $\mathcal{H}(\eta)$ or $\mathcal{C}_\alpha(\eta)$, so one can center a minimizing sequence. Second, the assumption that α is not supported on a hyperplane gives a coercive lower bound of the form $\mathcal{C}_\alpha(\eta) \geq c_\alpha \int \|x\| d\eta(x)$ for centered absolutely continuous η , with $c_\alpha > 0$. Together with the lower-semicontinuity estimates for entropy and maximal correlation, this yields a minimizer.

Let $\eta = r dx$ be a minimizer and let u be a convex optimizer in the dual formula (15.29). Keeping u fixed and varying η in (15.31) gives the Euler equation $\log r(x) + 1 + u(x) = \text{constant}$ on $\{r > 0\}$, so $\eta = Z_u^{-1} e^{-u} dx$. The optimality condition for the scalar-product transport problem says that an optimal coupling is supported on $\{(x, y) : y \in \partial u(x)\}$. Since η is absolutely continuous and u is convex, $\partial u(x) = \{\nabla u(x)\}$ for η -almost every x , hence $\alpha = (\nabla u)_\# \eta$.

Conversely, assume $\eta = Z_u^{-1} e^{-u} dx$ and $\alpha = (\nabla u)_\# \eta$. Let ν be a smooth compactly supported competitor, and let T be the Brenier map from η to ν . Along the geodesic $\eta_t = ((1-t)\operatorname{Id} + tT)_\# \eta$, the right derivative of the entropy at $t = 0$ is $\left. \frac{d}{dt} \mathcal{H}(\eta_t) \right|_{t=0^+} =$

– $\int \langle T(x) - x, \nabla u(x) \rangle d\eta(x)$, where the identity follows by differentiating the Jacobian formula and integrating by parts. The dual optimizer u gives the upper directional bound $\left. \frac{d}{dt} \mathcal{C}_\alpha(\eta_t) \right|_{t=0^+} \leq \int \langle T(x) - x, \nabla u(x) \rangle d\eta(x)$. Conversely, Santambrogio's derivative estimate gives $\left. \frac{d}{dt} \mathcal{C}_\alpha(\eta_t) \right|_{t=0^+} \geq \int \langle T(x) - x, \nabla u(x) \rangle d\eta(x)$, because $(\text{Id}, \nabla u)_\# \eta$ is optimal for the scalar-product problem. The two bounds coincide, so the first-order terms cancel. Hence the one-sided derivative of $\mathcal{H} + \mathcal{C}_\alpha$ at η in every such direction is zero. Displacement convexity implies global minimality, and approximation removes the smooth compact-support restriction. Strict displacement convexity of entropy gives uniqueness, except for translations; translations do not change $\mathcal{C}_\alpha$ because α is centered. The entropy term is displacement convex by McCann's theorem, recalled in Theorem 14.6. If α has finite second moment, identity (15.30) writes $\mathcal{C}_\alpha$ as the sum of the 1-convex moment term $\eta \mapsto \frac{1}{2} \int \|x\|^2 d\eta$ and the (-1) -convex term $\eta \mapsto -\frac{1}{2} \mathcal{W}_2^2(\eta, \alpha)$, hence $\mathcal{C}_\alpha$ is displacement convex. For a target with only a finite first moment, Santambrogio obtains the same convexity along $\mathcal{P}_2$ geodesics by approximation and proves the full variational characterization by lower semicontinuity. $\square$

Conjugate moment measures for generation.

$$\beta = (\nabla w^*)_\# \left(Z_w^{-1} e^{-w(z)} dz \right). \quad (15.32)$$

15.4 Evolution in Depth of Transformers

Attention as a context-dependent velocity. After tokenization, embedding, and positional encoding, each input is represented as a point cloud $(x_i)_{i=1}^n$ of n vectorized tokens. An attention layer with a skip connection and residual scale $1/T$, where T is the depth, transforms the tokens according to

$x_i \mapsto x_i + \frac{1}{T} \sum_j \frac{e^{\langle Qx_i, Kx_j \rangle} Vx_j}{\sum_\ell e^{\langle Qx_i, Kx_\ell \rangle}}$, where $\theta = (K, Q, V)$ denotes the three parameter matrices. The conventional factor $1/\sqrt{r}$, with r the query/key dimension, can be absorbed into Q or K and is omitted here.

Token measure evolution. To handle an arbitrary number of tokens, we define $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$ as the empirical measure of tokens and rewrite the transformer mapping as:

$$x_i \mapsto x_i + \frac{1}{T} \Gamma_\theta[\alpha](x_i), \quad \Gamma_\theta[\alpha](x) := \frac{\int e^{\langle Qx, Ky \rangle} Vy d\alpha(y)}{\int e^{\langle Qx, Ky \rangle} d\alpha(y)}. \quad \alpha_{t+\tau} = (\text{Id} + \tau \Gamma_{\theta_t}[\alpha_t])_\# \alpha_t. \quad \partial_t \alpha_t + \text{div}(\alpha_t \Gamma_{\theta_t}[\alpha_t]) = 0.$$

L^2 attention and mean shift.

$$K_\varepsilon(x, y) = \exp(-\|x - y\|^2 / (2\varepsilon)), \quad \rho_\varepsilon[\alpha](x) = \int K_\varepsilon(x, y) d\alpha(y), \quad m_\varepsilon[\alpha](x) = \frac{\int y K_\varepsilon(x, y) d\alpha(y)}{\rho_\varepsilon[\alpha](x)}.$$

$$M_\varepsilon[\alpha](x) := m_\varepsilon[\alpha](x) - x = \frac{\int (y - x) K_\varepsilon(x, y) d\alpha(y)}{\rho_\varepsilon[\alpha](x)} = \varepsilon \nabla \log \rho_\varepsilon[\alpha](x) \quad (15.33)$$

$$x_i^{k+1} = (1 - \tau) x_i^k + \tau m_\varepsilon[\alpha_k](x_i^k) = x_i^k + \tau M_\varepsilon[\alpha_k](x_i^k) \quad \partial_t \alpha_t + \text{div}(\alpha_t M_\varepsilon[\alpha_t]) = 0.$$

Consensus and Markov averaging.

$$M_K[\alpha](x) := \frac{\int (y - x) K(x, y) d\alpha(y)}{\int K(x, y) d\alpha(y)}. \quad (15.34)$$

For an empirical law $\alpha = \sum_{i=1}^n a_i \delta_{x_i}$, where $a_i > 0$ and $\sum_i a_i = 1$, let $X \in \mathbb{R}^{n \times d}$ have rows $x_i^\top$, set $\mathbf{a} = (a_i)_i$, and define

$$(K_X)_{ij} = K(x_i, x_j), \quad P_X := \text{diag}(K_X \mathbf{a})^{-1} K_X \text{diag}(\mathbf{a}). \quad (15.35)$$

$$\dot{x}_i = \frac{\sum_j a_j K(x_i, x_j)(x_j - x_i)}{\sum_j a_j K(x_i, x_j)}, \quad \text{or equivalently} \quad \dot{X} = (P_X - I_n)X. \quad (15.36)$$

Dobrushin contraction from Hilbert geometry. $d_{\mathcal{H}}(e^z, e^{z'}) = \|z - z'\|_V$. A row-stochastic matrix P is order preserving and satisfies $P(z + s\mathbb{1}_n) = Pz + s\mathbb{1}_n$. It is therefore a linear topical map, so Proposition 8.8 gives nonexpansiveness on the additive quotient.

Proposition 15.9 (Dobrushin contraction is the tangent form of Birkhoff contraction). *Let $P \in \mathbb{R}_+^{n \times n}$ be row-stochastic. Its Dobrushin coefficient is*

$$\delta(P) := \frac{1}{2} \max_{i, \ell} \sum_j |P_{ij} - P_{\ell j}| = 1 - \min_{i, \ell} \sum_j \min\{P_{ij}, P_{\ell j}\}. \quad (15.37)$$

It is the exact operator norm induced by $\|\cdot\|_V$ on $\mathbb{R}^n / \text{Span}(\mathbb{1}_n)$:

$$\delta(P) = \sup_{z \notin \text{Span}(\mathbb{1}_n)} \frac{\|Pz\|_V}{\|z\|_V}. \quad (15.38)$$

$$\|Pz\|_V \leq \delta(P) \|z\|_V \quad (z \in \mathbb{R}^n). \quad (15.39)$$

If $P > 0$, then this exact tangent factor is bounded by the Birkhoff coefficient of Theorem 8.18:

$$\delta(P) \leq \lambda(P) < 1. \quad (15.40)$$

Proof. Fix two rows $p = (P_{ij})_j$ and $p' = (P_{\ell j})_j$, and remove their common mass $r_j = \min\{p_j, p'_j\}$. The two residual nonnegative measures have the same mass $1 - \sum_j r_j = \frac{1}{2} \sum_j |p_j - p'_j|$. Their averages of z can differ by at most this mass times $\|z\|_V$. Maximizing over the two output rows proves the upper bound in (15.38). Conversely, for a pair of rows attaining the maximum, choose $z_j = 1$ when $p_j \geq p'_j$ and $z_j = 0$ otherwise. Then $\|z\|_V = 1$ and $(Pz)_i - (Pz)_\ell = \frac{1}{2} \sum_j |p_j - p'_j|$, proving equality and hence (15.39).

Assume now that $P > 0$. Apply Theorem 8.18 to $u_s = e^{sz}$ and $\mathbb{1}_n$. Since $P\mathbb{1}_n = \mathbb{1}_n$, $d_{\mathcal{H}}(Pe^{sz}, \mathbb{1}_n) \leq \lambda(P) d_{\mathcal{H}}(e^{sz}, \mathbb{1}_n) = \lambda(P) |s| \|z\|_V$. Moreover, $Pe^{sz} = \mathbb{1}_n + sPz + O(s^2)$, hence $\log(Pe^{sz}) = sPz + O(s^2)$. Divide by $|s|$ and let $s \rightarrow 0$ to obtain $\|Pz\|_V \leq \lambda(P) \|z\|_V$. Taking the exact quotient norm (15.38) yields (15.40). This Hilbert–Hopf–Dobrushin relation extends to Markov operators on general cones. $\square$

$\|\mathbf{P}^\top \mathbf{p} - \mathbf{P}^\top \mathbf{p}'\|_{\ell^1} \leq \delta(\mathbf{P})\|\mathbf{p} - \mathbf{p}'\|_{\ell^1}$. Thus (15.39) contracts observables modulo constants, while the adjoint inequality contracts probability laws in total variation. The proposition also separates two useful constants. The same coefficient applies to the measure-valued dynamics. For $\alpha \in \mathcal{P}(C)$, where $C \subset \mathbb{R}^d$ is compact, define the Markov operator

$$(\mathbf{P}_\alpha h)(x) := \int_C h(y) p_{\alpha,x}(y) d\alpha(y), \quad p_{\alpha,x}(y) := \frac{K(x,y)}{Z_\alpha(x)}, \quad Z_\alpha(x) := \int_C K(x,z) d\alpha(z). \quad (15.41)$$

$$\delta(\mathbf{P}_\alpha) := \frac{1}{2} \sup_{x,x' \in \text{supp } \alpha} \int_C |p_{\alpha,x}(y) - p_{\alpha,x'}(y)| d\alpha(y). \quad (15.42)$$

$$\|\mathbf{P}_\alpha h\|_{V, \text{supp } \alpha} \leq \delta(\mathbf{P}_\alpha) \|h\|_{V, \text{supp } \alpha}.$$

$$\eta_K(C) := \sup_{x,x',y,y' \in C} \frac{K(x,y)K(x',y')}{K(x',y)K(x,y')}, \quad \lambda_K(C) := \frac{\sqrt{\eta_K(C)} - 1}{\sqrt{\eta_K(C)} + 1}. \quad (15.43)$$

$$\frac{(\mathbf{P}_X)_{ij}(\mathbf{P}_X)_{\ell r}}{(\mathbf{P}_X)_{ir}(\mathbf{P}_X)_{\ell j}} = \frac{K(x_i, x_j)K(x_\ell, x_r)}{K(x_i, x_r)K(x_\ell, x_j)}. \quad (15.44)$$

$$\delta(\mathbf{P}_X) \leq \lambda(\mathbf{P}_X) = \lambda(K_X) \leq \lambda_K(C). \quad (15.45)$$

$$\bar{\delta}_K(C) := \sup_{\alpha \in \mathcal{P}(C)} \delta(\mathbf{P}_\alpha) \leq \lambda_K(C) < 1. \quad (15.46)$$

$$\lambda_K(C) \leq \frac{k_+ - k_-}{k_+ + k_-}. \quad (15.47)$$

Theorem 15.10 (Dobrushin consensus for positive mean shift). *Let $\alpha_0 \in \mathcal{P}(\mathbb{R}^d)$ have compact support, and set $C_0 = \text{conv}(\text{supp } \alpha_0)$, $D_0 = \text{diam}(\text{supp } \alpha_0)$, $\bar{\delta} = \bar{\delta}_K(C_0) < 1$. Assume that $K : C_0 \times C_0 \rightarrow (0, +\infty)$ is Lipschitz. For an empirical initial measure $\alpha_0 = \sum_i a_i \delta_{x_i^0}$, let X^k follow the damped blurring iteration $X^{k+1} = ((1-\tau)I_n + \tau \mathbf{P}_{X^k})X^k$, $0 < \tau \leq 1$, define $\delta_k = \delta(\mathbf{P}_{X^k})$, $q_k = 1 - \tau(1 - \delta_k)$, $q_\tau = 1 - \tau(1 - \bar{\delta}) < 1$. Then*

$$D(X^k) := \max_{i,j} \|x_i^k - x_j^k\| \leq D_0 \prod_{r=0}^{k-1} q_r \leq q_\tau^k D_0. \quad (15.48)$$

All particles converge to a common $x_\infty^d \in C_0$, and $\max_i \|x_i^k - x_\infty^d\| \leq \frac{D_0}{1-\bar{\delta}} q_\tau^k$. The characteristic solution of

$$\partial_t \alpha_t + \text{div}(\alpha_t M_K[\alpha_t]) = 0, \quad \alpha_{t=0} = \alpha_0, \quad (15.49)$$

is globally defined and has nested convex hulls, $\text{conv}(\text{supp } \alpha_t) \subset \text{conv}(\text{supp } \alpha_s)$ ($t \geq s \geq 0$). Writing $D(t) = \text{diam}(\text{supp } \alpha_t)$, one has the adaptive estimate

$$D(t) \leq D(s) \exp\left(-\int_s^t [1 - \delta(\mathbf{P}_{\alpha_r})] dr\right) \leq e^{-(1-\bar{\delta})(t-s)} D(s) \quad (t \geq s \geq 0). \quad (15.50)$$

$$\mathcal{W}_\infty(\alpha_t, \delta_{x_\infty^c}) \leq \frac{D_0}{1-\bar{\delta}} e^{-(1-\bar{\delta})t}. \quad (15.51)$$

Proof. Because K is positive and Lipschitz on the compact set $C_0 \times C_0$, its minimum is positive. Hence the denominator in (15.34) is uniformly bounded away from zero, and the characteristic vector field is uniformly Lipschitz in x and Lipschitz in α for $\mathcal{W}_1$. Moreover, $x + M_K[\alpha](x) = \mathbf{P}_\alpha \text{Id}(x)$ lies in $\text{conv}(\text{supp } \alpha)$. No characteristic can cross an outward supporting hyperplane of the current convex hull, so the convex hulls are nested and the dynamics is globally well posed.

For any compact set $S \subset \mathbb{R}^d$, the support-function identity

$$\text{diam}(S) = \sup_{\theta \in \mathbb{S}^{d-1}} \left(\sup_{x \in S} \langle \theta, x \rangle - \inf_{x \in S} \langle \theta, x \rangle \right) \quad (15.52)$$

reduces Euclidean diameter contraction to scalar variation contraction. For a finite configuration, set $z_\theta = X\theta \in \mathbb{R}^n$. Proposition 15.9 gives

$$\begin{aligned} \|z_\theta^{k+1}\|_V &\leq (1-\tau)\|z_\theta^k\|_V + \tau\|\mathbf{P}_{X^k} z_\theta^k\|_V \\ &\leq [1 - \tau(1 - \delta(\mathbf{P}_{X^k}))]\|z_\theta^k\|_V \leq q_\tau \|z_\theta^k\|_V. \end{aligned}$$

Taking the supremum over θ gives the one-step factor q_k ; multiplying these factors and using $q_k \leq q_\tau$ proves (15.48). Since $\|x_i^{k+1} - x_i^k\| \leq \tau D(X^k)$, every particle path is Cauchy; the vanishing diameter forces a common limit, and summing the uniform geometric tail yields the displayed pointwise estimate.

For the measure-valued flow, let $f_\theta(x) = \langle \theta, x \rangle$ and denote by $w_\theta(t) = \|f_\theta\|_{V, \text{supp } \alpha_t}$ the width of the support in direction θ . The upper Dini derivative of the extrema along characteristics satisfies

$$D^+ w_\theta(t) \leq \|\mathbf{P}_{\alpha_t} f_\theta\|_{V, \text{supp } \alpha_t} - w_\theta(t) \leq -[1 - \delta(\mathbf{P}_{\alpha_t})] w_\theta(t).$$

Gronwall's lemma, followed by the support-function identity (15.52), proves (15.50). Finally, every characteristic satisfies $\|\dot{X}_t\| \leq D(t)$. It is therefore Cauchy, all characteristic limits coincide because $D(t) \rightarrow 0$, and integrating the uniform exponential bound from t to $+\infty$ gives (15.51). $\square$

For the Gaussian kernel $K_\varepsilon(x, y) = e^{-\|x-y\|^2/(2\varepsilon)}$, the cross-ratio has the closed form $\log \frac{K_\varepsilon(x, y)K_\varepsilon(x', y')}{K_\varepsilon(x', y)K_\varepsilon(x, y')} = \frac{\langle x-x', y-y' \rangle}{\varepsilon}$. Hence, for a compact convex set C of diameter D ,

$$\eta_{K_\varepsilon}(C) = e^{D^2/\varepsilon}, \quad \lambda_{K_\varepsilon}(C) = \tanh\left(\frac{D^2}{4\varepsilon}\right). \quad (15.53)$$

$$D(X^{k+1}) \leq \tanh\left(\frac{D(X^k)^2}{4\varepsilon}\right) D(X^k) \leq \frac{D(X^k)^3}{4\varepsilon},$$

$$D(t) \leq D(s) \exp\left(-\int_s^t \left[1 - \tanh\left(\frac{D(r)^2}{4\varepsilon}\right)\right] dr\right). \quad (15.54)$$

Scope of the consensus result. The projective argument also clarifies what is not conserved and when clustering replaces consensus. Because row normalization makes the pairwise coefficients asymmetric, the consensus point need not equal the barycenter of α_0 ; that barycenter is conserved for the constant kernel, but not in general.

Gradient structure and limitations. When the token space has dimension d and the query/key space has dimension r , take $Q, K \in \mathbb{R}^{r \times d}$ and $V \in \mathbb{R}^{d \times d}$. If $V = Q^\top K$, the field $\Gamma_\theta[\alpha]$ is a gradient vector field in the token variable. Indeed, define the log-partition potential

$\Phi_\alpha(x) = \int \exp(\langle Qx, Ky \rangle) d\alpha(y)$, $U_\alpha(x) = \log \Phi_\alpha(x)$. $\nabla_x U_\alpha(x) = \frac{\int Q^\top Ky \exp(\langle Qx, Ky \rangle) d\alpha(y)}{\int \exp(\langle Qx, Ky \rangle) d\alpha(y)} = \Gamma_\theta[\alpha](x)$. $f_{\text{att}}(\alpha) = \int U_\alpha(x) d\alpha(x)$. $\delta f_{\text{att}}(\alpha)(z) = U_\alpha(z) + \int \frac{\exp(\langle Qx, Kz \rangle)}{\Phi_\alpha(x)} d\alpha(x)$, up to an additive constant. The second term is the response of every query normalization to a perturbation of the key distribution, and its spatial gradient is absent from $\Gamma_\theta[\alpha]$.

15.5 Flows over the Gaussian Manifold

Gaussianity preservation.

Proposition 15.11 (Affine velocities preserve Gaussianity). *Let $\alpha_0 = \mathcal{N}(m_0, \Sigma_0)$, with $\Sigma_0 \succ 0$. Let $b_t \in \mathbb{R}^d$ and $A_t \in \mathbb{R}^{d \times d}$ be locally integrable on a time interval, and let (m_t, Σ_t) solve $\dot{m}_t = b_t$, $\dot{\Sigma}_t = A_t \Sigma_t + \Sigma_t A_t^\top$, $(m_{t=0}, \Sigma_{t=0}) = (m_0, \Sigma_0)$. Then, as long as this matrix ODE is defined, $\Sigma_t \succ 0$ and $\alpha_t = \mathcal{N}(m_t, \Sigma_t)$ is the solution of the continuity equation $\partial_t \alpha_t + \text{div}(\alpha_t v_t) = 0$, $v_t(x) = b_t + A_t(x - m_t)$. In particular, if a smooth functional f has a Wasserstein gradient on Gaussian measures of the affine form $\nabla_{\mathcal{W}} f(\mathcal{N}(m, \Sigma))(x) = b_f(m, \Sigma) + A_f(m, \Sigma)(x - m)$, with $A_f(m, \Sigma)$ symmetric, then the Wasserstein gradient flow of f , initialized from any non-degenerate Gaussian, stays Gaussian and satisfies $\dot{m}_t = h_f(m_t, \Sigma_t)$, $\dot{\Sigma}_t = H_f(m_t, \Sigma_t)$, where $h_f(m, \Sigma) = -b_f(m, \Sigma)$, $H_f(m, \Sigma) = -(A_f(m, \Sigma)\Sigma + \Sigma A_f(m, \Sigma))$. Conversely, fix a non-degenerate Gaussian $\mathcal{N}(m, \Sigma)$. Suppose that a finite-energy Wasserstein tangent field $V_{m, \Sigma}$ is represented by a gradient in $L^2(\mathcal{N}(m, \Sigma))$ and is tangent to the Gaussian manifold at this Gaussian. Then there exist $b(m, \Sigma)$ and a symmetric matrix $A(m, \Sigma)$ such that $V_{m, \Sigma}(x) = b(m, \Sigma) + A(m, \Sigma)(x - m)$ $\mathcal{N}(m, \Sigma)$ -a.e. Without the Wasserstein gradient representative, this converse is false because one may add velocity fields with zero $\mathcal{N}(m, \Sigma)$ -weighted divergence. Finally, any smooth Gaussian curve with positive definite covariance can be generated by an affine velocity. If one wants the velocity to be a Wasserstein tangent gradient, one chooses the unique symmetric solution of the Lyapunov equation $A_t \Sigma_t + \Sigma_t A_t = \dot{\Sigma}_t$.*

Proof. Let X_t follow the characteristic ODE $\dot{X}_t = b_t + A_t(X_t - m_t)$ with $X_0 \sim \mathcal{N}(m_0, \Sigma_0)$. Since $\dot{m}_t = b_t$, the centered variable $\tilde{X}_t = X_t - m_t$ solves the homogeneous linear ODE $\dot{\tilde{X}}_t = A_t \tilde{X}_t$. If Φ_t is the fundamental matrix $\dot{\Phi}_t = A_t \Phi_t$, $\Phi_{t=0} = \text{Id}$, then $X_t = m_t + \Phi_t(X_0 - m_0)$, $\Sigma_t = \Phi_t \Sigma_0 \Phi_t^\top$. Hence X_t is Gaussian and $\Sigma_t \succ 0$, and $\dot{\Sigma}_t = \frac{d}{dt} \mathbb{E}(\tilde{X}_t \tilde{X}_t^\top) = A_t \Sigma_t + \Sigma_t A_t^\top$. This proves Gaussian preservation and the moment ODE. The Wasserstein gradient-flow statement follows by inserting the descent velocity $v_t = -\nabla_{\mathcal{W}} f(\alpha_t)$.

For the converse, fix $\alpha = \mathcal{N}(m, \Sigma)$ and denote its density by ρ . Tangency to the Gaussian manifold means that the density variation $-\text{div}(\rho V)$ is generated by some moment variation $(\dot{m}, \dot{\Sigma})$, with $\dot{\Sigma}$ symmetric. Set $b = \dot{m}$, and let $A = A^\top$ be the unique solution of $A\Sigma + \Sigma A = \dot{\Sigma}$. By the first part of the proposition, the affine gradient field $V_0(x) = b + A(x - m)$ generates exactly the same infinitesimal Gaussian variation. Hence $\text{div}(\rho(V - V_0)) = 0$ in the distributional sense. Both V and V_0 belong to the $L^2(\alpha)$ closure of gradient fields. The weighted Helmholtz decomposition therefore makes $V - V_0$ simultaneously a tangent gradient and orthogonal to every tangent gradient, so $V = V_0$ in $L^2(\alpha)$. Equivalently, for a smooth representative $V - V_0 = \nabla \psi$, integration by parts gives $\int \|\nabla \psi\|^2 d\alpha = 0$. This proves that the Wasserstein tangent representative is affine. The qualification is essential: without selecting the gradient representative, one can add a nonzero field with zero α -weighted divergence without changing the Gaussian curve.

For a prescribed smooth Gaussian curve, set $b_t = \dot{m}_t$ and choose any matrix A_t satisfying $A_t \Sigma_t + \Sigma_t A_t^\top = \dot{\Sigma}_t$. Since Σ_t is positive definite, the Lyapunov map $A \mapsto A\Sigma_t + \Sigma_t A$ is invertible on symmetric matrices, which gives the unique symmetric choice when a gradient velocity is required. In that case v_t is the gradient of the quadratic potential $x \mapsto \langle b_t, x \rangle + \langle A_t(x - m_t), x - m_t \rangle/2$. $\square$

Gaussian-preserving gradient flows.

Proposition 15.12 (Gaussian closure catalogue). *Let $\gamma = \mathcal{N}(\bar{m}, \bar{\Sigma})$ on $\mathbb{R}^d$, with $\bar{\Sigma} \succ 0$, and let the initial condition be $\alpha_0 = \mathcal{N}(m_0, \Sigma_0)$, with $\Sigma_0 \succ 0$. For each functional displayed below, let α_t be the usual Wasserstein gradient flow on $\mathcal{P}_2(\mathbb{R}^d)$, initialized at α_0 . Then the Gaussian family is invariant for these dynamics: as long as the solution exists and $\Sigma_t \succ 0$, $\alpha_t = \mathcal{N}(m_t, \Sigma_t)$, and the mean and covariance satisfy $\dot{m}_t = h(m_t, \Sigma_t)$, $\dot{\Sigma}_t = H(m_t, \Sigma_t)$. Write $\delta_m = m - \bar{m}$, $A = \bar{\Sigma}^{-1}$, and $M_{\Sigma, \bar{\Sigma}} := \Sigma^{-1/2}(\Sigma^{1/2} \bar{\Sigma} \Sigma^{1/2})^{1/2} \Sigma^{-1/2}$. With the normalizations displayed in the first column, the mean vector field h and covariance vector field H are listed in the following table. Gradients with respect to Σ are symmetric Frobenius gradients on the cone of covariance matrices.*

Gaussian-preserving Wasserstein gradient flows.

Here, in the MMD row, $R = \Sigma + m m^\top - \bar{\Sigma} - \bar{m} \bar{m}^\top$. The debiased Sinkhorn row uses the notation of Corollary 8.22: for Gaussian inputs, $\tilde{\mathcal{L}}_{\parallel, \dots, \parallel}^\varepsilon(\alpha, \gamma) = \|\delta_m\|^2 + \mathcal{B}_\varepsilon(\Sigma, \bar{\Sigma})^2$, with the closed-form covariance gradient

$$G_\varepsilon(\Sigma, \bar{\Sigma}) = \tau_\varepsilon(\Sigma) - \bar{\Sigma}^{1/2} \tau_\varepsilon(B_{\Sigma, \bar{\Sigma}}^{1/2}) B_{\Sigma, \bar{\Sigma}}^{-1/2} \bar{\Sigma}^{1/2}, \quad B_{\Sigma, \bar{\Sigma}} = \bar{\Sigma}^{1/2} \Sigma \bar{\Sigma}^{1/2}.$$

Here τ_ε is the scalar function $\tau_\varepsilon(r) := \frac{\sqrt{\varepsilon^2 + 16r^2} - \varepsilon}{4r}$, $r > 0$, applied to positive matrices by spectral calculus. Equivalently, for $M \succ 0$, $\tau_\varepsilon(M) = (\sqrt{\varepsilon^2 \text{Id} + 16M^2} - \varepsilon \text{Id})(4M)^{-1}$. With this convention, $G_\varepsilon = \nabla_\Sigma \mathcal{B}_\varepsilon(\Sigma, \bar{\Sigma})^2$. In the sliced row, σ is the normalized spherical measure on $\mathbb{S}^{d-1}$, and

$$G_{\text{sw}}(\Sigma, \bar{\Sigma}) = \int_{\mathbb{S}^{d-1}} \left(1 - \sqrt{\frac{\theta^\top \bar{\Sigma} \theta}{\theta^\top \Sigma \theta}} \right) \theta \theta^\top d\sigma(\theta).$$

Here

$$\mathcal{I}(\alpha|\gamma) = \int \left| \nabla \log \frac{\rho(x)}{\rho_\gamma(x)} \right|^2 \rho(x) dx = \int |\nabla \log \rho(x) + A(x - \bar{m})|^2 \rho(x) dx \quad (\alpha = \rho dx),$$

where ρ_γ is the density of γ .

Proof. For the moment-functional row, the formula is exact on the full Wasserstein space. Indeed, write $m_\alpha = \int x d\alpha(x)$, $\Sigma_\alpha = \int (x - m_\alpha)(x - m_\alpha)^\top d\alpha(x)$. If η is a signed perturbation with zero total mass, then $dm_\alpha[\eta] = \int x d\eta(x)$, $d\Sigma_\alpha[\eta] = \int (x - m_\alpha)(x - m_\alpha)^\top d\eta(x)$. Thus, up to an irrelevant additive constant, the first variation is $\delta f(\alpha)(x) = \langle \nabla_m g, x \rangle + \langle \nabla_\Sigma g, (x - m_\alpha)(x - m_\alpha)^\top \rangle$, and its spatial gradient is the affine field $\nabla_x \delta f(\alpha)(x) = \nabla_m g + 2\nabla_\Sigma g (x - m_\alpha)$.

For the quadratic potential row, the ambient first variation is $\delta f(\alpha)(x) = \frac{1}{2} x^\top Bx + \langle \ell, x \rangle$, $\nabla_x \delta f(\alpha)(x) = Bm + \ell + B(x - m)$, which gives the displayed affine velocity. For the quadratic interaction row, the ambient first variation is $\delta f(\alpha)(x) = \frac{1}{2} \int (x - y)^\top G(x - y) d\alpha(y)$, $\nabla_x \delta f(\alpha)(x) = G(x - m_\alpha)$. These first variations are quadratic in x , hence the affine Gaussian flow coincides with the full Wasserstein gradient flow.

For the KL row, write $\alpha = \rho dx$ and let ρ_γ denote the density of γ . The ambient first variation is $\log(\rho/\rho_\gamma)$ up to an additive constant. If $\alpha = \mathcal{N}(m, \Sigma)$, then $\nabla \log(\rho/\rho_\gamma)(x) = -\Sigma^{-1}(x - m) + A(x - \bar{m})$, so the descent velocity is $v(x) = -A\delta_m + (\Sigma^{-1} - A)(x - m)$. Proposition 15.11 gives $h(m, \Sigma) = -A\delta_m$ and $H(m, \Sigma) = 2\text{Id} - \Sigma A - A\Sigma$. For the Fisher row, writing $u = \log(\rho/\rho_\gamma)$, the ambient first variation of $\frac{1}{2}\mathcal{I}(\alpha|\gamma)$ is, up to constants, $-\Delta u - \frac{1}{2}|\nabla u|^2 - \langle \nabla u, \nabla \log \rho_\gamma \rangle$. For Gaussian ρ and Gaussian γ , this is quadratic in x . Multiplying by two for $\mathcal{I}(\alpha|\gamma)$ gives the descent velocity $v(x) = -2A^2\delta_m + 2(\Sigma^{-2} - A^2)(x - m)$. Hence the Wasserstein velocity is affine and the Gaussian family is closed, with the displayed h and H . Although the ambient Fisher flow is a fourth-order PDE for a generic density, it is therefore an exact finite-dimensional closure in this quadratic-reference case.

For $\mathcal{W}_2^2(\alpha, \gamma)$, the Brenier map from $\mathcal{N}(m, \Sigma)$ to γ is $T(x) = \bar{m} + M_{\Sigma, \bar{\Sigma}}(x - m)$. The descent velocity for the unhalved squared distance is $2(T - \text{Id})$, which gives the mean and covariance equations through Proposition 15.11. For the MMD row, $k(x, y) = \langle x, y \rangle^2$ identifies the kernel mean embedding with the raw second moment. Thus $\text{MMD}_k^2(\alpha, \gamma) = \|\Sigma + m m^\top - \bar{\Sigma} - \bar{m} \bar{m}^\top\|_F^2 = \|R\|_F^2$, and the ambient first variation is $2x^\top R x$, whose descent velocity is $-4R x$.

For the Sinkhorn row, Corollary 8.22 gives the closed Gaussian formula as a squared mean displacement plus the smoothed Bures term $\mathcal{B}_\varepsilon^2$. This is not only a formula on the Gaussian submanifold. The first variation of the entropic value with respect to the first marginal is a Sinkhorn dual potential, and for Gaussian marginals the cross and self dual potentials are quadratic. Their debiased combination is therefore quadratic, so its spatial gradient is affine. Since the restriction of the functional to Gaussians is $(m, \Sigma) \mapsto \|m - \bar{m}\|^2 + \mathcal{B}_\varepsilon(\Sigma, \bar{\Sigma})^2$, the quadratic coefficient is $G_\varepsilon = \nabla_\Sigma \mathcal{B}_\varepsilon(\Sigma, \bar{\Sigma})^2$, and the moment-functional computation gives the displayed h and H . To compute this gradient, set $B_{\Sigma, \bar{\Sigma}} = \bar{\Sigma}^{1/2} \Sigma \bar{\Sigma}^{1/2}$. Since $\psi'_\varepsilon(r) = -2\tau_\varepsilon(r)$, differentiating $\text{tr} \psi_\varepsilon(B_{\Sigma, \bar{\Sigma}}^{1/2})$ gives $-\bar{\Sigma}^{1/2} \tau_\varepsilon(B_{\Sigma, \bar{\Sigma}}^{1/2}) B_{\Sigma, \bar{\Sigma}}^{-1/2} \bar{\Sigma}^{1/2}$, whereas differentiating the self term $-\frac{1}{2} \text{tr} \psi_\varepsilon(\Sigma)$ gives $\tau_\varepsilon(\Sigma)$. This proves the displayed closed form for G_ε . When $\bar{\Sigma} = \text{Id}$, this covariance contribution is a spectral function of Σ . If λ is an eigenvalue of Σ , the corresponding scalar velocity is $4\sqrt{\lambda + \varepsilon^2/16} - 4\sqrt{\lambda^2 + \varepsilon^2/16}$, which gives the displayed covariance-eigenvalue equation by functional calculus.

For sliced Wasserstein, the exact ambient Wasserstein gradient is obtained by averaging the one-dimensional gradients. Each projection is Gaussian:

$$(P_\theta)_\# \alpha = \mathcal{N}(\langle \theta, m \rangle, \theta^\top \Sigma \theta), \quad (P_\theta)_\# \gamma = \mathcal{N}(\langle \theta, \bar{m} \rangle, \theta^\top \bar{\Sigma} \theta).$$

Hence

$$\text{SW}_2^2(\alpha, \gamma) = \int_{\mathbb{S}^{d-1}} \left[\langle \theta, \delta_m \rangle^2 + \left(\sqrt{\theta^\top \Sigma \theta} - \sqrt{\theta^\top \bar{\Sigma} \theta} \right)^2 \right] d\sigma(\theta).$$

If T_θ is the monotone transport from $(P_\theta)_\# \alpha$ to $(P_\theta)_\# \gamma$, then the descent velocity for the unhalved sliced objective is $2 \int_{\mathbb{S}^{d-1}} (T_\theta(P_\theta x) - P_\theta x) \theta d\sigma(\theta)$. For Gaussian marginals, T_θ is affine. Thus this velocity is affine in x , and Proposition 15.11 applies. The spherical identity $\int \theta \theta^\top d\sigma(\theta) = \text{Id}/d$ gives the mean equation, and differentiating the covariance term gives G_{sw} . $\square$

One-dimensional case. Fix a Gaussian target $\gamma = \mathcal{N}(\bar{m}, \bar{\sigma}^2)$, with $\bar{\sigma} > 0$, and write $\alpha = \mathcal{N}(m, \sigma^2)$, with $\sigma > 0$. By Proposition 2.35, the parametrization $(m, \sigma) \mapsto \mathcal{N}(m, \sigma^2)$ identifies the one-dimensional Gaussian family with the open Euclidean half-plane.

$g_f(m, \sigma) := f(\mathcal{N}(m, \sigma^2))$. Set $\delta = m - \bar{m}$. For the Sinkhorn row below, with $\varepsilon > 0$, abbreviate

$$\psi_\varepsilon(r) := -\frac{\sqrt{\varepsilon^2 + 16r^2} - \varepsilon}{2} - \frac{\varepsilon}{2} \log \left(\frac{2\varepsilon}{\sqrt{\varepsilon^2 + 16r^2} + \varepsilon} \right), \quad r > 0.$$

Constrained evolution on the Gaussian manifold. $\mathcal{G} = \{\mathcal{N}(m, \Sigma) : m \in \mathbb{R}^d, \Sigma \succ 0\}$ be the Gaussian submanifold of $\mathcal{P}_2(\mathbb{R}^d)$. The Wasserstein gradient of a functional constrained to a smooth submanifold $\mathcal{M} \subset \mathcal{P}_2$ is defined as the Riesz representative of the differential restricted to tangent velocities of $\mathcal{M}$.

$\alpha^{k+1} \in \argmin_{\alpha \in \mathcal{M}} \frac{1}{2\tau} \mathcal{W}_2^2(\alpha, \alpha^k) + f(\alpha)$. For $\mathcal{M} = \mathcal{G}$, tangent velocities are affine gradient fields $v(x) = b + A(x - m)$ with $A = A^\top$.

The constrained gradient is therefore the $L^2(\mathcal{N}(m, \Sigma))$ projection of the ambient Wasserstein gradient onto this finite-dimensional affine space, whenever the ambient gradient exists.

Proposition 15.13 (Gaussian-constrained Wasserstein gradients). *Let f be a smooth functional and assume that its restriction to nondegenerate Gaussian measures can be written as $f(\mathcal{N}(m, \Sigma)) = F(m, \Sigma)$. Then the Wasserstein gradient constrained to the Gaussian family is the affine vector field $v_F(x) = \nabla_m F(m, \Sigma) + 2\nabla_\Sigma F(m, \Sigma)(x - m)$, where $\nabla_\Sigma F$ denotes the symmetric*

matrix derivative. Equivalently, v_F is the $L^2(\mathcal{N}(m, \Sigma))$ projection of the ambient Wasserstein gradient onto affine gradient fields, whenever the ambient gradient exists. Hence the gradient descent flow constrained to Gaussian measures satisfies

$$\dot{m}_t = -\nabla_m F(m_t, \Sigma_t), \quad \dot{\Sigma}_t = -2(\Sigma_t \nabla_\Sigma F(m_t, \Sigma_t) + \nabla_\Sigma F(m_t, \Sigma_t) \Sigma_t), \quad (15.55)$$

and the descent velocity is affine.

Proof. Test the functional along a Gaussian tangent vector, represented by an affine gradient field $v(x) = b + A(x - m)$ with A symmetric. The induced first-order variations are $\dot{m} = b$ and $\dot{\Sigma} = A\Sigma + \Sigma A$. Therefore

$$dF(m, \Sigma)[b, A\Sigma + \Sigma A] = \langle \nabla_m F, b \rangle + \text{tr}(\nabla_\Sigma F(A\Sigma + \Sigma A)).$$

Since A , Σ and $\nabla_\Sigma F$ are symmetric, the second term equals $2 \text{tr}(\nabla_\Sigma F A \Sigma) = \int \langle 2\nabla_\Sigma F(x - m), A(x - m) \rangle d\mathcal{N}(m, \Sigma)(x)$. Together with the mean term, this gives $dF(m, \Sigma)[\dot{m}, \dot{\Sigma}] = \int \langle v_F(x), v(x) \rangle d\mathcal{N}(m, \Sigma)(x)$ for all affine gradient fields v . This identifies the constrained Wasserstein gradient in the induced $L^2(\alpha)$ metric, or equivalently the projection of the ambient gradient when it exists. Substituting the descent velocity $-v_F$ in Proposition 15.11 gives (15.55). $\square$

Non-variational Gaussian-preserving flows.

Contractive Gaussian projection.

Theorem 15.14 (Gelbrich theorem). *For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, let $\text{Gauss}(\alpha) := \mathcal{N}(m_\alpha, \Sigma_\alpha)$ be the Gaussian with the same mean and covariance as α . Then, for every $\beta \in \mathcal{P}_2(\mathbb{R}^d)$, $\mathcal{W}_2^2(\text{Gauss}(\alpha), \text{Gauss}(\beta)) = \|m_\alpha - m_\beta\|^2 + \mathcal{B}^2(\Sigma_\alpha, \Sigma_\beta) \leq \mathcal{W}_2^2(\alpha, \beta)$.*

Proof. Take any coupling (X, Y) of α and β , center the variables, and write $C = \mathbb{E}[(X - m_\alpha)(Y - m_\beta)^\top]$. In the positive definite case, positivity of the block covariance matrix implies the factorization $C = \Sigma_\alpha^{1/2} K \Sigma_\beta^{1/2}$ with $\|K\|_{\text{op}} \leq 1$, and therefore, by operator/nuclear norm duality,

$$\text{tr } C \leq \text{tr} \left((\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2})^{1/2} \right).$$

The semidefinite case follows by adding ηId to both covariance matrices and letting $\eta \downarrow 0$. Expanding $\mathbb{E}\|X - Y\|^2$ gives the lower bound $\mathbb{E}\|X - Y\|^2 \geq \|m_\alpha - m_\beta\|^2 + \mathcal{B}^2(\Sigma_\alpha, \Sigma_\beta)$. Taking the infimum over couplings proves the inequality, while equality for Gaussian laws is Proposition 2.35. $\square$

Theorem 15.15 (Hugo Lavenant Gaussian-preservation criterion). *Let $f : \mathcal{P}_2(\mathbb{R}^d) \rightarrow (-\infty, +\infty]$ satisfy $f(\text{Gauss}(\alpha)) \leq f(\alpha) \quad \forall \alpha \in \mathcal{P}_2(\mathbb{R}^d)$, with Gauss defined in Theorem 15.14. If γ is Gaussian and α^* minimizes the JKO step $\eta \mapsto f(\eta) + \frac{1}{2\tau} \mathcal{W}_2^2(\gamma, \eta)$, then $\text{Gauss}(\alpha^*)$ is also a minimizer. If this JKO minimizer is unique, it is Gaussian.*

Proof. For the JKO claim, $\text{Gauss}(\gamma) = \gamma$ because γ is Gaussian. Hence, for any competitor η , $f(\text{Gauss}(\eta)) + \frac{1}{2\tau} \mathcal{W}_2^2(\gamma, \text{Gauss}(\eta)) \leq f(\eta) + \frac{1}{2\tau} \mathcal{W}_2^2(\gamma, \eta)$. Applying this to a minimizer $\eta = \alpha^*$ shows that $\text{Gauss}(\alpha^*)$ is again a minimizer. Uniqueness forces $\alpha^* = \text{Gauss}(\alpha^*)$. $\square$